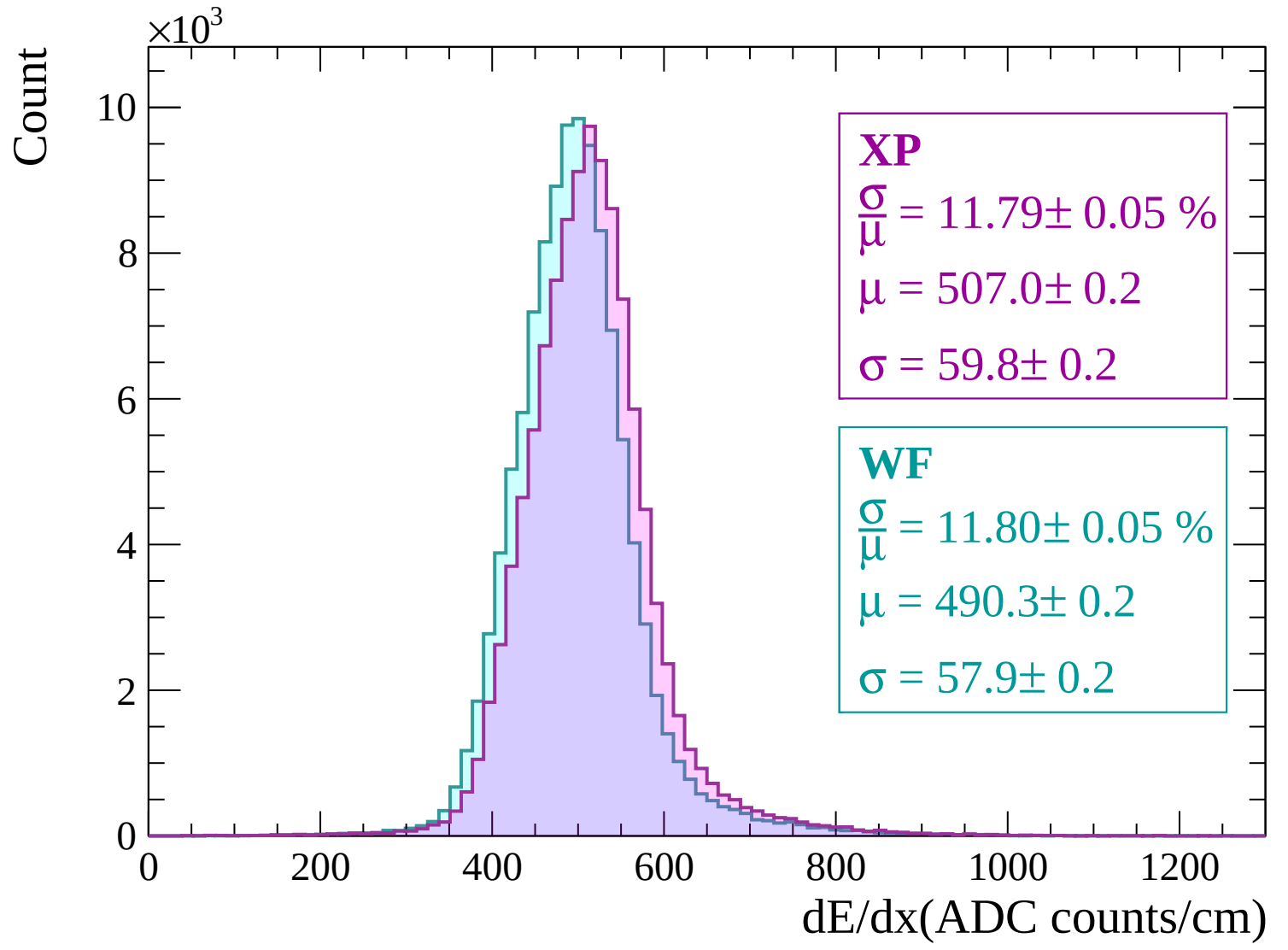
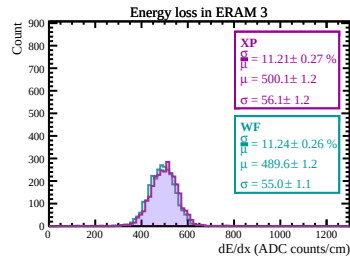
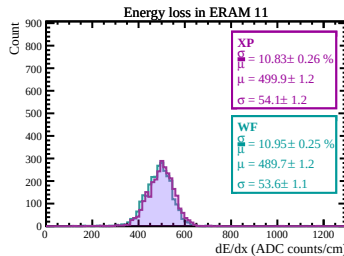
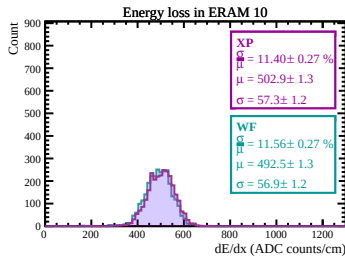
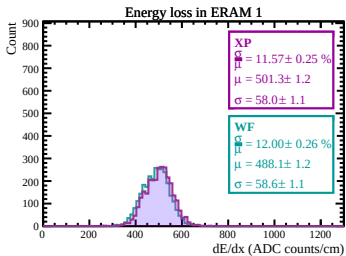
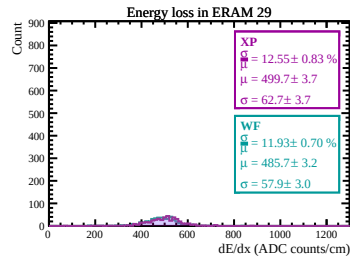
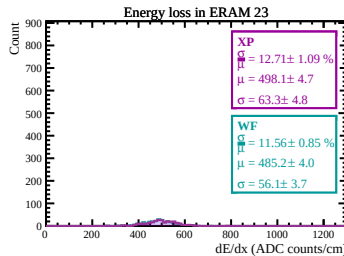
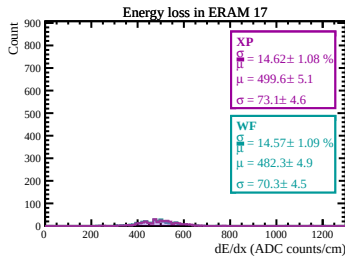
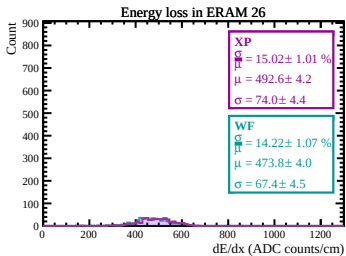
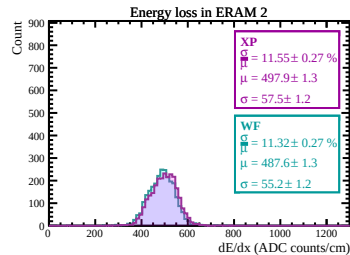
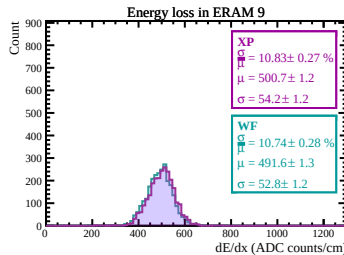
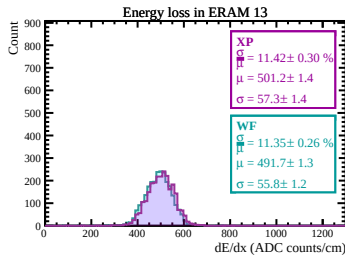
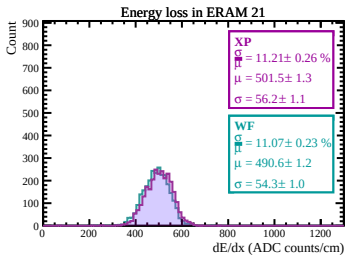
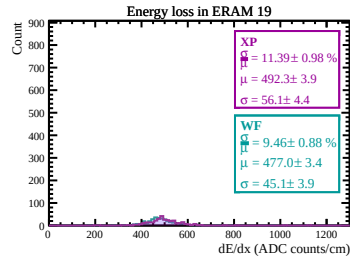
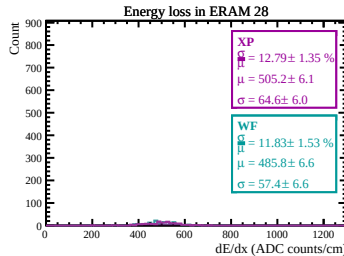
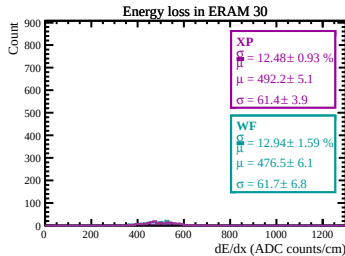
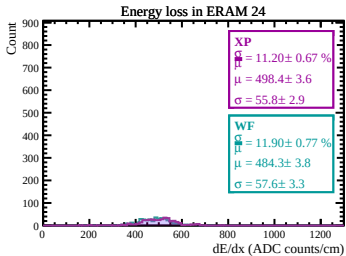
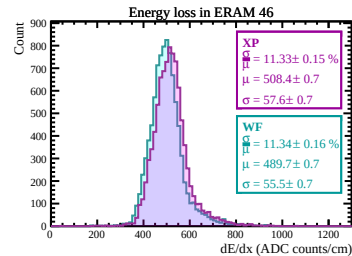
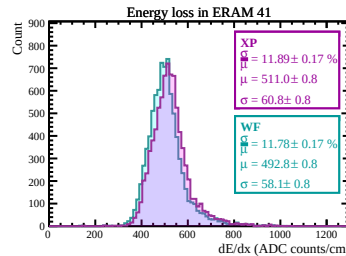
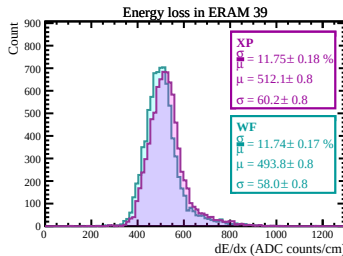
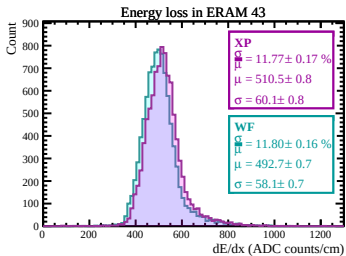
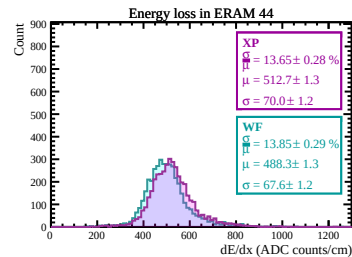
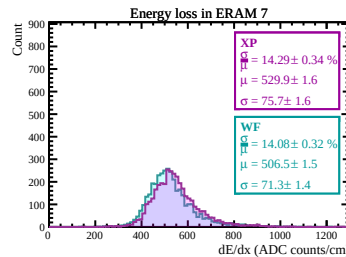
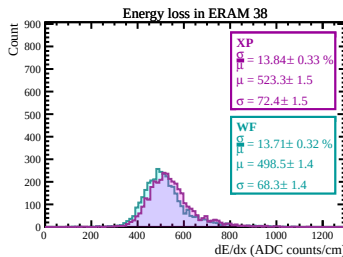
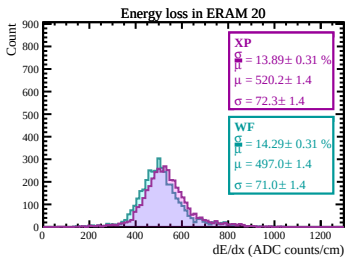
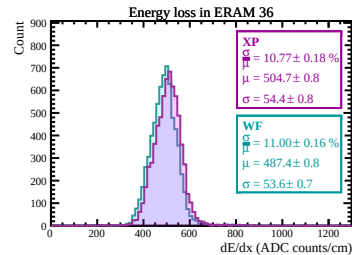
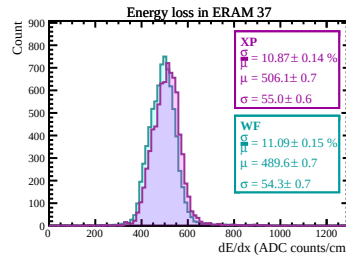
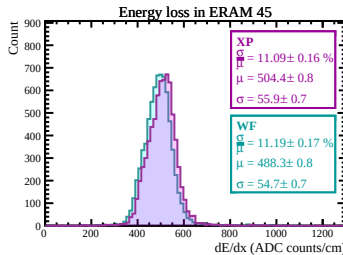
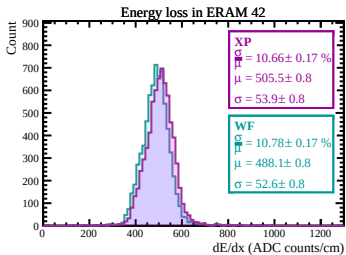
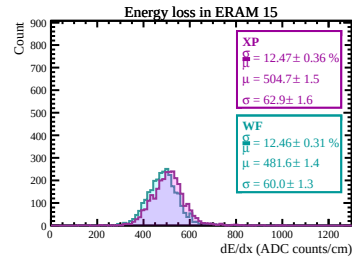
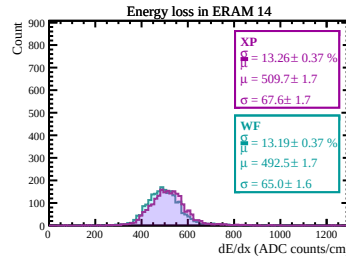
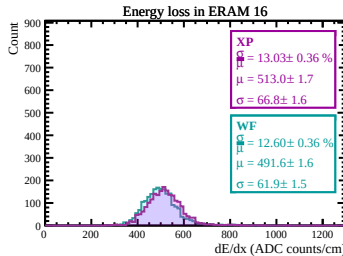
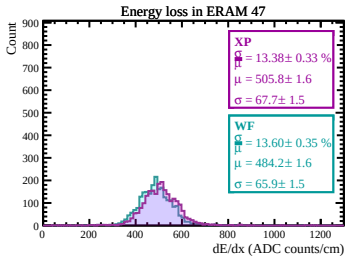
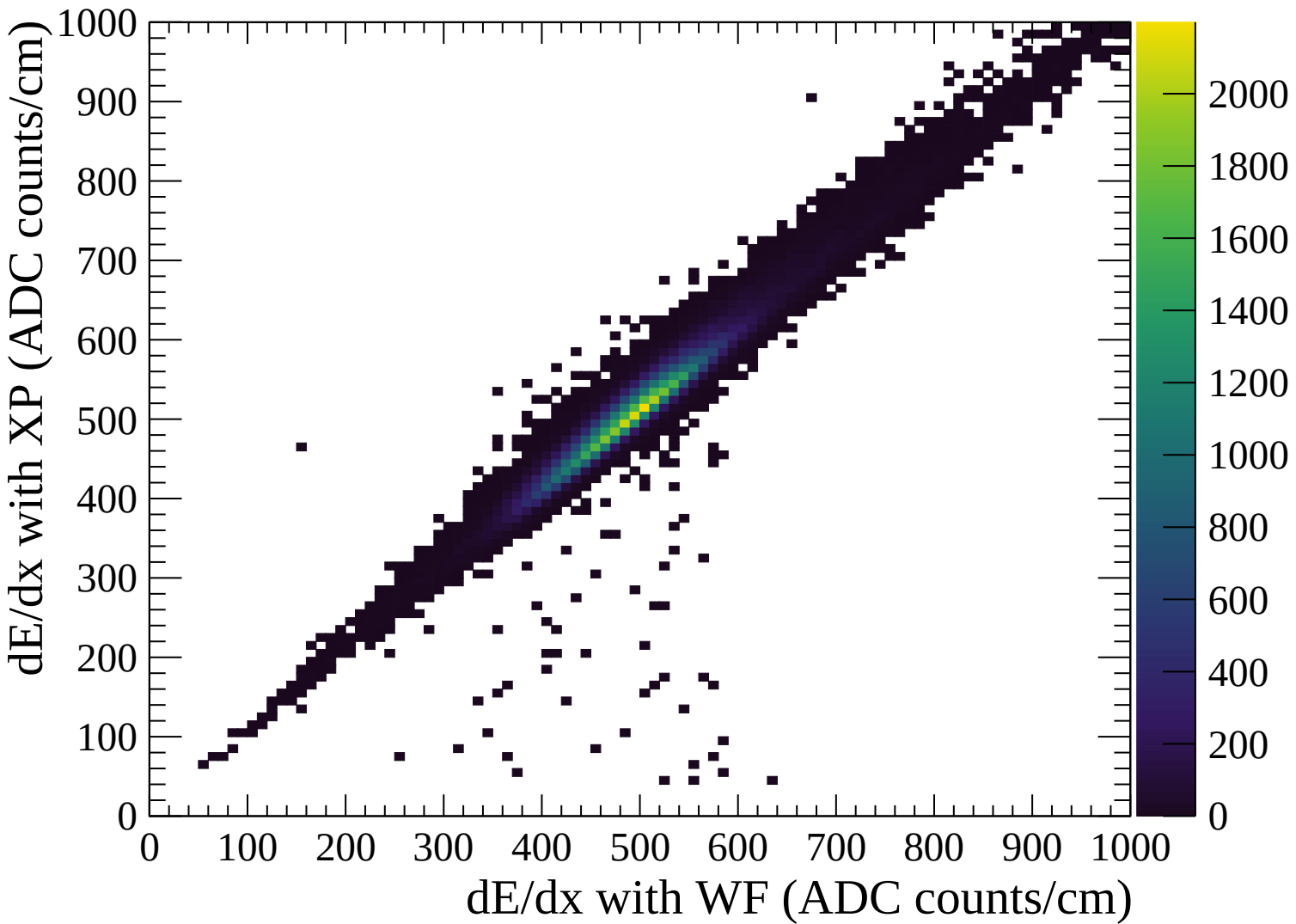
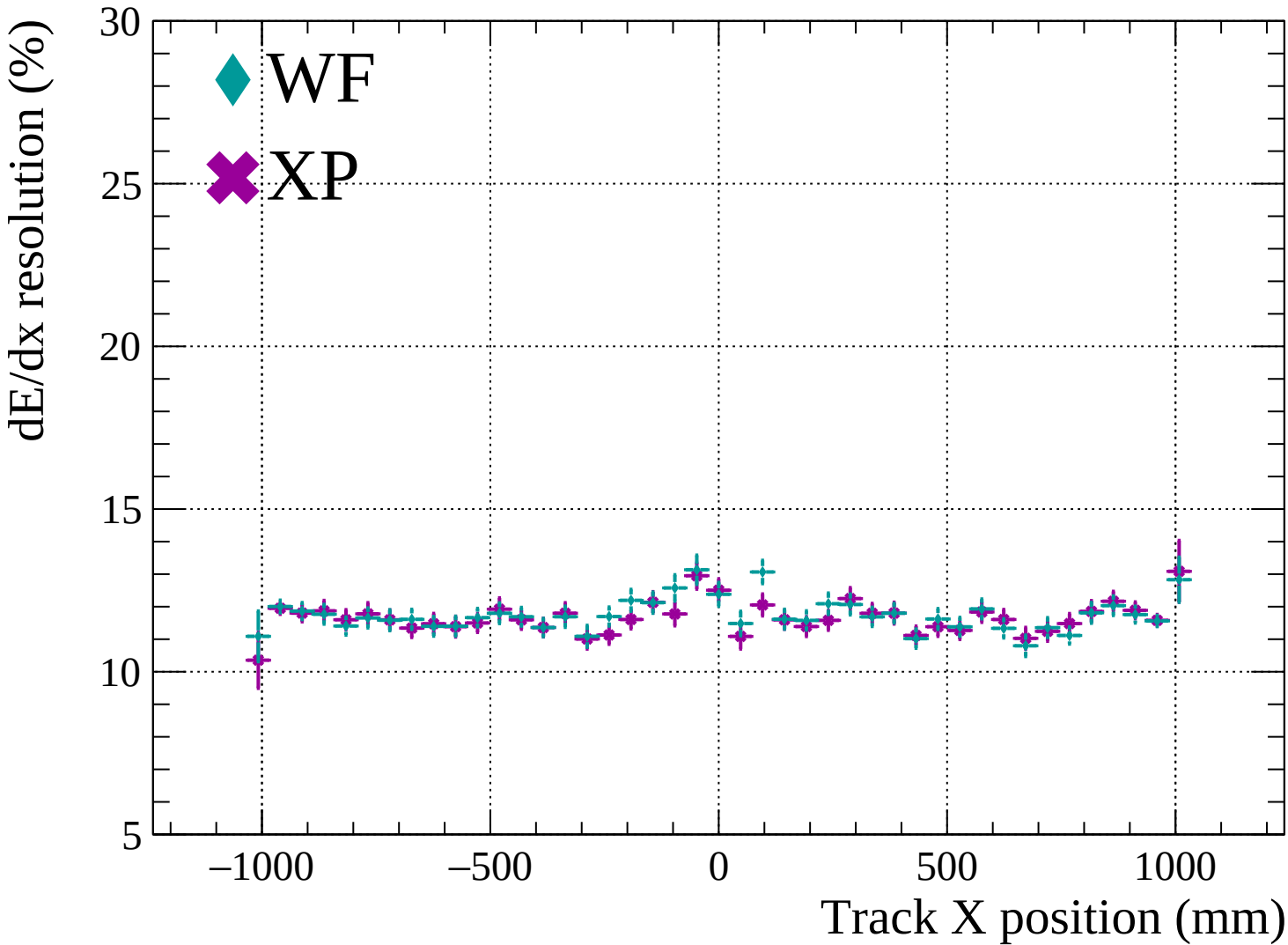


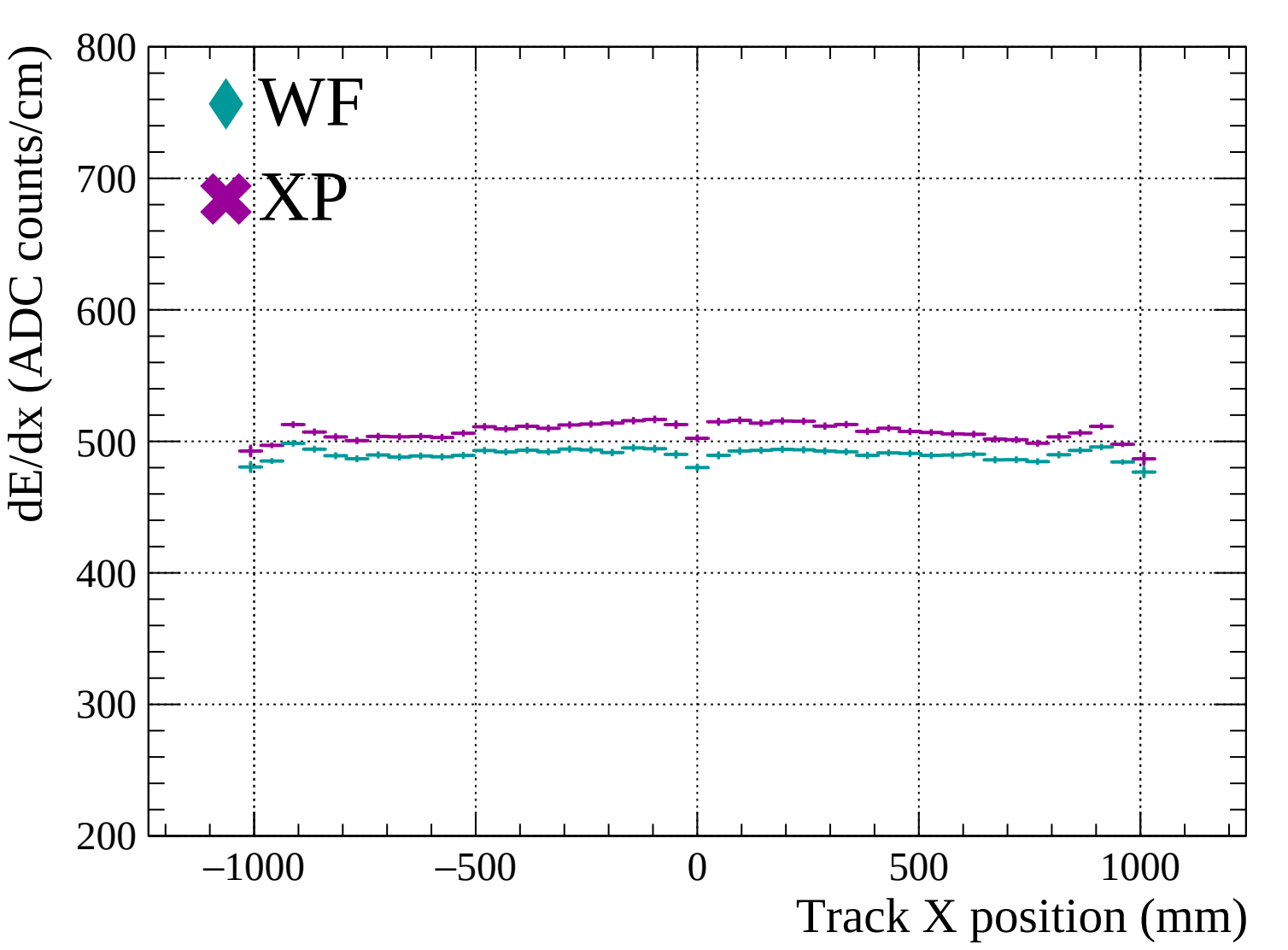


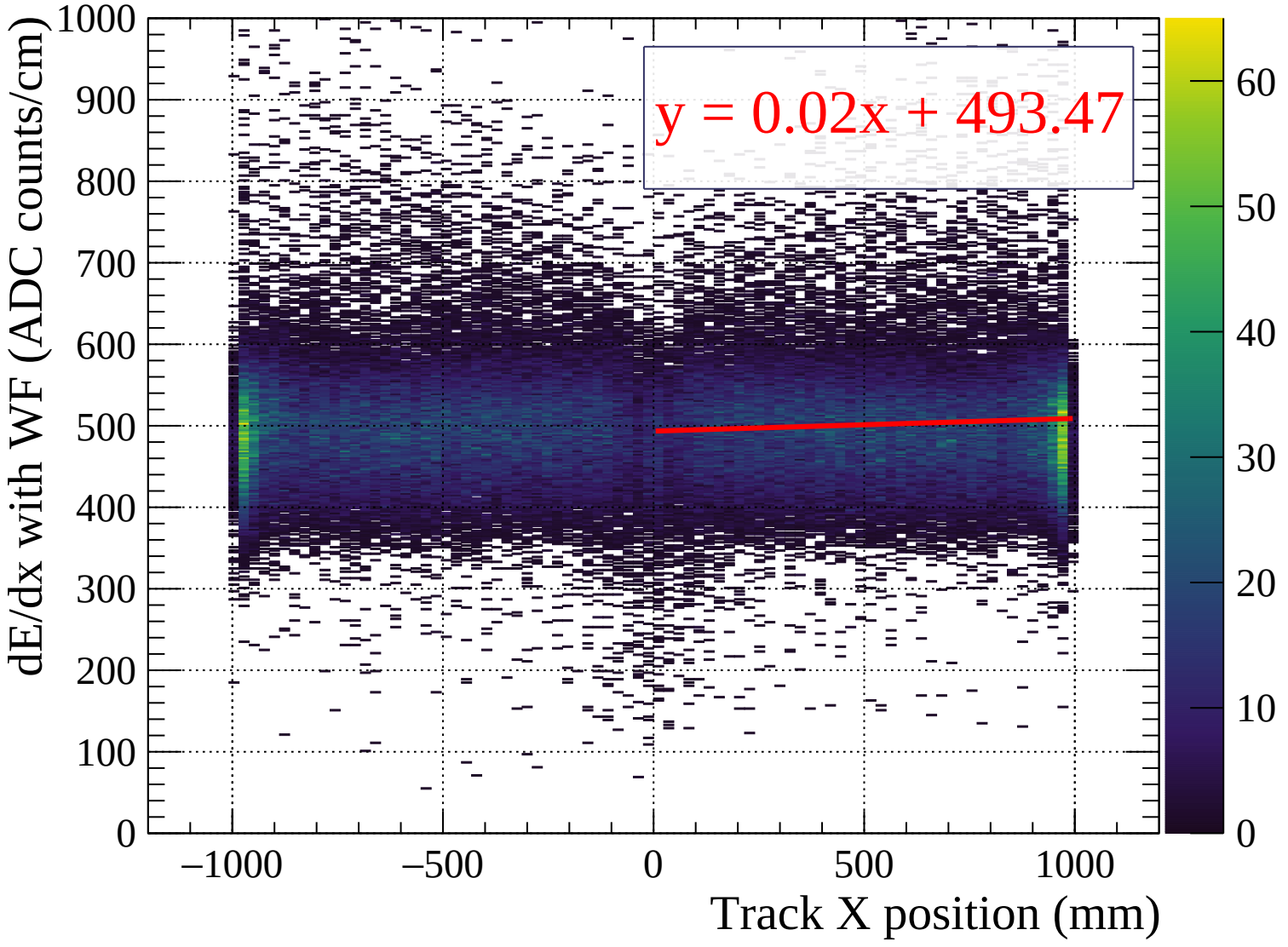


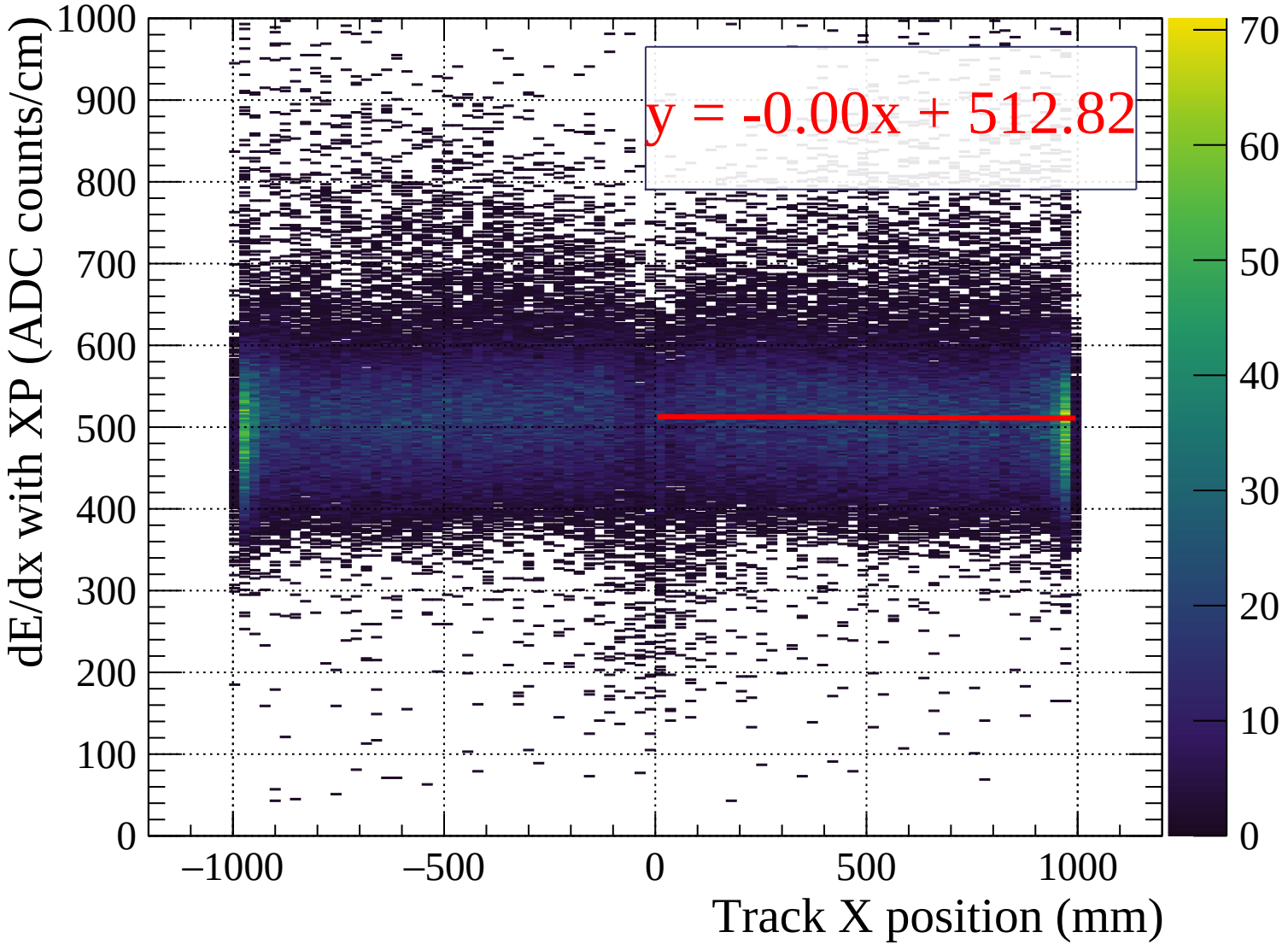


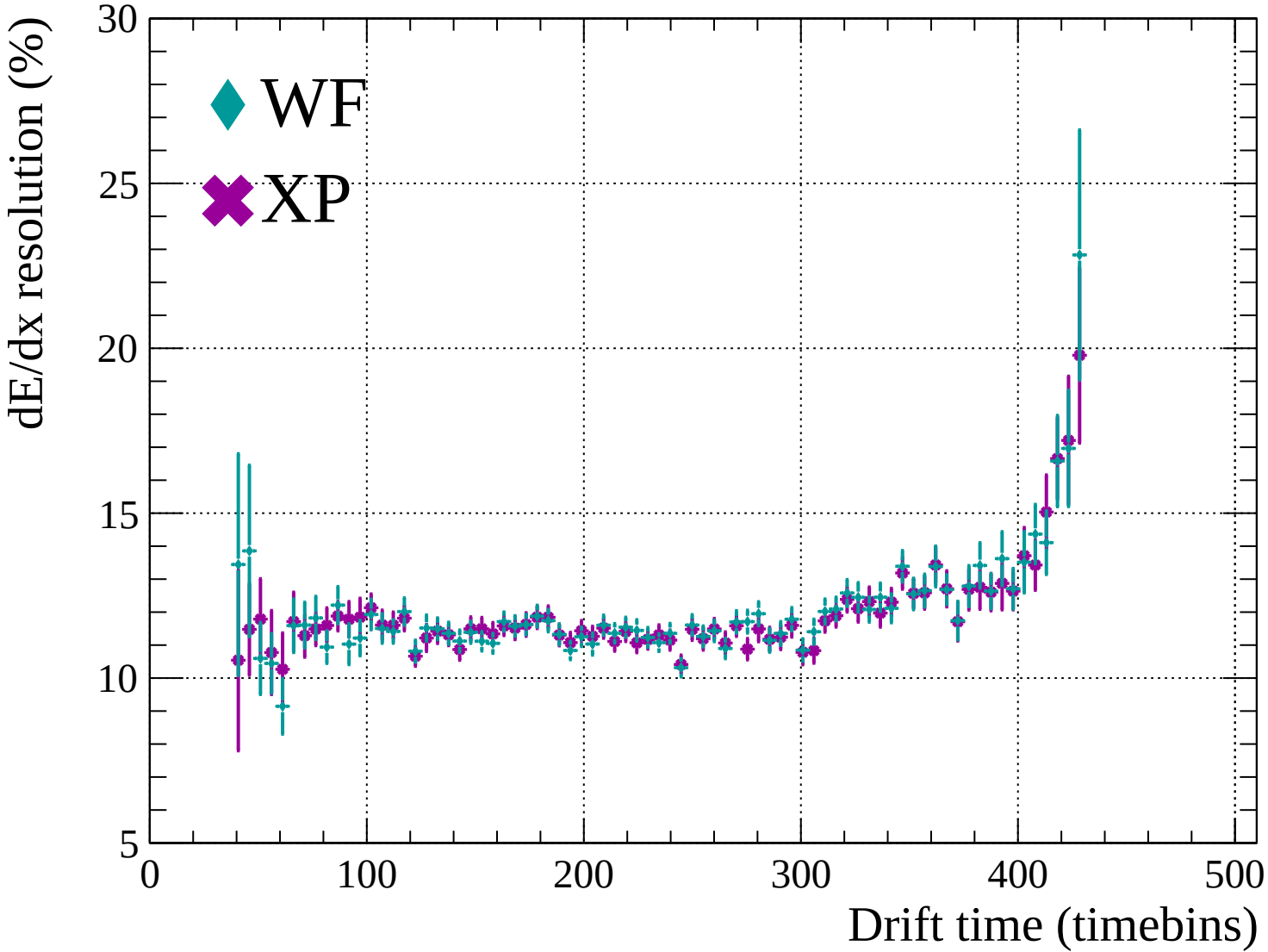


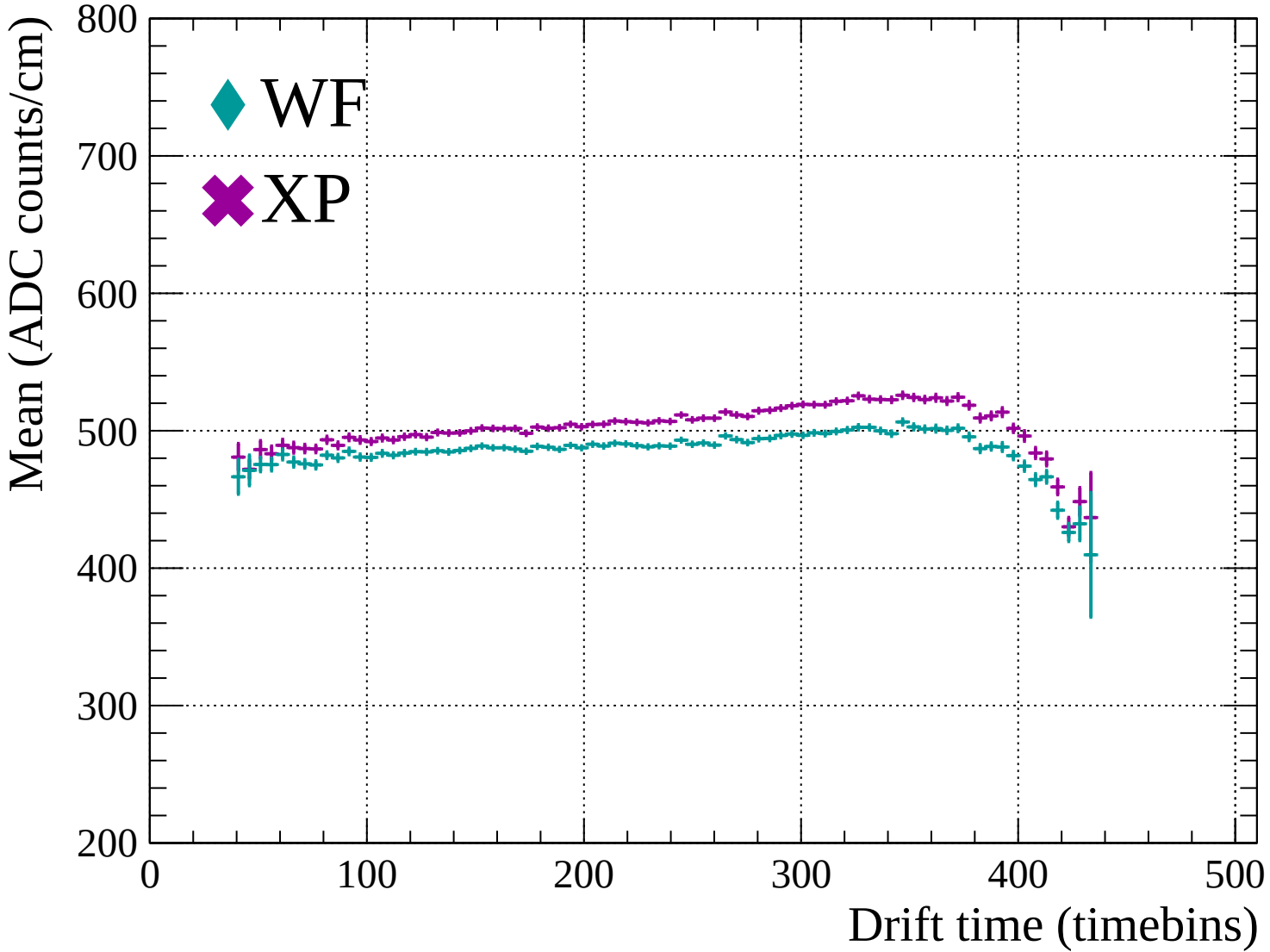


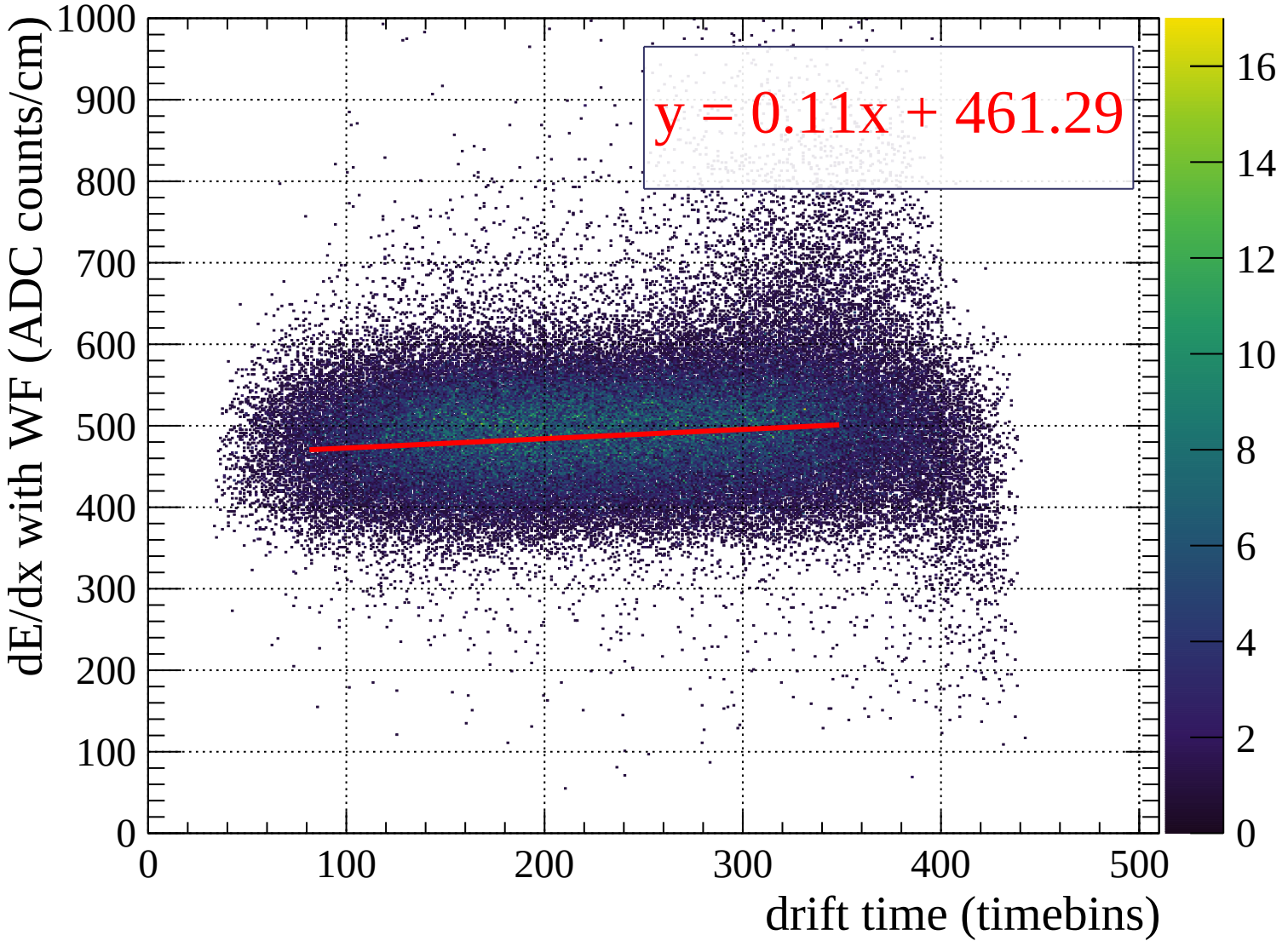


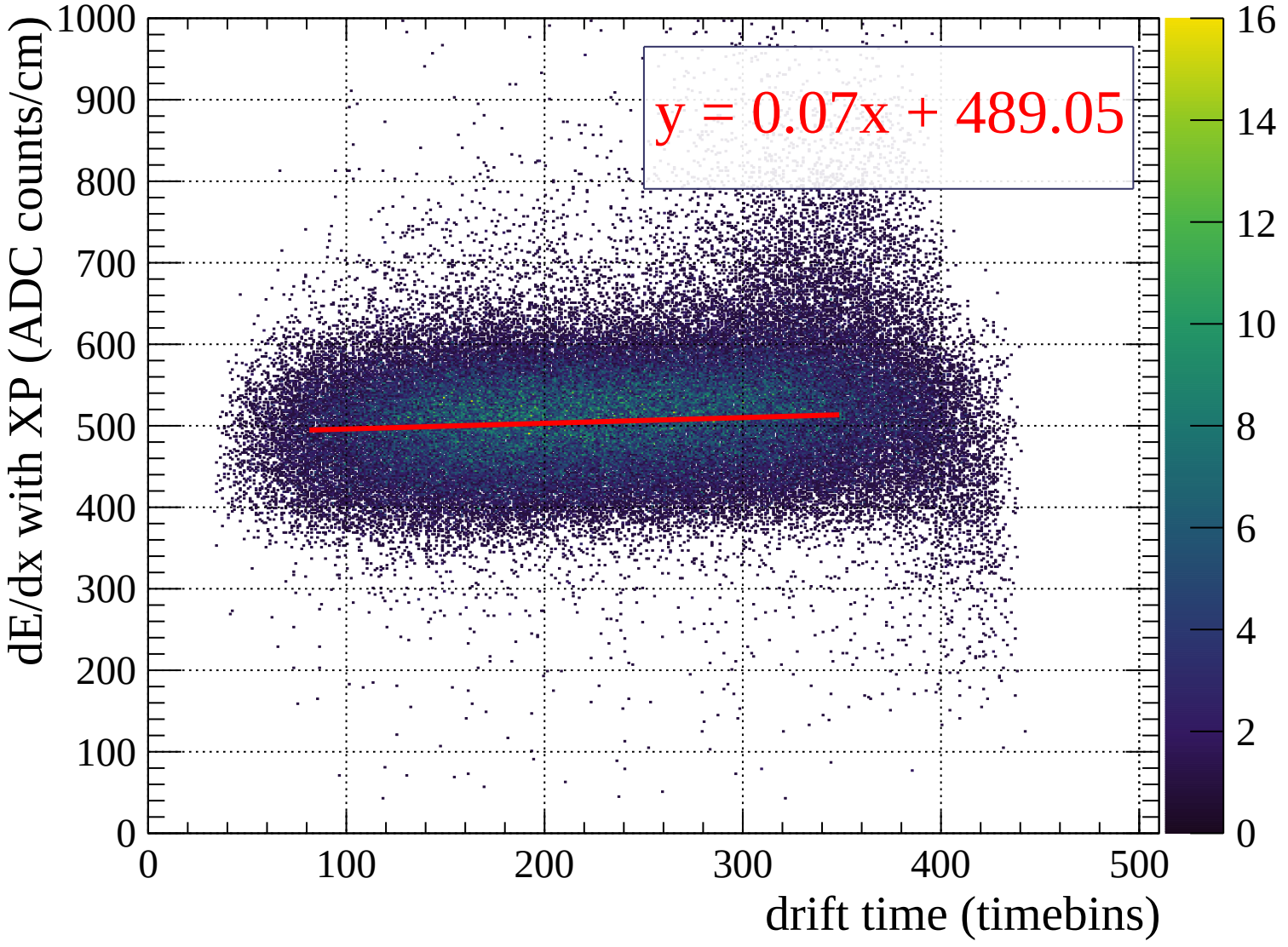


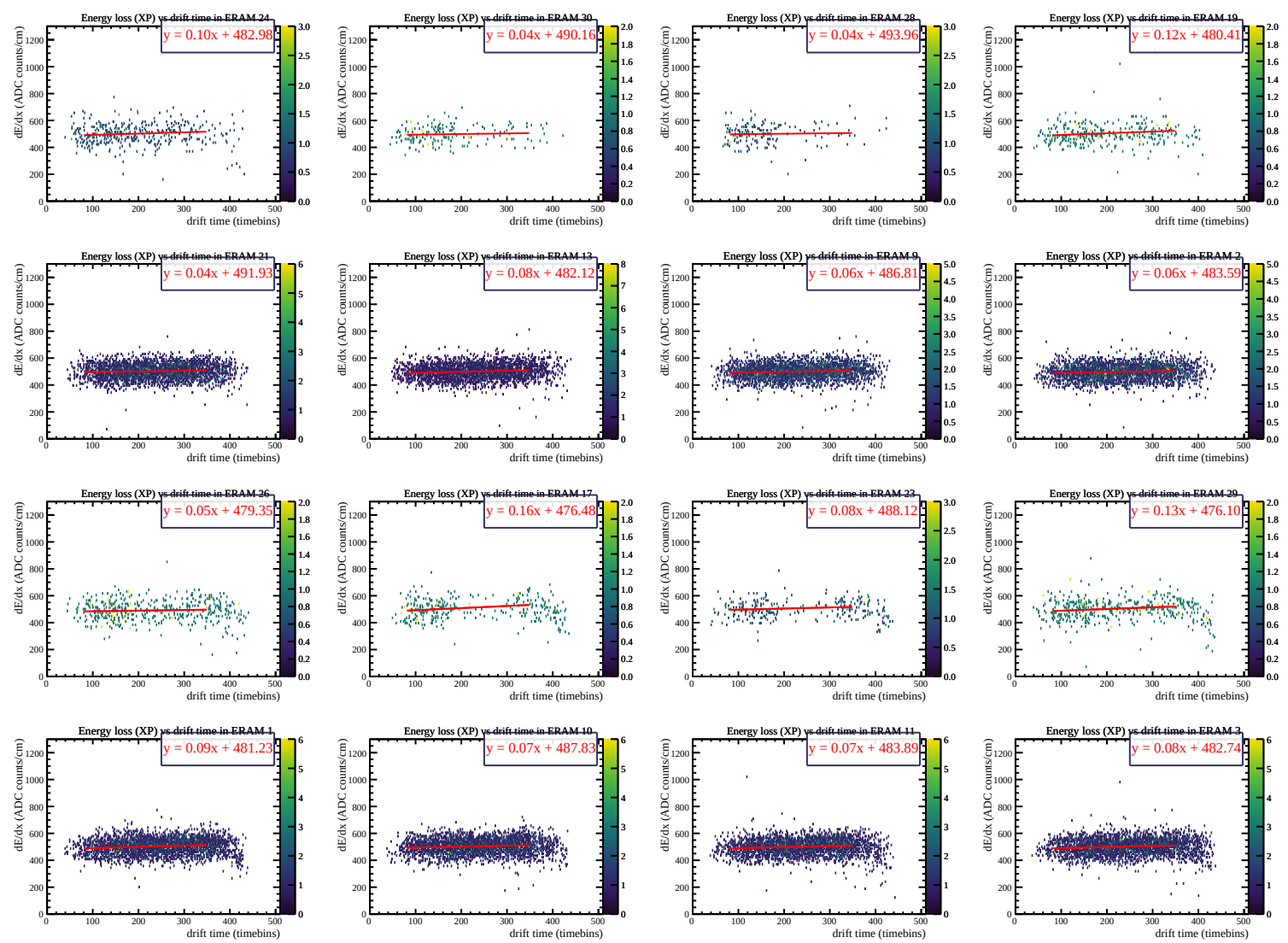


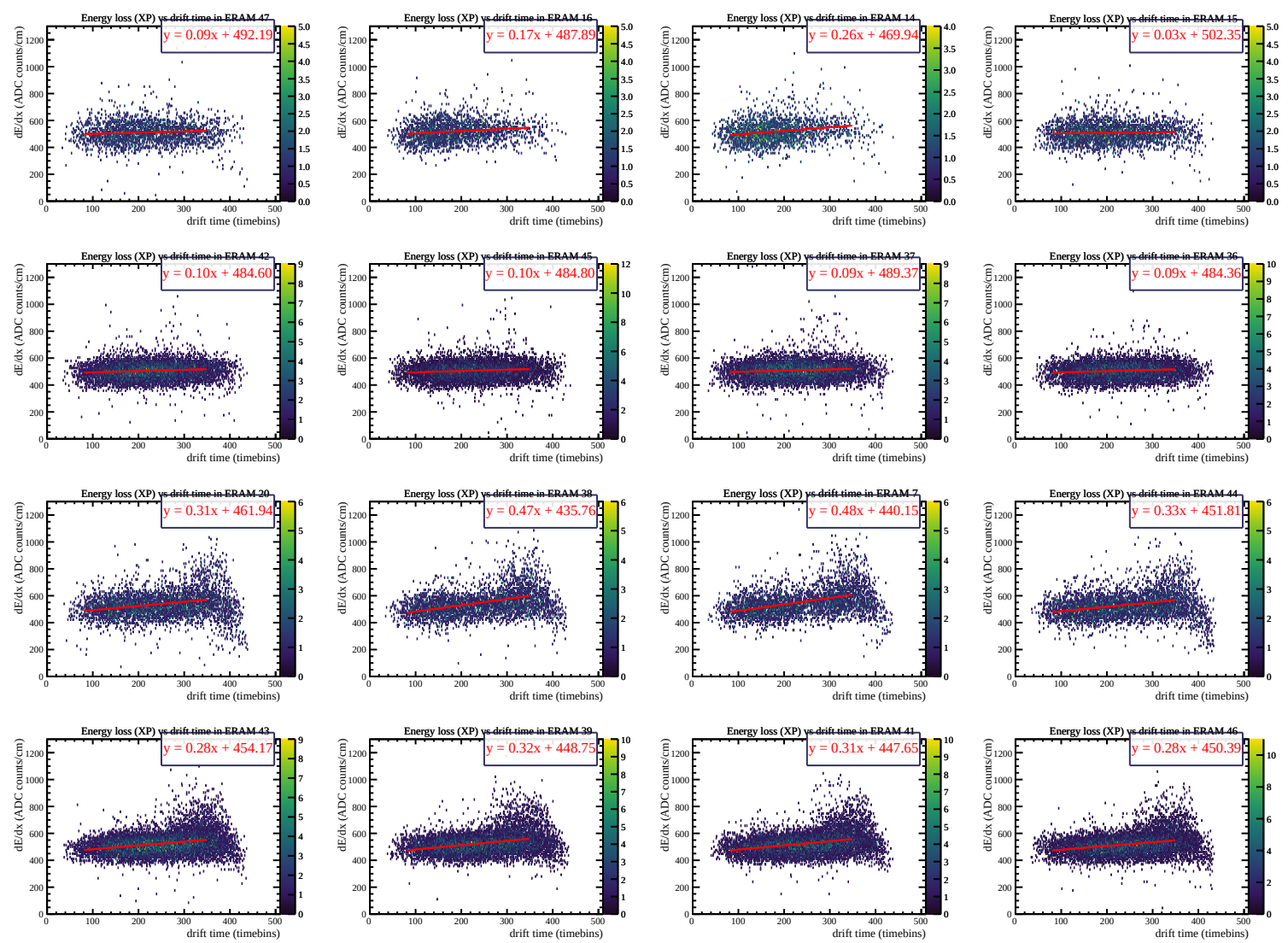


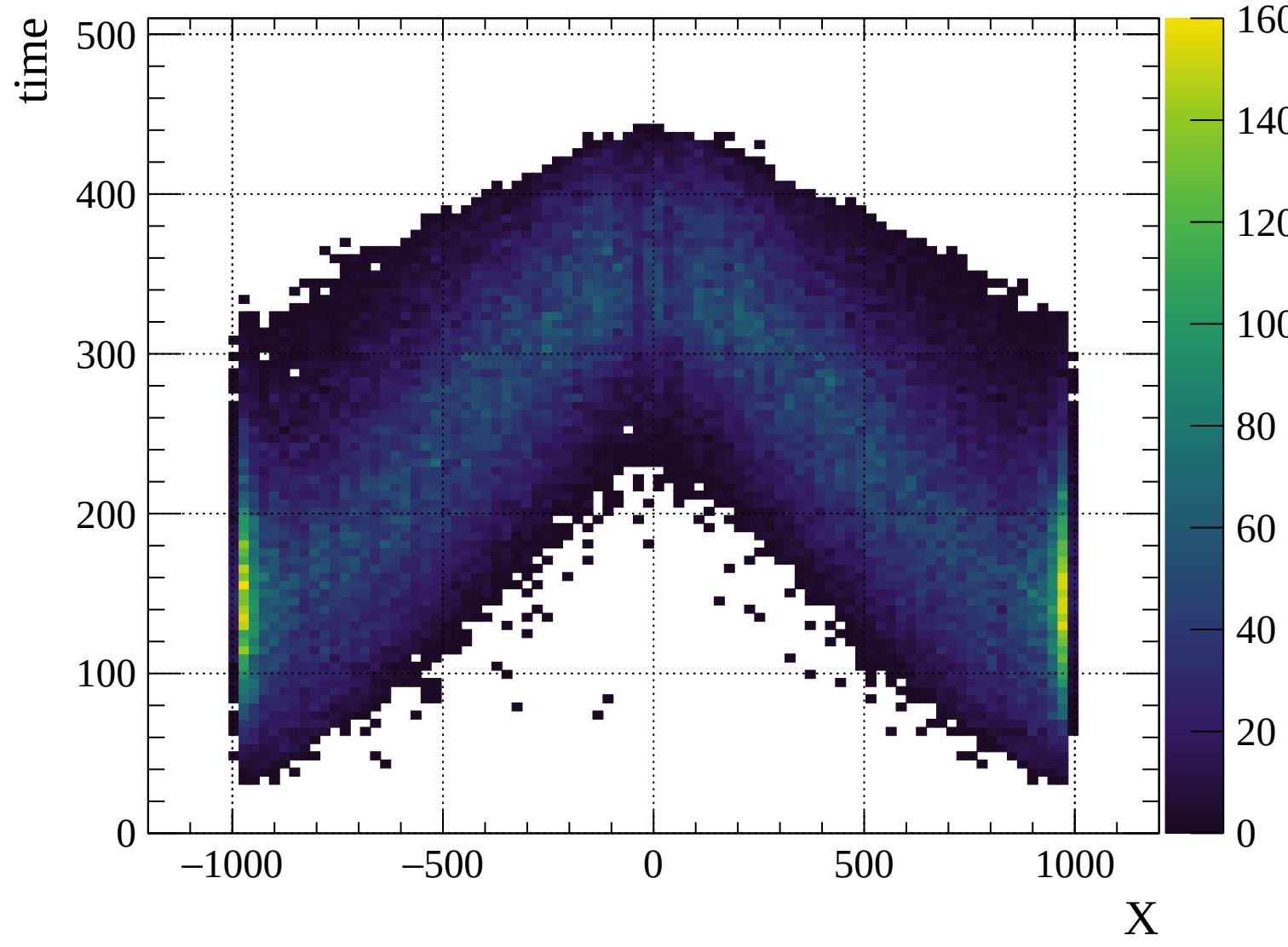


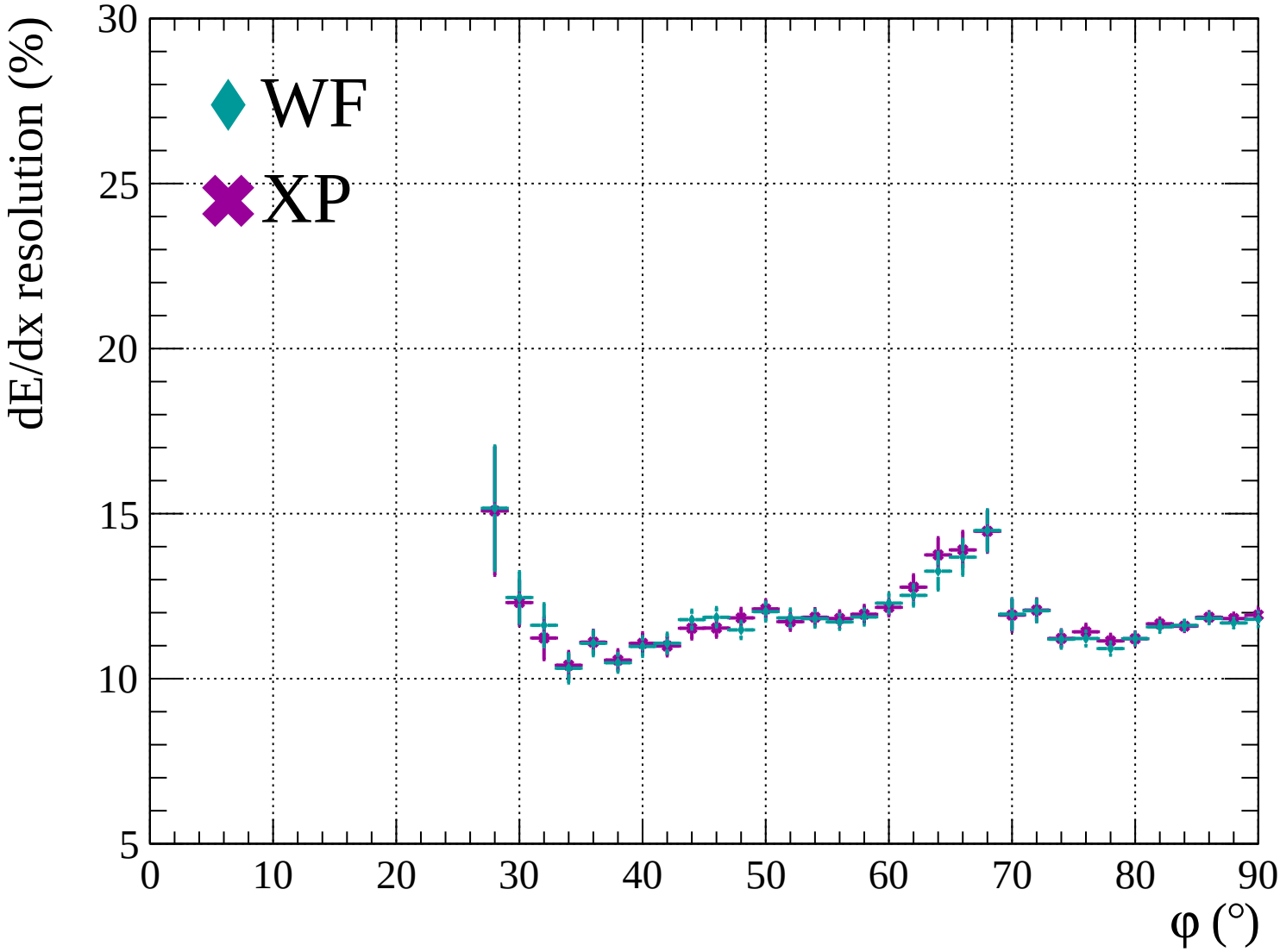


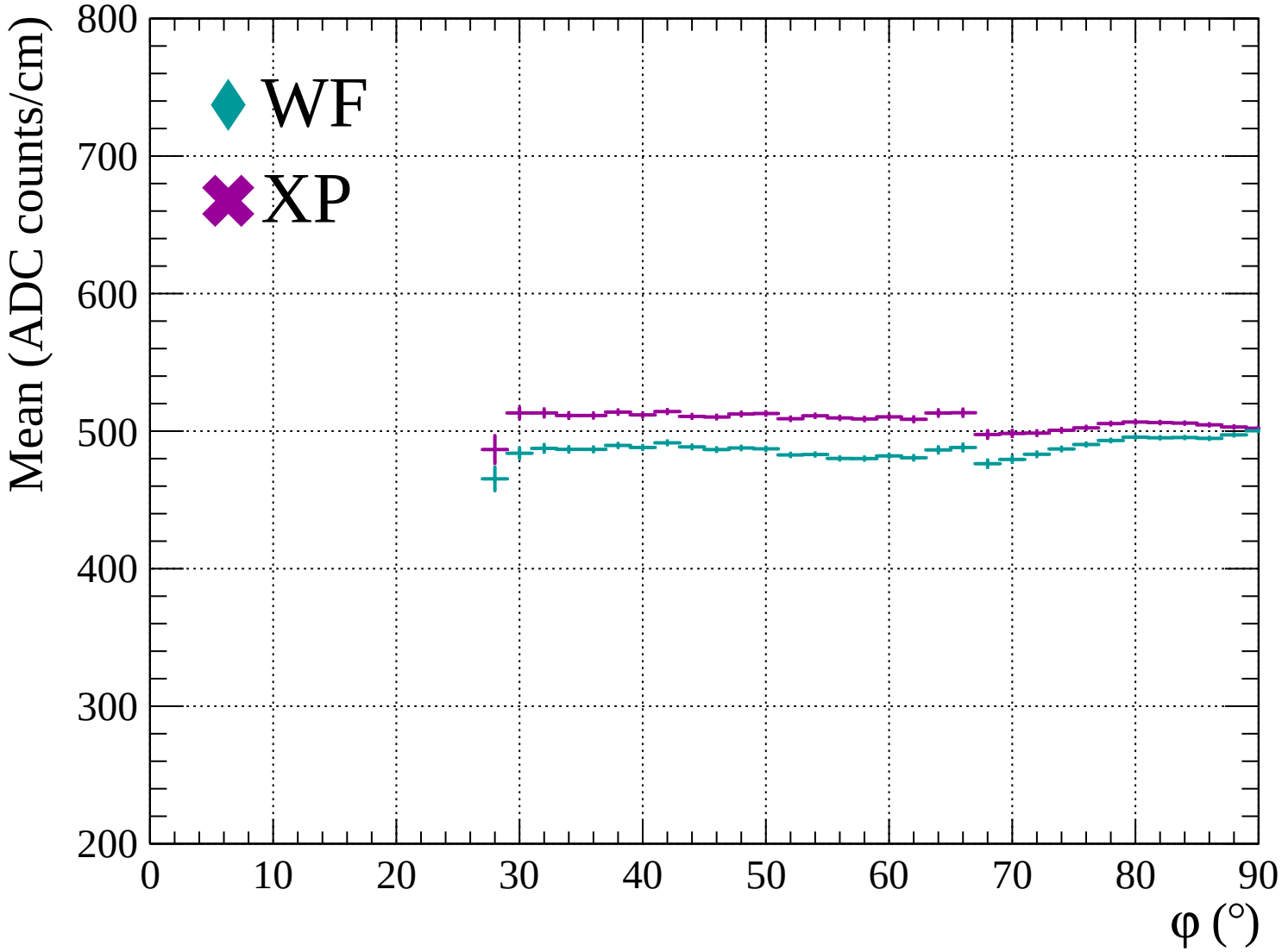


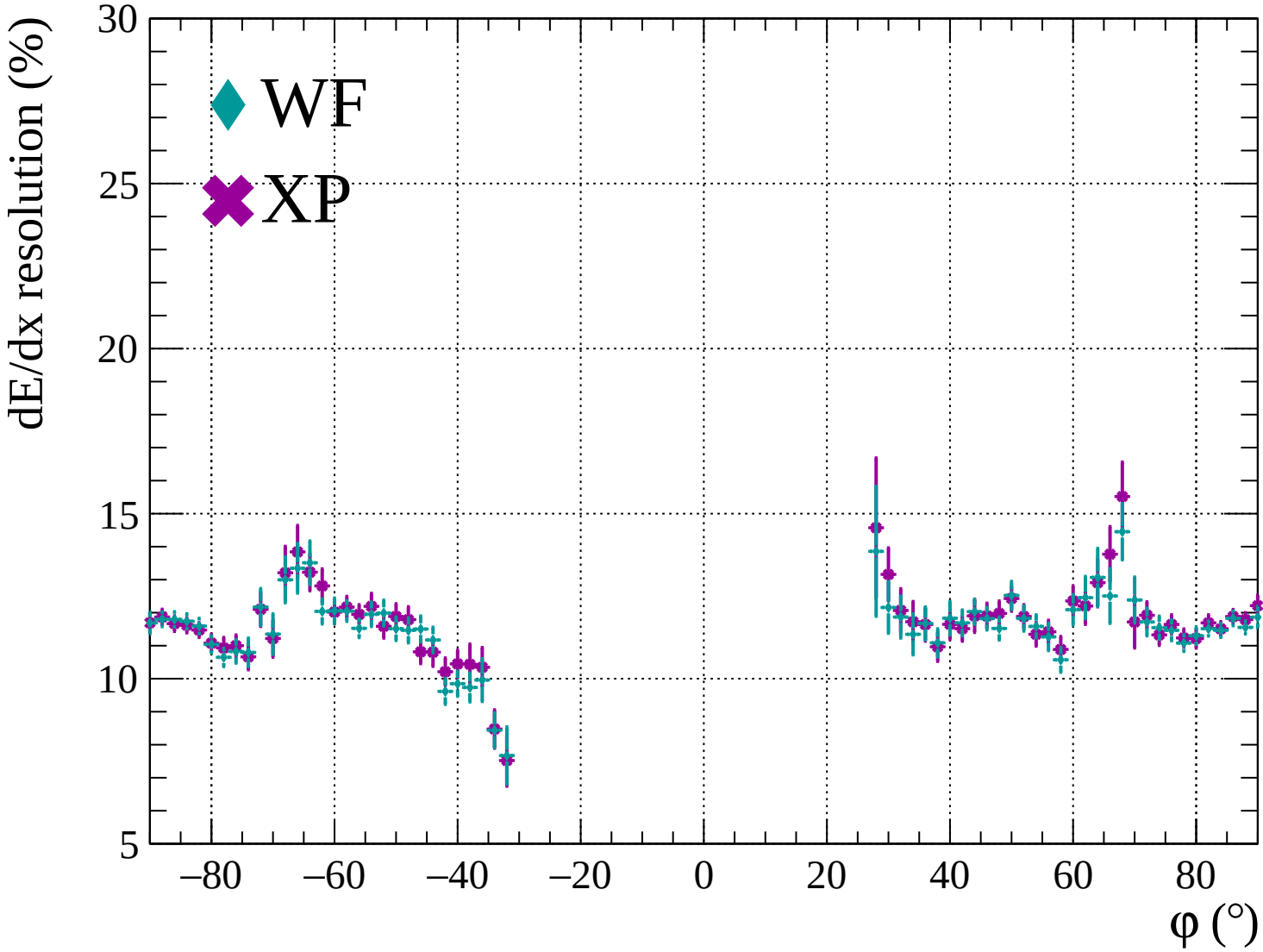


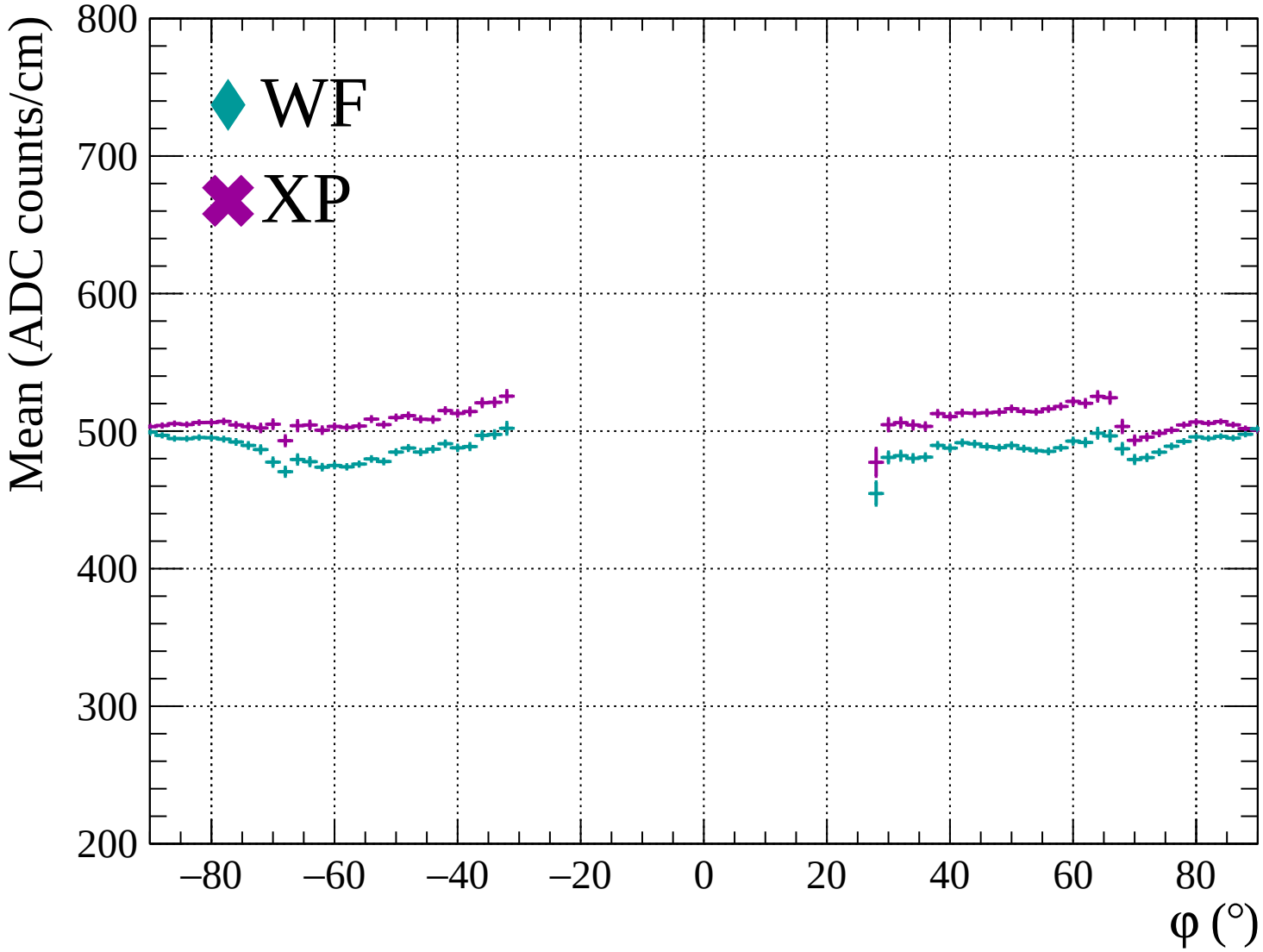


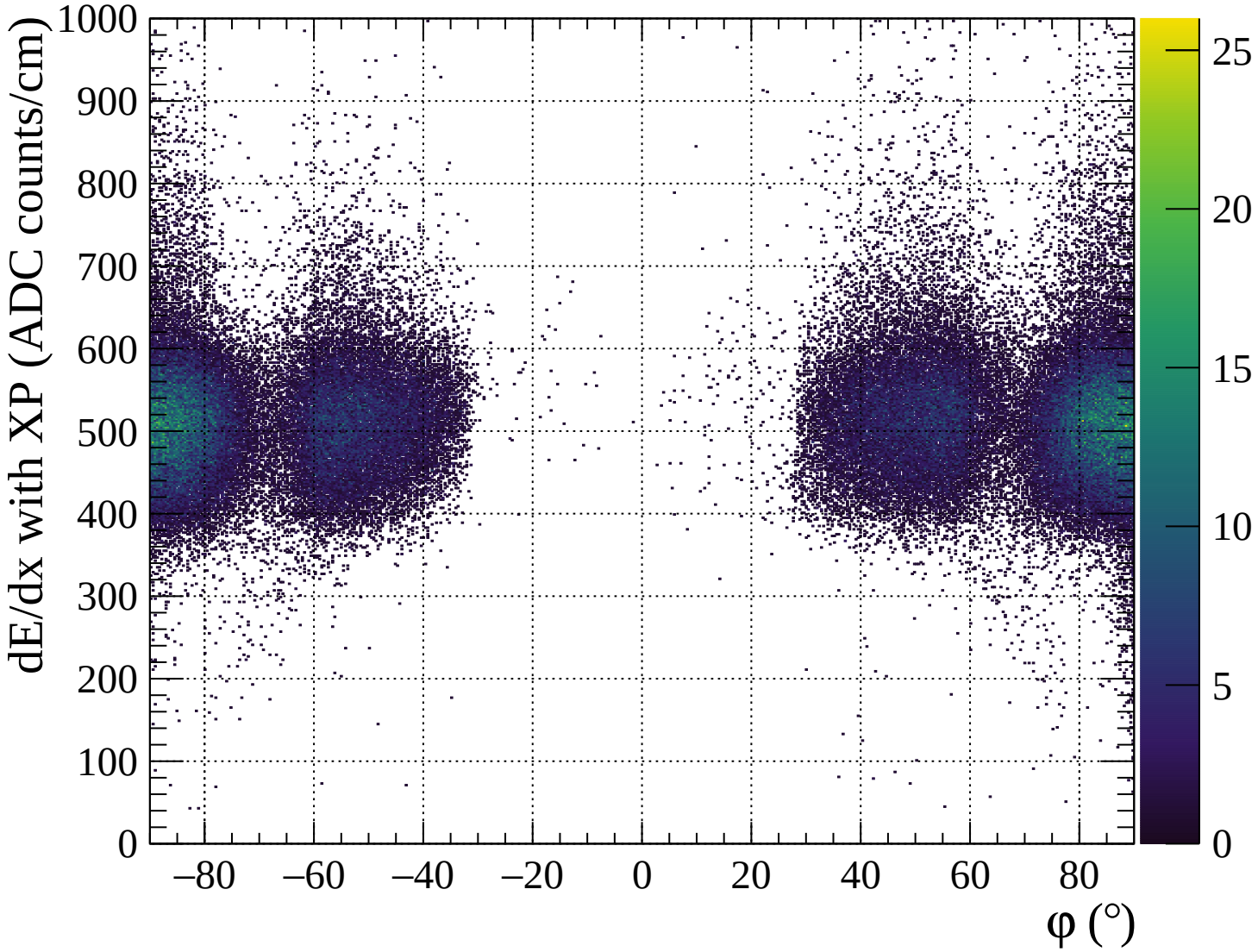


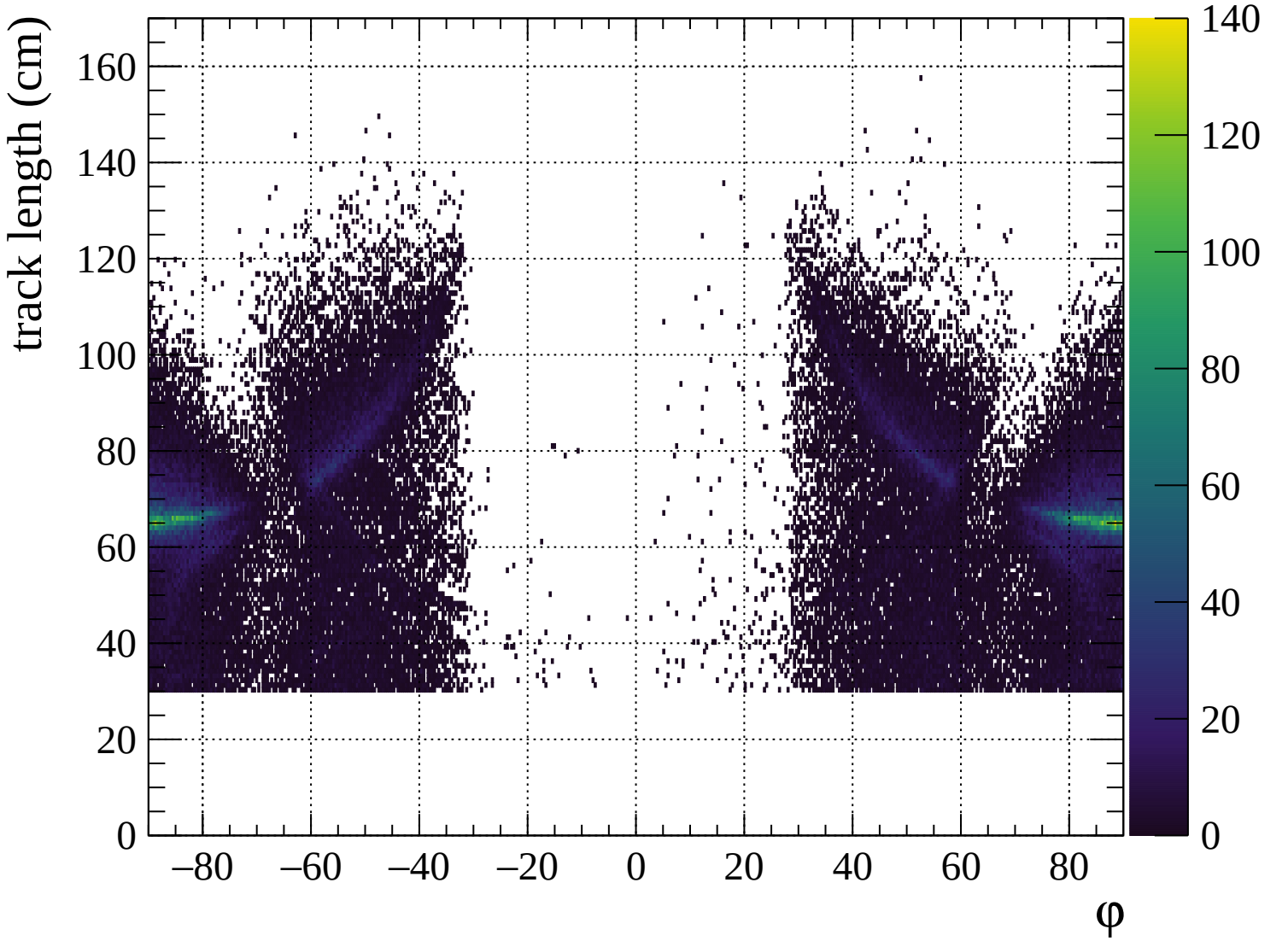


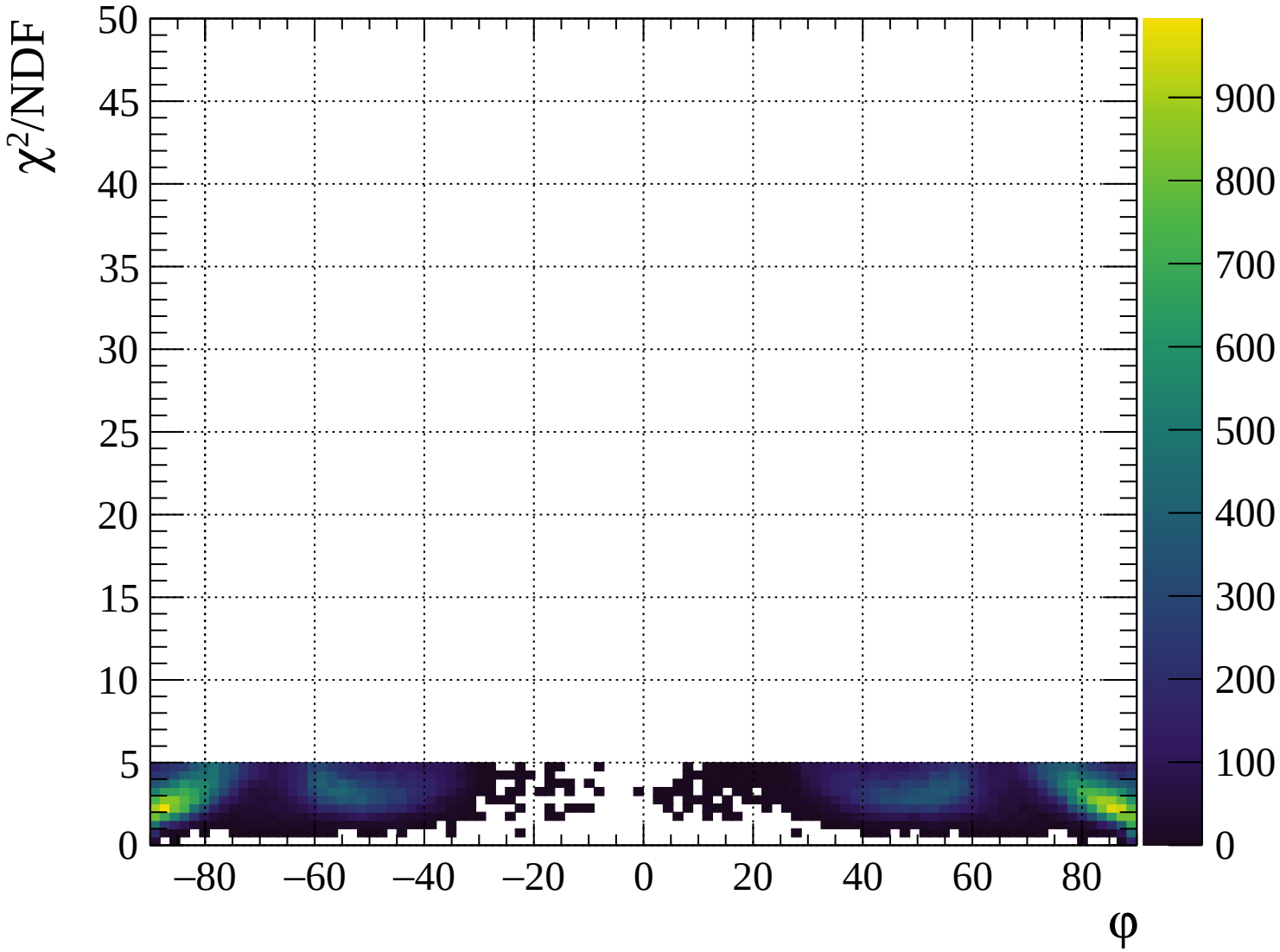


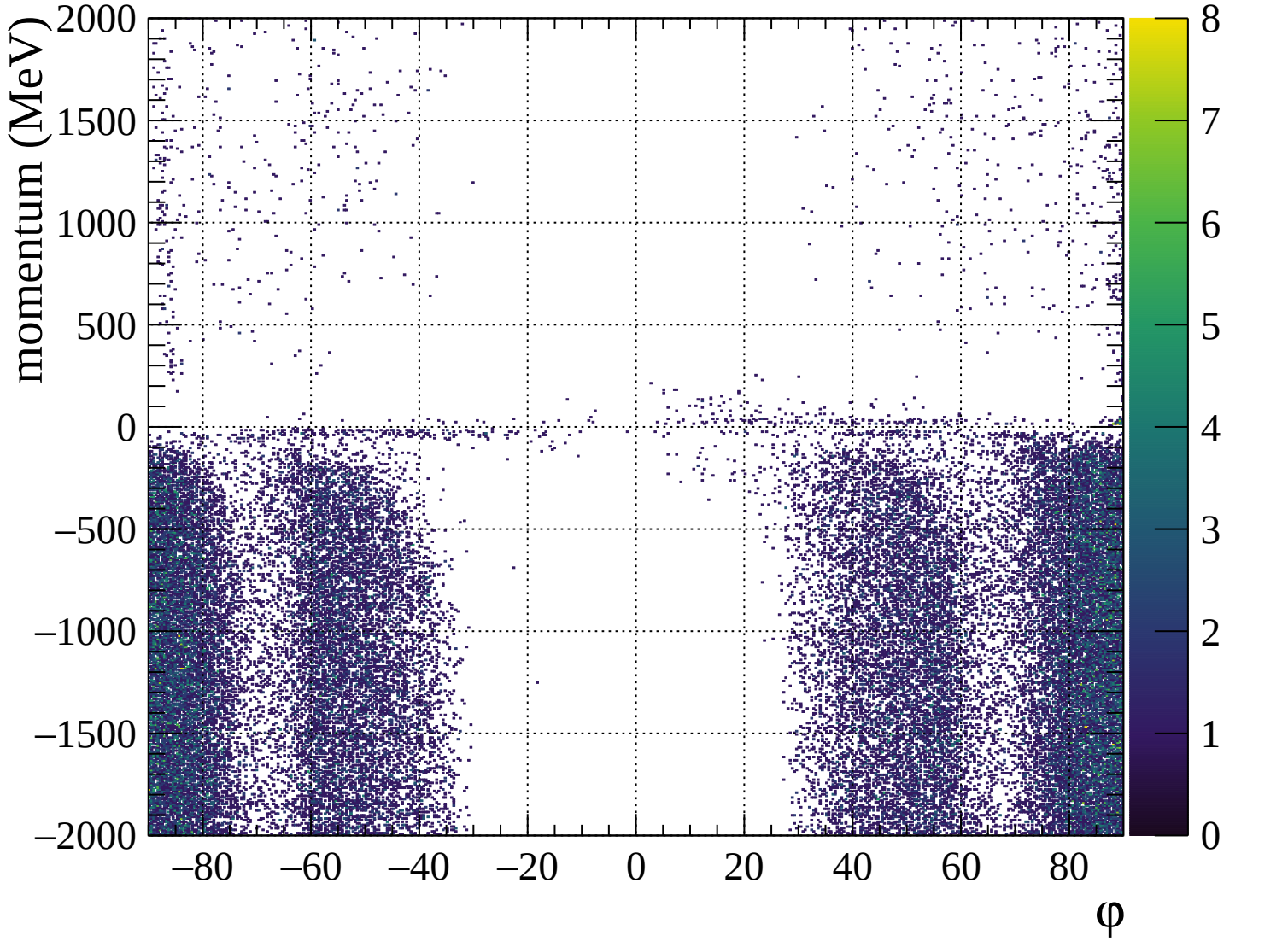


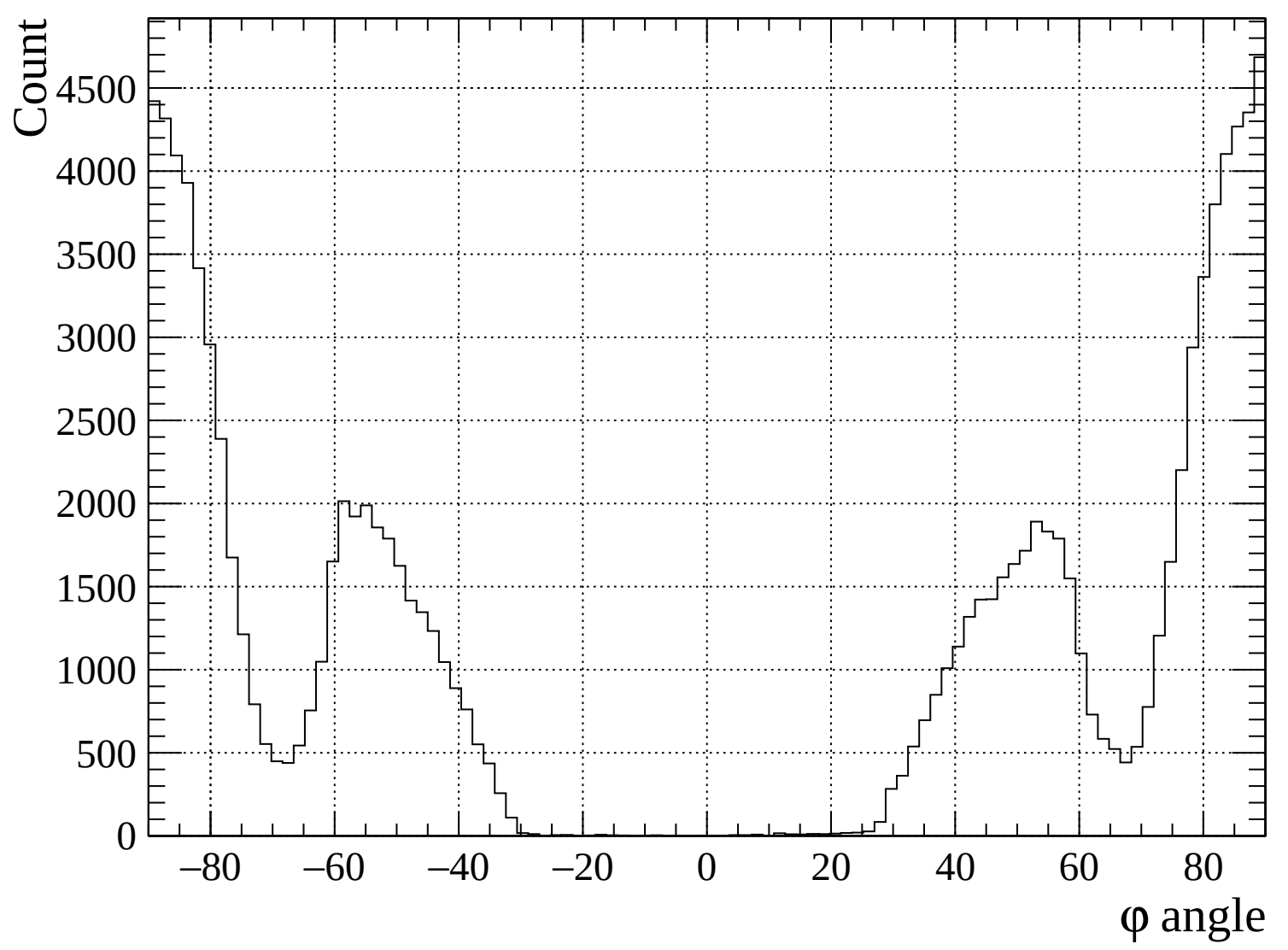


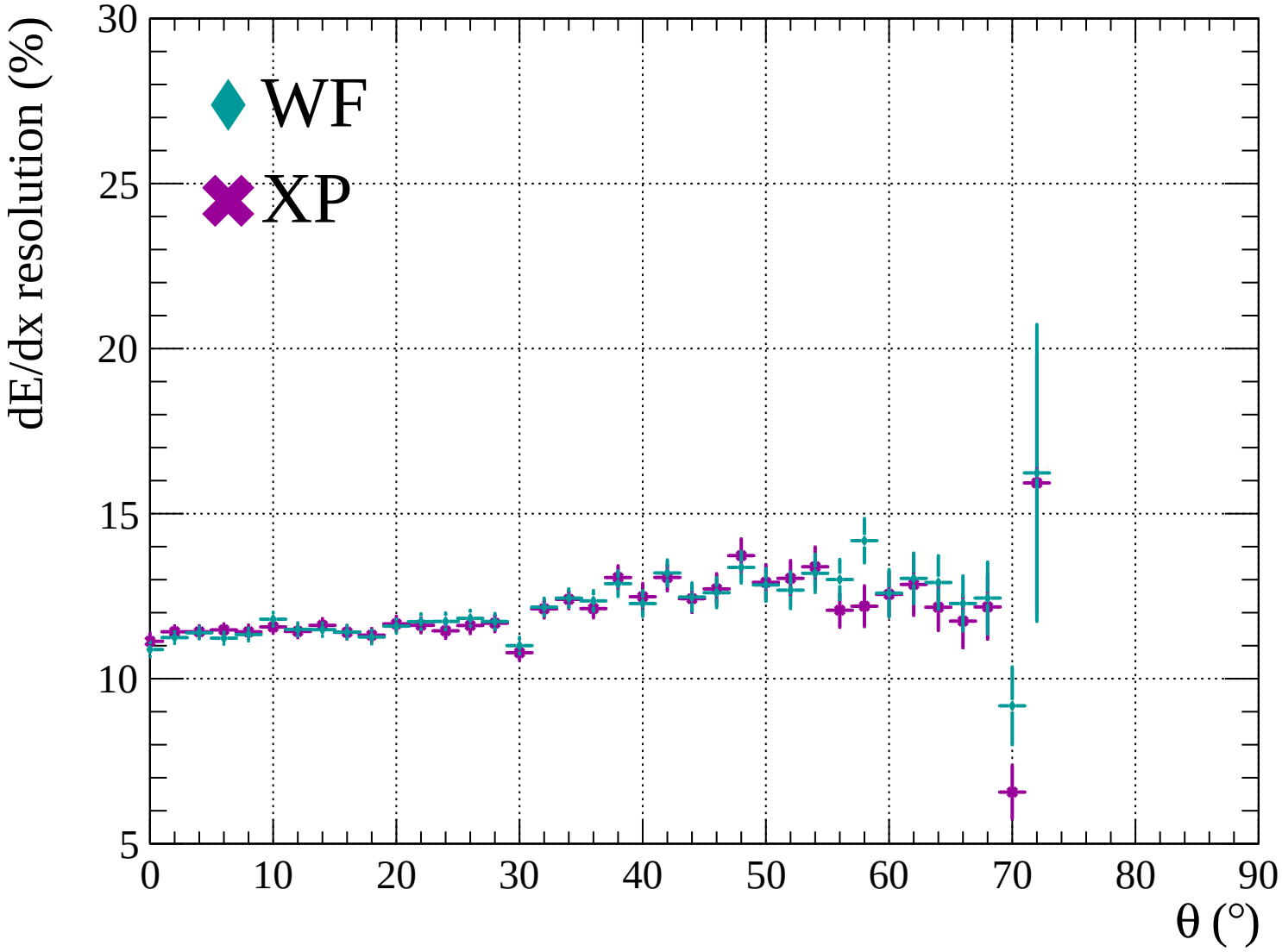


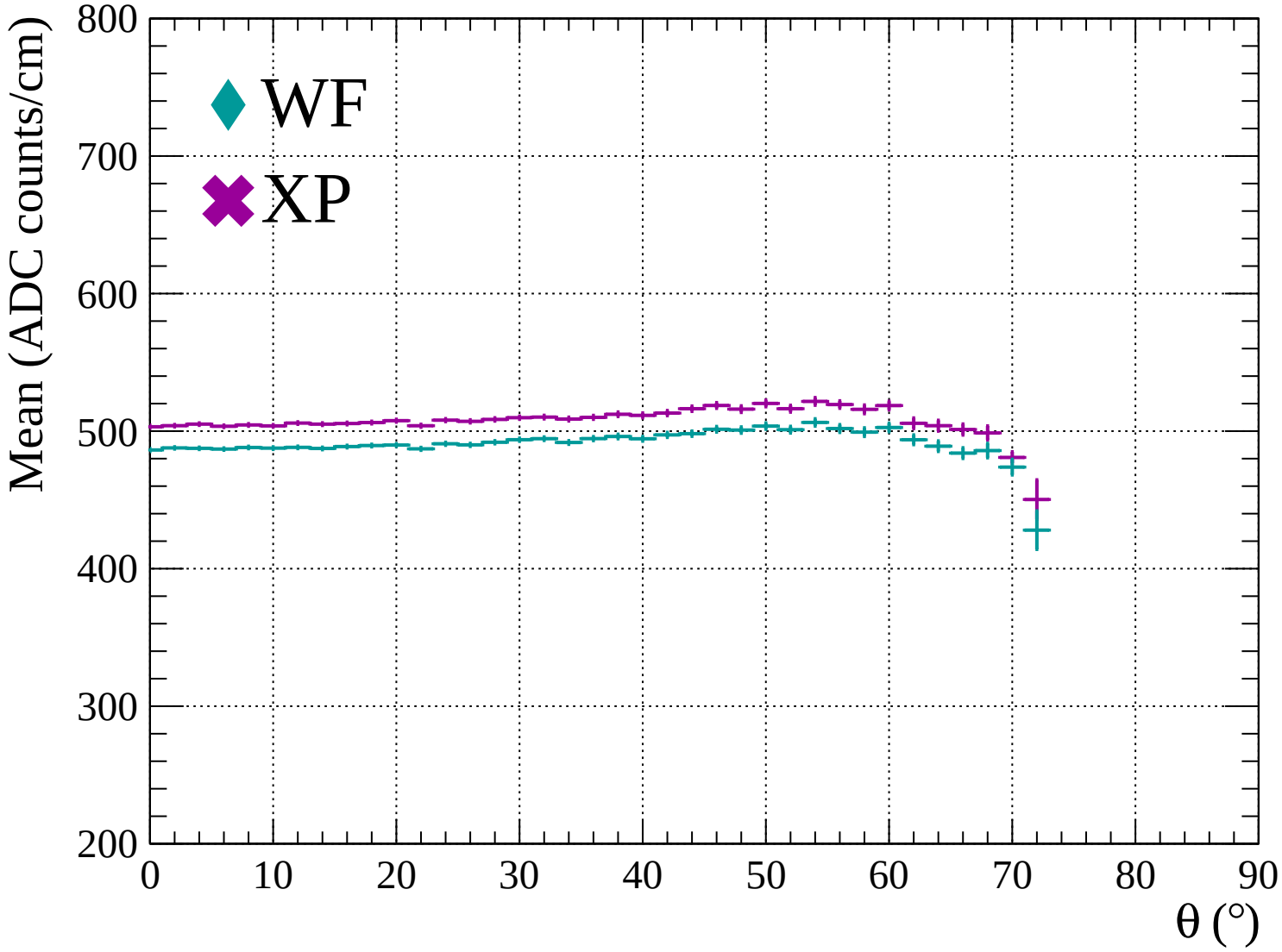


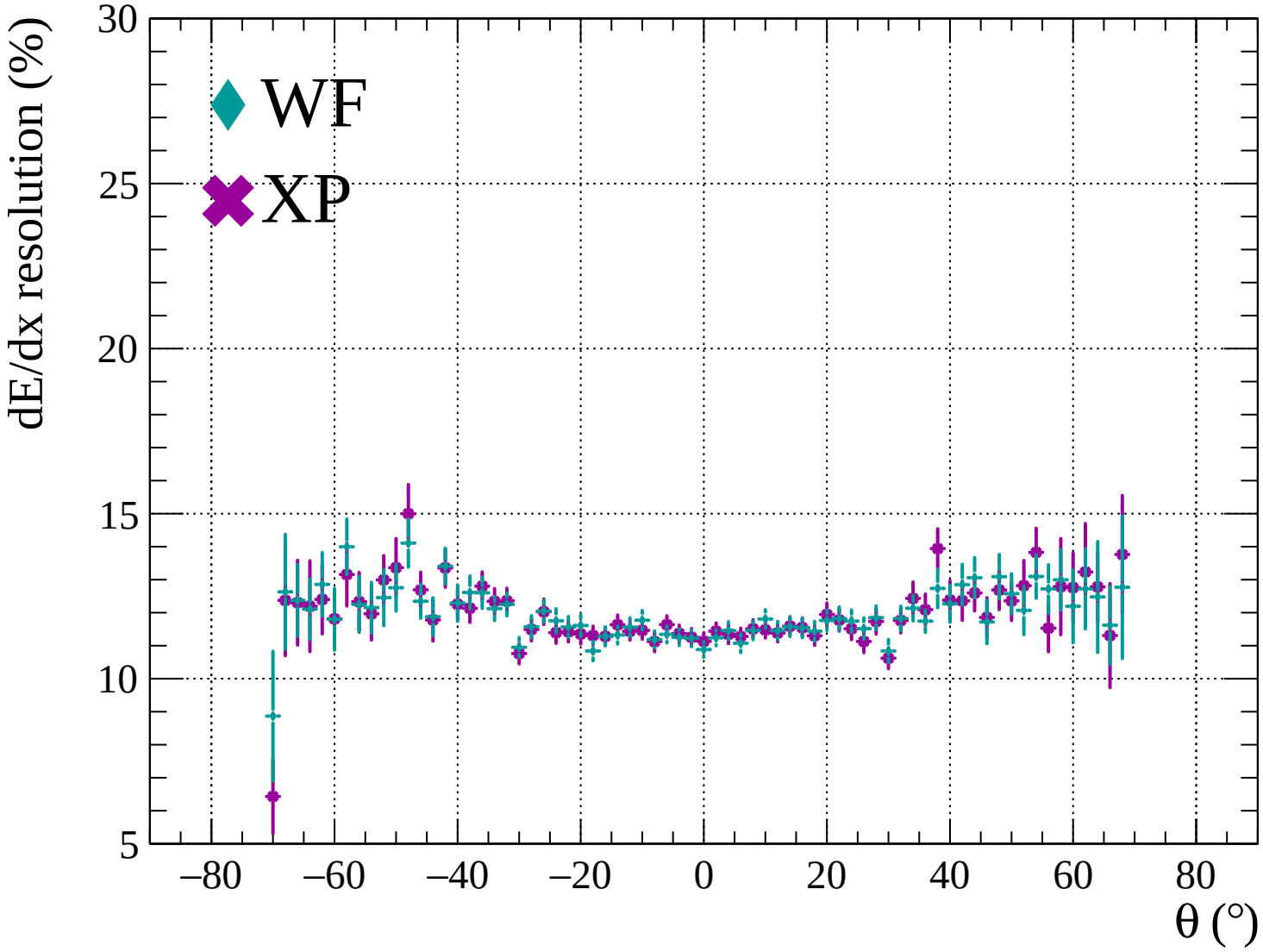


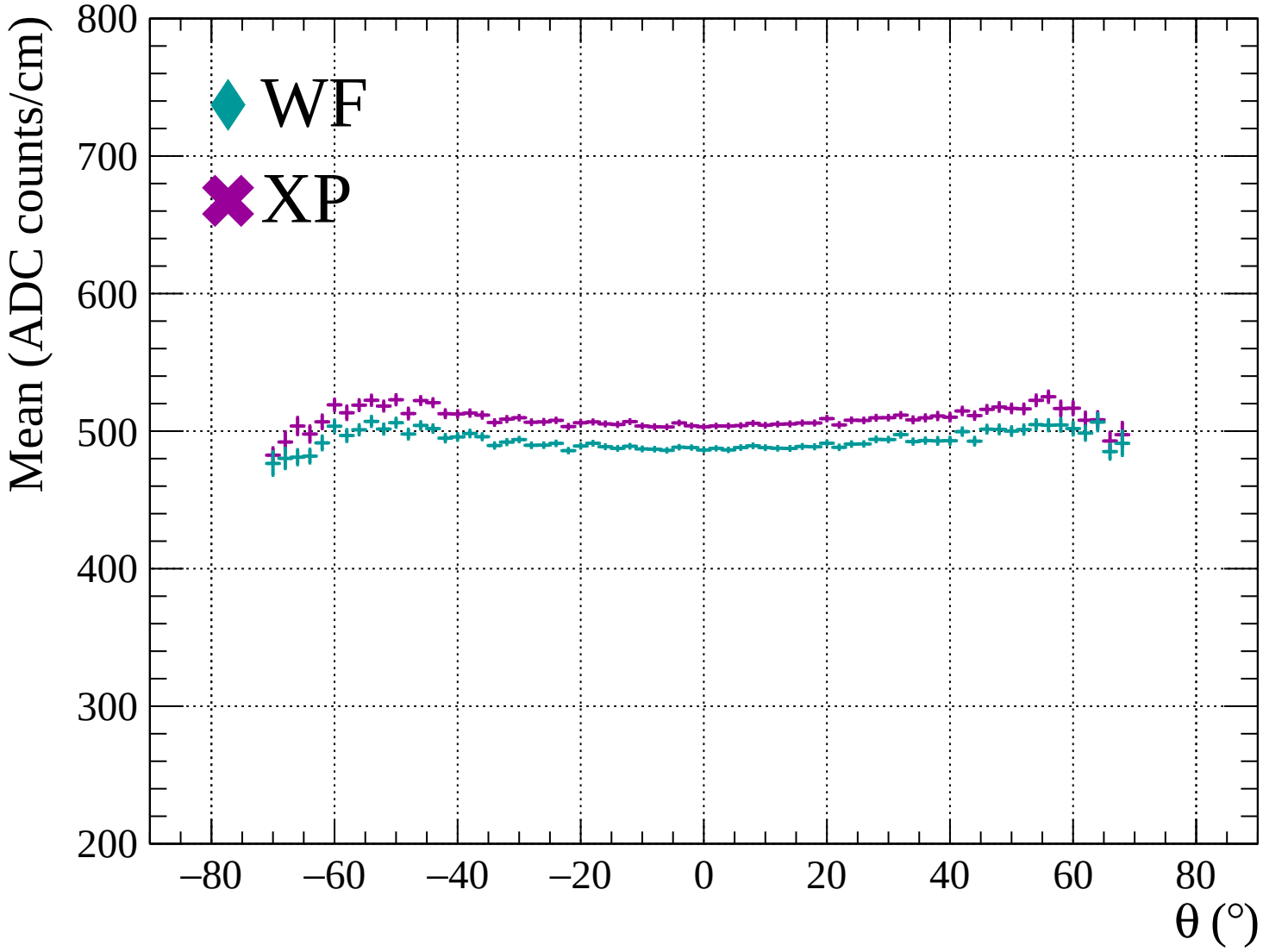


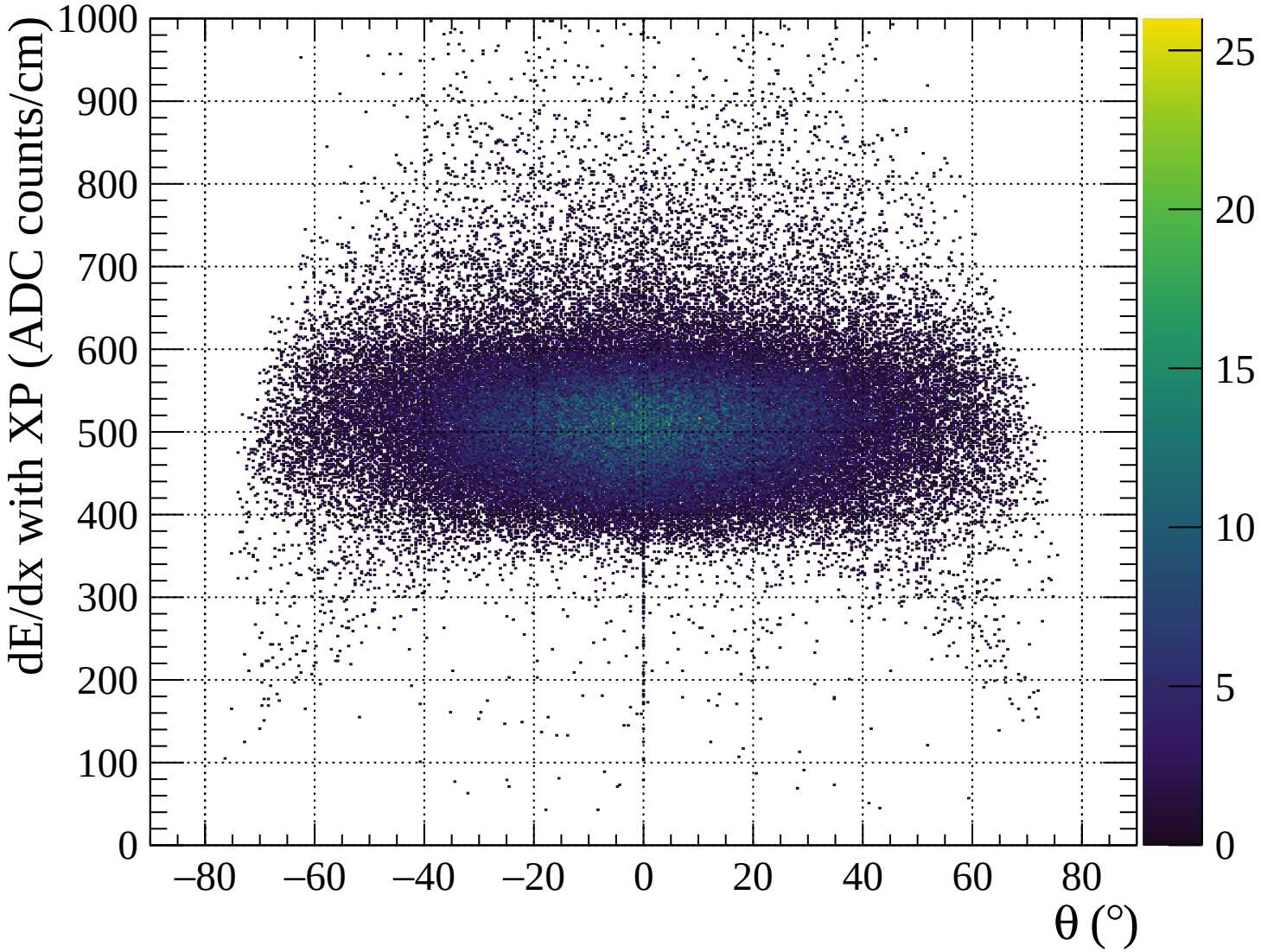


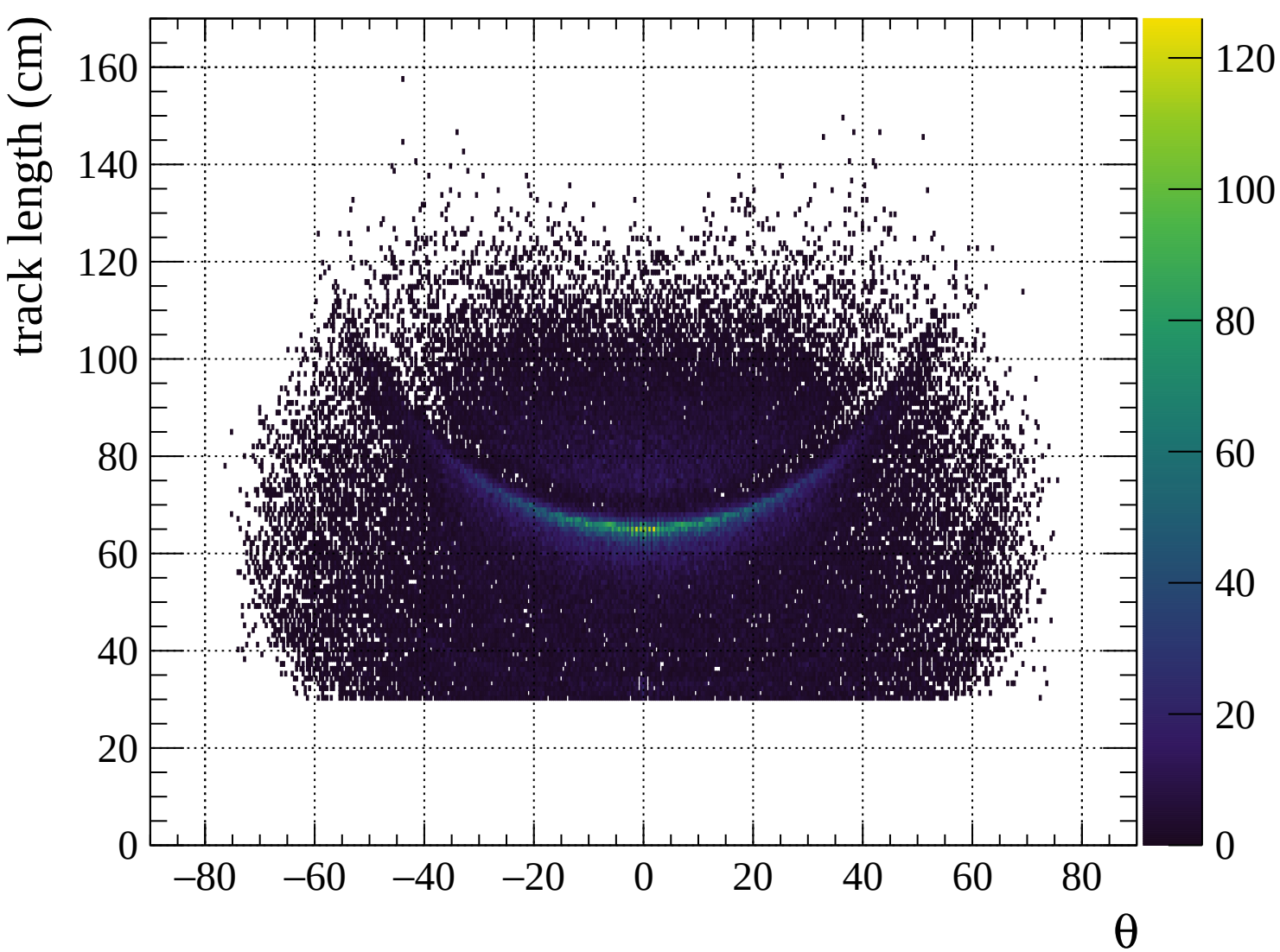


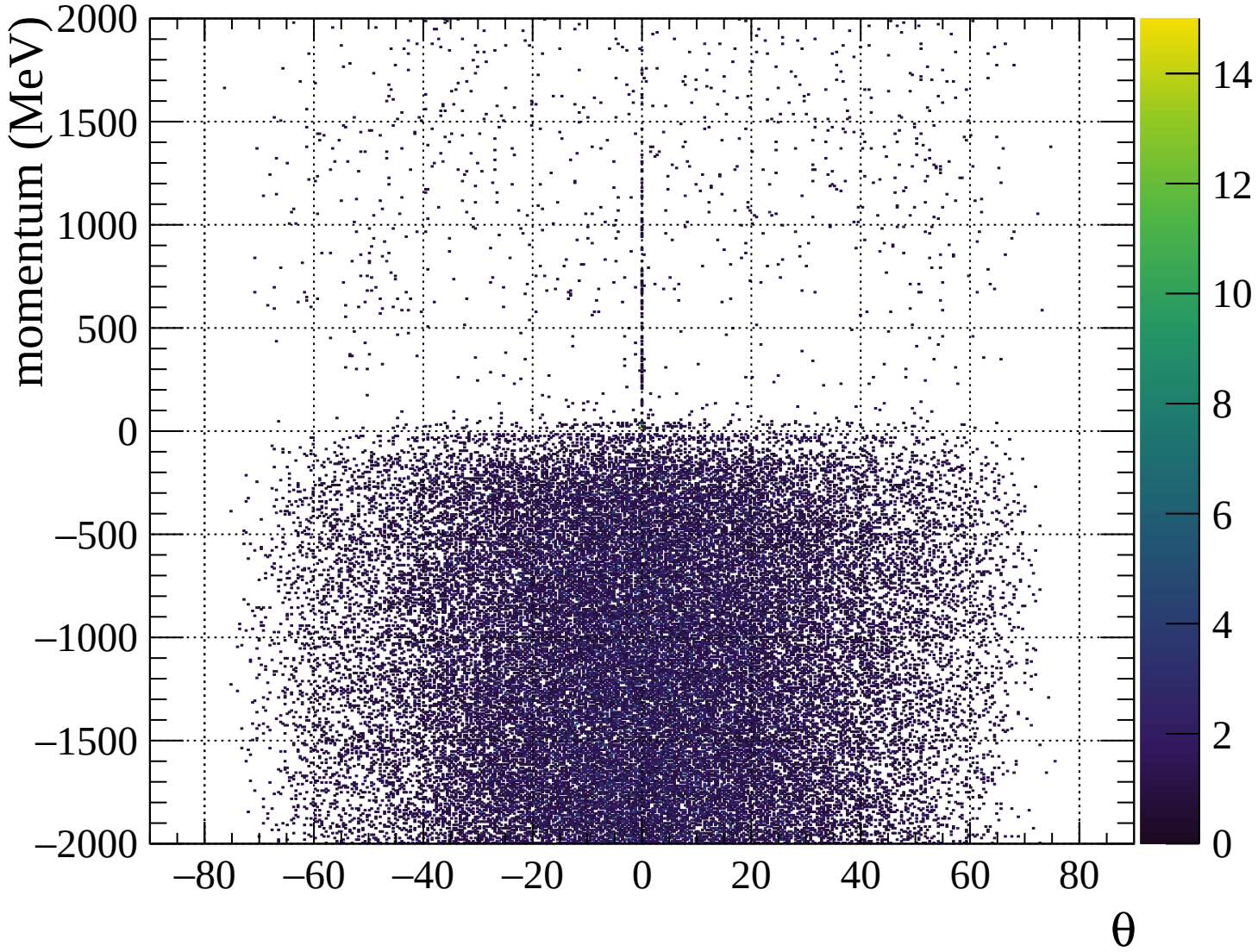


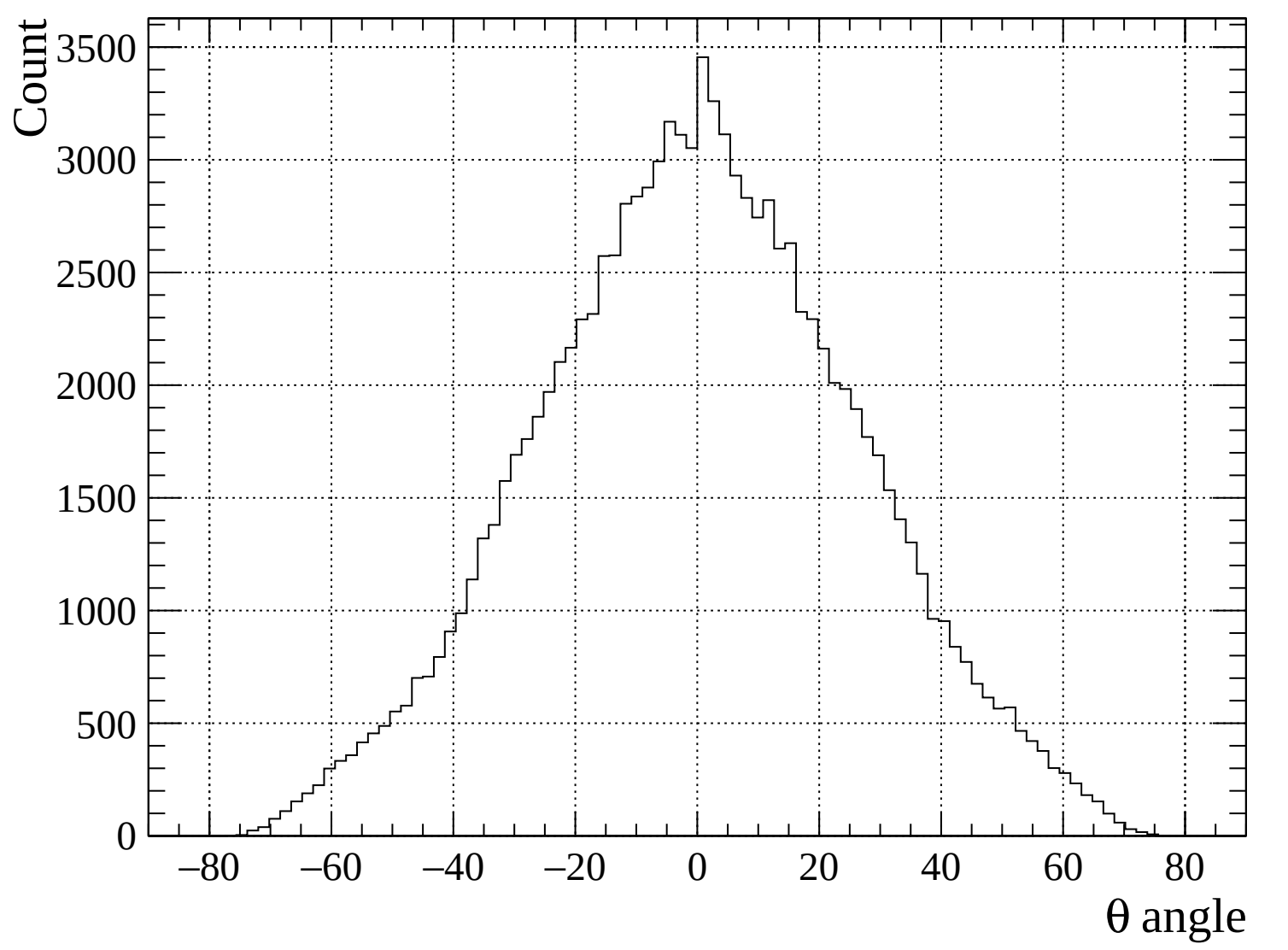


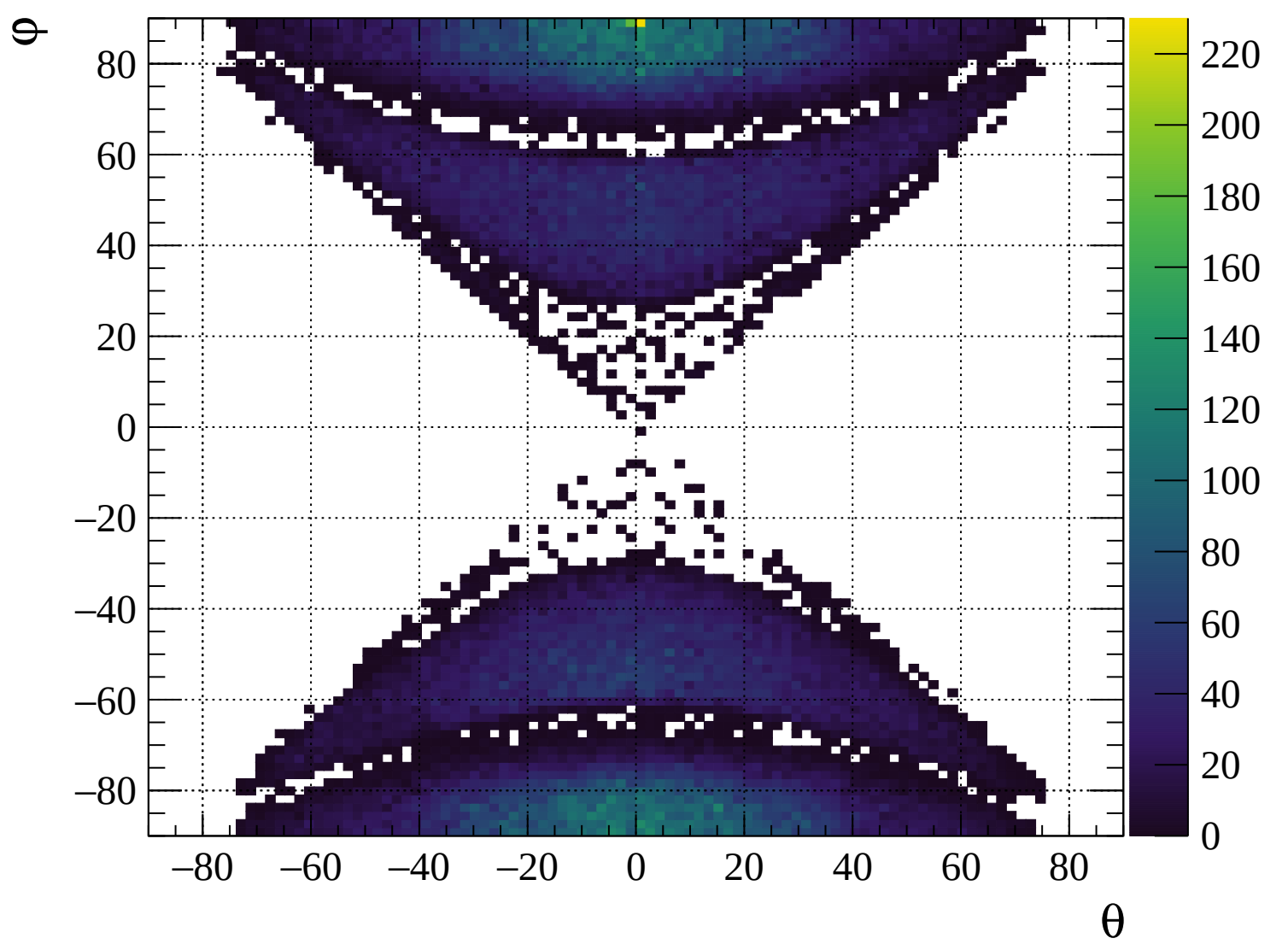


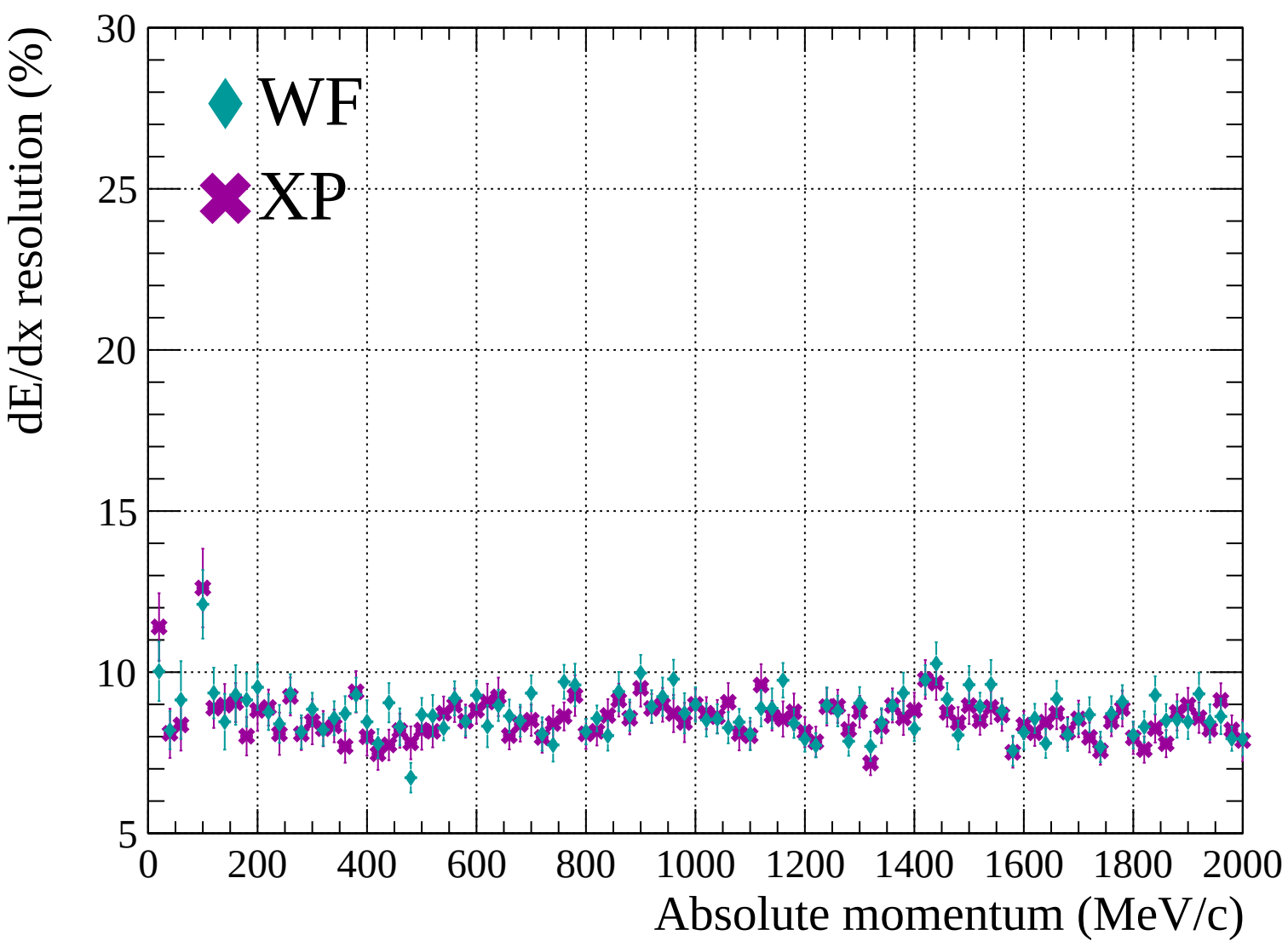


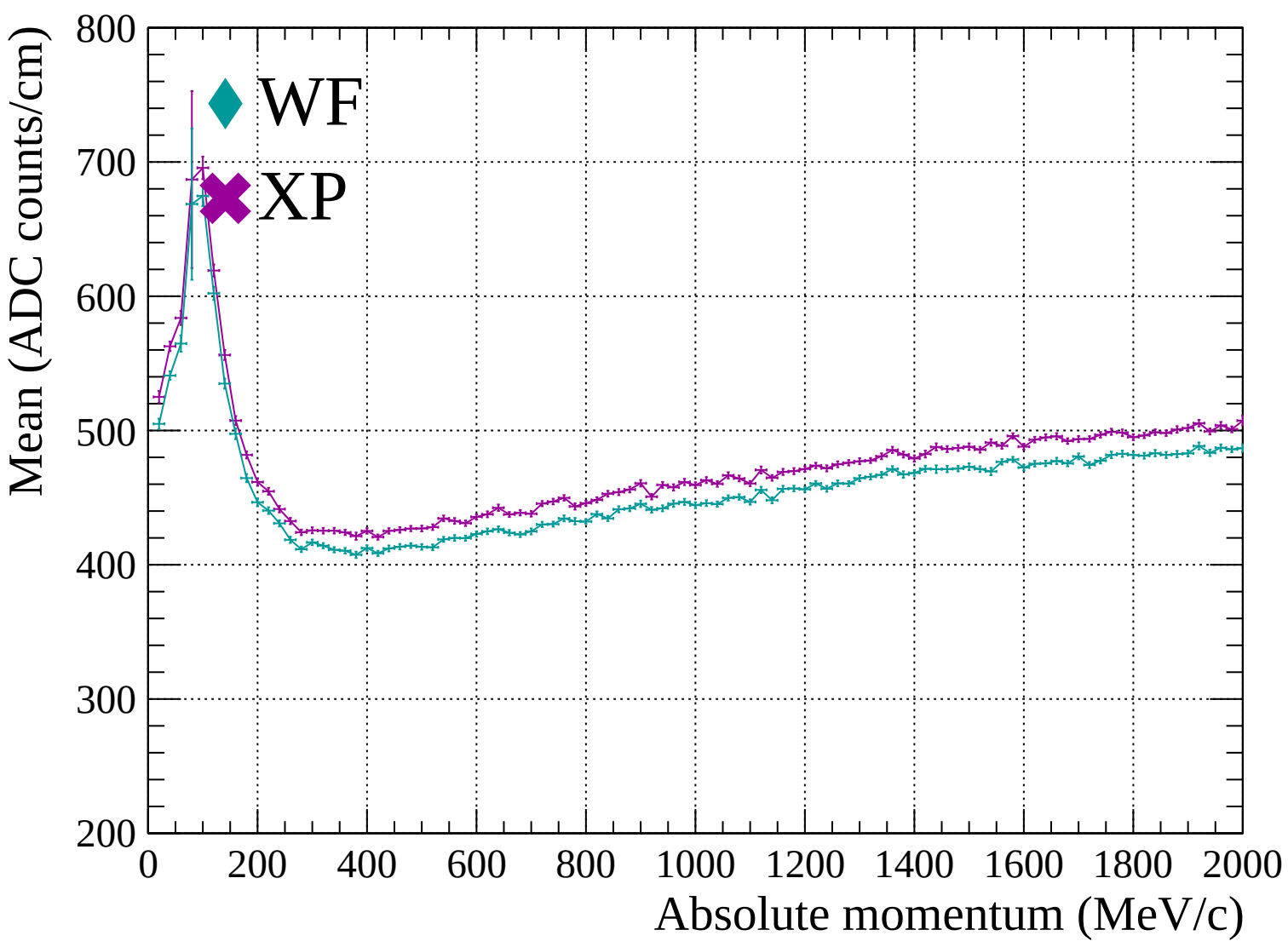


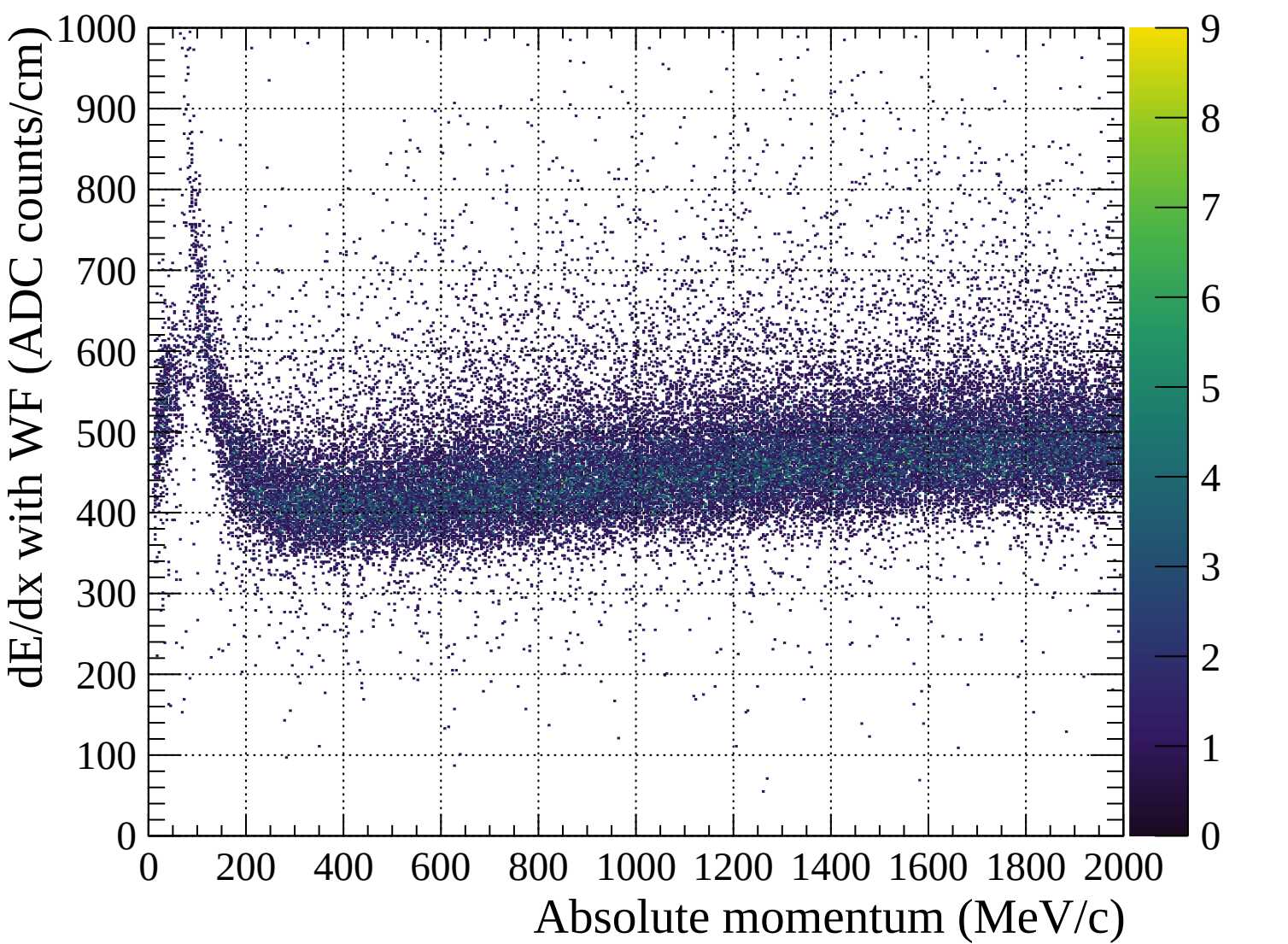


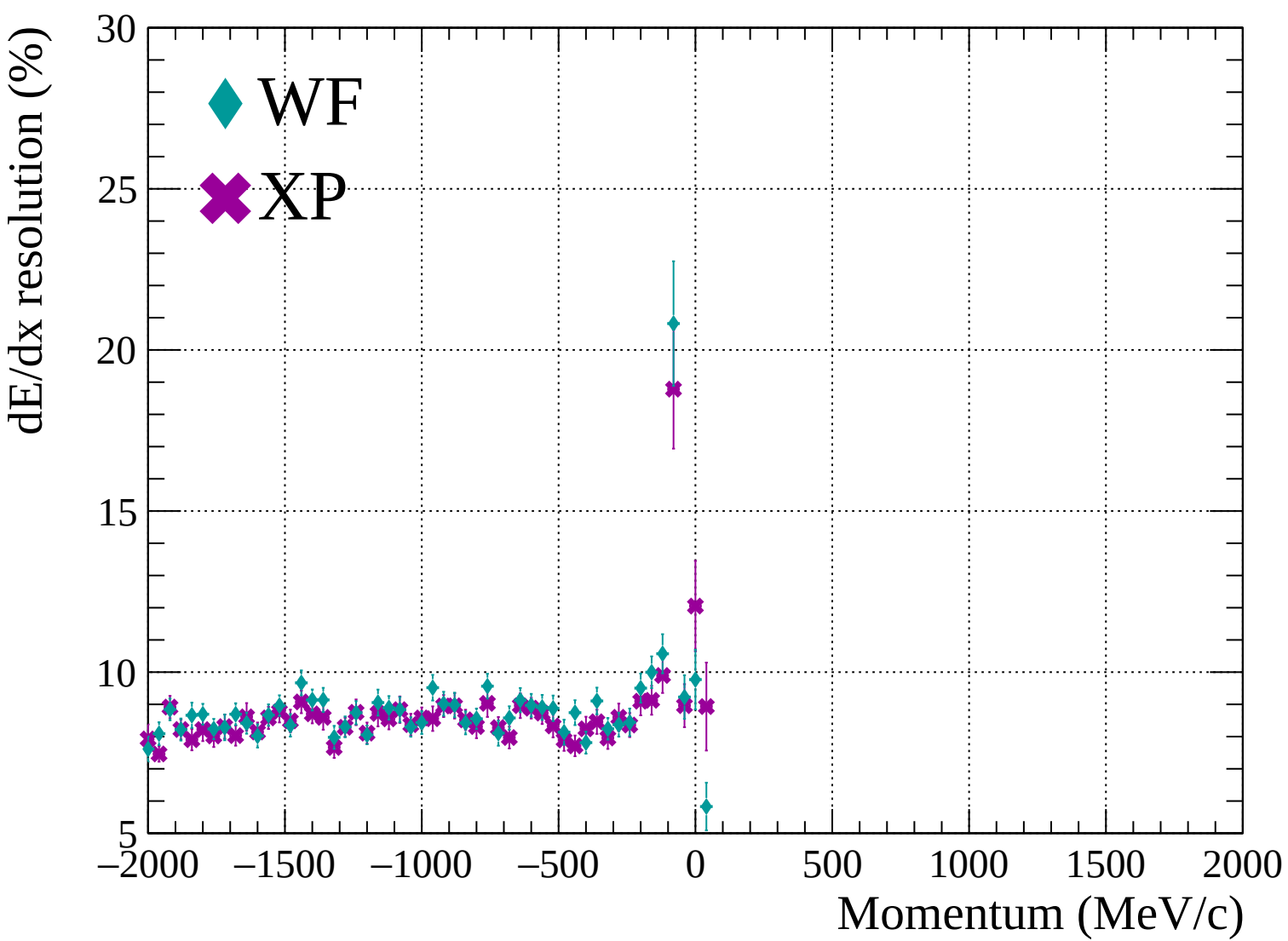


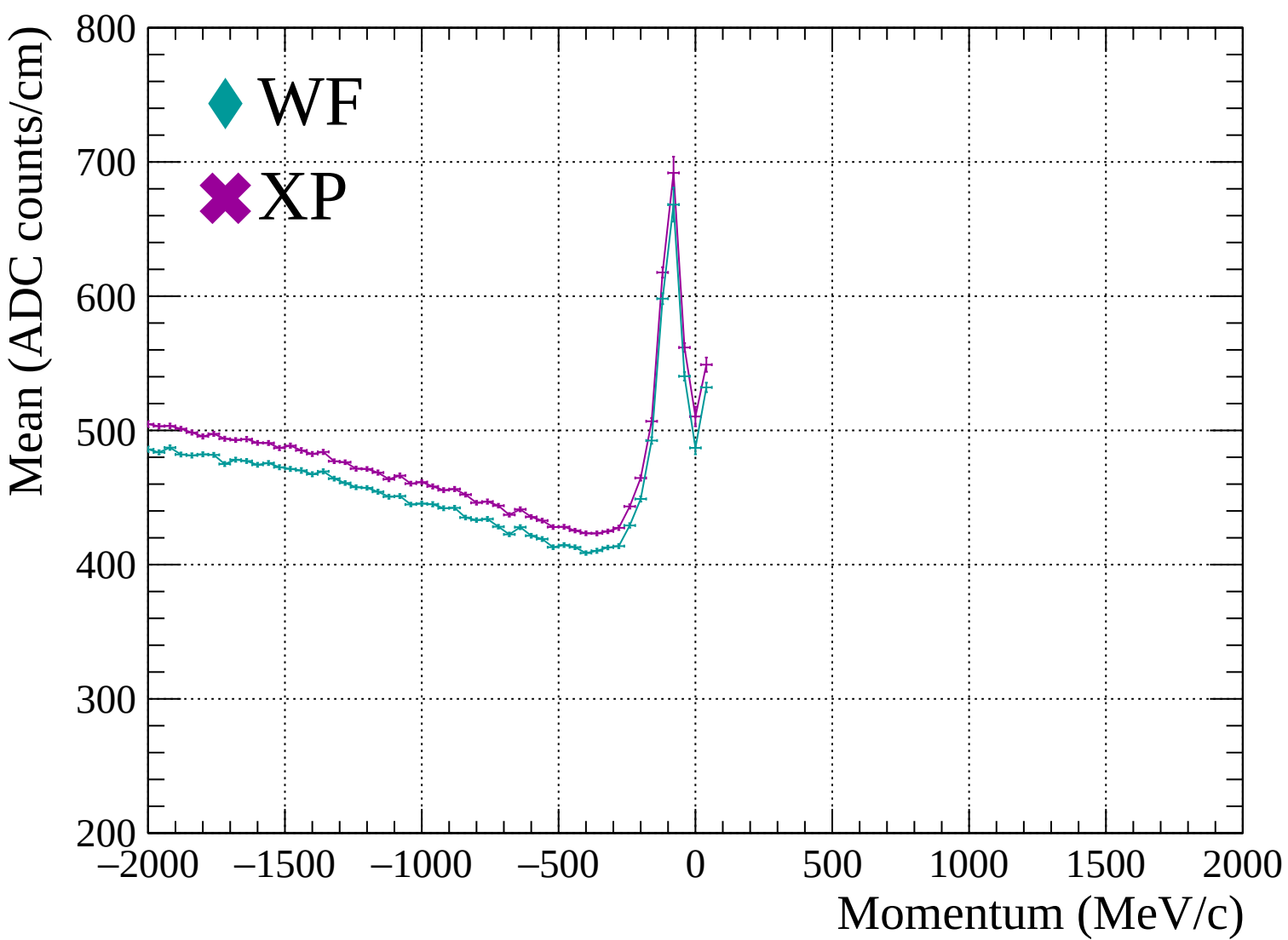


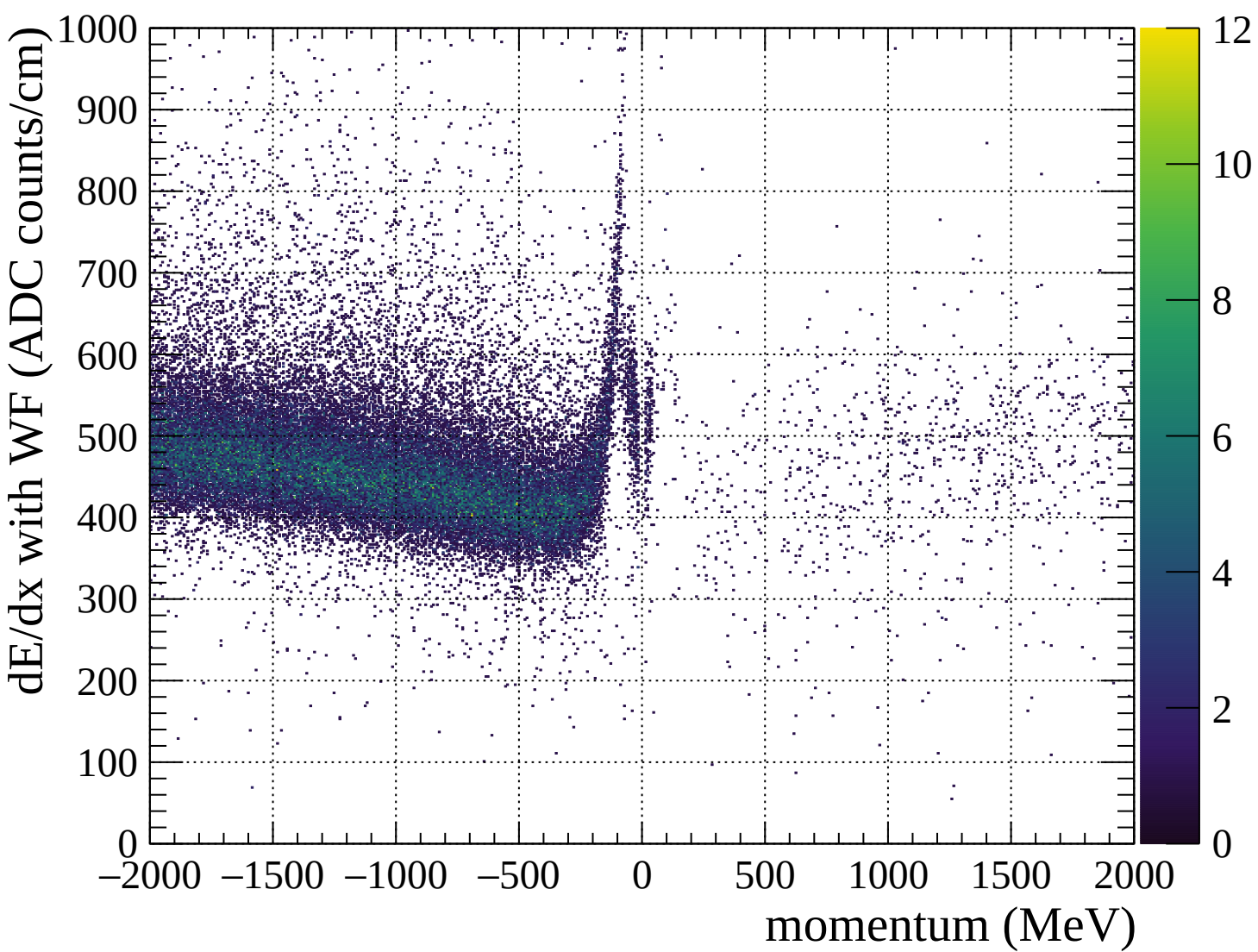


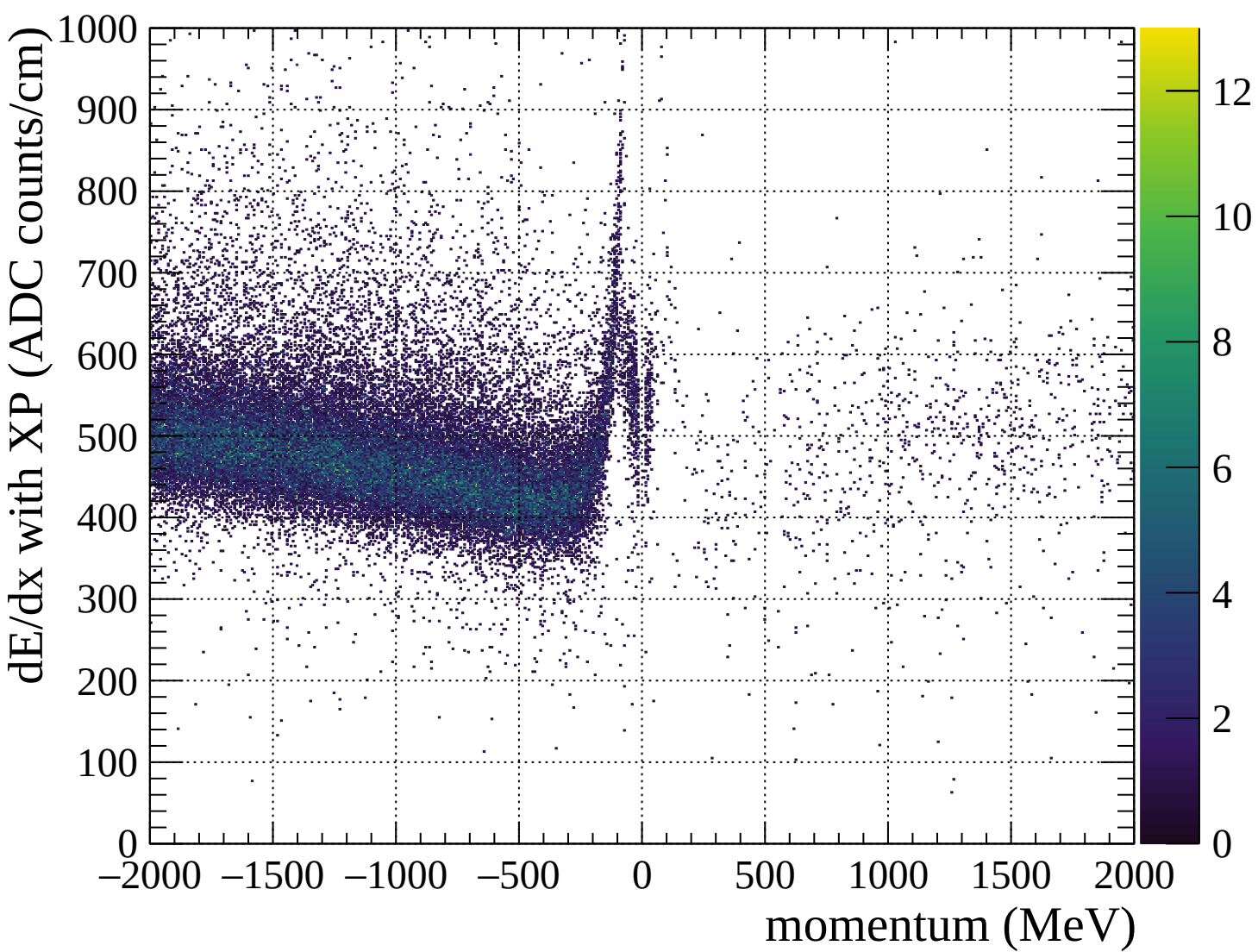


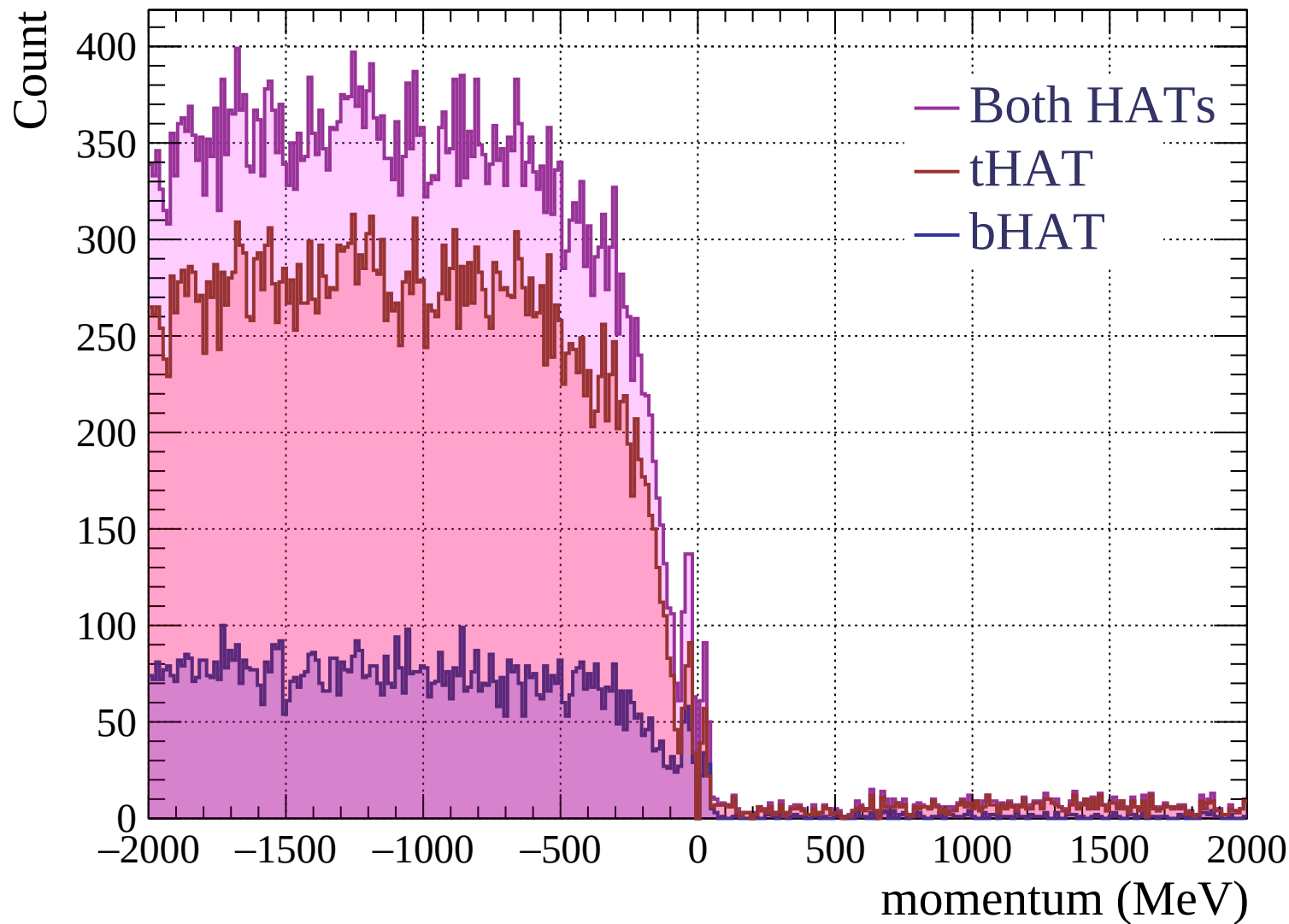


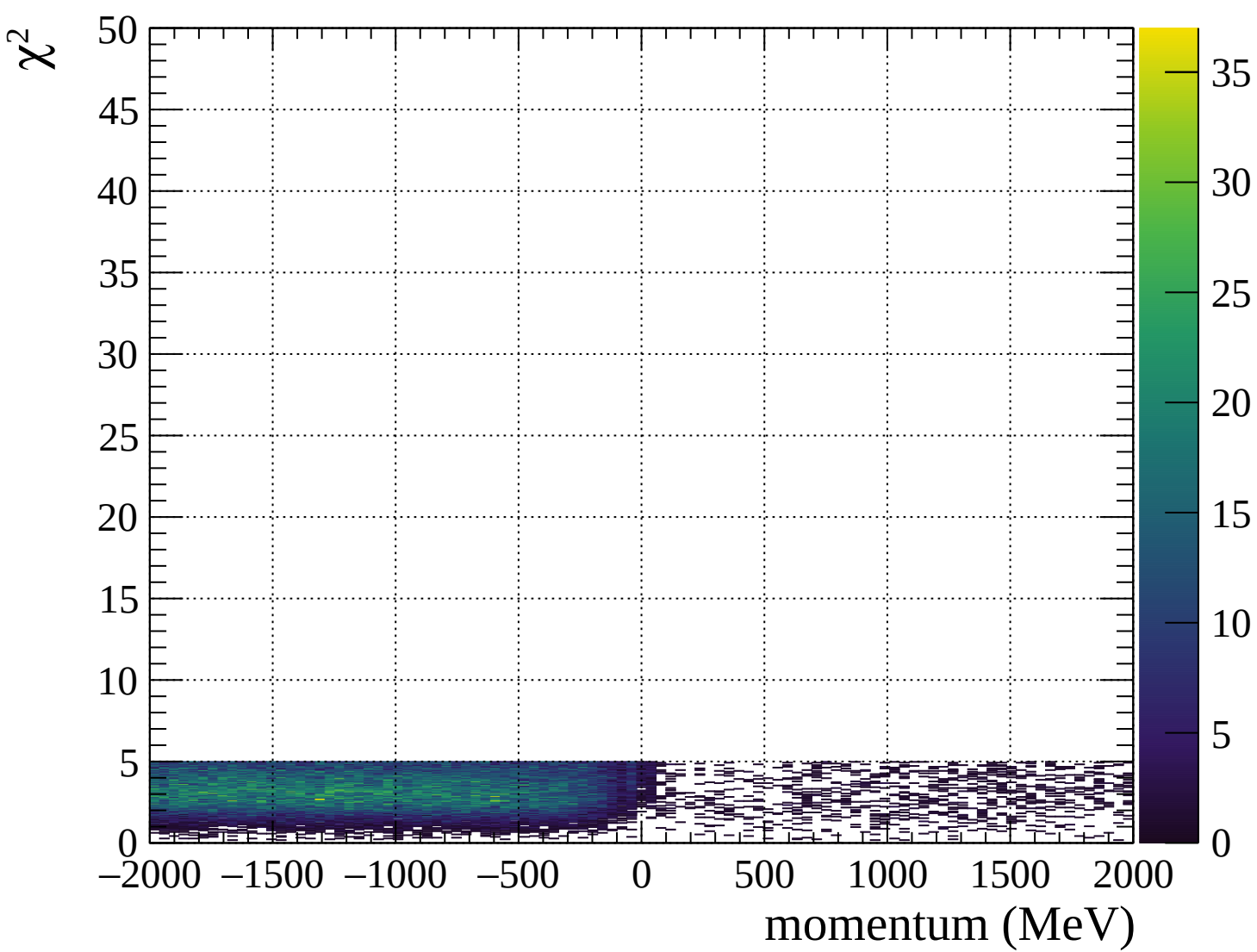




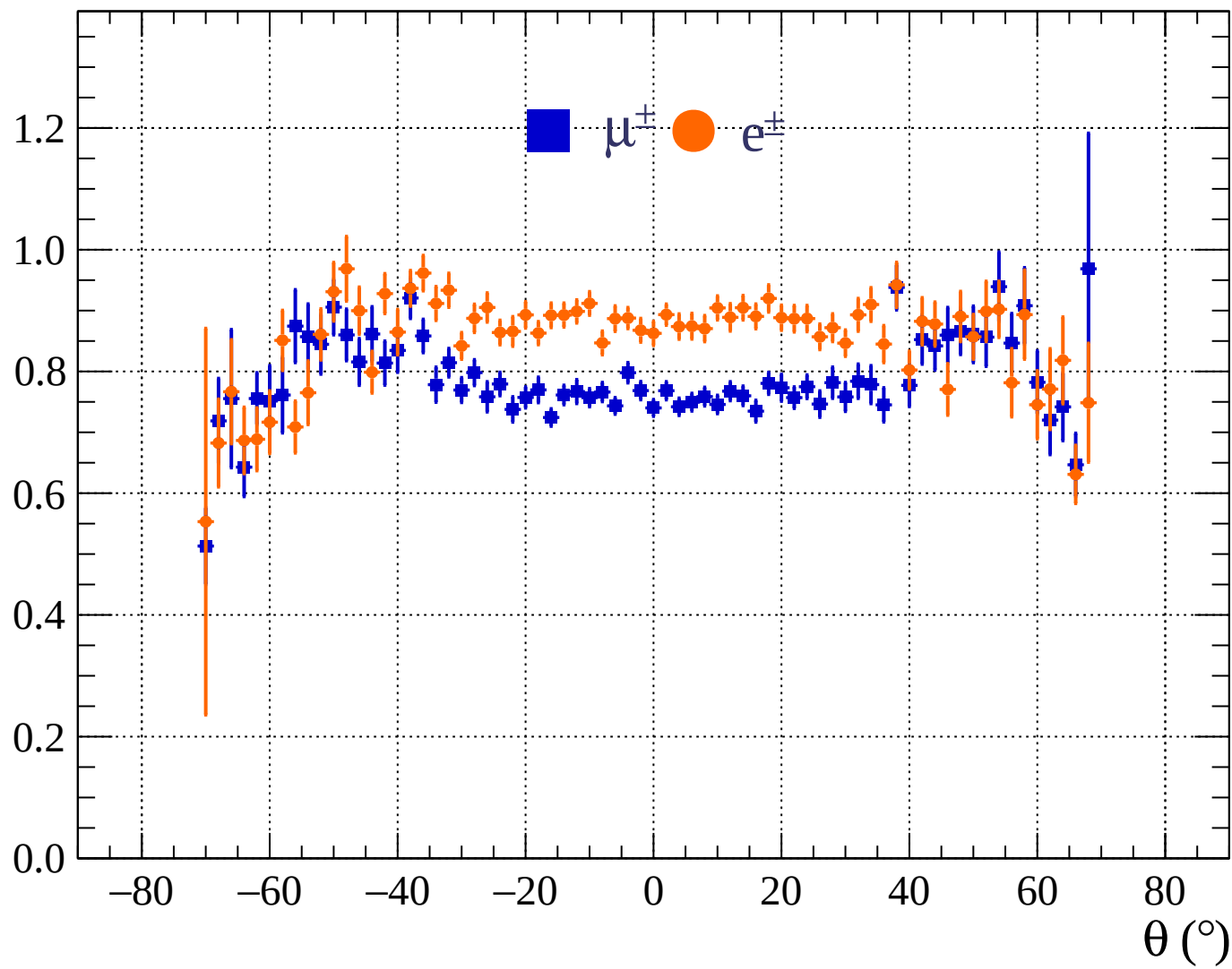


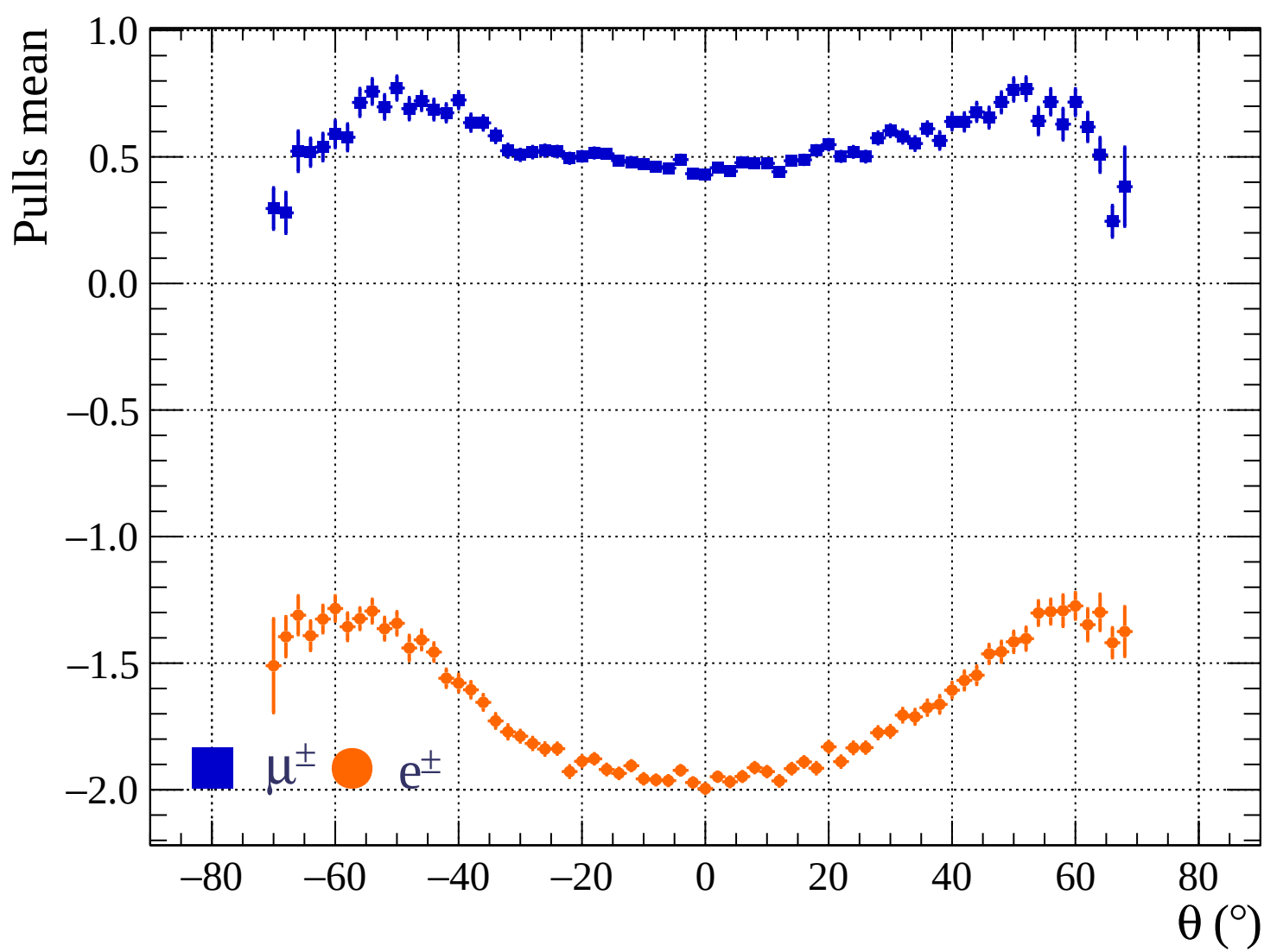






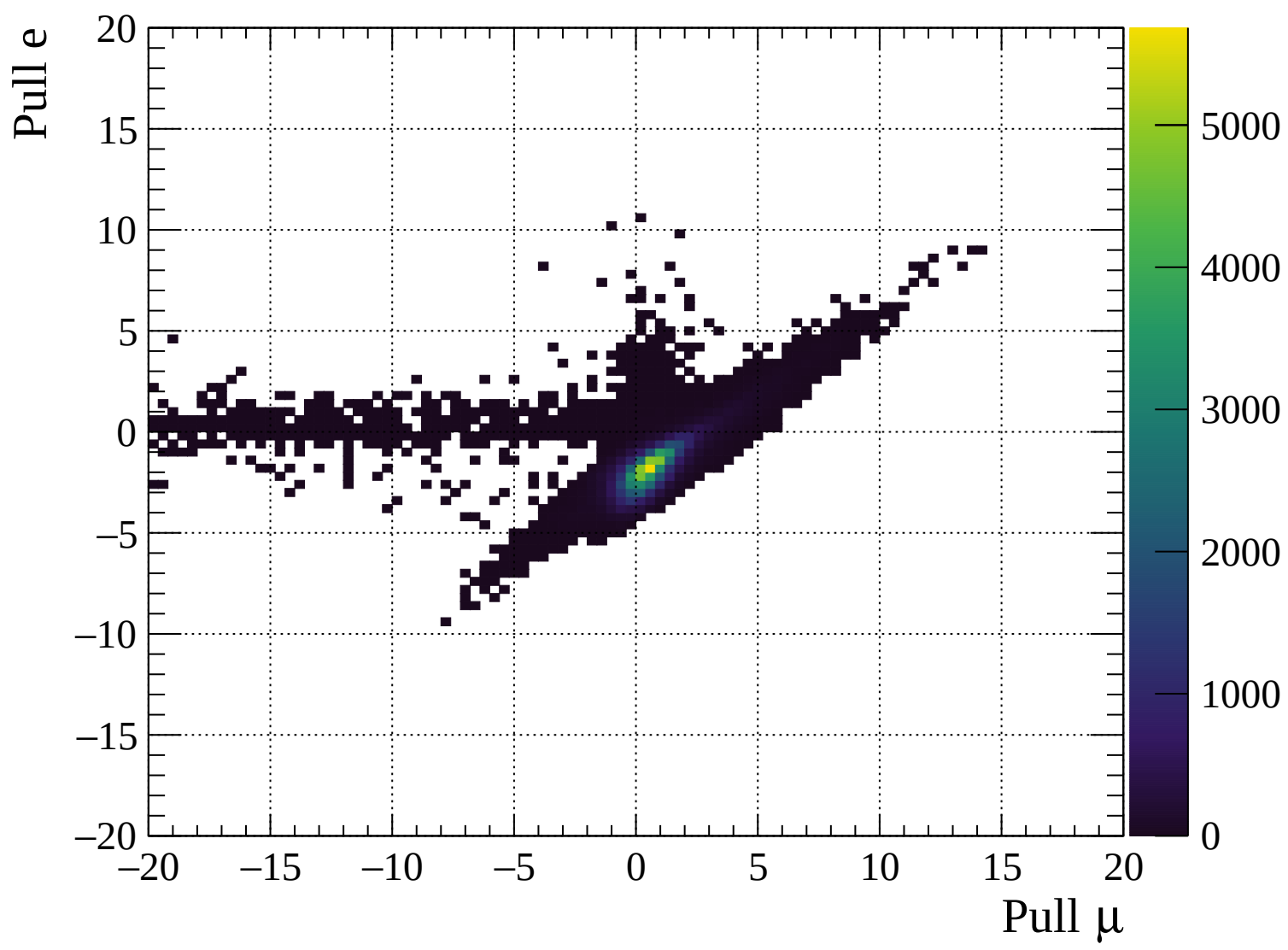
Pulls standard deviation

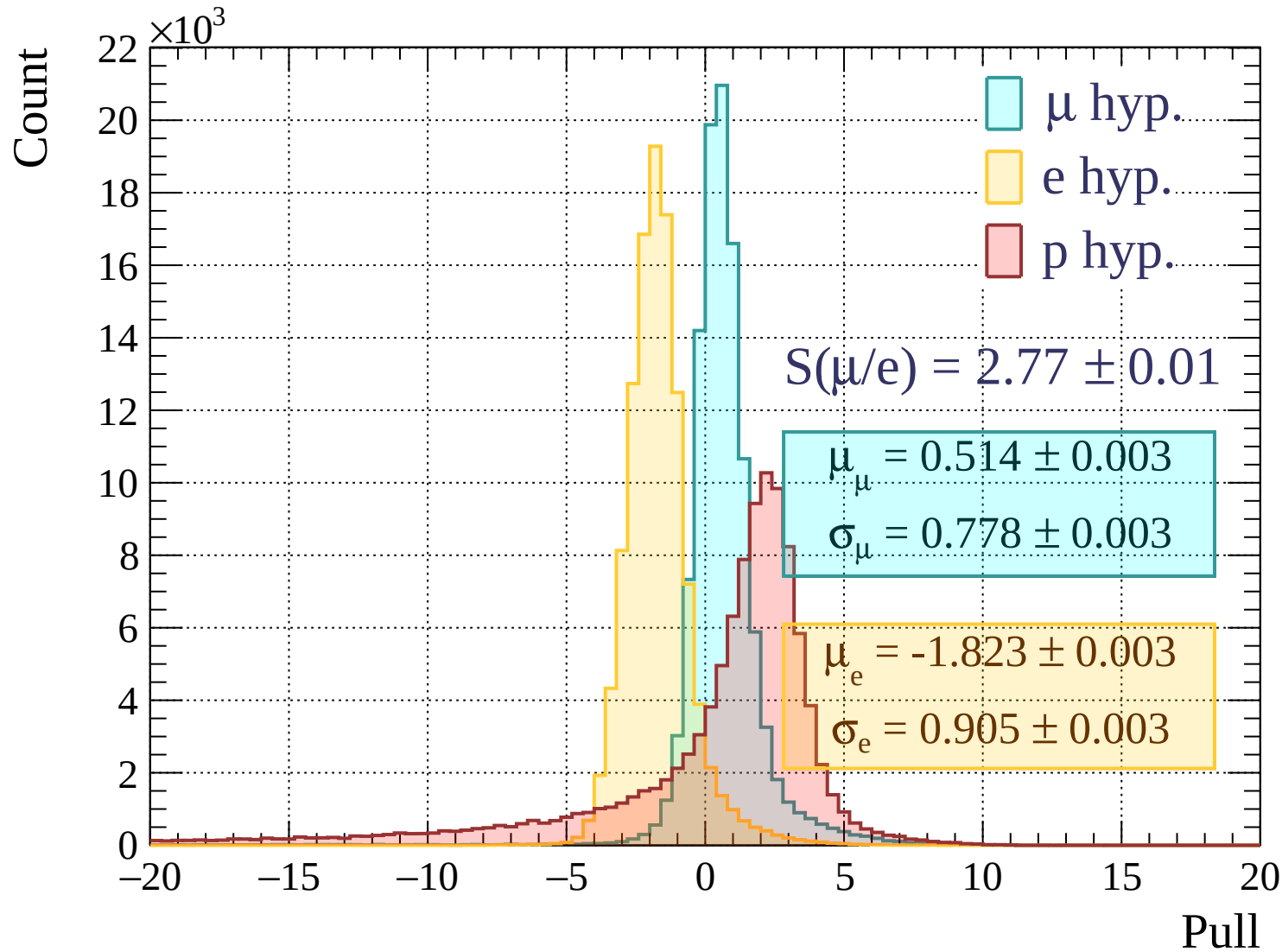


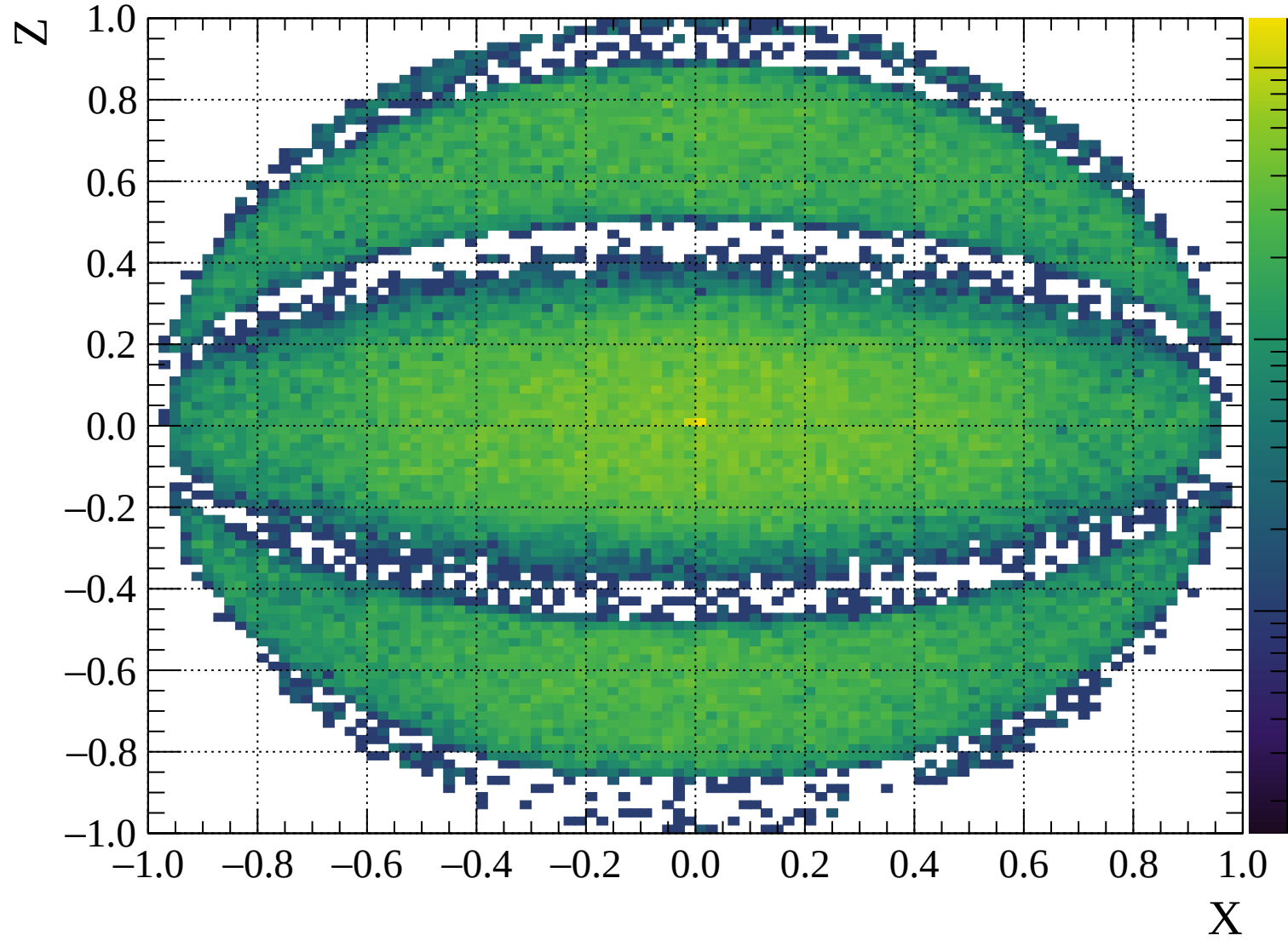


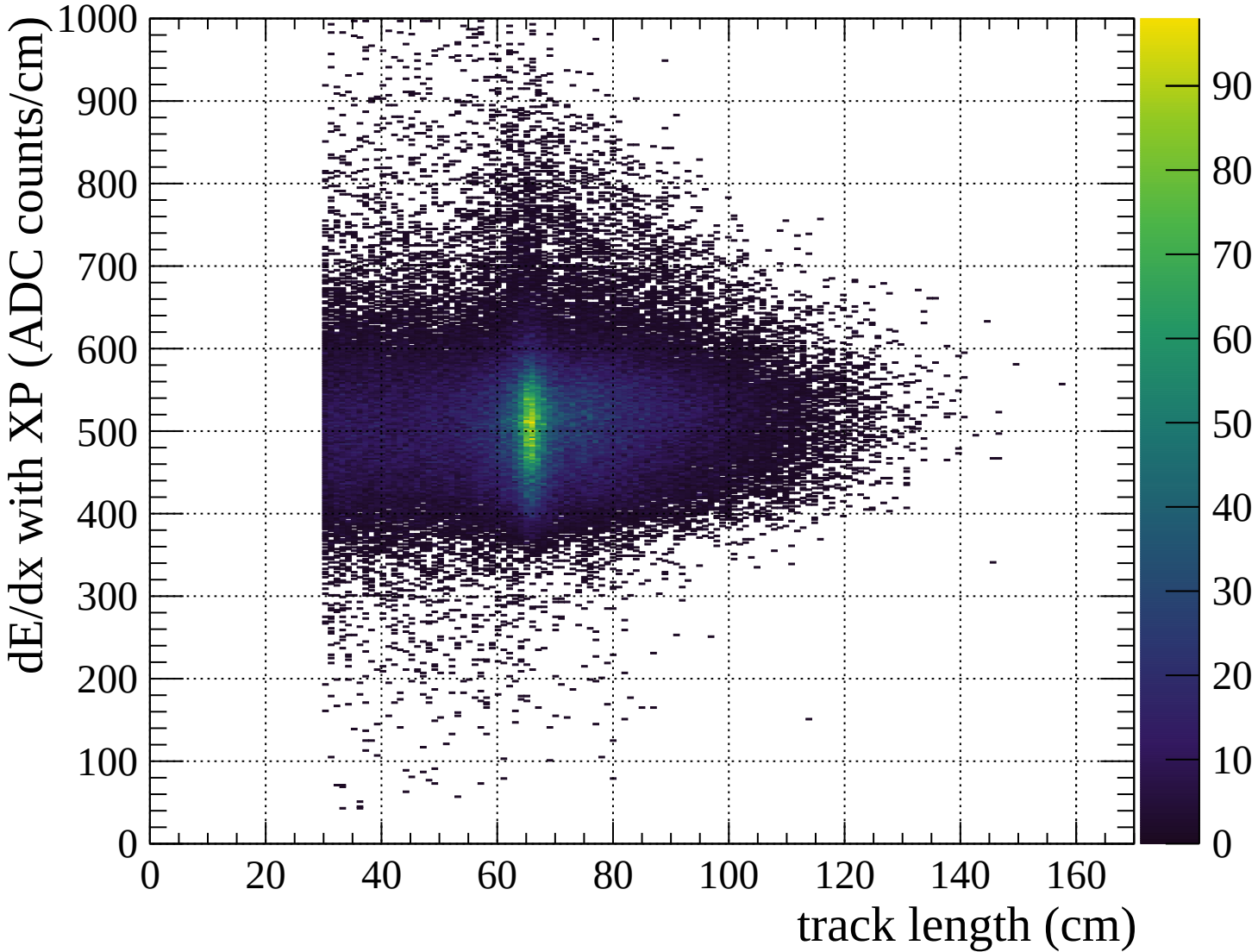
■ $\mu^\pm$ ● $e^\pm$

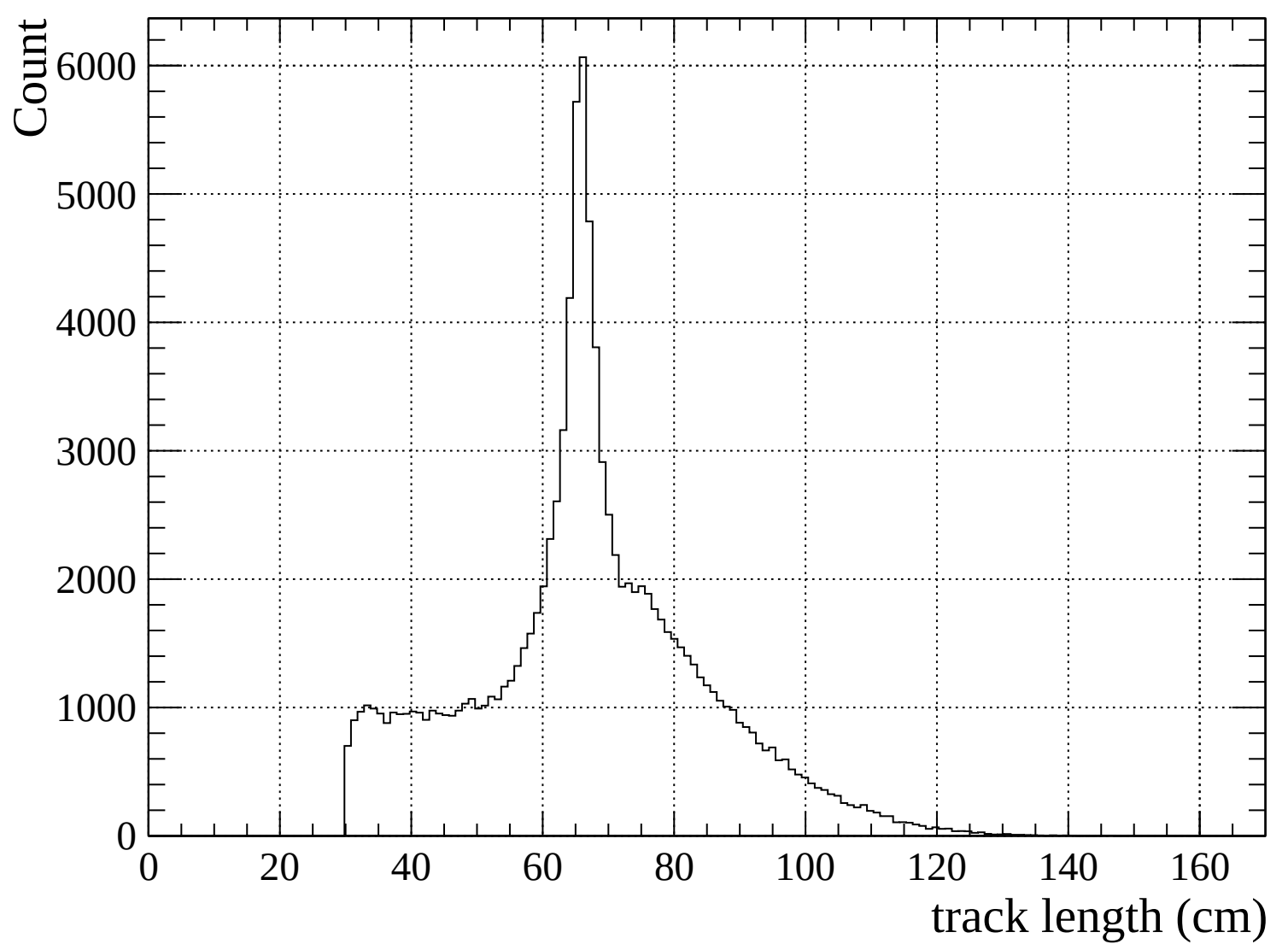
■ $\mu^\pm$ ● $e^\pm$

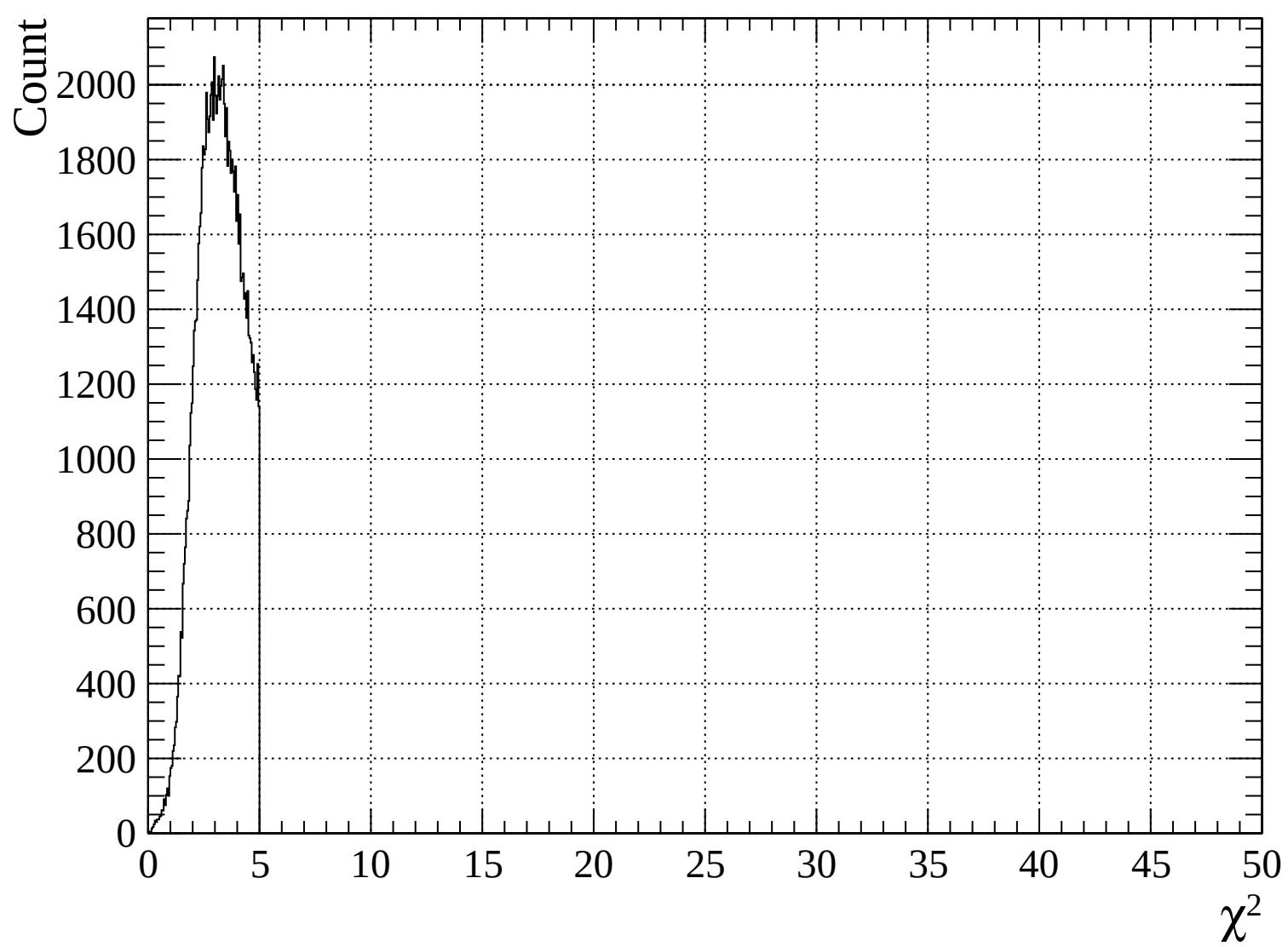


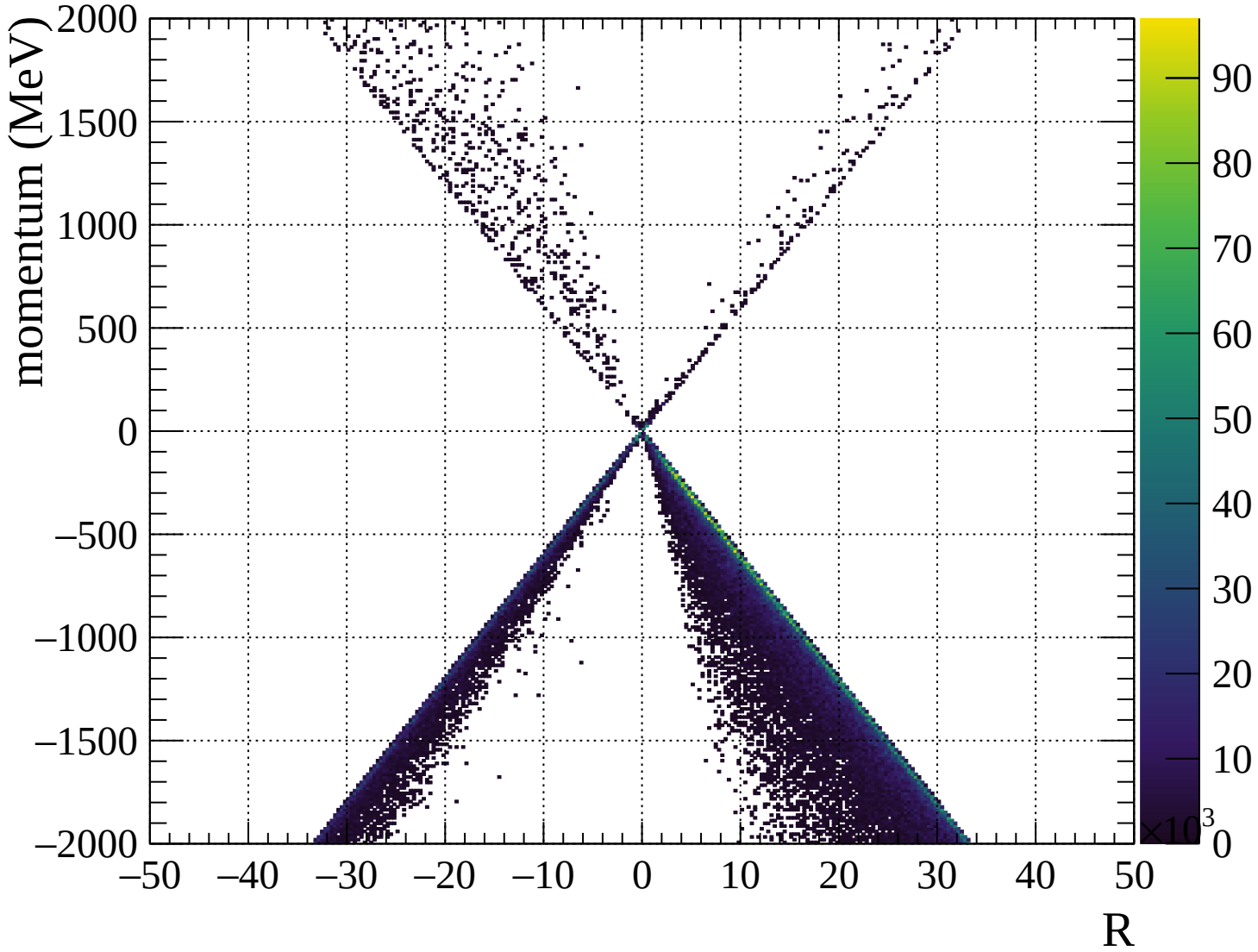


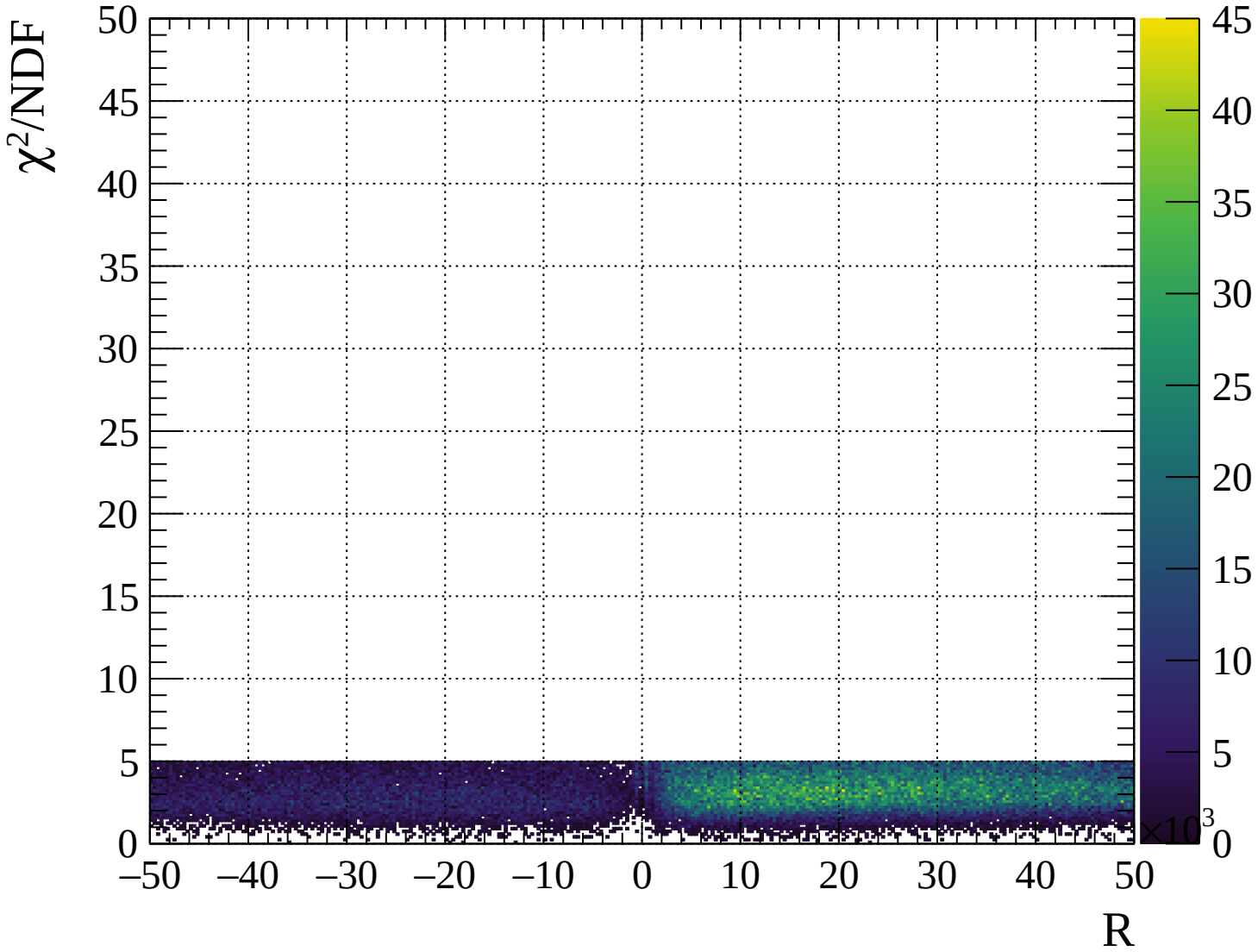






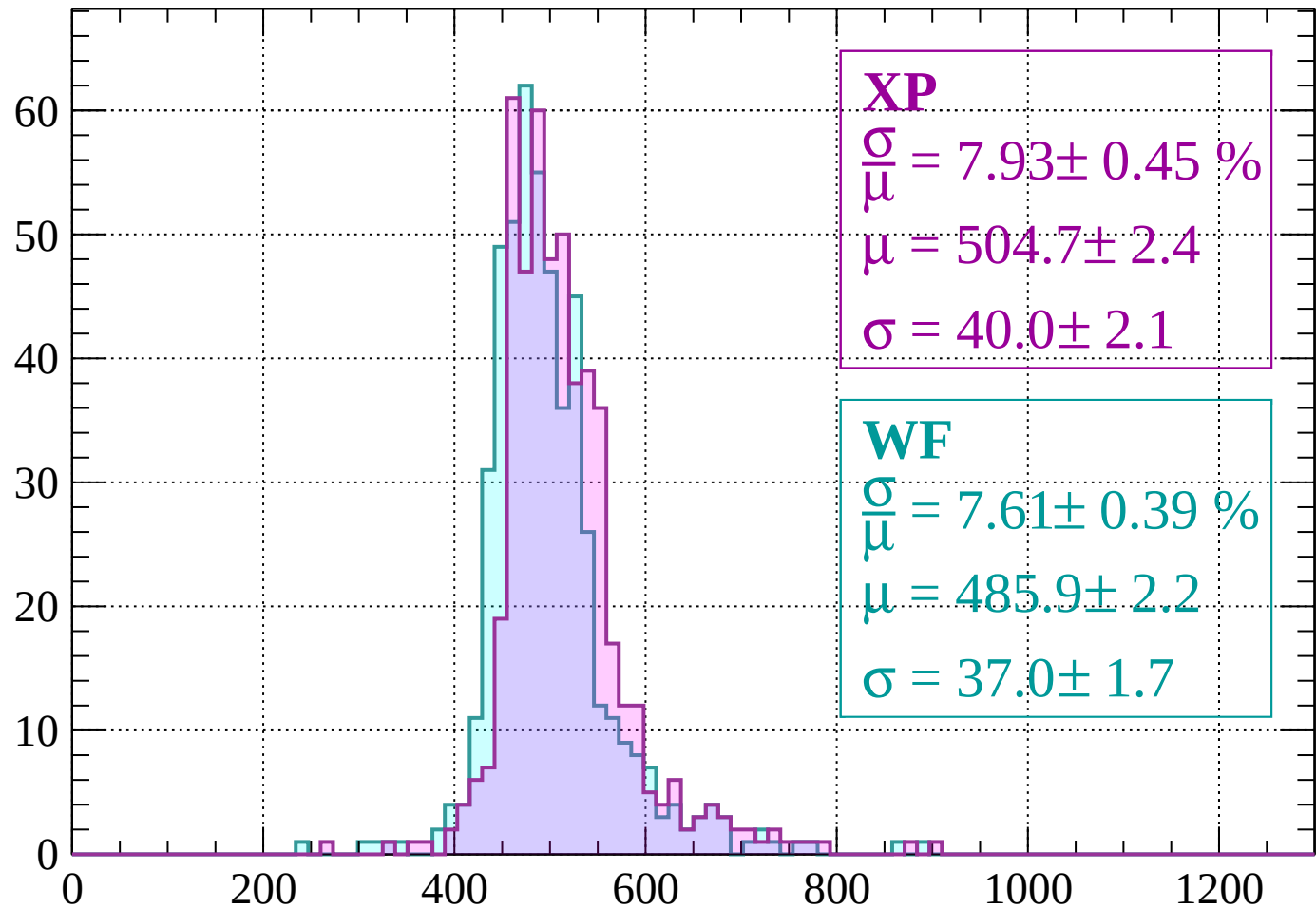






Energy loss | $-2000 < p < -1960$

Count



XP

$$\frac{\sigma}{\mu} = 7.93 \pm 0.45 \%$$

$$\mu = 504.7 \pm 2.4$$

$$\sigma = 40.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.39 \%$$

$$\mu = 485.9 \pm 2.2$$

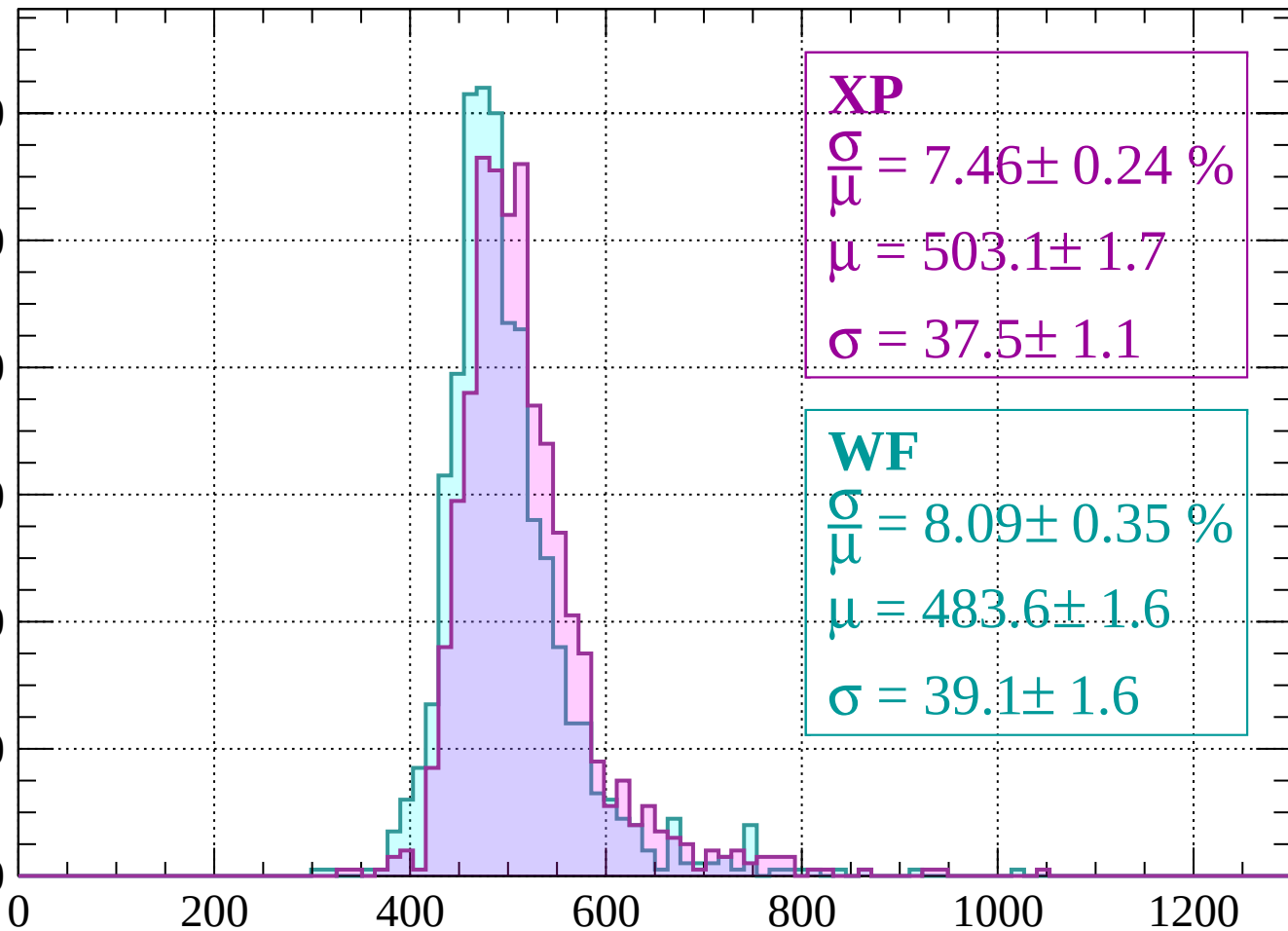
$$\sigma = 37.0 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.24 \%$$

$$\mu = 503.1 \pm 1.7$$

$$\sigma = 37.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.09 \pm 0.35 \%$$

$$\mu = 483.6 \pm 1.6$$

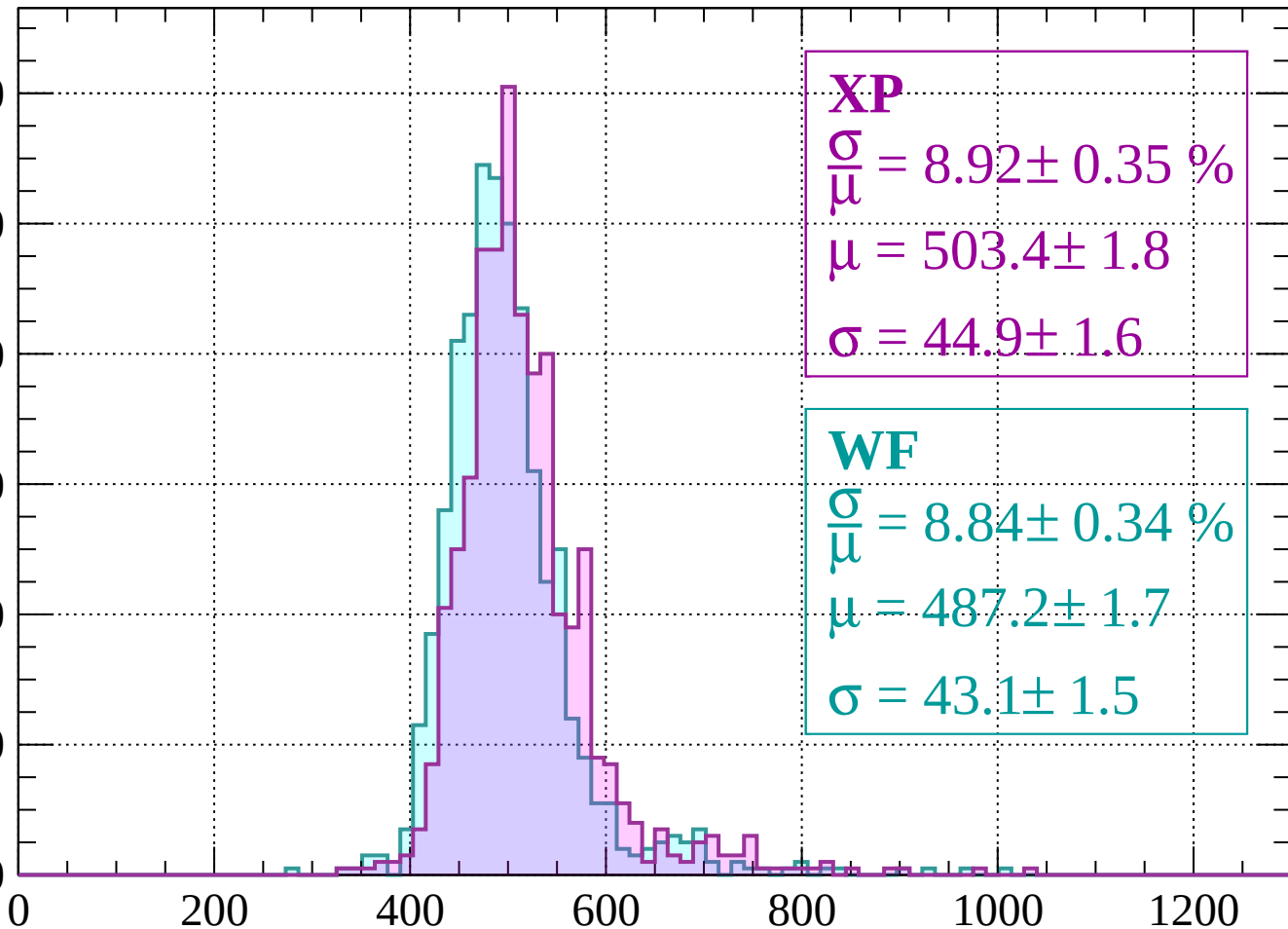
$$\sigma = 39.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.35 \%$$

$$\mu = 503.4 \pm 1.8$$

$$\sigma = 44.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.34 \%$$

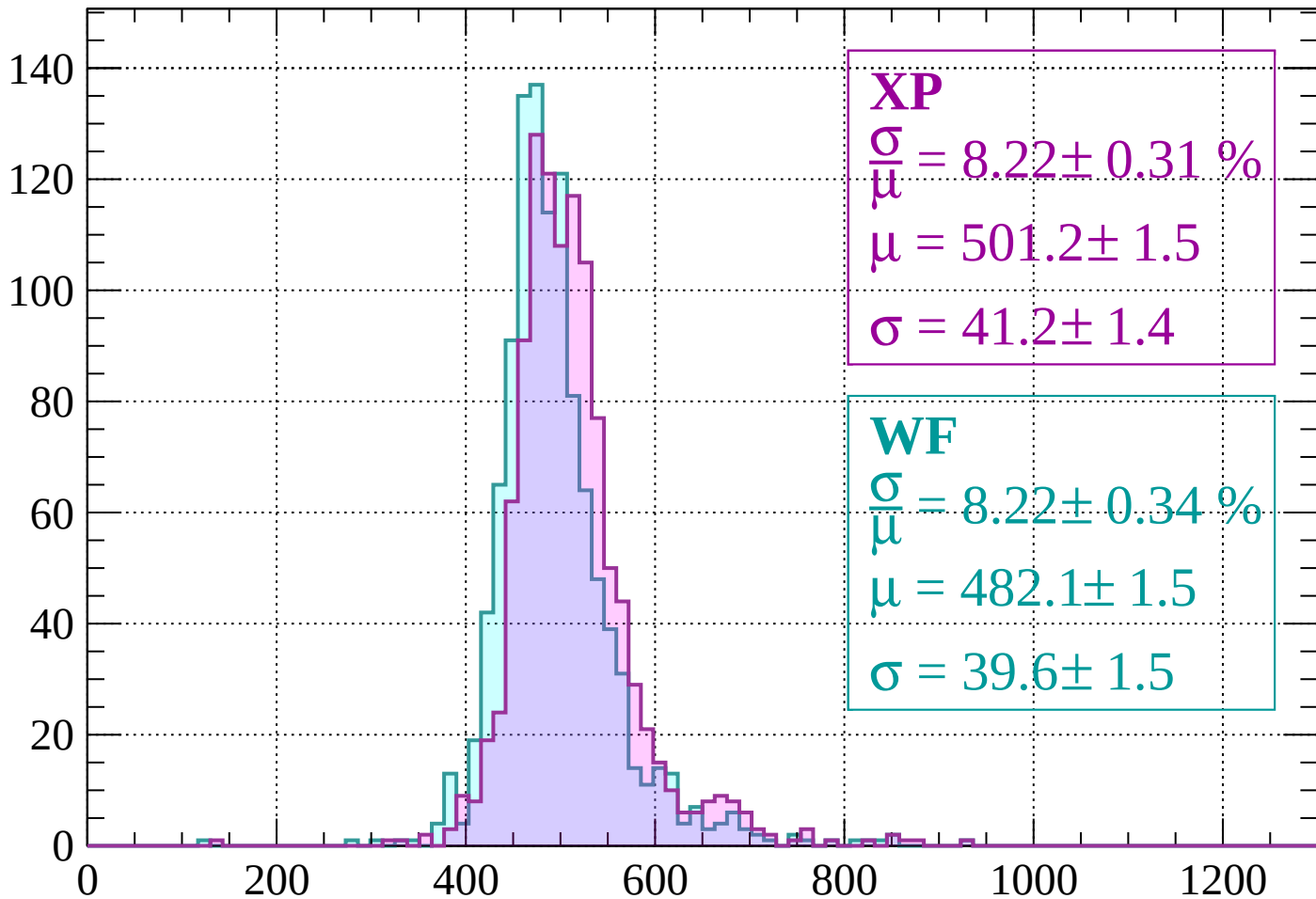
$$\mu = 487.2 \pm 1.7$$

$$\sigma = 43.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.31 \%$$

$$\mu = 501.2 \pm 1.5$$

$$\sigma = 41.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.34 \%$$

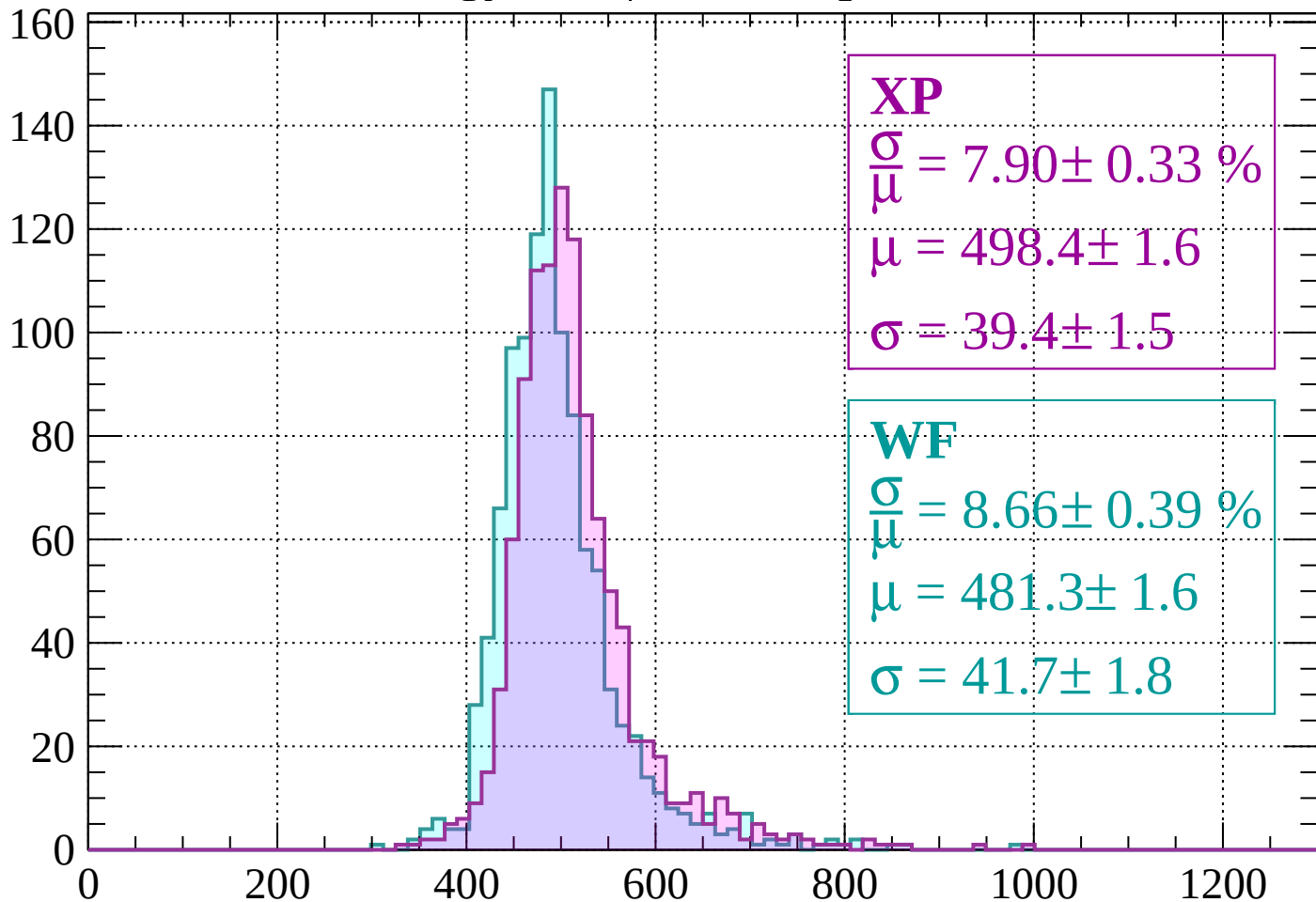
$$\mu = 482.1 \pm 1.5$$

$$\sigma = 39.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.33 \%$$

$$\mu = 498.4 \pm 1.6$$

$$\sigma = 39.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.39 \%$$

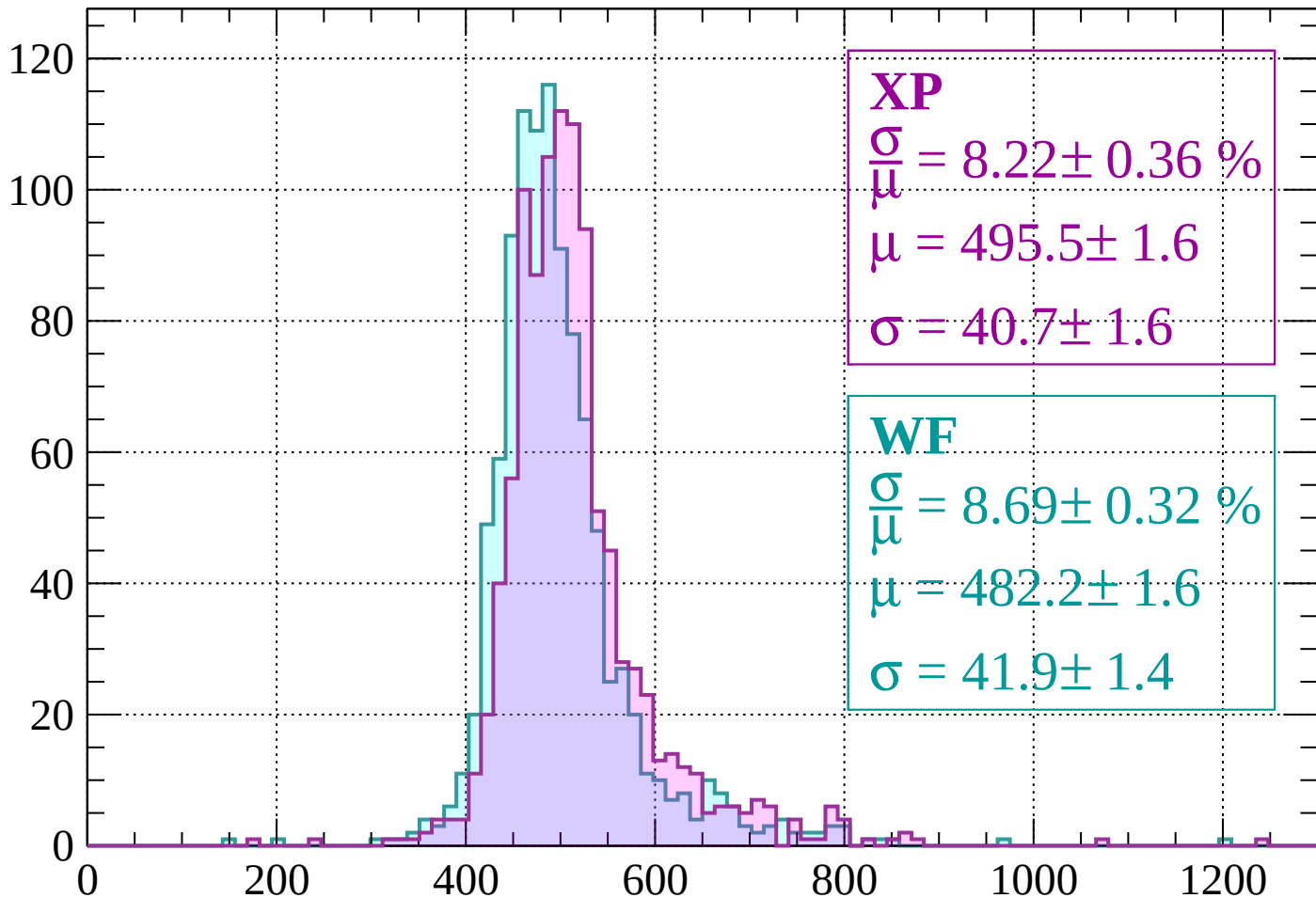
$$\mu = 481.3 \pm 1.6$$

$$\sigma = 41.7 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.36 \%$$

$$\mu = 495.5 \pm 1.6$$

$$\sigma = 40.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.69 \pm 0.32 \%$$

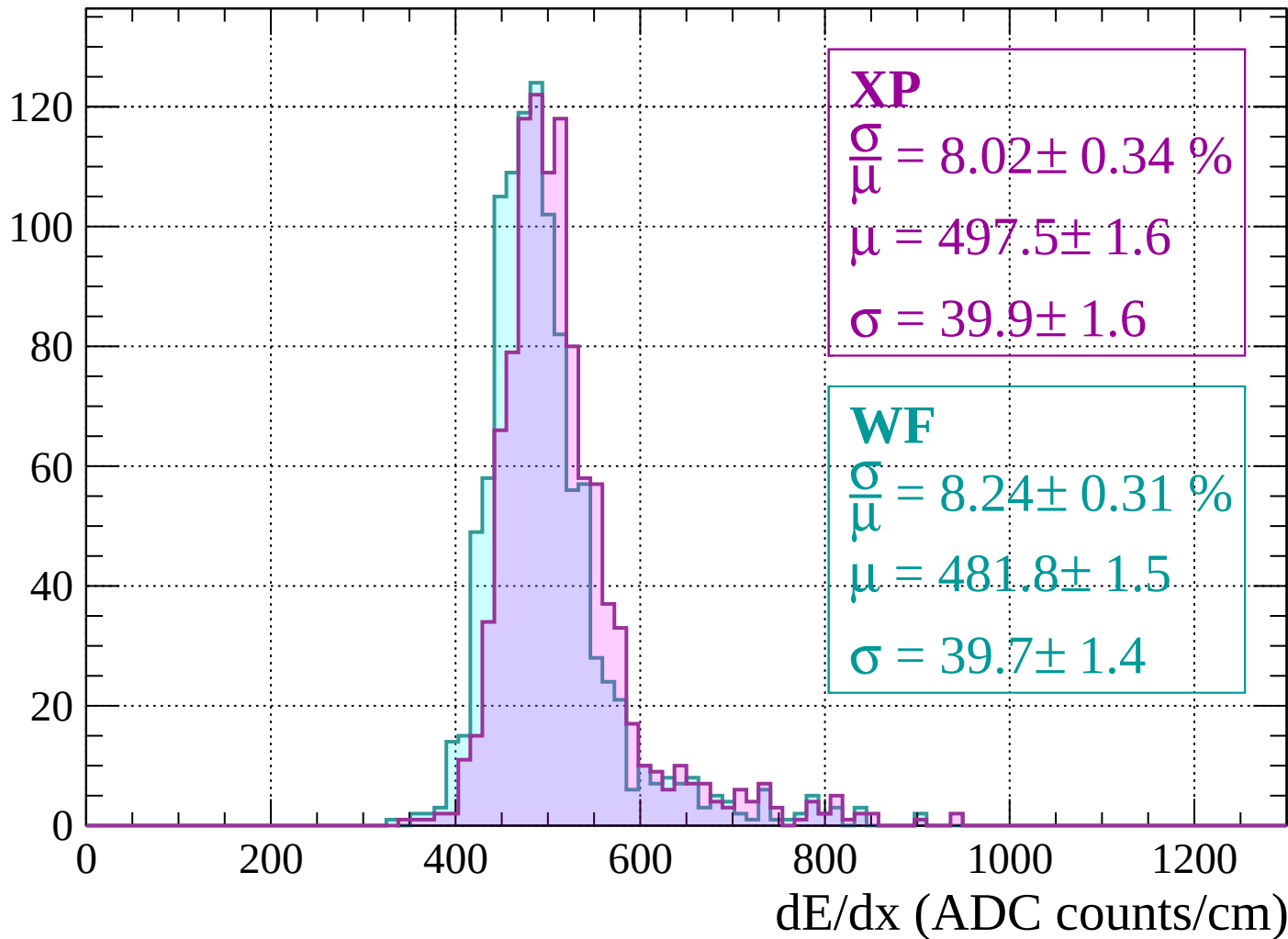
$$\mu = 482.2 \pm 1.6$$

$$\sigma = 41.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

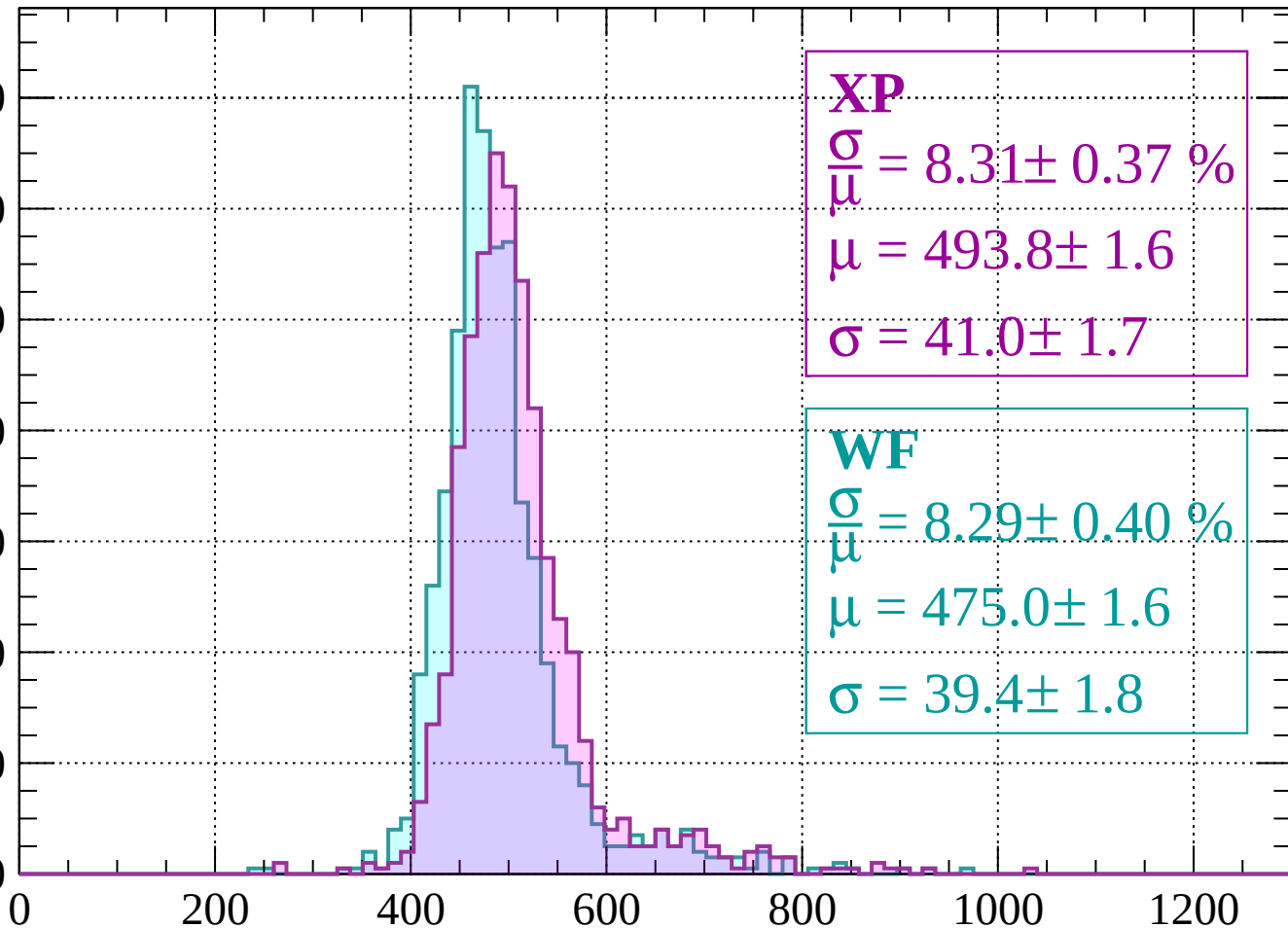
Count



Energy loss | $-1720 < p < -1680$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.31 \pm 0.37 \%$$

$$\mu = 493.8 \pm 1.6$$

$$\sigma = 41.0 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.40 \%$$

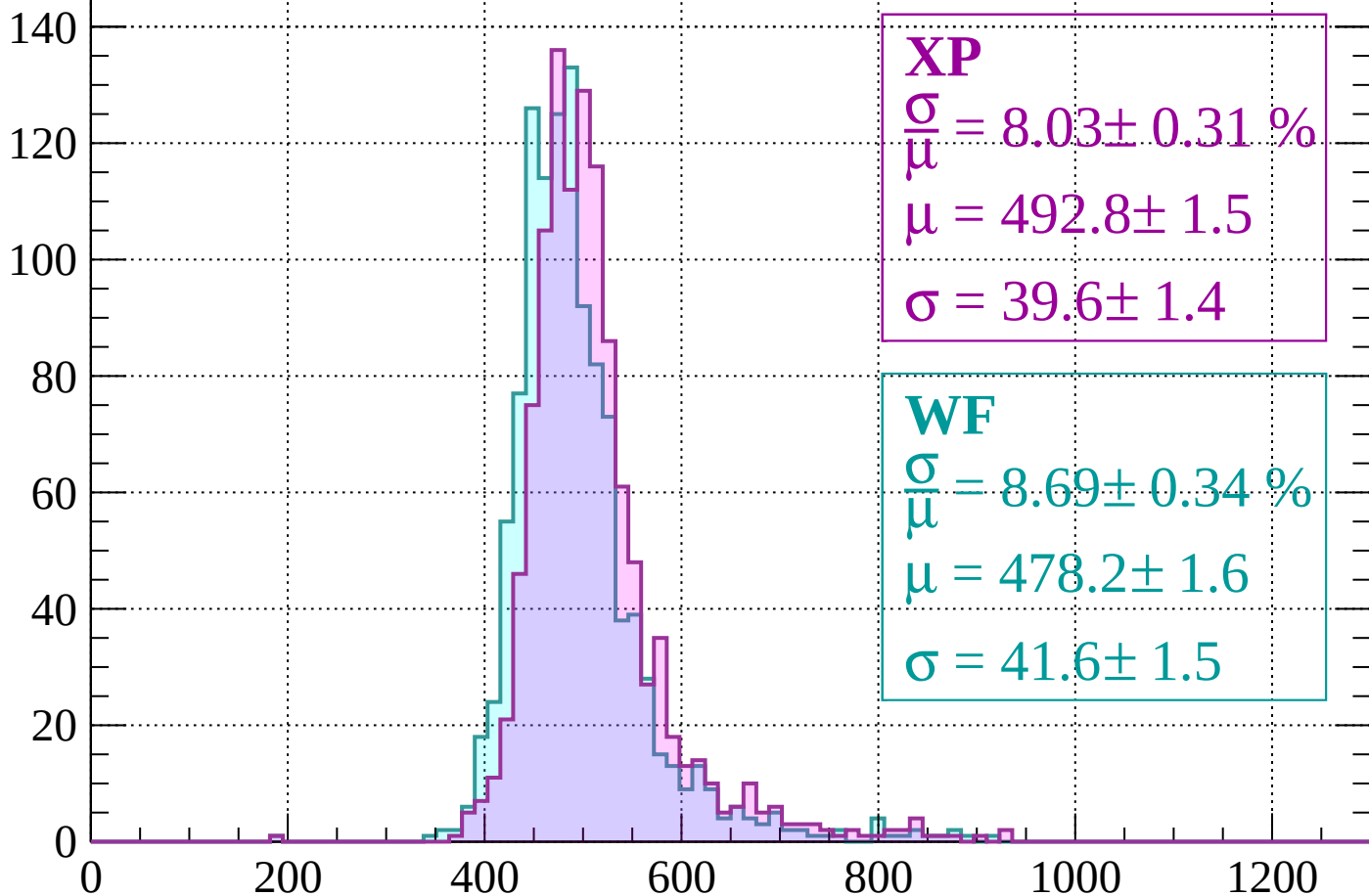
$$\mu = 475.0 \pm 1.6$$

$$\sigma = 39.4 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.31 \%$$

$$\mu = 492.8 \pm 1.5$$

$$\sigma = 39.6 \pm 1.4$$

WF

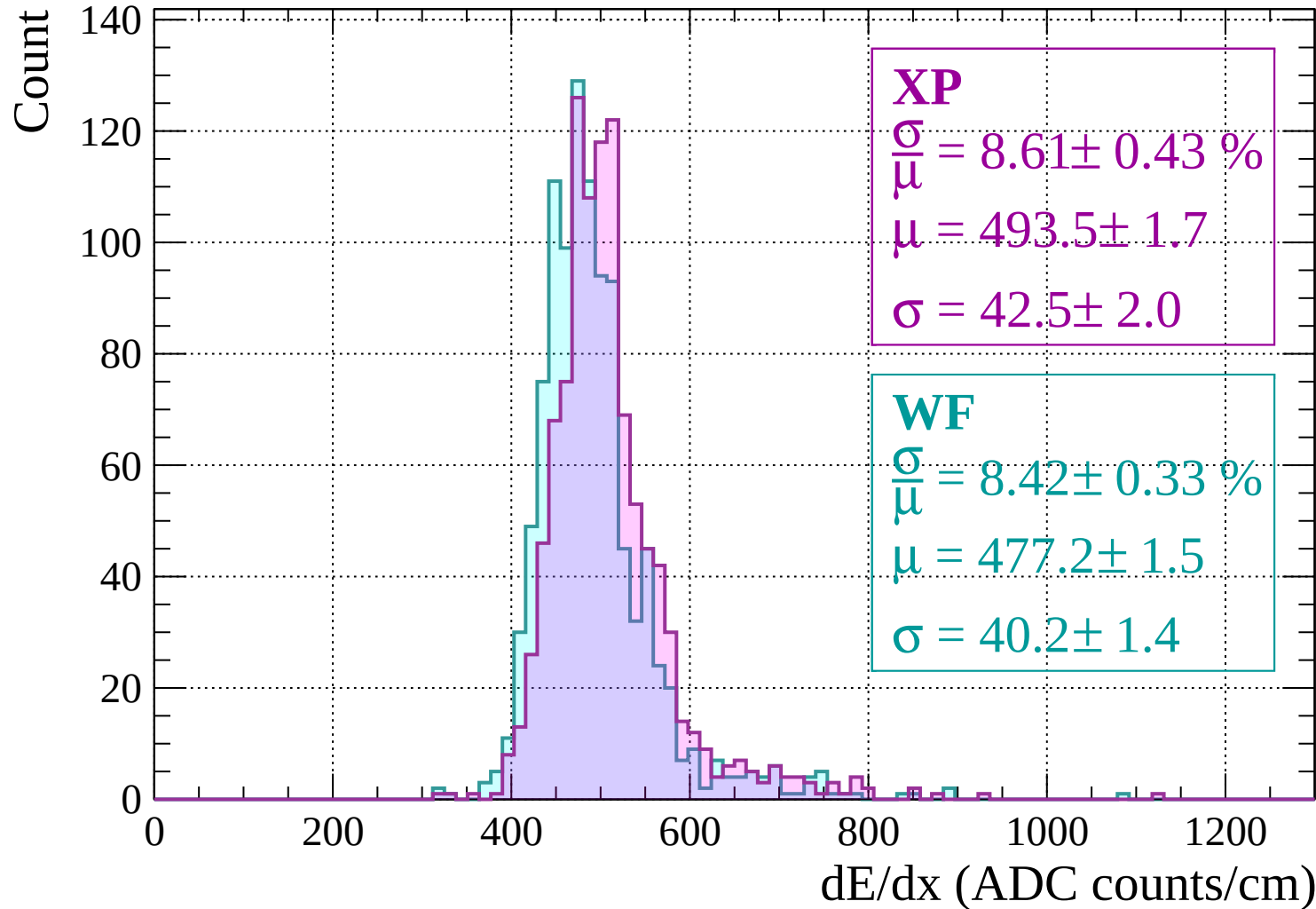
$$\frac{\sigma}{\mu} = 8.69 \pm 0.34 \%$$

$$\mu = 478.2 \pm 1.6$$

$$\sigma = 41.6 \pm 1.5$$

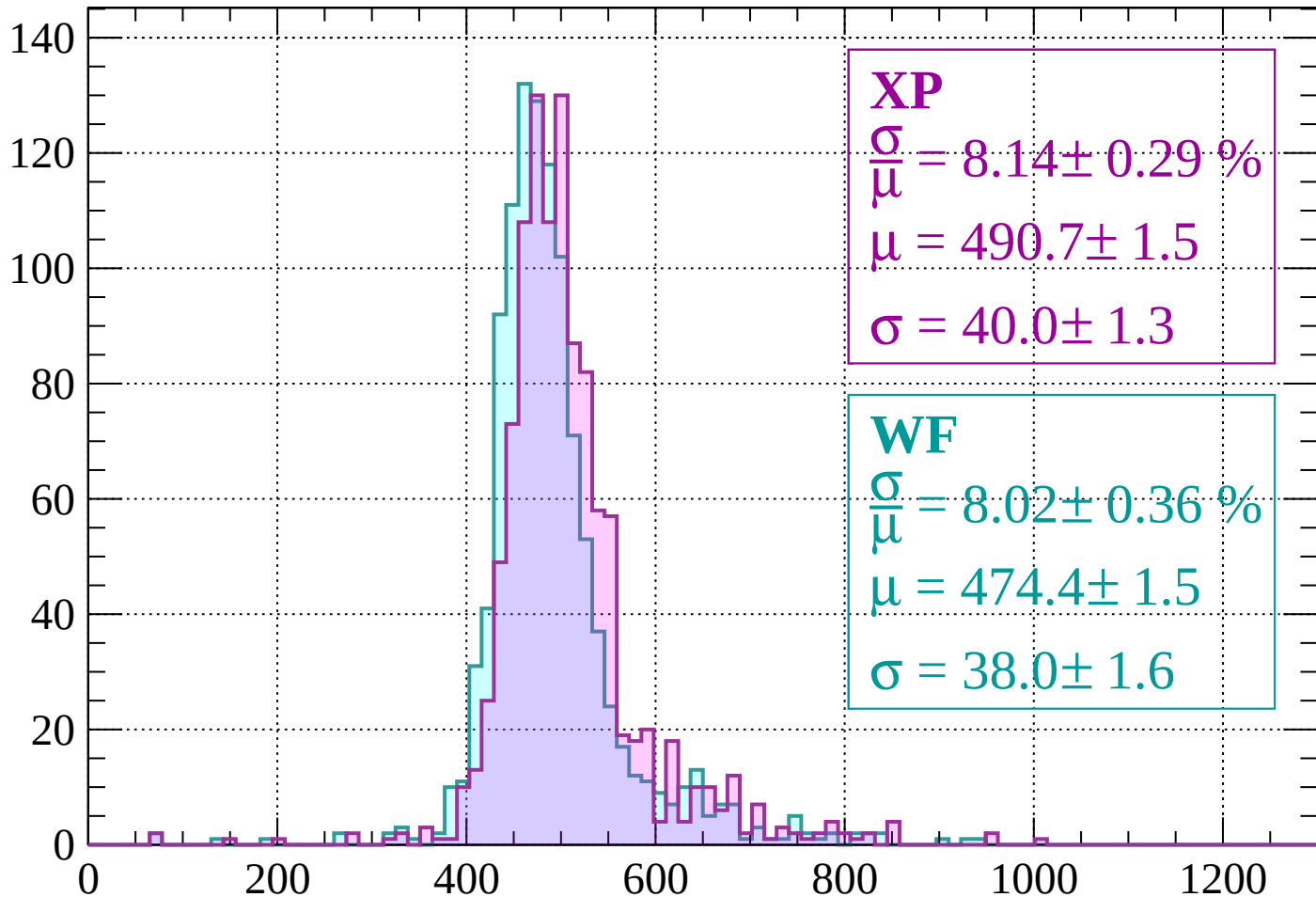
dE/dx (ADC counts/cm)

Energy loss | -1640 < p < -1600



Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.14 \pm 0.29 \%$$

$$\mu = 490.7 \pm 1.5$$

$$\sigma = 40.0 \pm 1.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.02 \pm 0.36 \%$$

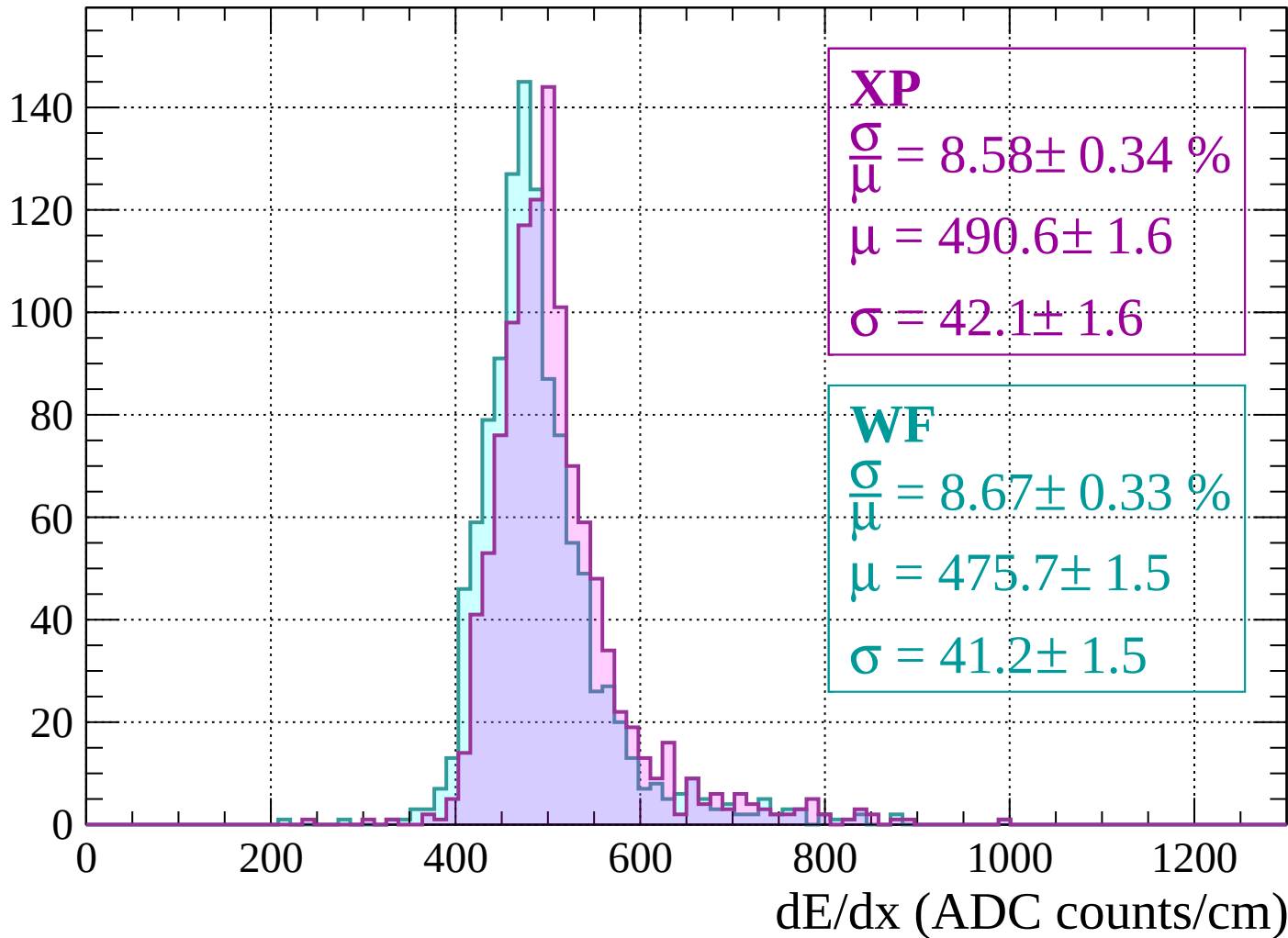
$$\mu = 474.4 \pm 1.5$$

$$\sigma = 38.0 \pm 1.6$$

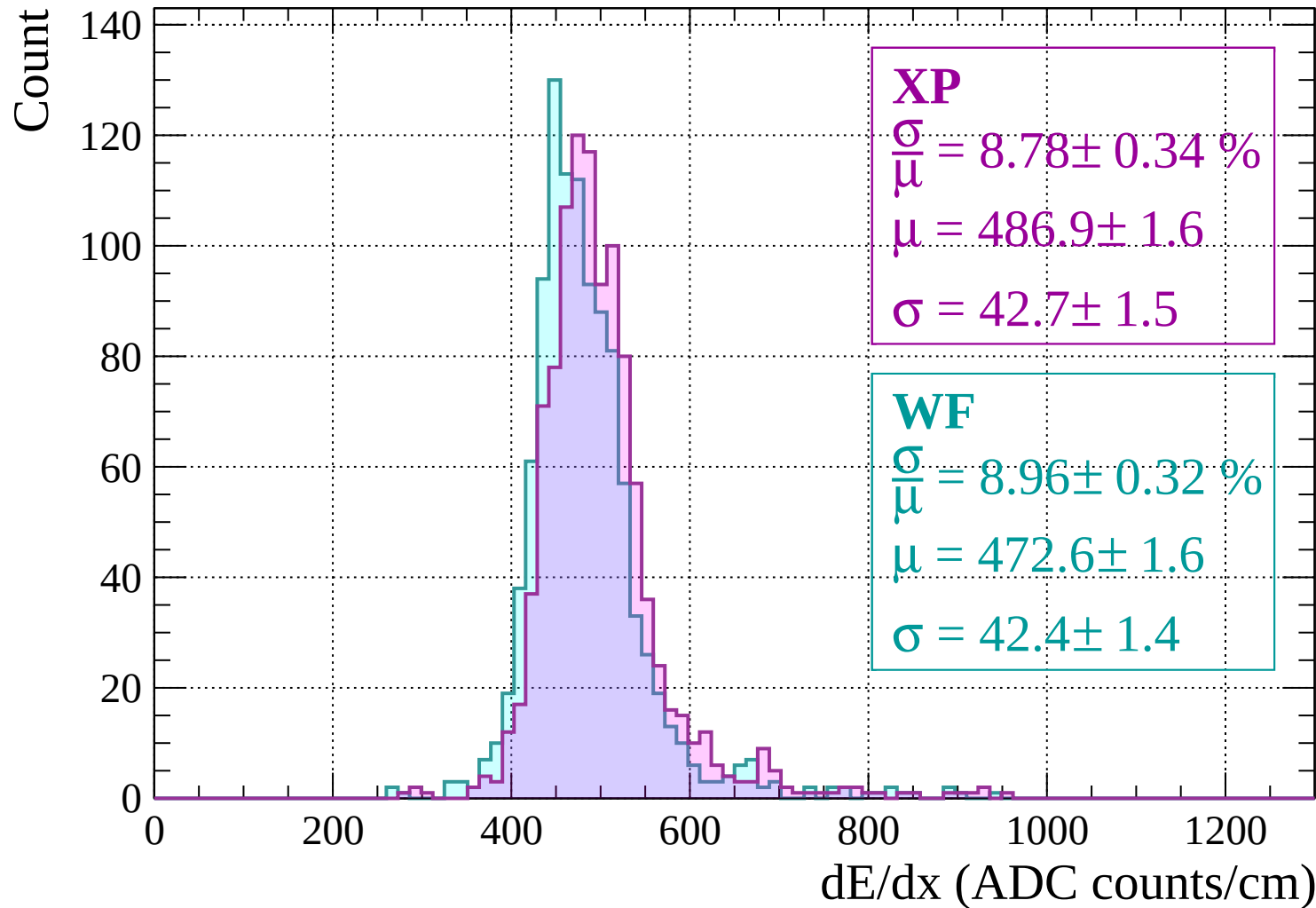
dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$

Count

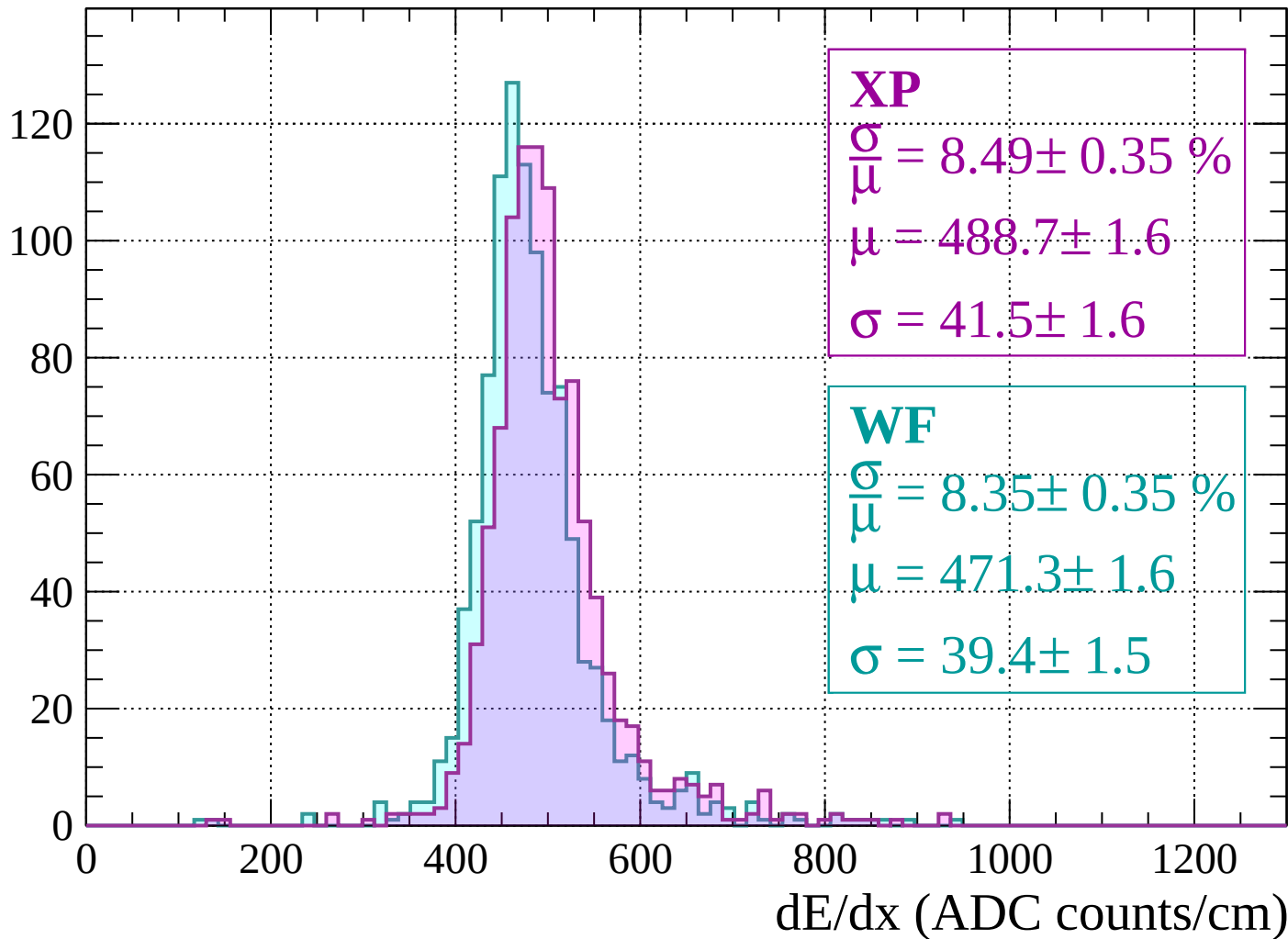


Energy loss | -1520 < p < -1480



Energy loss | $-1480 < p < -1440$

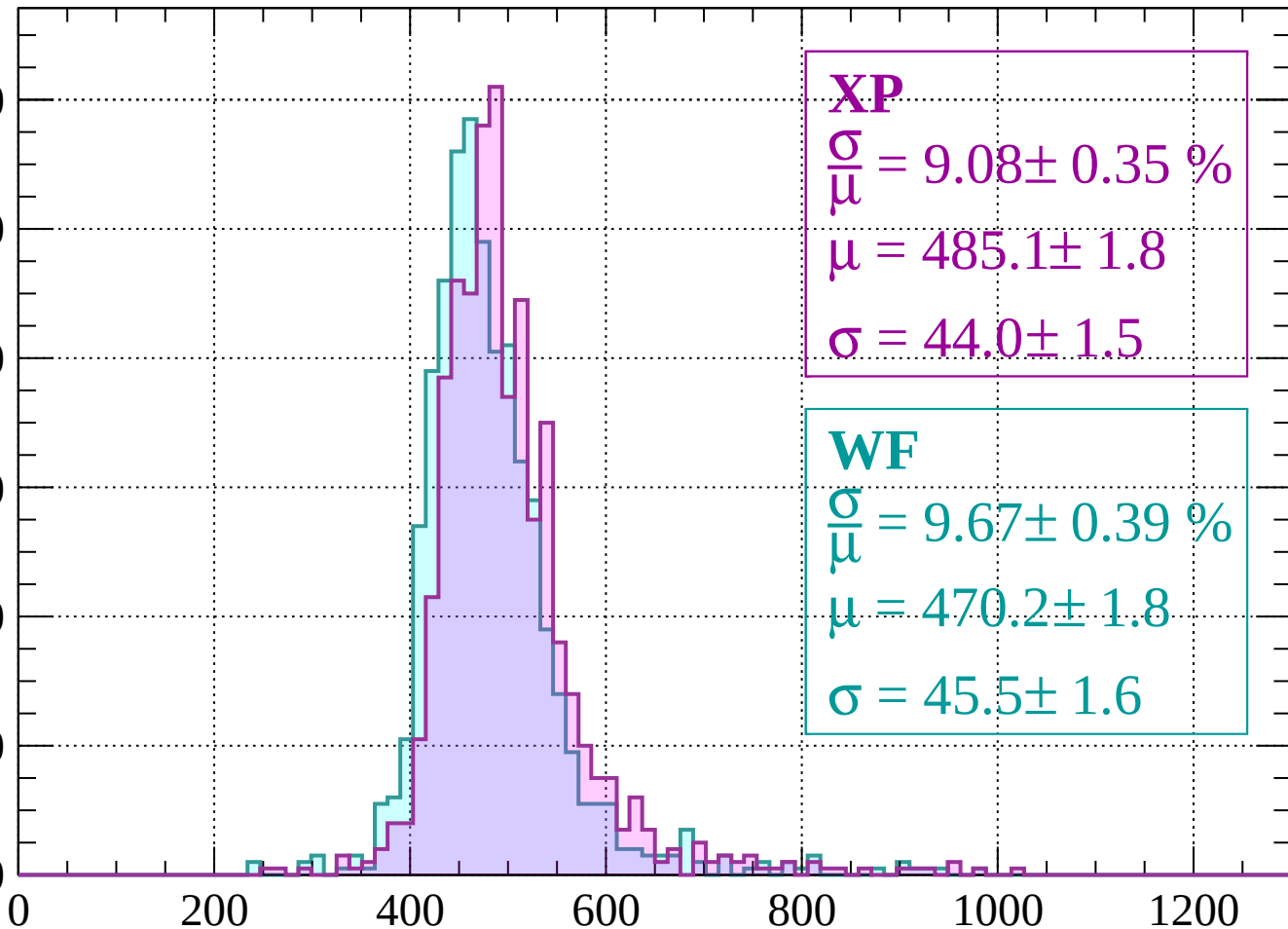
Count



Energy loss | $-1440 < p < -1400$

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.35 \%$$

$$\mu = 485.1 \pm 1.8$$

$$\sigma = 44.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.39 \%$$

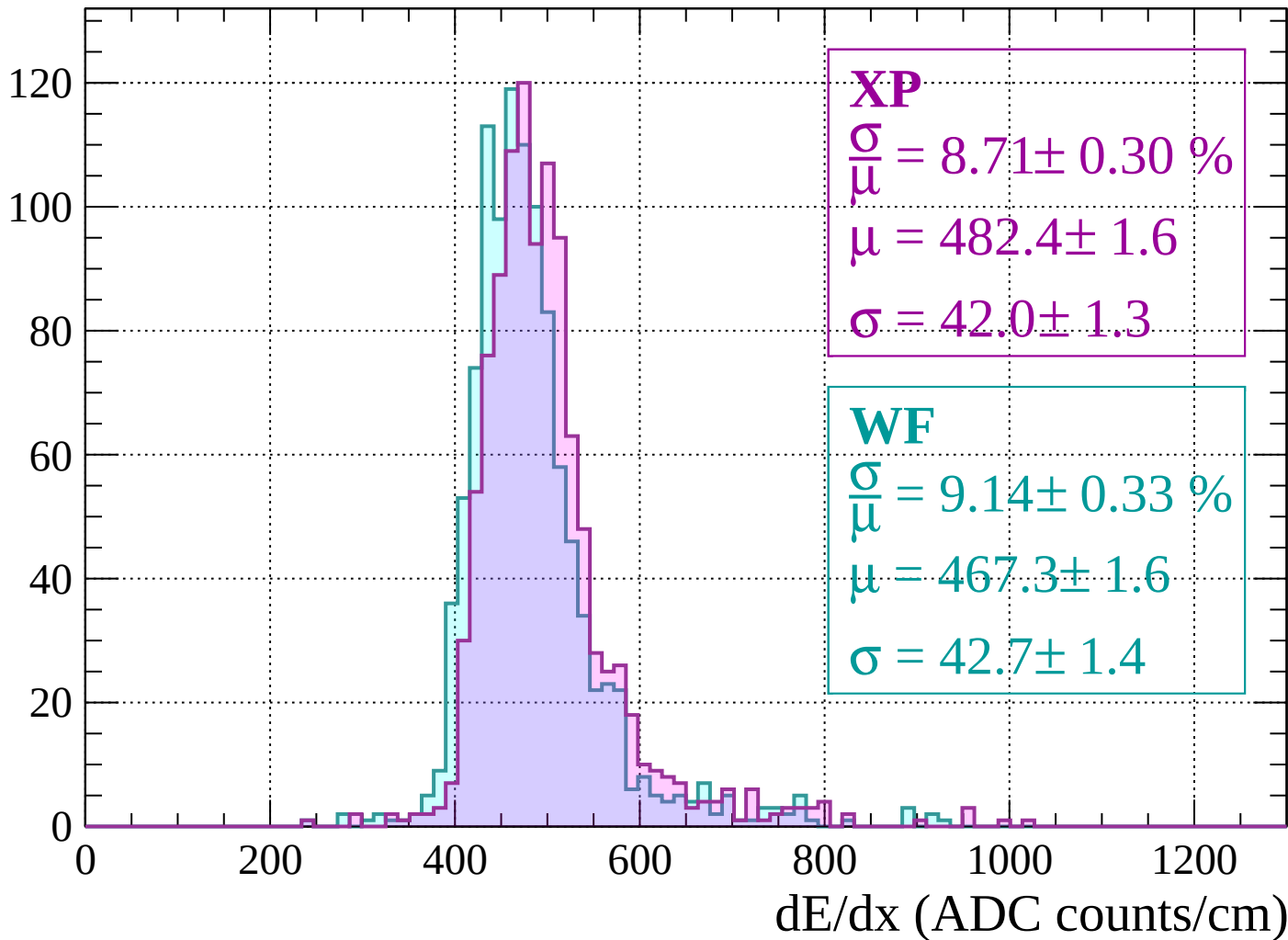
$$\mu = 470.2 \pm 1.8$$

$$\sigma = 45.5 \pm 1.6$$

dE/dx (ADC counts/cm)

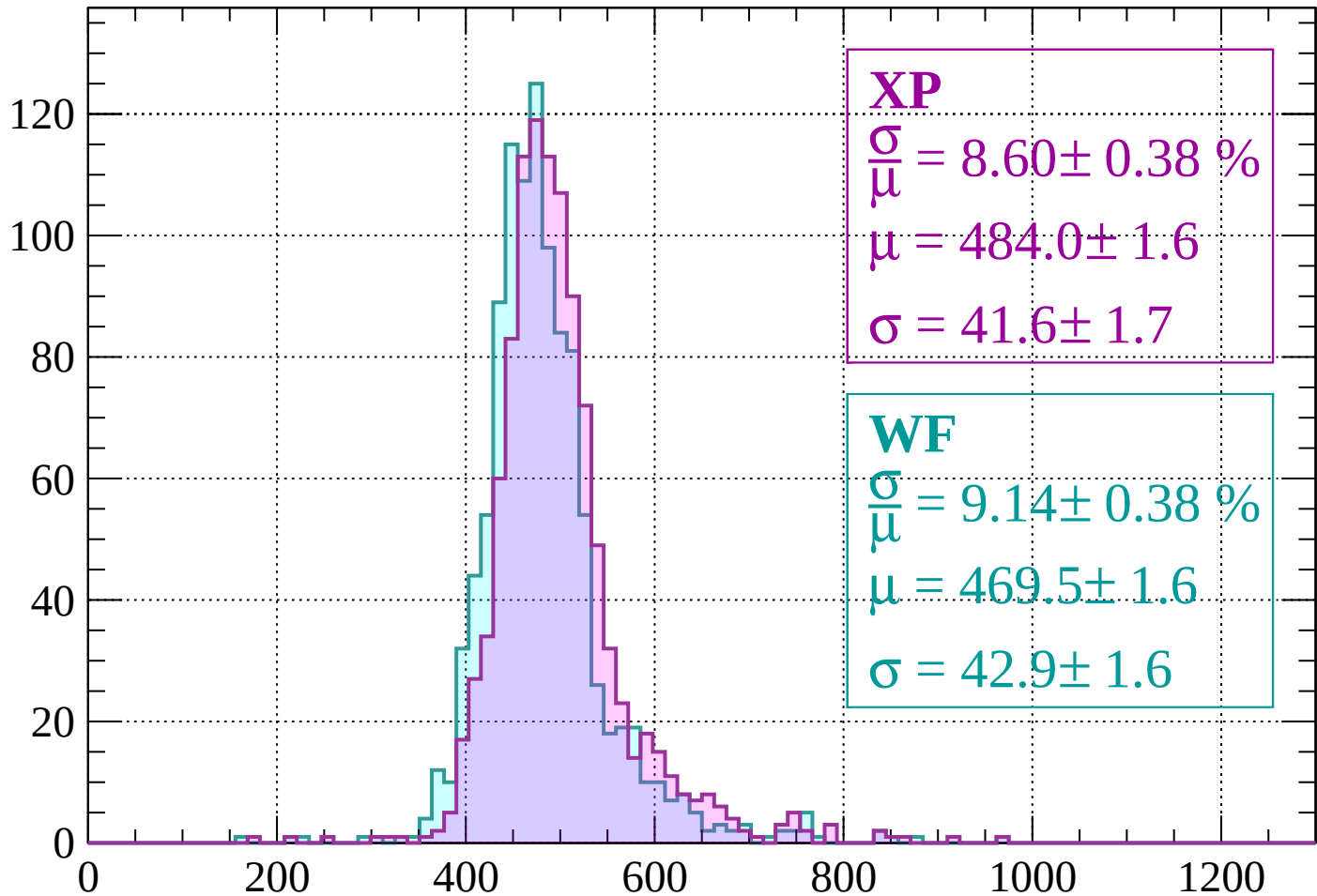
Energy loss | $-1400 < p < -1360$

Count



Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 8.60 \pm 0.38 \%$$

$$\mu = 484.0 \pm 1.6$$

$$\sigma = 41.6 \pm 1.7$$

WF

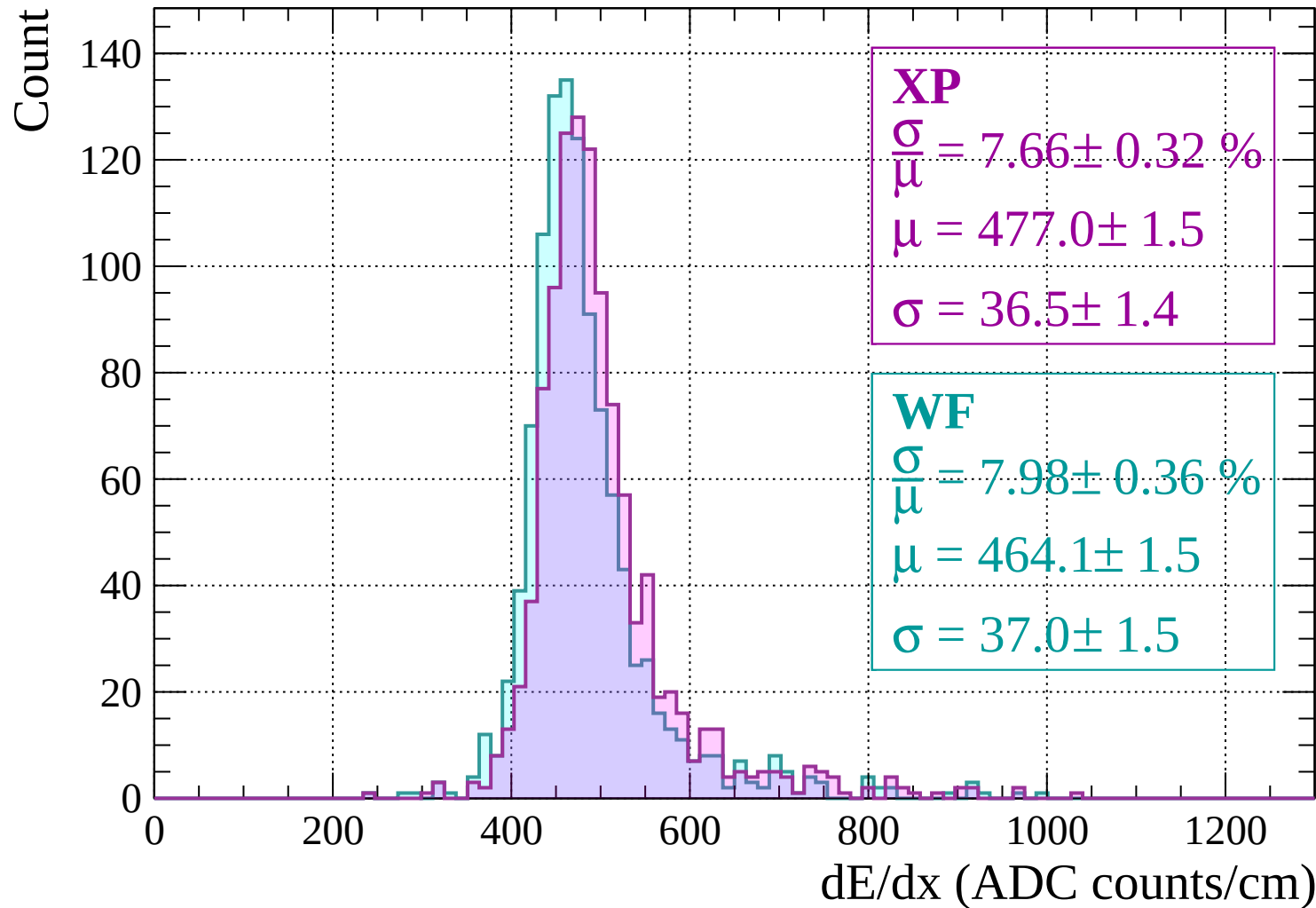
$$\frac{\sigma}{\mu} = 9.14 \pm 0.38 \%$$

$$\mu = 469.5 \pm 1.6$$

$$\sigma = 42.9 \pm 1.6$$

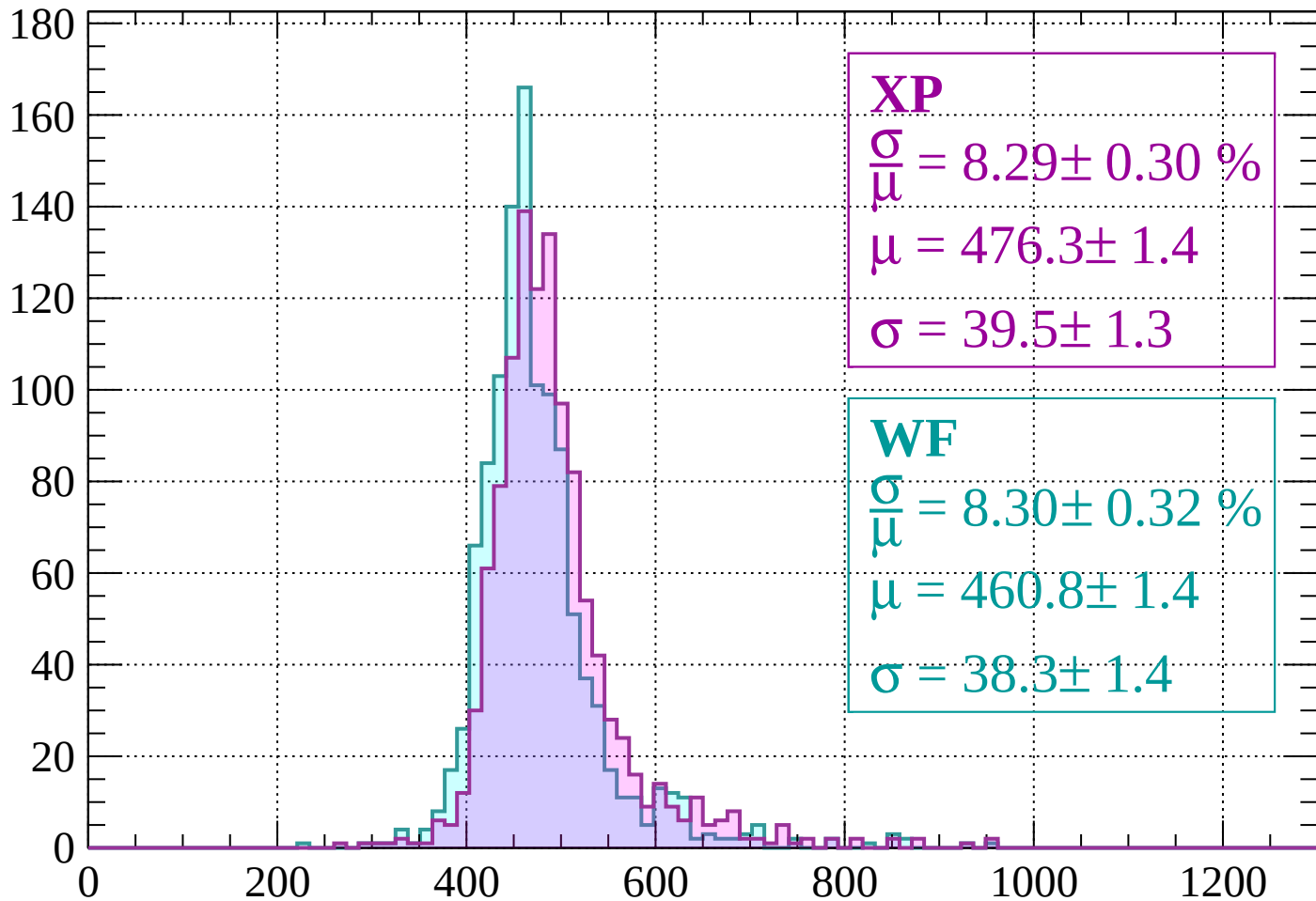
dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1280



Energy loss | -1280 < p < -1240

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.30 \%$$

$$\mu = 476.3 \pm 1.4$$

$$\sigma = 39.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.32 \%$$

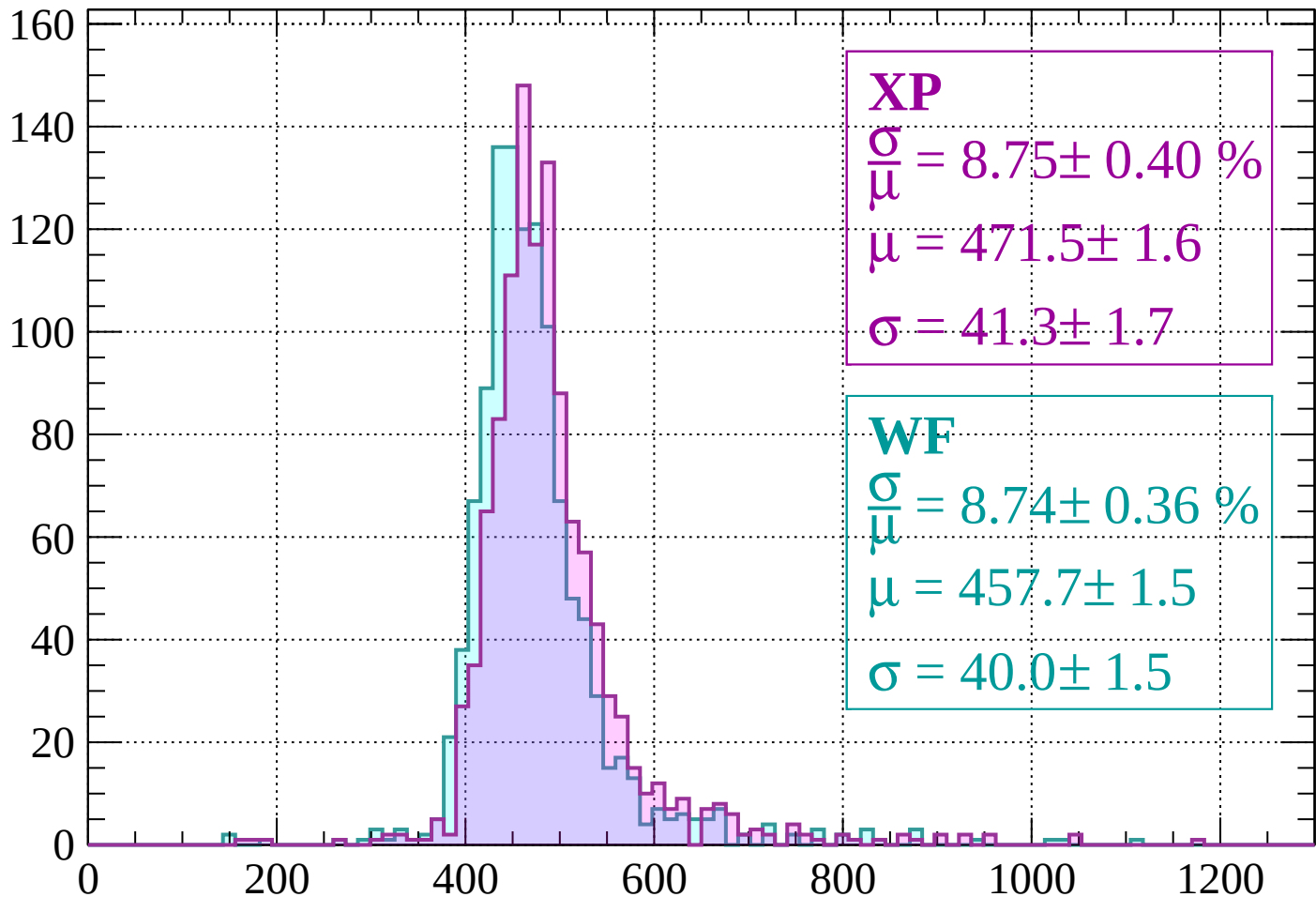
$$\mu = 460.8 \pm 1.4$$

$$\sigma = 38.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -1240 < p < -1200

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.40 \%$$

$$\mu = 471.5 \pm 1.6$$

$$\sigma = 41.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.74 \pm 0.36 \%$$

$$\mu = 457.7 \pm 1.5$$

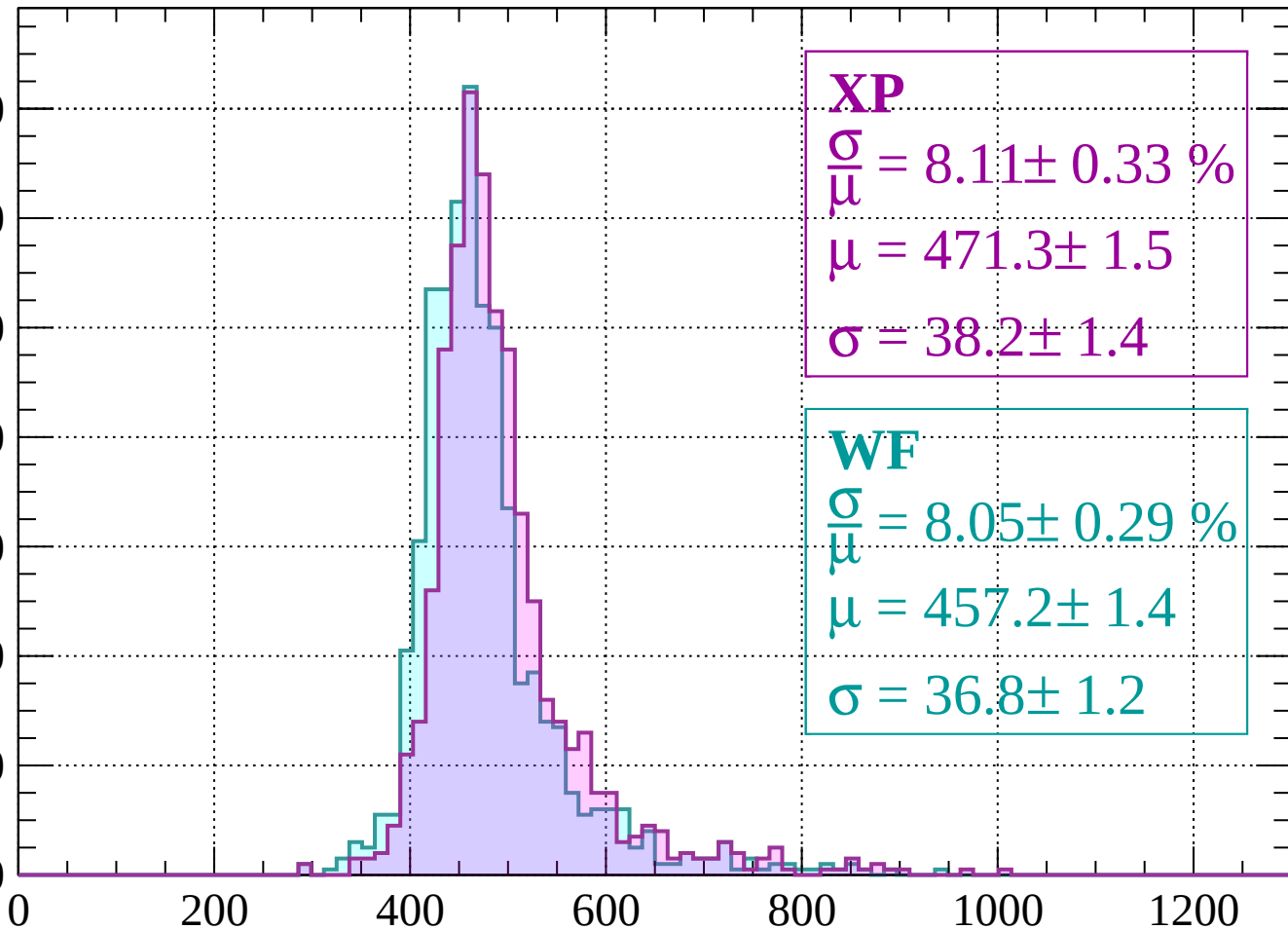
$$\sigma = 40.0 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.11 \pm 0.33 \%$$

$$\mu = 471.3 \pm 1.5$$

$$\sigma = 38.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.05 \pm 0.29 \%$$

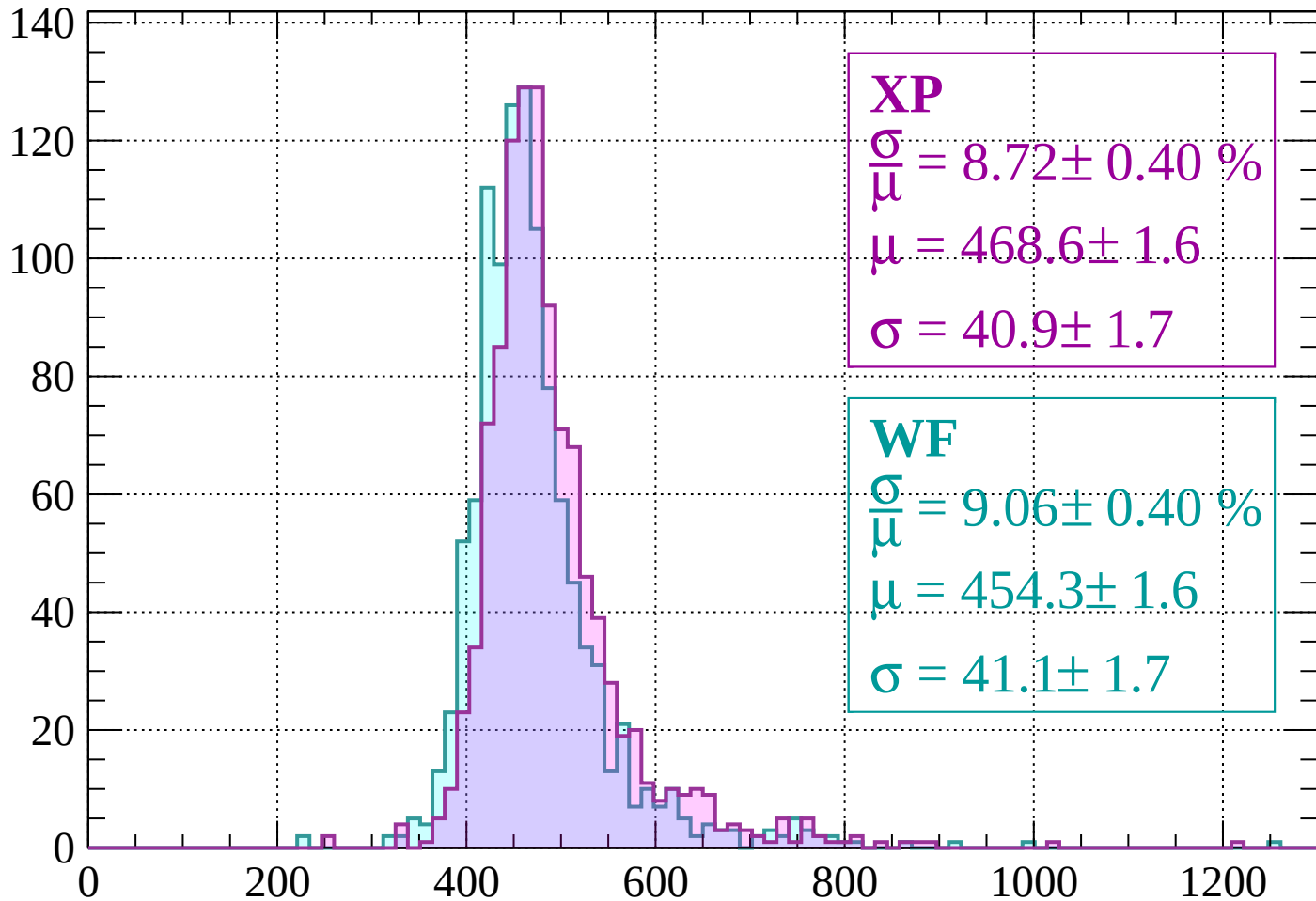
$$\mu = 457.2 \pm 1.4$$

$$\sigma = 36.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | -1160 < p < -1120

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.40 \%$$

$$\mu = 468.6 \pm 1.6$$

$$\sigma = 40.9 \pm 1.7$$

WF

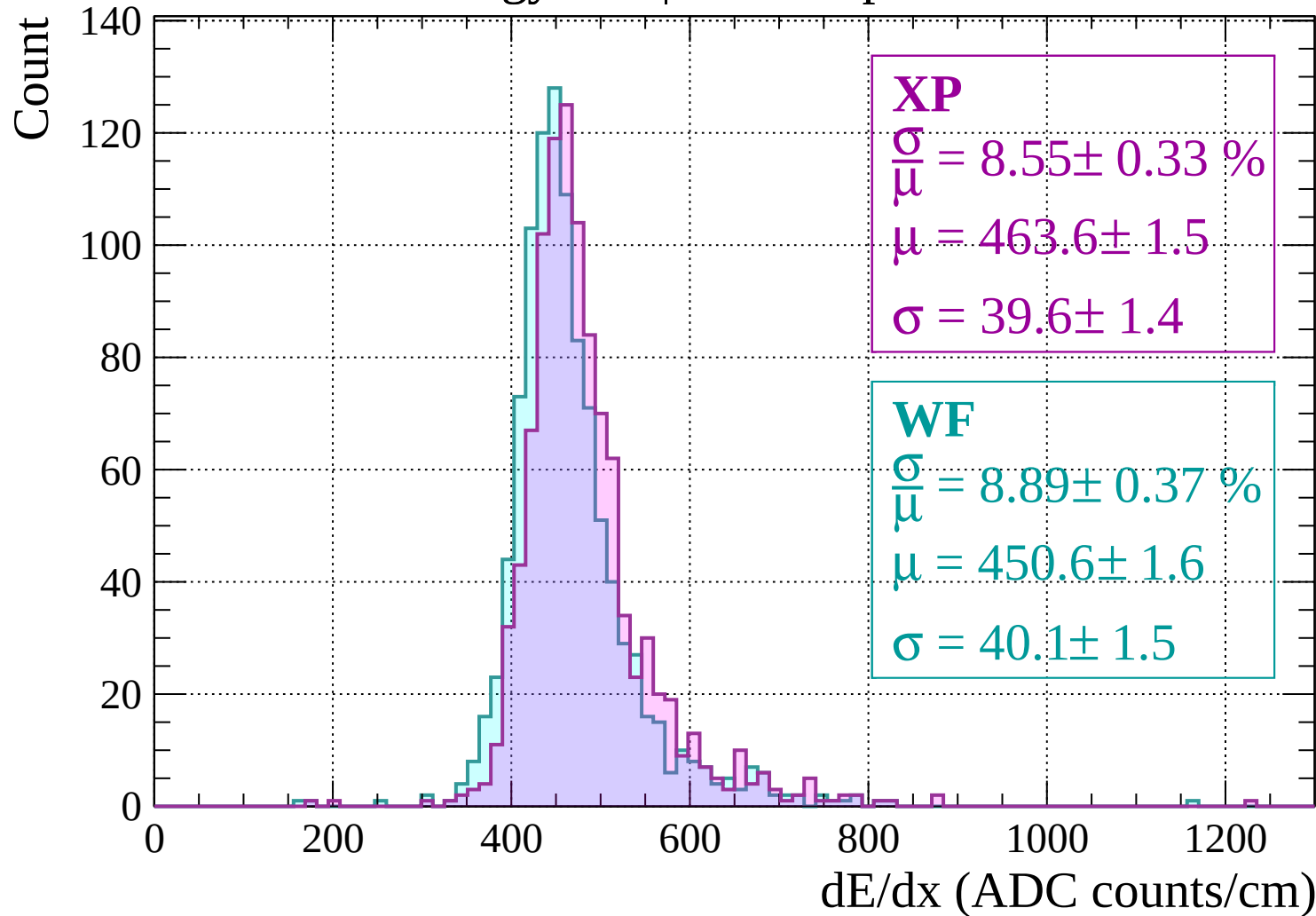
$$\frac{\sigma}{\mu} = 9.06 \pm 0.40 \%$$

$$\mu = 454.3 \pm 1.6$$

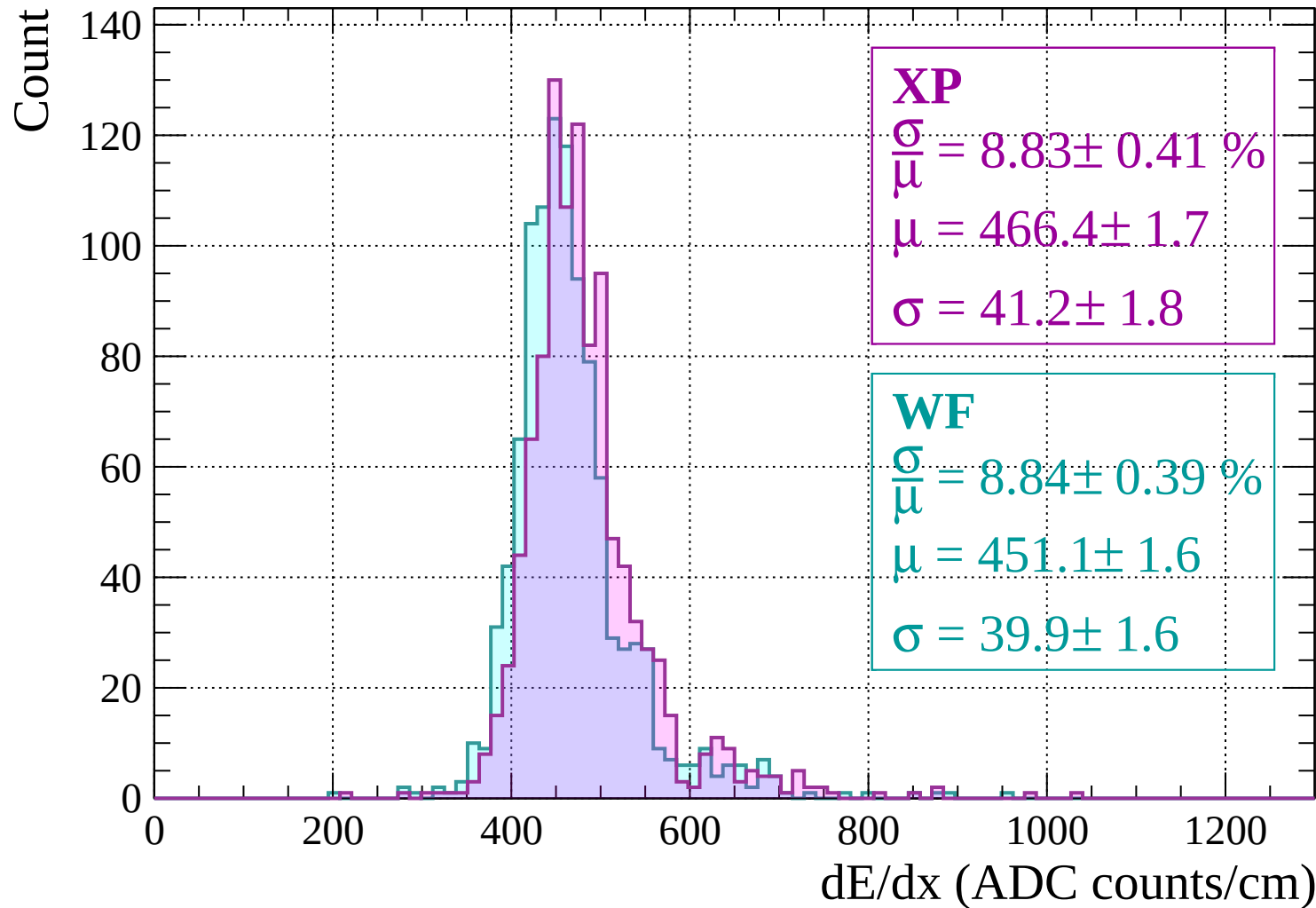
$$\sigma = 41.1 \pm 1.7$$

dE/dx (ADC counts/cm)

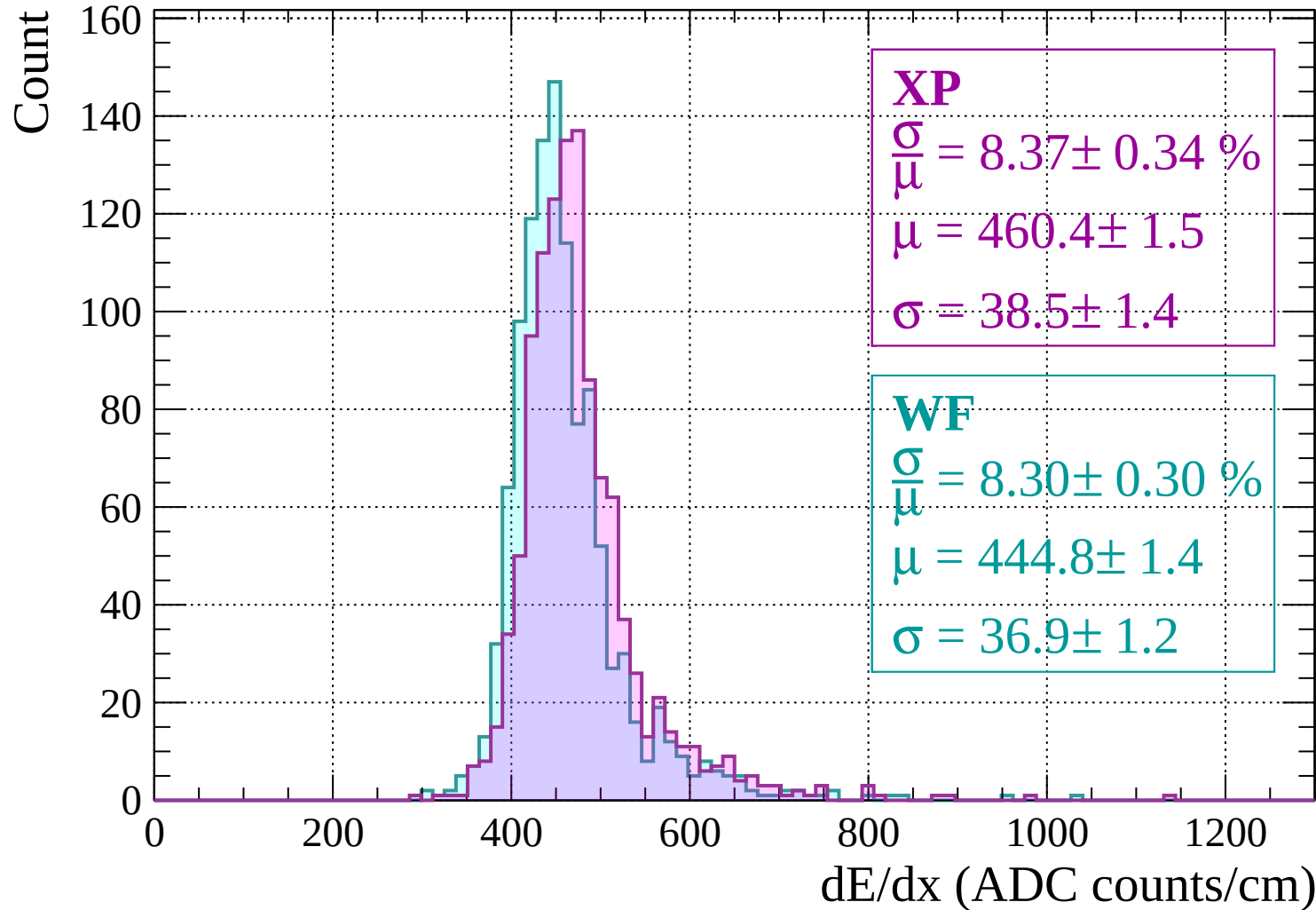
Energy loss | -1120 < p < -1080



Energy loss | -1080 < p < -1040



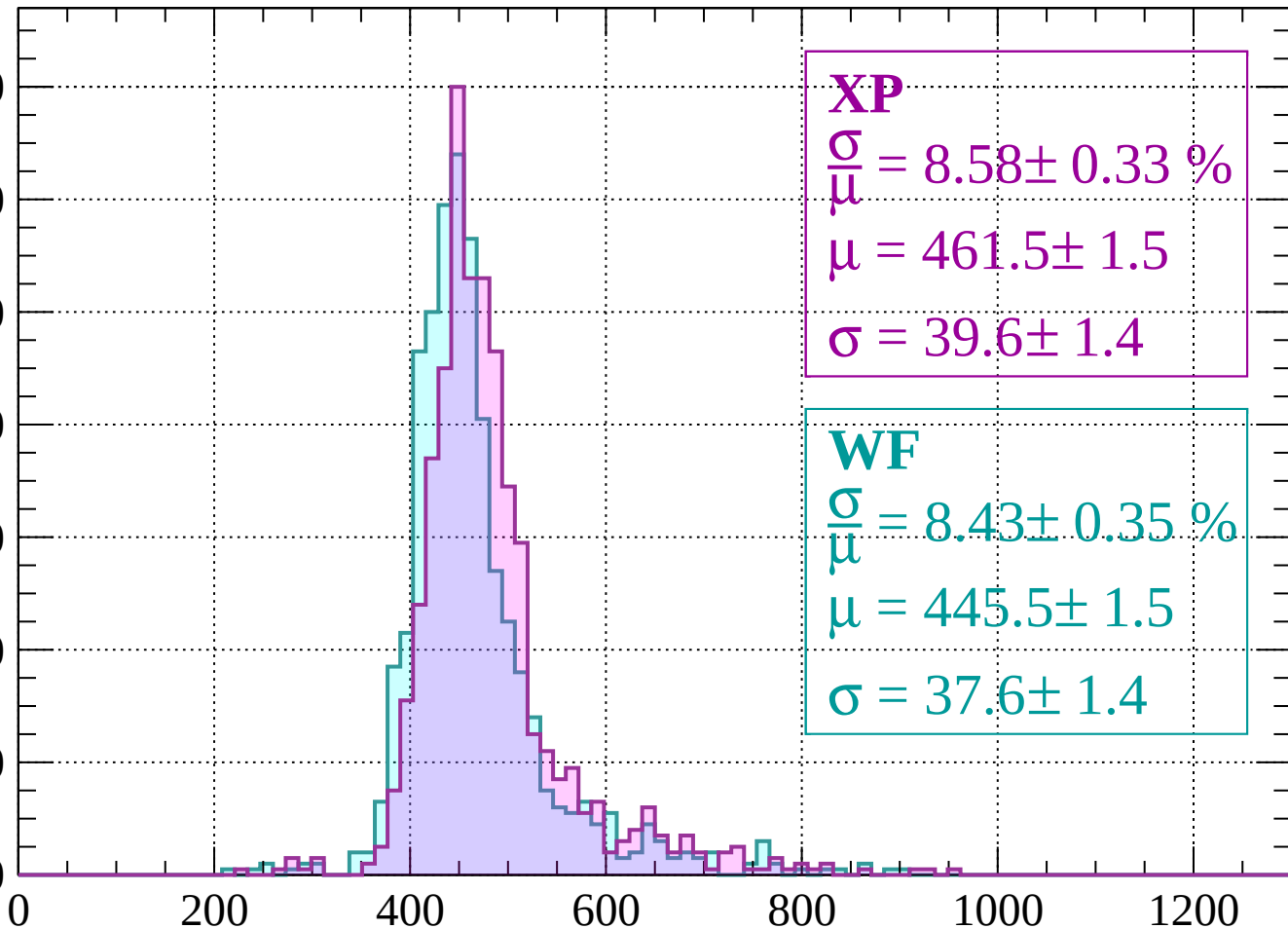
Energy loss | -1040 < p < -1000



Energy loss | $-1000 < p < -960$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.58 \pm 0.33 \%$$

$$\mu = 461.5 \pm 1.5$$

$$\sigma = 39.6 \pm 1.4$$

WF

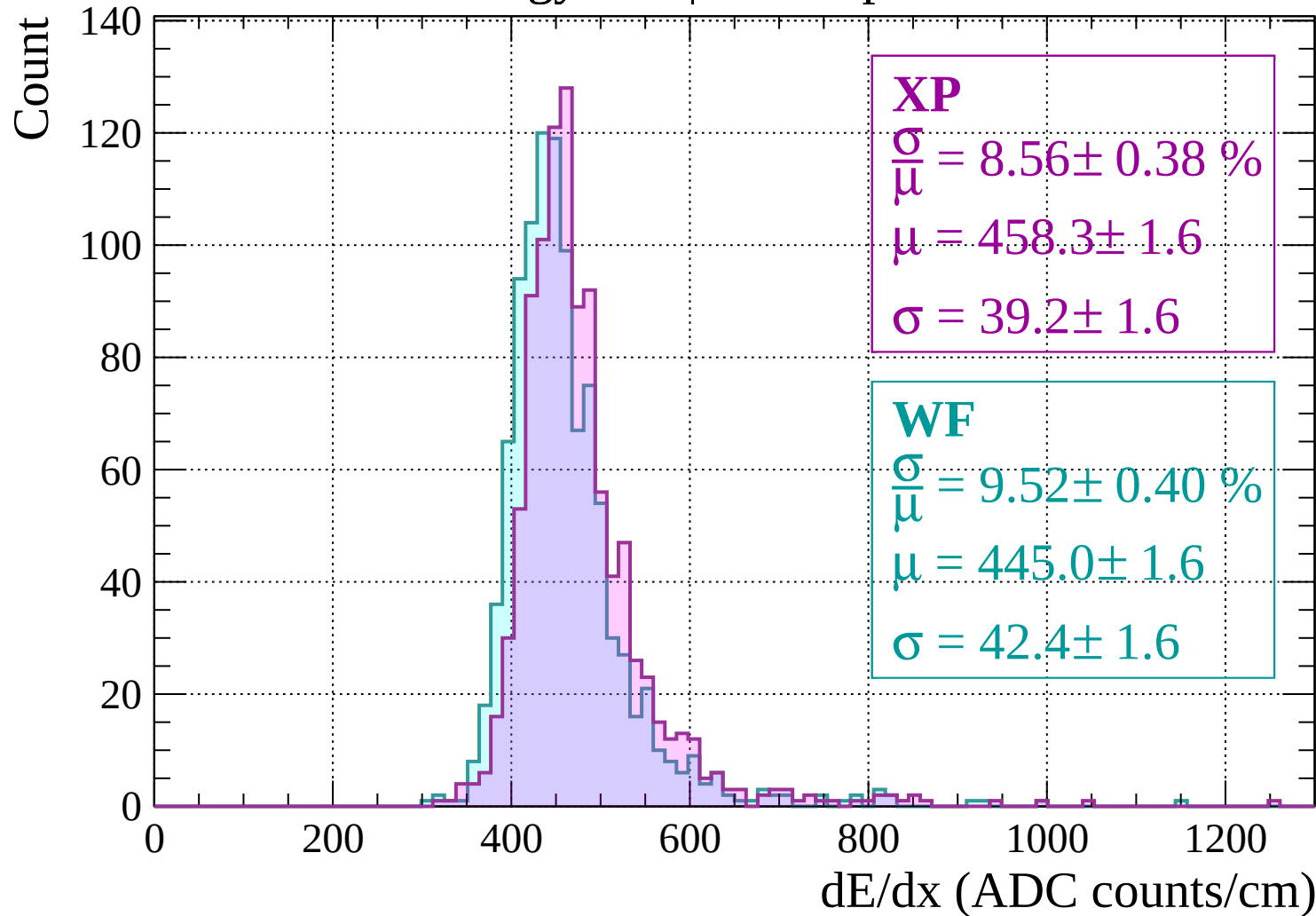
$$\frac{\sigma}{\mu} = 8.43 \pm 0.35 \%$$

$$\mu = 445.5 \pm 1.5$$

$$\sigma = 37.6 \pm 1.4$$

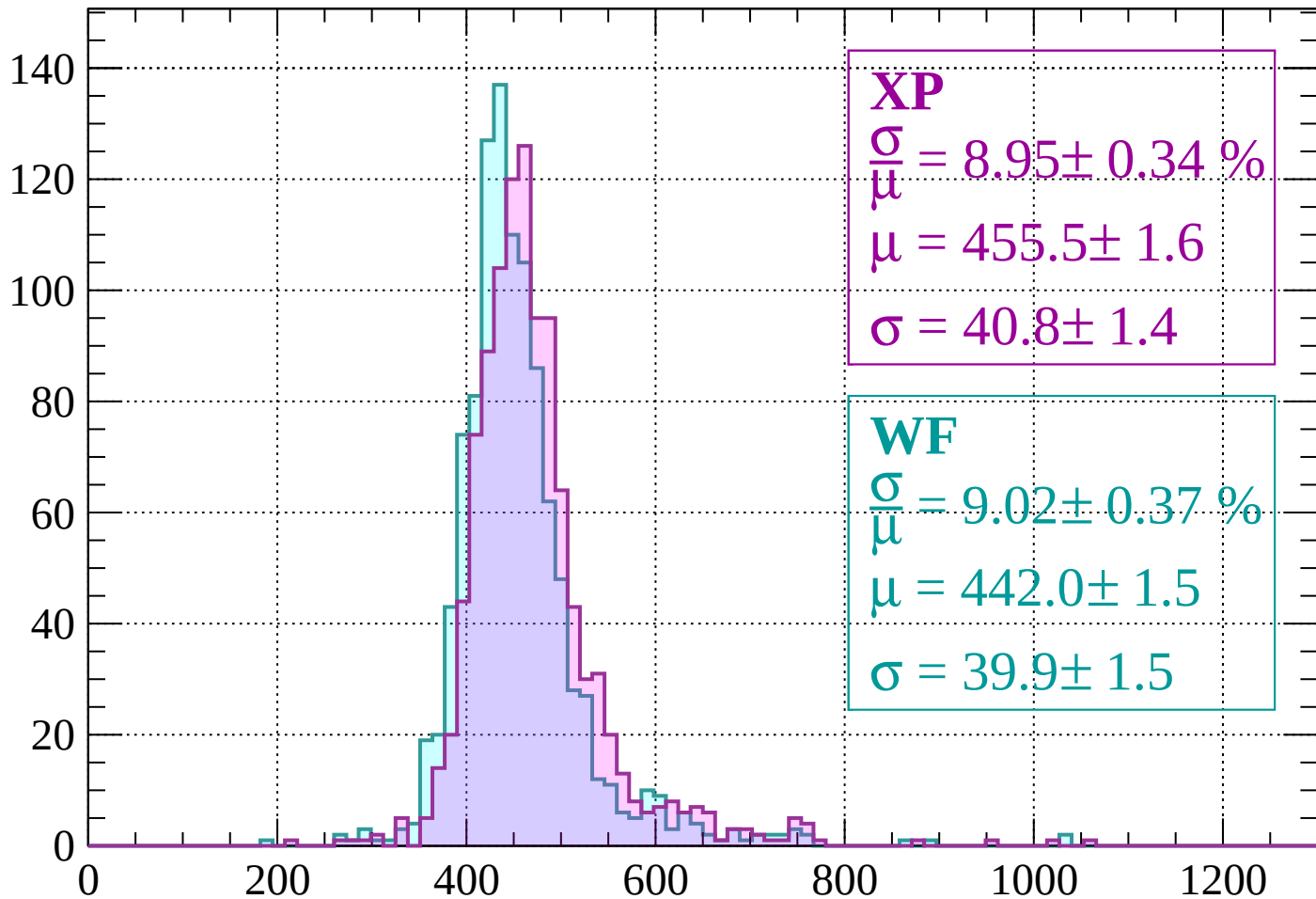
dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$



Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.34 \%$$

$$\mu = 455.5 \pm 1.6$$

$$\sigma = 40.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.02 \pm 0.37 \%$$

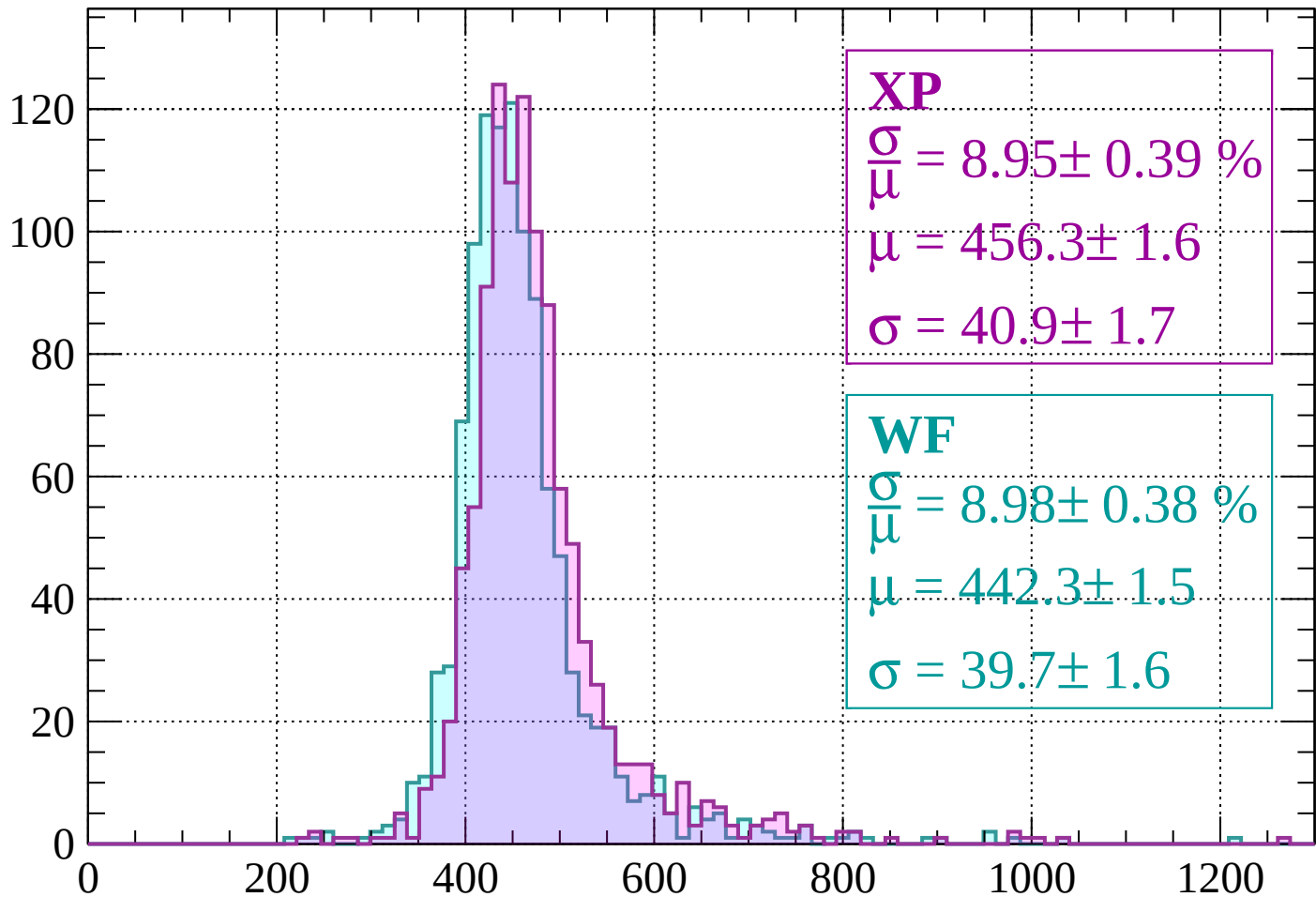
$$\mu = 442.0 \pm 1.5$$

$$\sigma = 39.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.39 \%$$

$$\mu = 456.3 \pm 1.6$$

$$\sigma = 40.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.38 \%$$

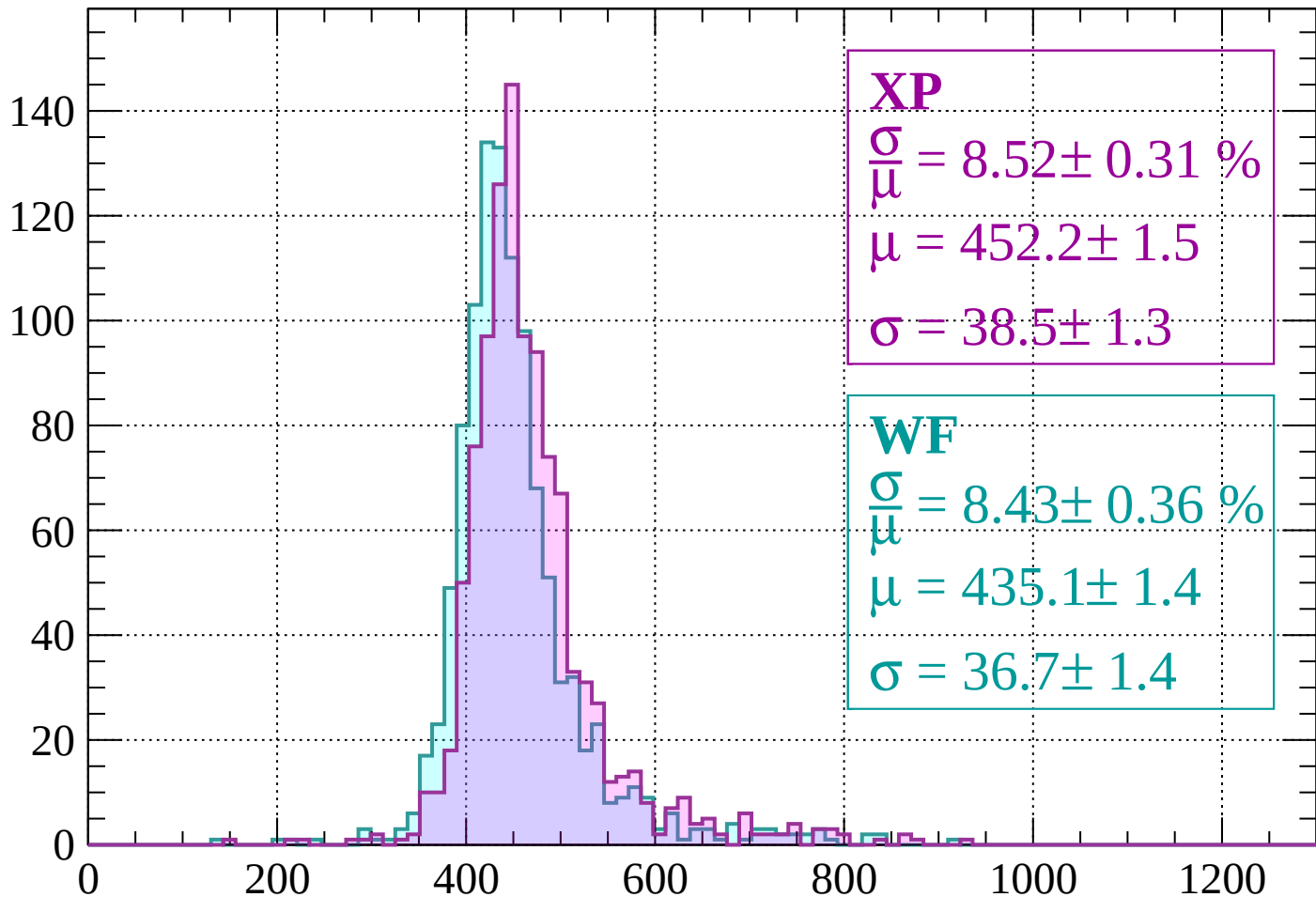
$$\mu = 442.3 \pm 1.5$$

$$\sigma = 39.7 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 8.52 \pm 0.31 \%$$

$$\mu = 452.2 \pm 1.5$$

$$\sigma = 38.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.43 \pm 0.36 \%$$

$$\mu = 435.1 \pm 1.4$$

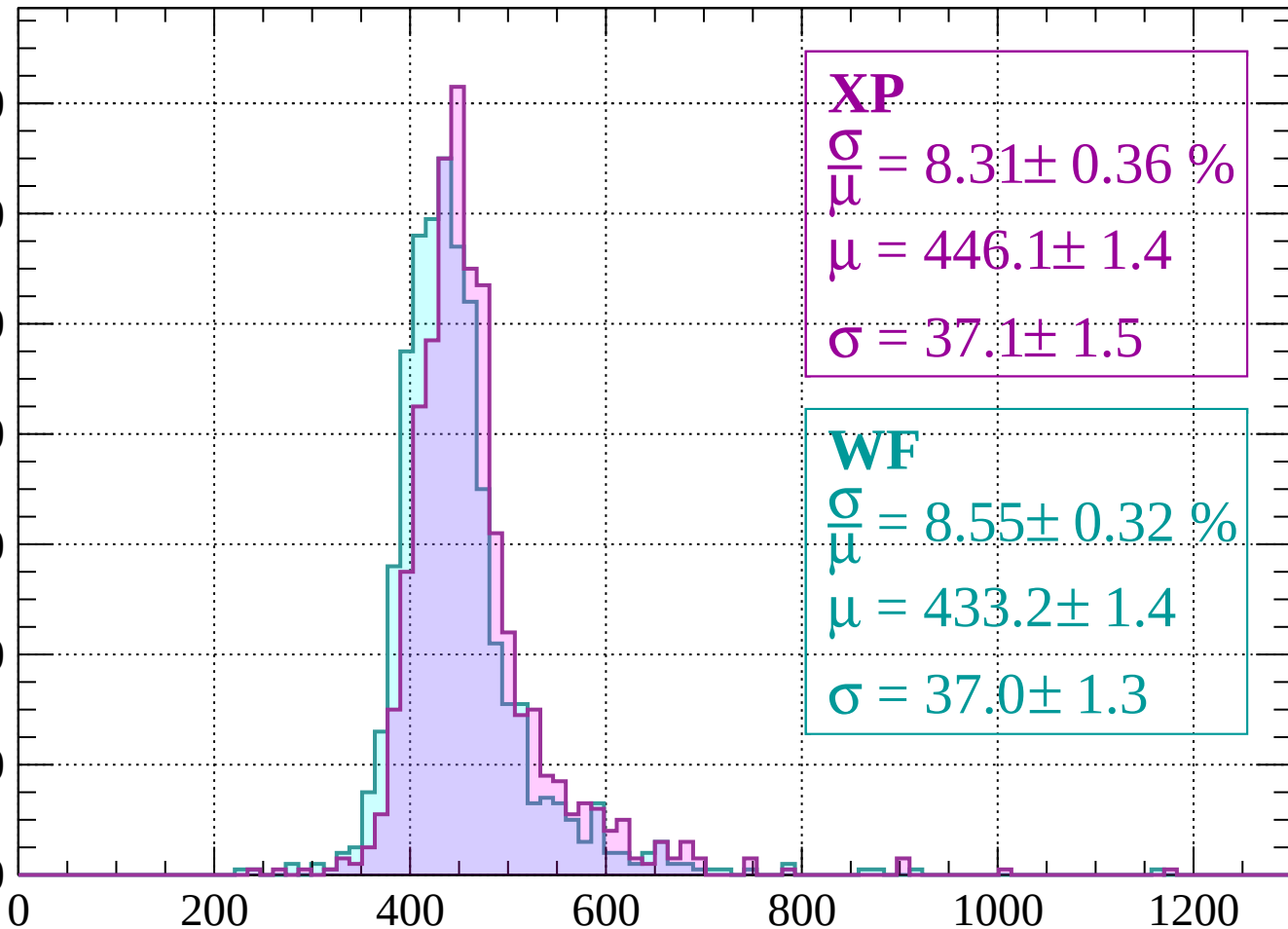
$$\sigma = 36.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.31 \pm 0.36 \%$$

$$\mu = 446.1 \pm 1.4$$

$$\sigma = 37.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.32 \%$$

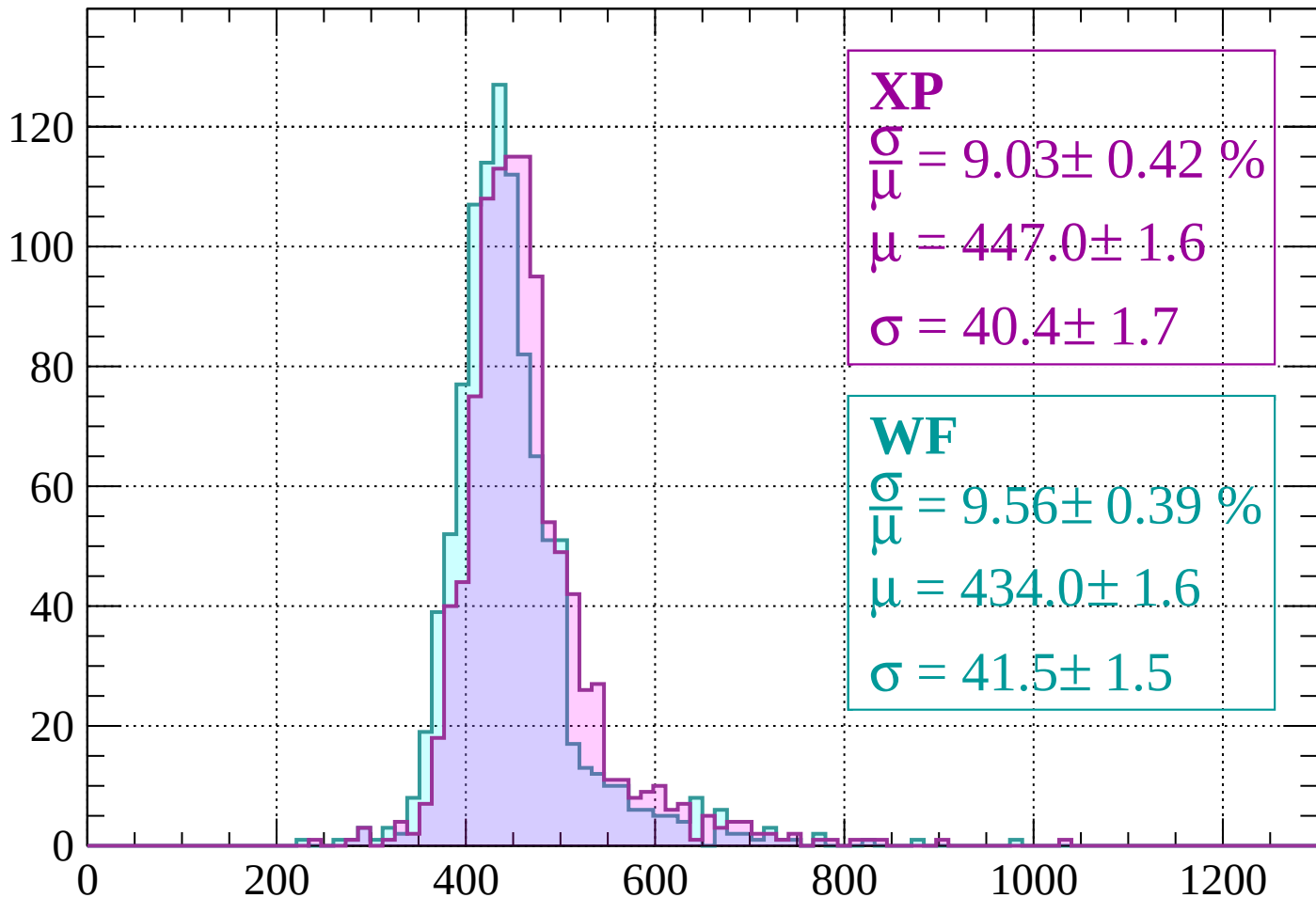
$$\mu = 433.2 \pm 1.4$$

$$\sigma = 37.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.42 \%$$

$$\mu = 447.0 \pm 1.6$$

$$\sigma = 40.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.39 \%$$

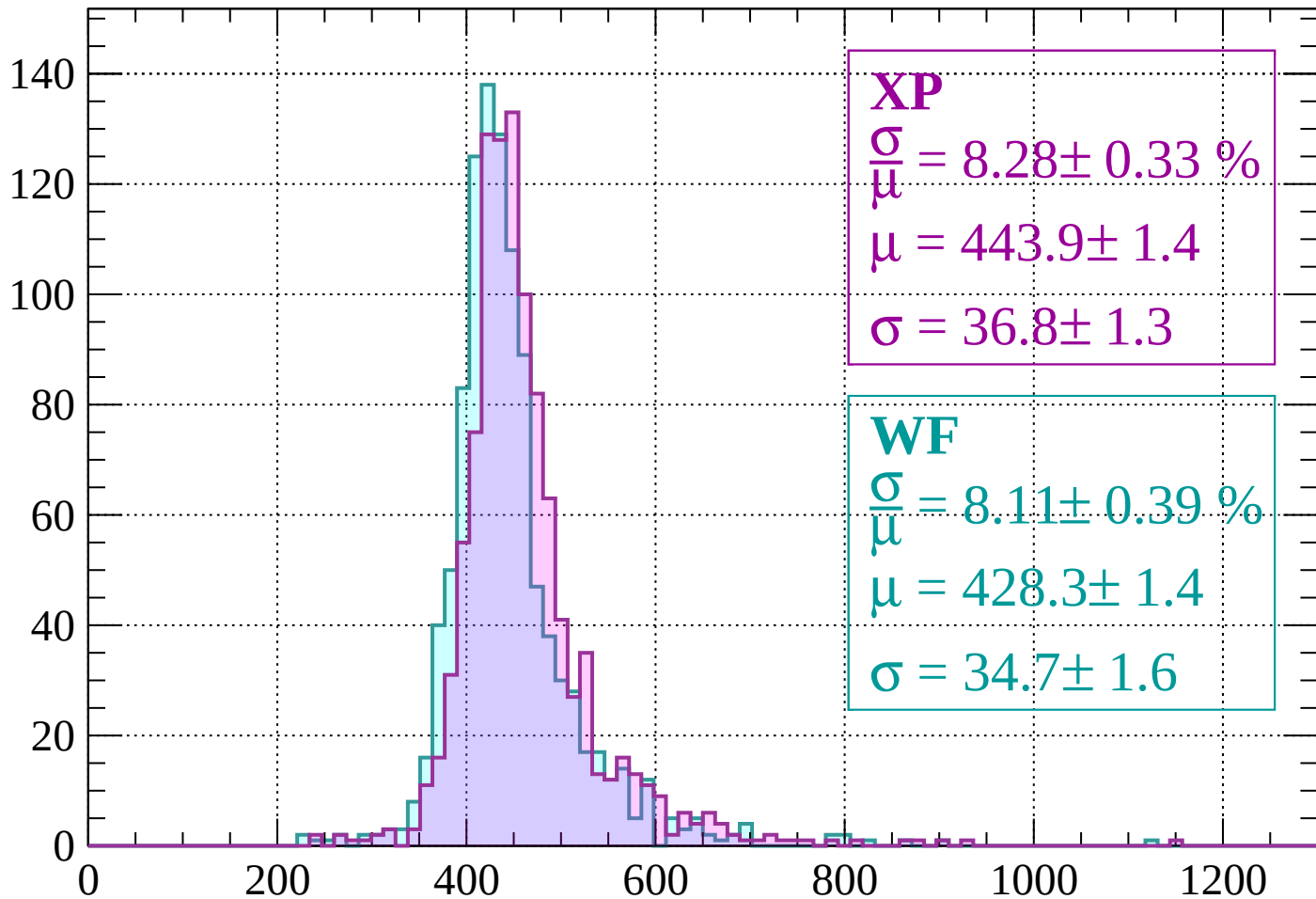
$$\mu = 434.0 \pm 1.6$$

$$\sigma = 41.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.28 \pm 0.33 \%$$

$$\mu = 443.9 \pm 1.4$$

$$\sigma = 36.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.39 \%$$

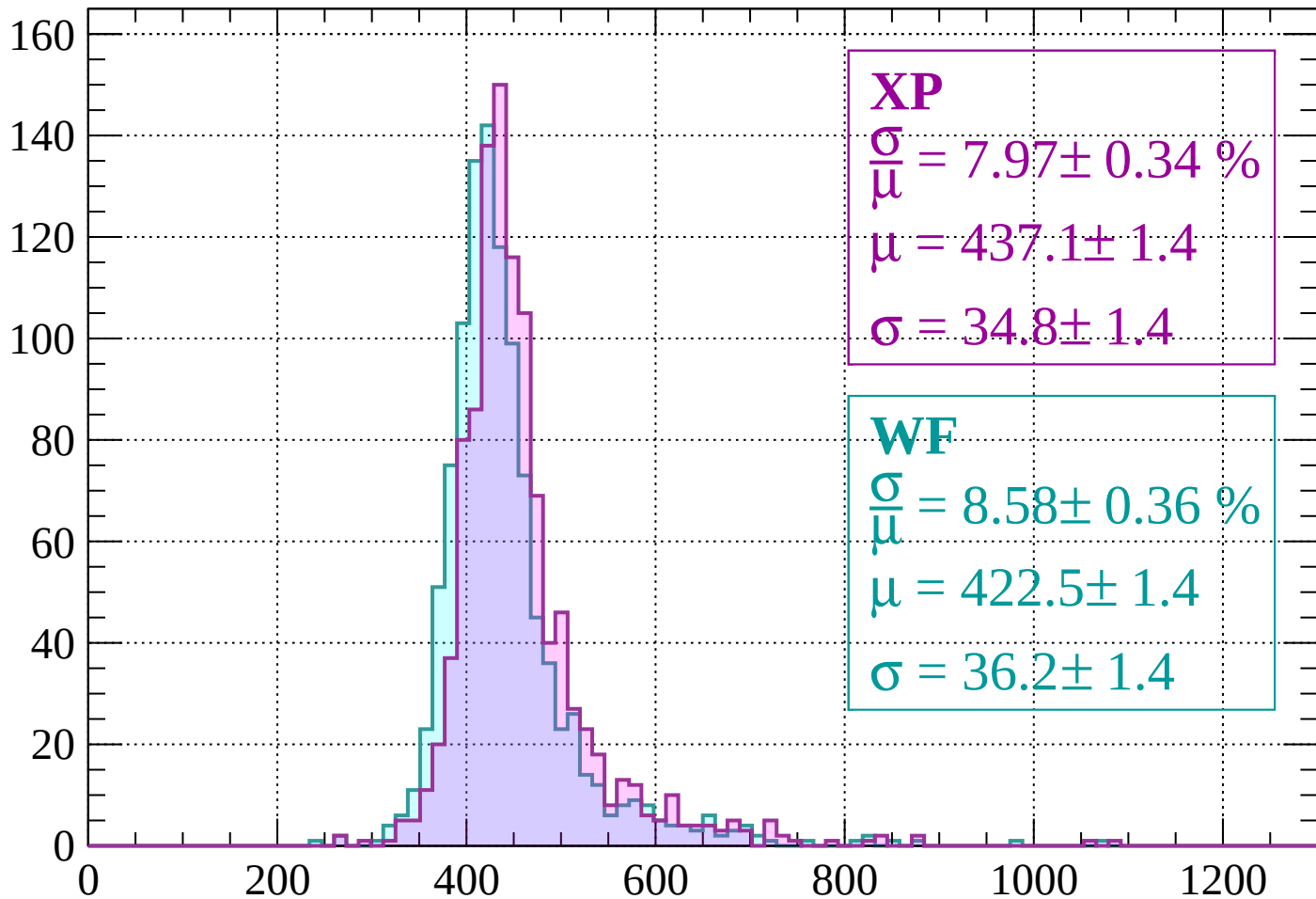
$$\mu = 428.3 \pm 1.4$$

$$\sigma = 34.7 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 7.97 \pm 0.34 \%$$

$$\mu = 437.1 \pm 1.4$$

$$\sigma = 34.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.58 \pm 0.36 \%$$

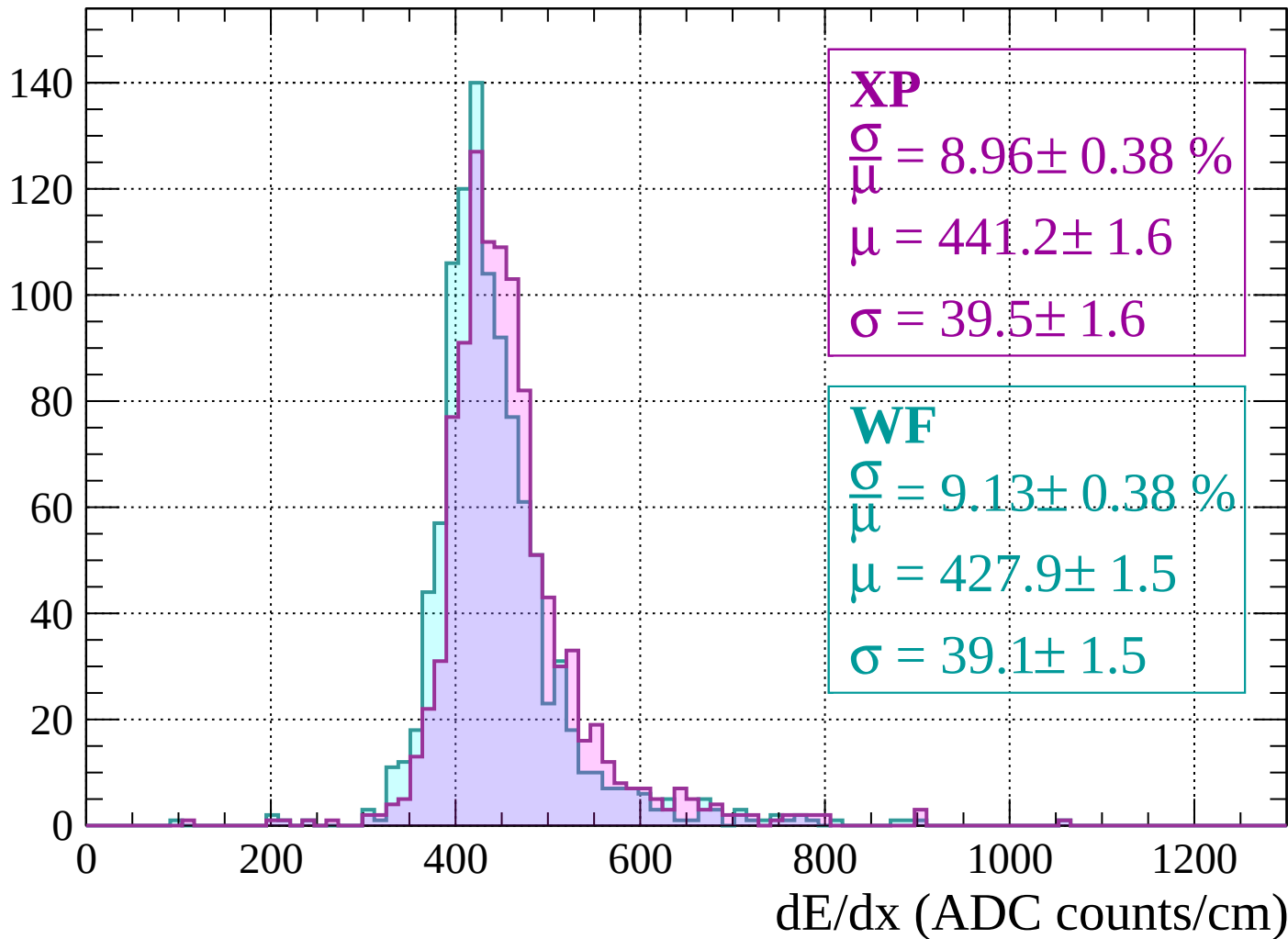
$$\mu = 422.5 \pm 1.4$$

$$\sigma = 36.2 \pm 1.4$$

dE/dx (ADC counts/cm)

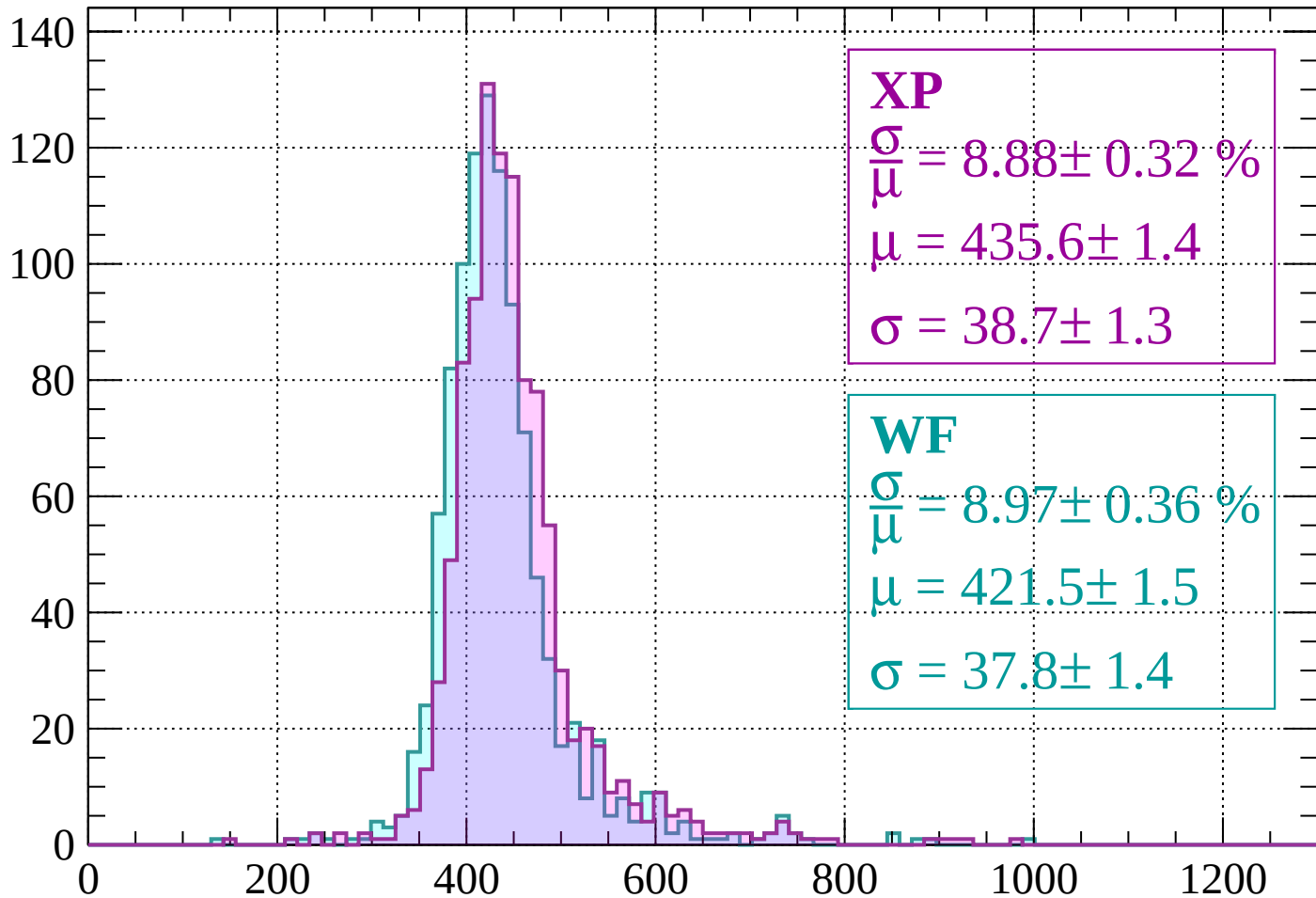
Energy loss | $-640 < p < -600$

Count



Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.88 \pm 0.32 \%$$

$$\mu = 435.6 \pm 1.4$$

$$\sigma = 38.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 0.36 \%$$

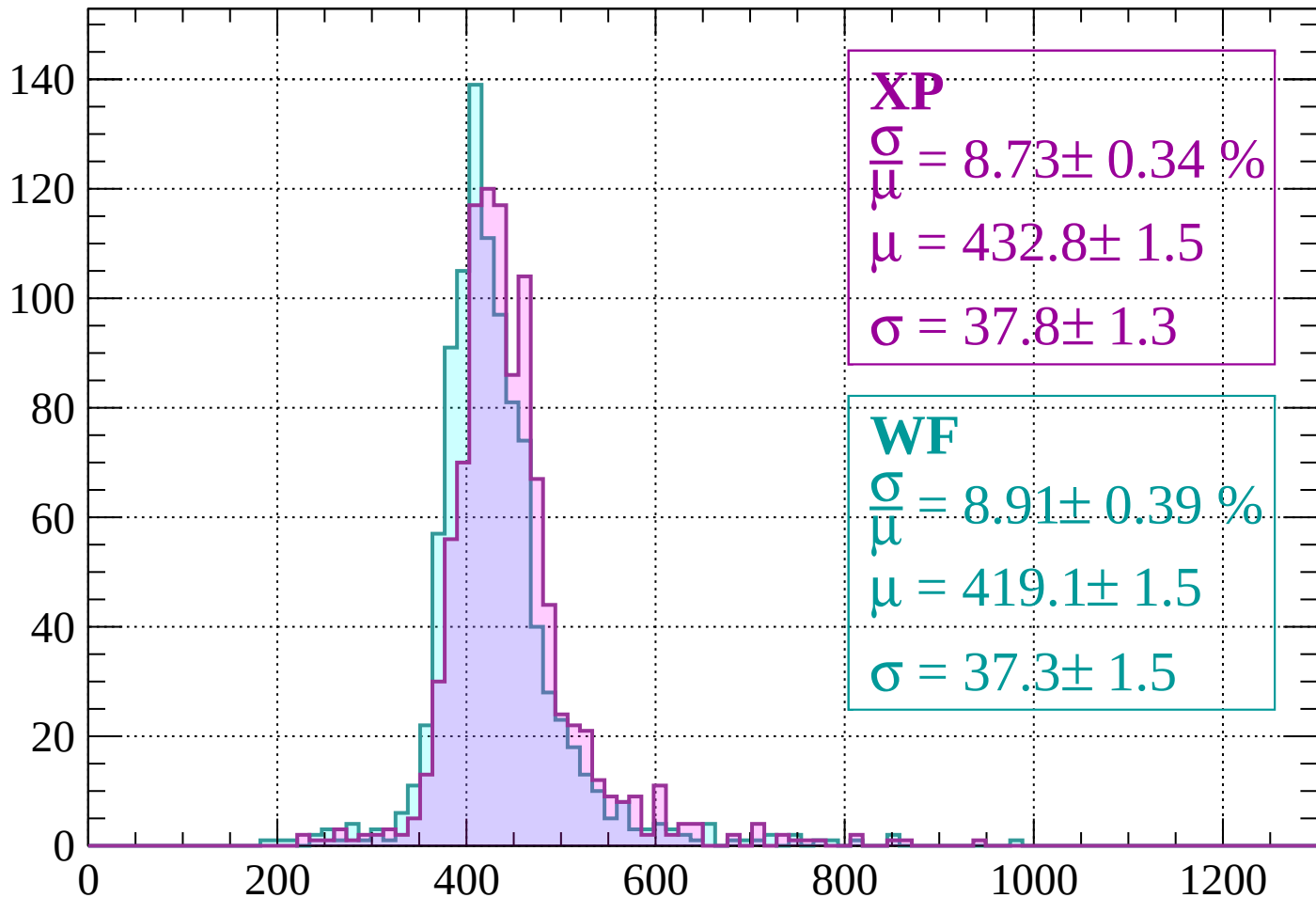
$$\mu = 421.5 \pm 1.5$$

$$\sigma = 37.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.73 \pm 0.34 \%$$

$$\mu = 432.8 \pm 1.5$$

$$\sigma = 37.8 \pm 1.3$$

WF

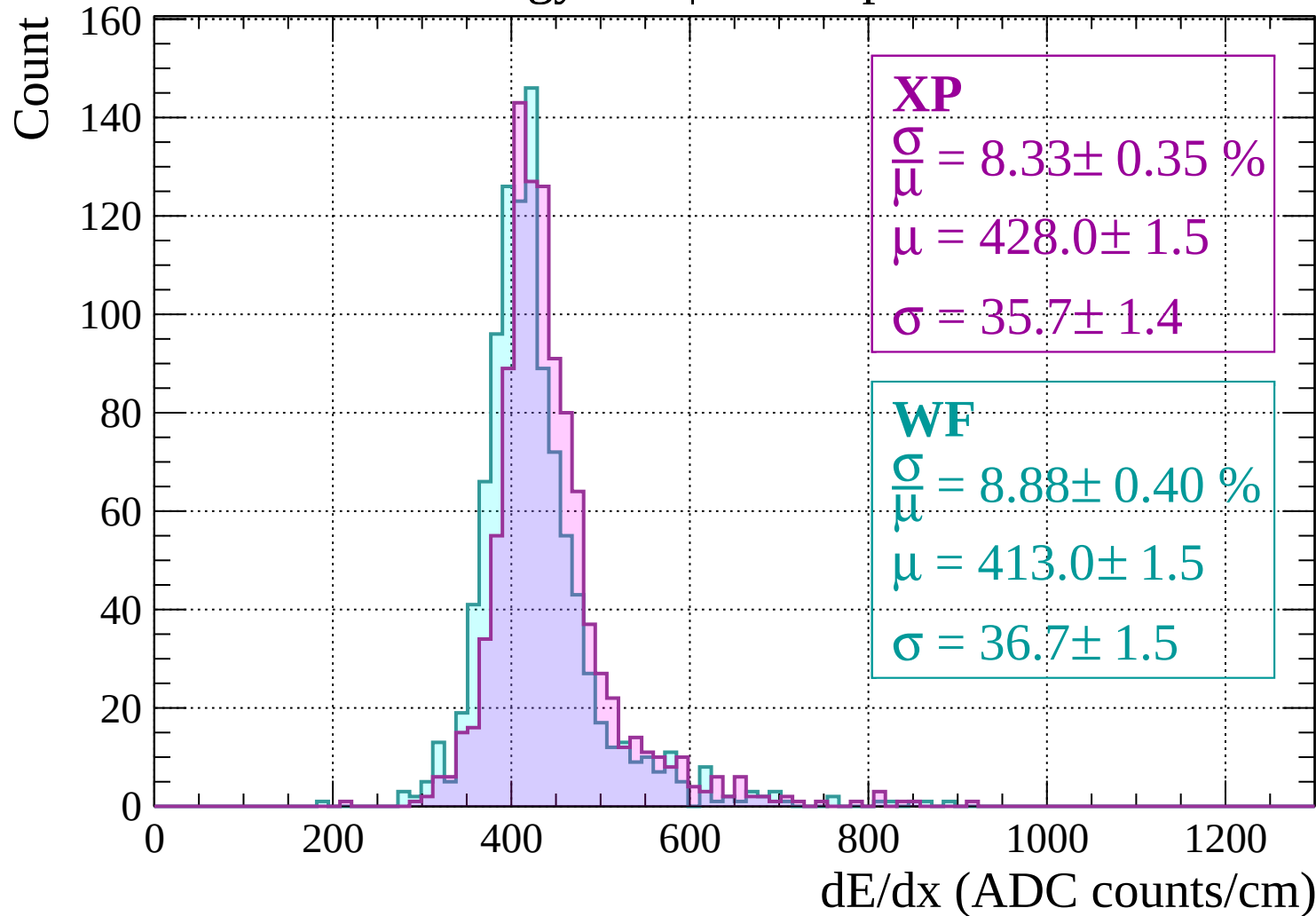
$$\frac{\sigma}{\mu} = 8.91 \pm 0.39 \%$$

$$\mu = 419.1 \pm 1.5$$

$$\sigma = 37.3 \pm 1.5$$

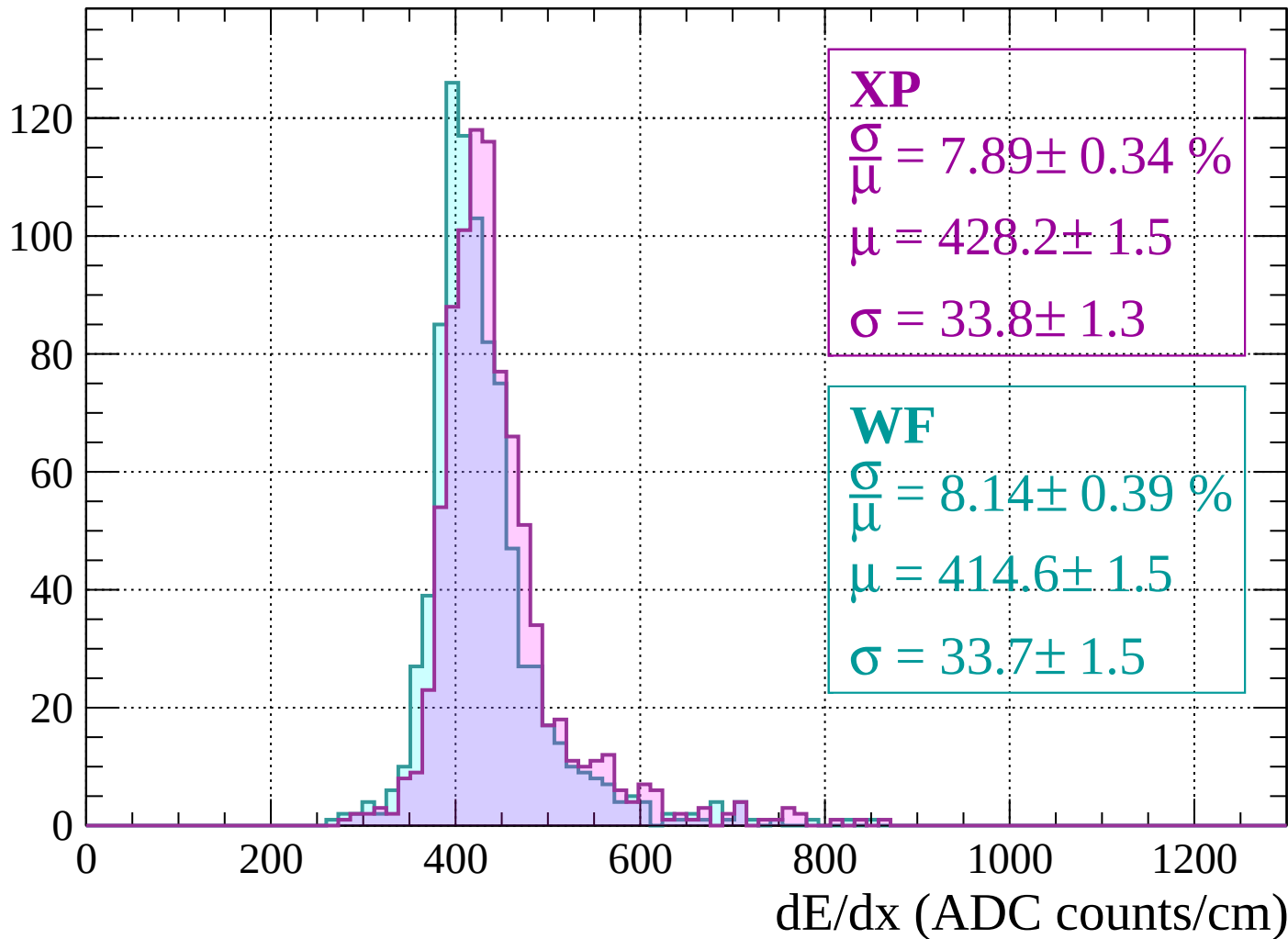
dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$



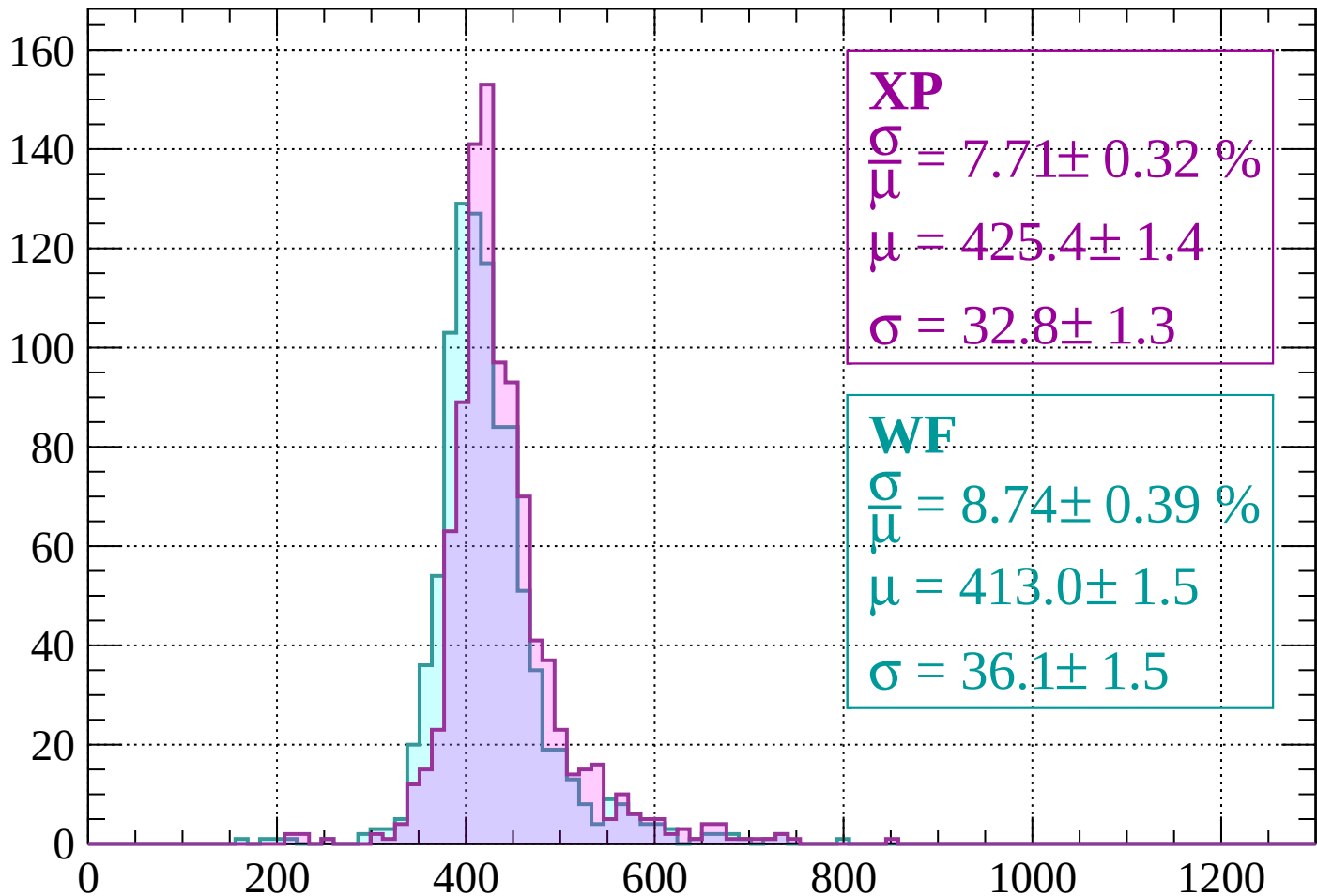
Energy loss | $-480 < p < -440$

Count



Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 7.71 \pm 0.32 \%$$

$$\mu = 425.4 \pm 1.4$$

$$\sigma = 32.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.74 \pm 0.39 \%$$

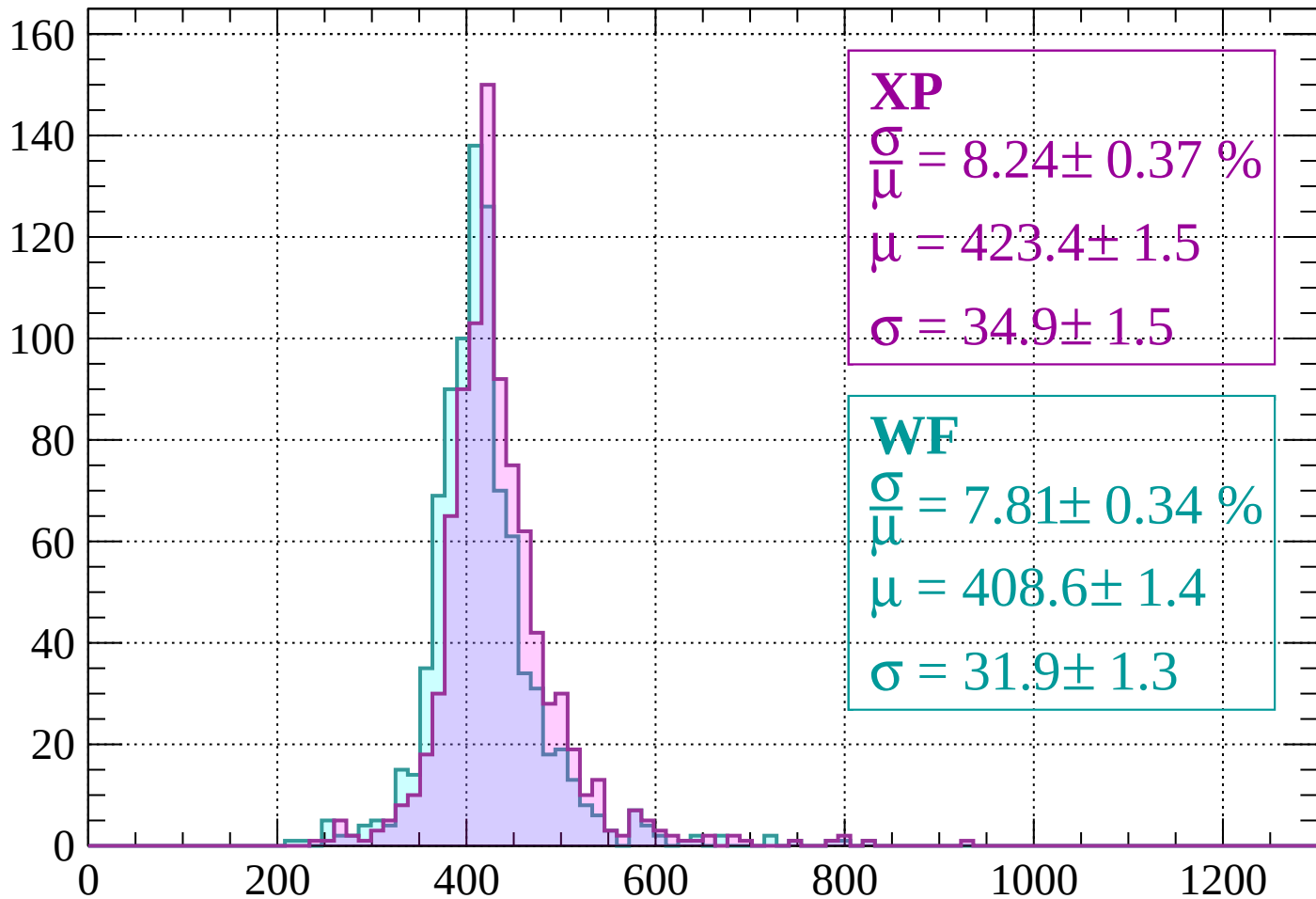
$$\mu = 413.0 \pm 1.5$$

$$\sigma = 36.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.24 \pm 0.37 \%$$

$$\mu = 423.4 \pm 1.5$$

$$\sigma = 34.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.81 \pm 0.34 \%$$

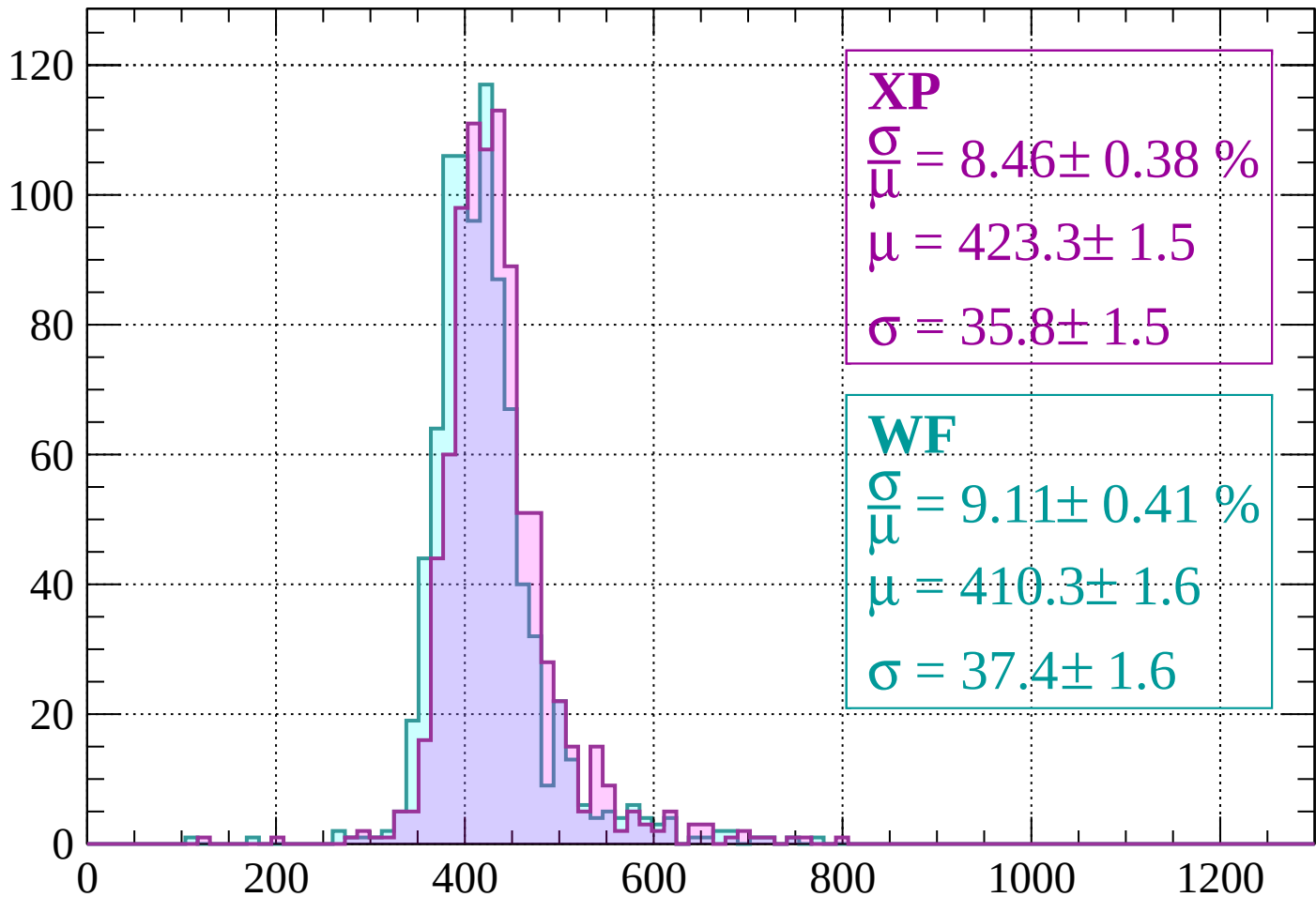
$$\mu = 408.6 \pm 1.4$$

$$\sigma = 31.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.38 \%$$

$$\mu = 423.3 \pm 1.5$$

$$\sigma = 35.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.11 \pm 0.41 \%$$

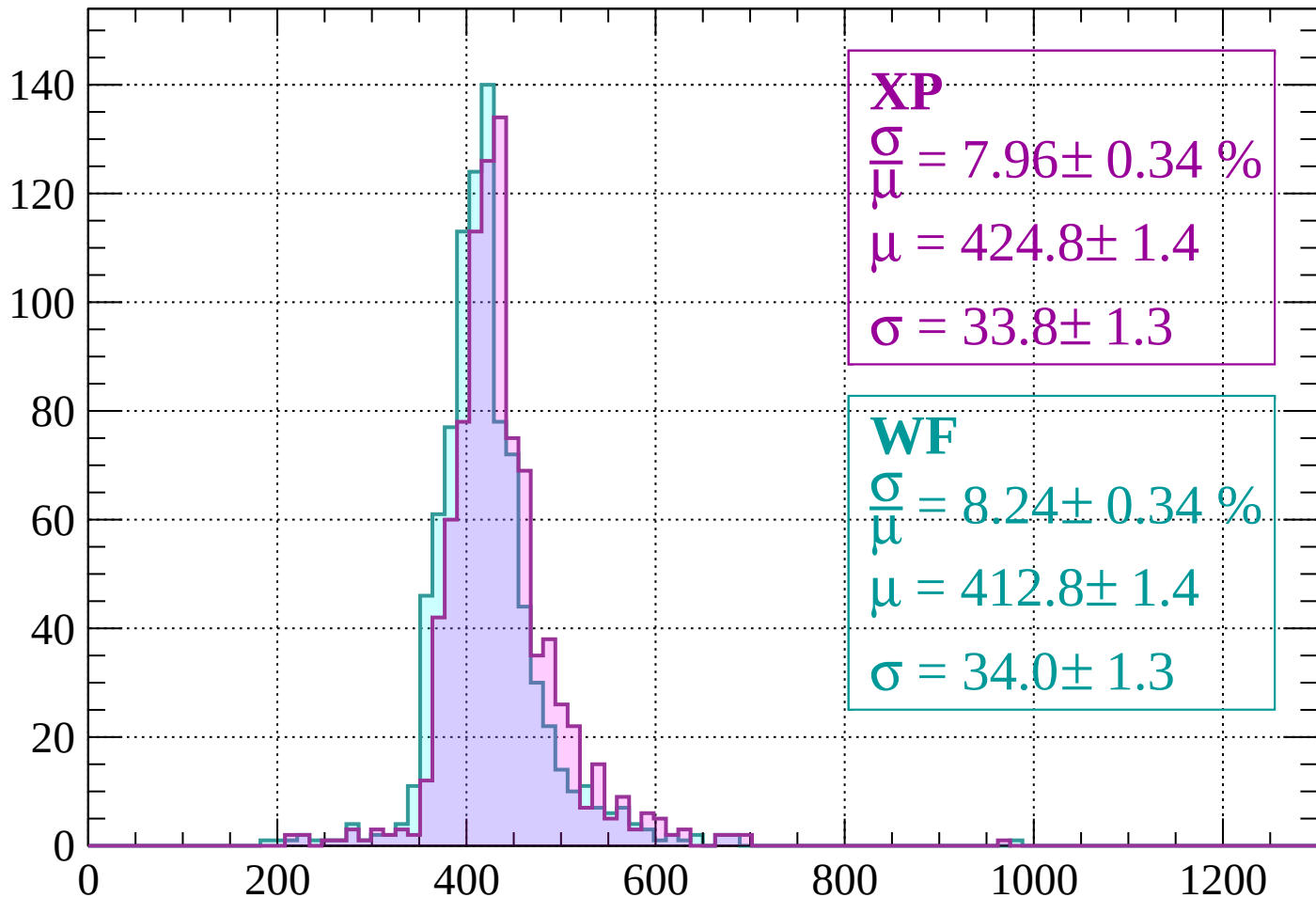
$$\mu = 410.3 \pm 1.6$$

$$\sigma = 37.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.96 \pm 0.34 \%$$

$$\mu = 424.8 \pm 1.4$$

$$\sigma = 33.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.34 \%$$

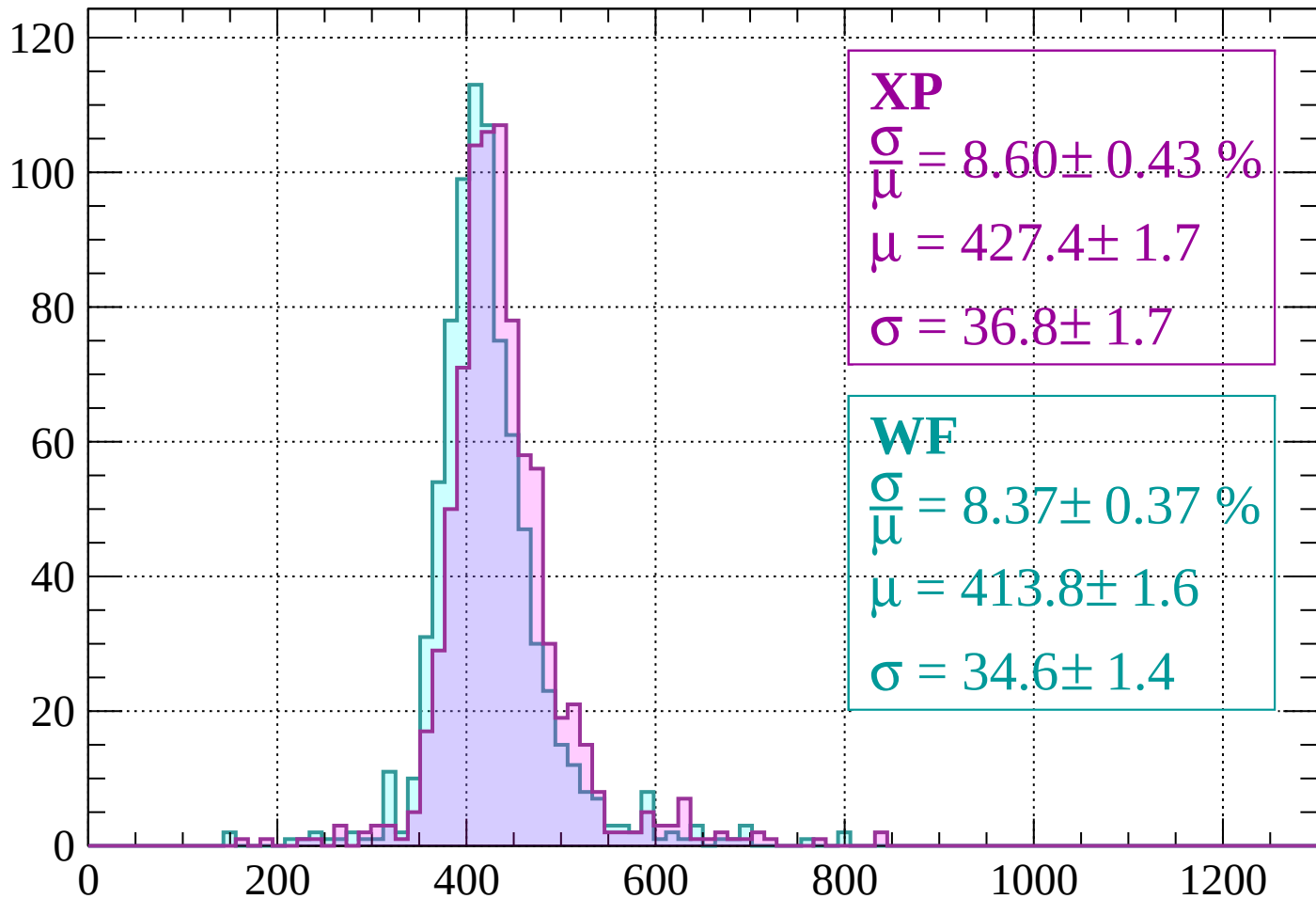
$$\mu = 412.8 \pm 1.4$$

$$\sigma = 34.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.60 \pm 0.43 \%$$

$$\mu = 427.4 \pm 1.7$$

$$\sigma = 36.8 \pm 1.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.37 \pm 0.37 \%$$

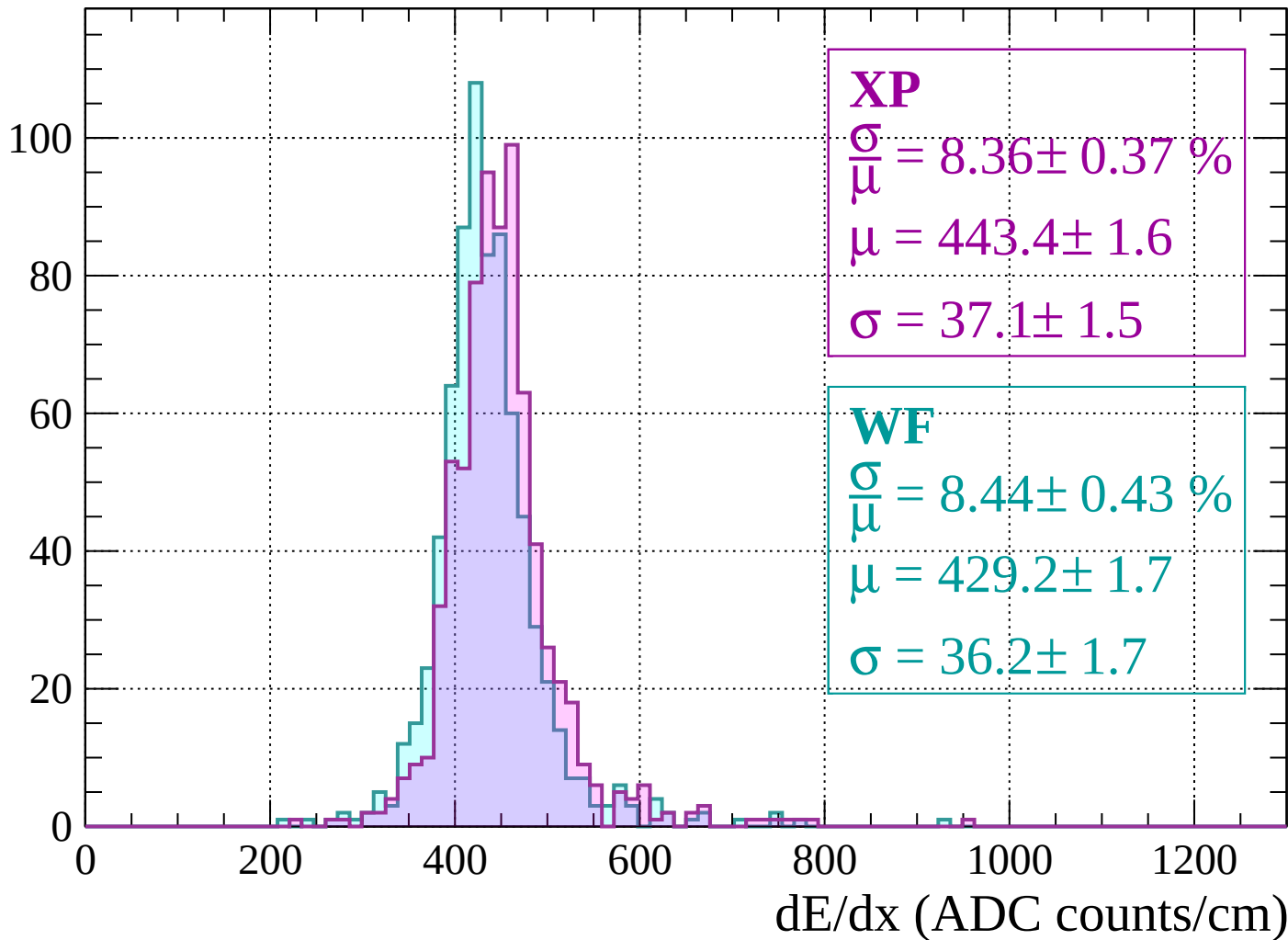
$$\mu = 413.8 \pm 1.6$$

$$\sigma = 34.6 \pm 1.4$$

dE/dx (ADC counts/cm)

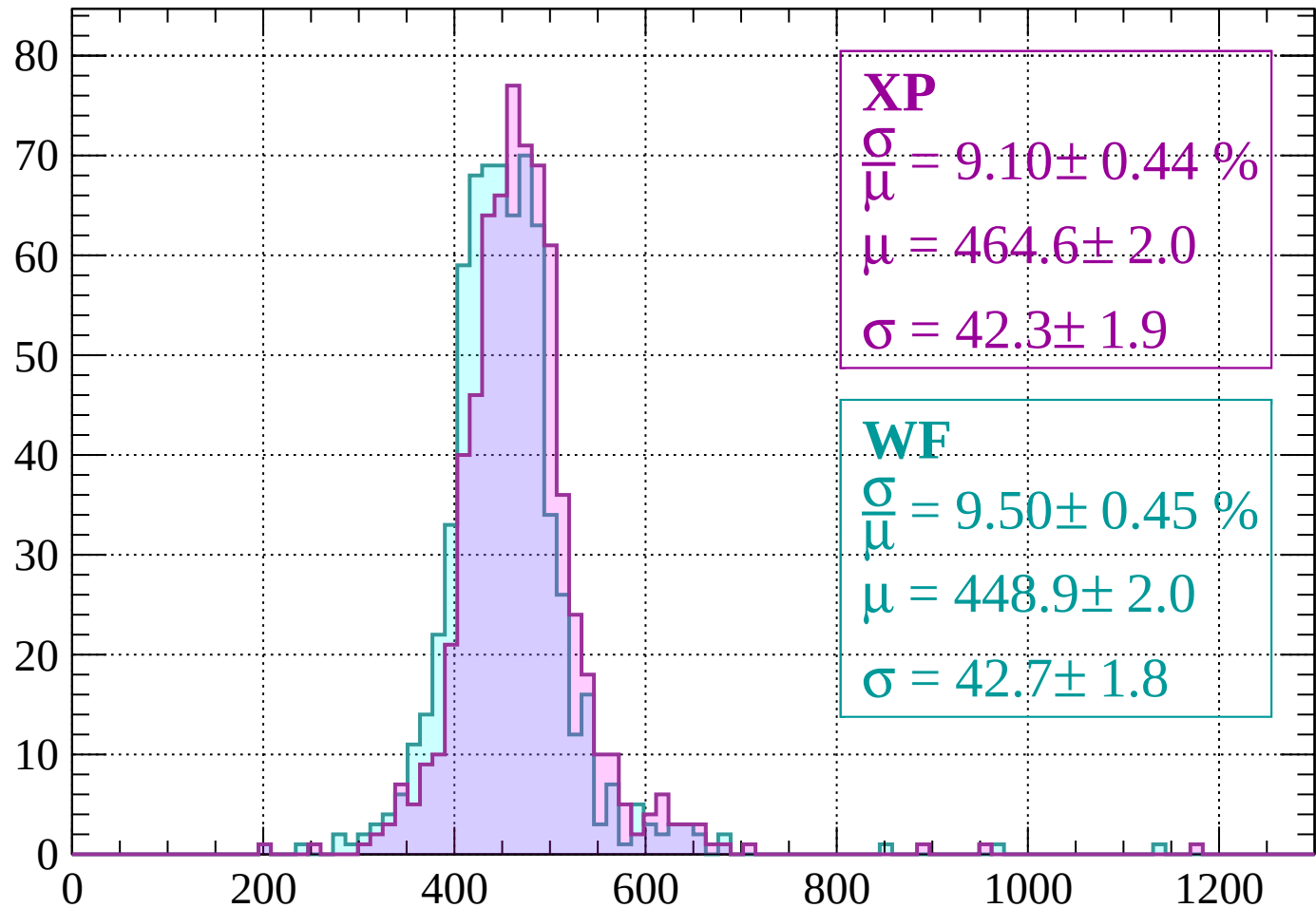
Energy loss | $-240 < p < -200$

Count



Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.44 \%$$

$$\mu = 464.6 \pm 2.0$$

$$\sigma = 42.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.45 \%$$

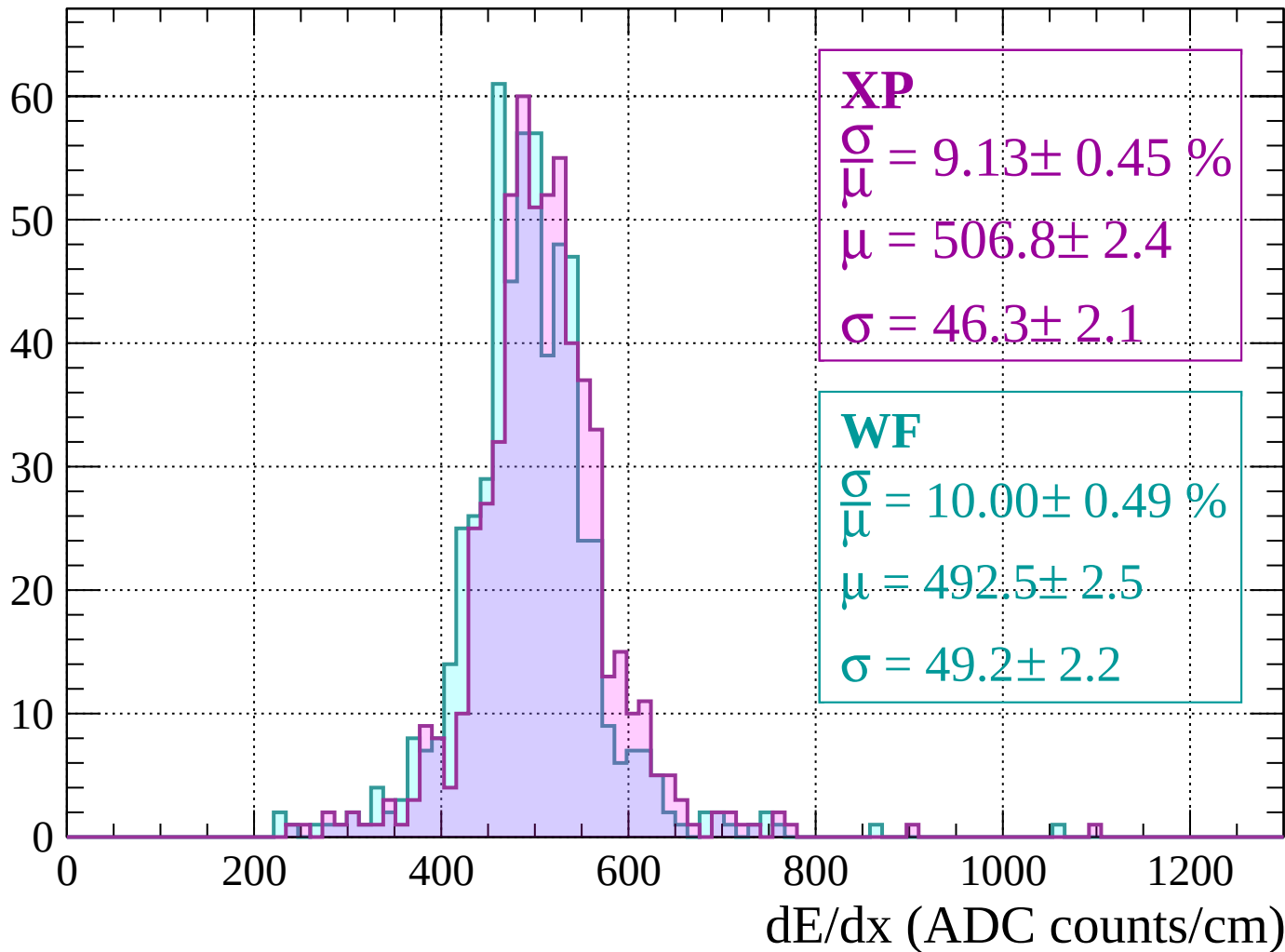
$$\mu = 448.9 \pm 2.0$$

$$\sigma = 42.7 \pm 1.8$$

dE/dx (ADC counts/cm)

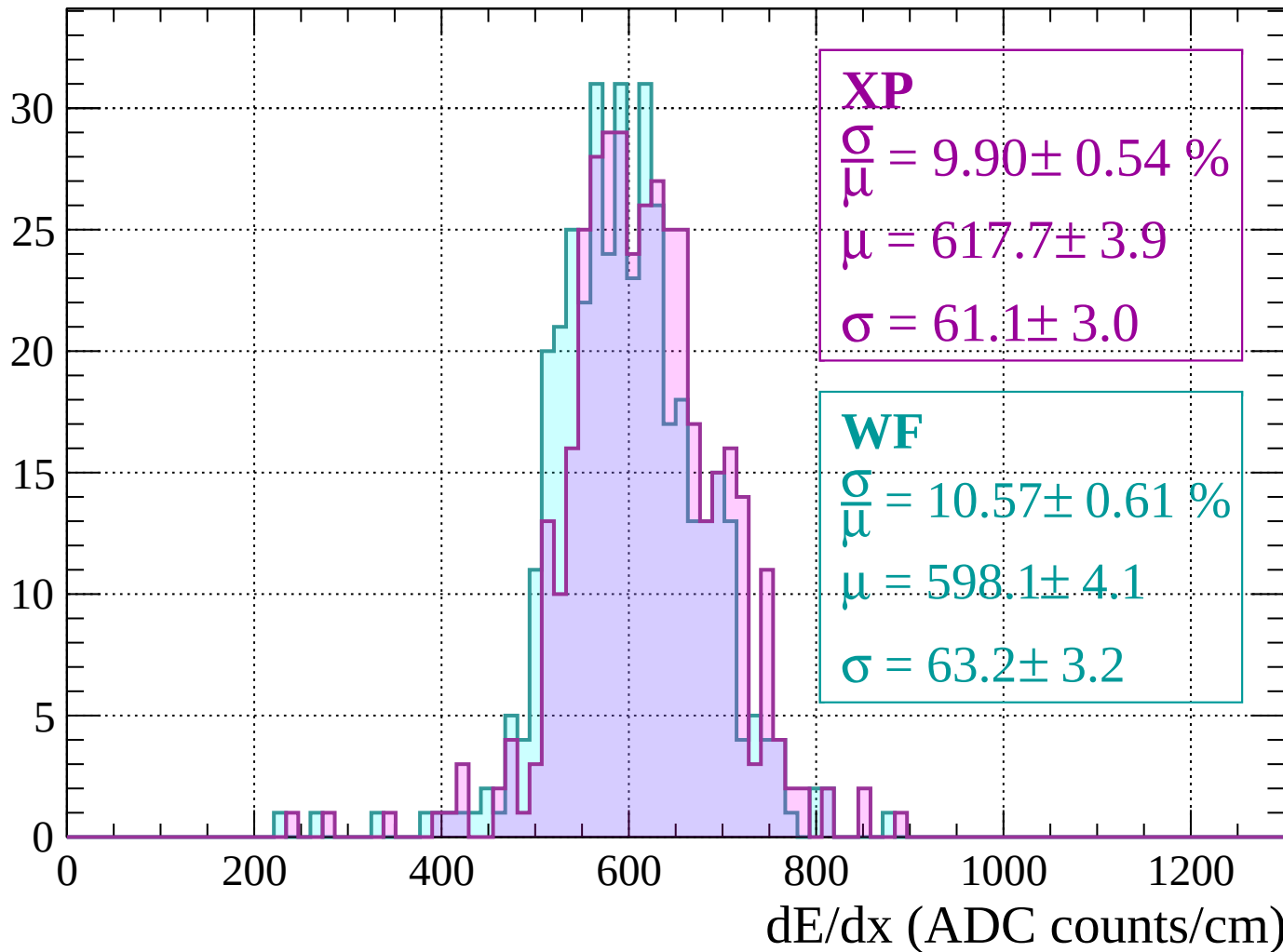
Energy loss | $-160 < p < -120$

Count



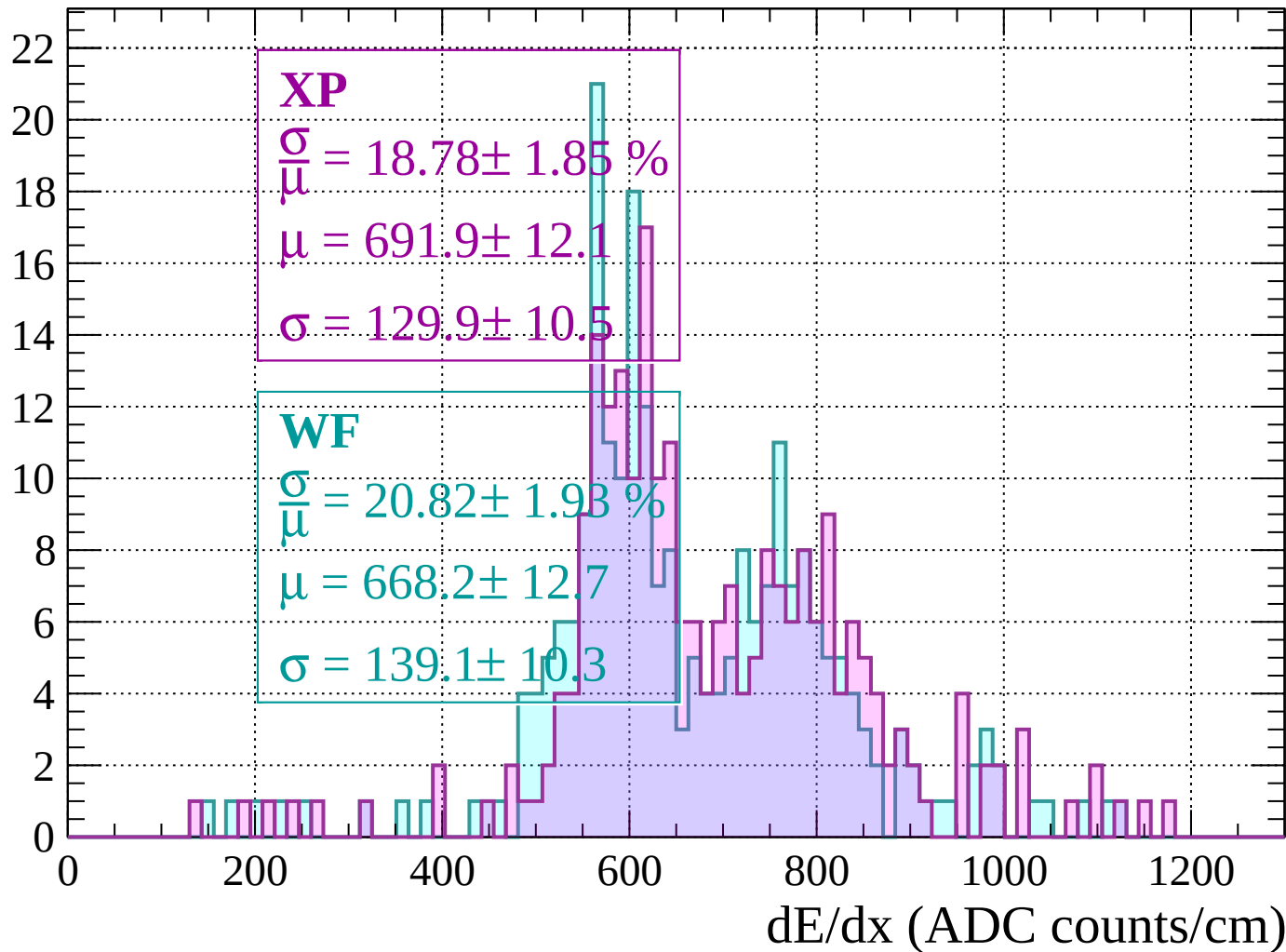
Energy loss | $-120 < p < -80$

Count



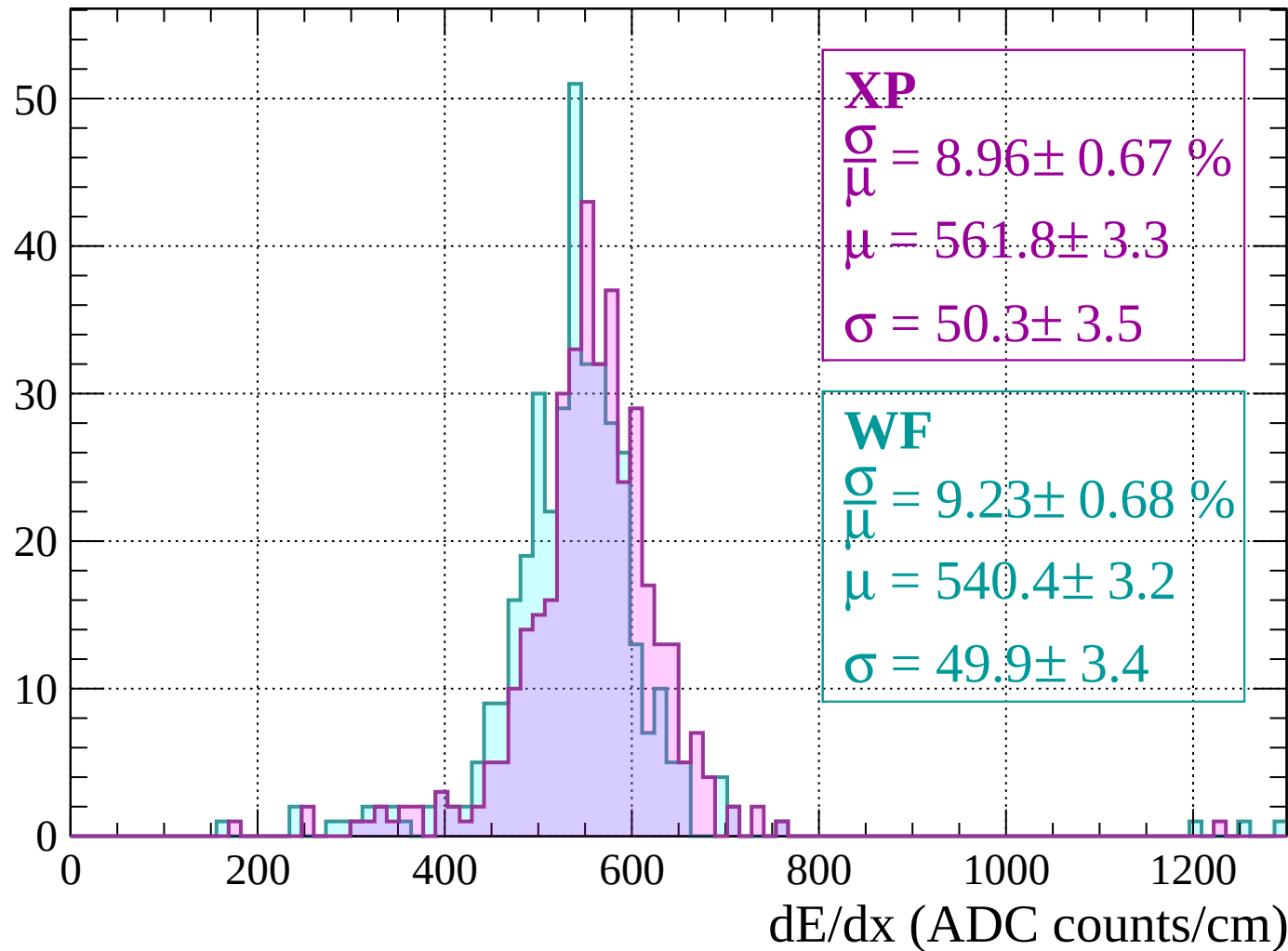
Energy loss | $-80 < p < -40$

Count



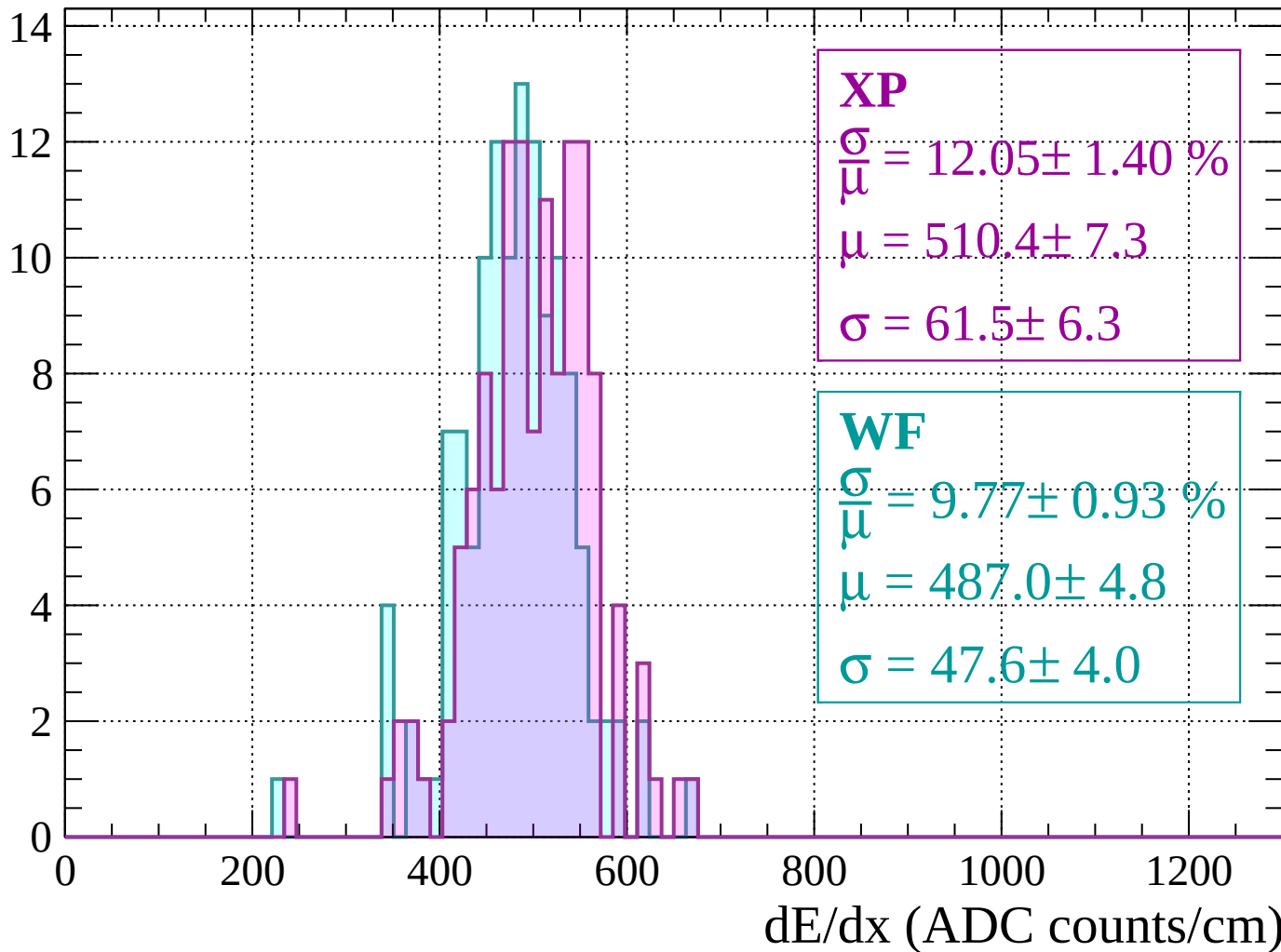
Energy loss | $-40 < p < 0$

Count



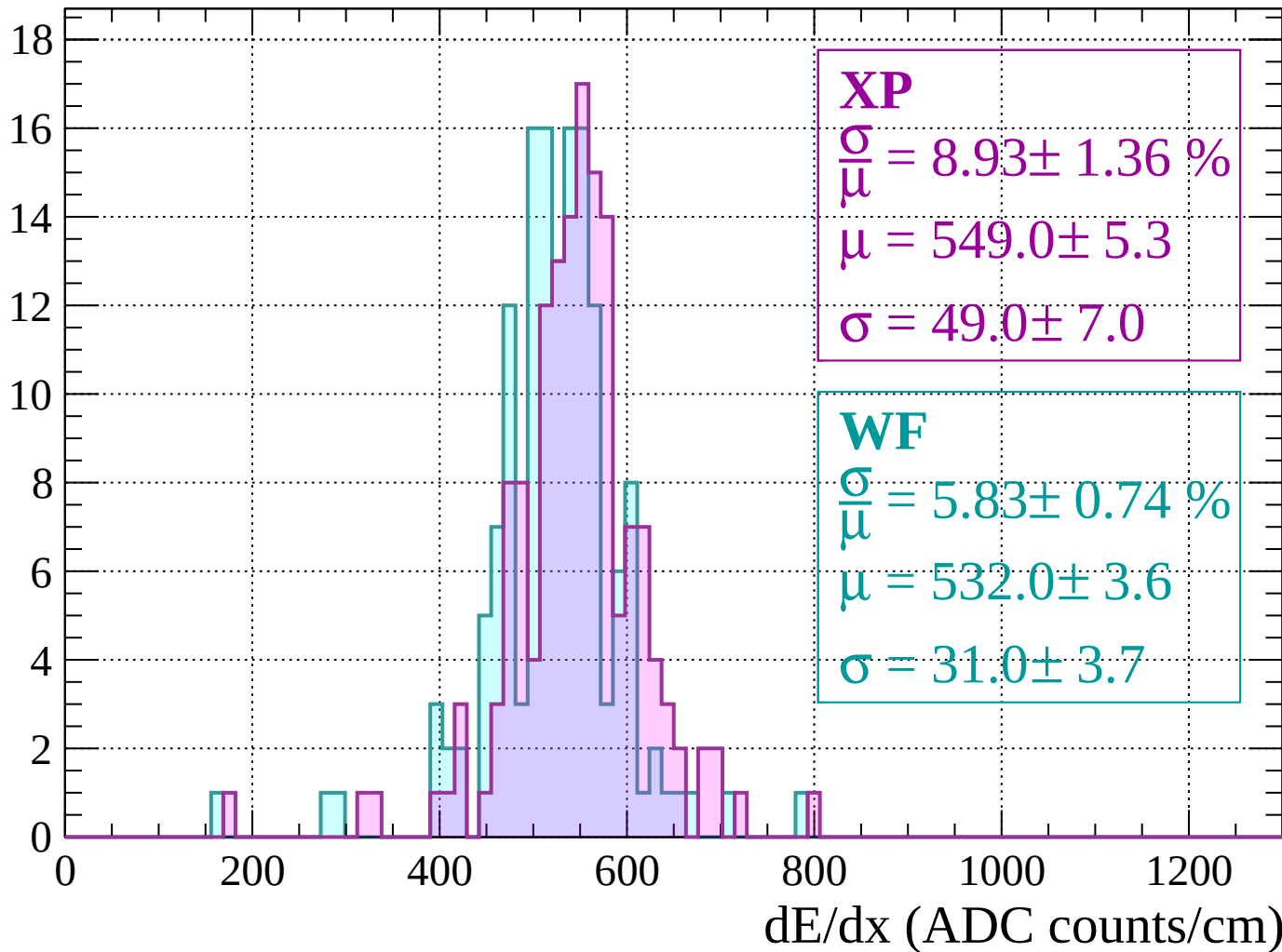
Energy loss | $0 < p < 40$

Count



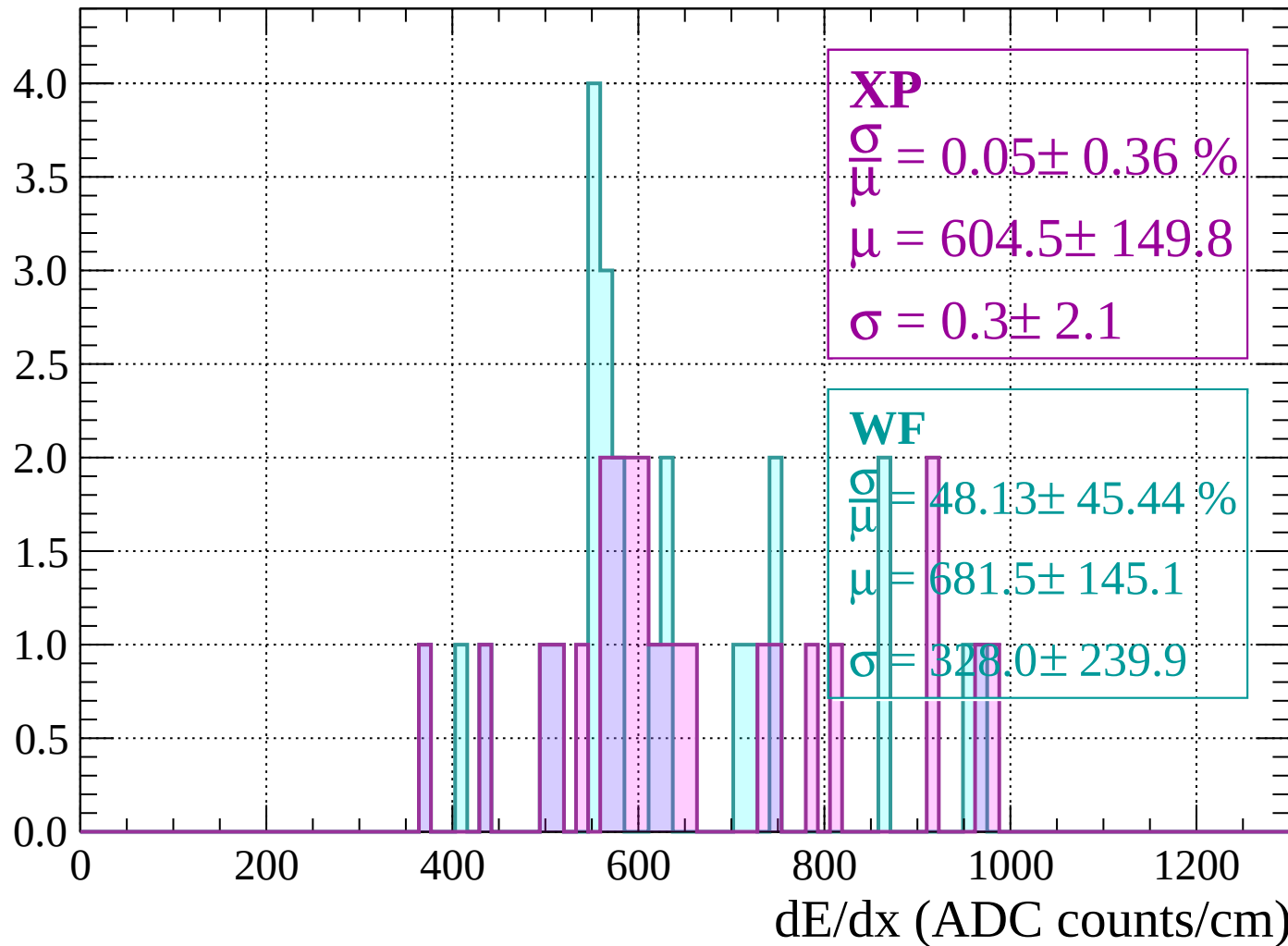
Energy loss | $40 < p < 80$

Count



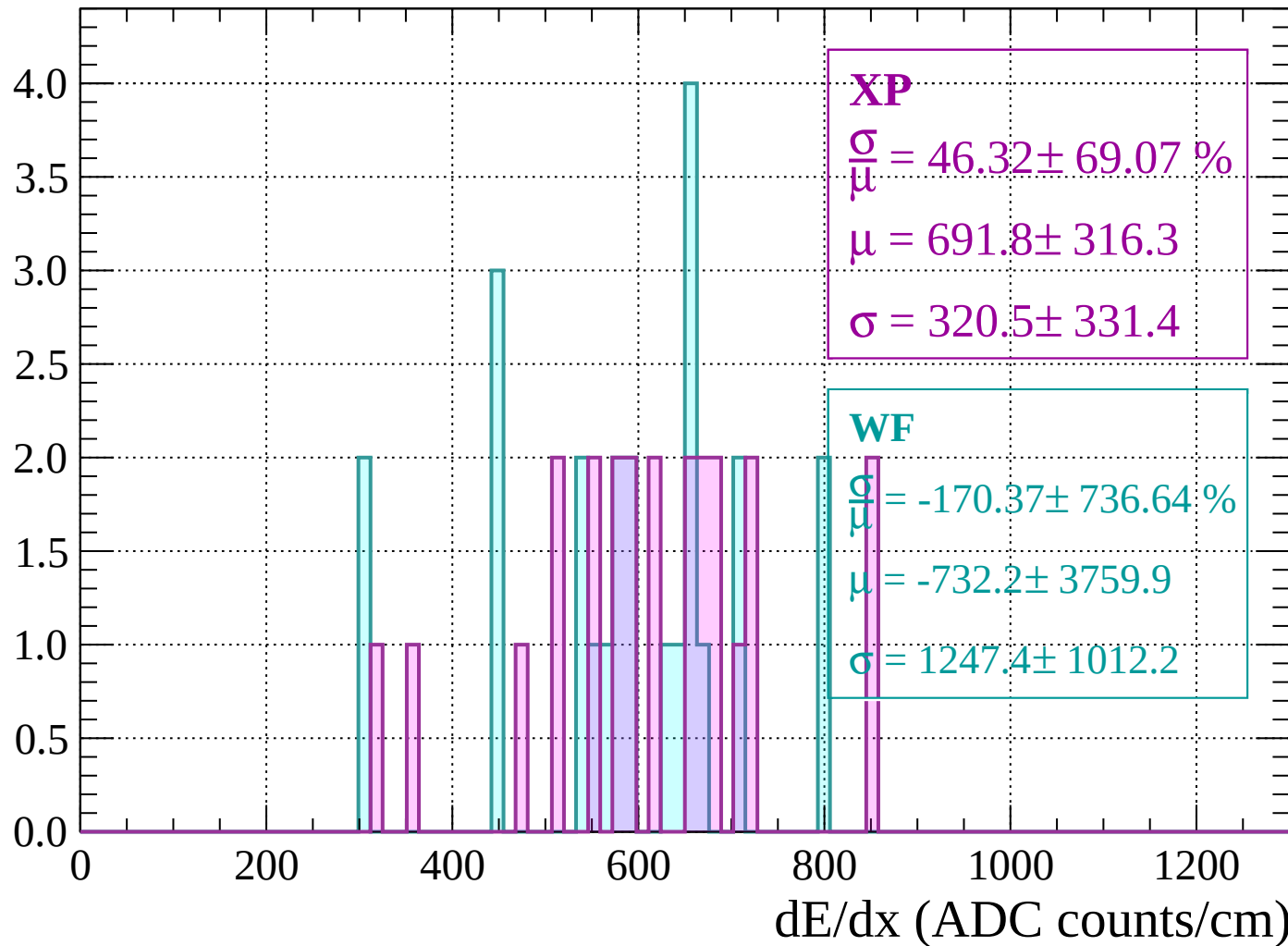
Energy loss | $80 < p < 120$

Count



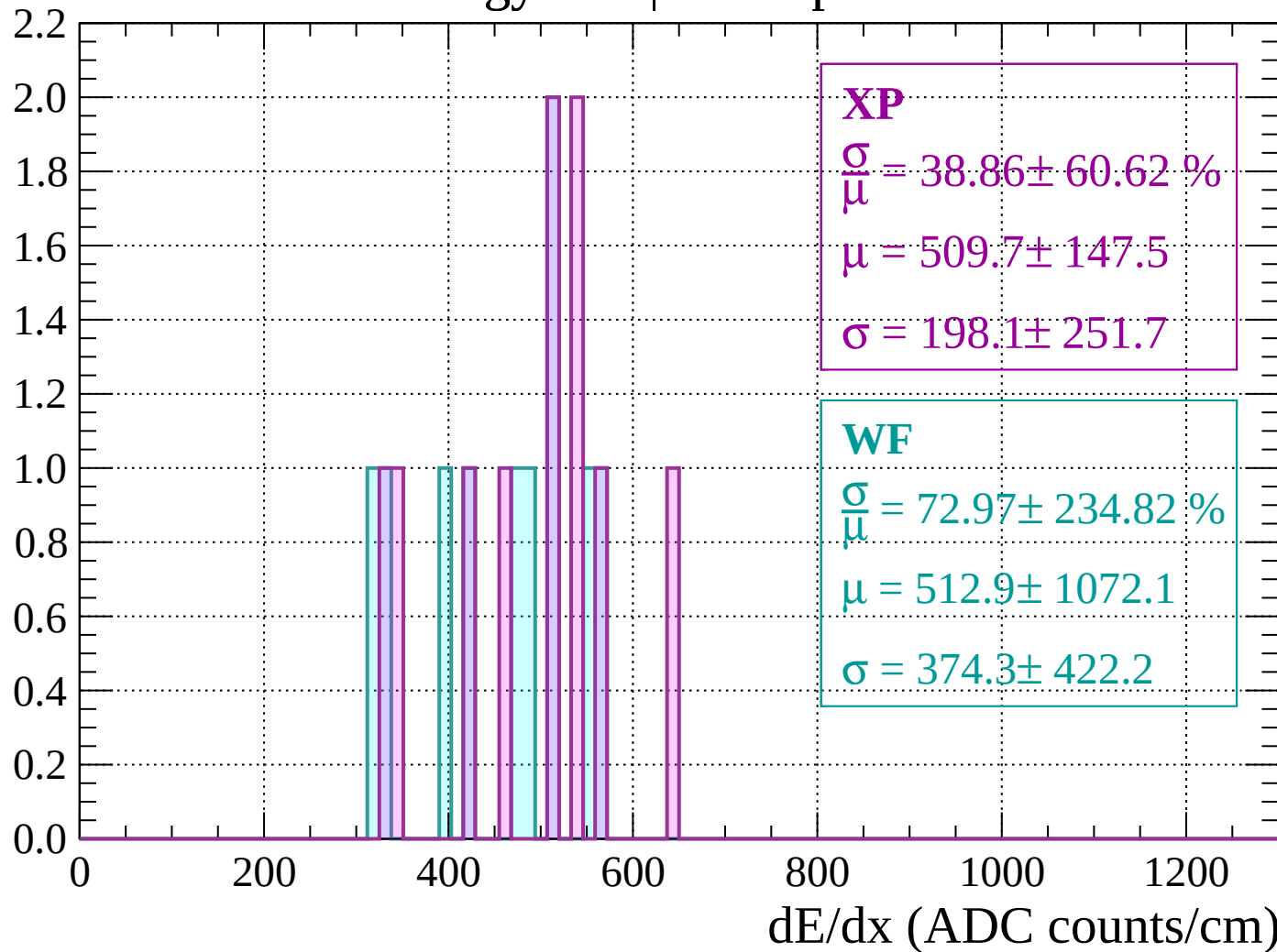
Energy loss | $120 < p < 160$

Count



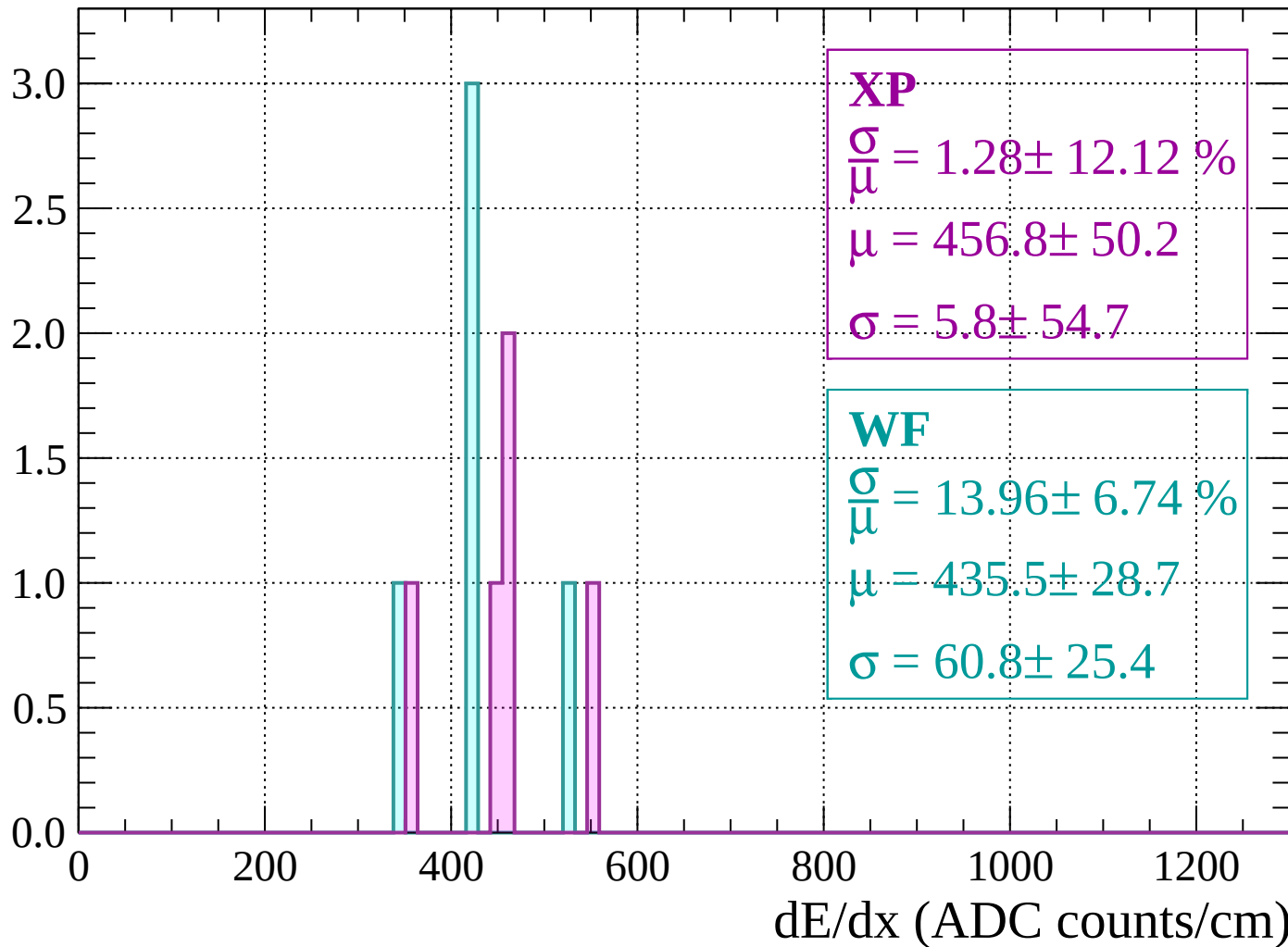
Energy loss | $160 < p < 200$

Count



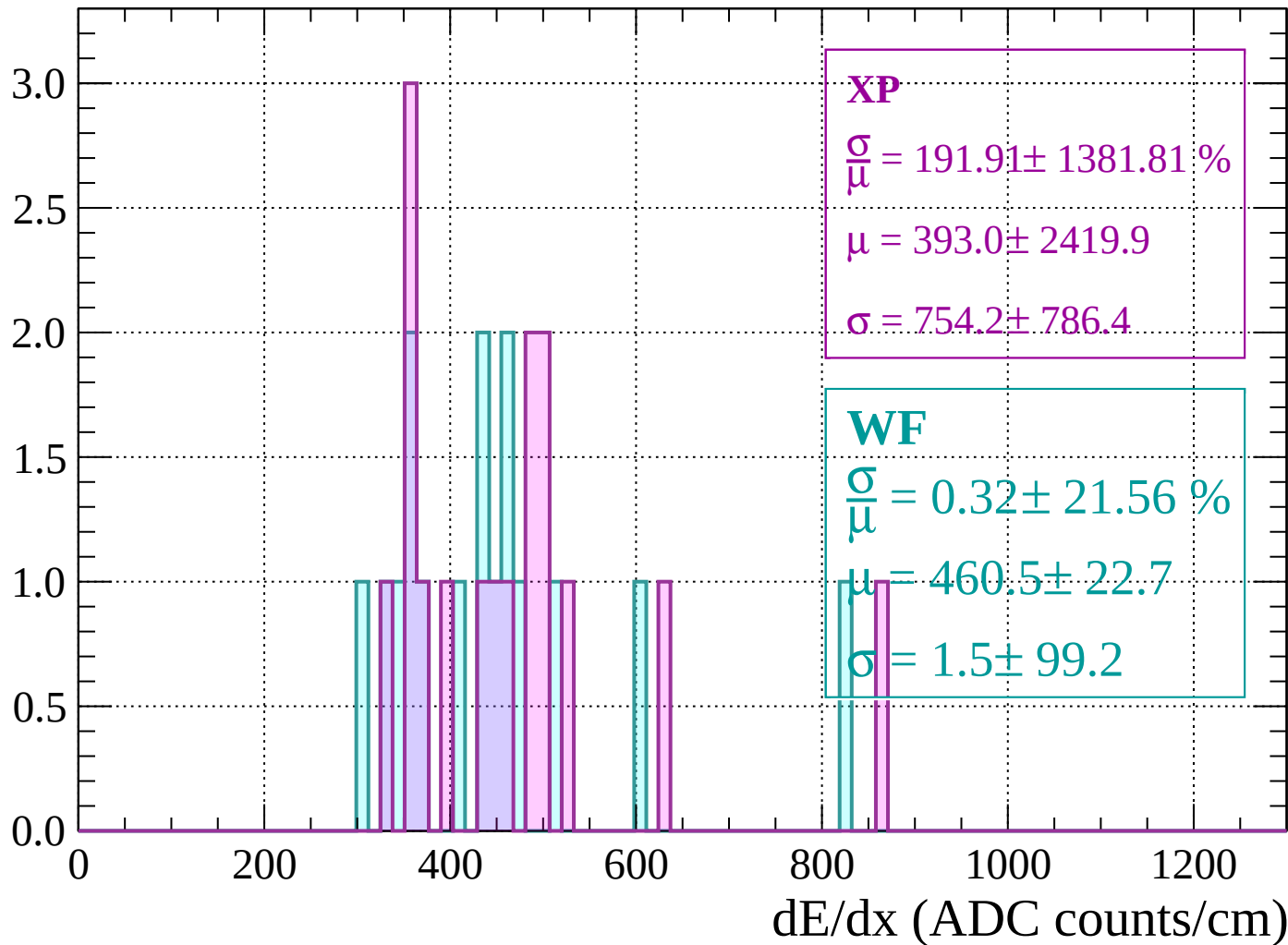
Energy loss | $200 < p < 240$

Count



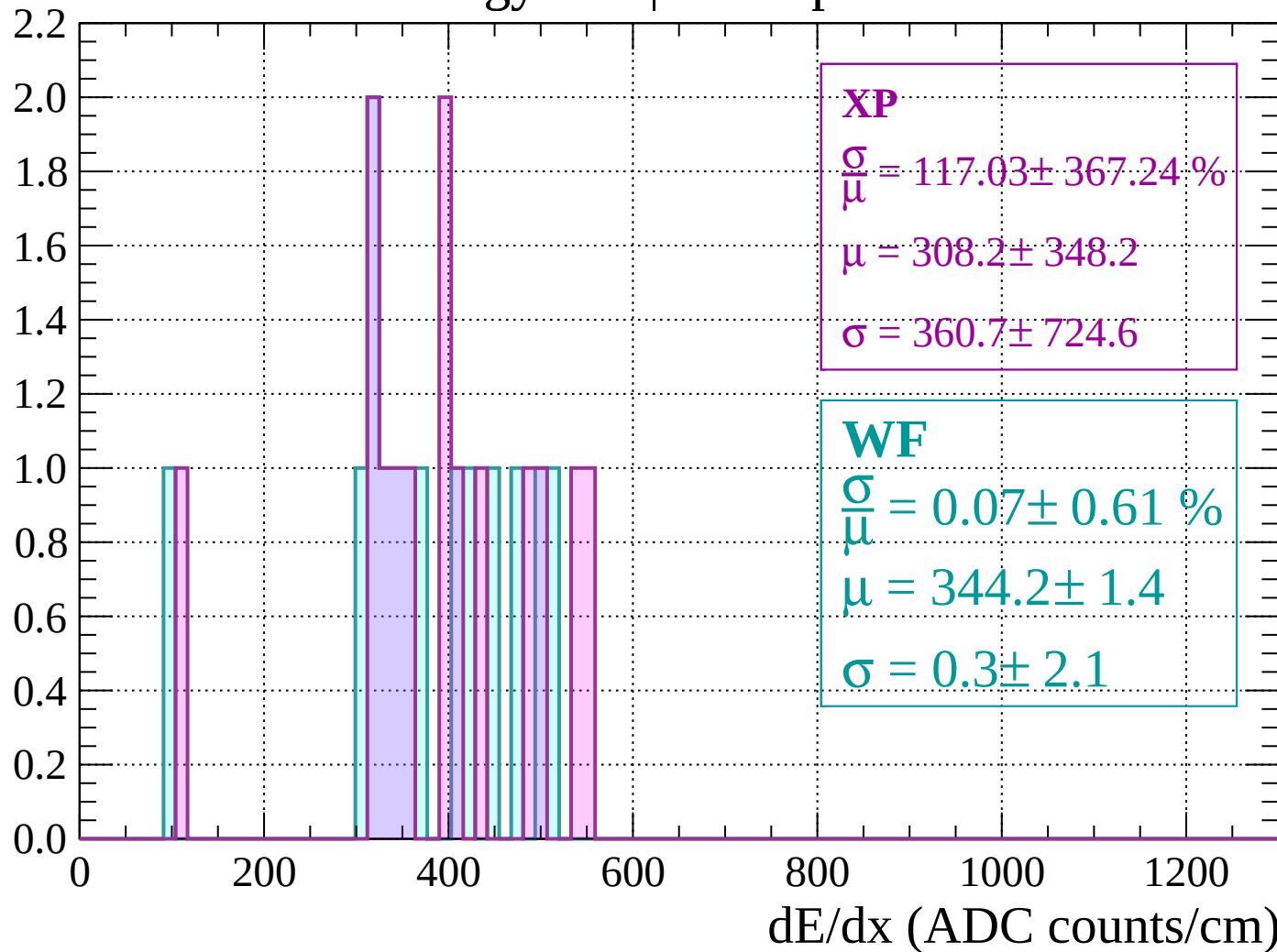
Energy loss | $240 < p < 280$

Count



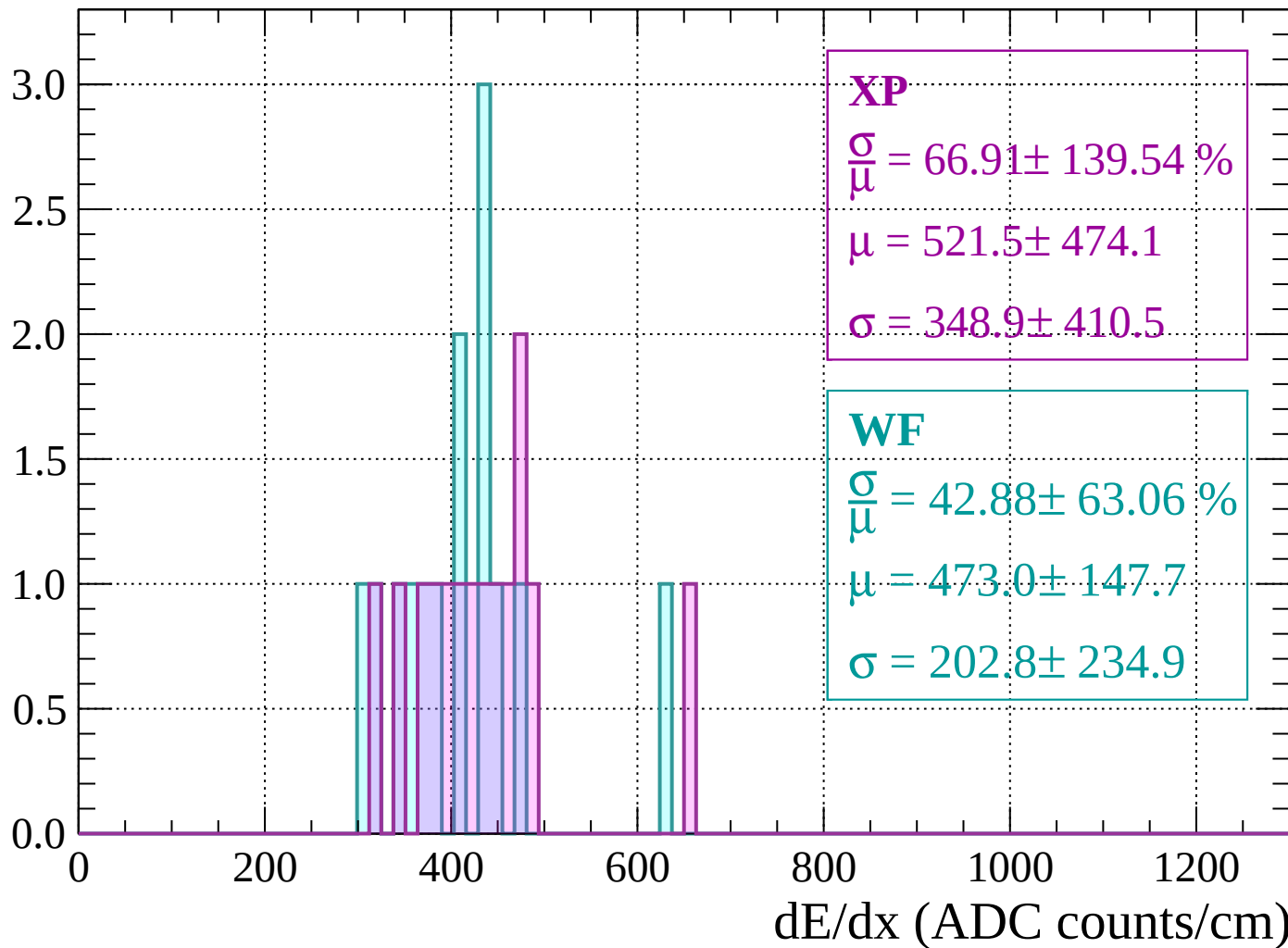
Energy loss | $280 < p < 320$

Count



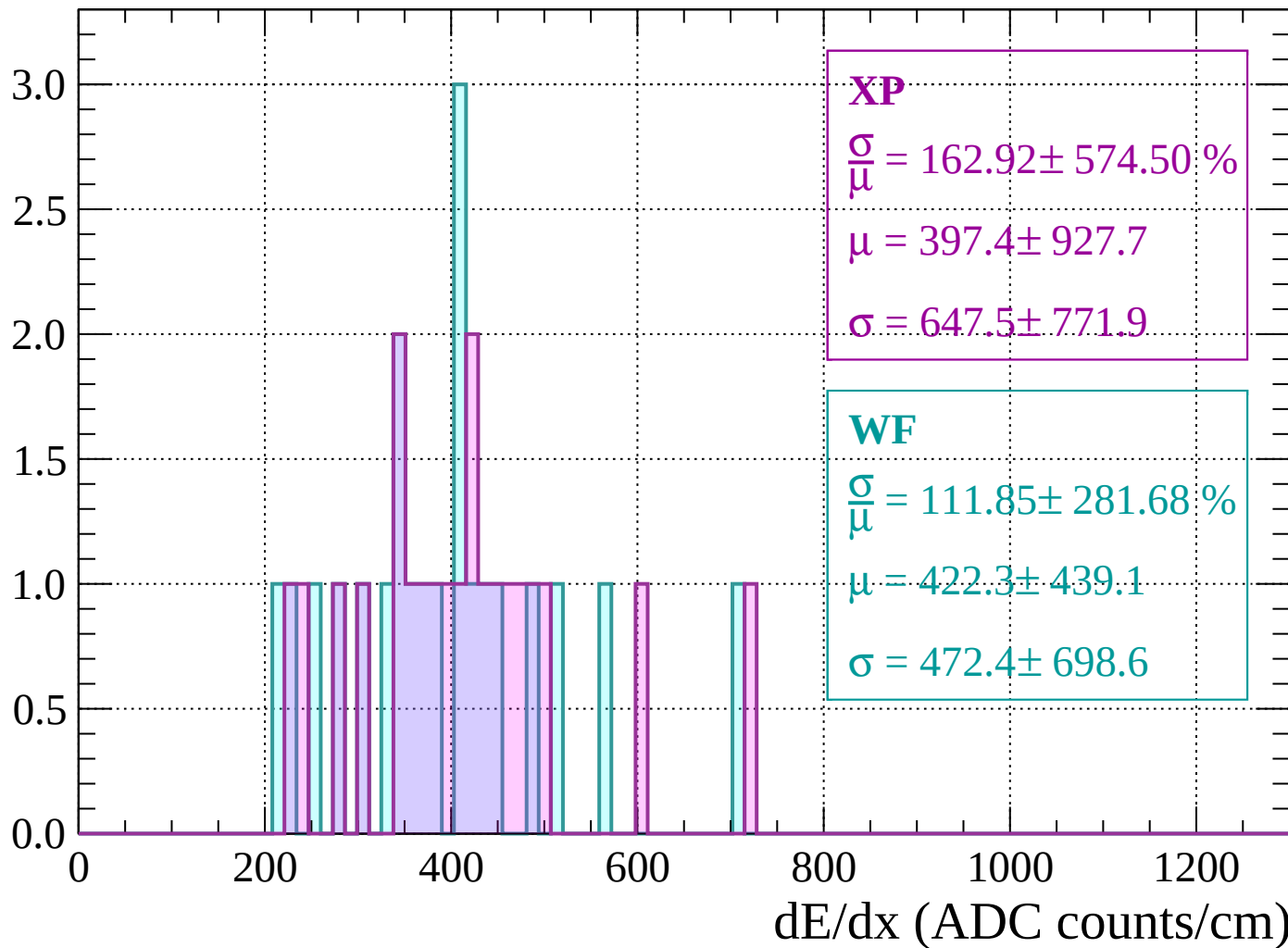
Energy loss | $320 < p < 360$

Count



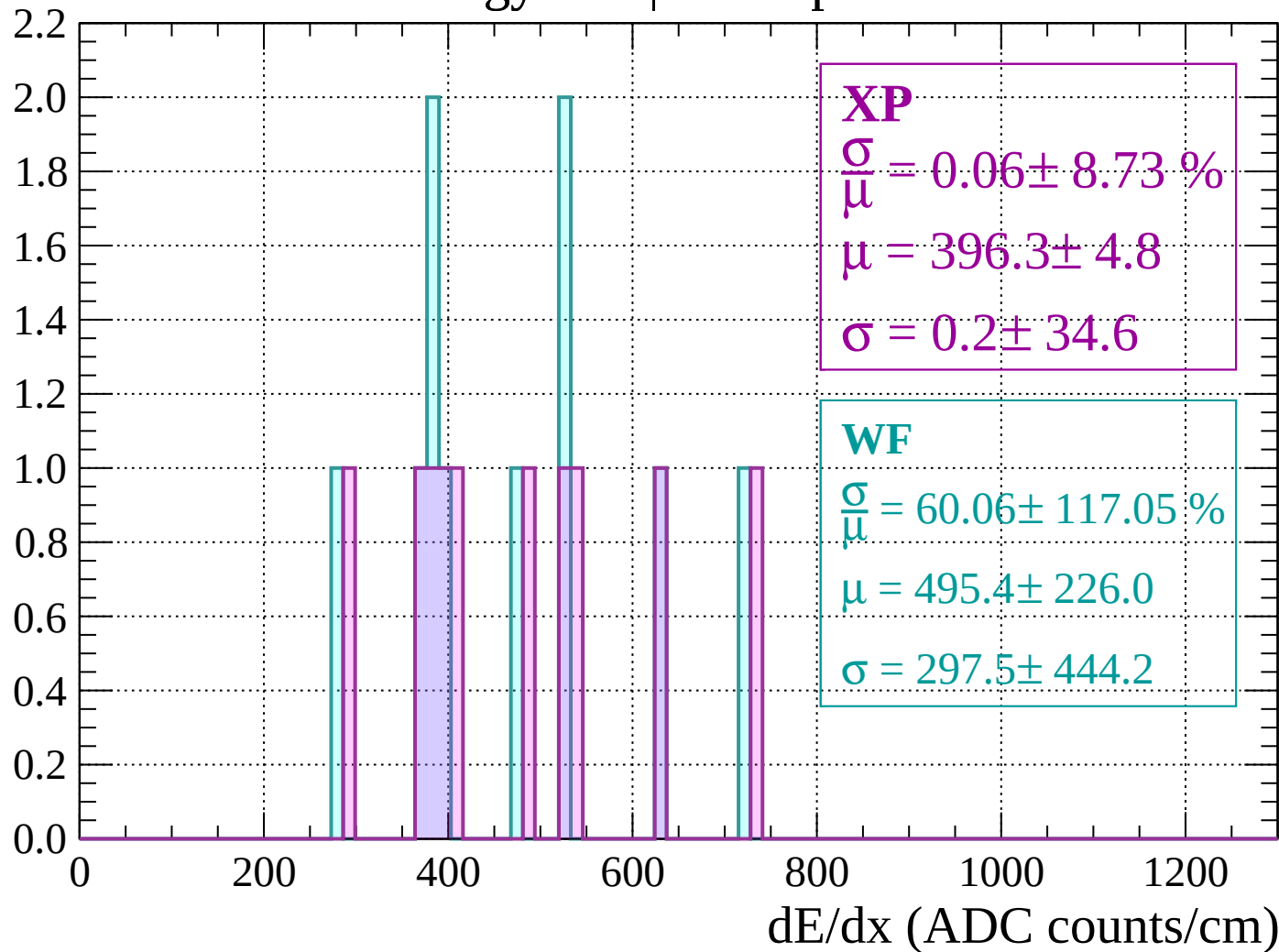
Energy loss | $360 < p < 400$

Count



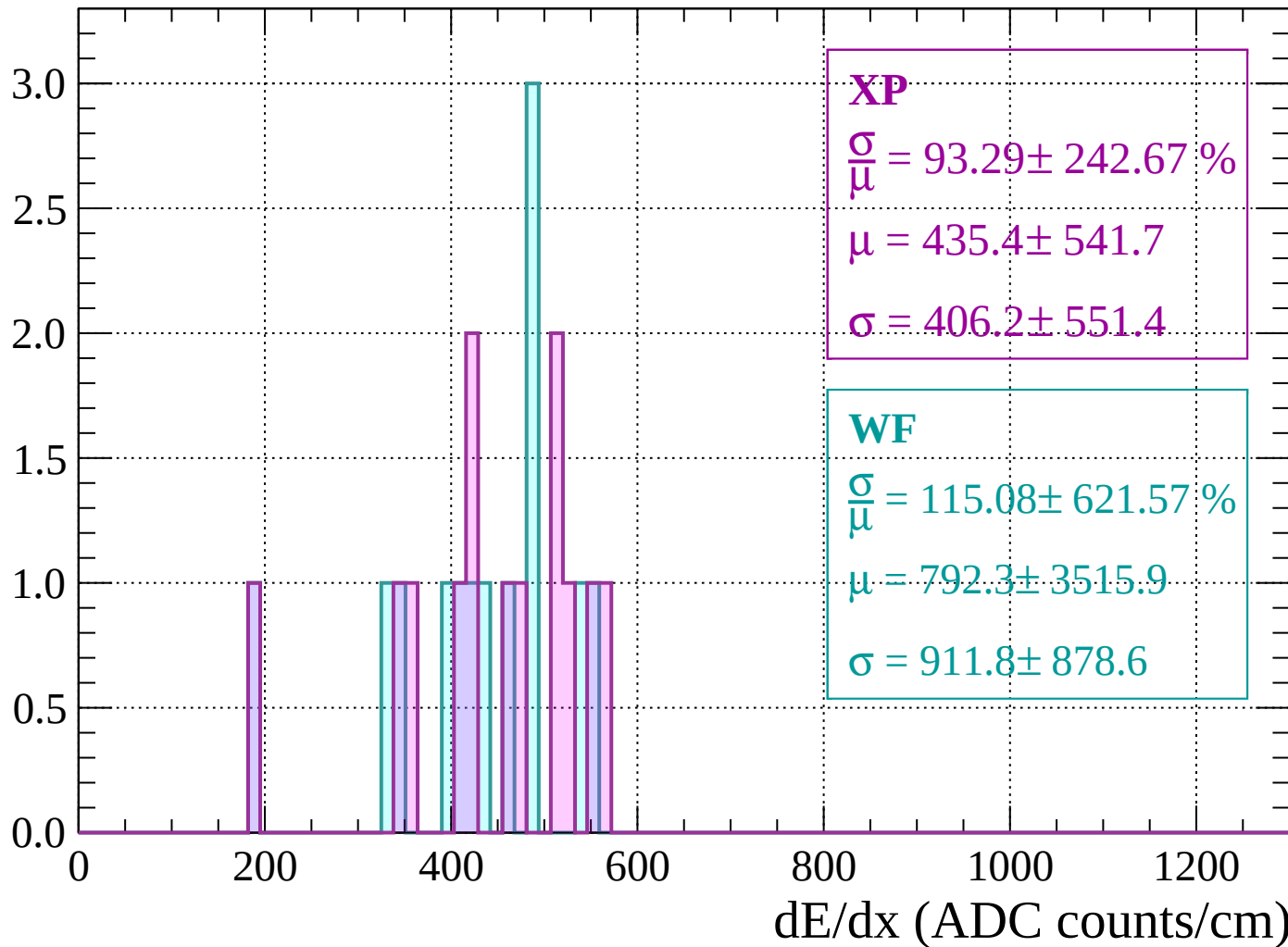
Energy loss | $400 < p < 440$

Count



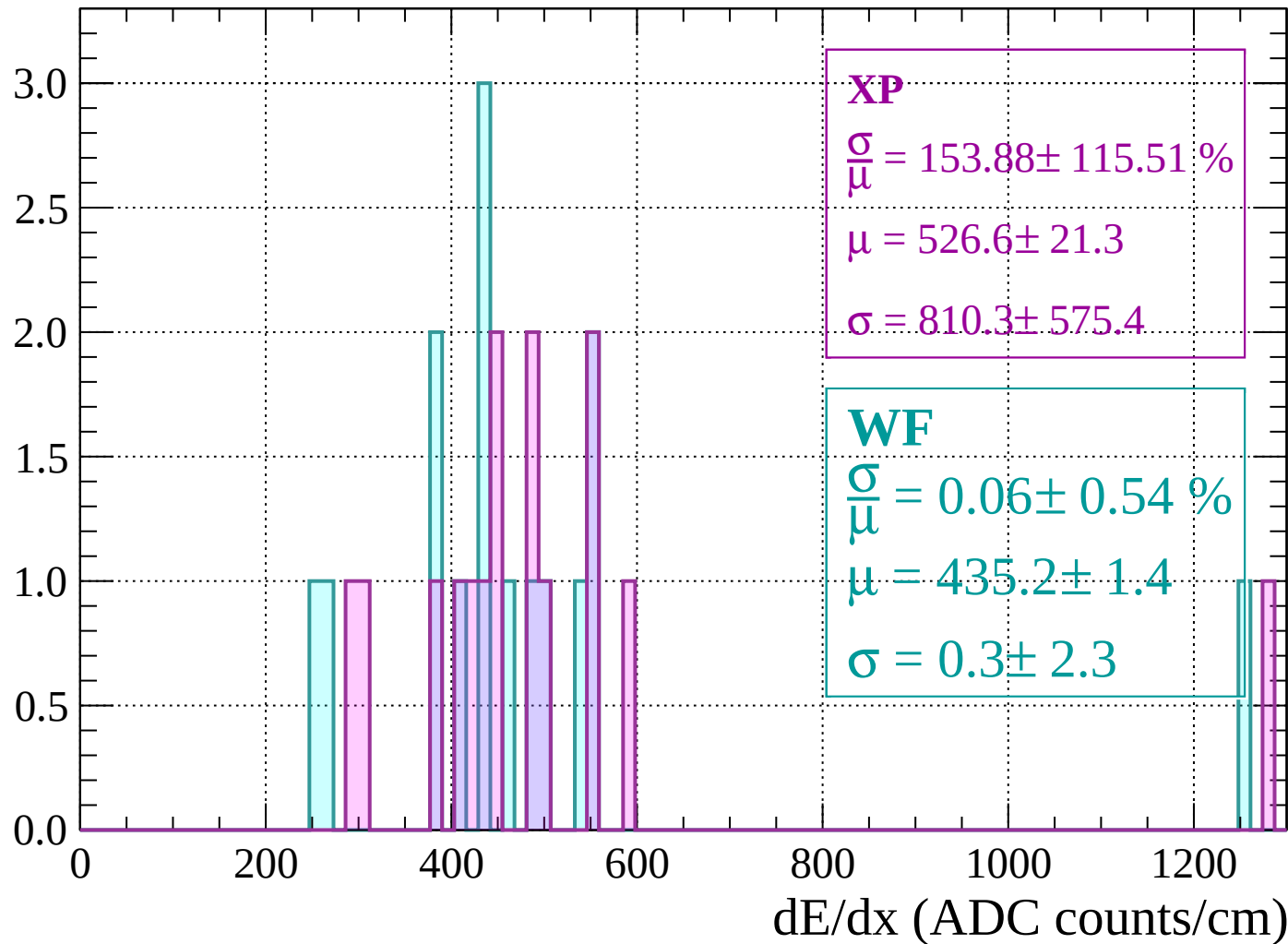
Energy loss | $440 < p < 480$

Count



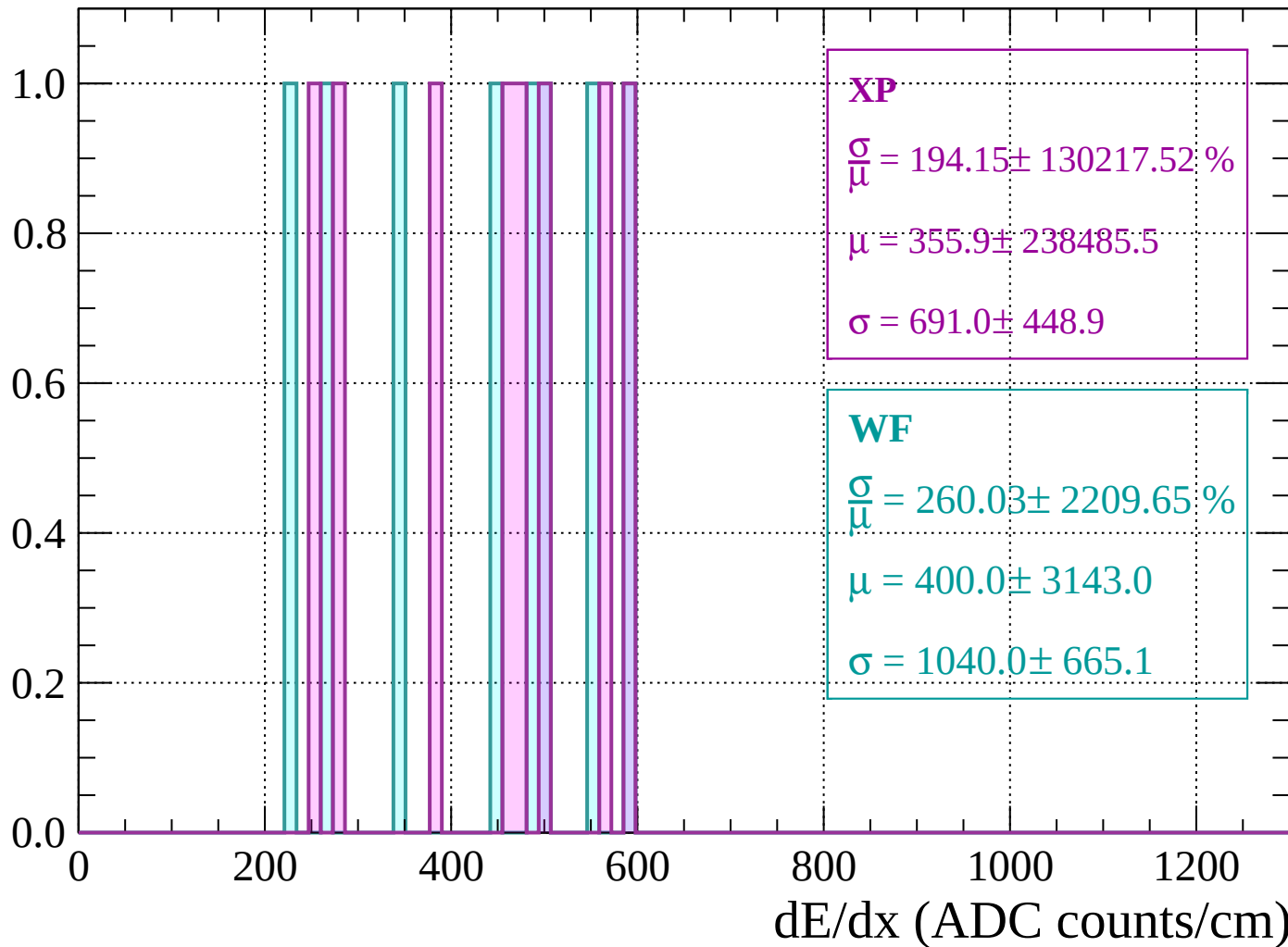
Energy loss | $480 < p < 520$

Count



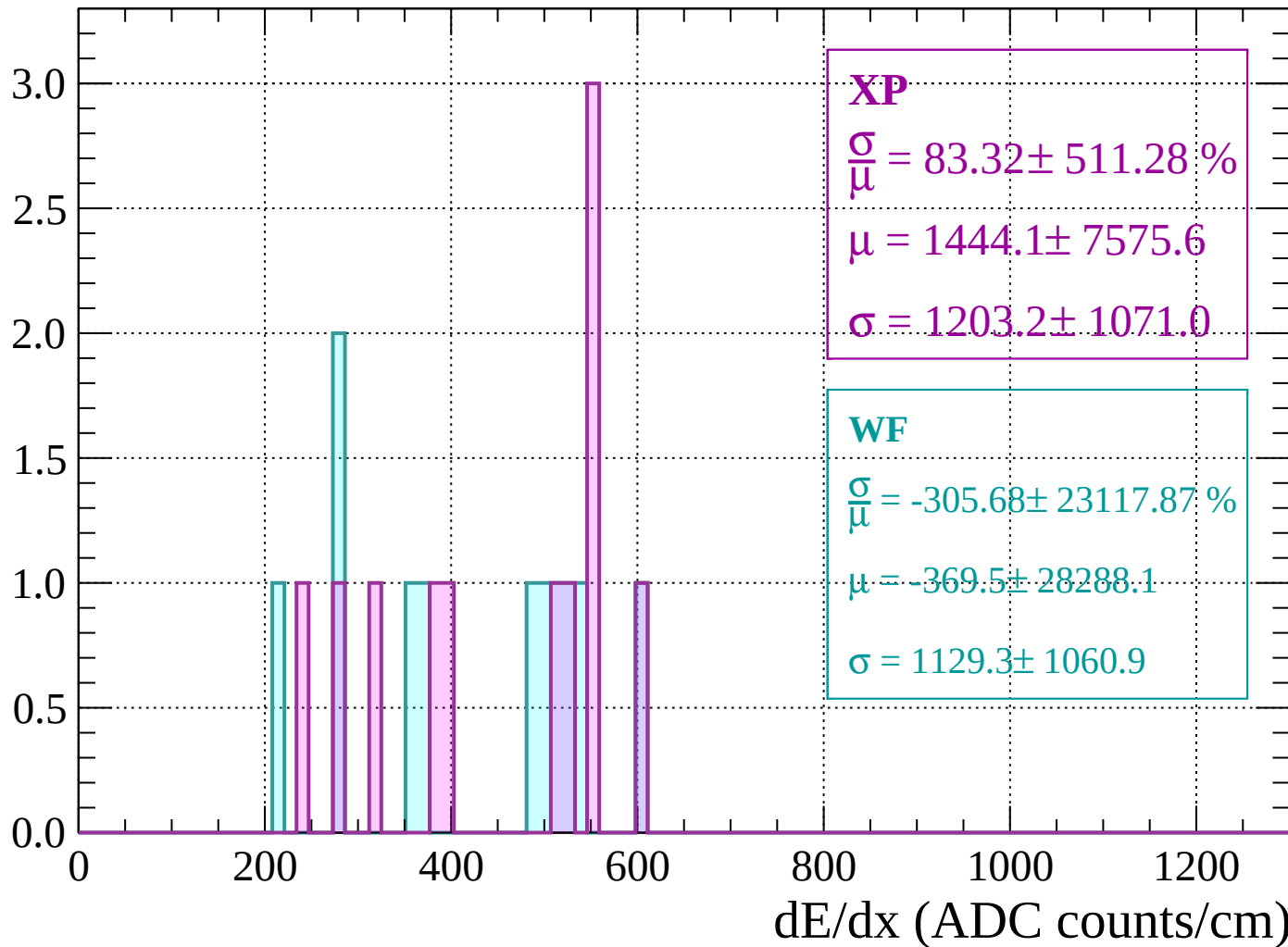
Energy loss | $520 < p < 560$

Count



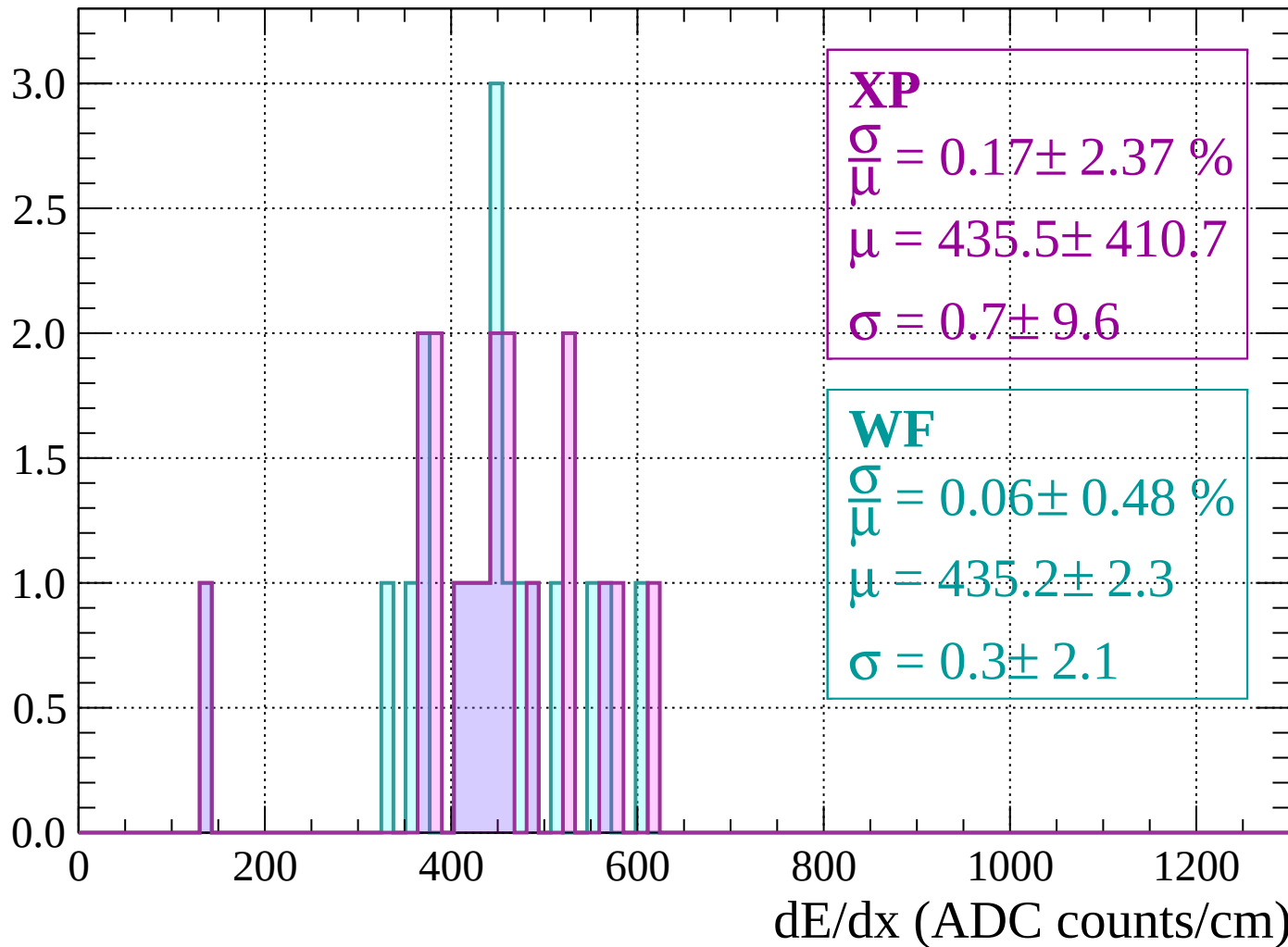
Energy loss | $560 < p < 600$

Count



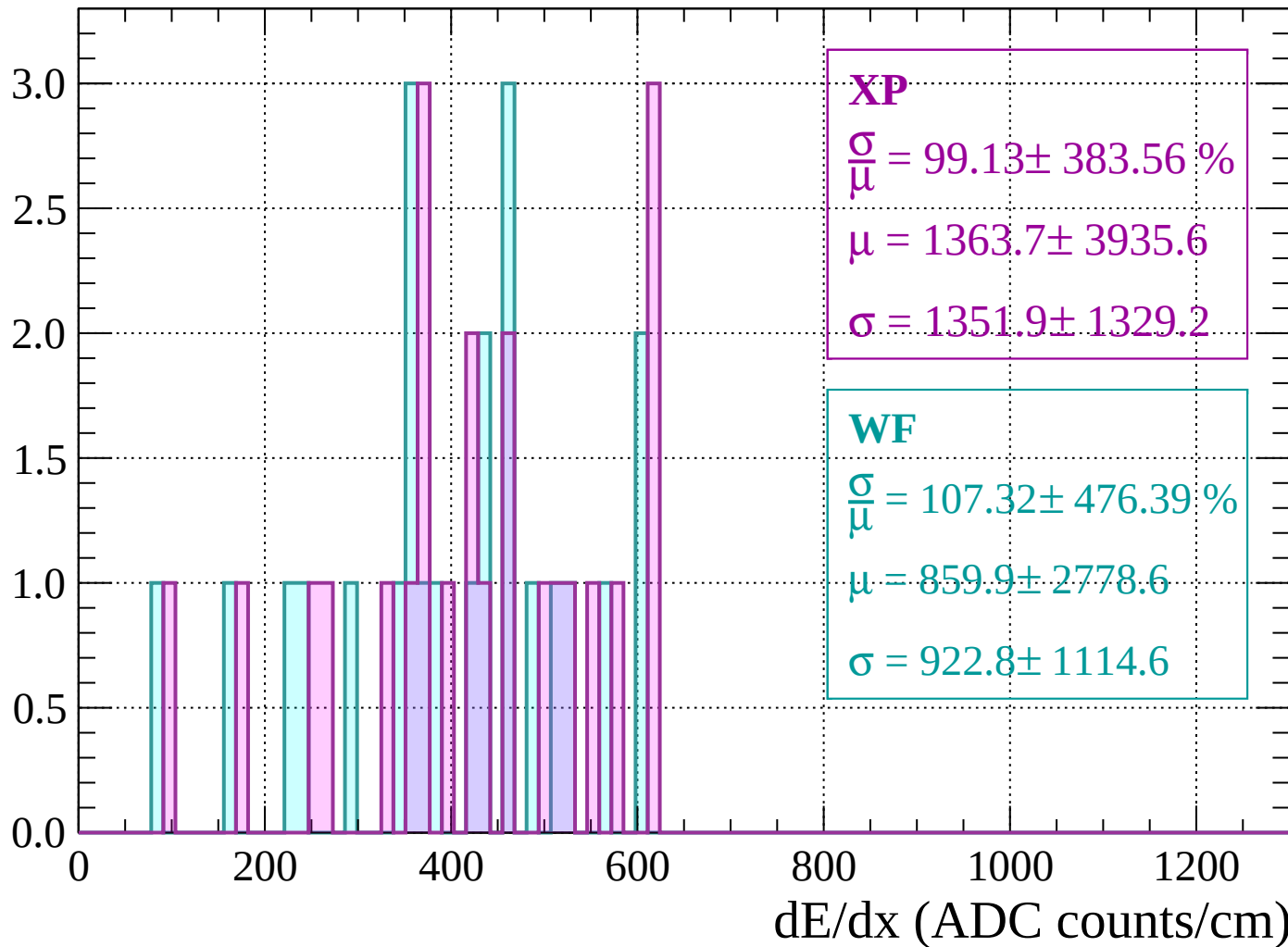
Energy loss | $600 < p < 640$

Count



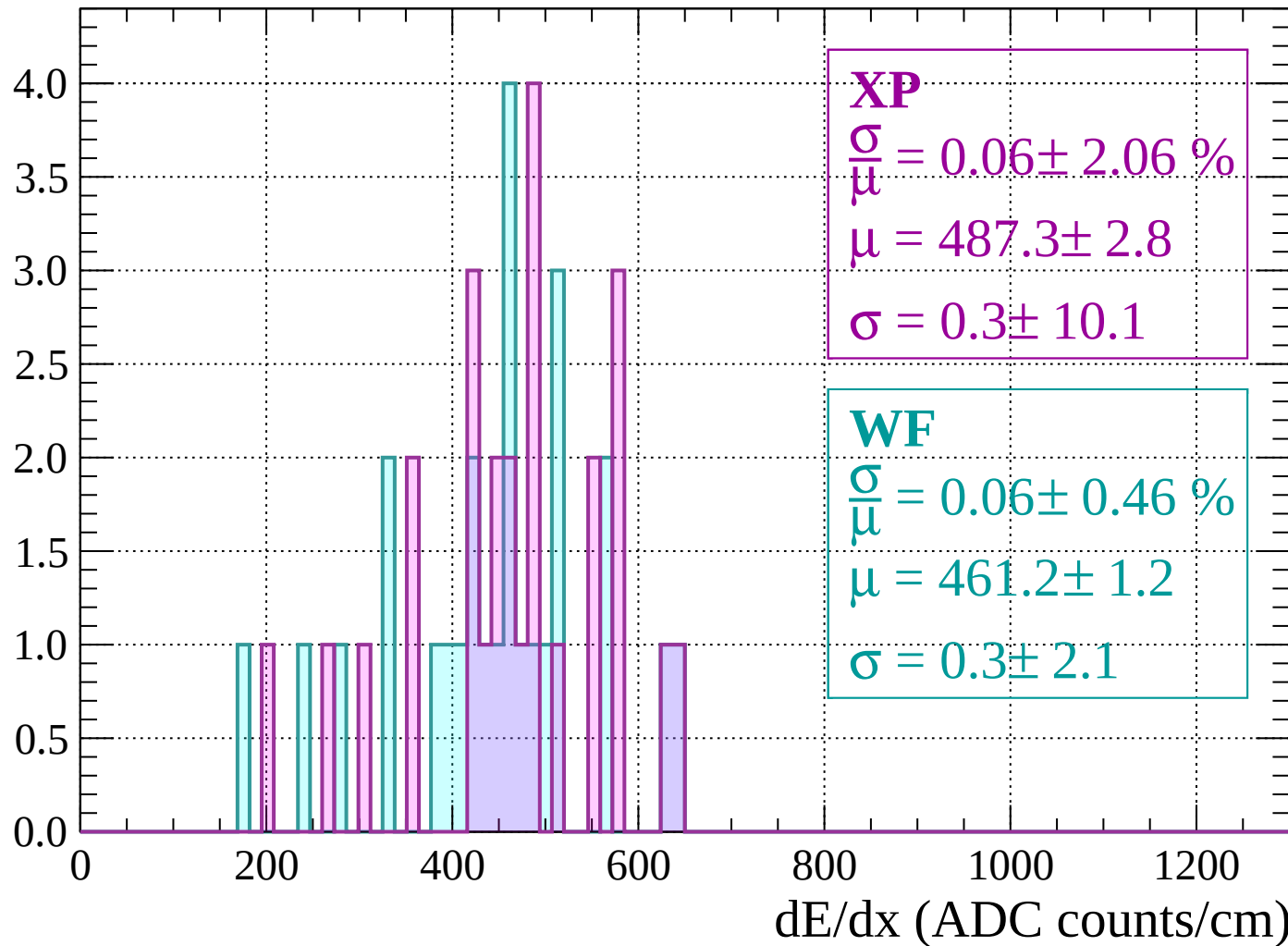
Energy loss | $640 < p < 680$

Count



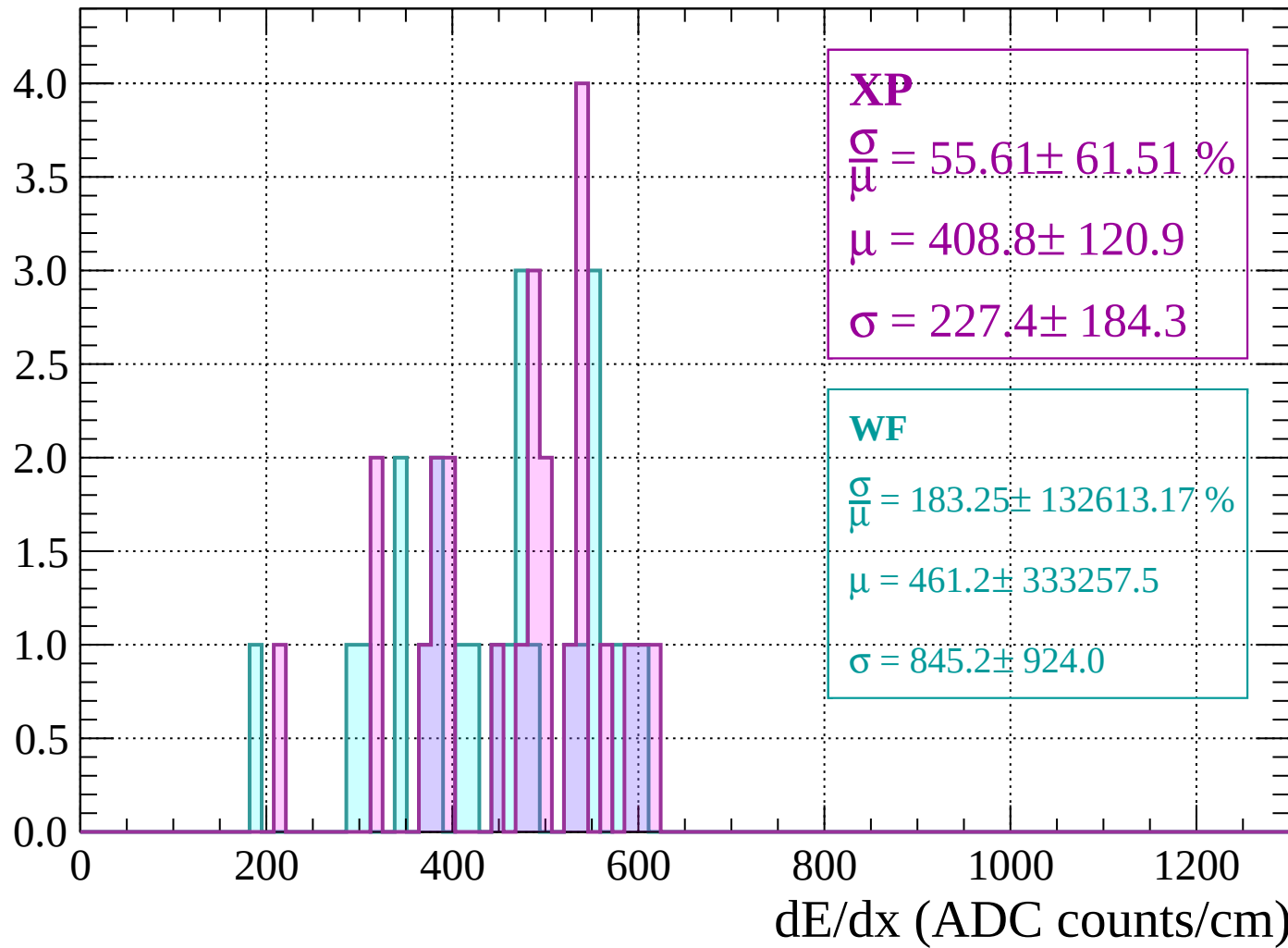
Energy loss | $680 < p < 720$

Count



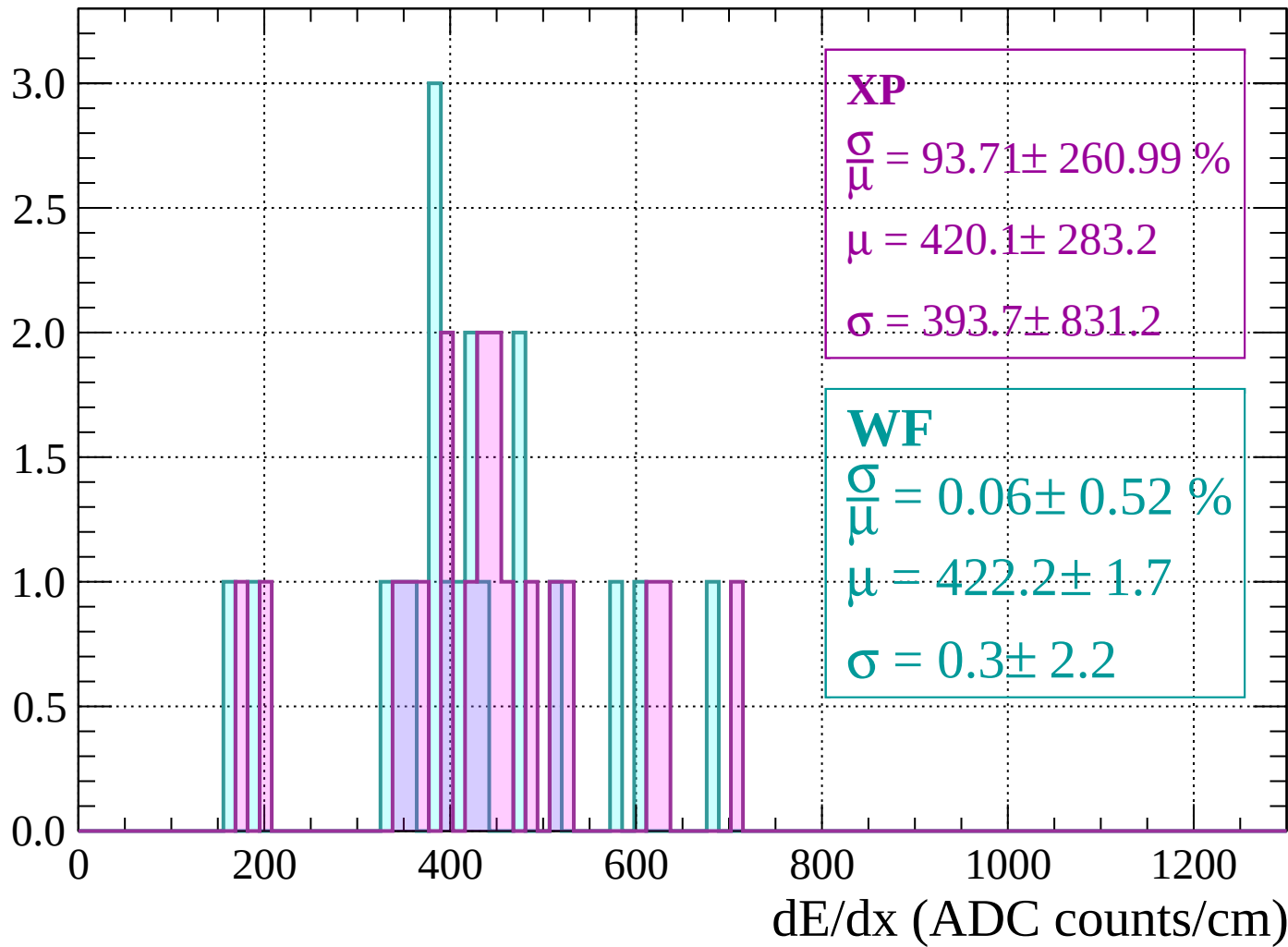
Energy loss | $720 < p < 760$

Count



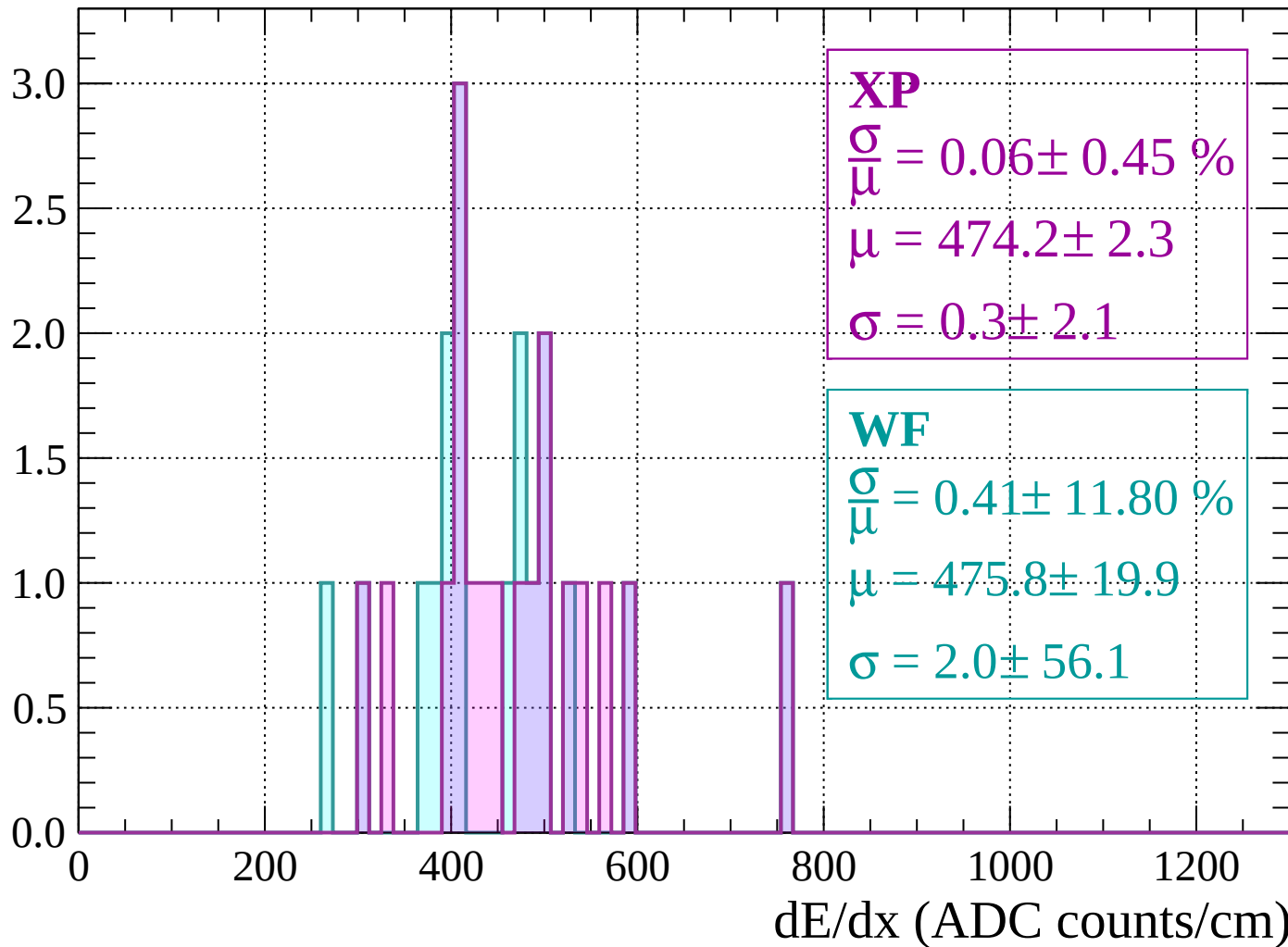
Energy loss | $760 < p < 800$

Count



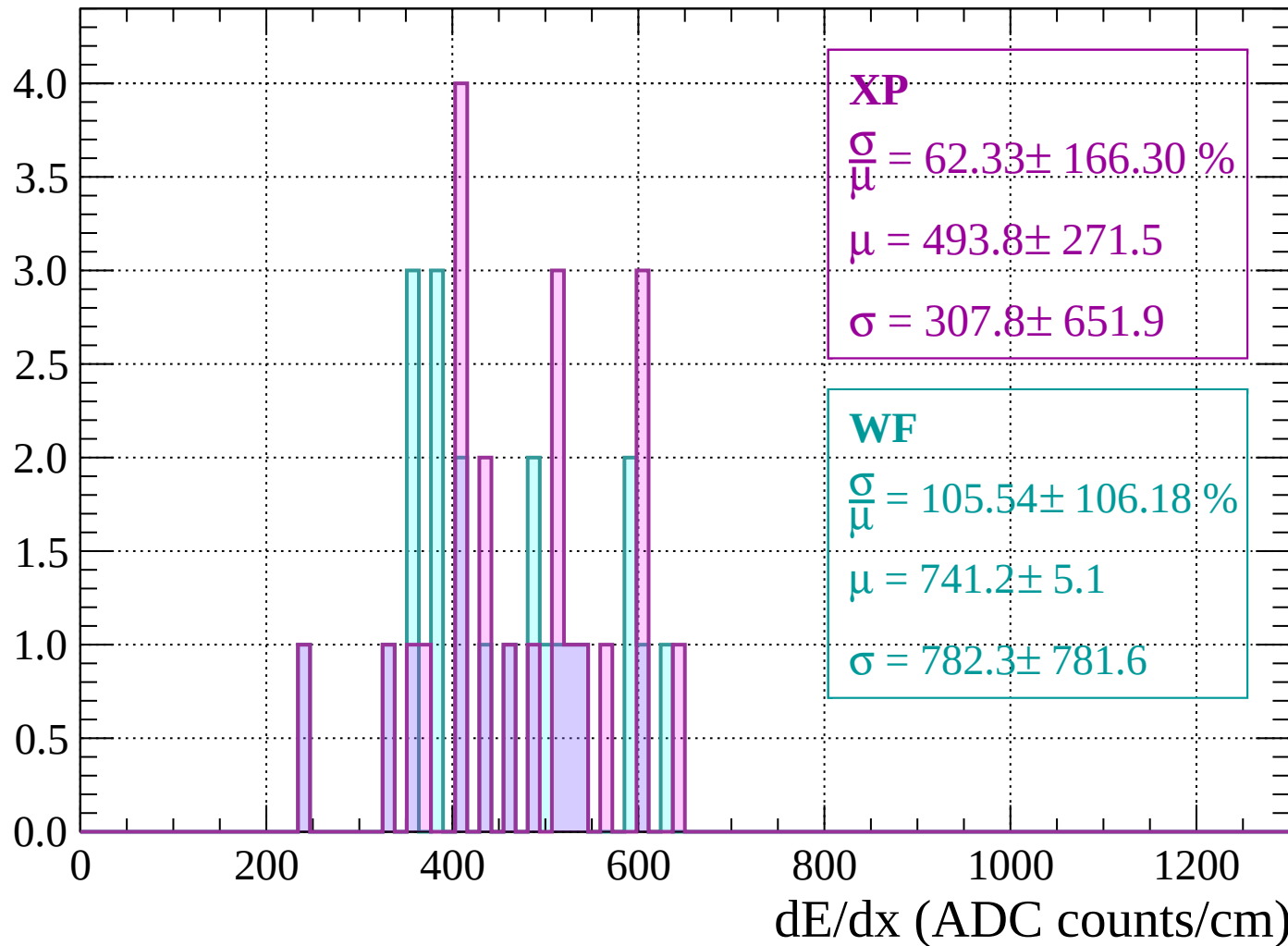
Energy loss | $800 < p < 840$

Count



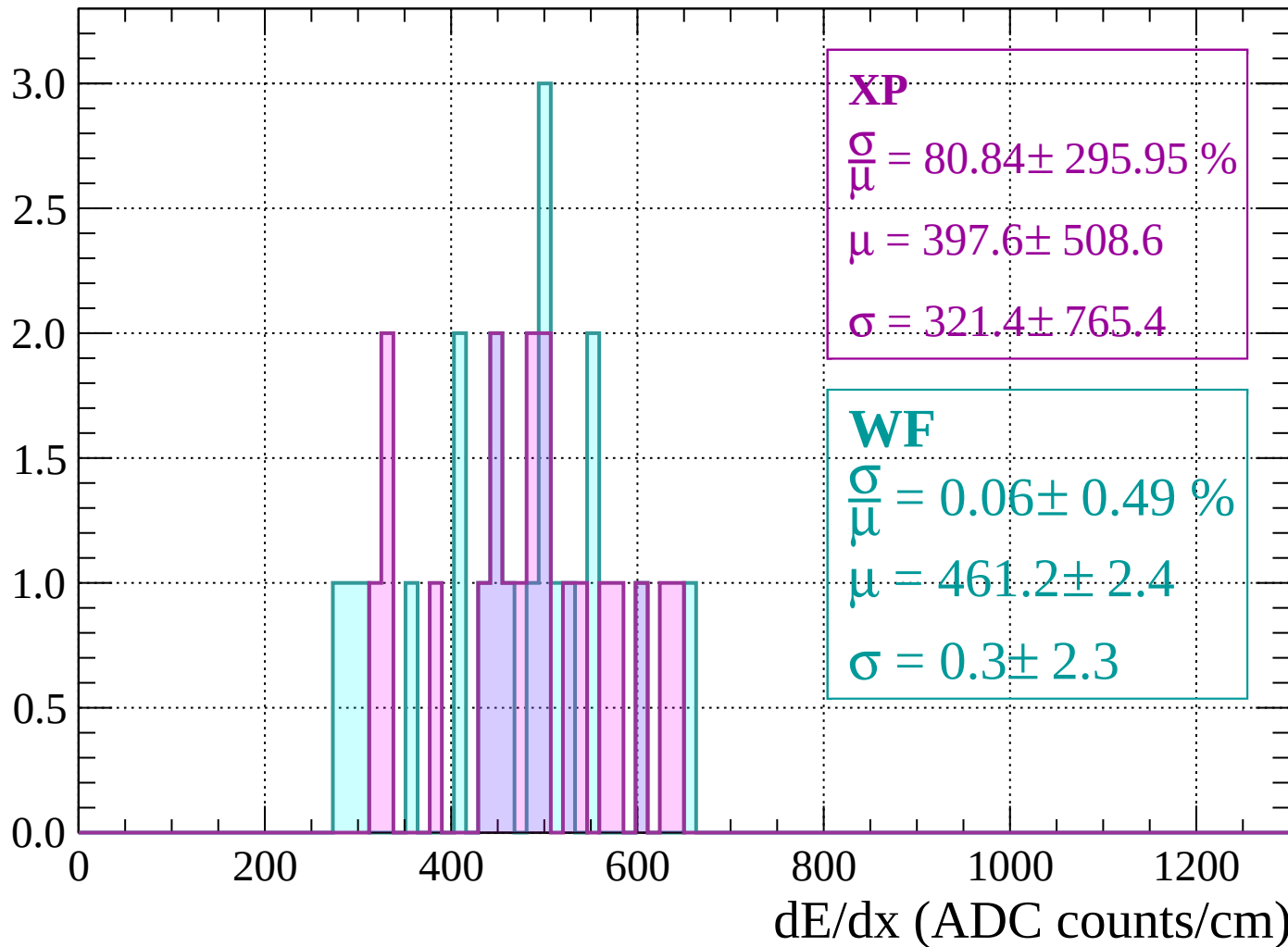
Energy loss | $840 < p < 880$

Count



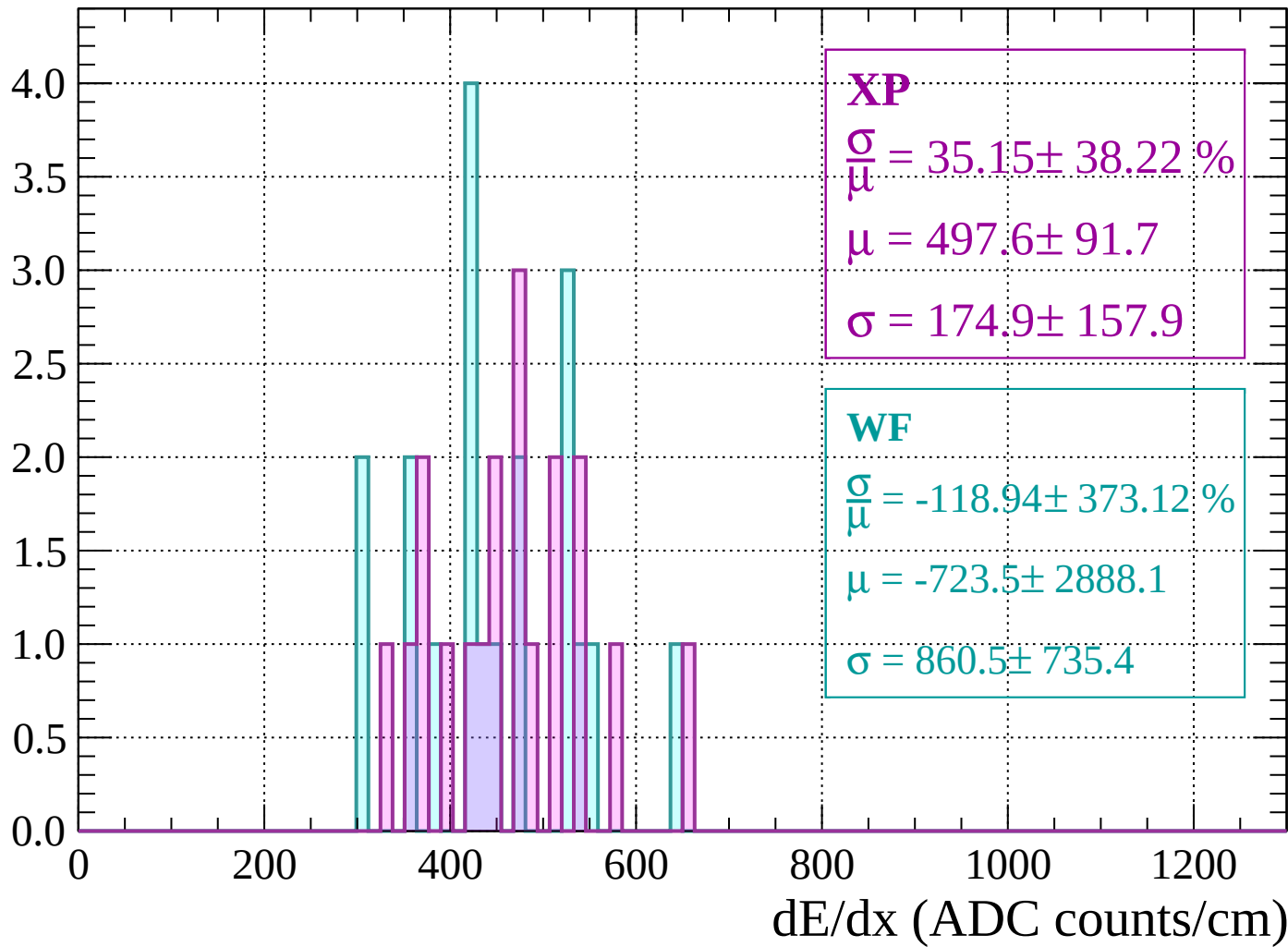
Energy loss | $880 < p < 920$

Count



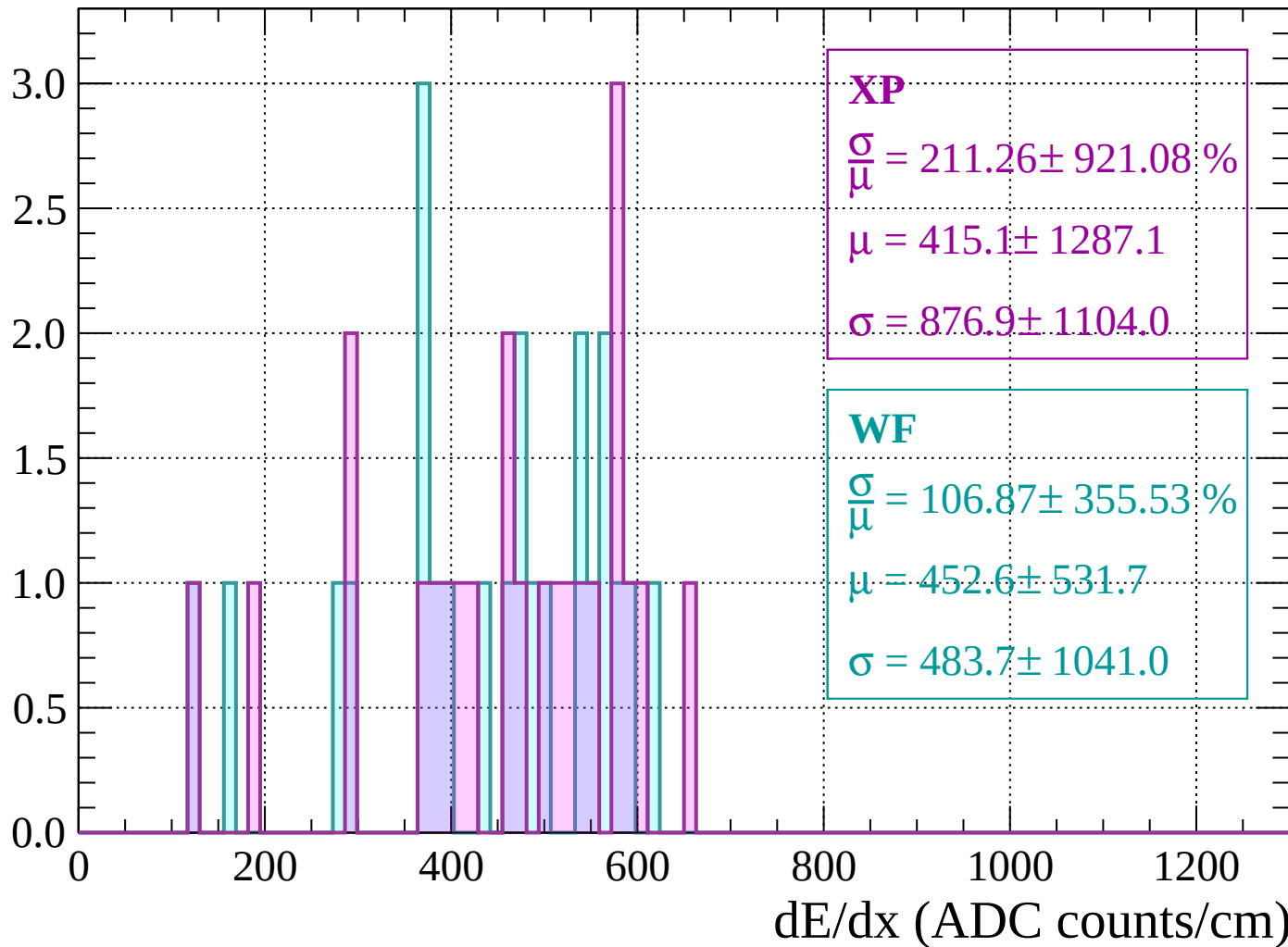
Energy loss | $920 < p < 960$

Count



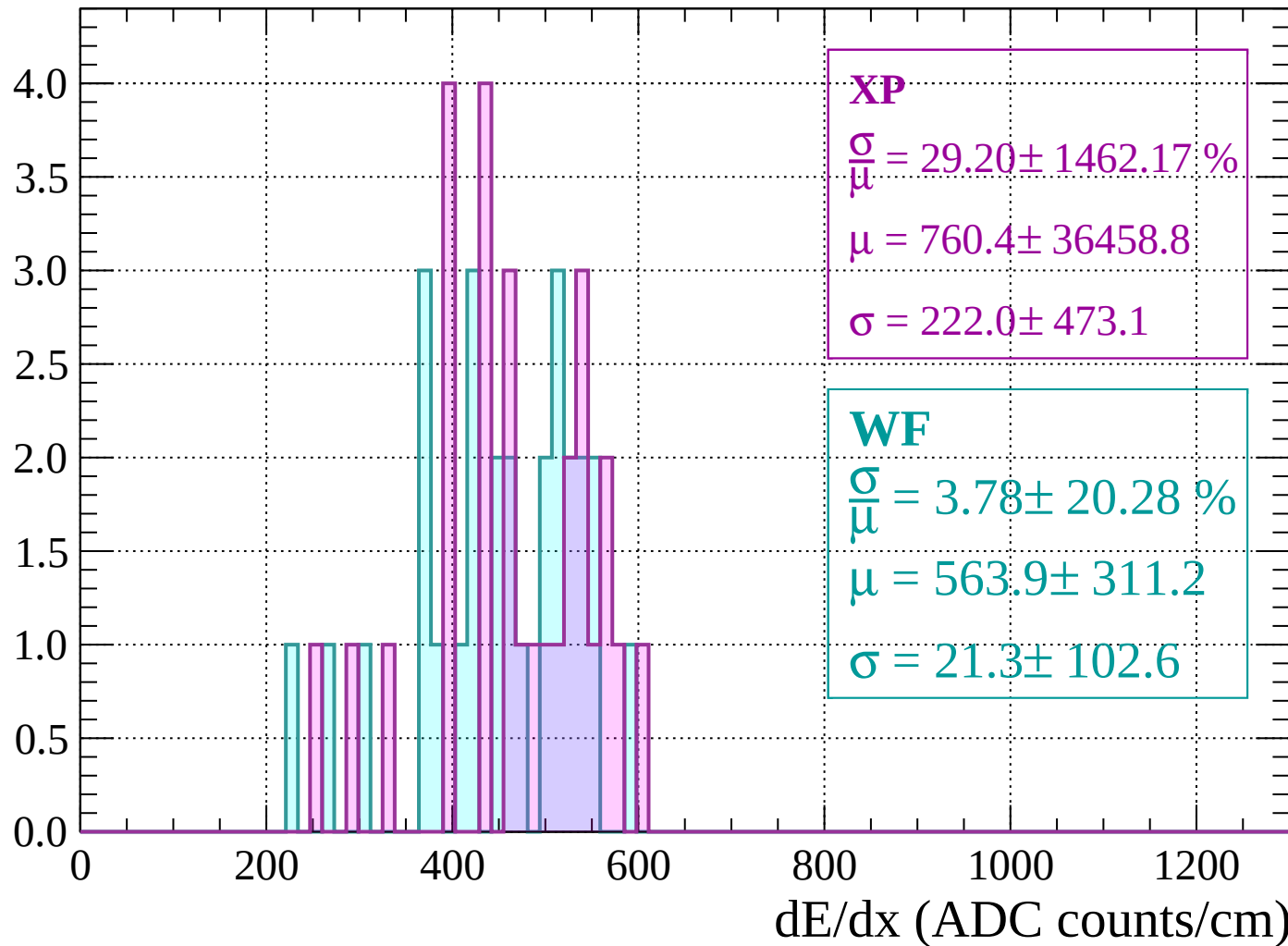
Energy loss | $960 < p < 1000$

Count



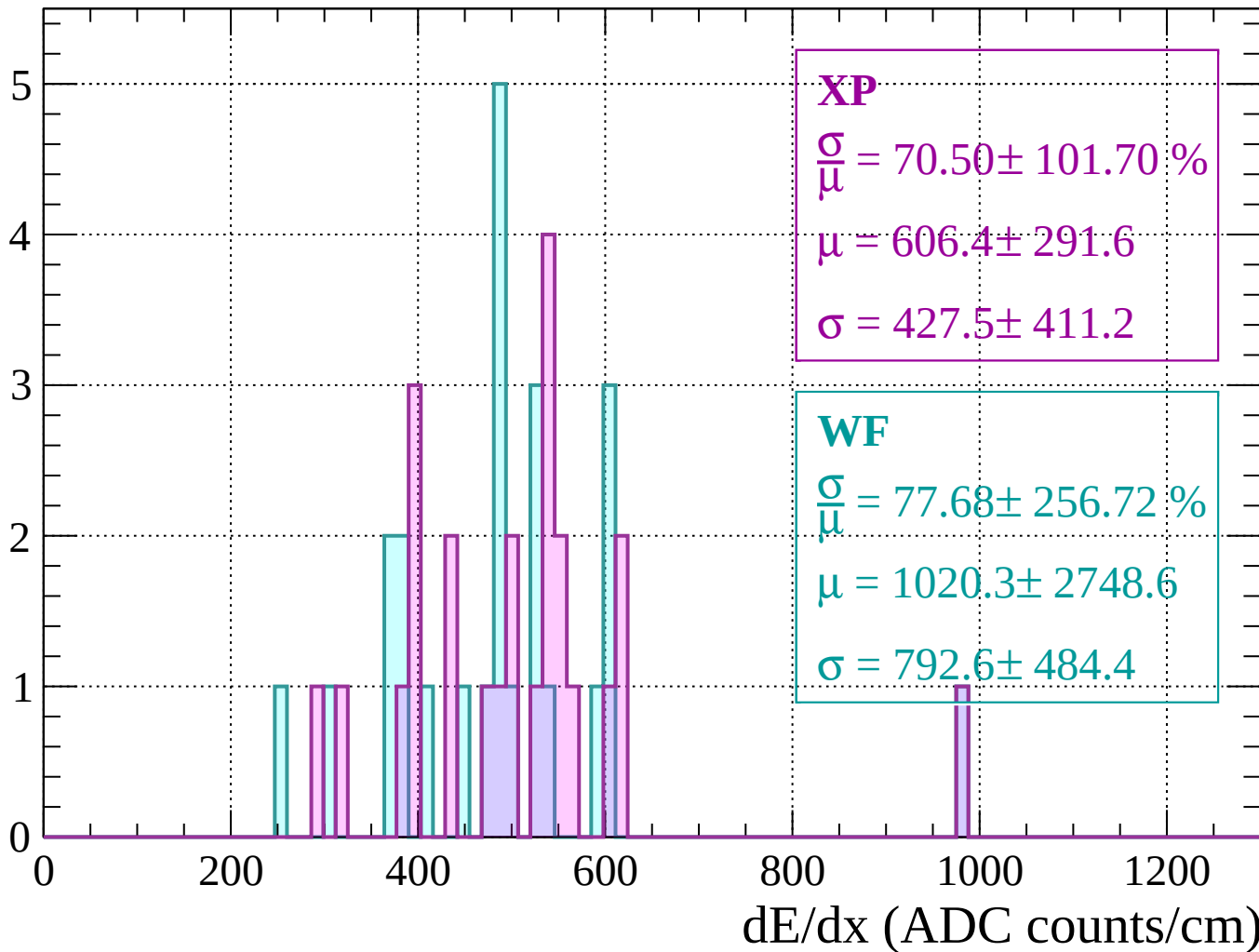
Energy loss | $1000 < p < 1040$

Count



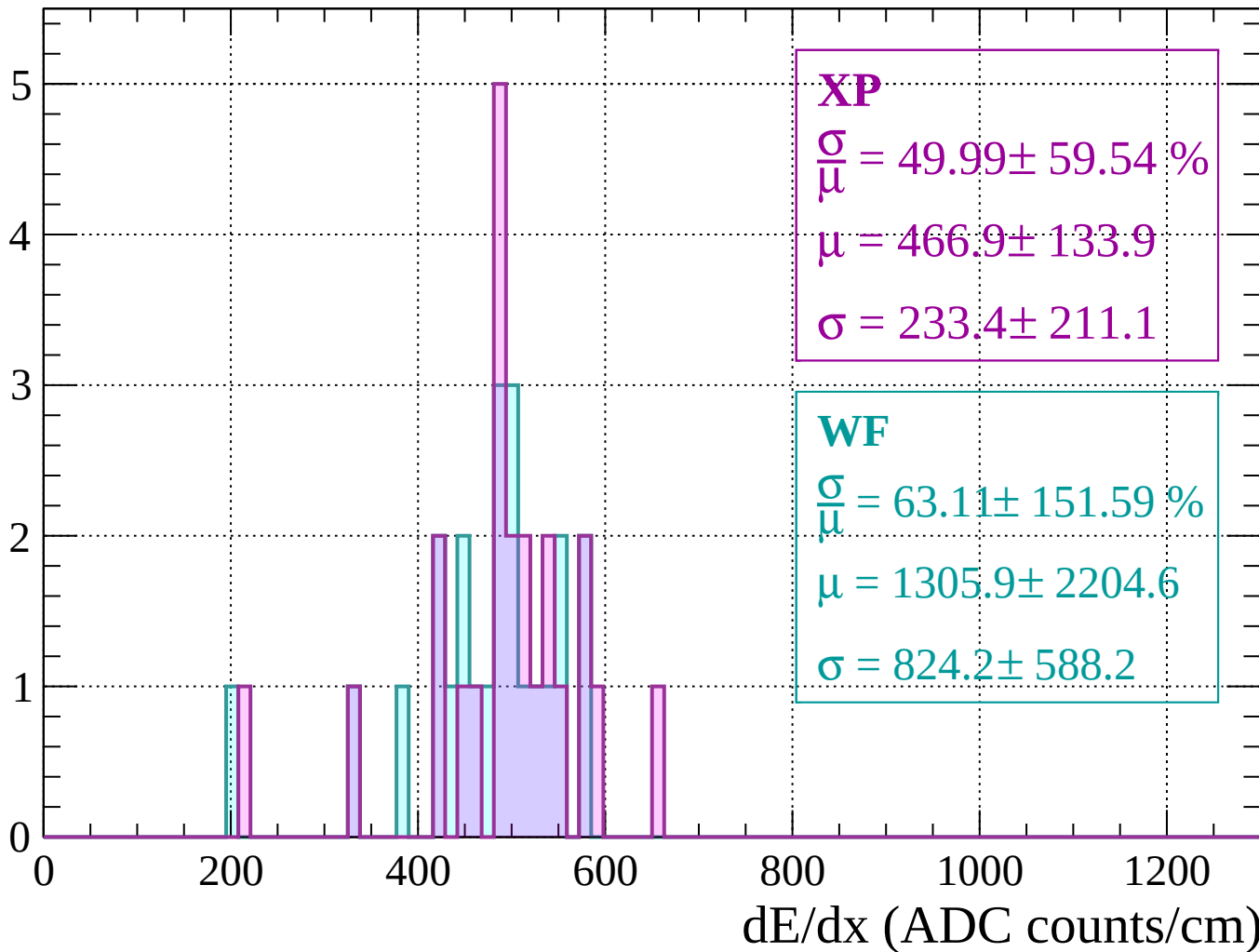
Energy loss | $1040 < p < 1080$

Count



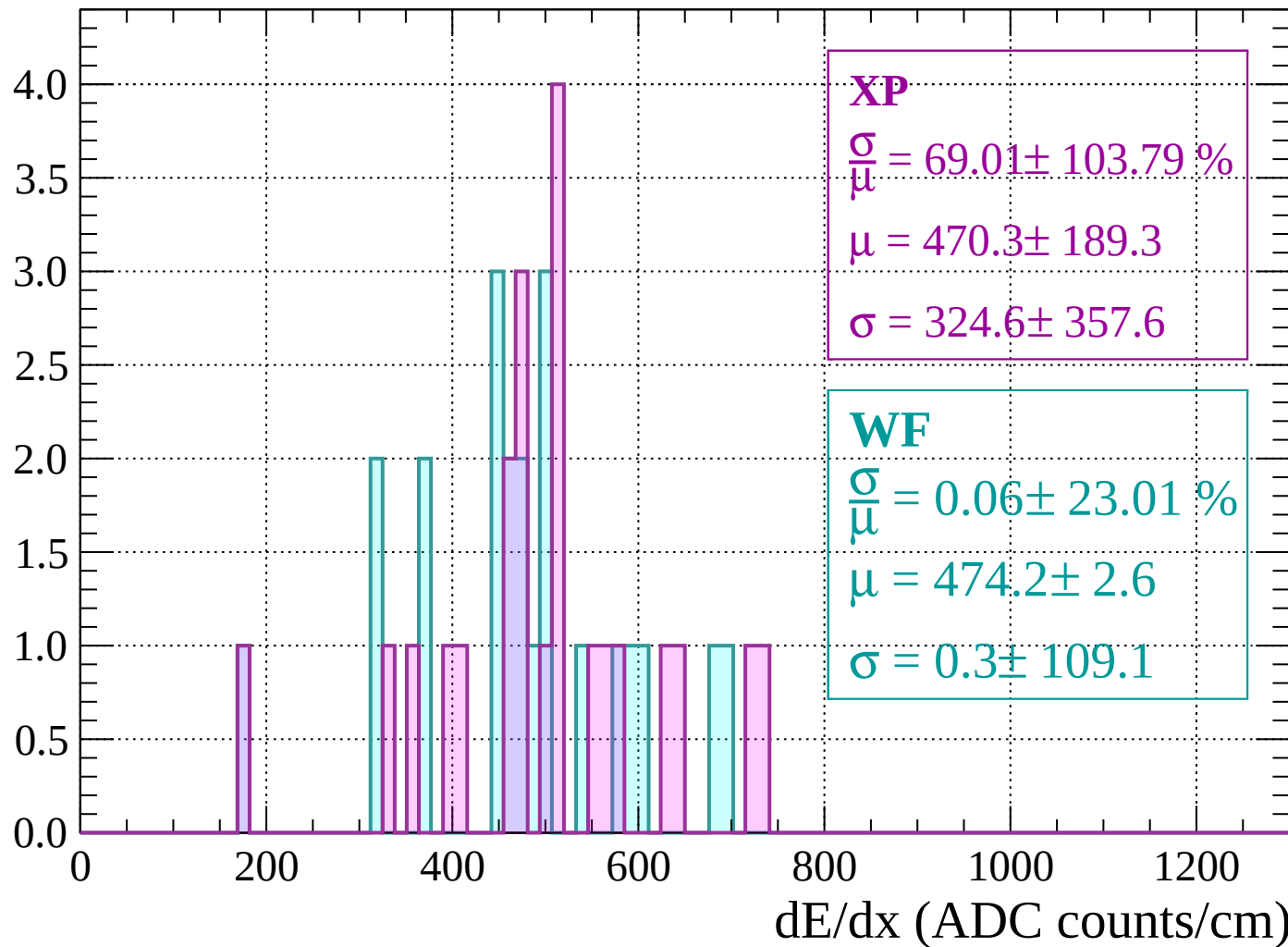
Energy loss | $1080 < p < 1120$

Count



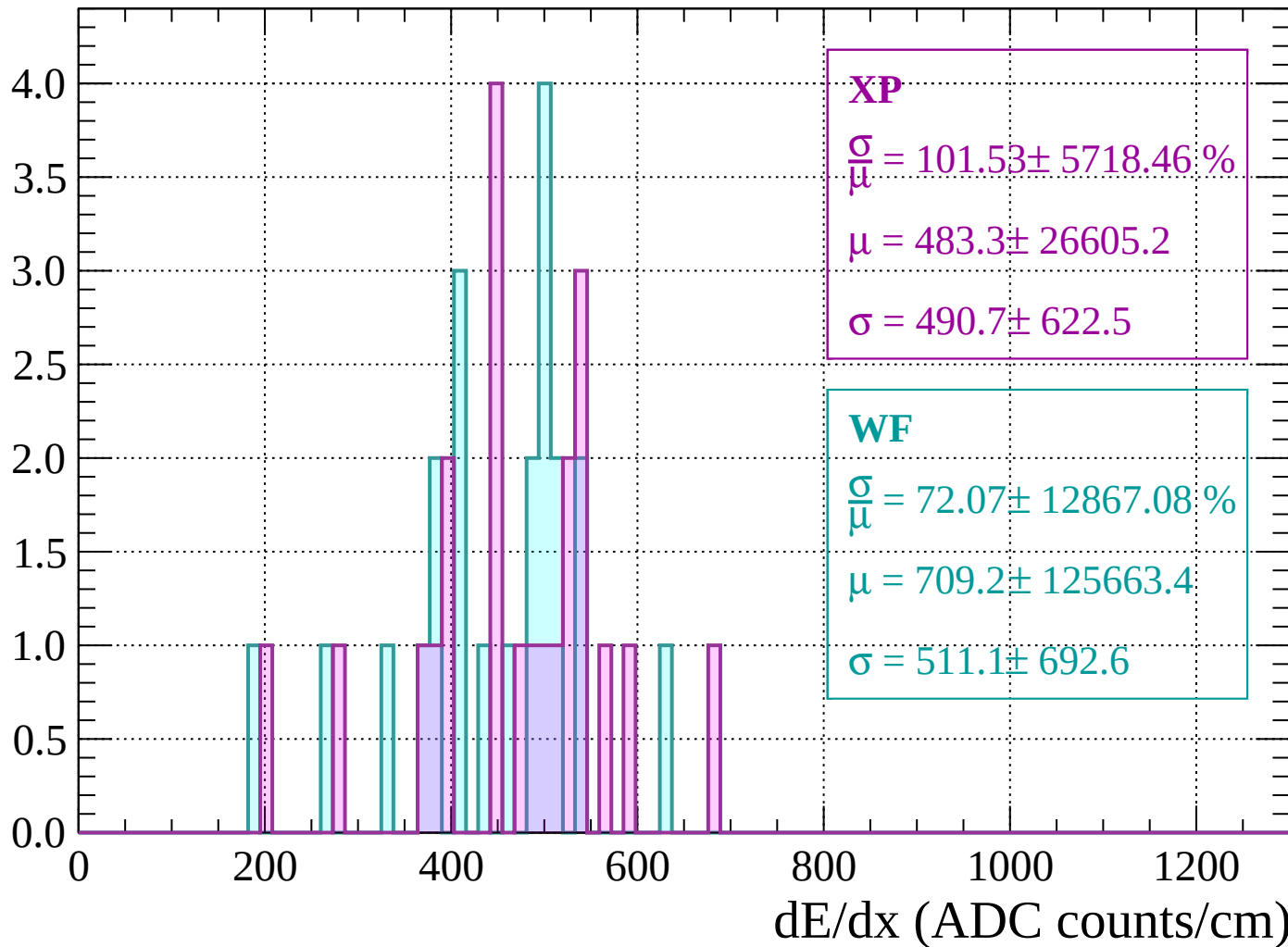
Energy loss | $1120 < p < 1160$

Count



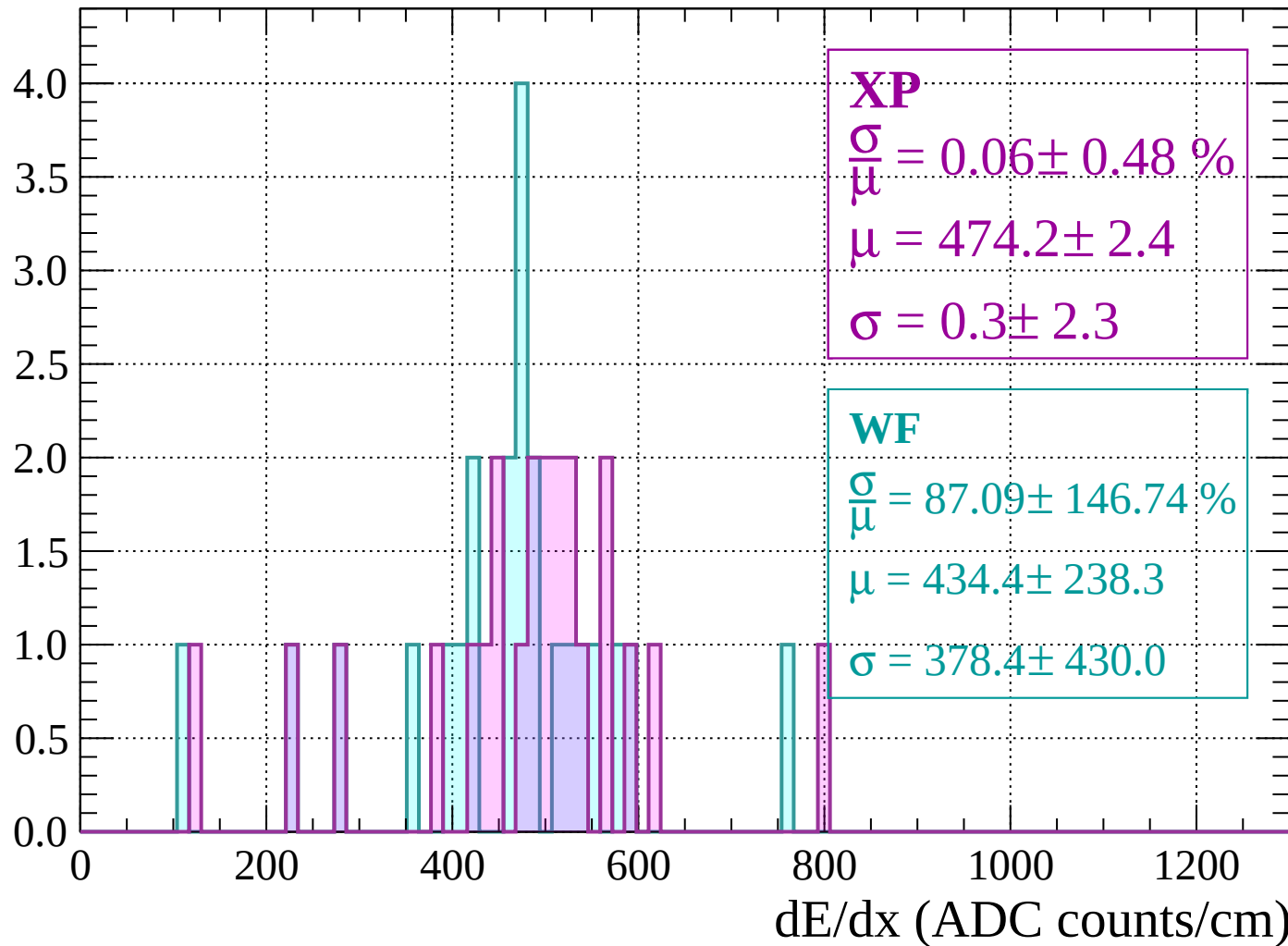
Energy loss | $1160 < p < 1200$

Count



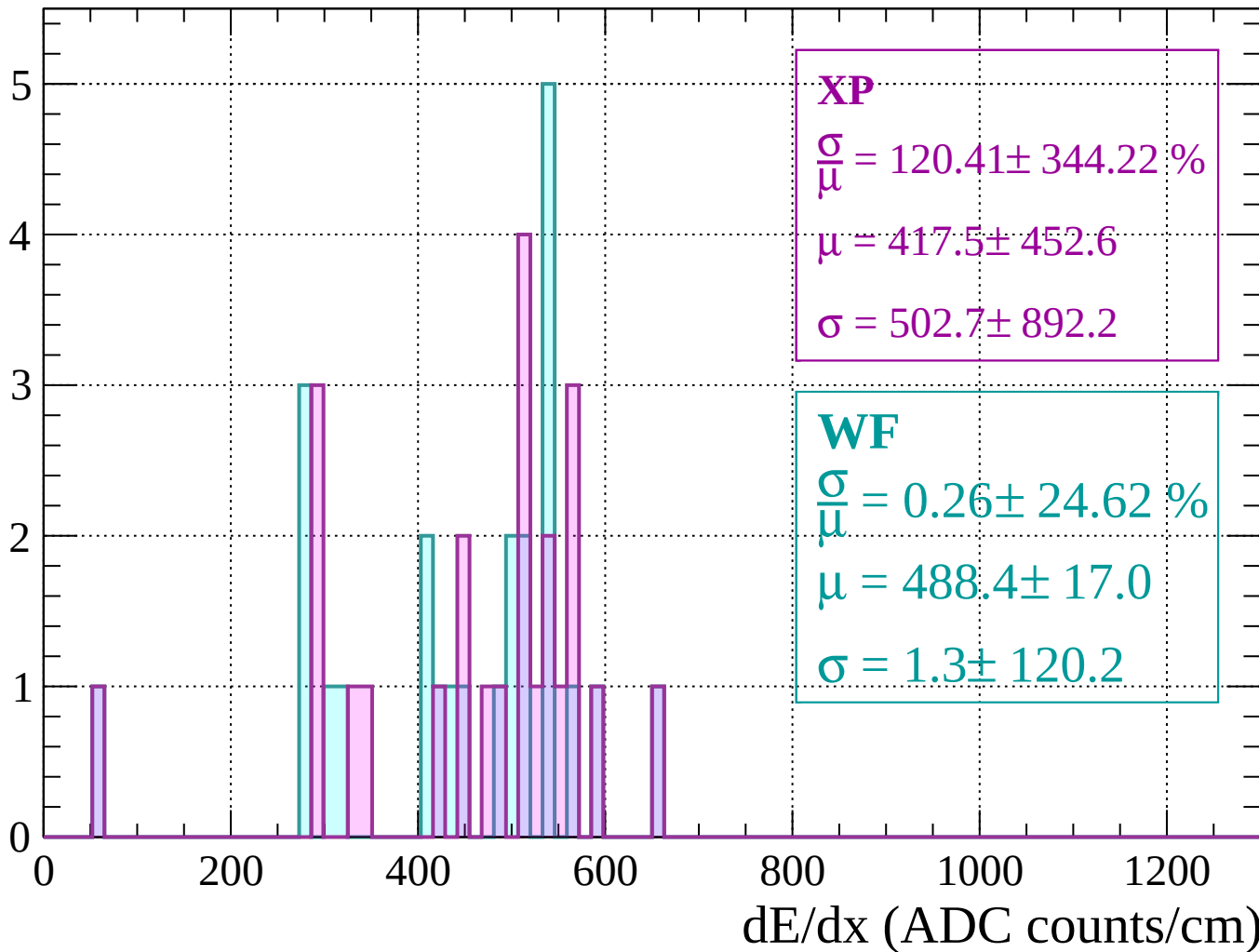
Energy loss | $1200 < p < 1240$

Count



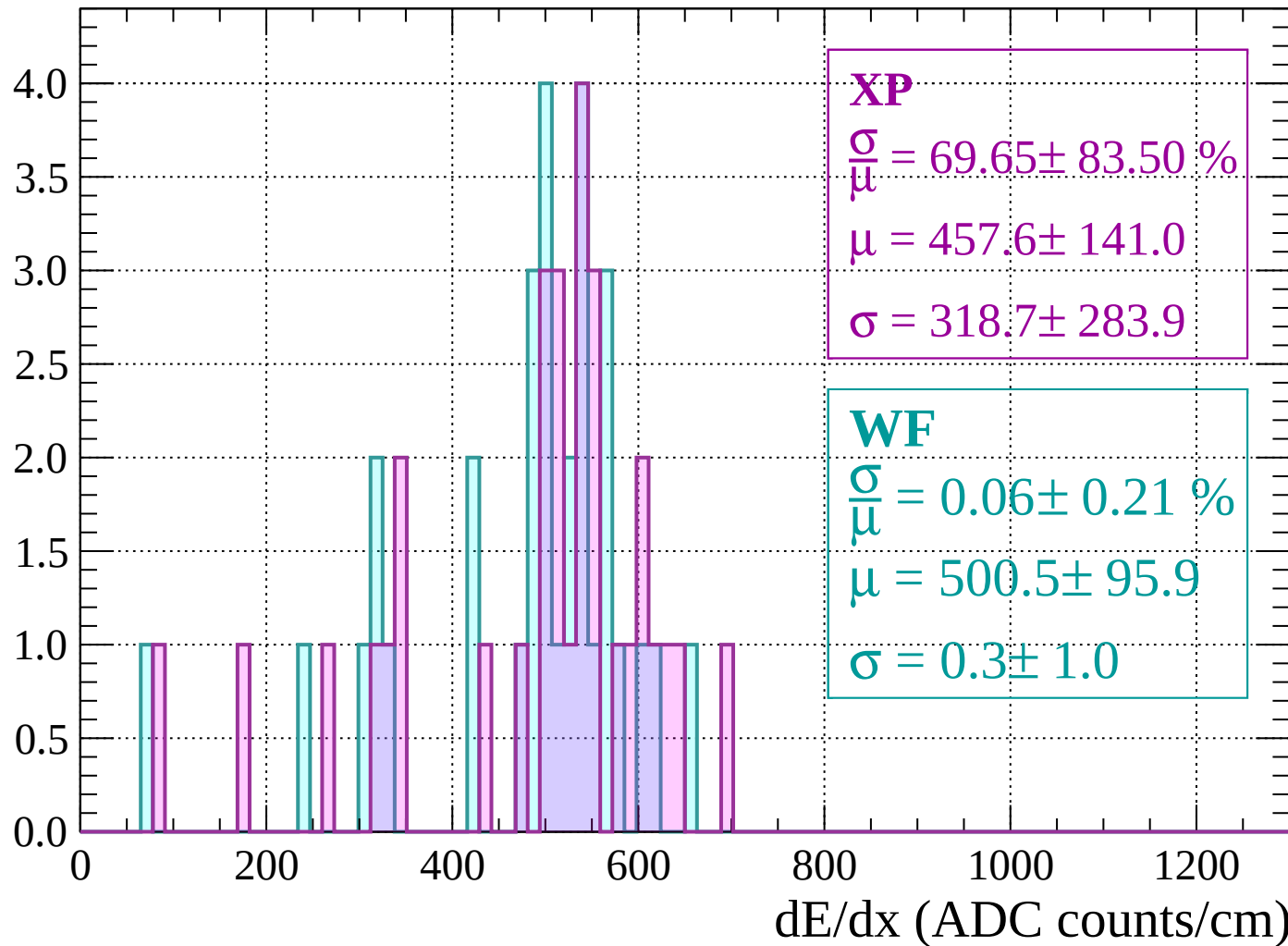
Energy loss | $1240 < p < 1280$

Count



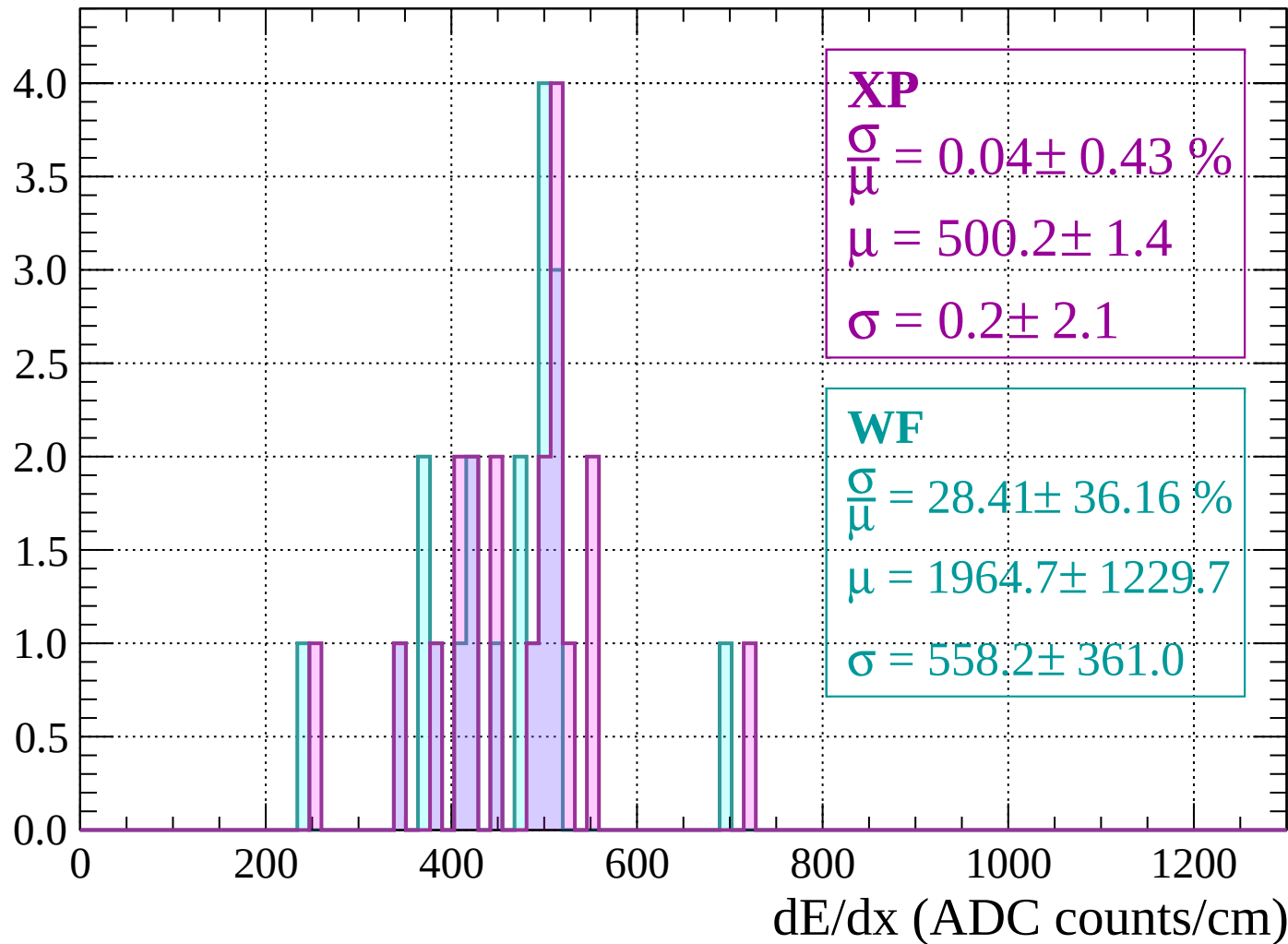
Energy loss | $1280 < p < 1320$

Count



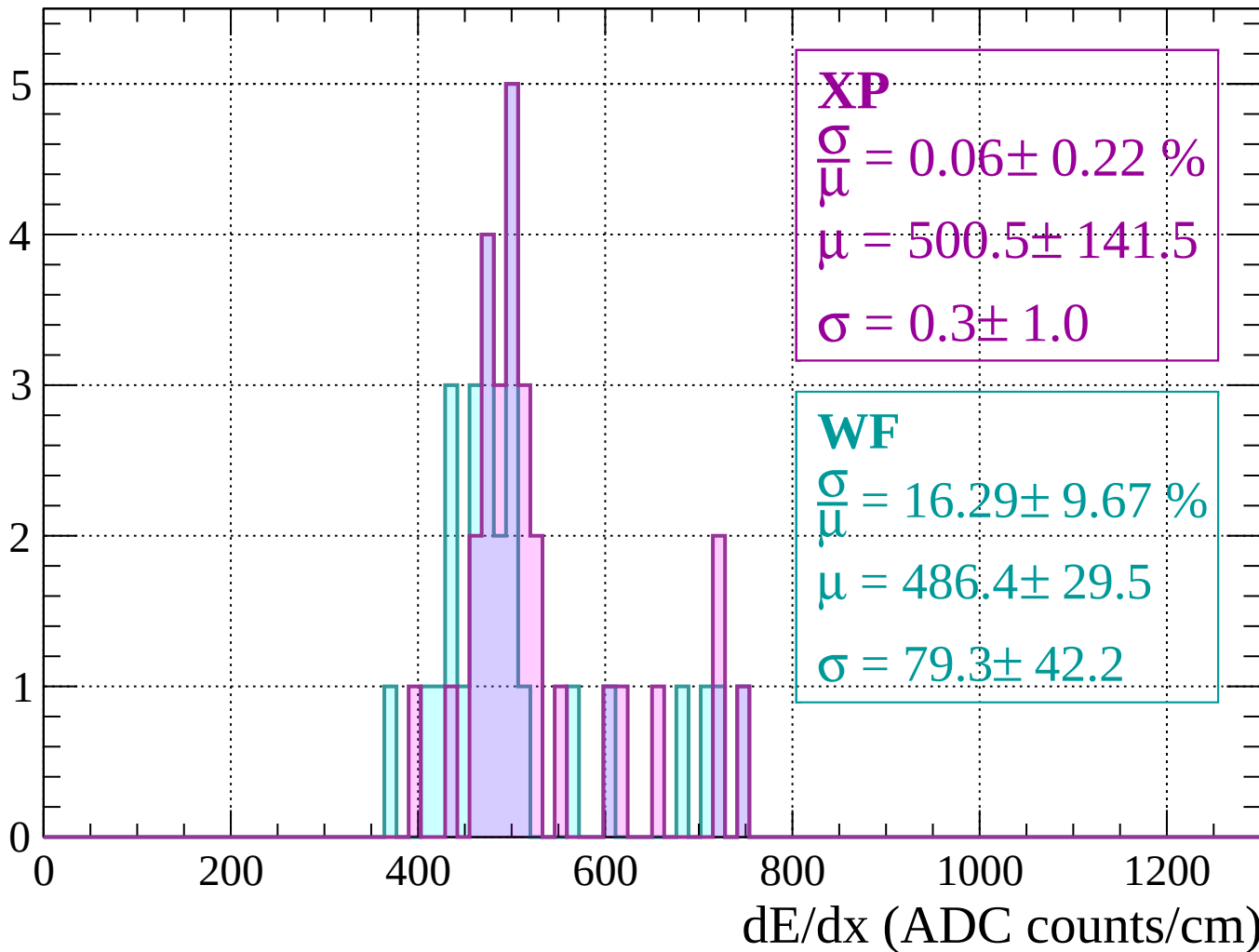
Energy loss | $1320 < p < 1360$

Count



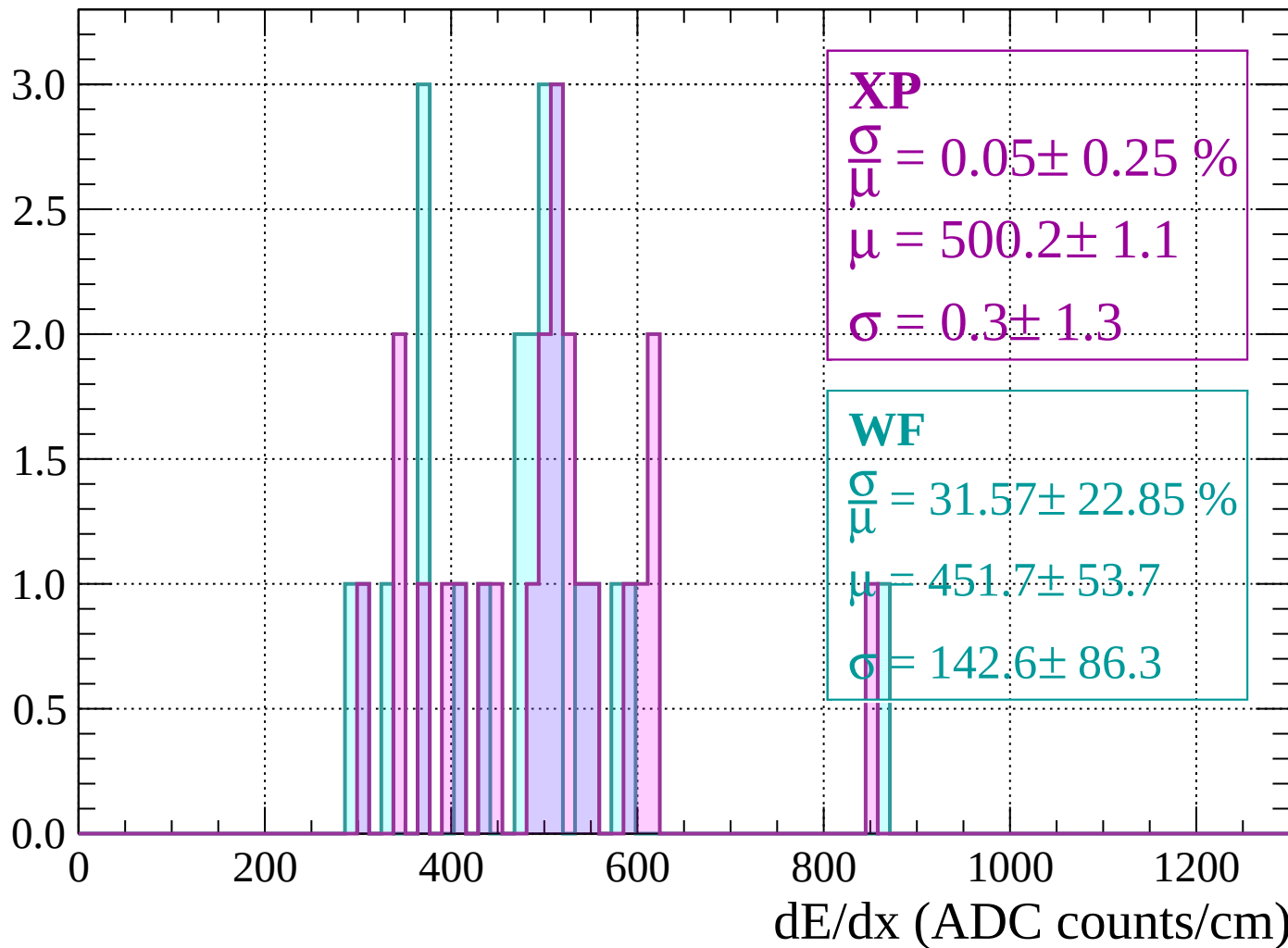
Energy loss | $1360 < p < 1400$

Count



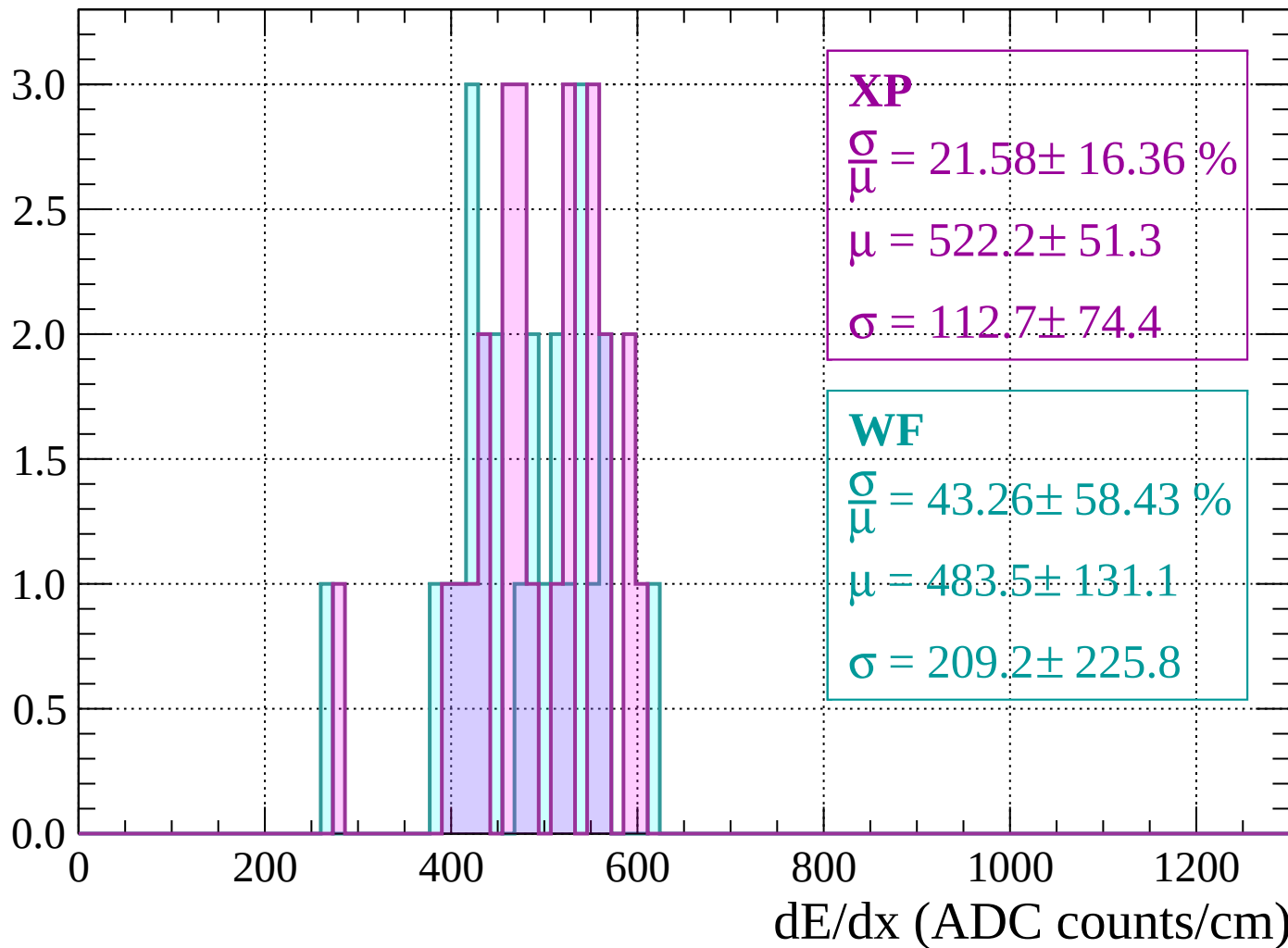
Energy loss | $1400 < p < 1440$

Count



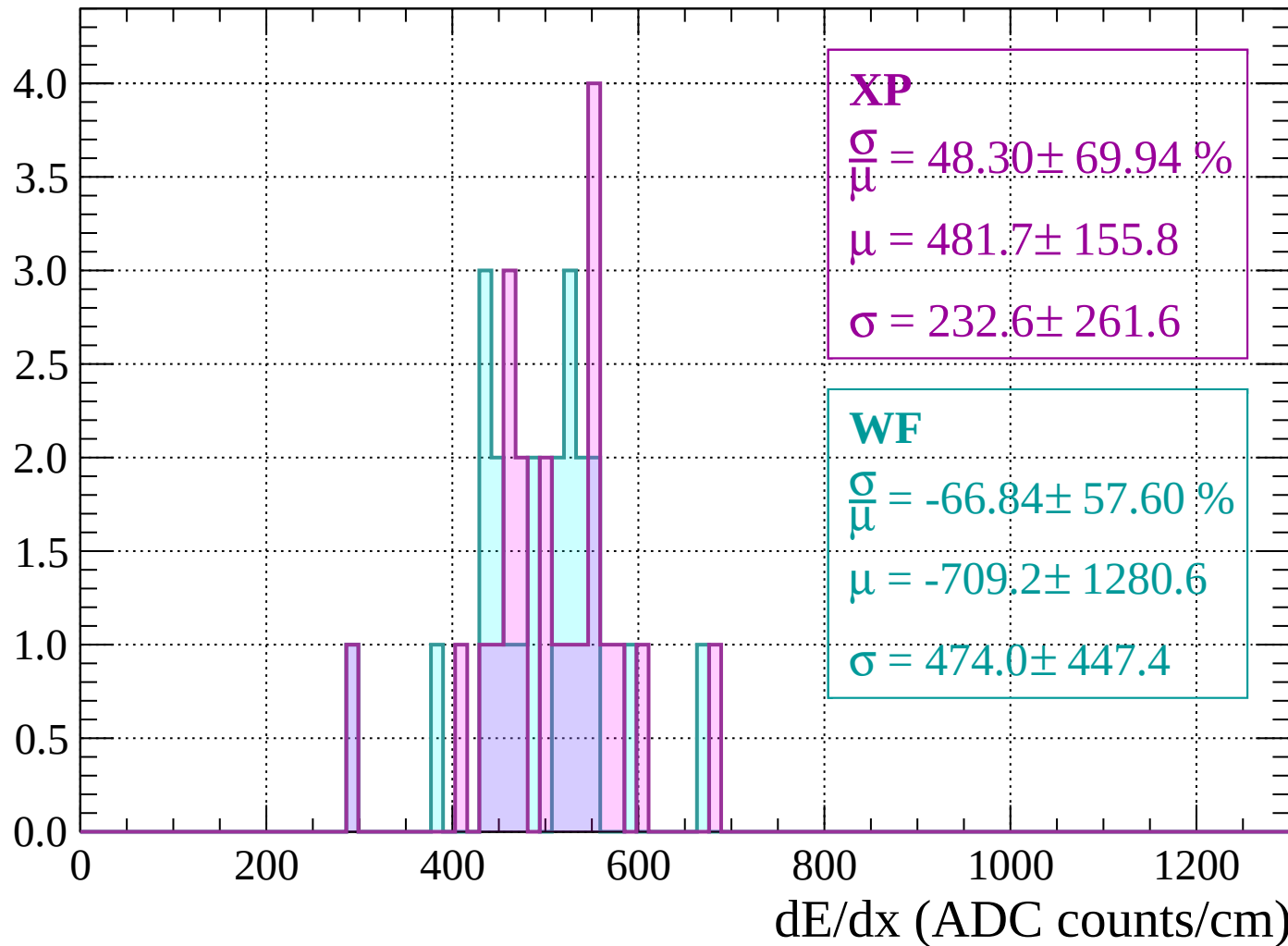
Energy loss | $1440 < p < 1480$

Count



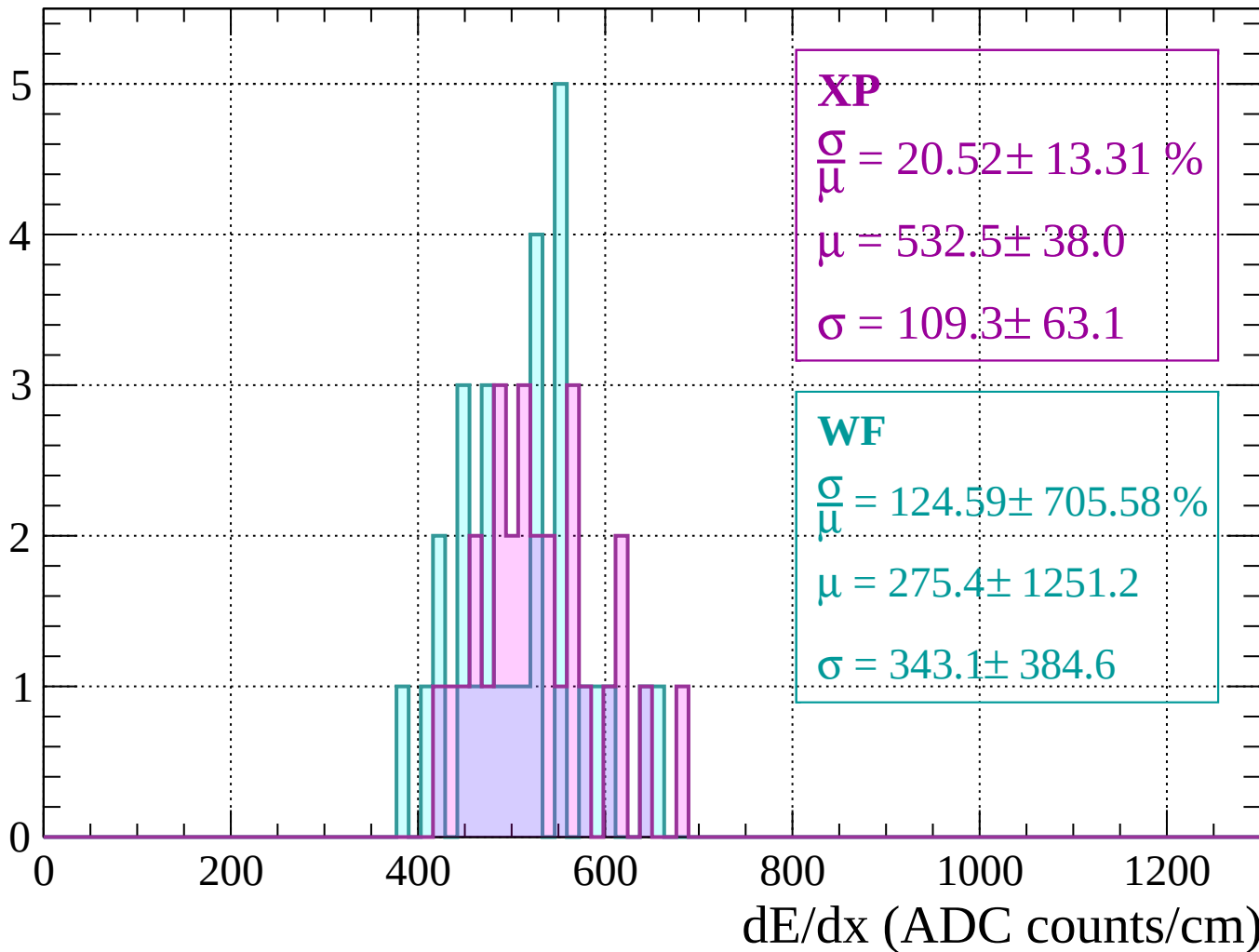
Energy loss | $1480 < p < 1520$

Count



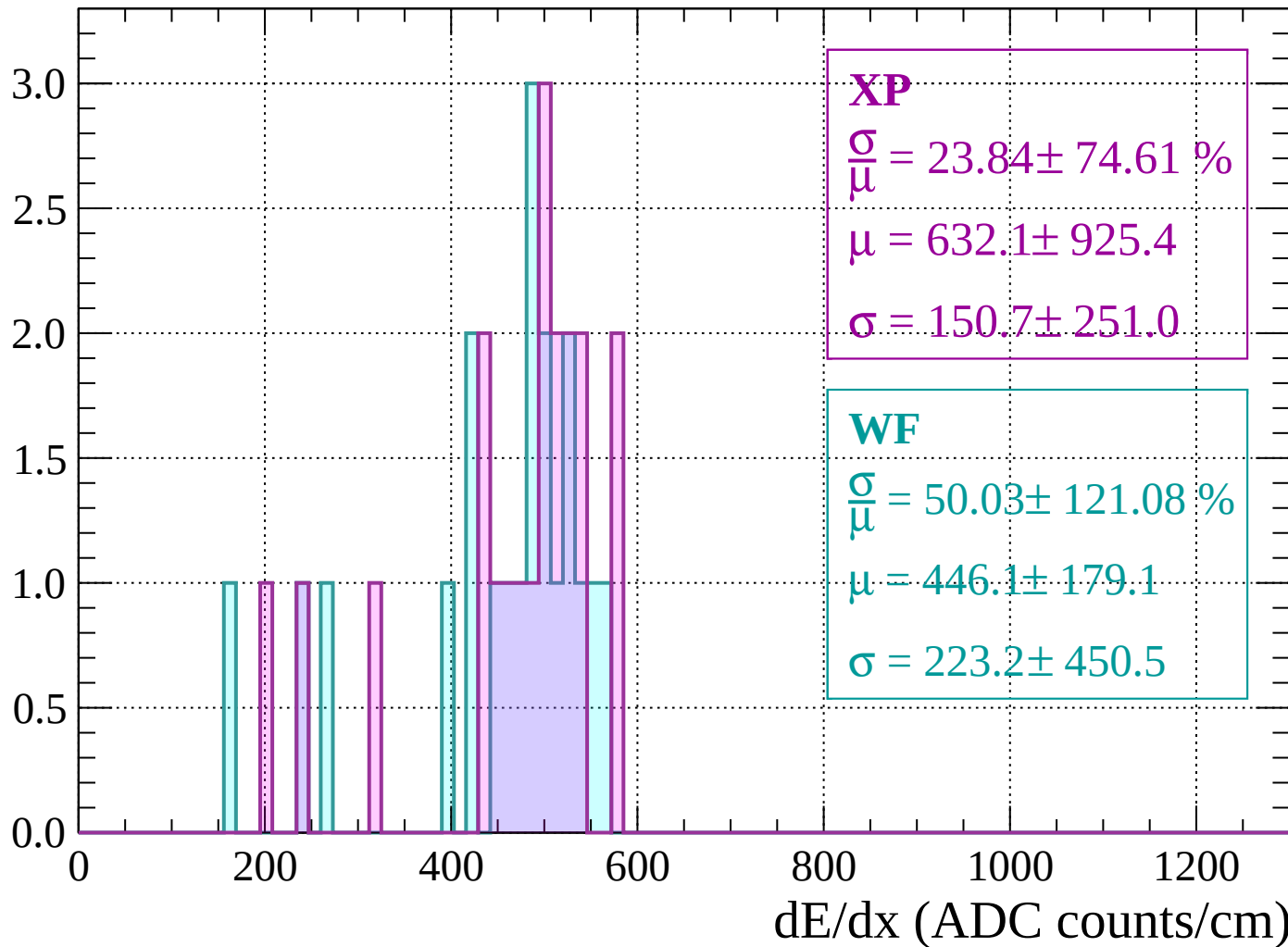
Energy loss | $1520 < p < 1560$

Count



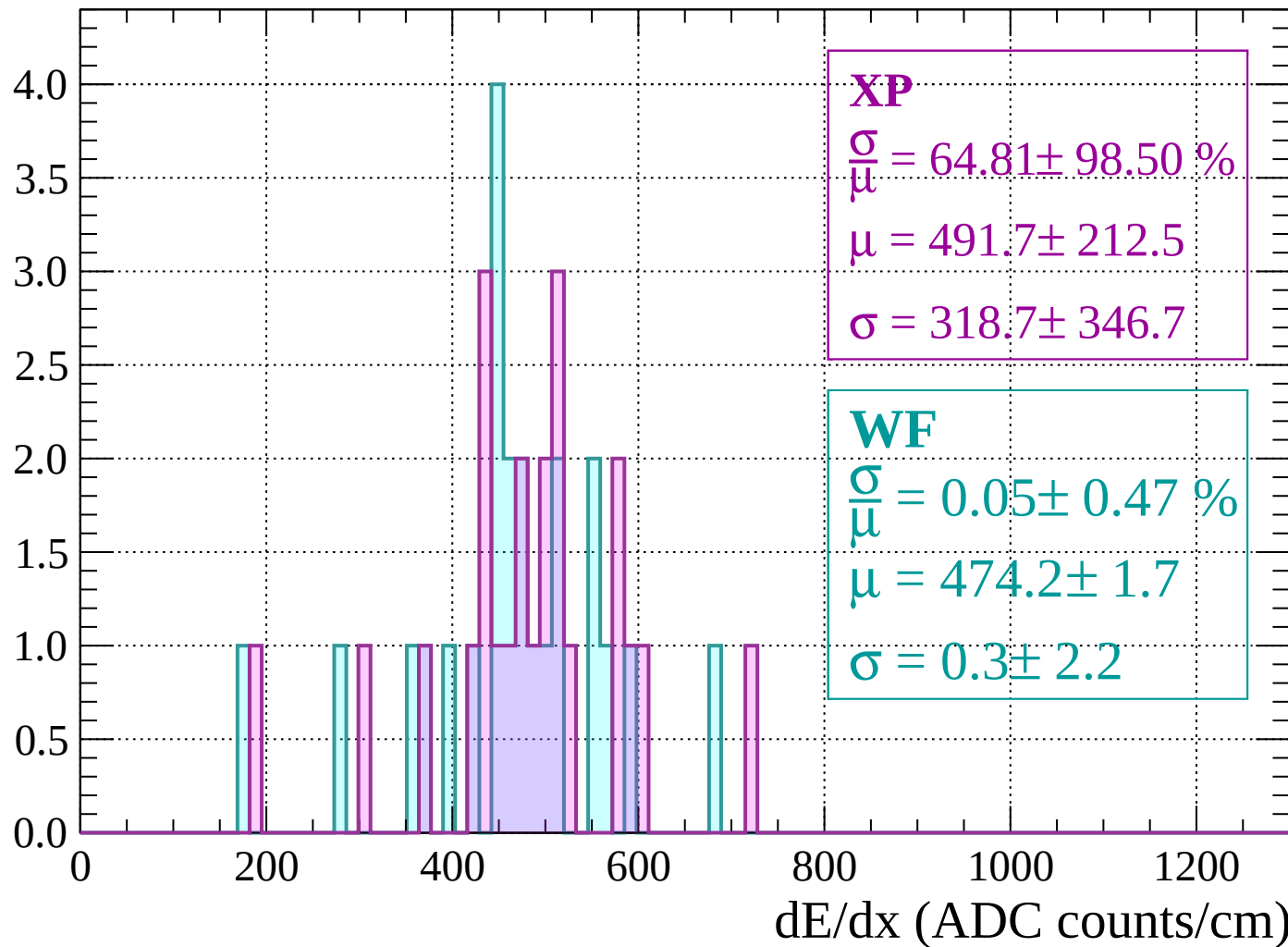
Energy loss | $1560 < p < 1600$

Count



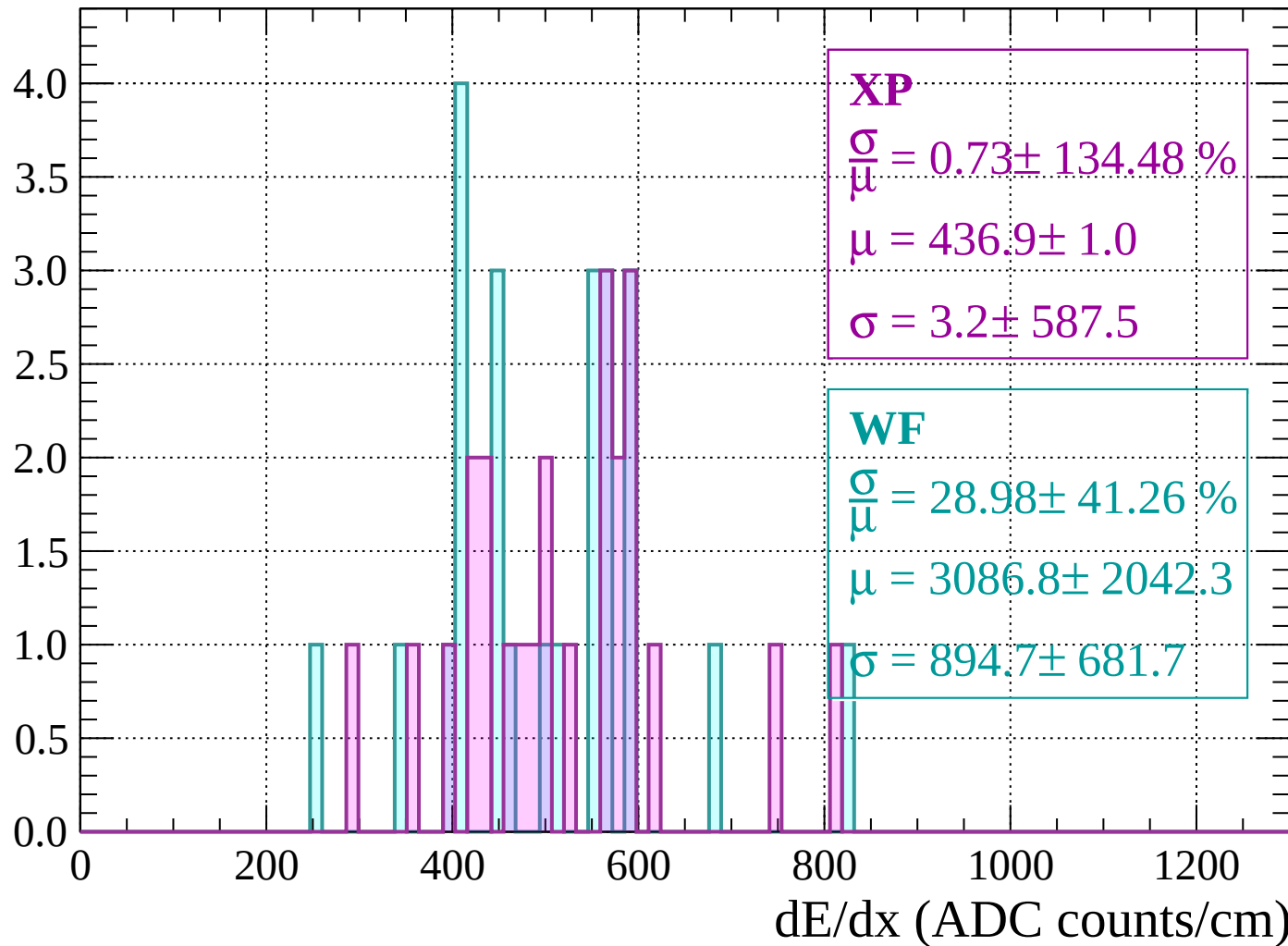
Energy loss | $1600 < p < 1640$

Count



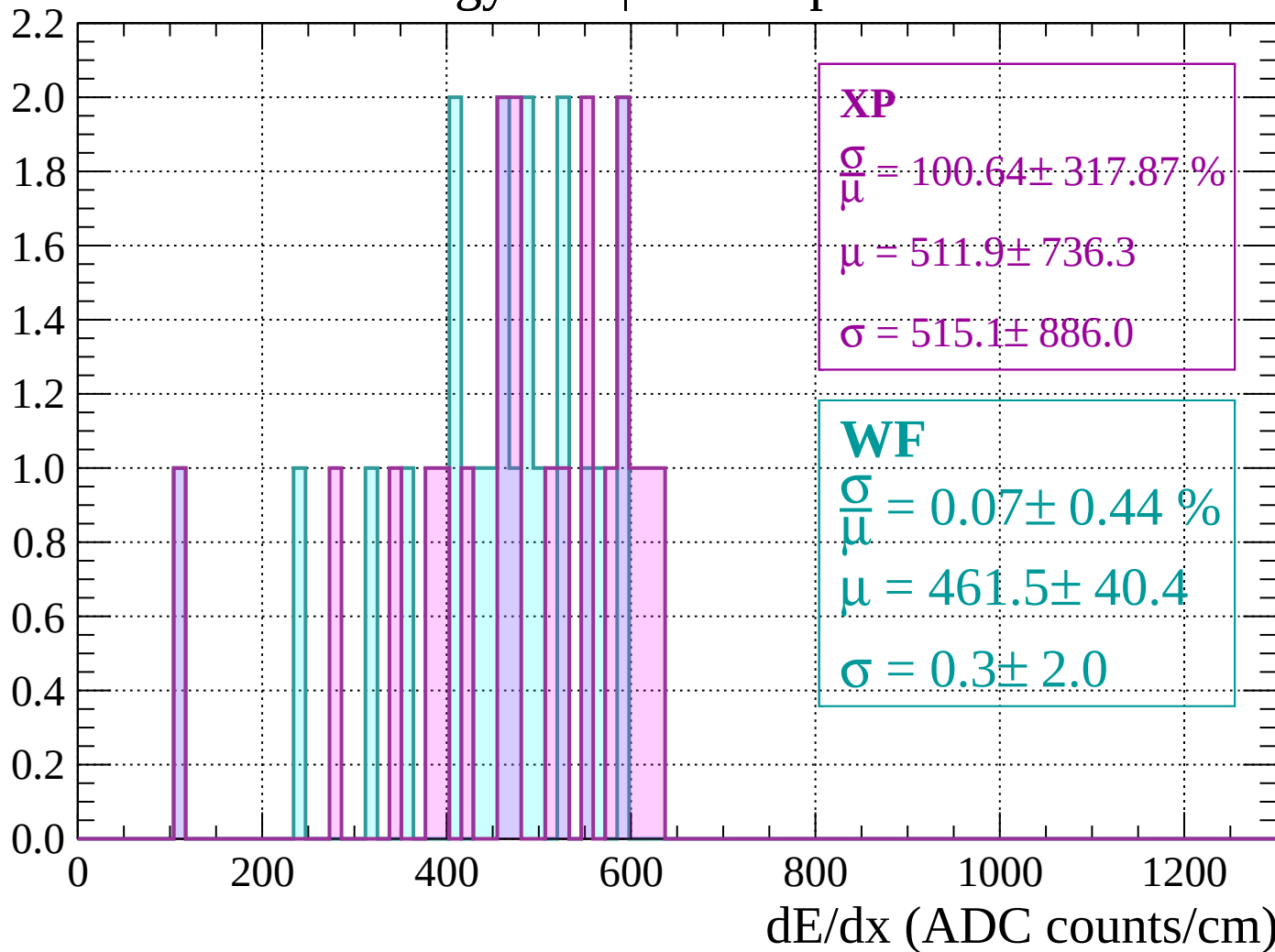
Energy loss | $1640 < p < 1680$

Count



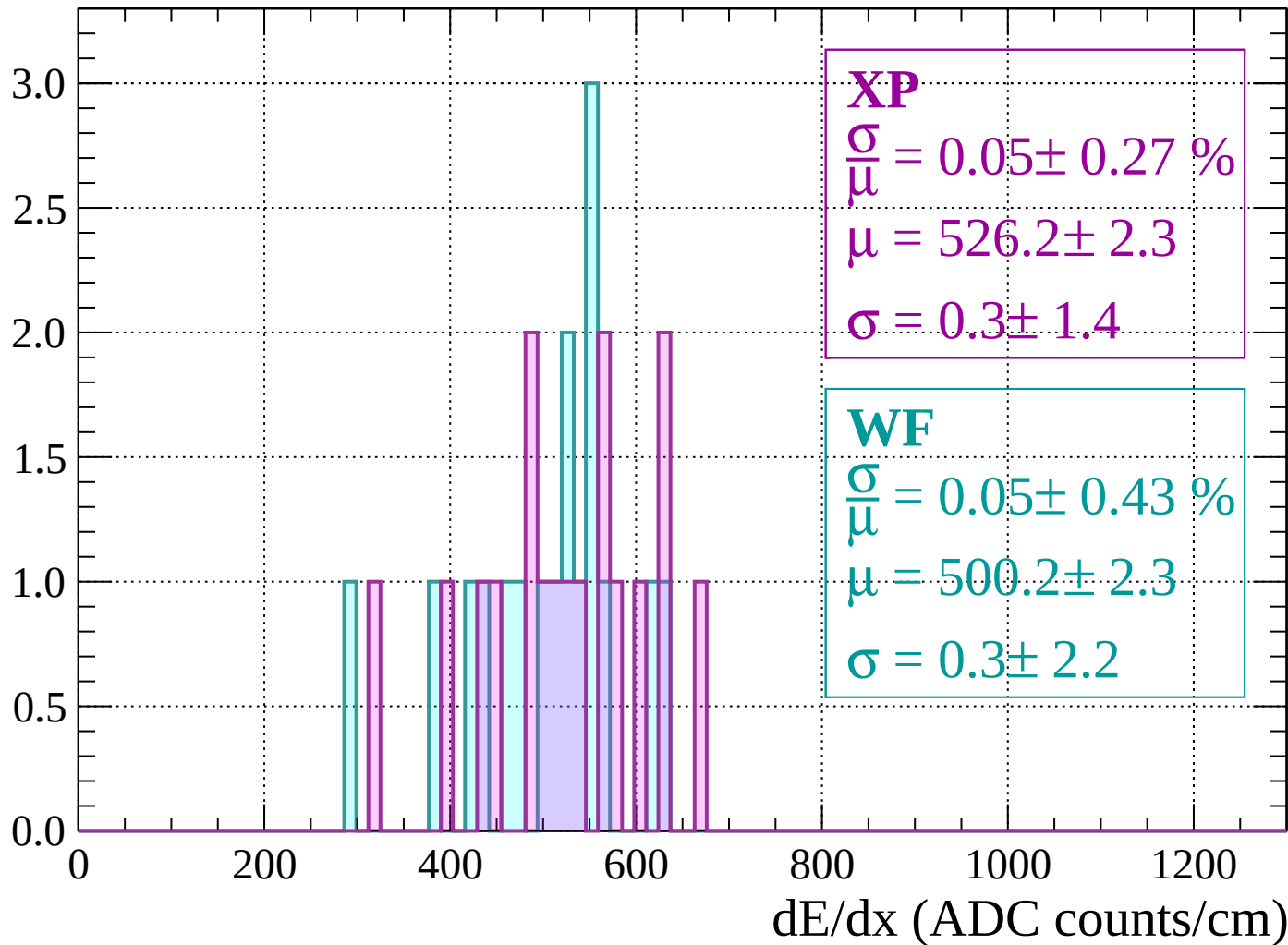
Energy loss | $1680 < p < 1720$

Count



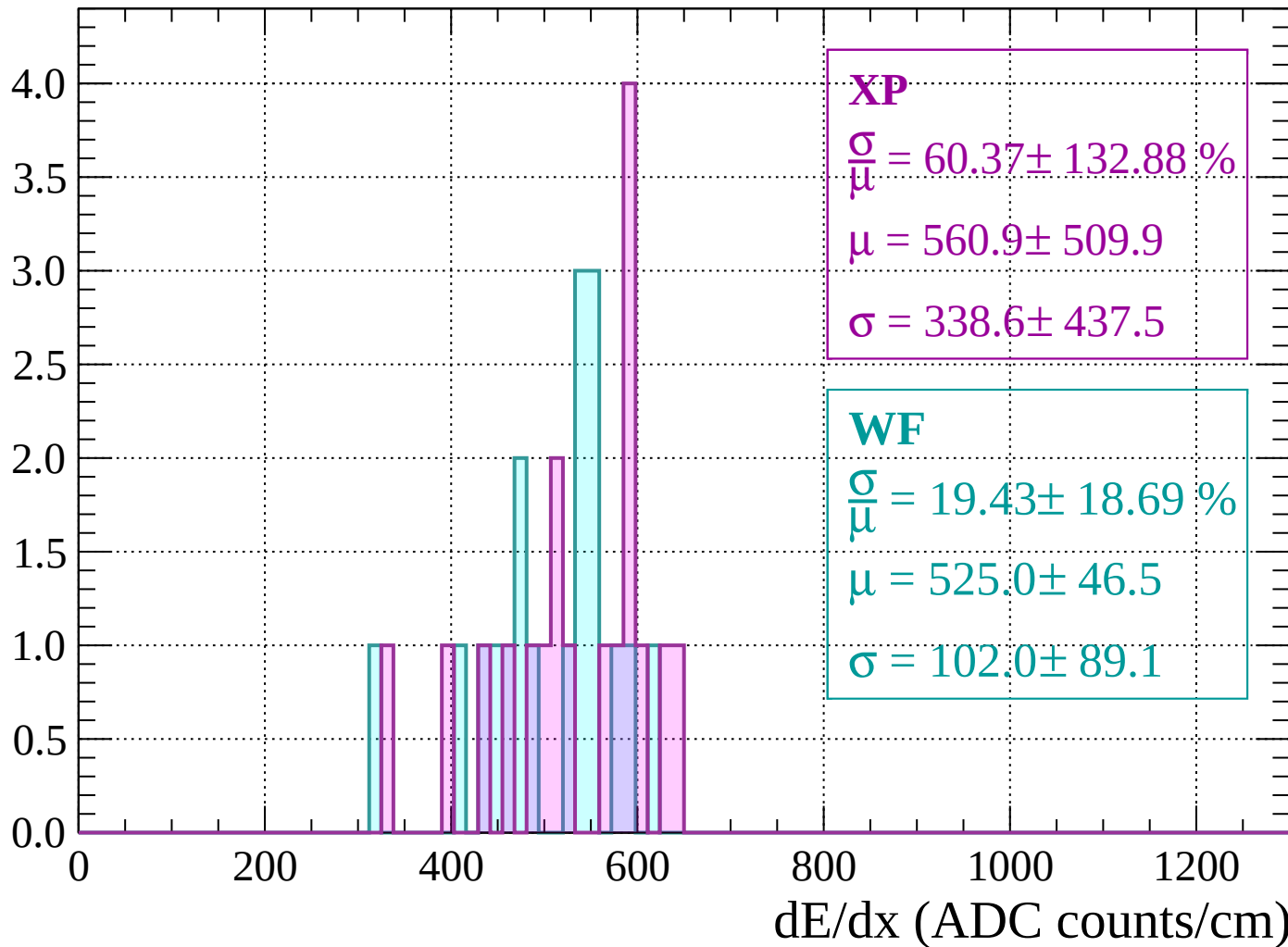
Energy loss | $1720 < p < 1760$

Count



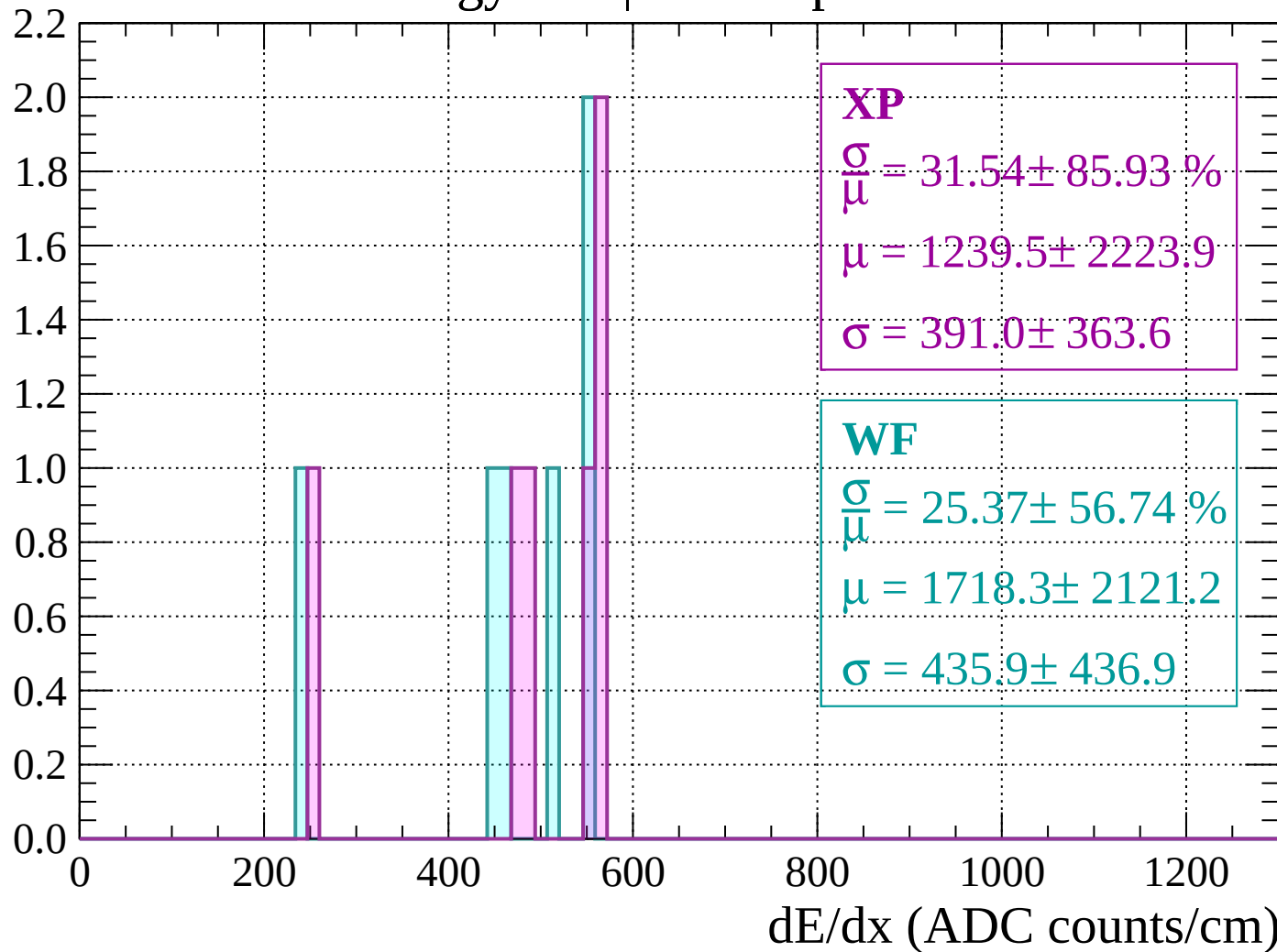
Energy loss | $1760 < p < 1800$

Count



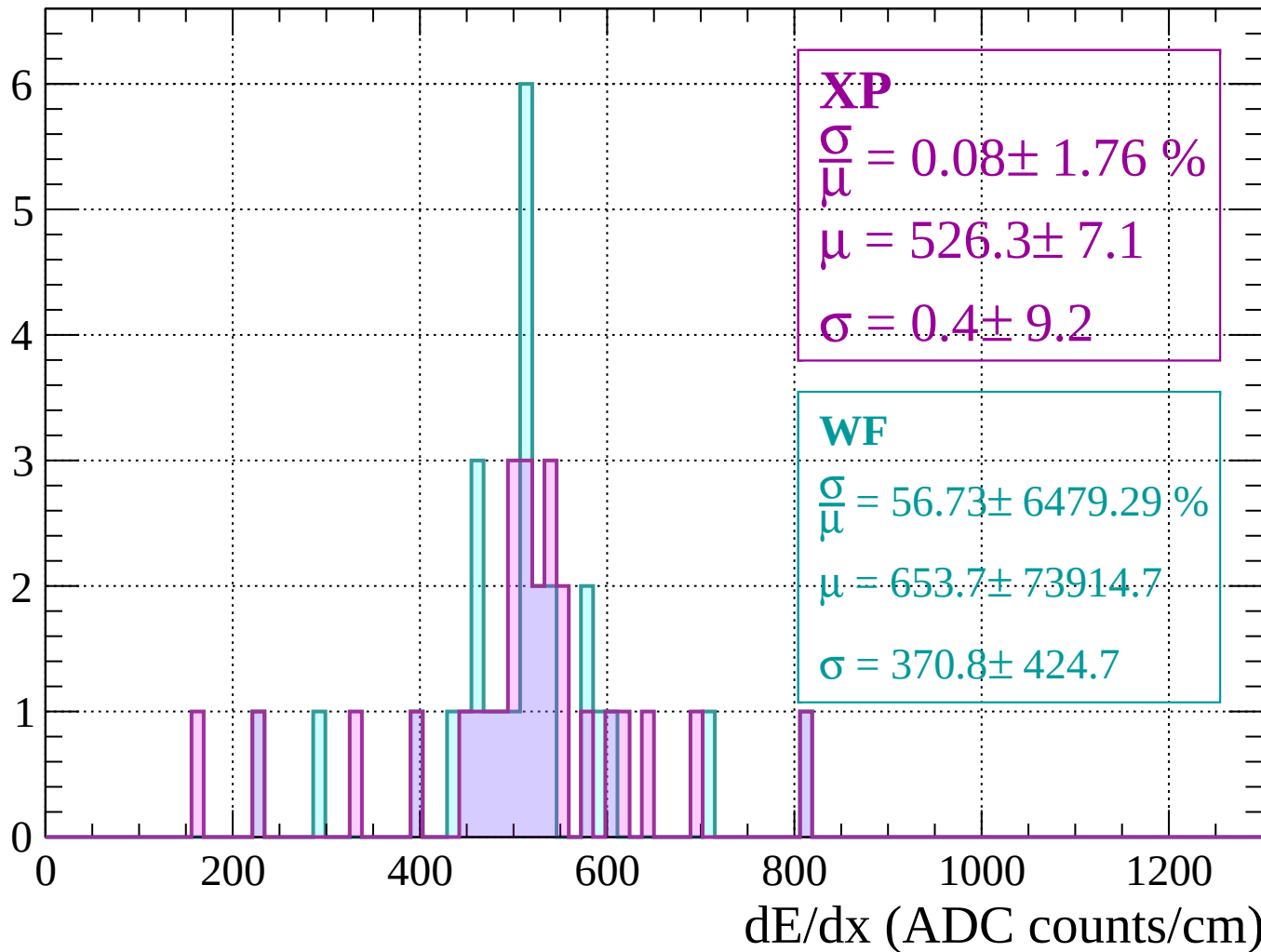
Energy loss | $1800 < p < 1840$

Count



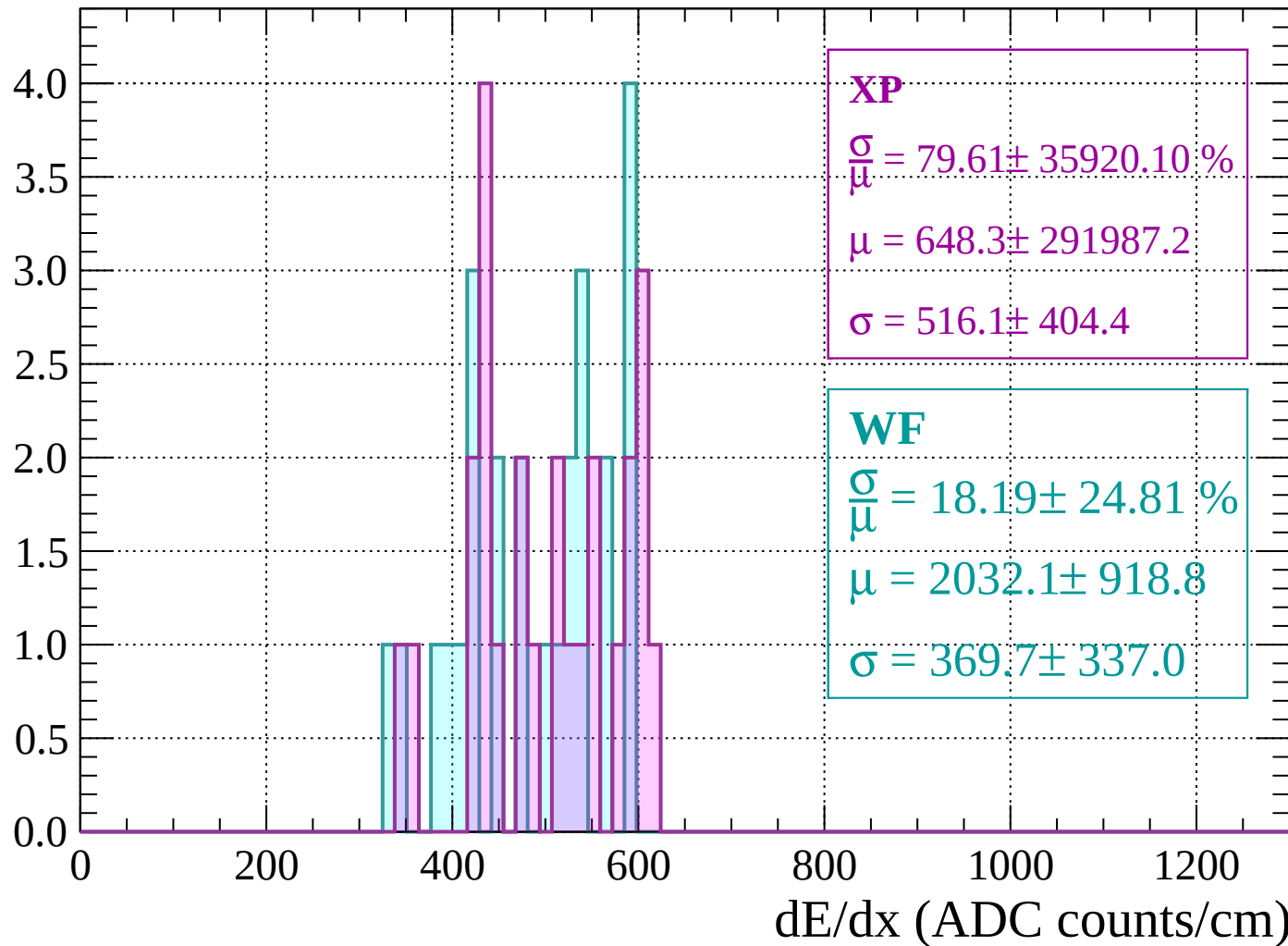
Energy loss | $1840 < p < 1880$

Count



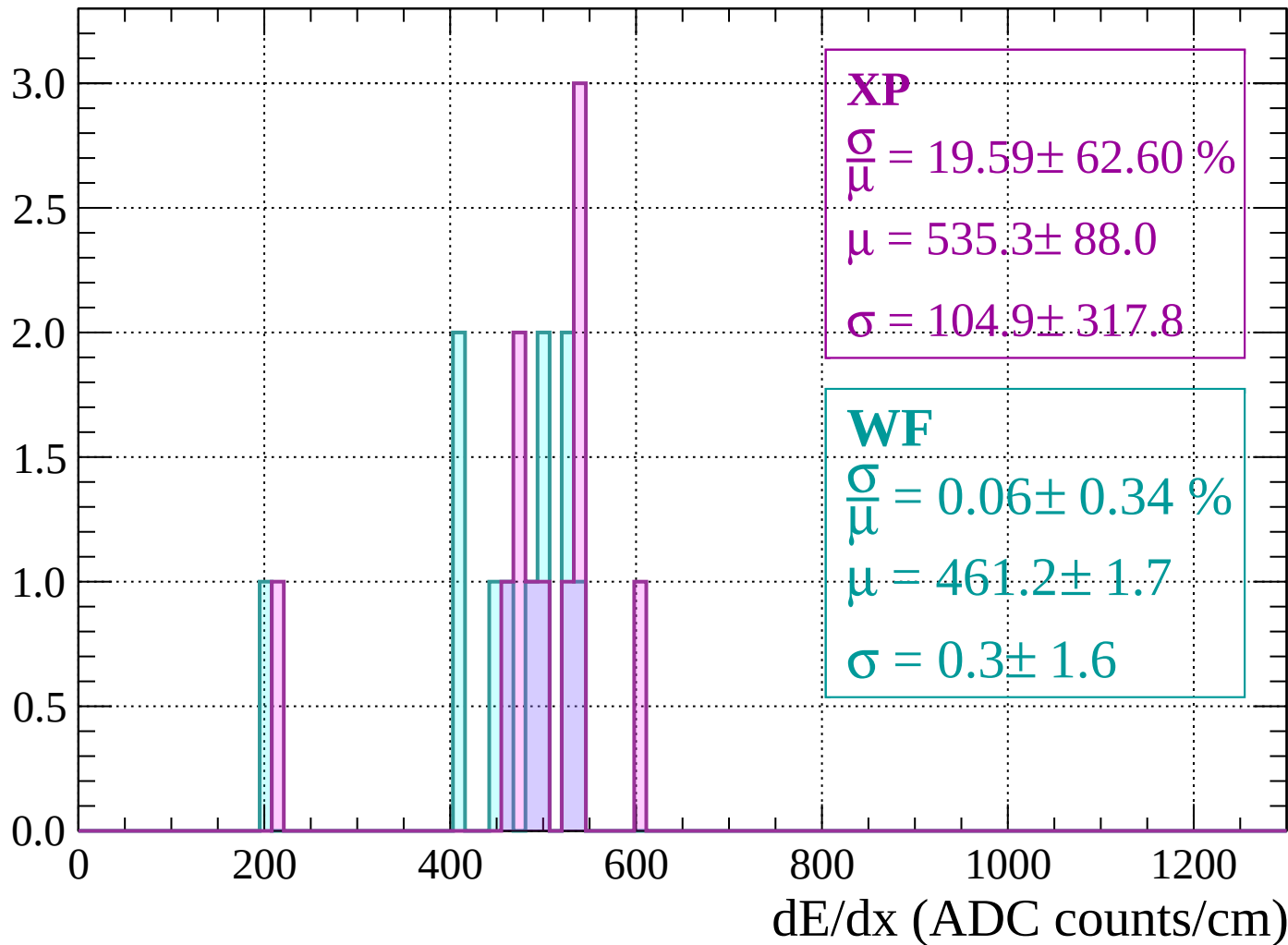
Energy loss | $1880 < p < 1920$

Count



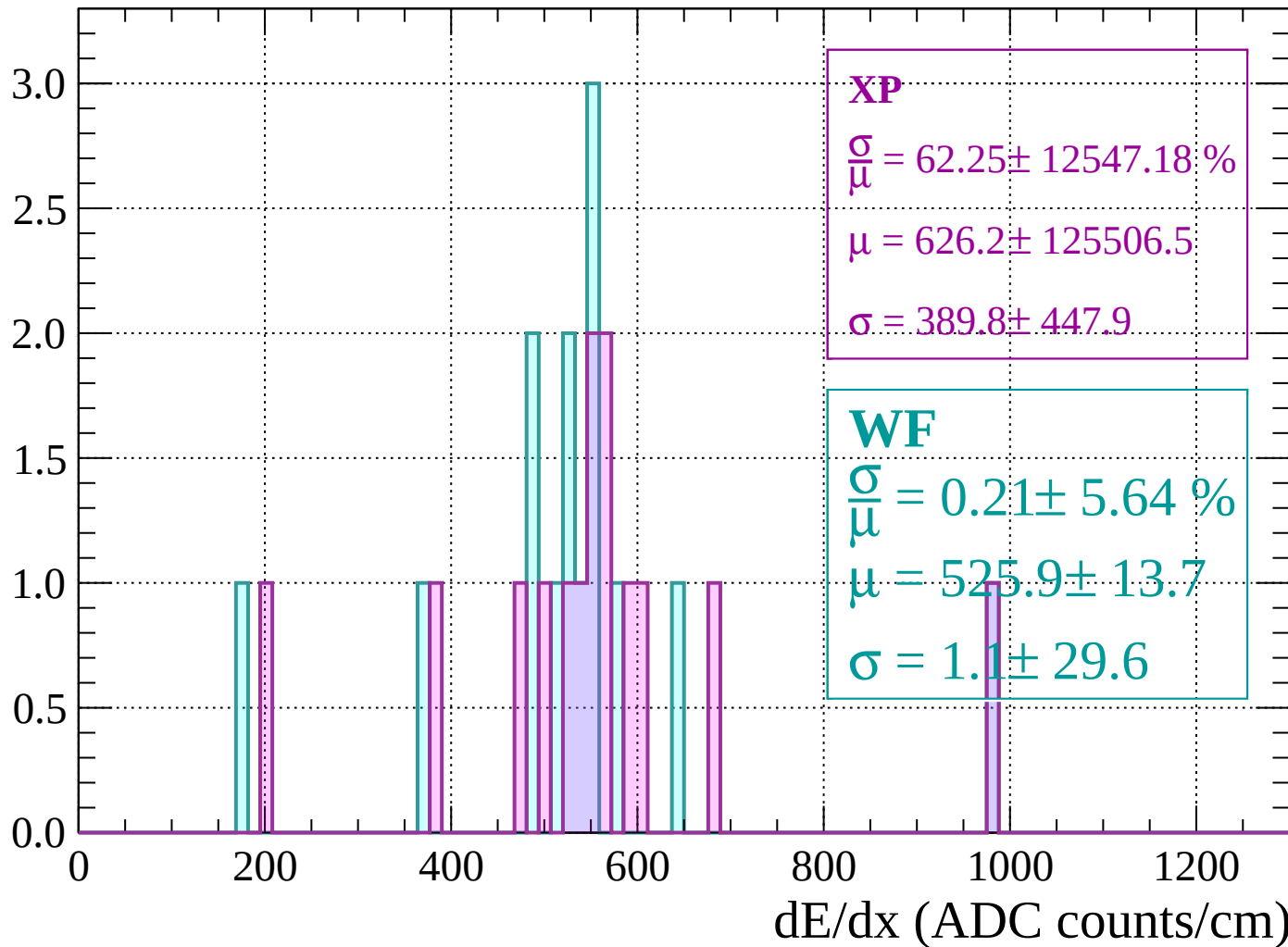
Energy loss | $1920 < p < 1960$

Count



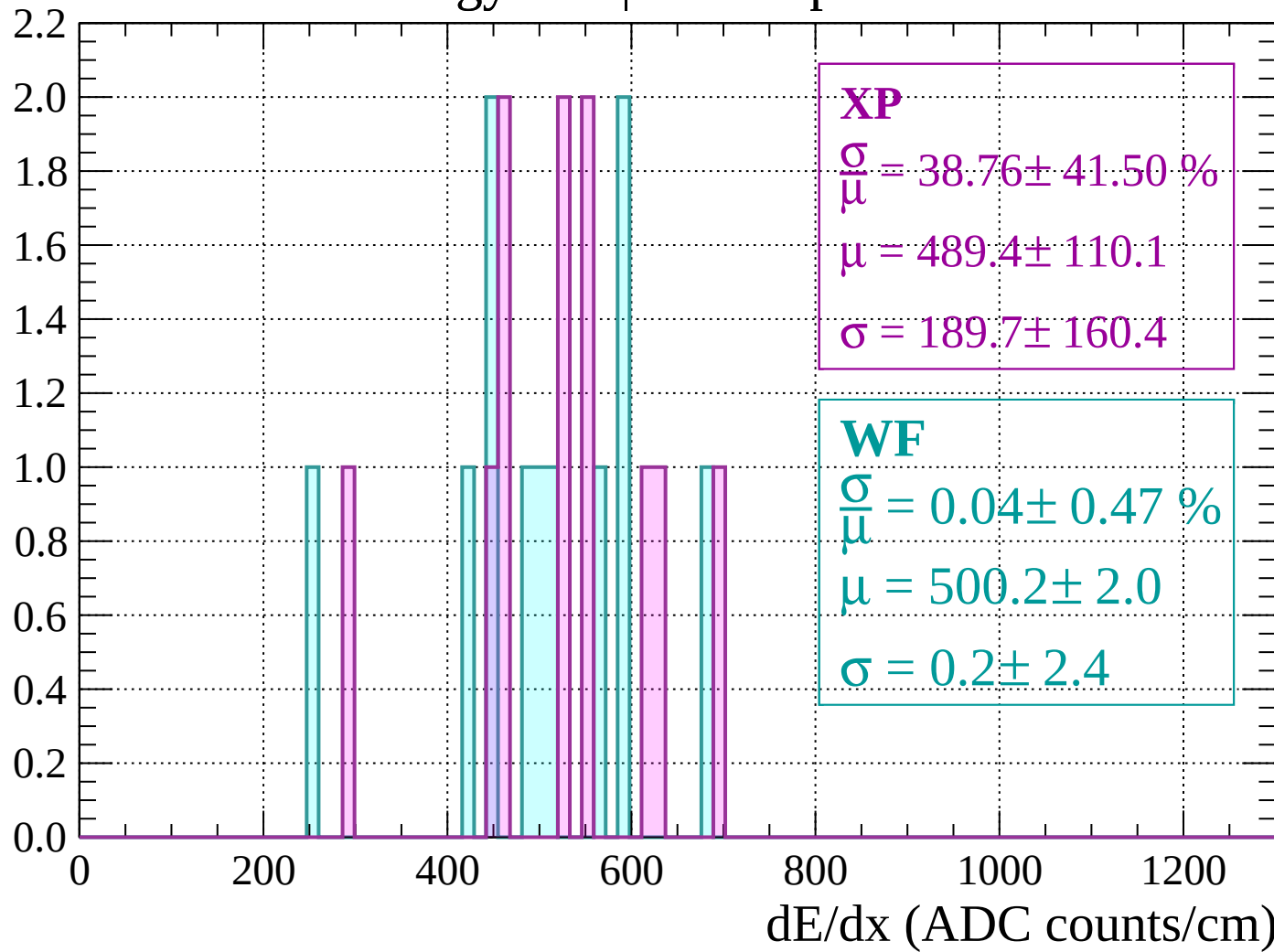
Energy loss | 1960 < p < 2000

Count



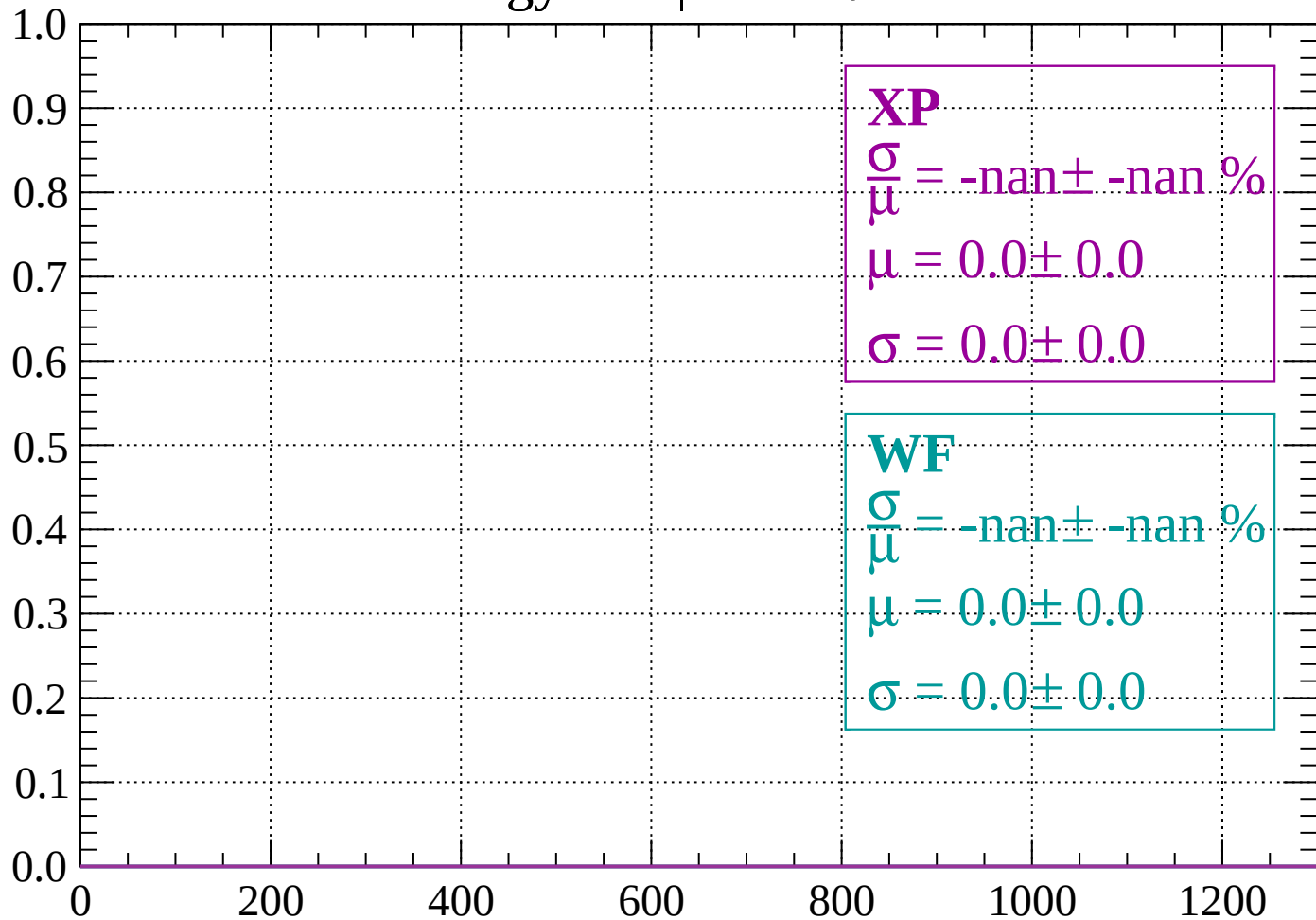
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

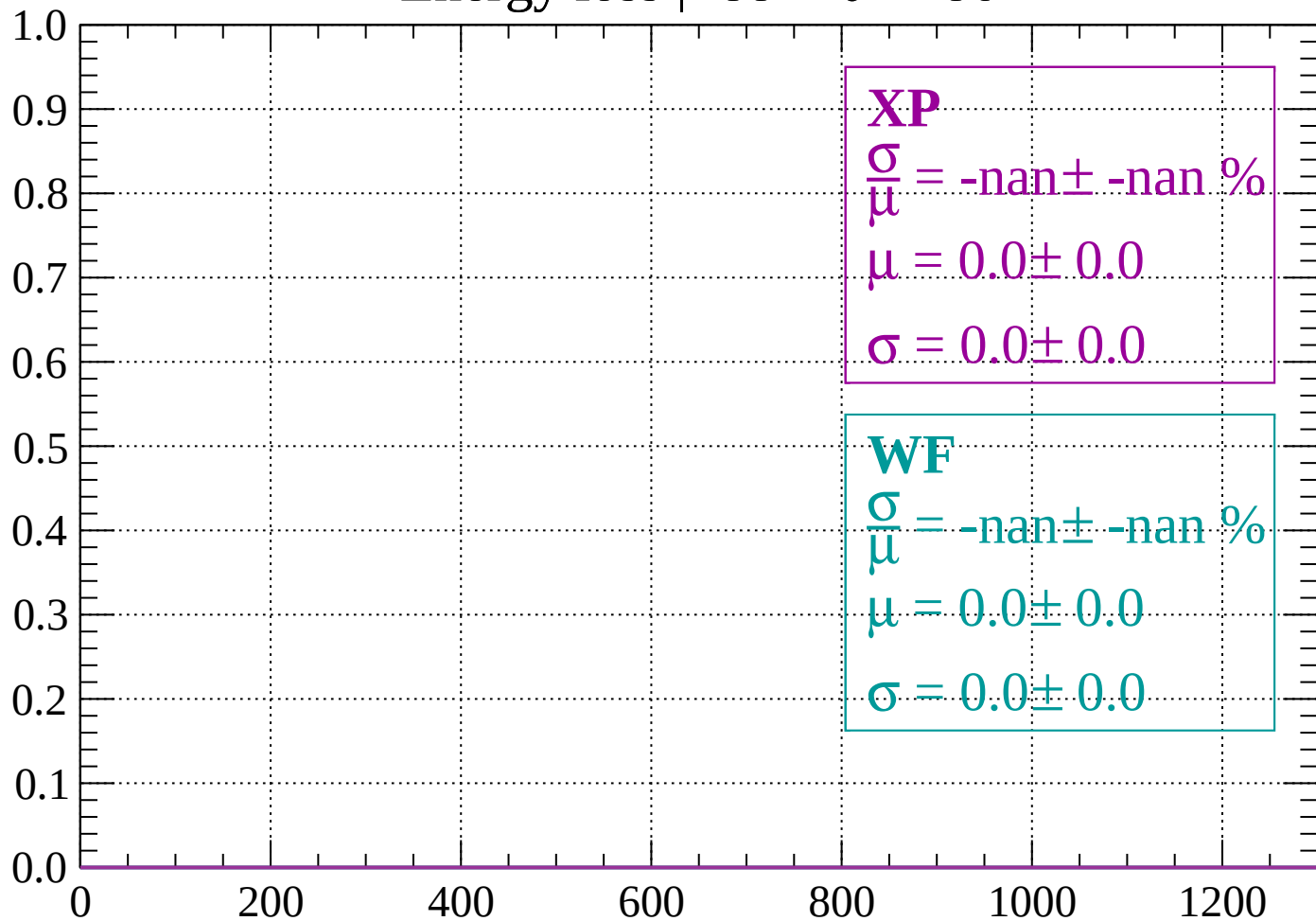
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

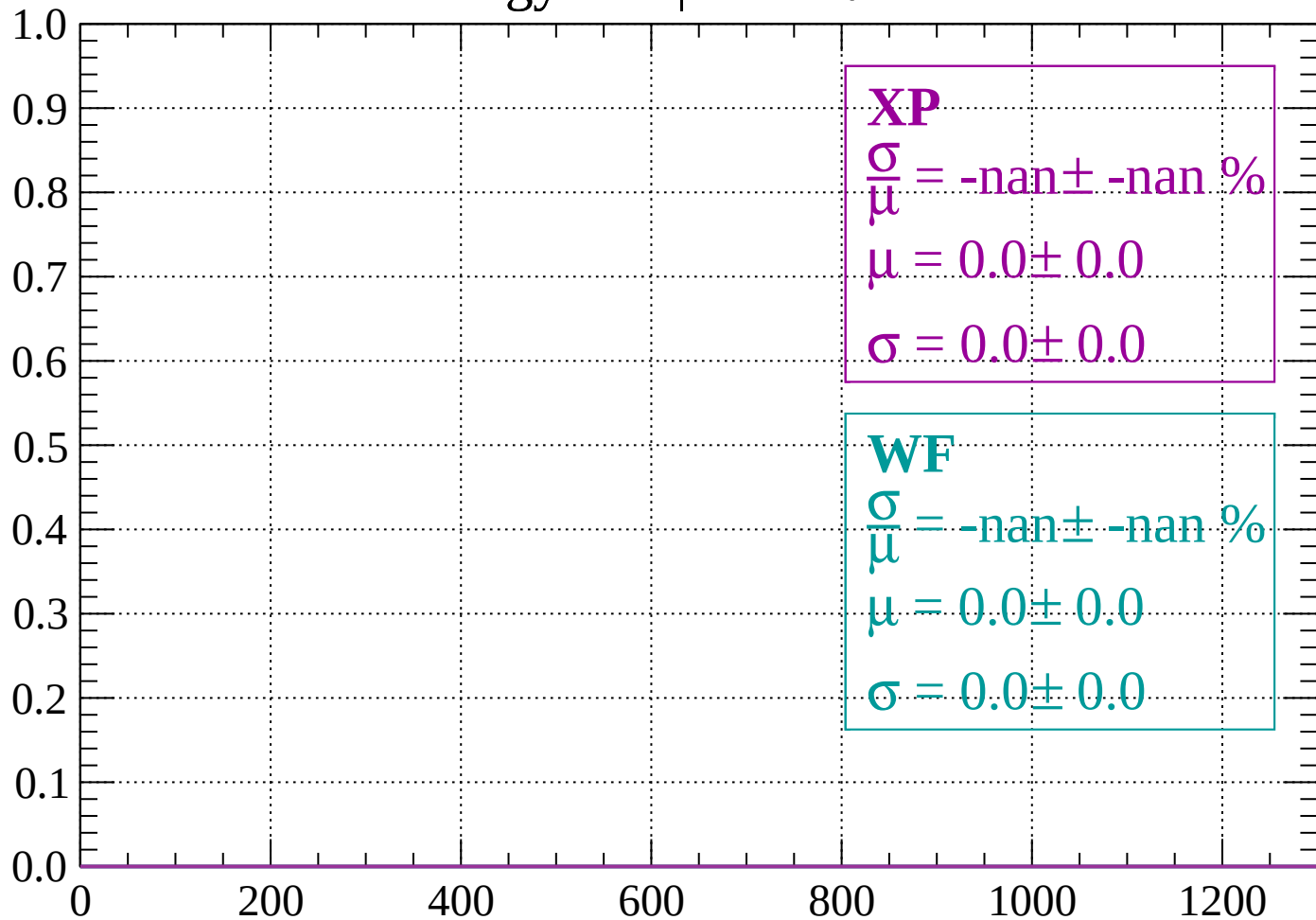
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

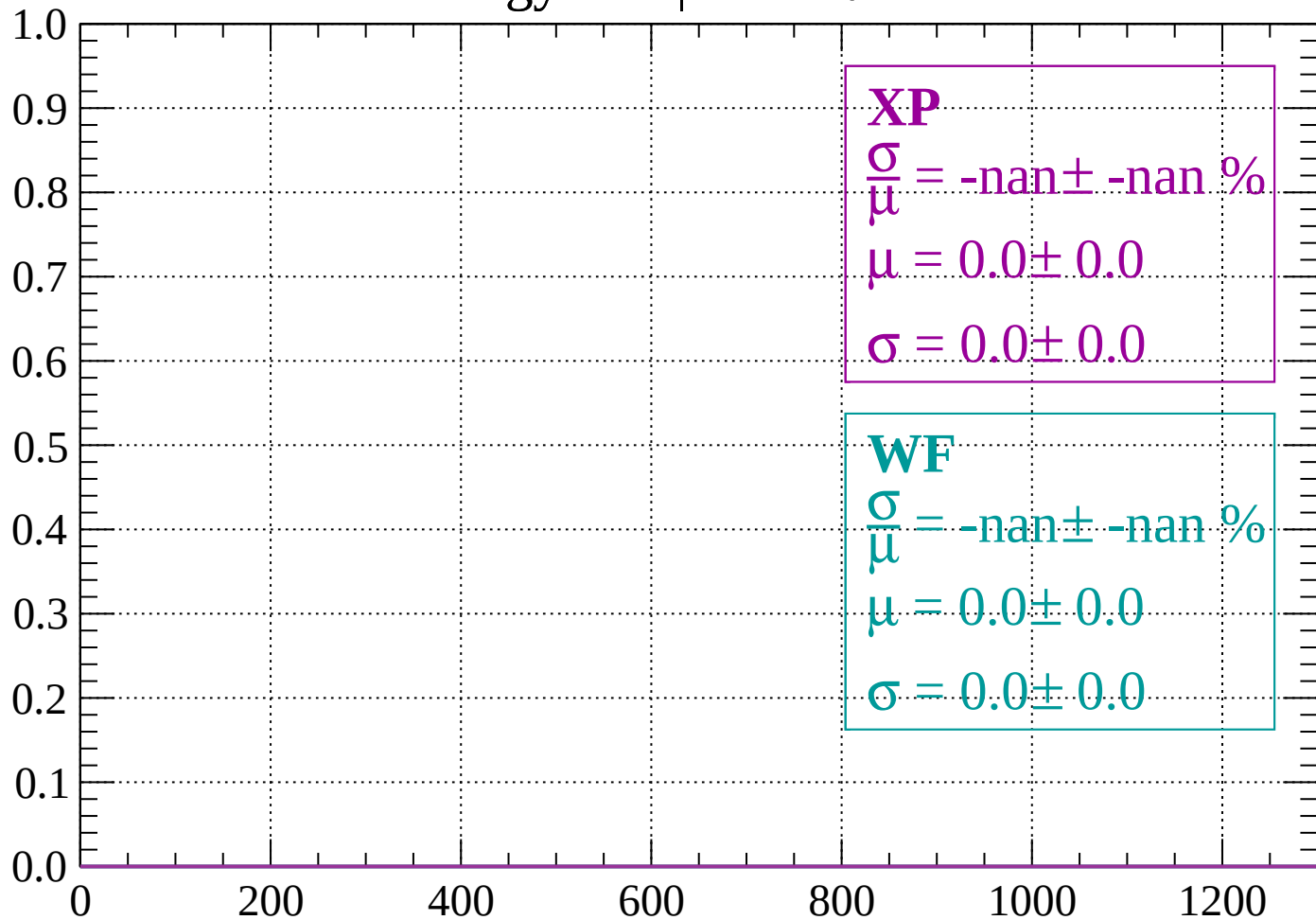
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

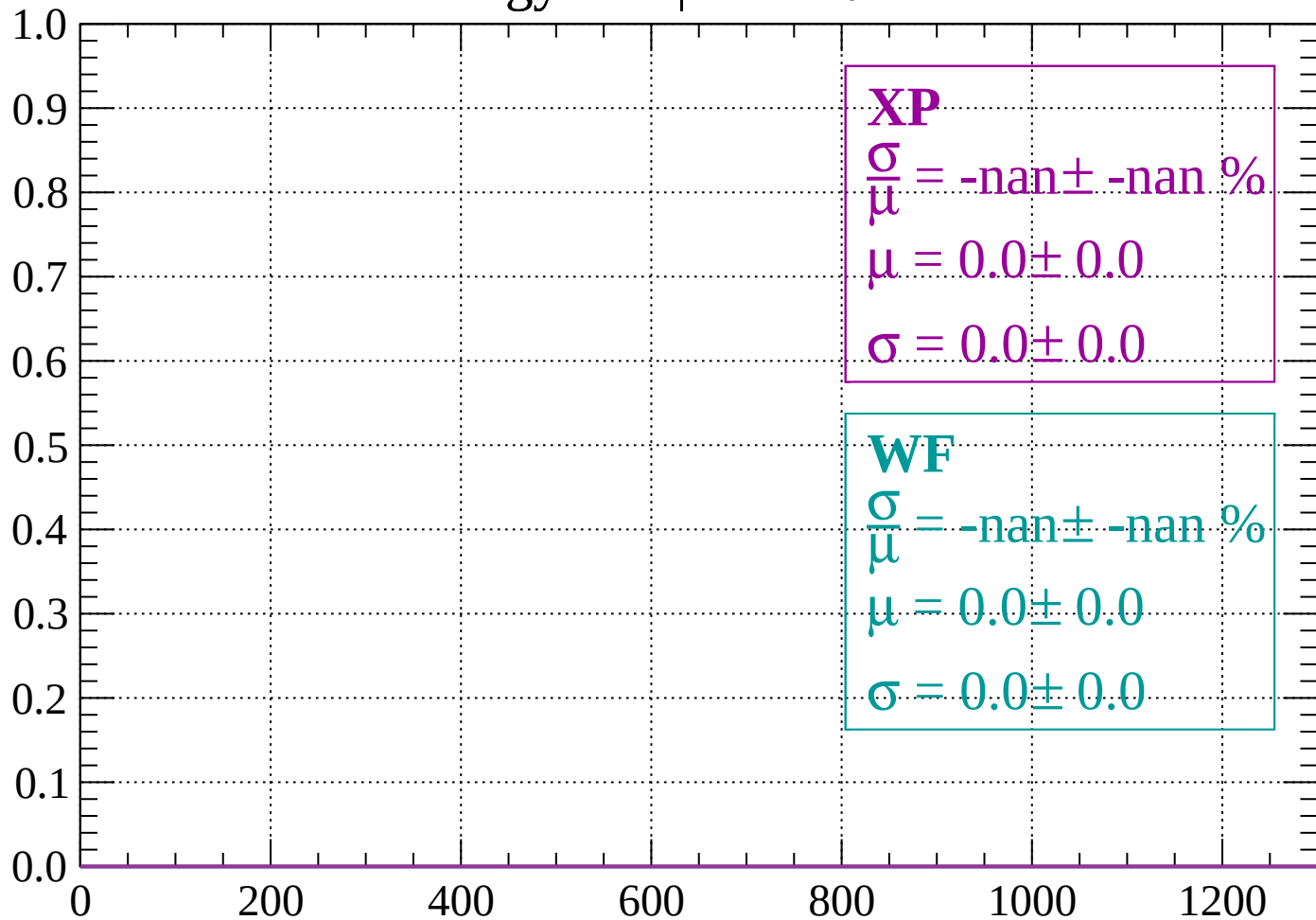
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

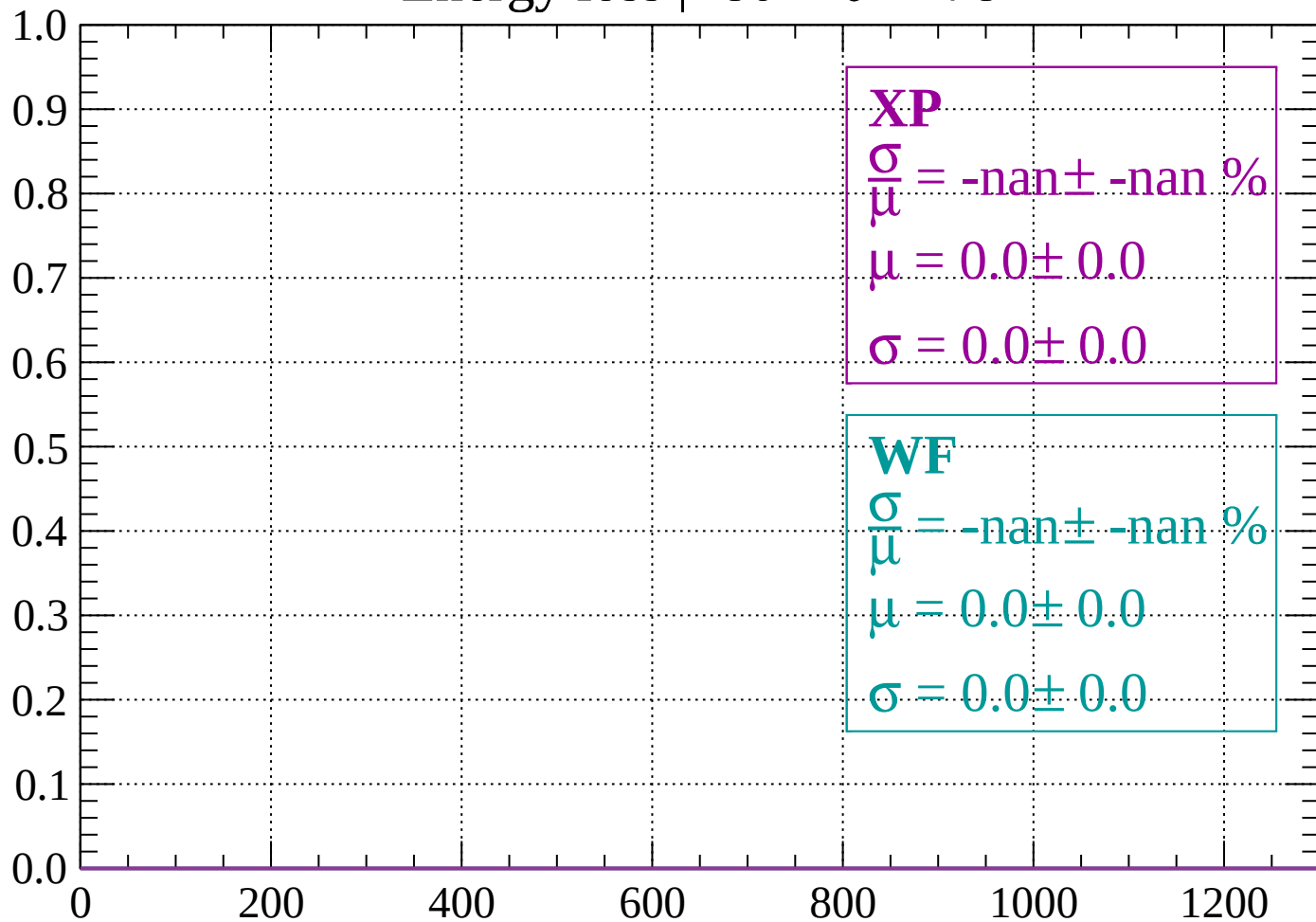
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

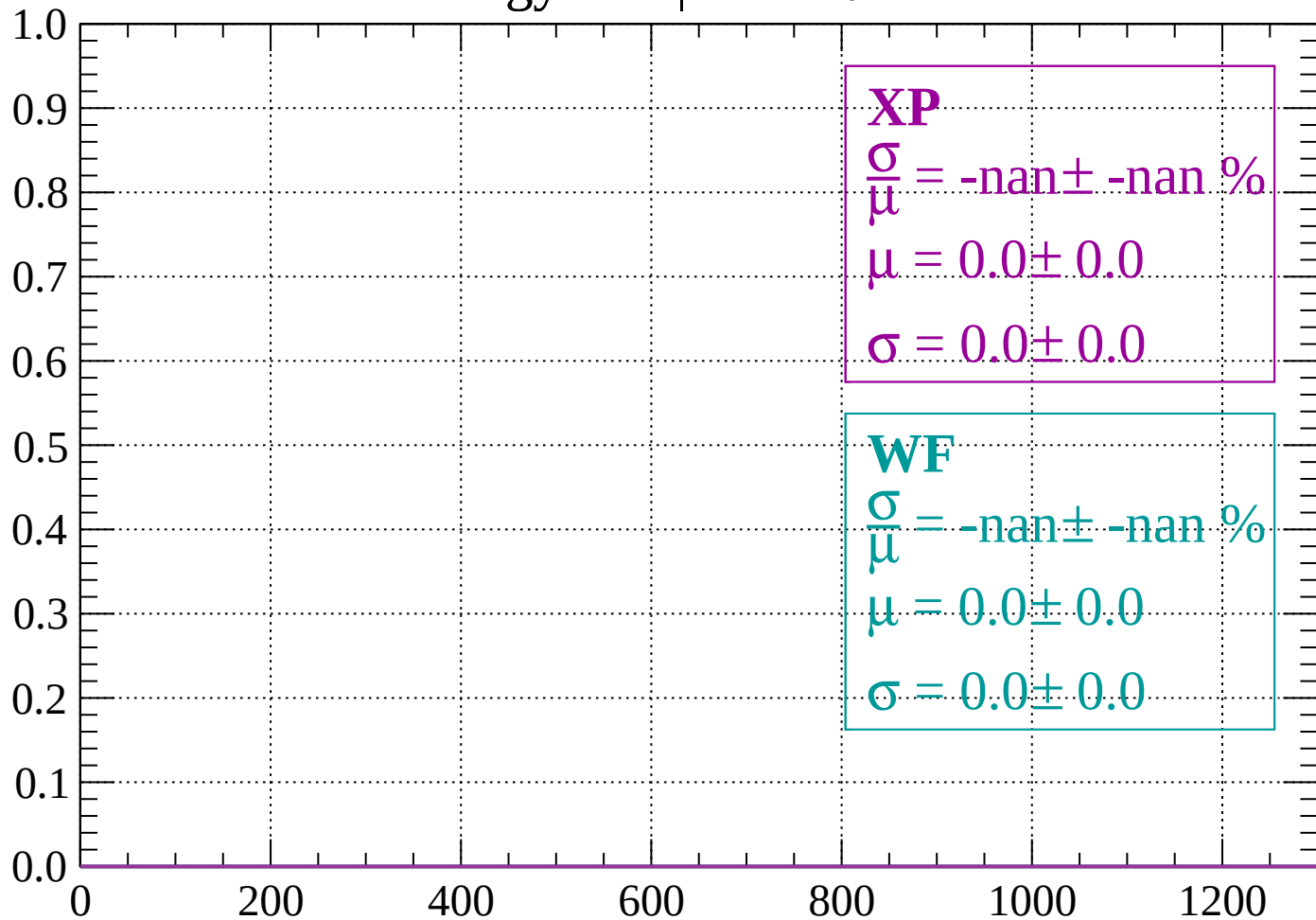
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

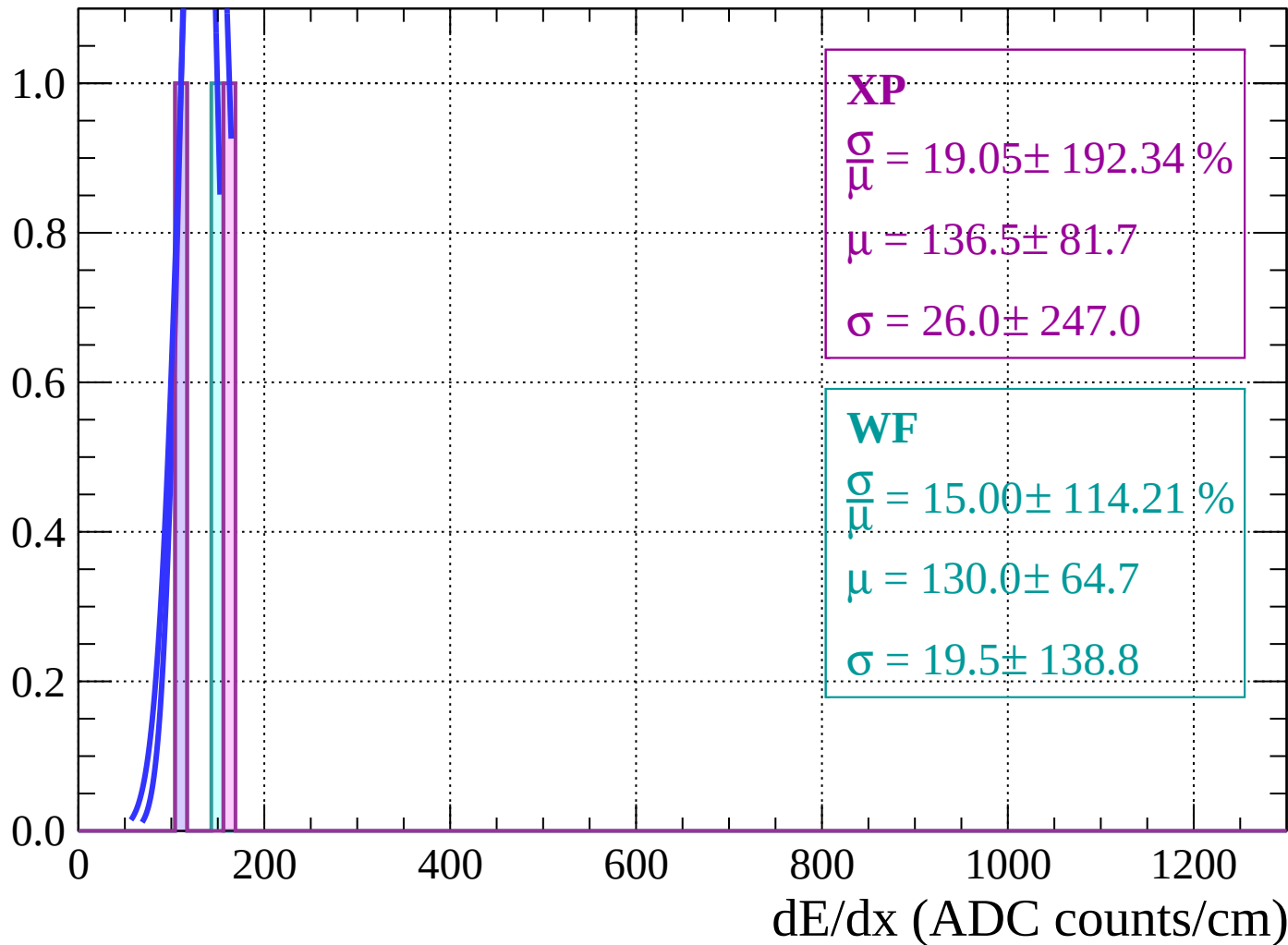
Count



dE/dx (ADC counts/cm)

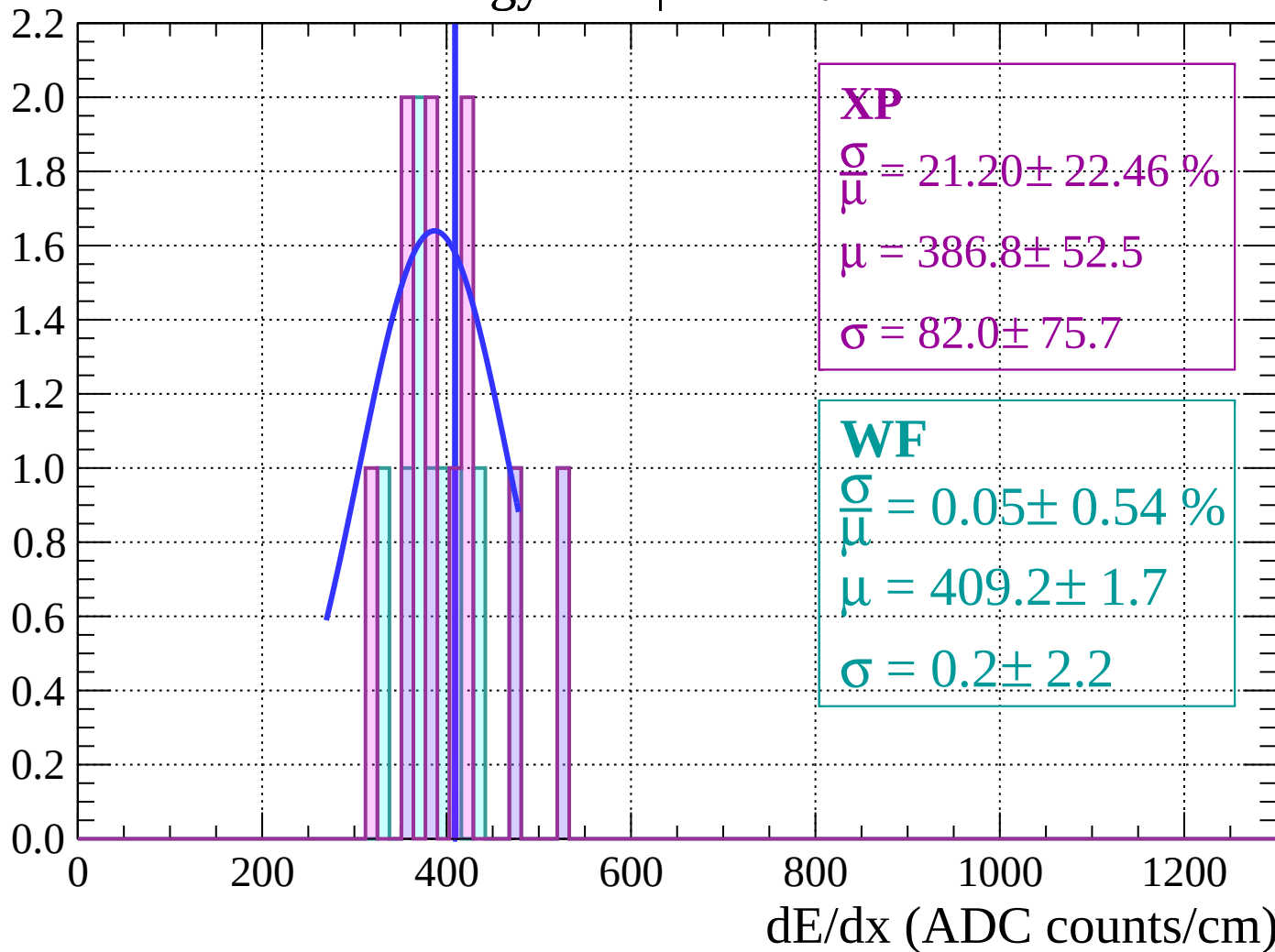
Energy loss | $-76 < \theta < -74$

Count



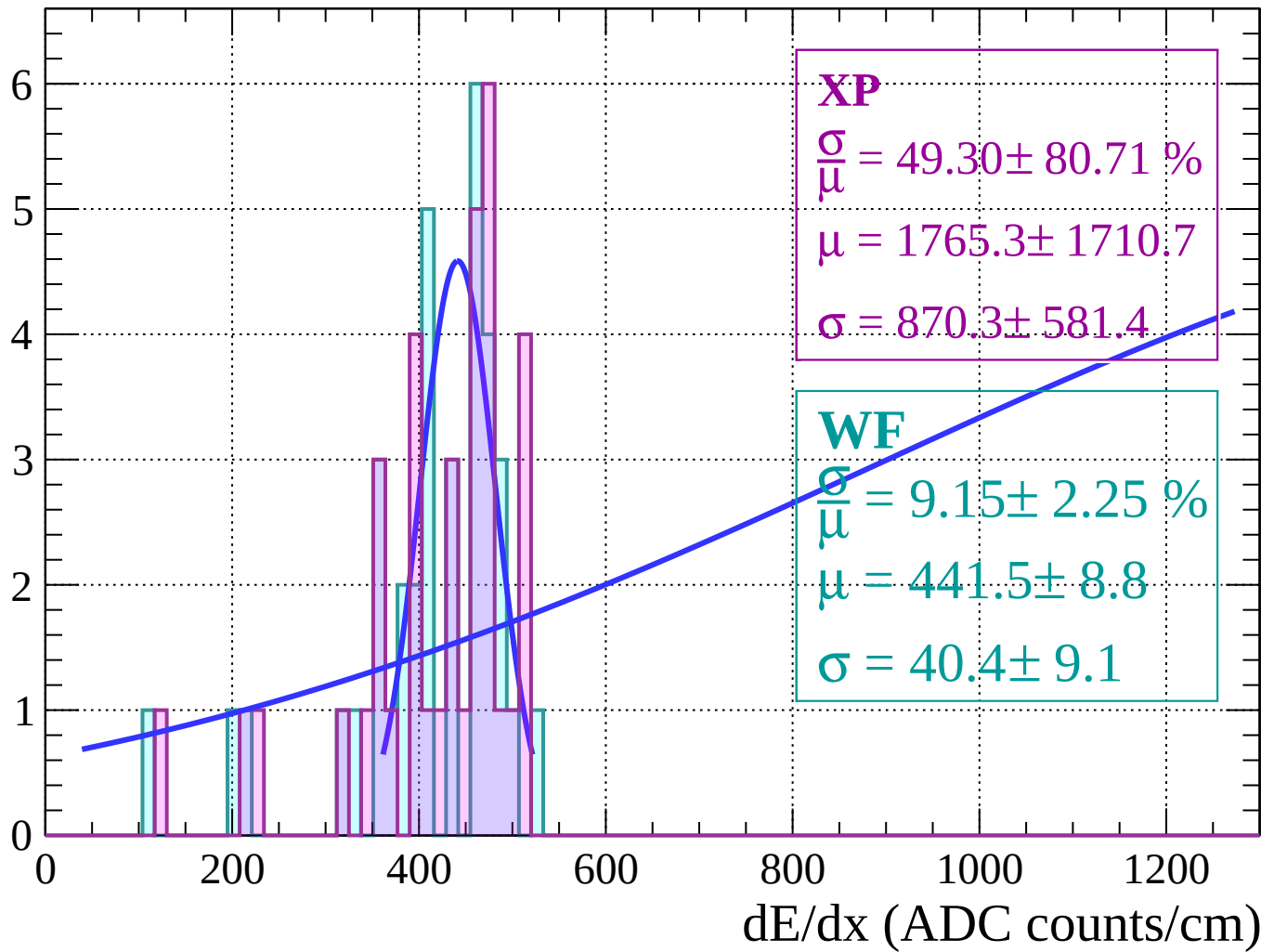
Energy loss $|-74 < \theta < -72$

Count



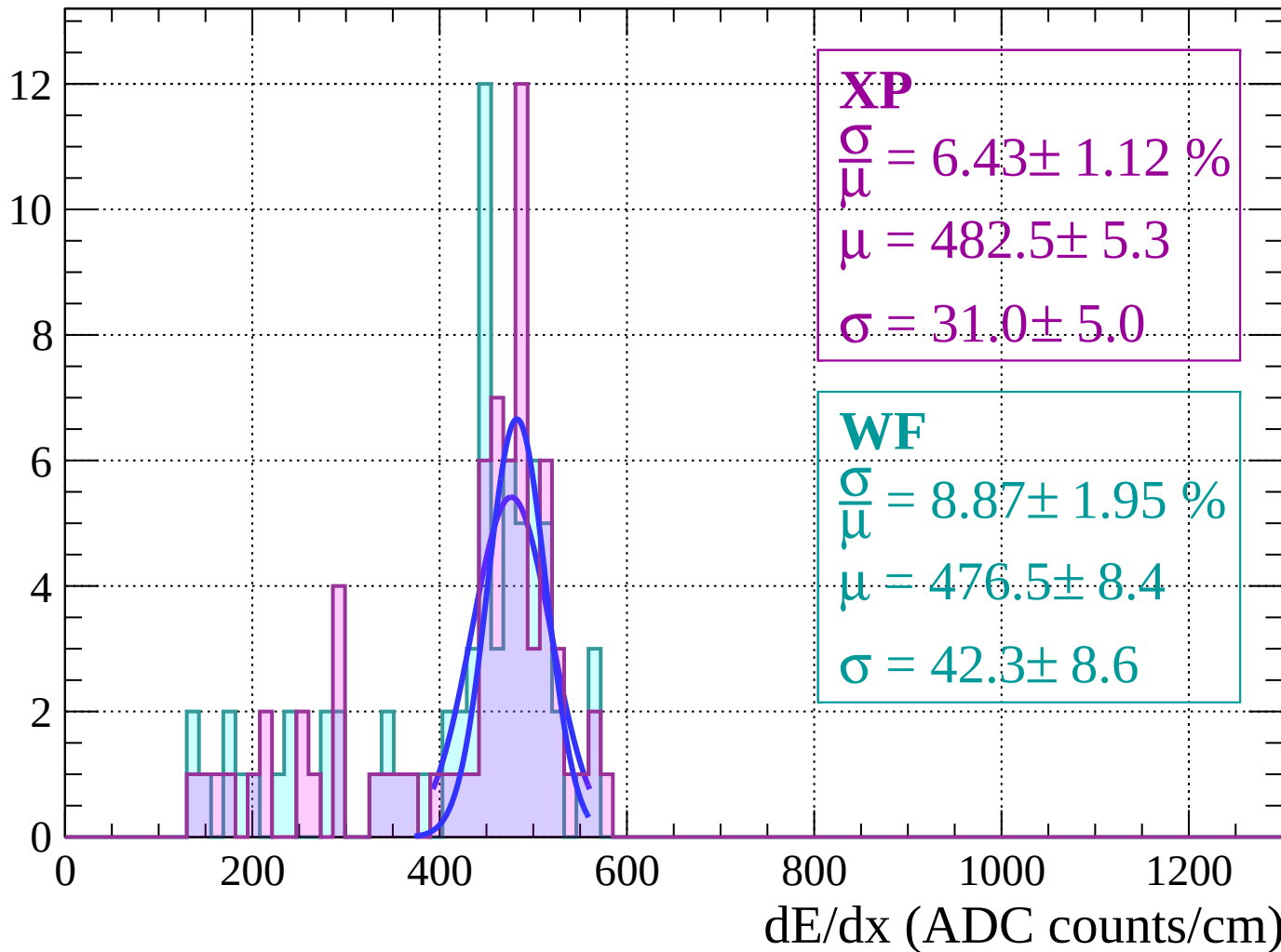
Energy loss | $-72 < \theta < -70$

Count



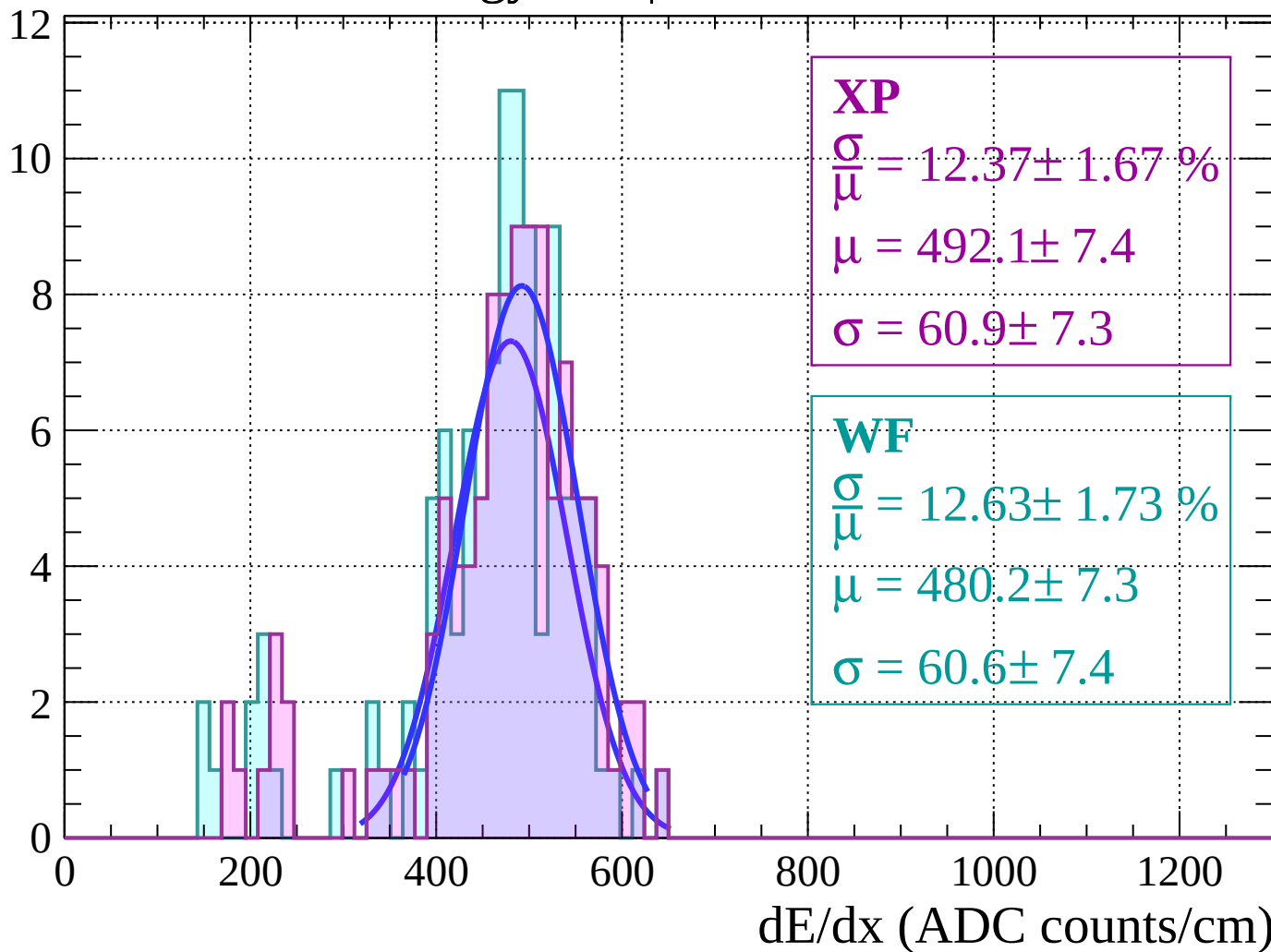
Energy loss | $-70 < \theta < -68$

Count



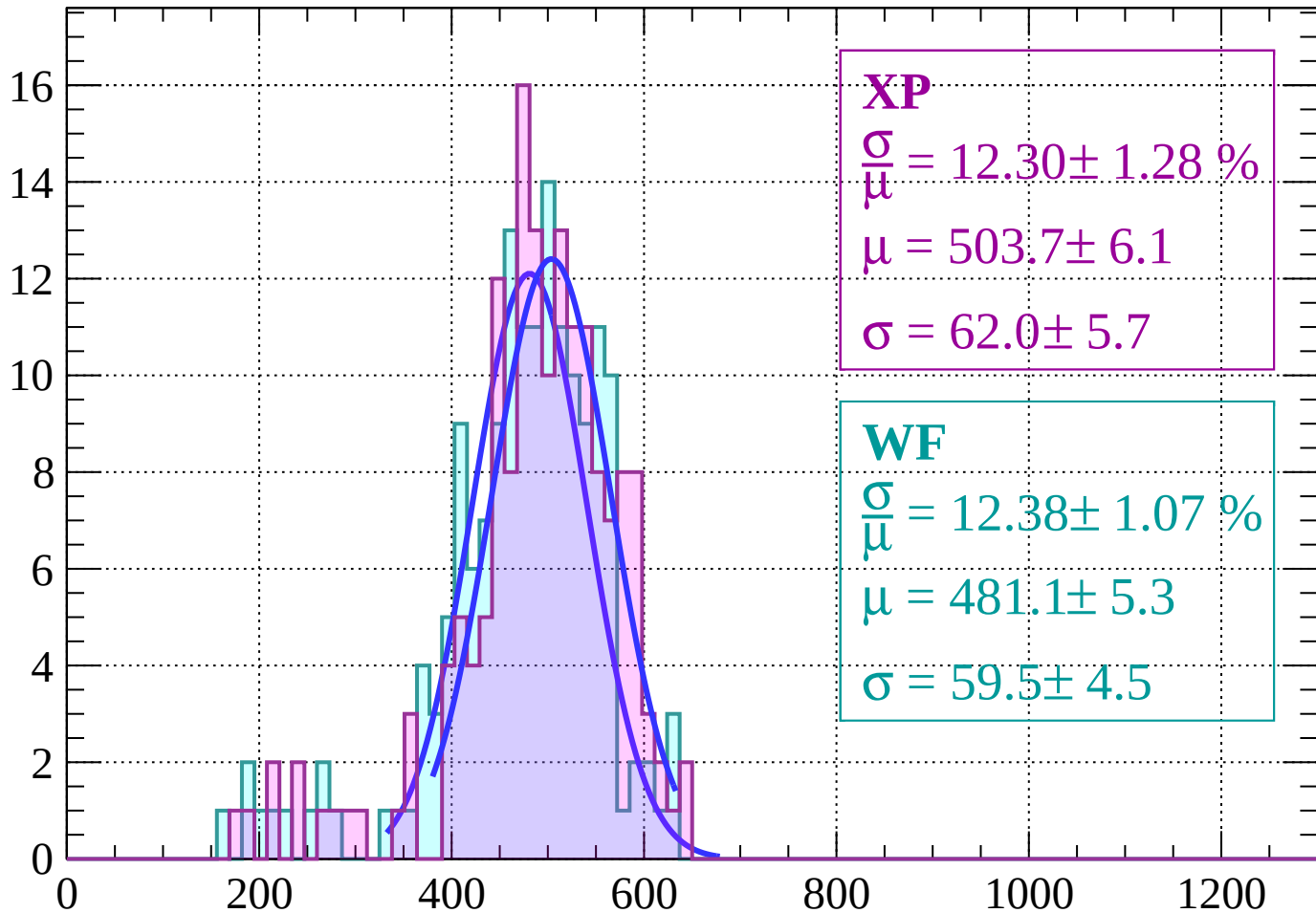
Energy loss $|-68 < \theta < -66$

Count



Energy loss | $-66 < \theta < -64$

Count



XP

$$\frac{\sigma}{\mu} = 12.30 \pm 1.28 \%$$

$$\mu = 503.7 \pm 6.1$$

$$\sigma = 62.0 \pm 5.7$$

WF

$$\frac{\sigma}{\mu} = 12.38 \pm 1.07 \%$$

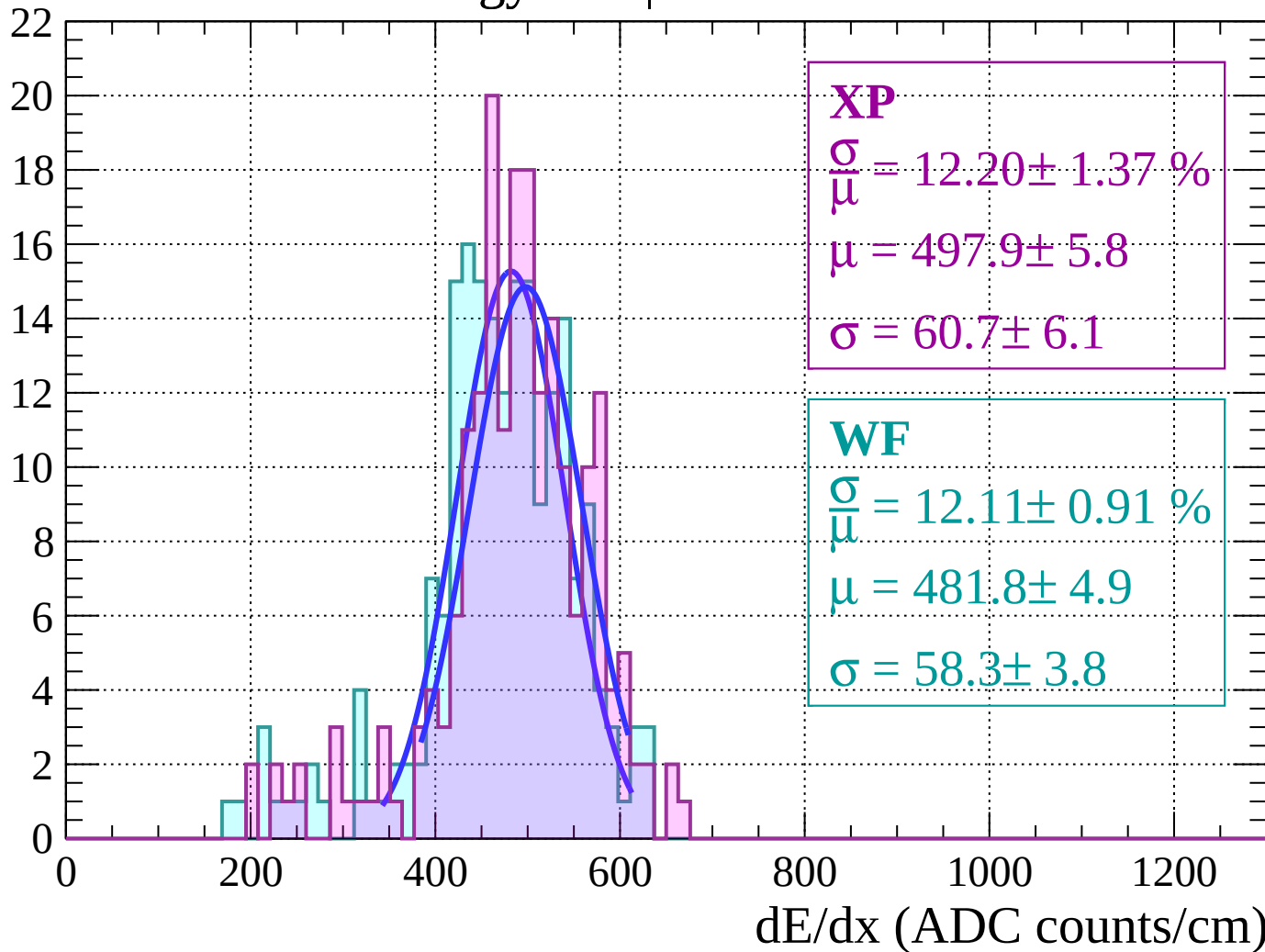
$$\mu = 481.1 \pm 5.3$$

$$\sigma = 59.5 \pm 4.5$$

dE/dx (ADC counts/cm)

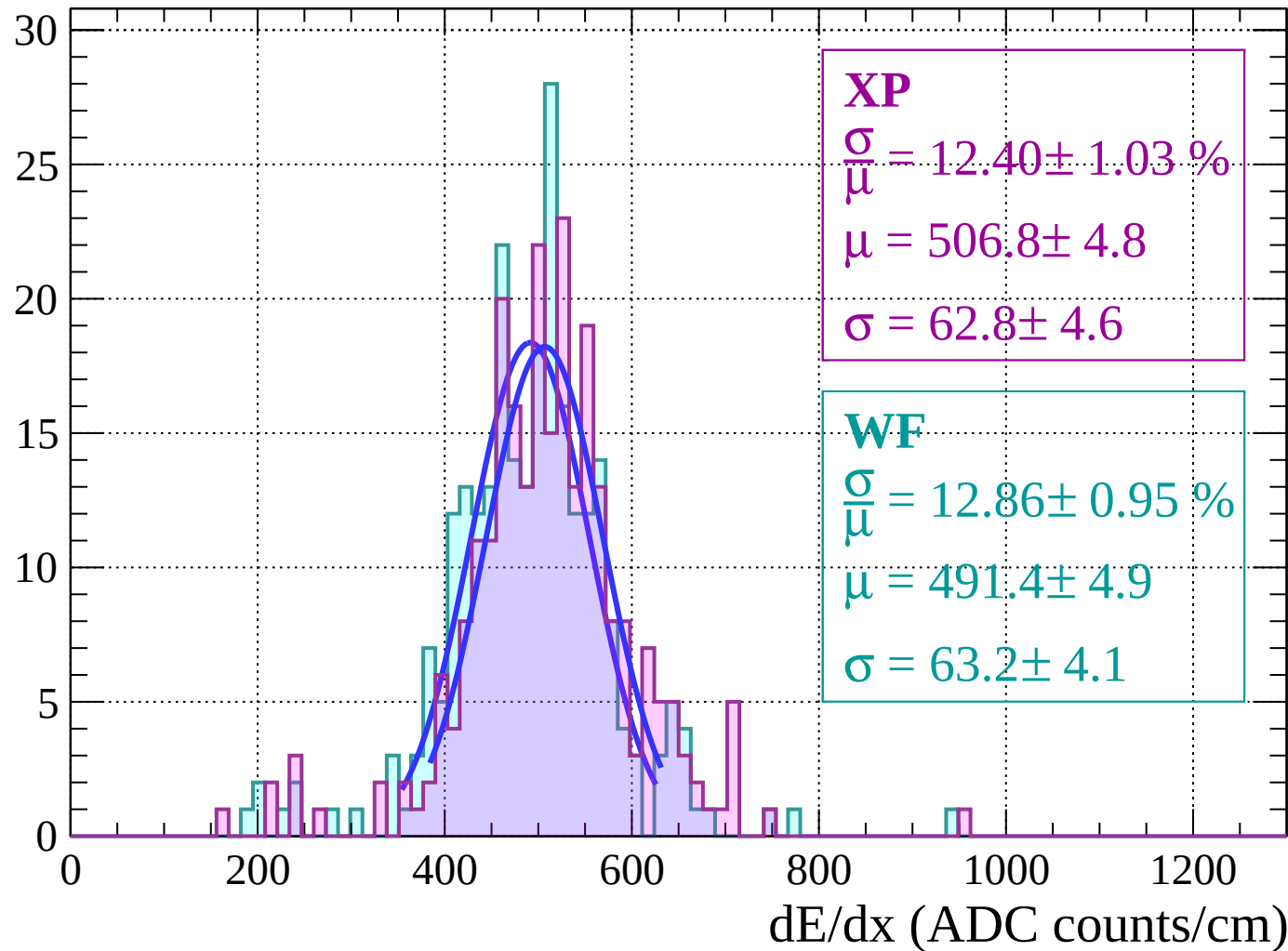
Energy loss | $-64 < \theta < -62$

Count



Energy loss | $-62 < \theta < -60$

Count



Energy loss | $-60 < \theta < -58$

Count

XP

$$\frac{\sigma}{\mu} = 11.81 \pm 0.78 \%$$

$$\mu = 519.1 \pm 4.1$$

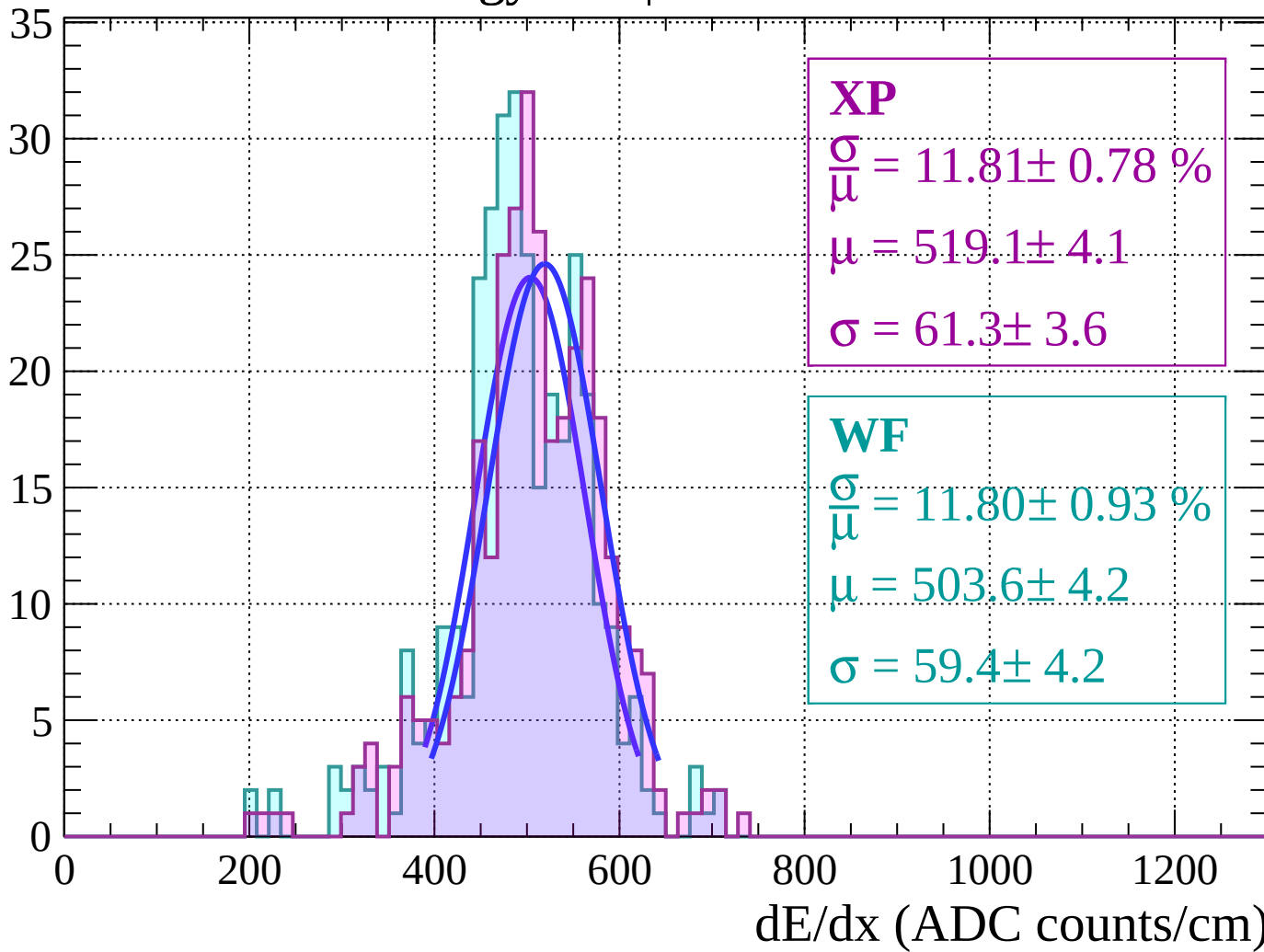
$$\sigma = 61.3 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 11.80 \pm 0.93 \%$$

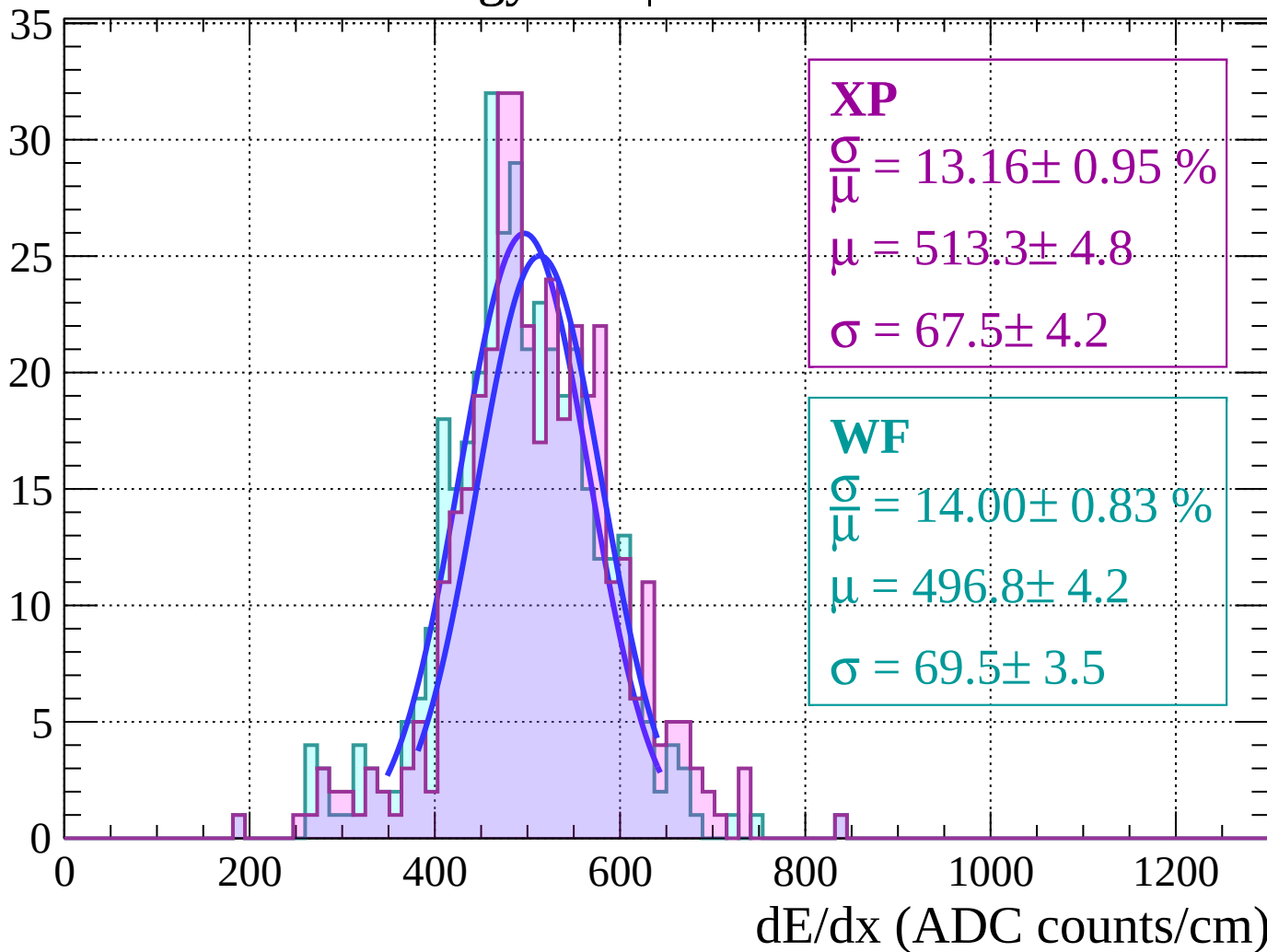
$$\mu = 503.6 \pm 4.2$$

$$\sigma = 59.4 \pm 4.2$$



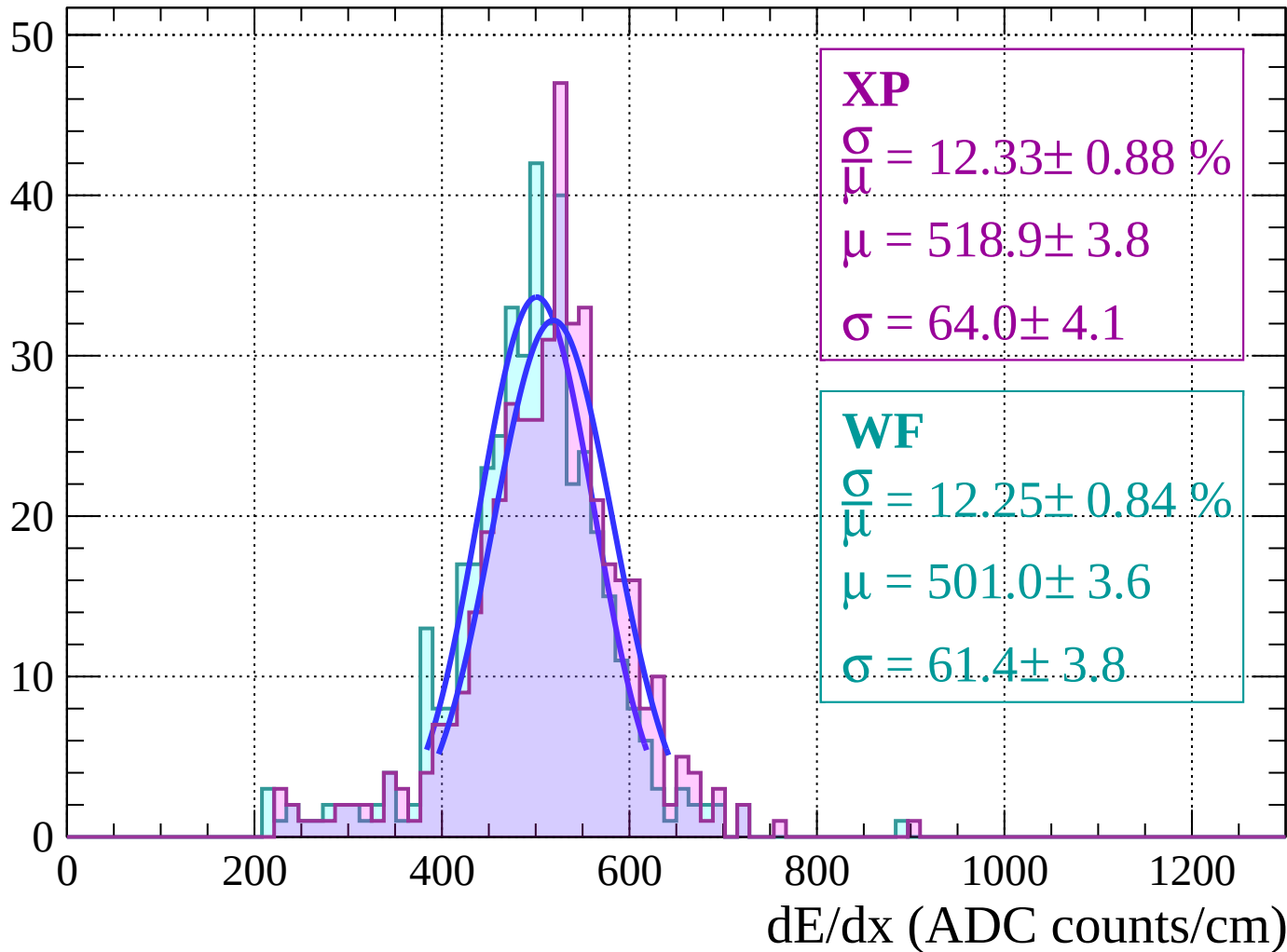
Energy loss $|-58 < \theta < -56$

Count



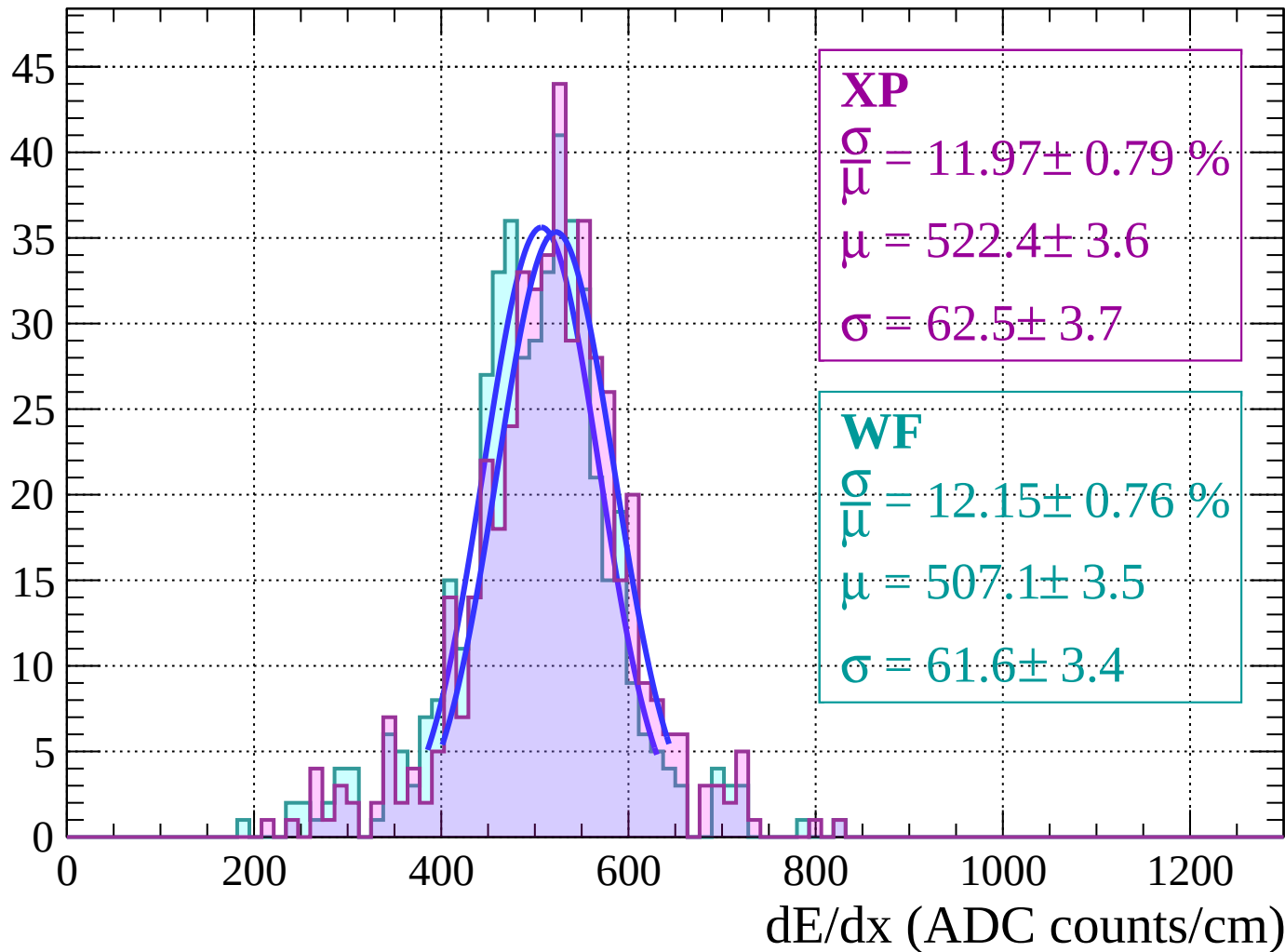
Energy loss $|-56 < \theta < -54$

Count



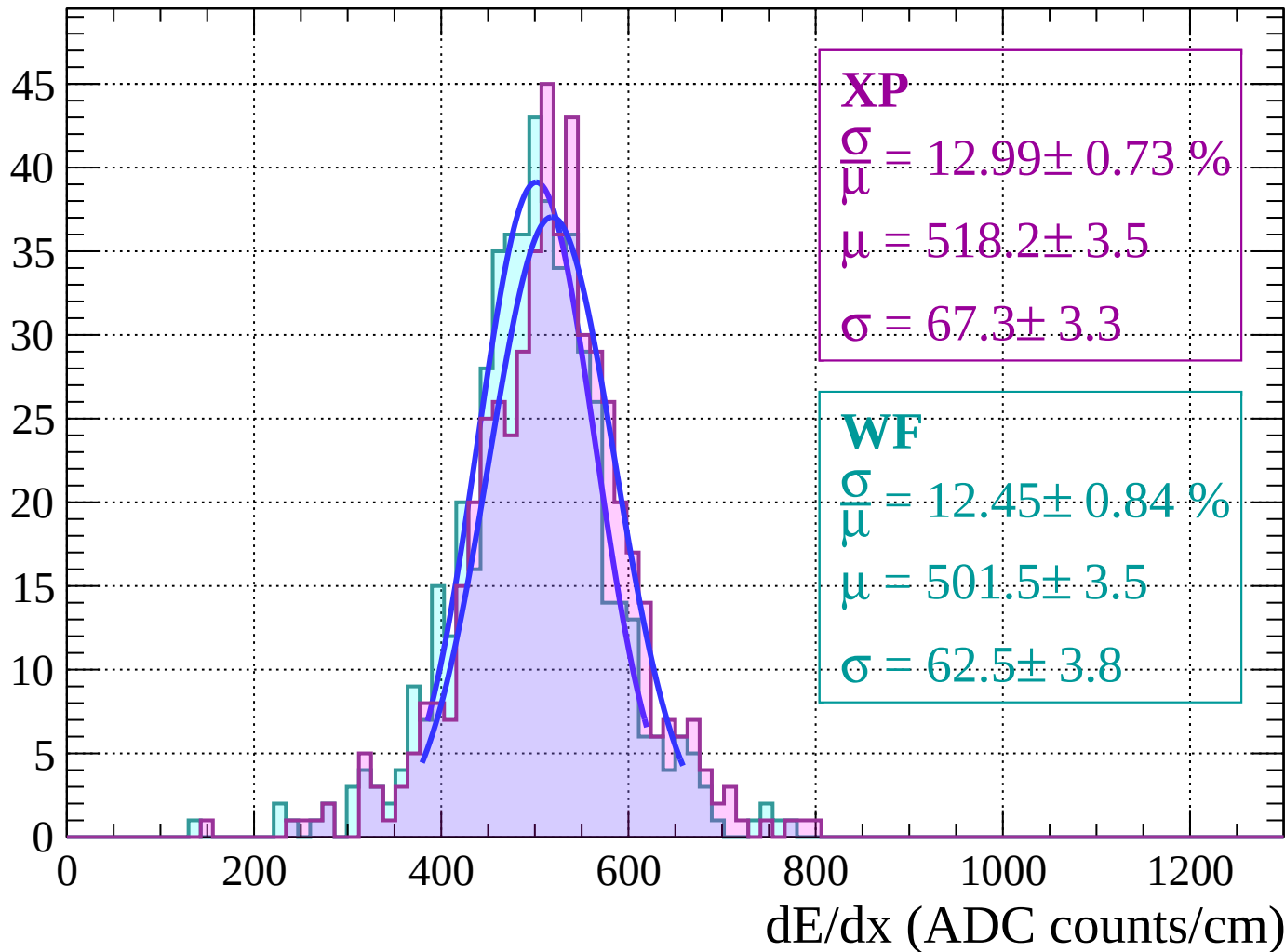
Energy loss | $-54 < \theta < -52$

Count



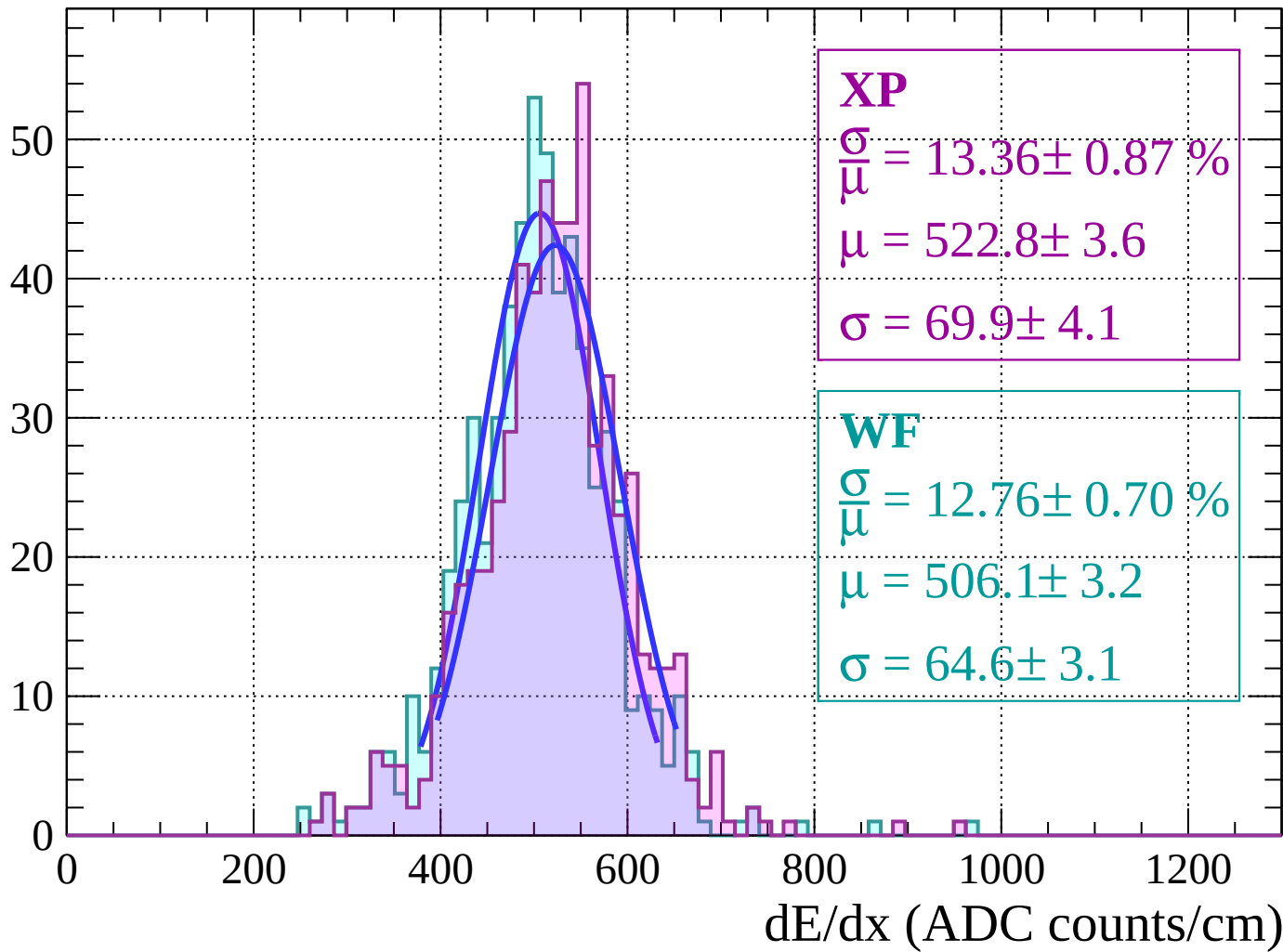
Energy loss | $-52 < \theta < -50$

Count



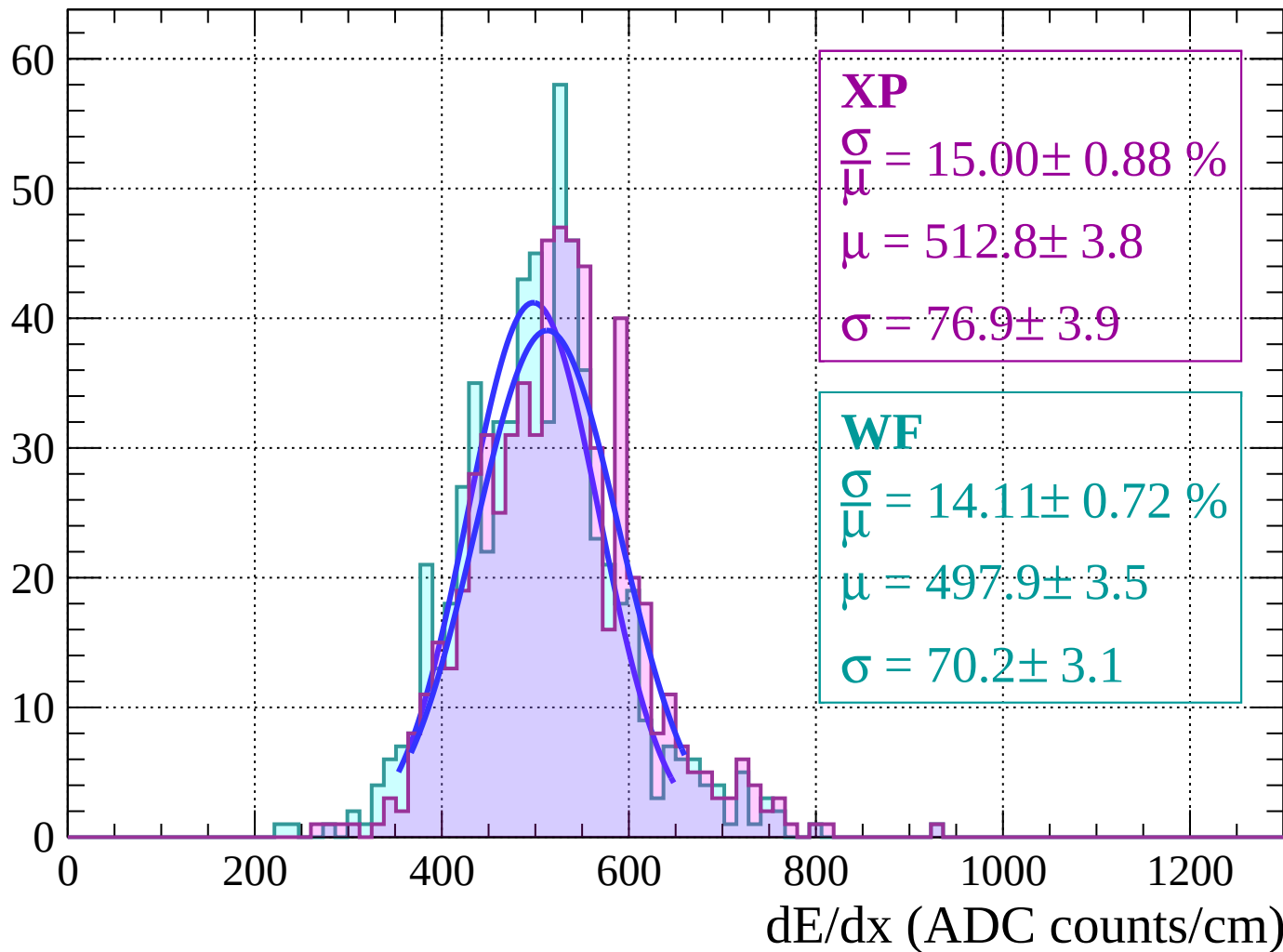
Energy loss | $-50 < \theta < -48$

Count



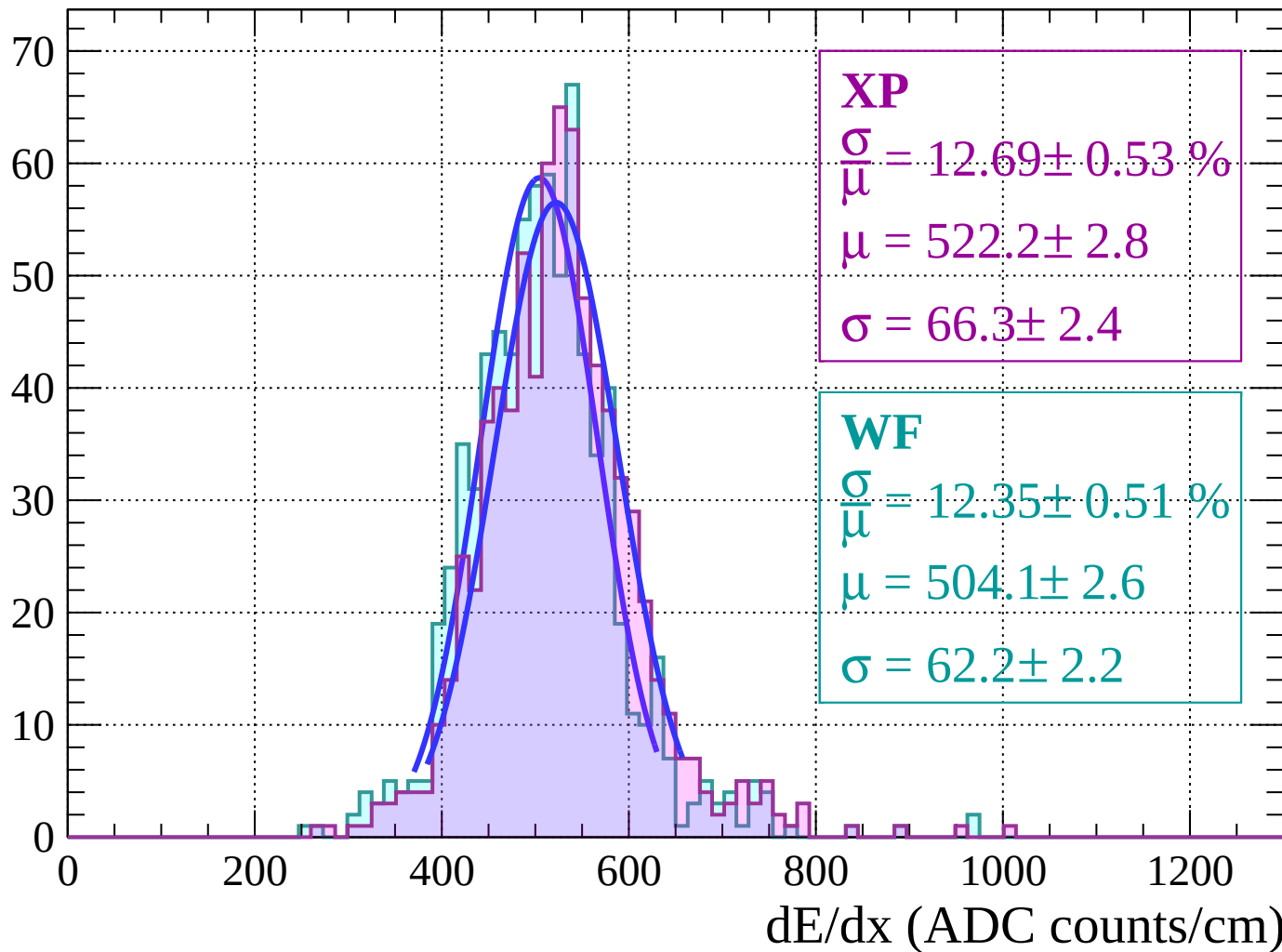
Energy loss $|-48 < \theta < -46$

Count



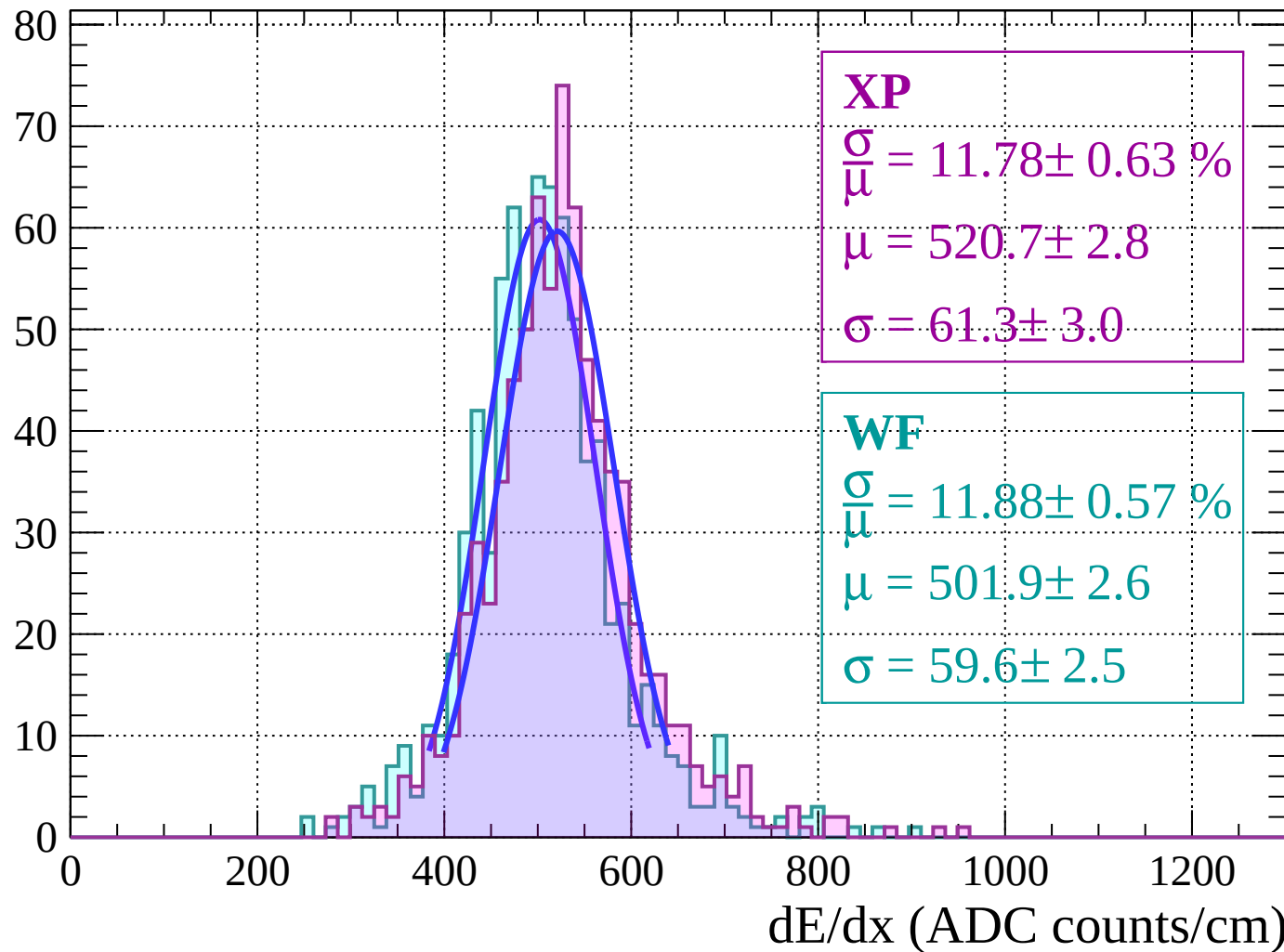
Energy loss | $-46 < \theta < -44$

Count



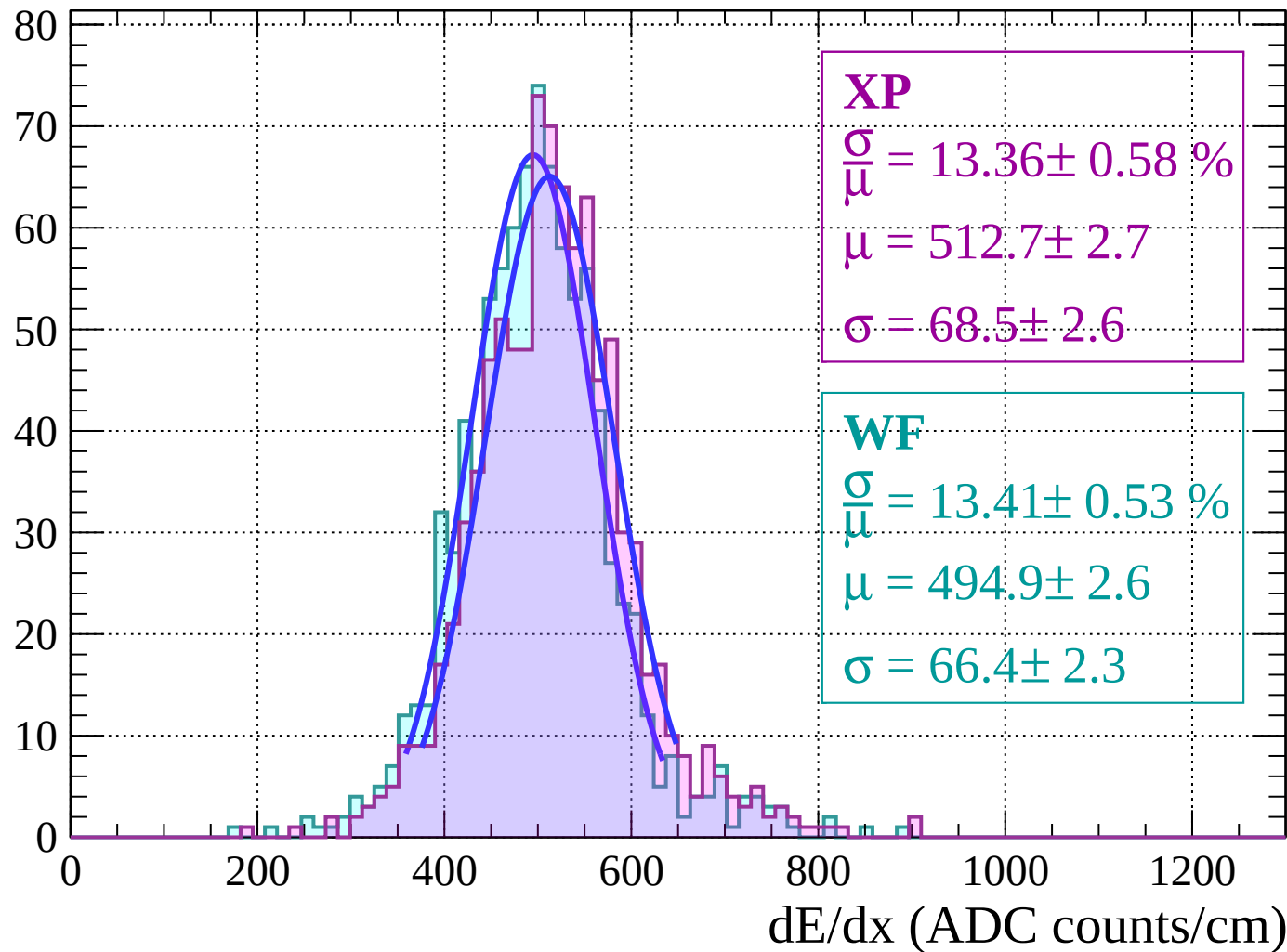
Energy loss | $-44 < \theta < -42$

Count



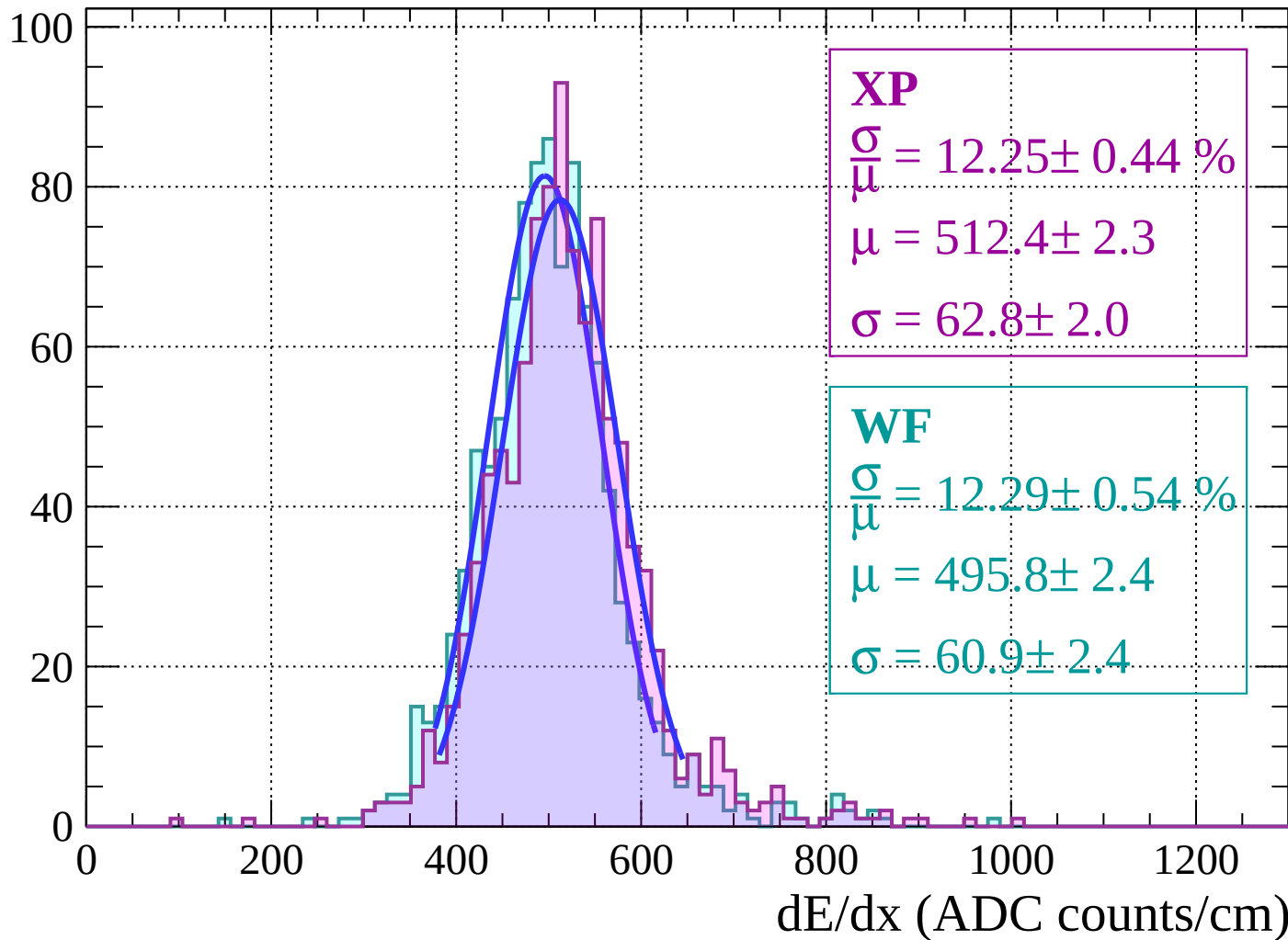
Energy loss | $-42 < \theta < -40$

Count



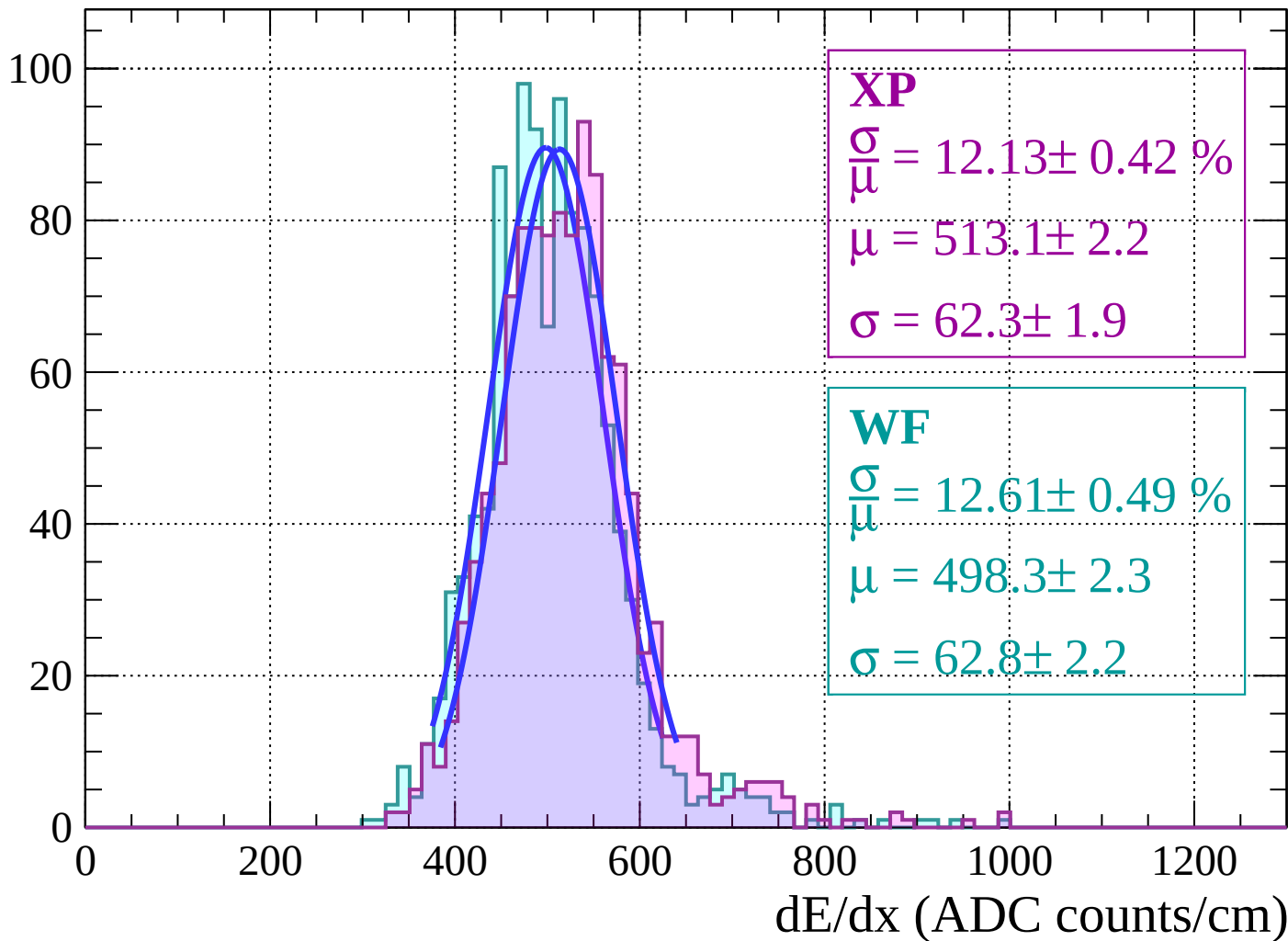
Energy loss | $-40 < \theta < -38$

Count



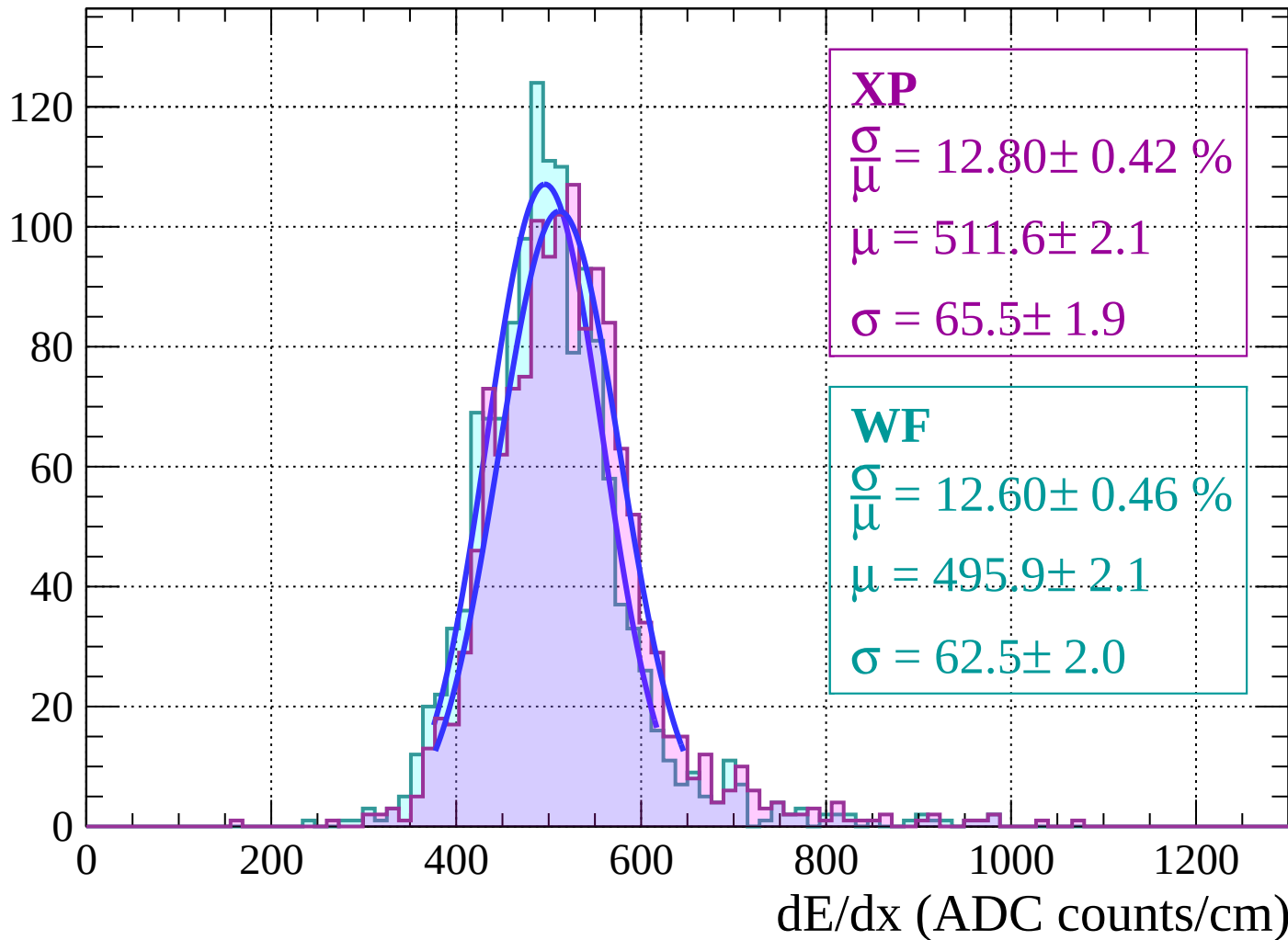
Energy loss $|-38 < \theta < -36$

Count



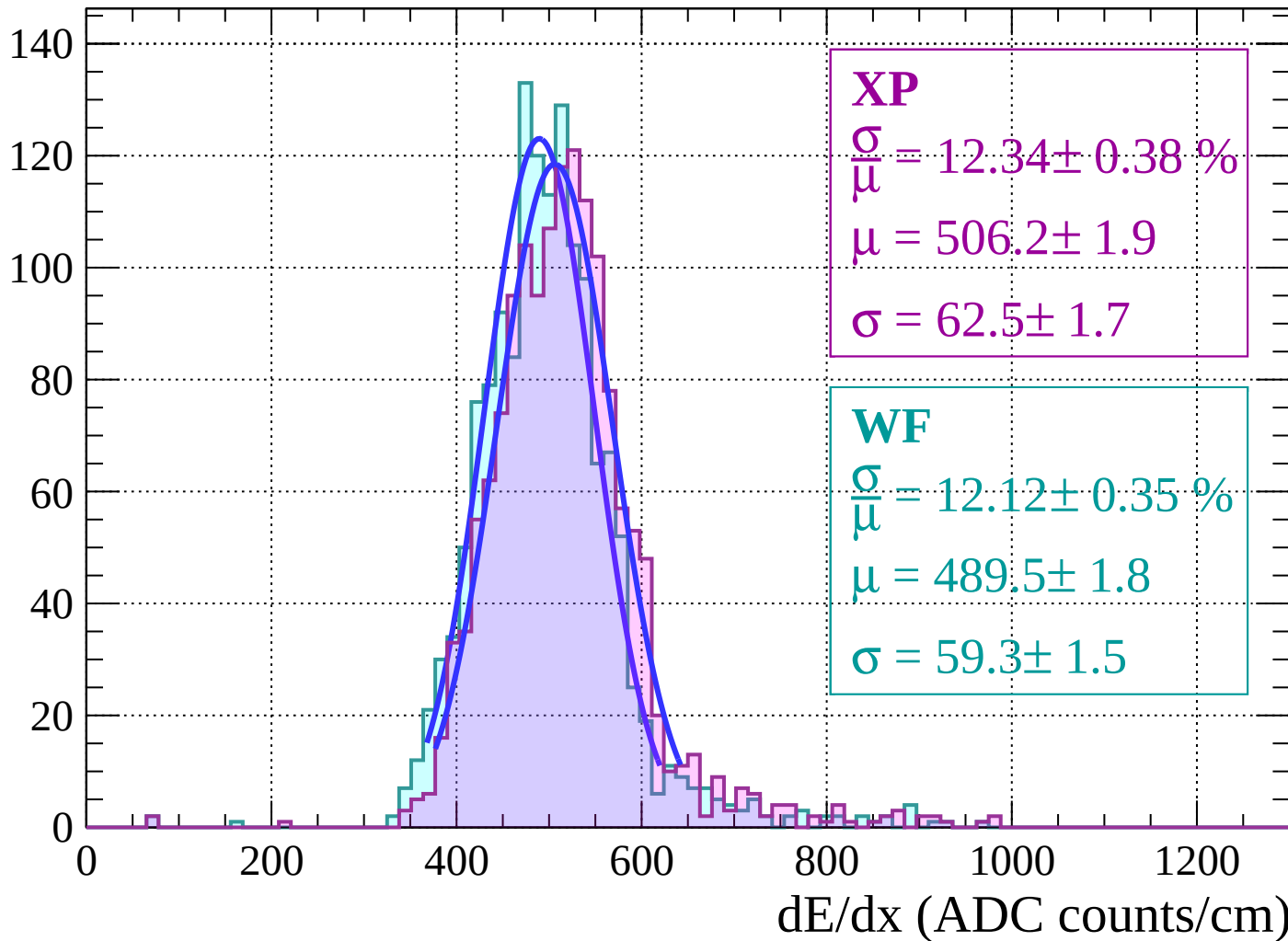
Energy loss $|-36 < \theta < -34$

Count



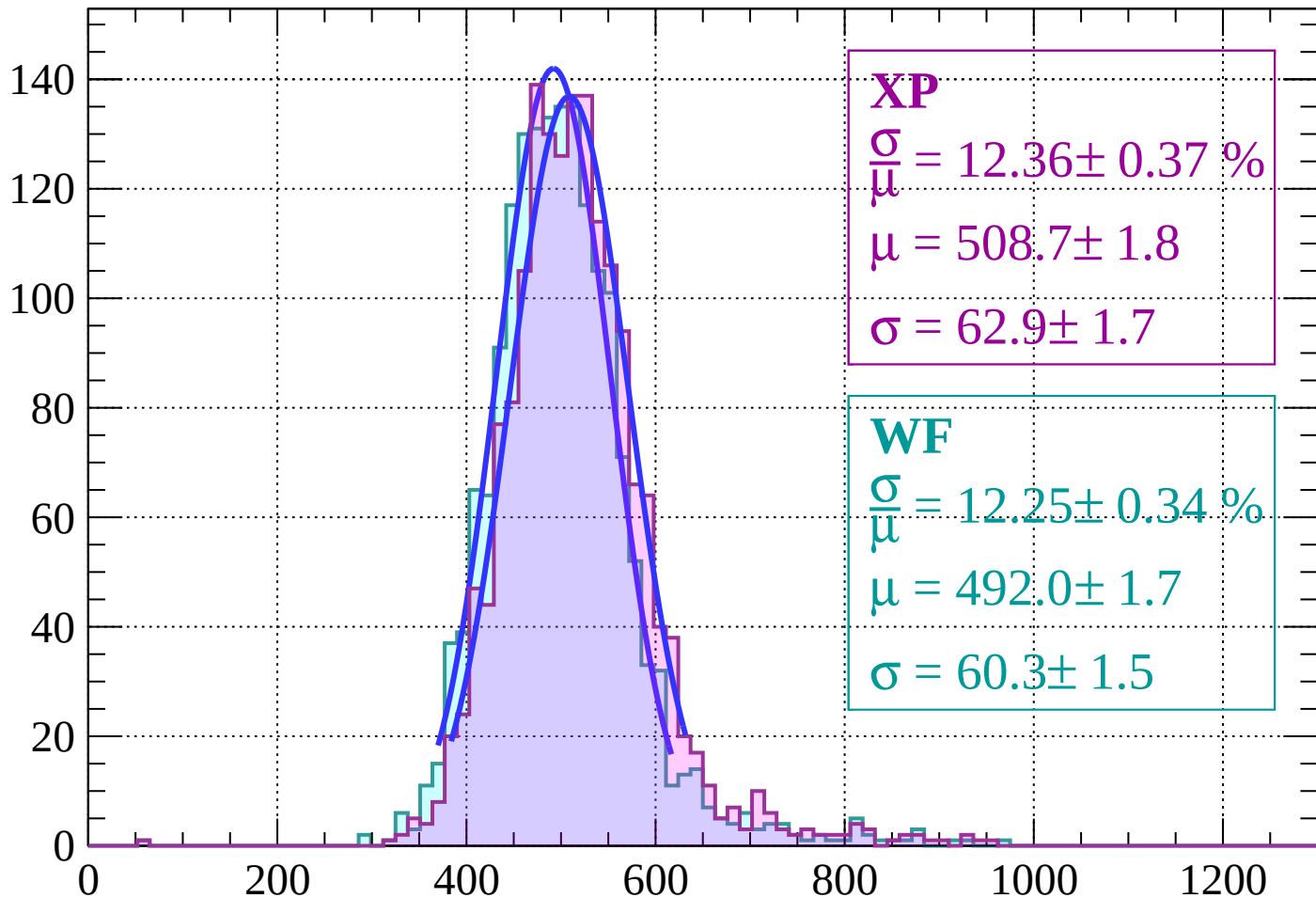
Energy loss $|-34 < \theta < -32$

Count



Energy loss $|-32 < \theta < -30$

Count



XP

$$\frac{\sigma}{\mu} = 12.36 \pm 0.37 \%$$

$$\mu = 508.7 \pm 1.8$$

$$\sigma = 62.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 12.25 \pm 0.34 \%$$

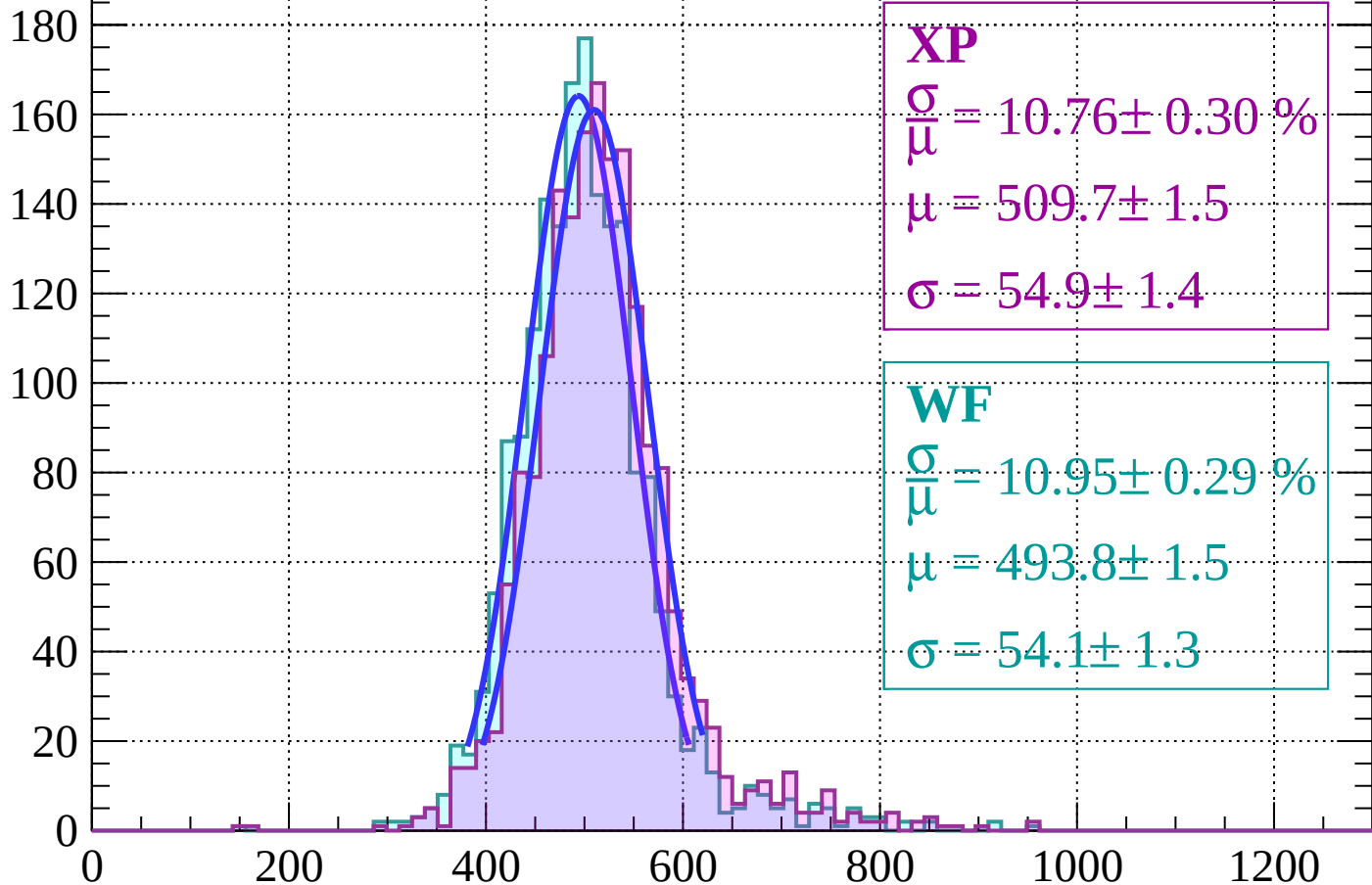
$$\mu = 492.0 \pm 1.7$$

$$\sigma = 60.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -28$

Count



XP

$$\frac{\sigma}{\mu} = 10.76 \pm 0.30 \%$$

$$\mu = 509.7 \pm 1.5$$

$$\sigma = 54.9 \pm 1.4$$

WF

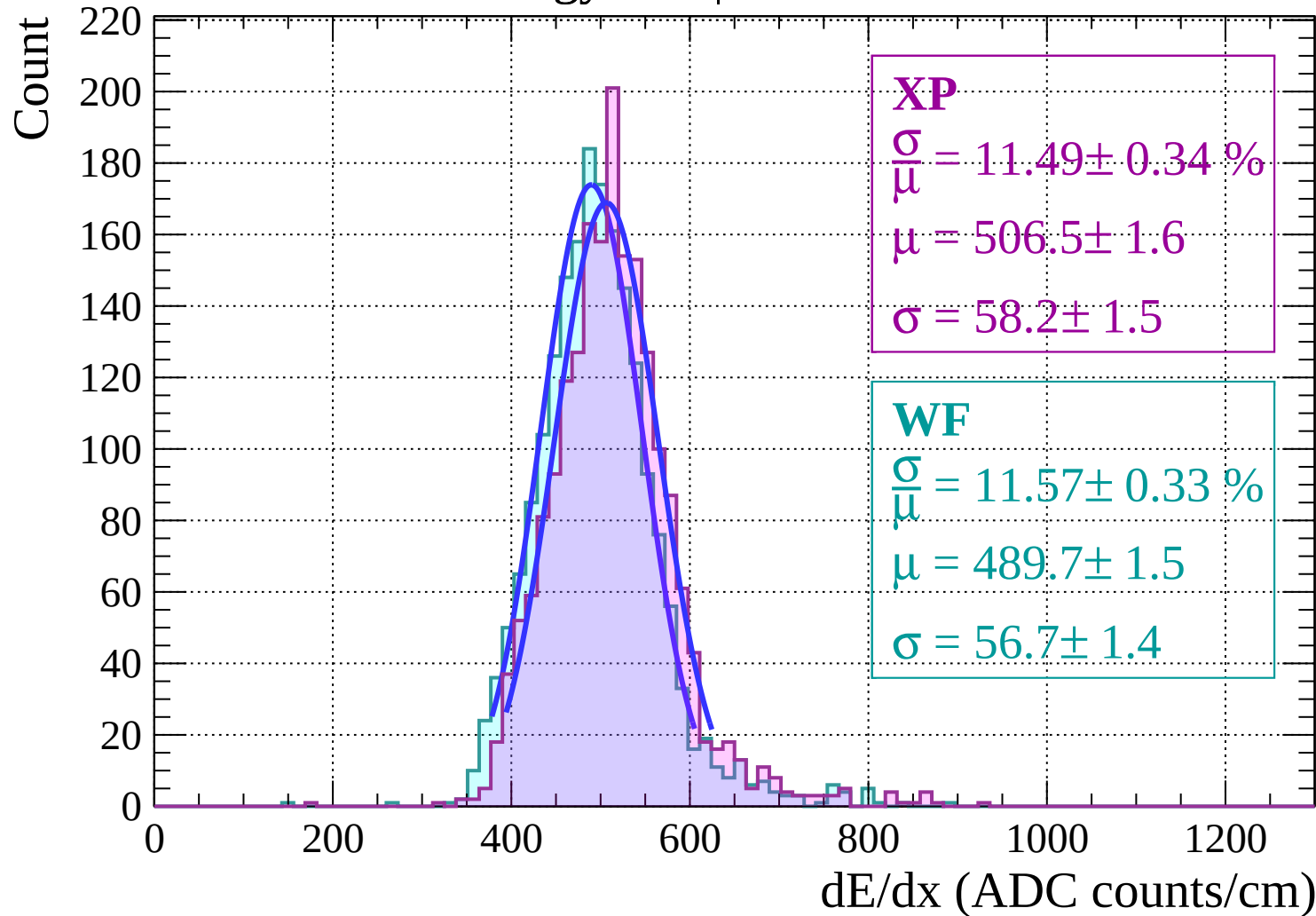
$$\frac{\sigma}{\mu} = 10.95 \pm 0.29 \%$$

$$\mu = 493.8 \pm 1.5$$

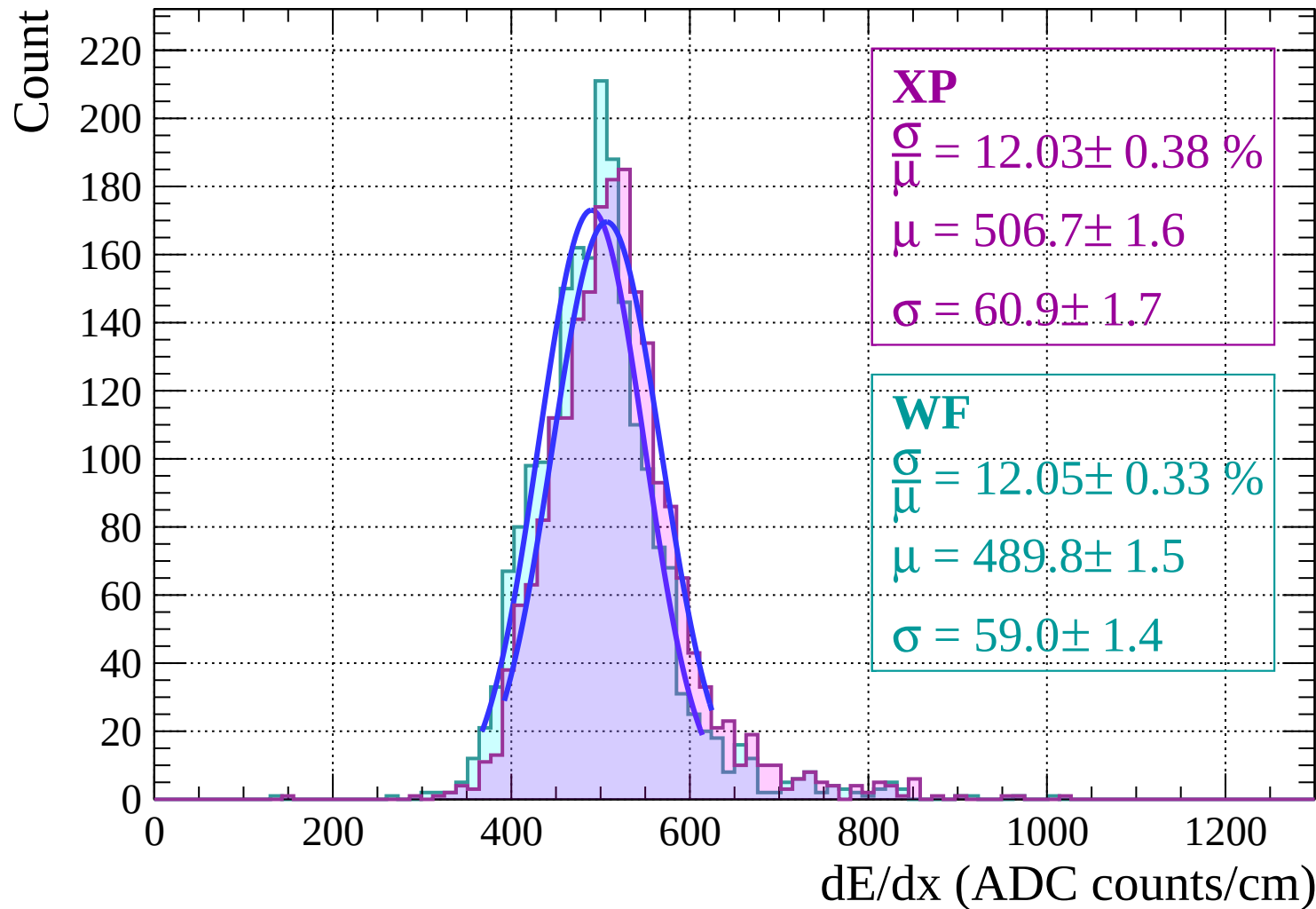
$$\sigma = 54.1 \pm 1.3$$

dE/dx (ADC counts/cm)

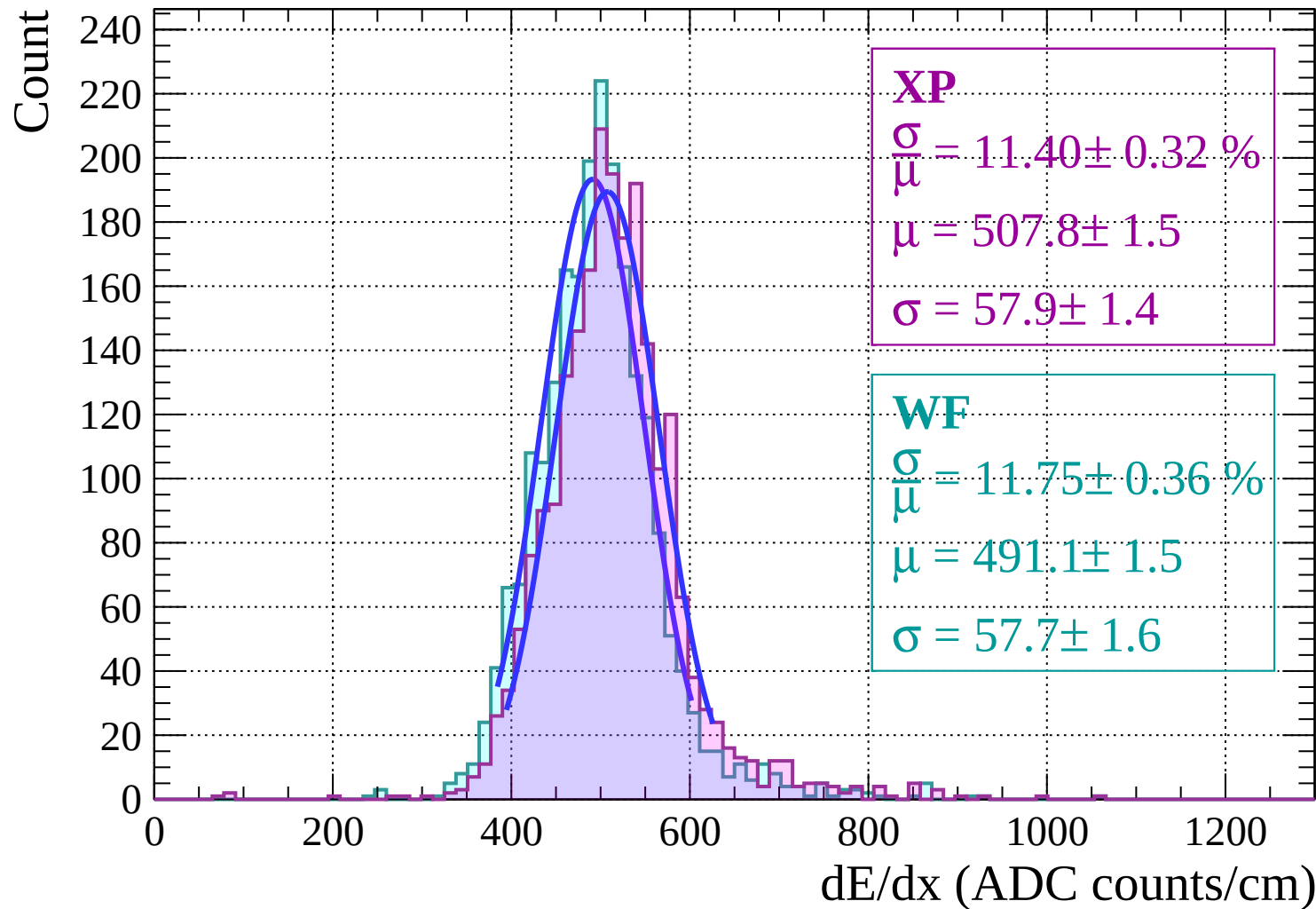
Energy loss | $-28 < \theta < -26$



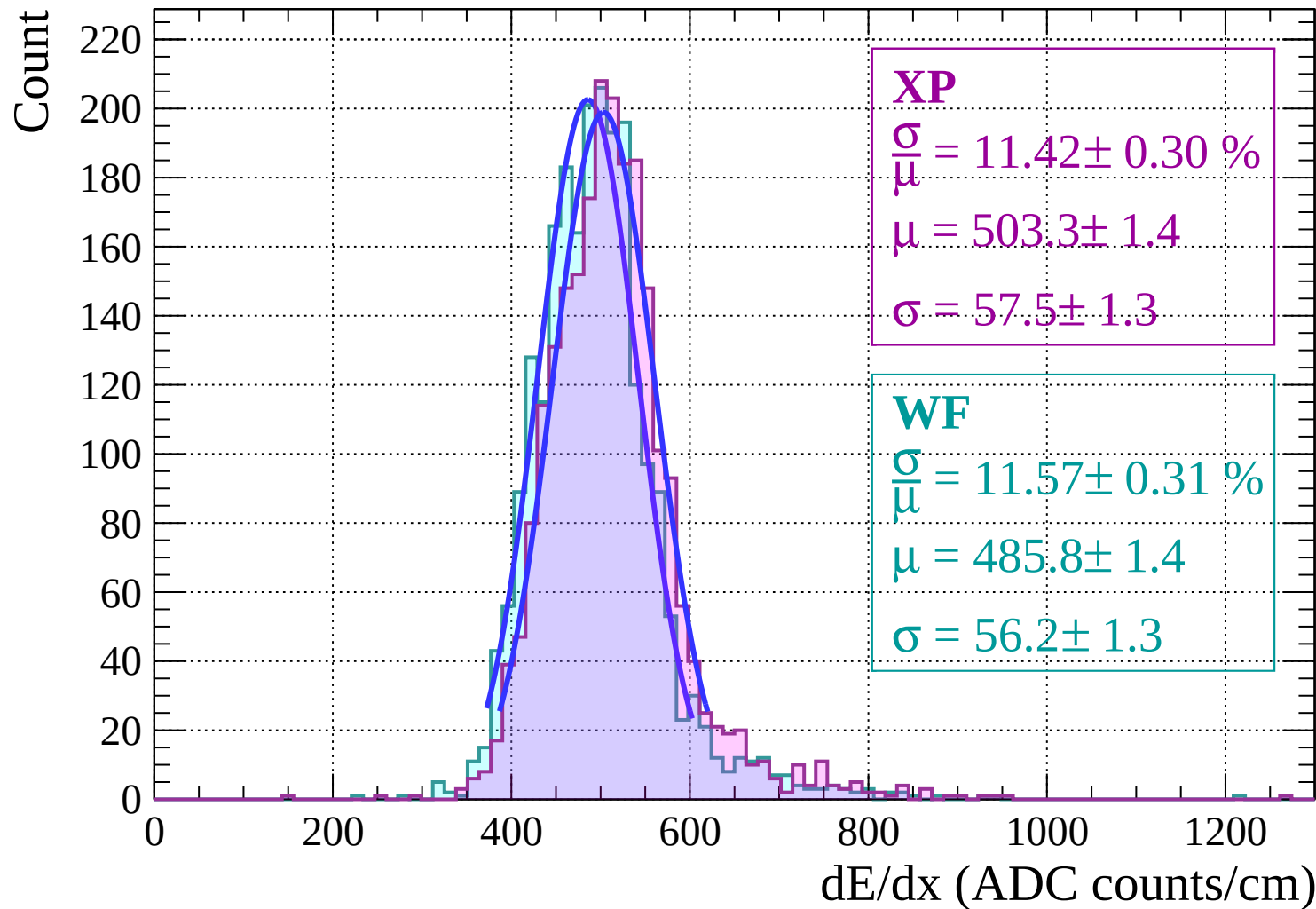
Energy loss | $-26 < \theta < -24$



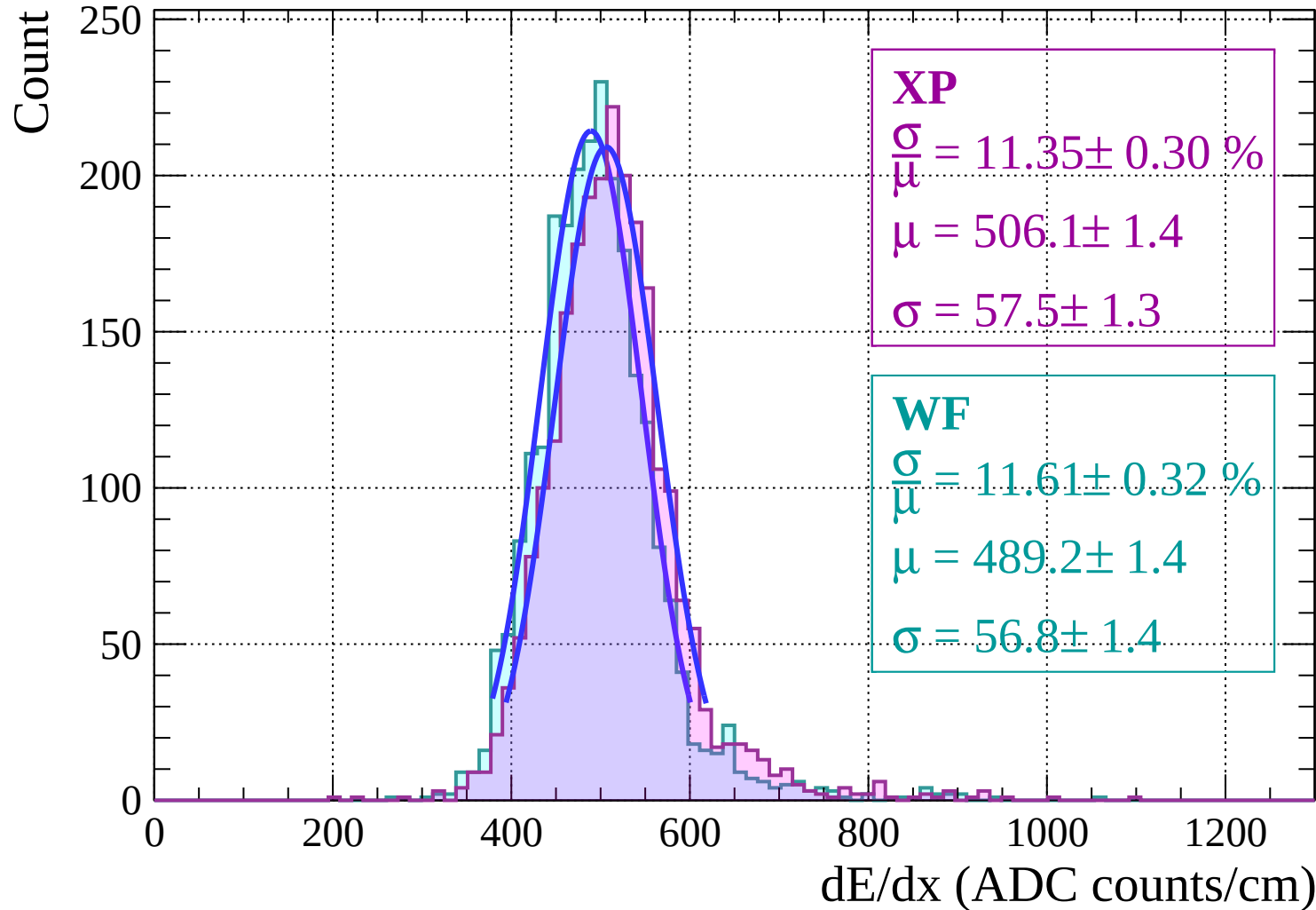
Energy loss | $-24 < \theta < -22$



Energy loss | $-22 < \theta < -20$

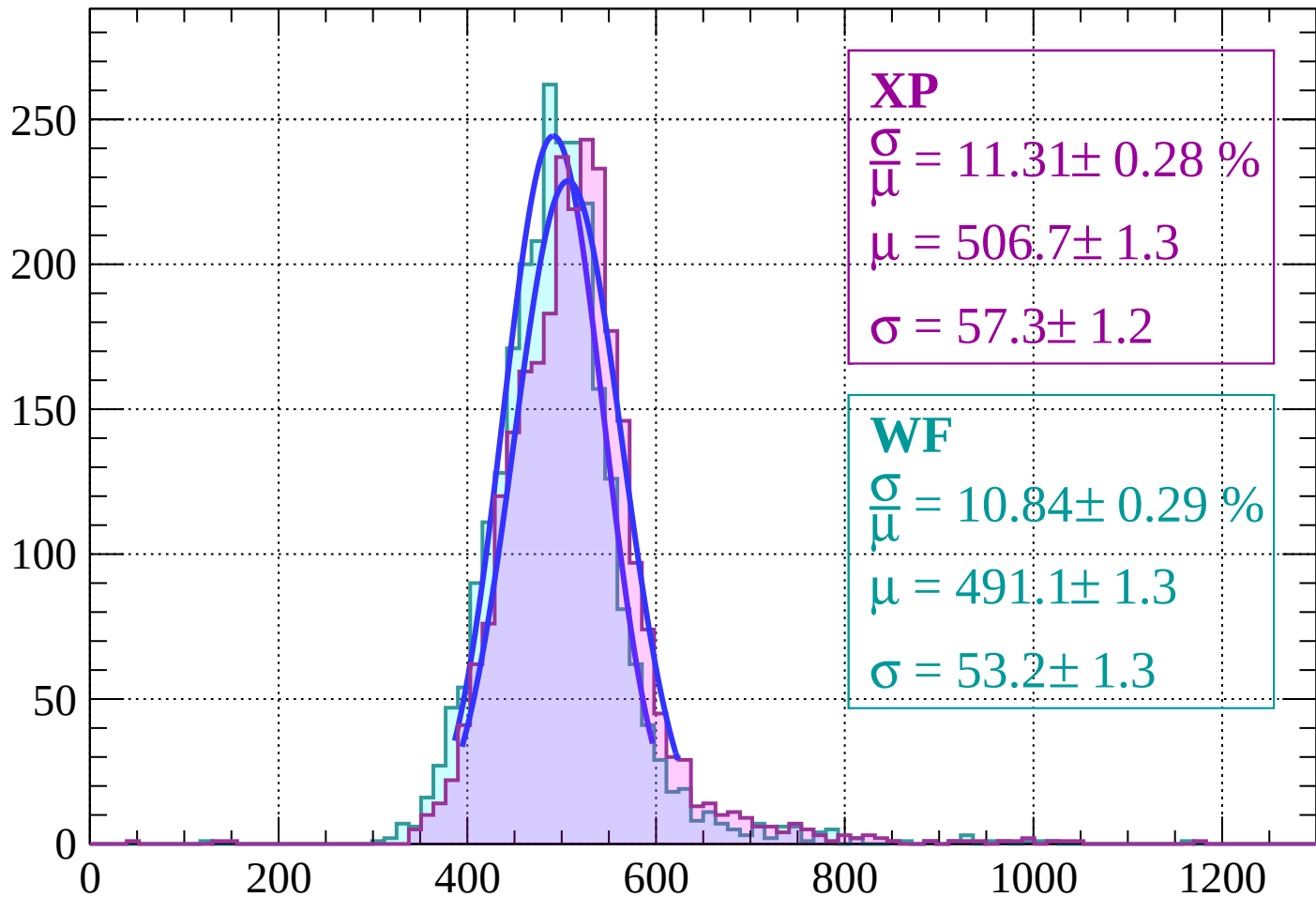


Energy loss | $-20 < \theta < -18$



Energy loss | $-18 < \theta < -16$

Count



XP

$$\frac{\sigma}{\mu} = 11.31 \pm 0.28 \%$$

$$\mu = 506.7 \pm 1.3$$

$$\sigma = 57.3 \pm 1.2$$

WF

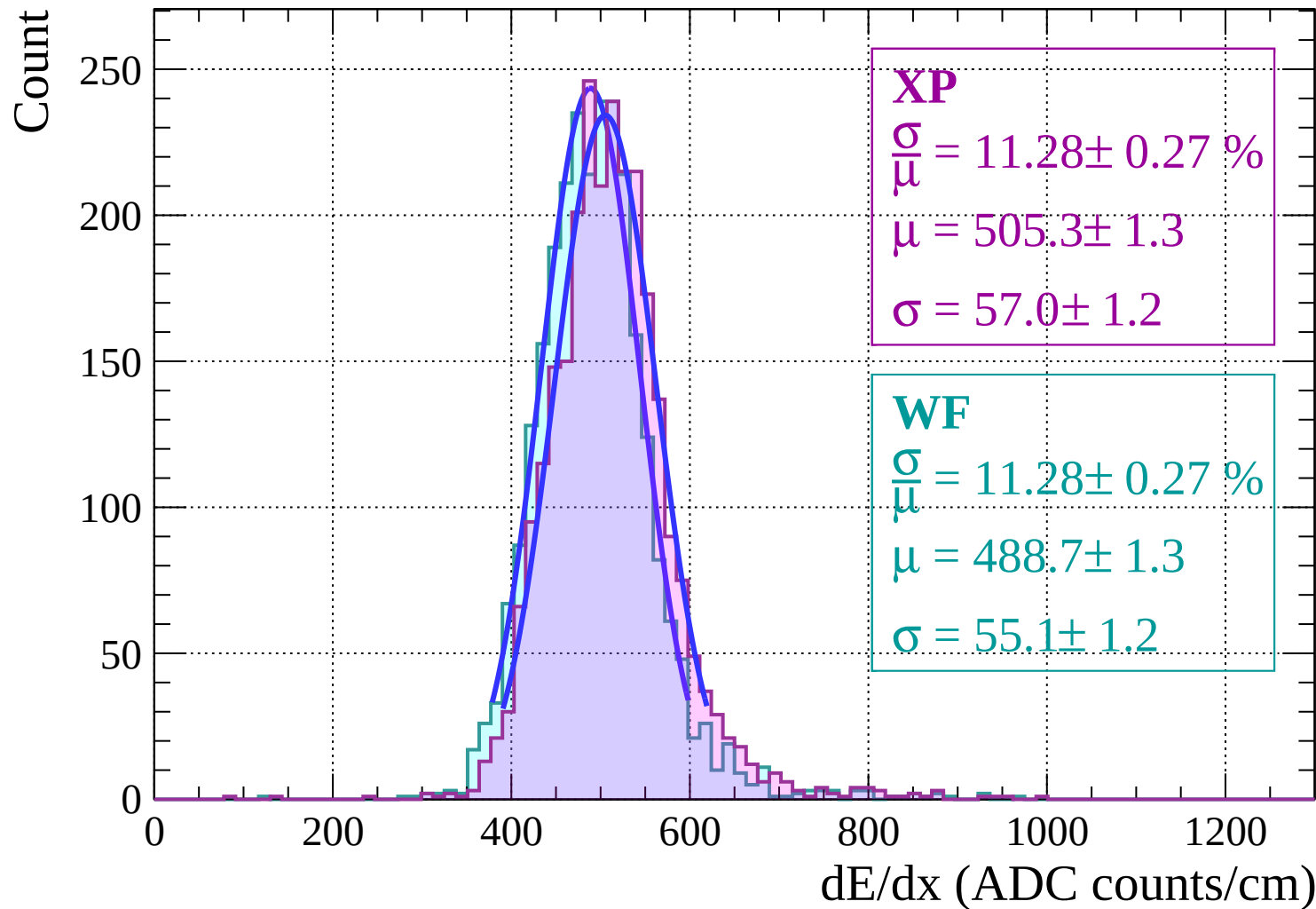
$$\frac{\sigma}{\mu} = 10.84 \pm 0.29 \%$$

$$\mu = 491.1 \pm 1.3$$

$$\sigma = 53.2 \pm 1.3$$

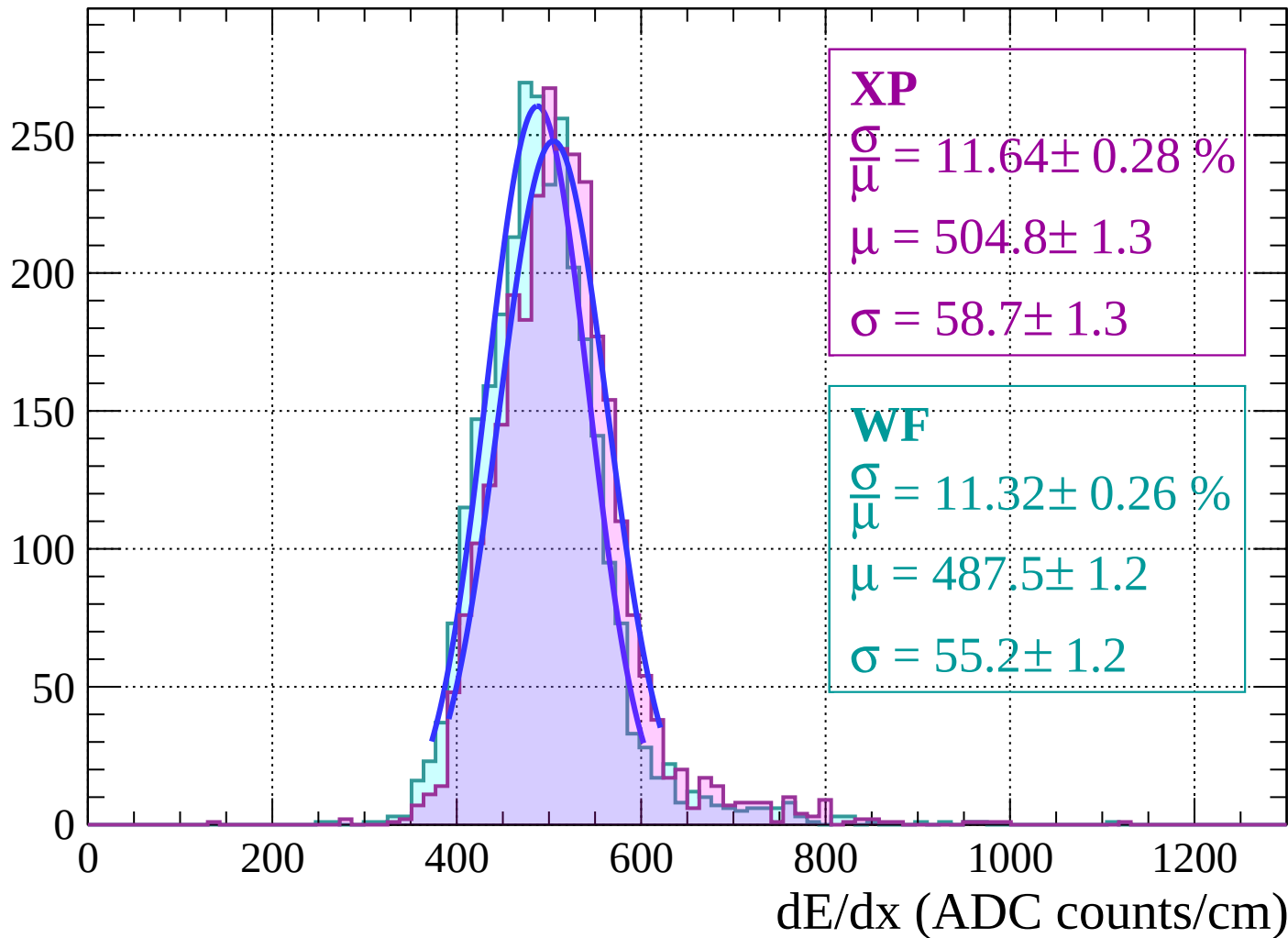
dE/dx (ADC counts/cm)

Energy loss $|-16 < \theta < -14$



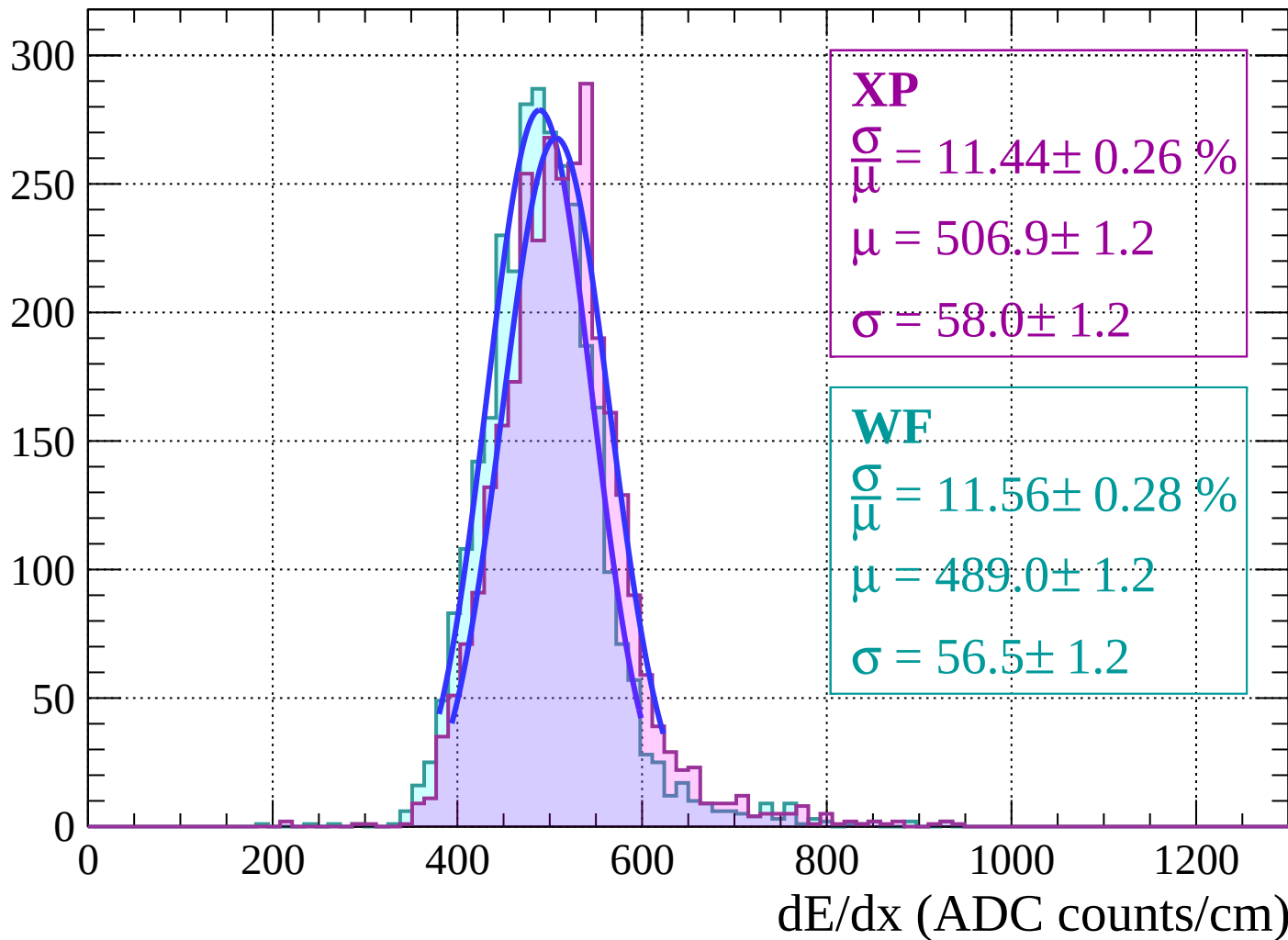
Energy loss $|-14 < \theta < -12$

Count



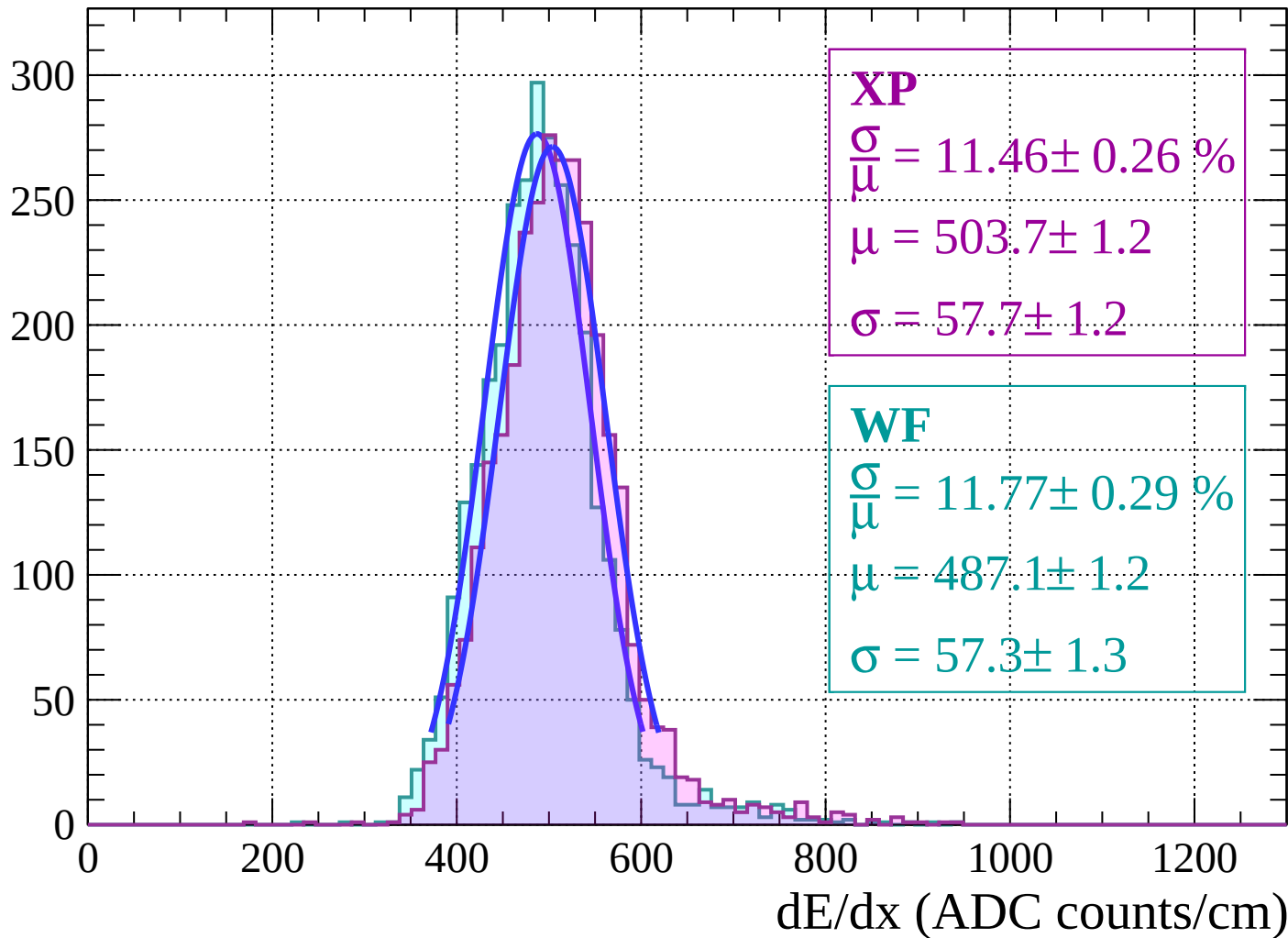
Energy loss $|-12 < \theta < -10$

Count



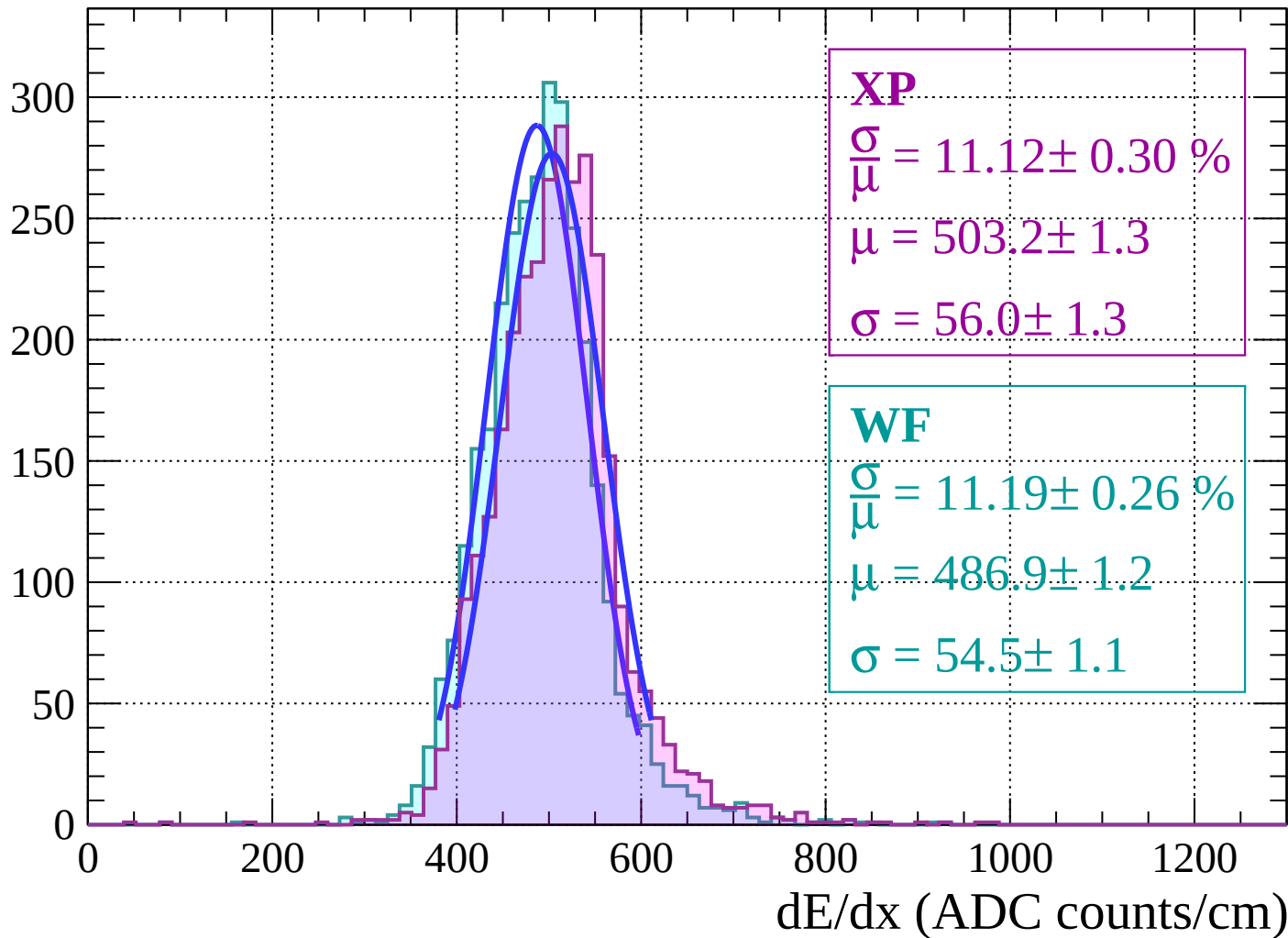
Energy loss | $-10 < \theta < -8$

Count

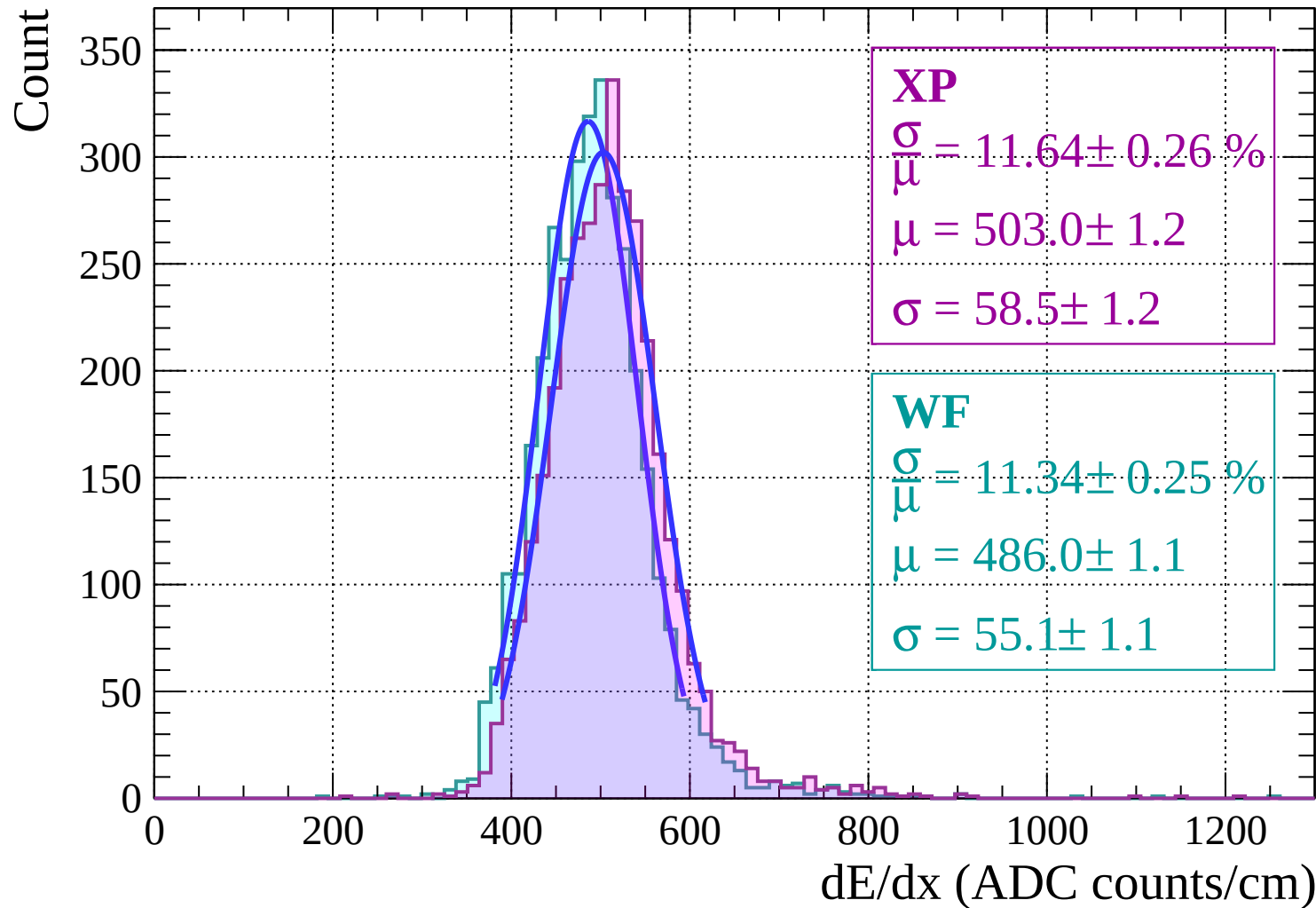


Energy loss $|-8 < \theta < -6$

Count

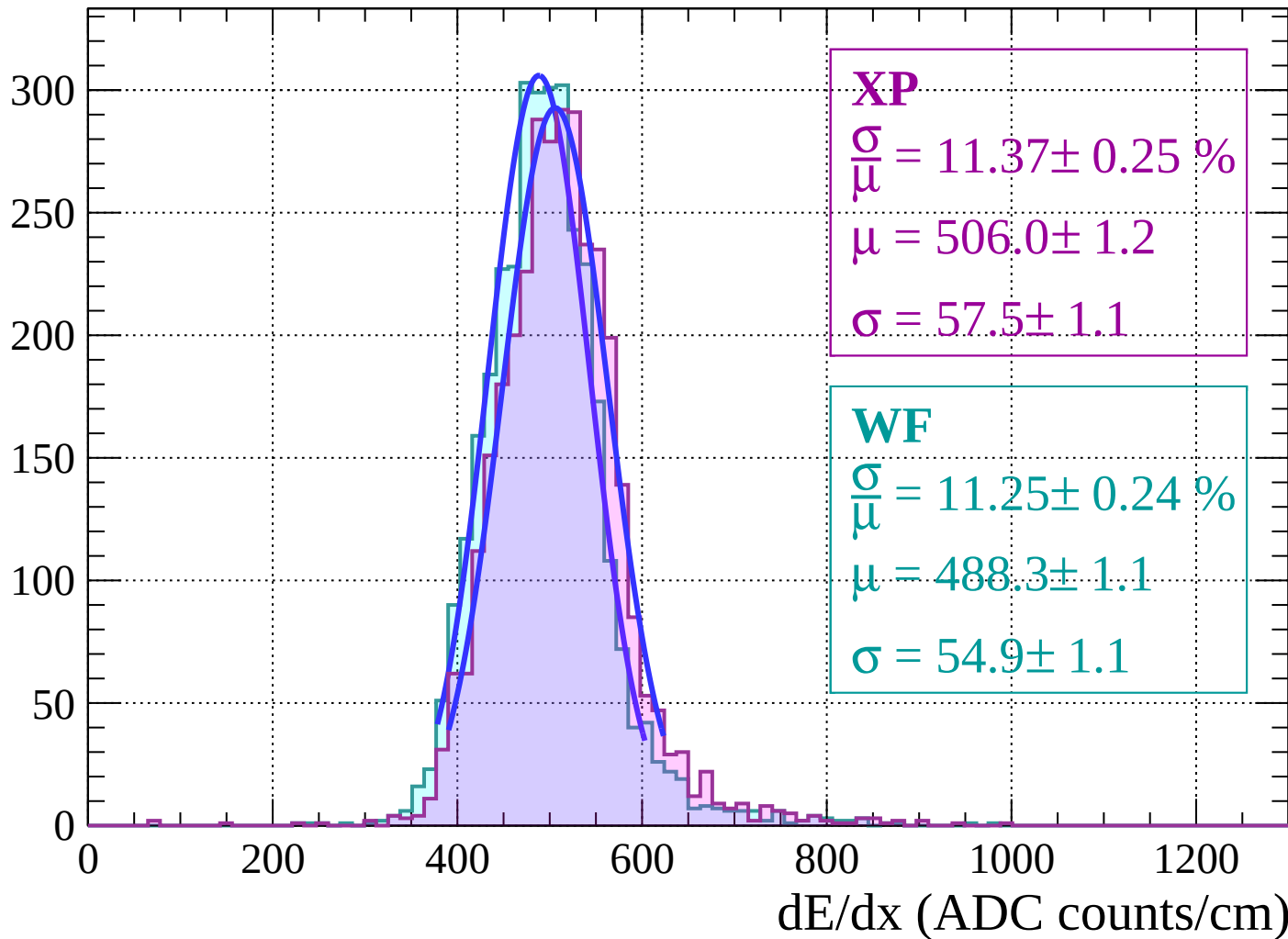


Energy loss $|-6 < \theta < -4|$



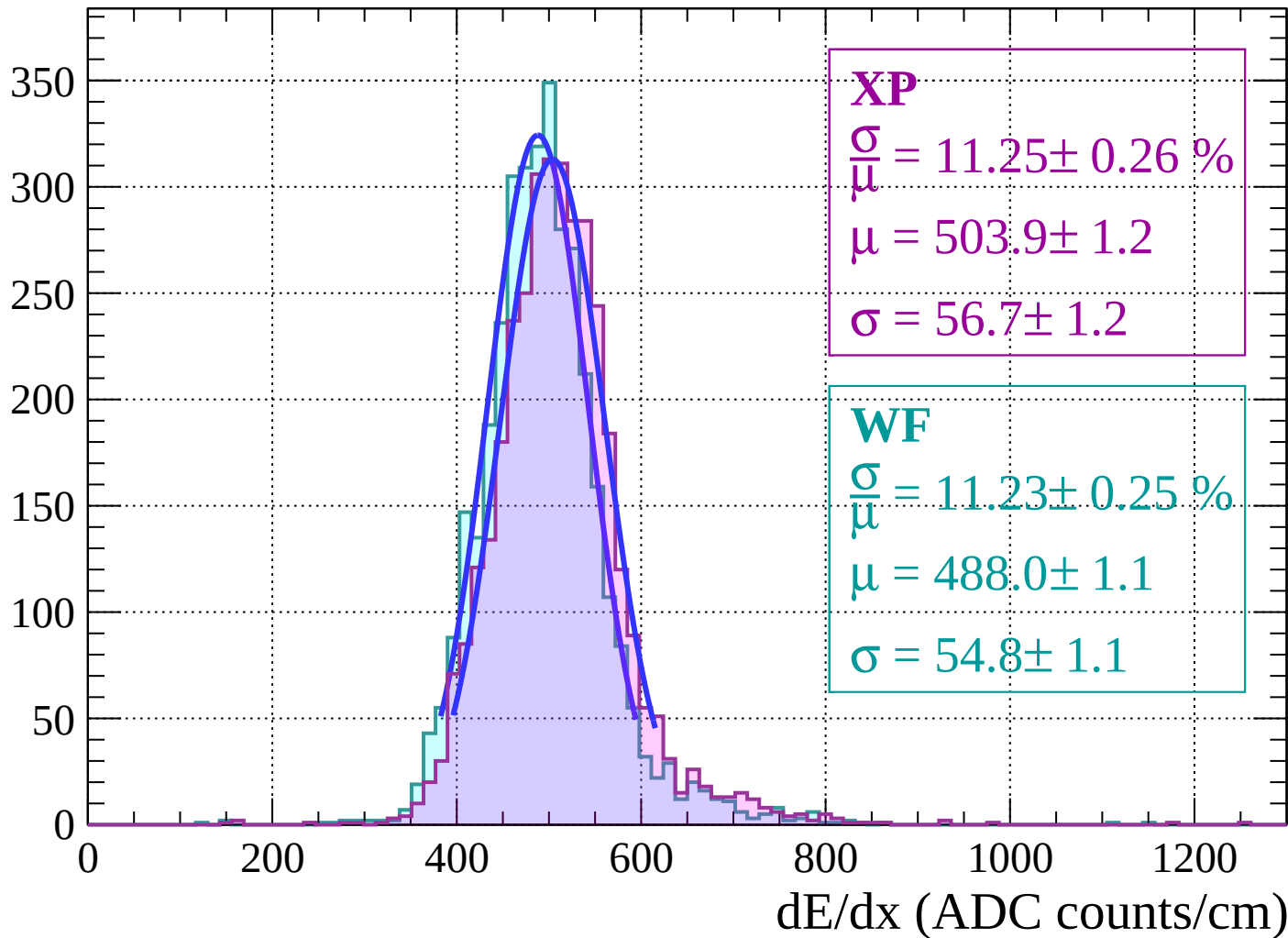
Energy loss $|-4 < \theta < -2$

Count

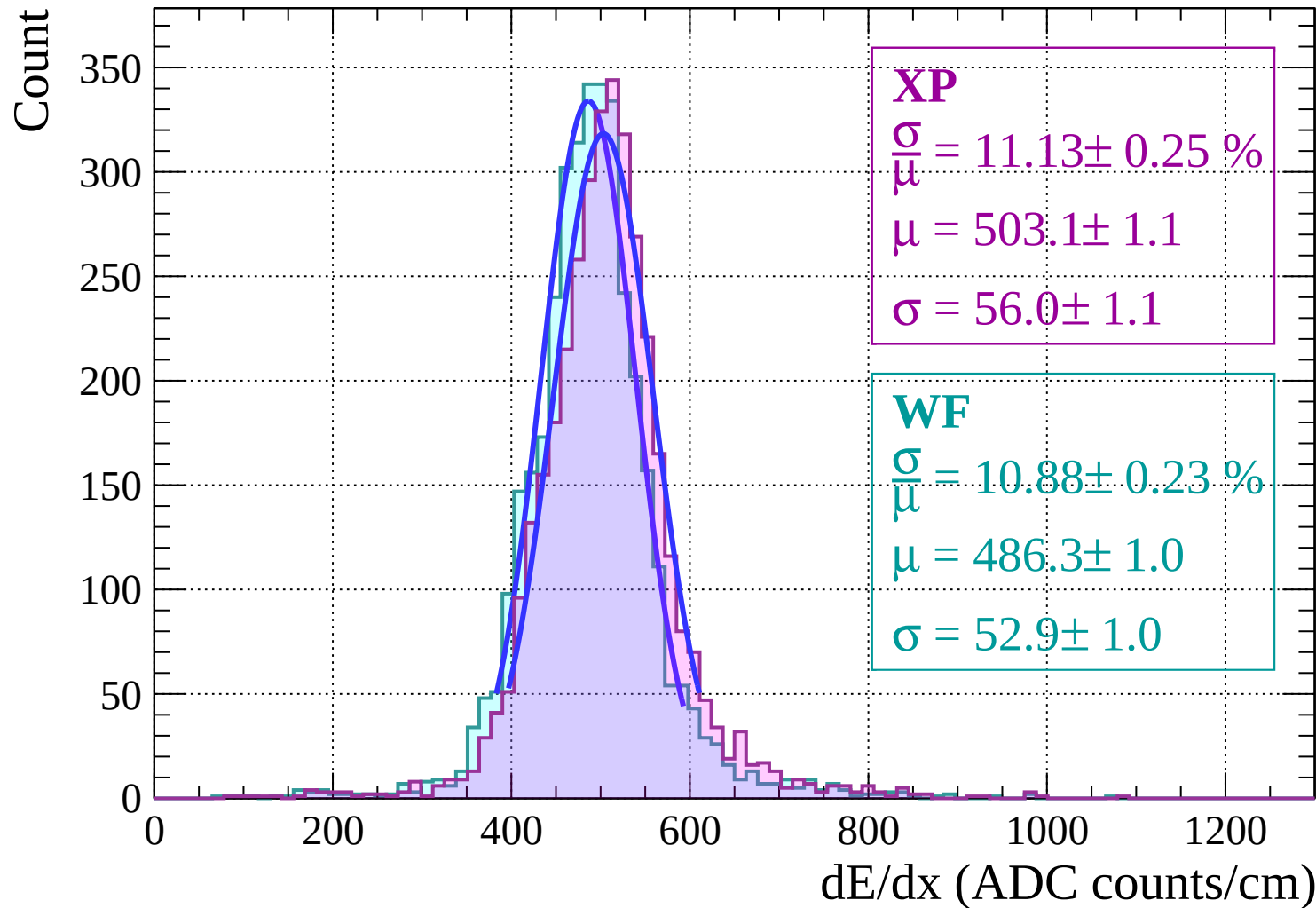


Energy loss | $-2 < \theta < 0$

Count



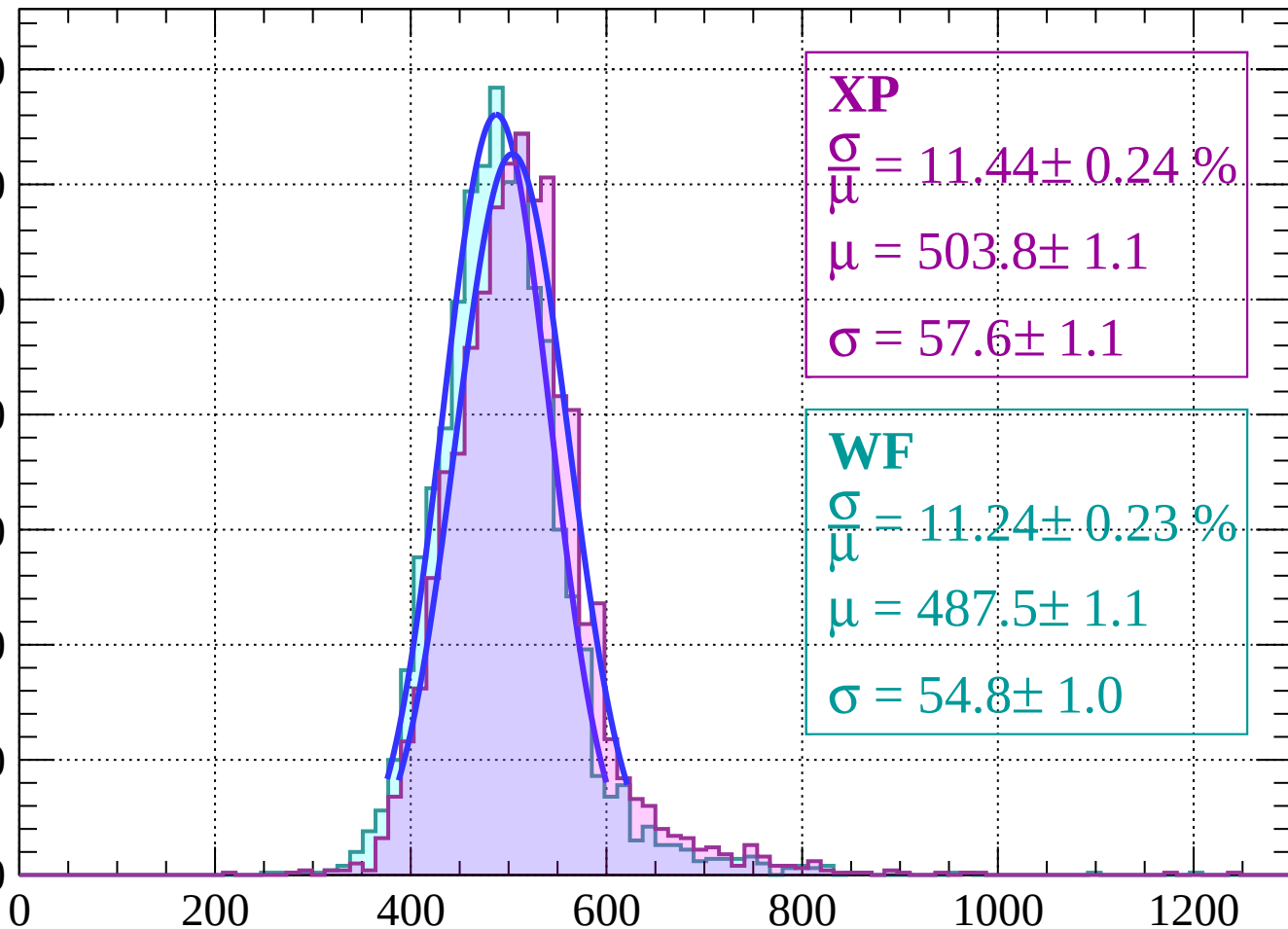
Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.44 \pm 0.24 \%$$

$$\mu = 503.8 \pm 1.1$$

$$\sigma = 57.6 \pm 1.1$$

WF

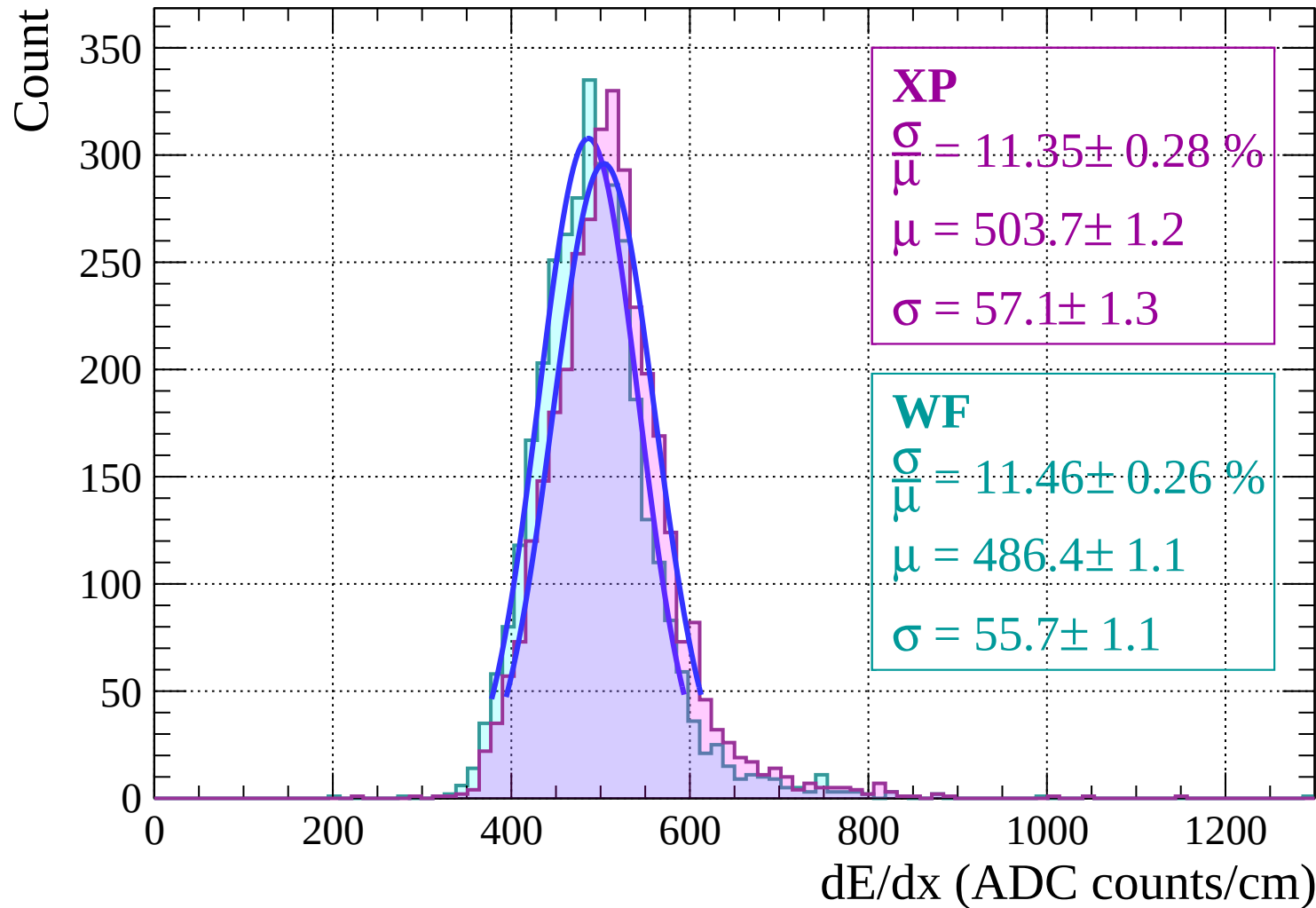
$$\frac{\sigma}{\mu} = 11.24 \pm 0.23 \%$$

$$\mu = 487.5 \pm 1.1$$

$$\sigma = 54.8 \pm 1.0$$

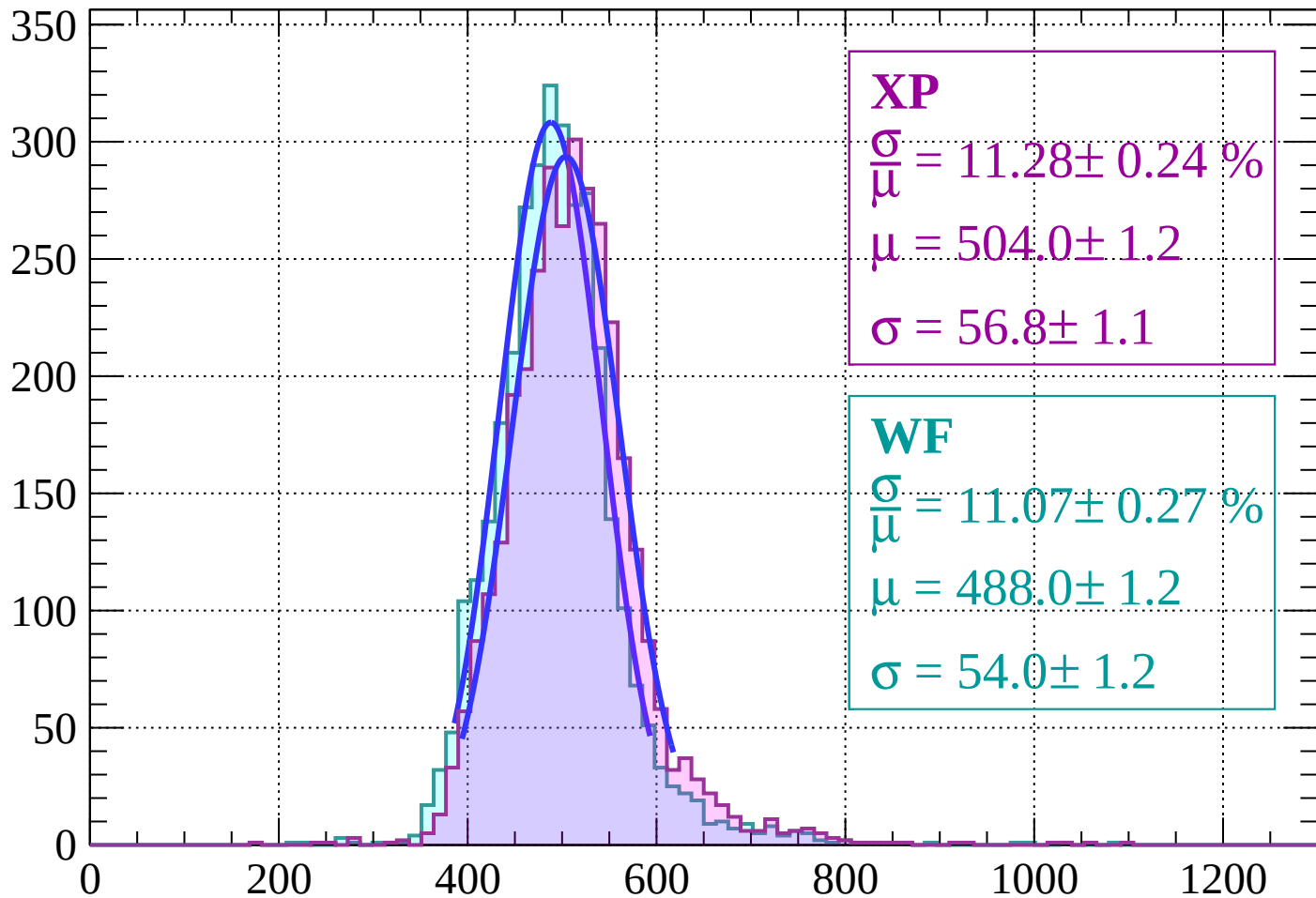
dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$



Energy loss | $6 < \theta < 8$

Count



XP

$$\frac{\sigma}{\mu} = 11.28 \pm 0.24 \%$$

$$\mu = 504.0 \pm 1.2$$

$$\sigma = 56.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.07 \pm 0.27 \%$$

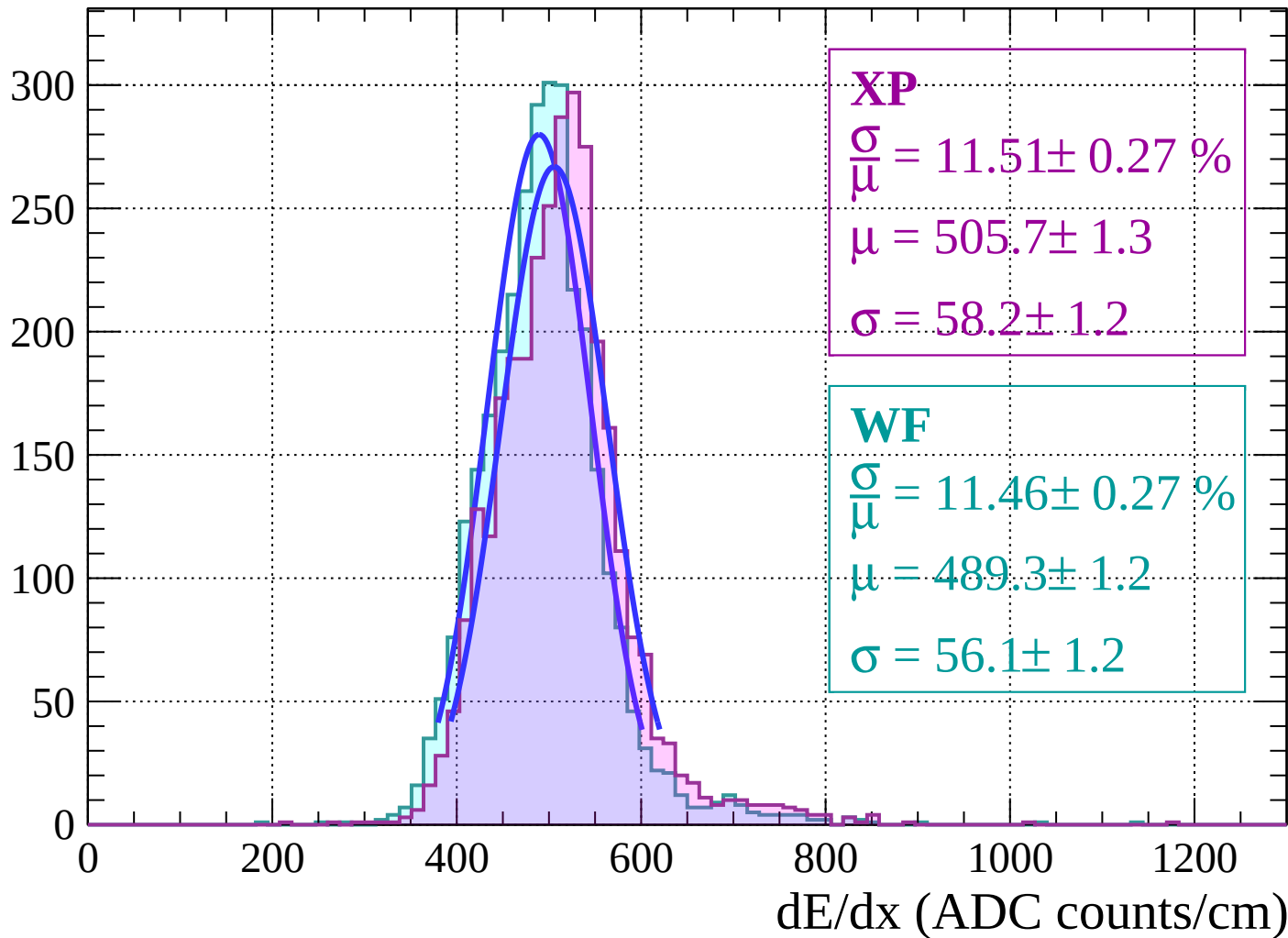
$$\mu = 488.0 \pm 1.2$$

$$\sigma = 54.0 \pm 1.2$$

dE/dx (ADC counts/cm)

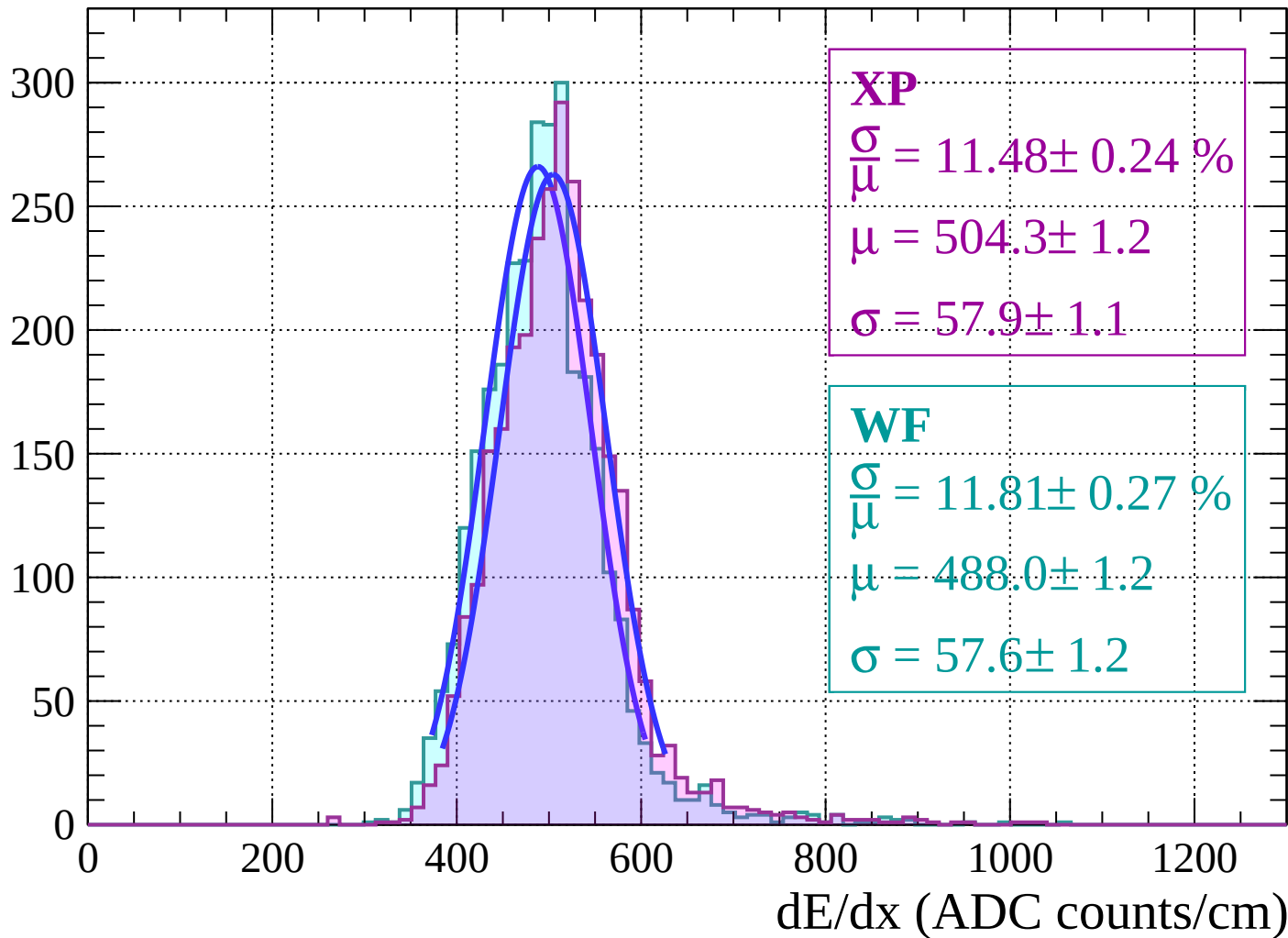
Energy loss | $8 < \theta < 10$

Count



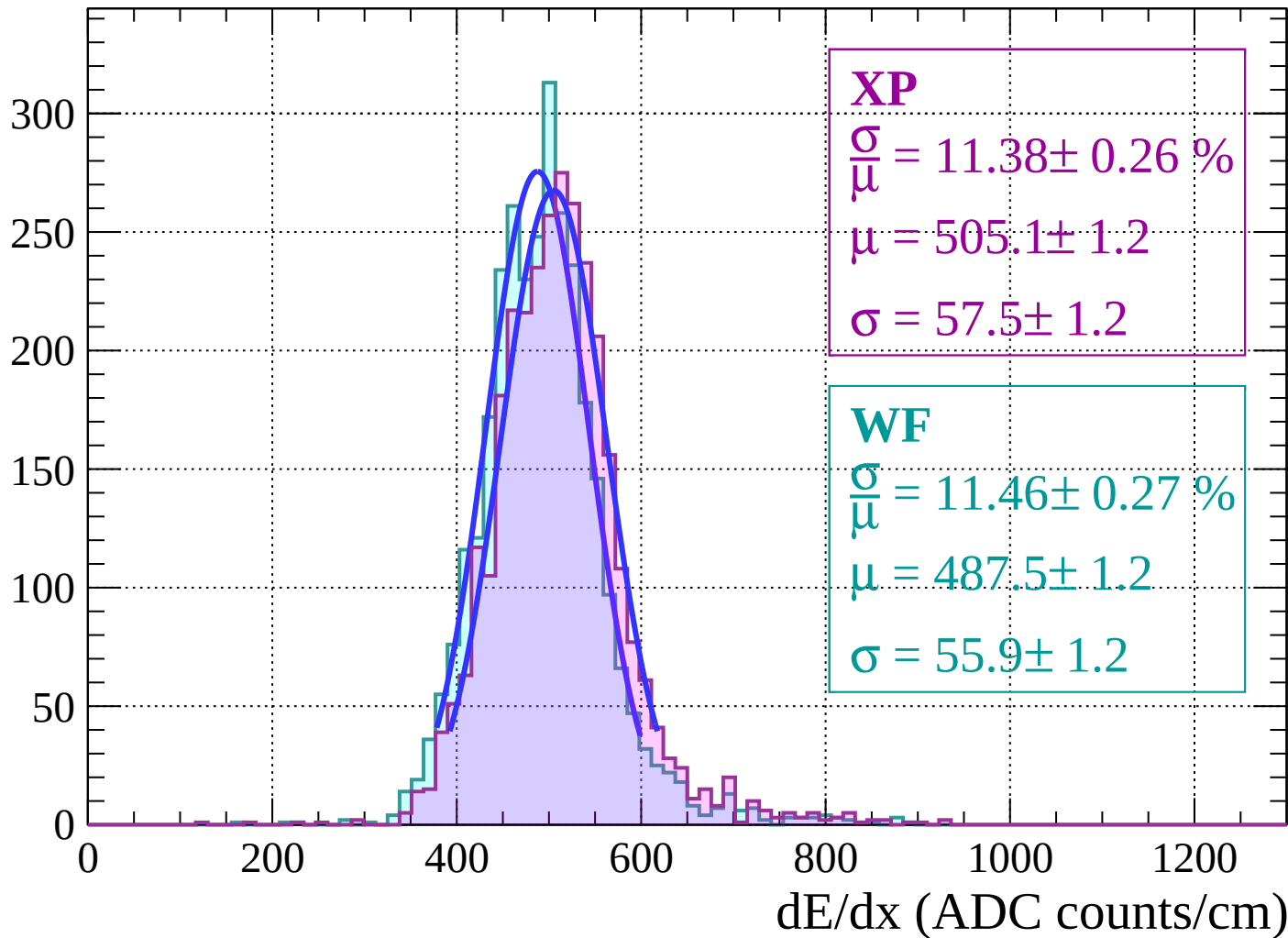
Energy loss | $10 < \theta < 12$

Count



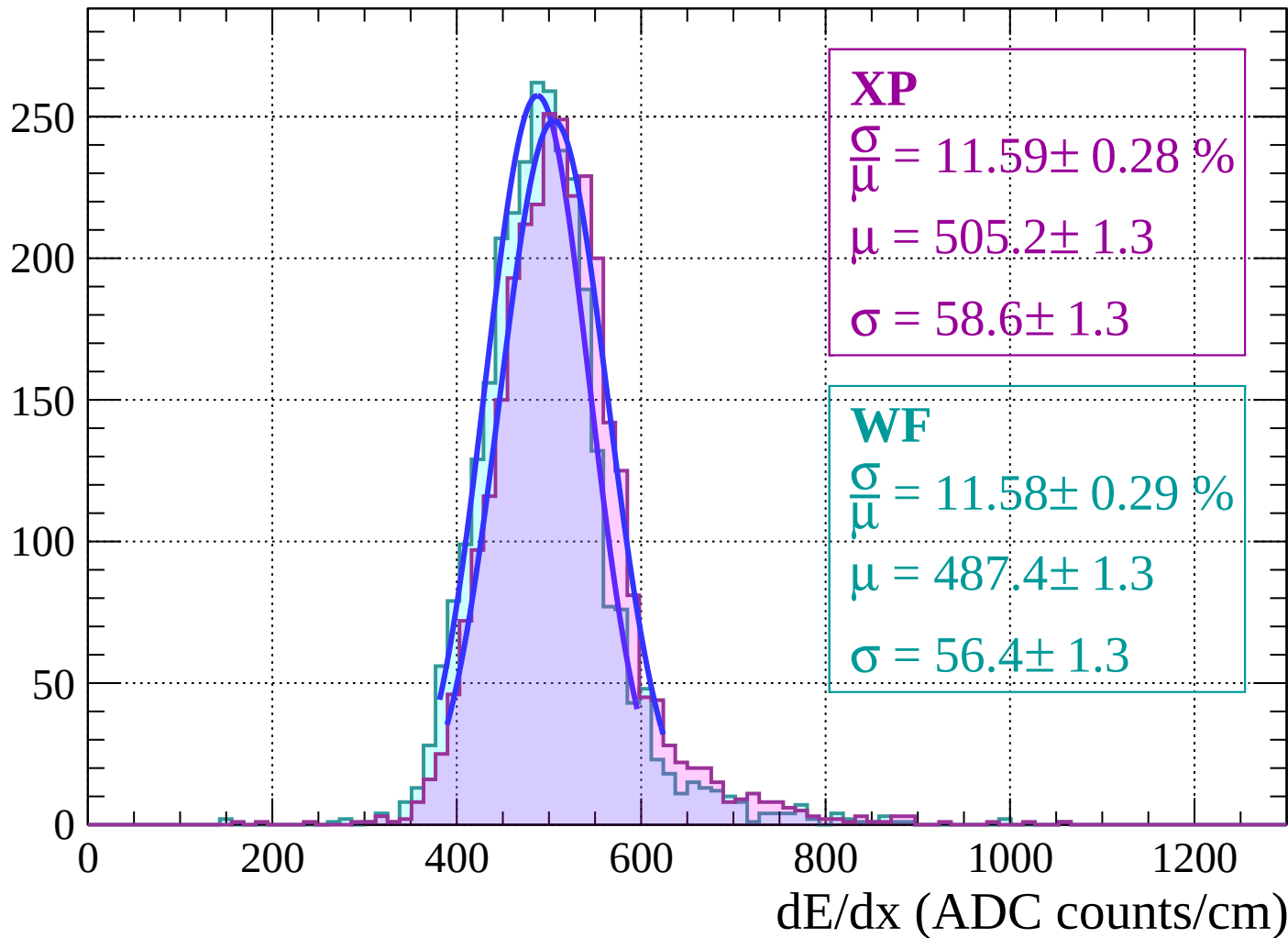
Energy loss | $12 < \theta < 14$

Count



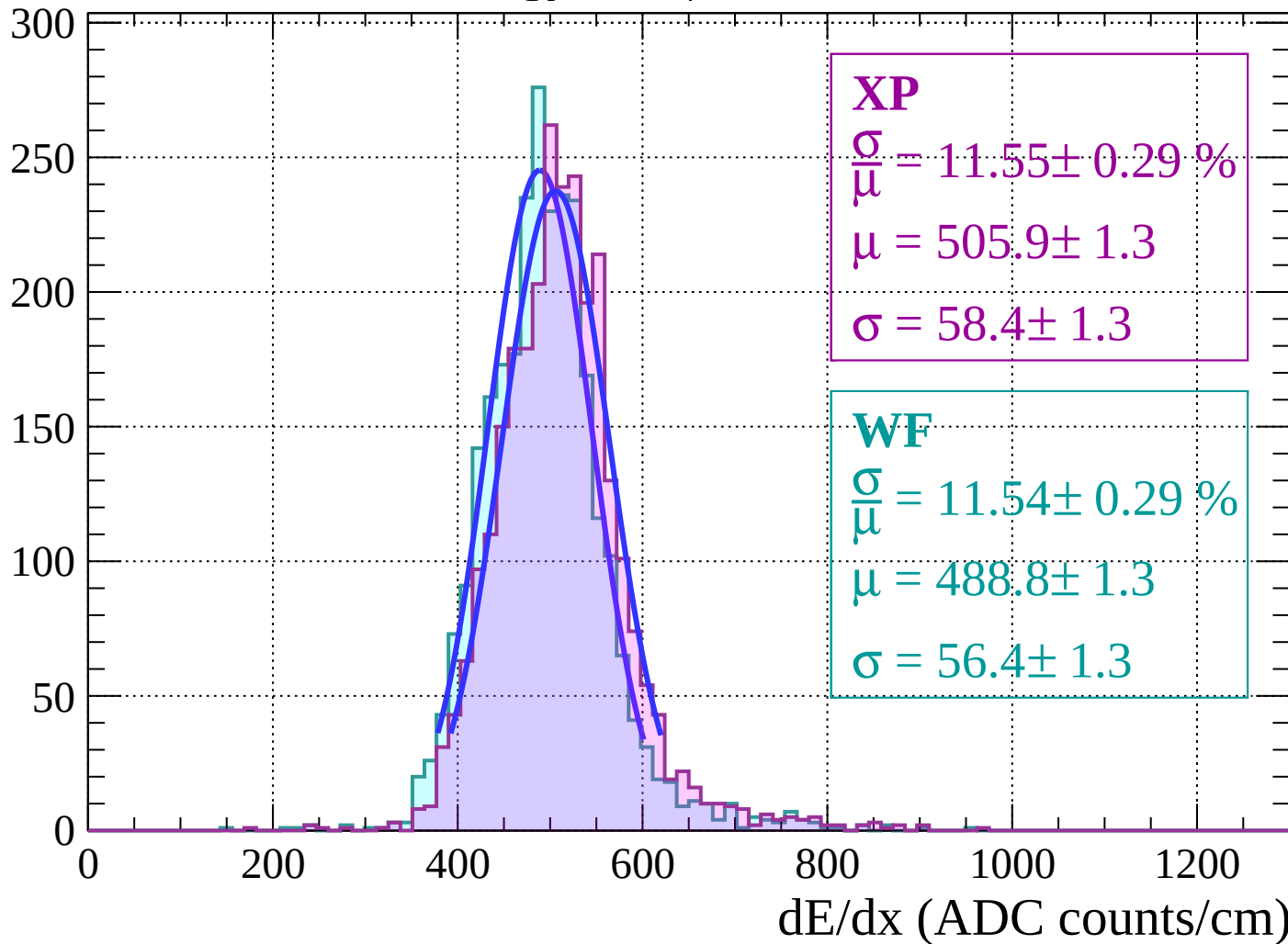
Energy loss | $14 < \theta < 16$

Count



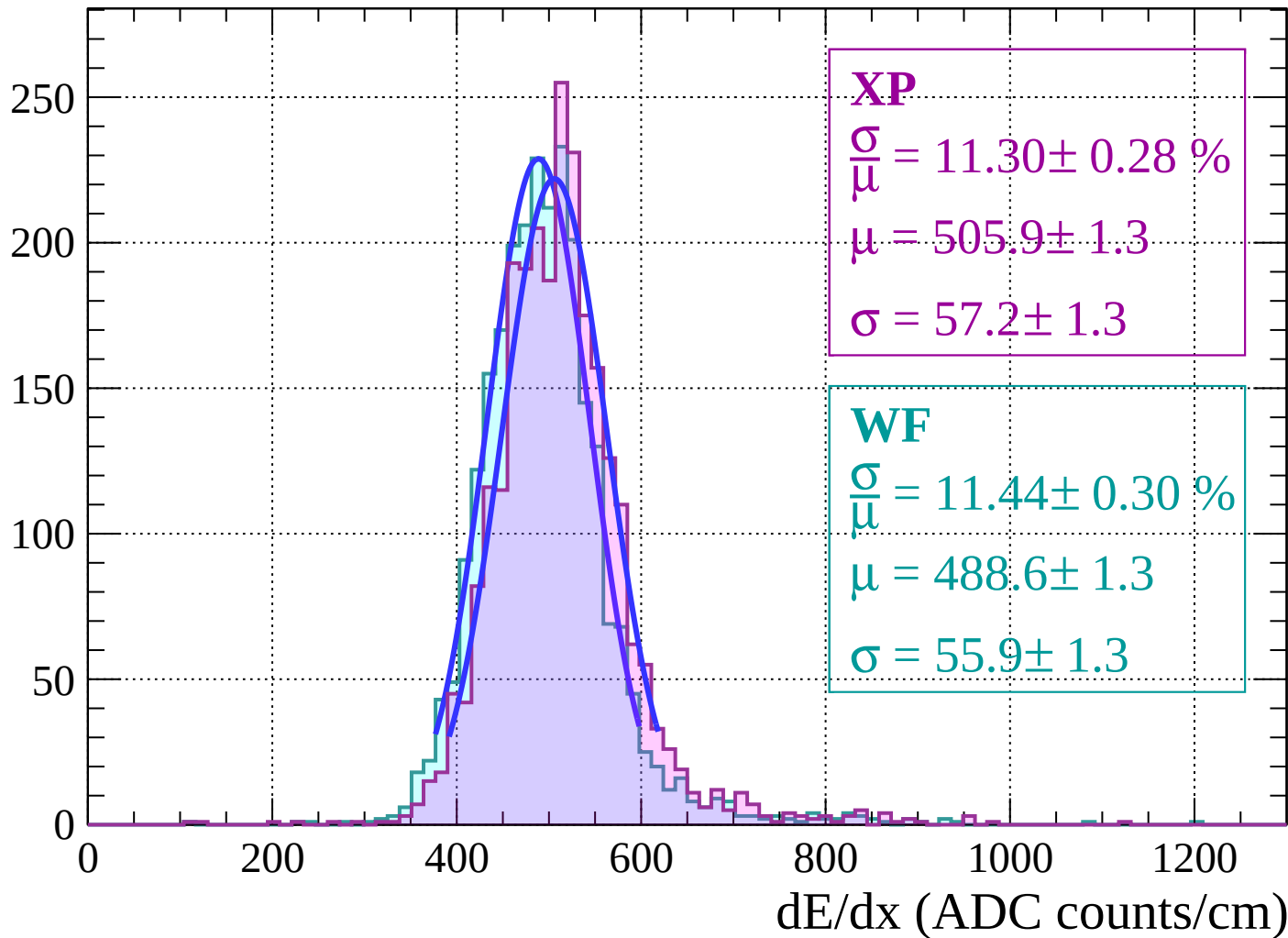
Energy loss | $16 < \theta < 18$

Count

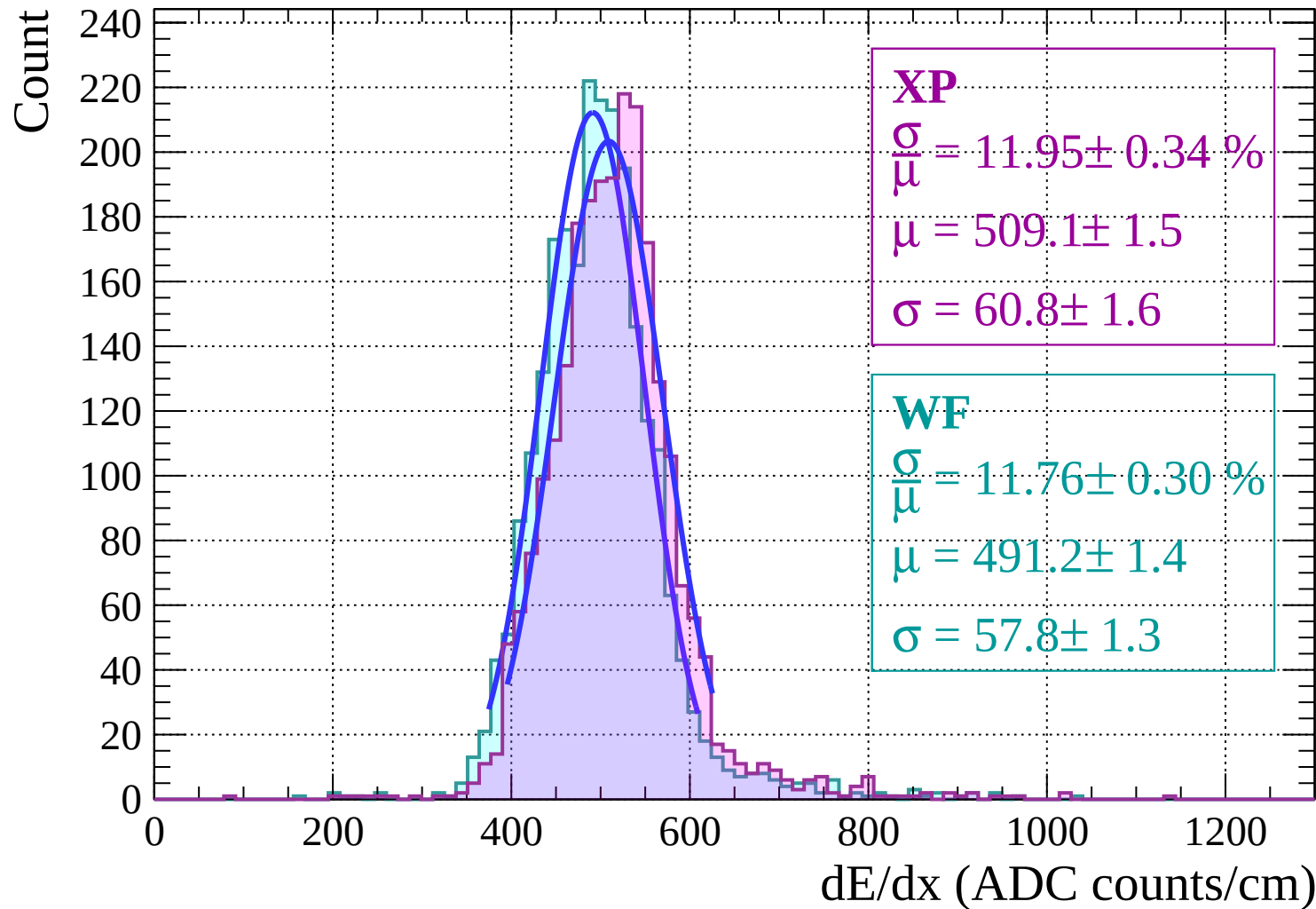


Energy loss | $18 < \theta < 20$

Count

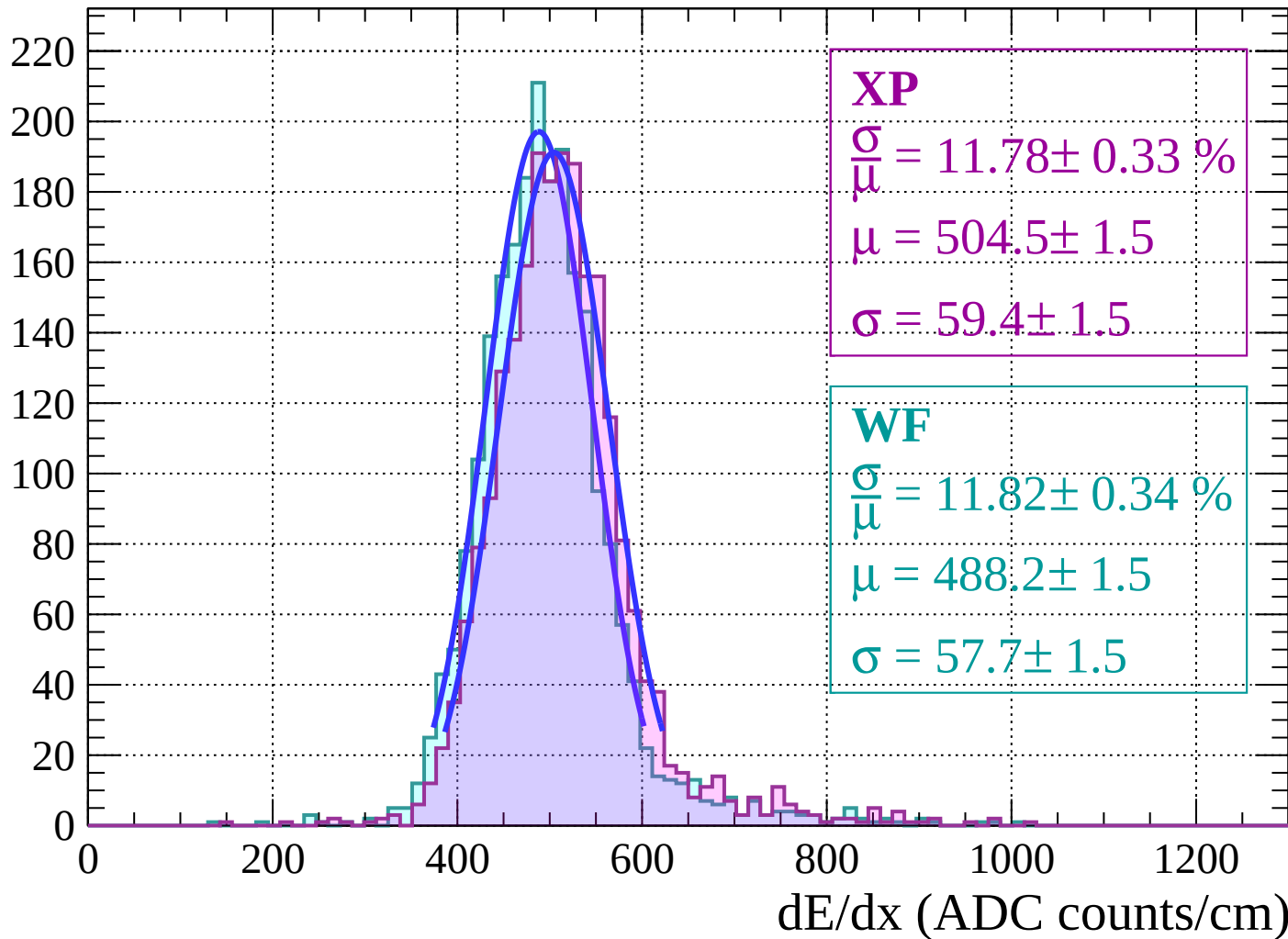


Energy loss | $20 < \theta < 22$

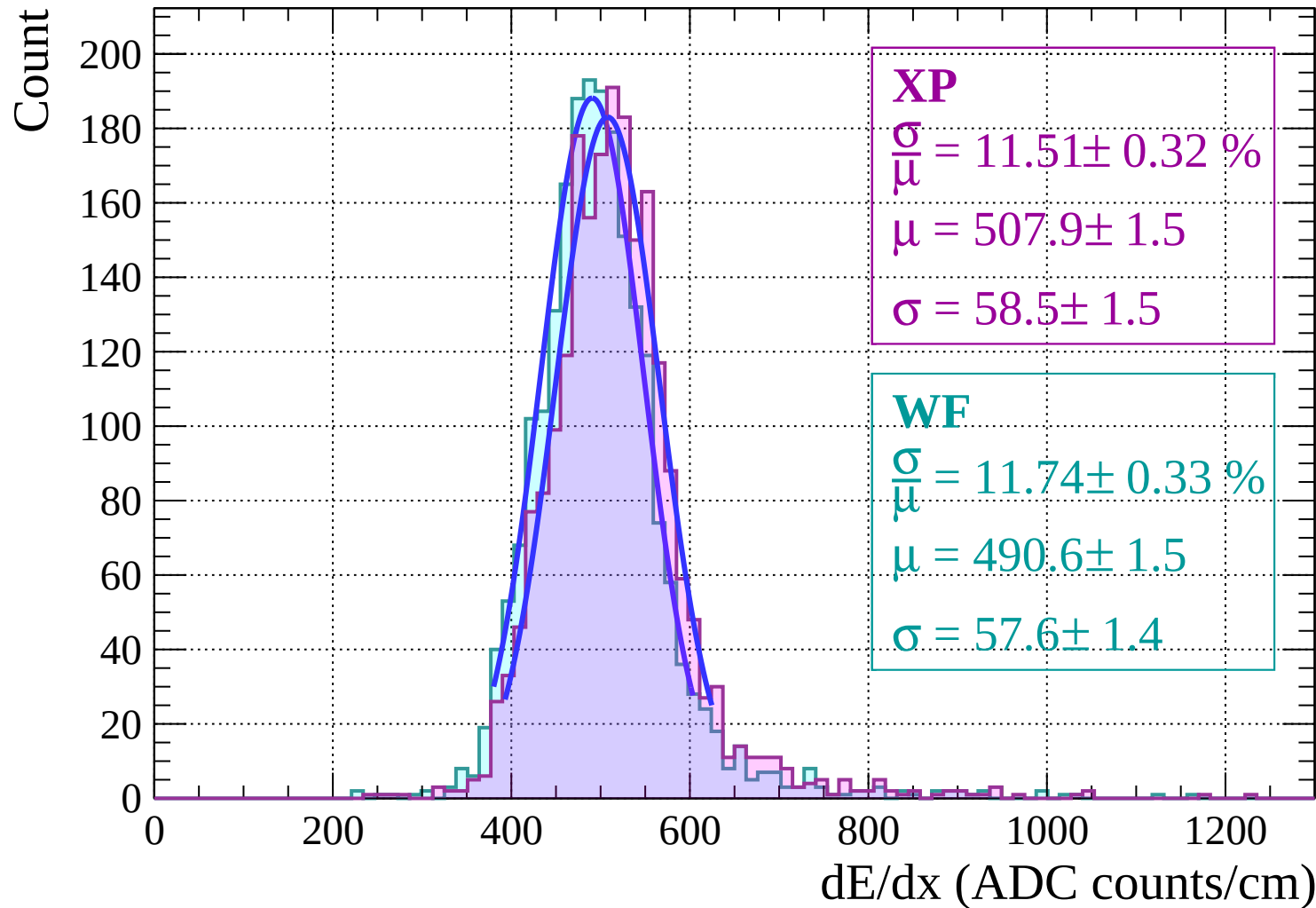


Energy loss | $22 < \theta < 24$

Count

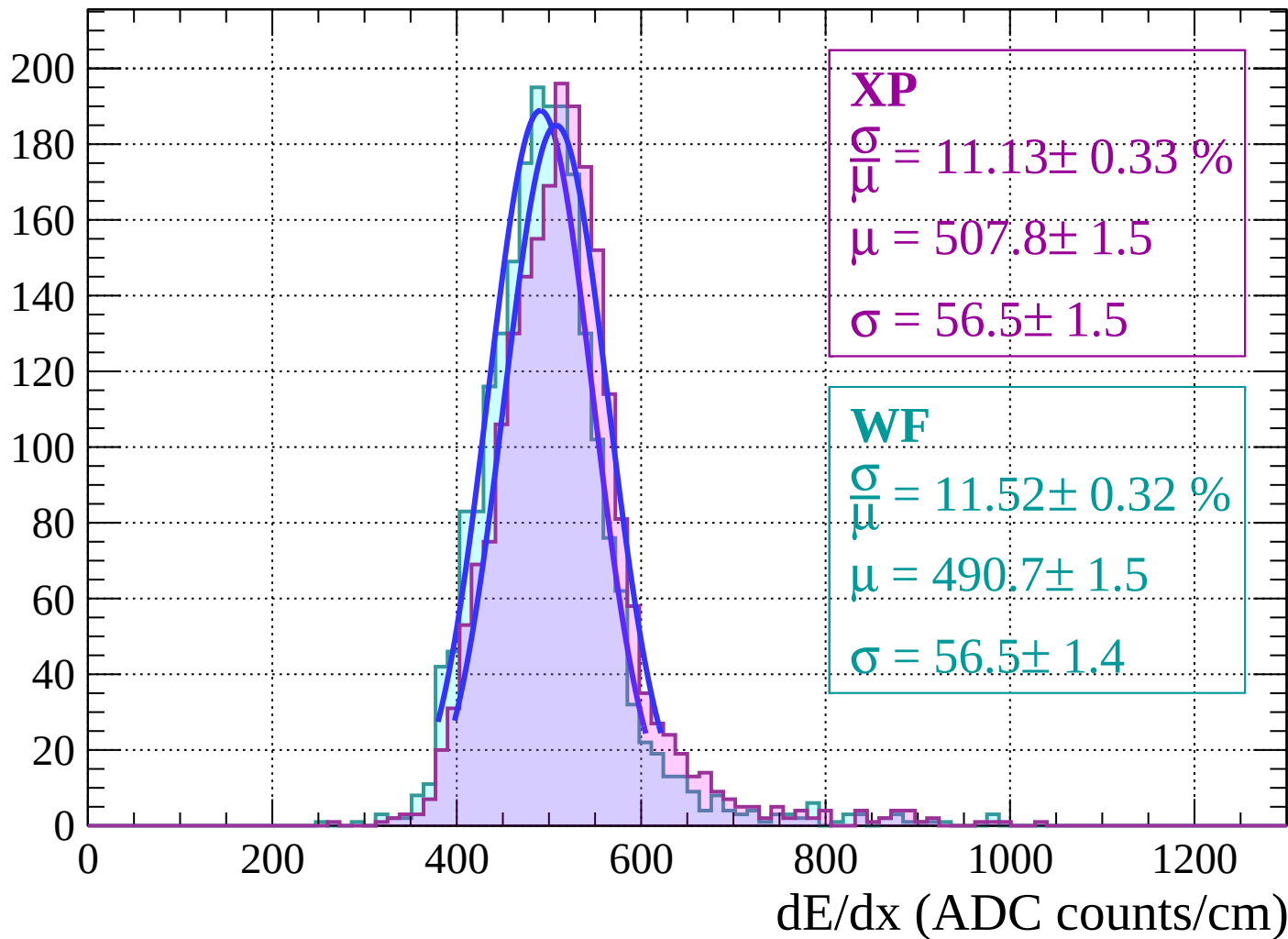


Energy loss | $24 < \theta < 26$



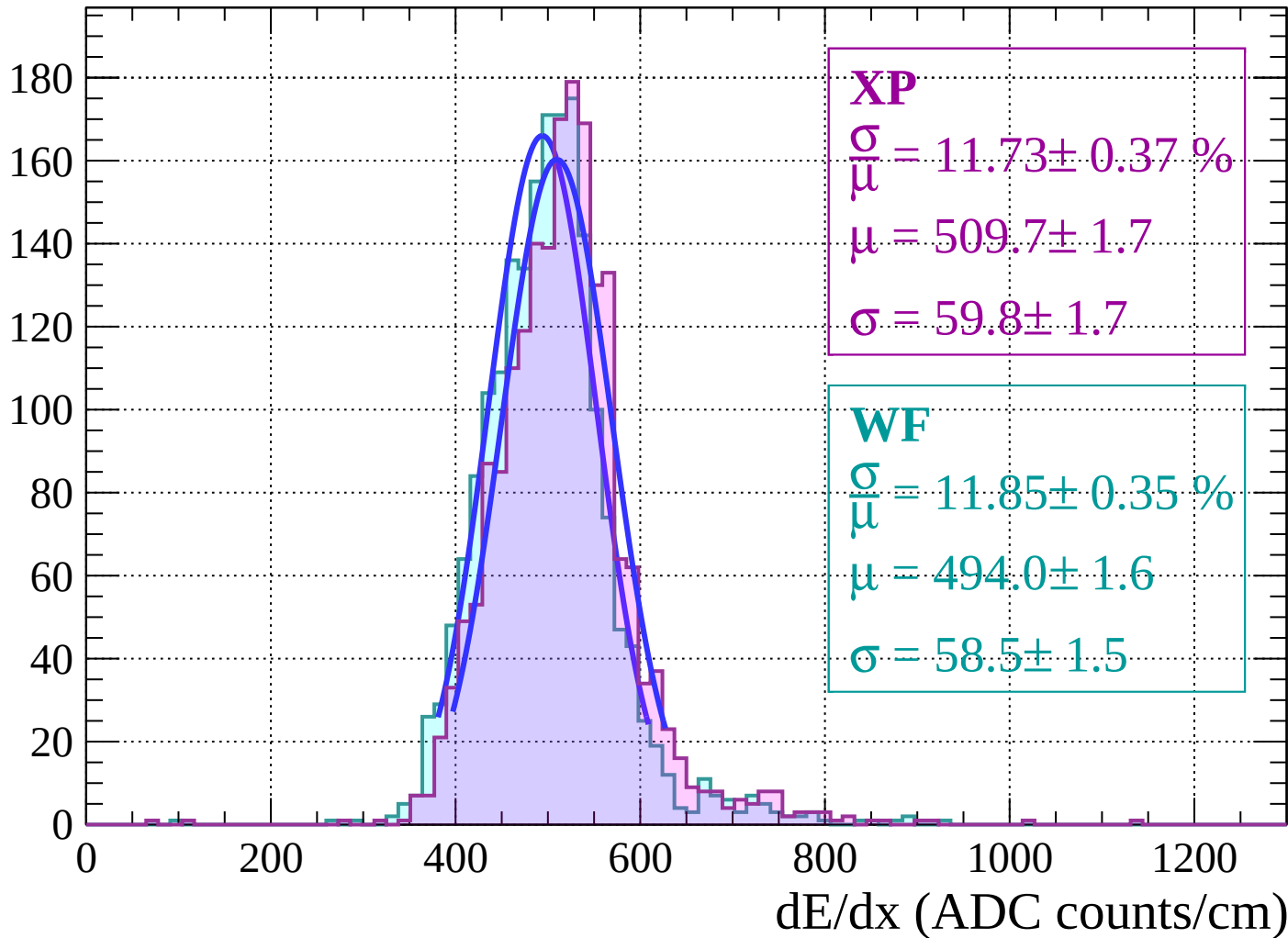
Energy loss | $26 < \theta < 28$

Count



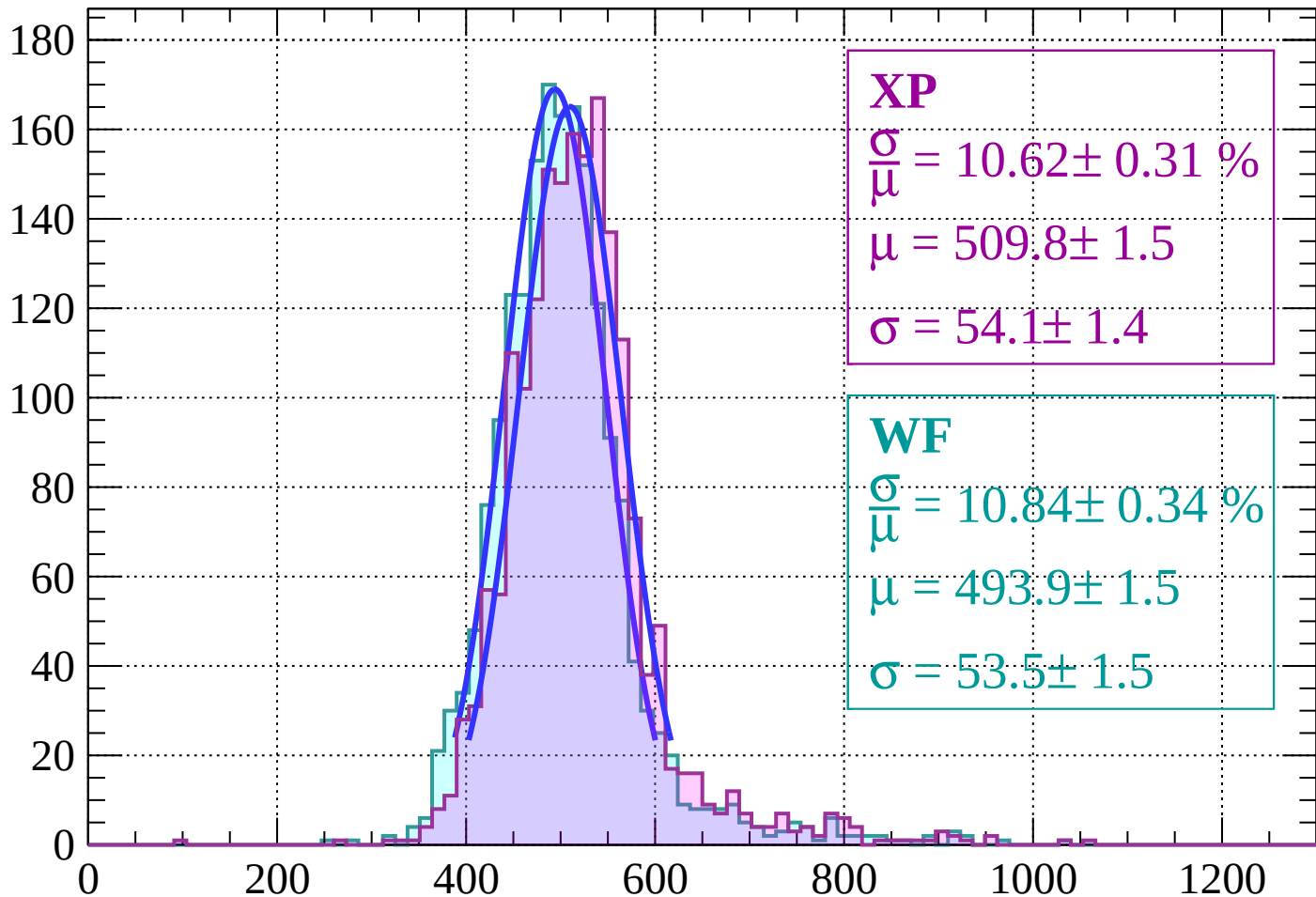
Energy loss | $28 < \theta < 30$

Count



Energy loss | $30 < \theta < 32$

Count



XP

$$\frac{\sigma}{\mu} = 10.62 \pm 0.31 \%$$

$$\mu = 509.8 \pm 1.5$$

$$\sigma = 54.1 \pm 1.4$$

WF

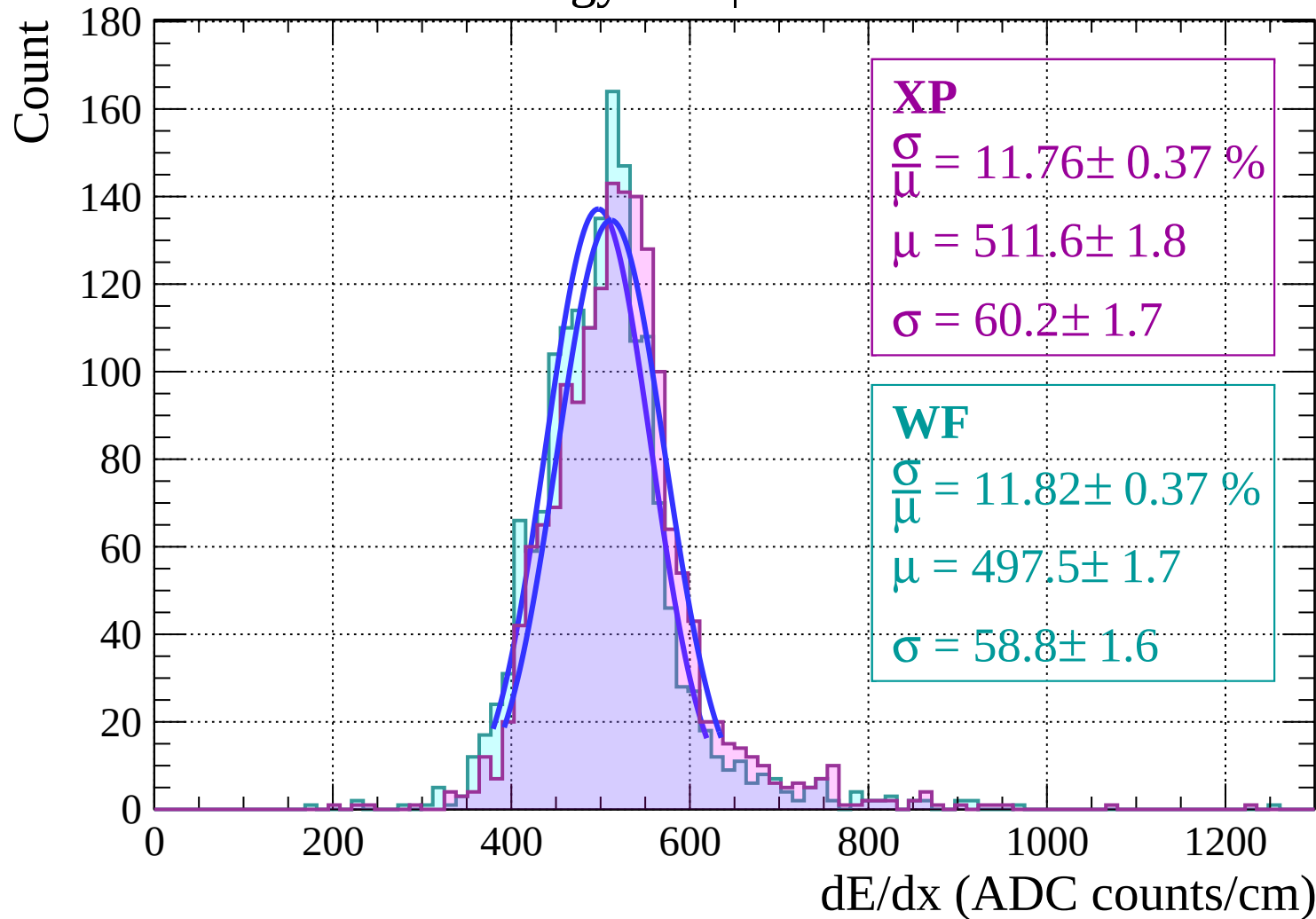
$$\frac{\sigma}{\mu} = 10.84 \pm 0.34 \%$$

$$\mu = 493.9 \pm 1.5$$

$$\sigma = 53.5 \pm 1.5$$

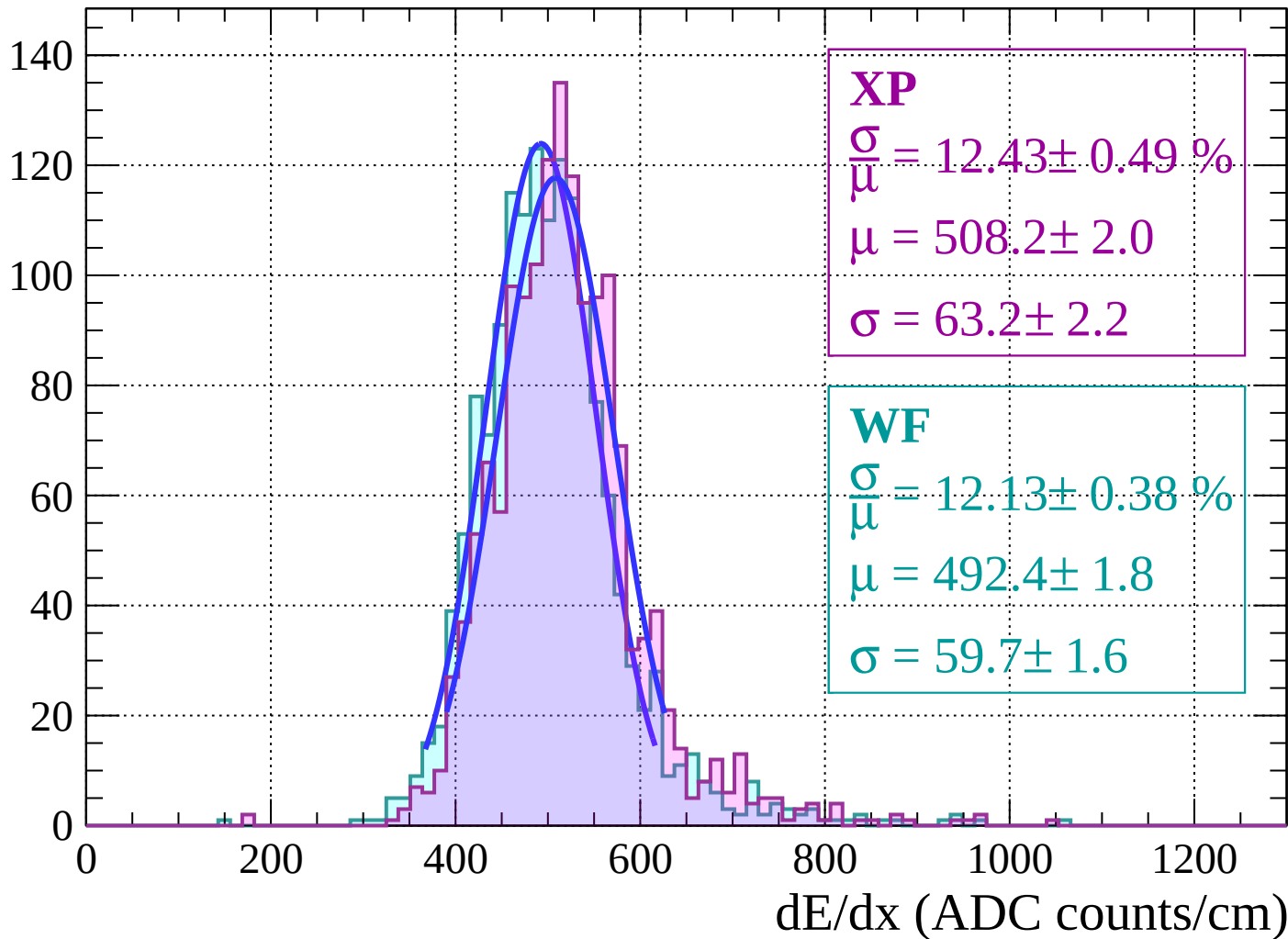
dE/dx (ADC counts/cm)

Energy loss | $32 < \theta < 34$



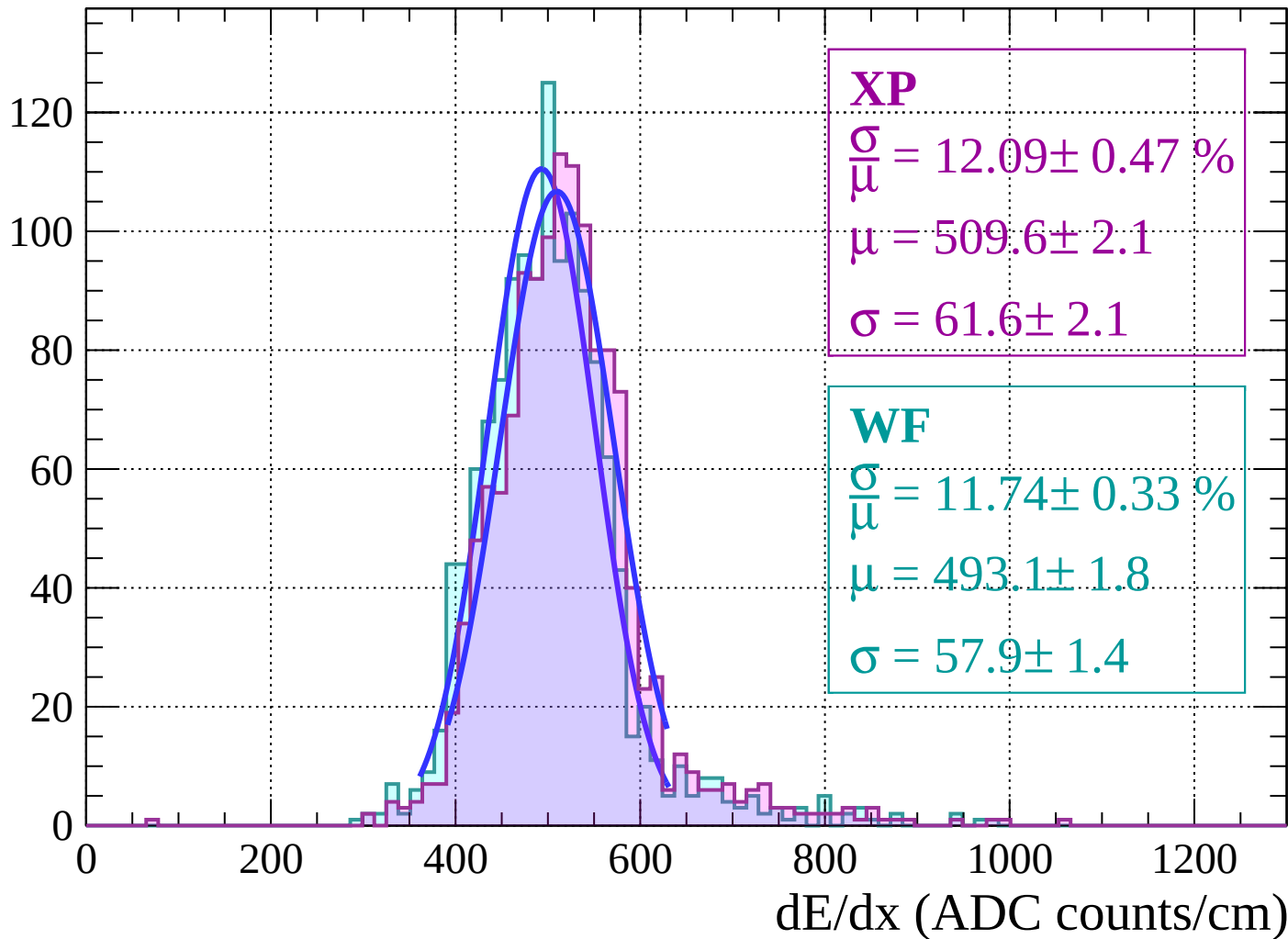
Energy loss | $34 < \theta < 36$

Count



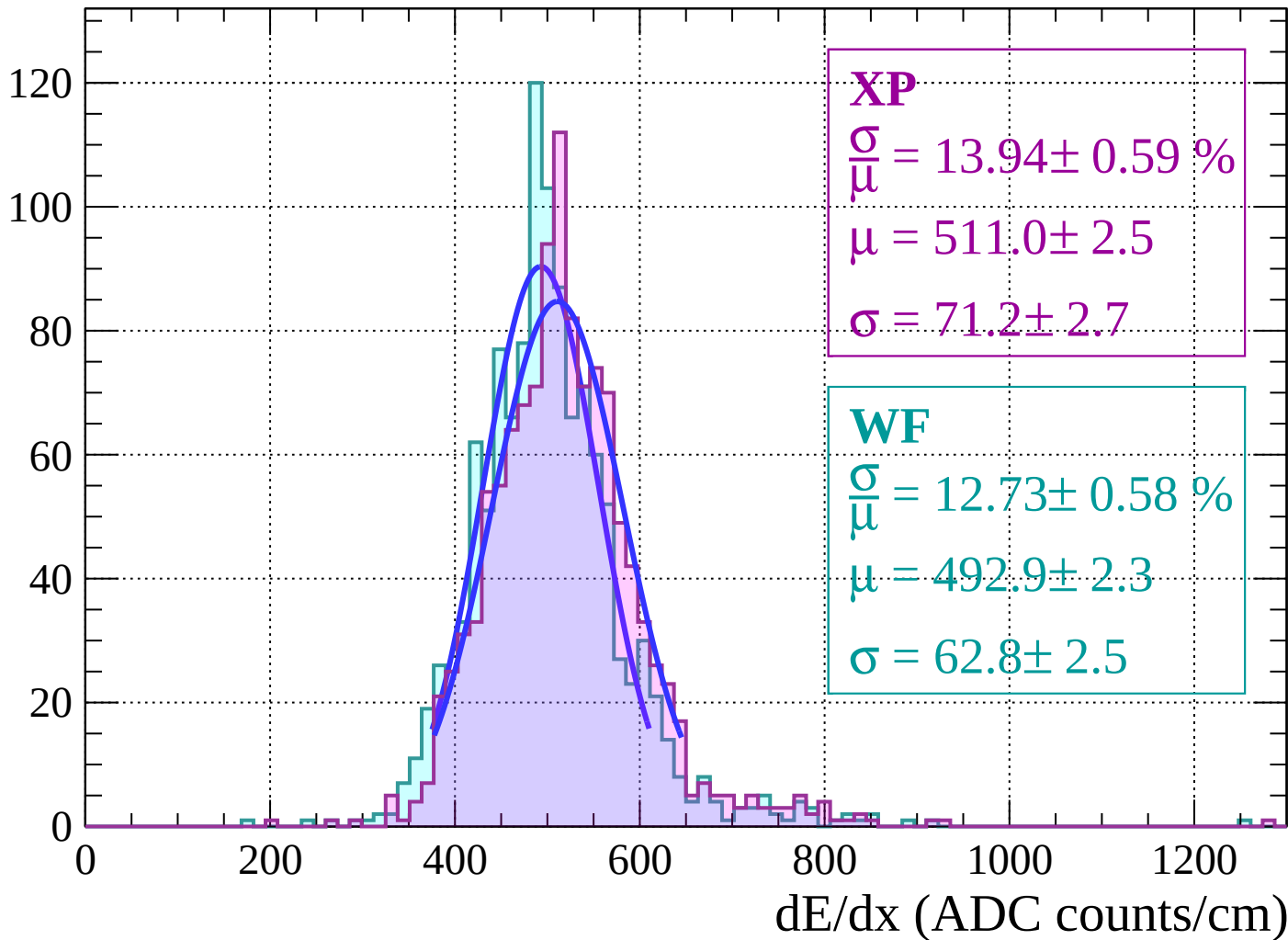
Energy loss | $36 < \theta < 38$

Count



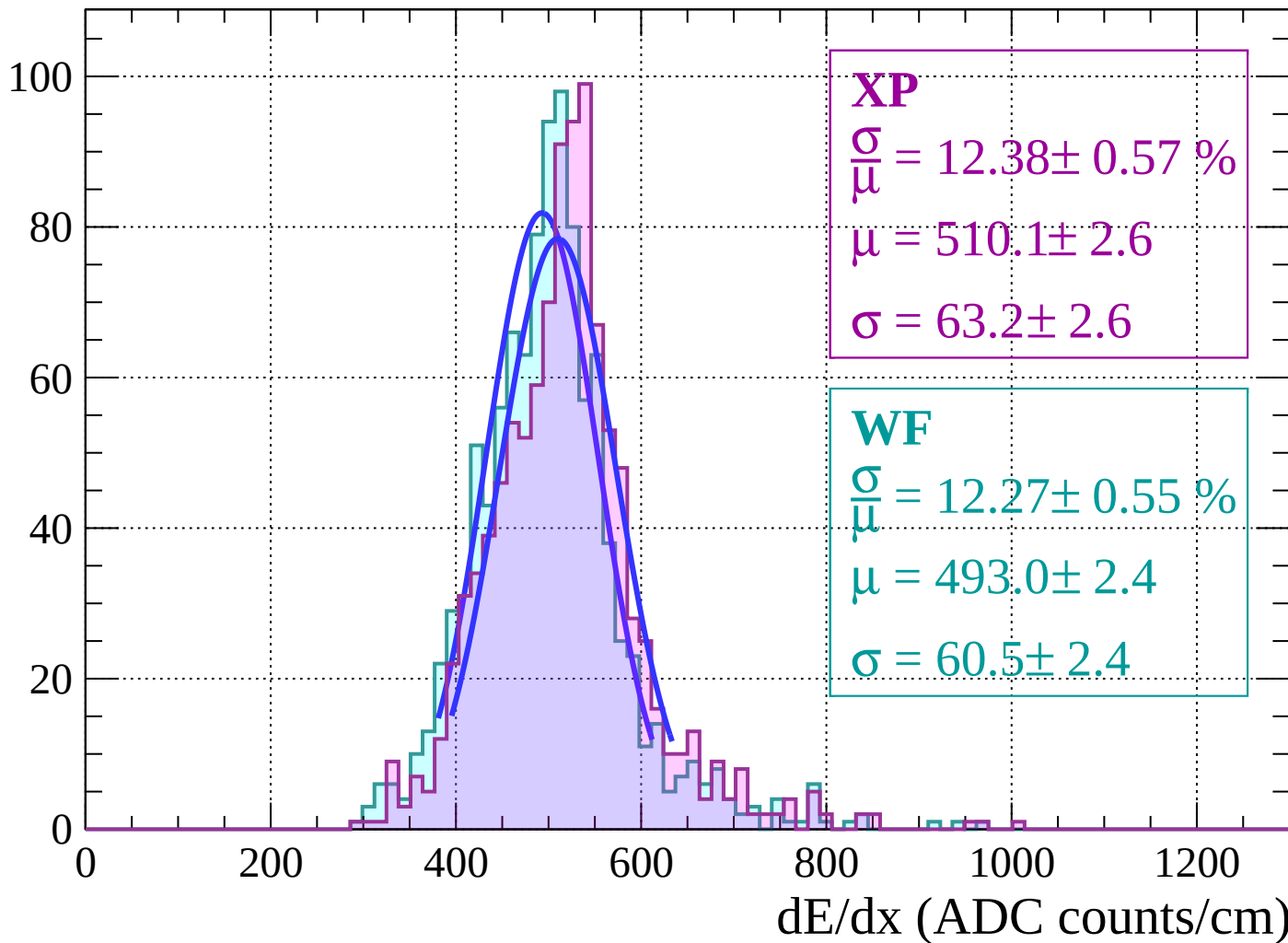
Energy loss | $38 < \theta < 40$

Count



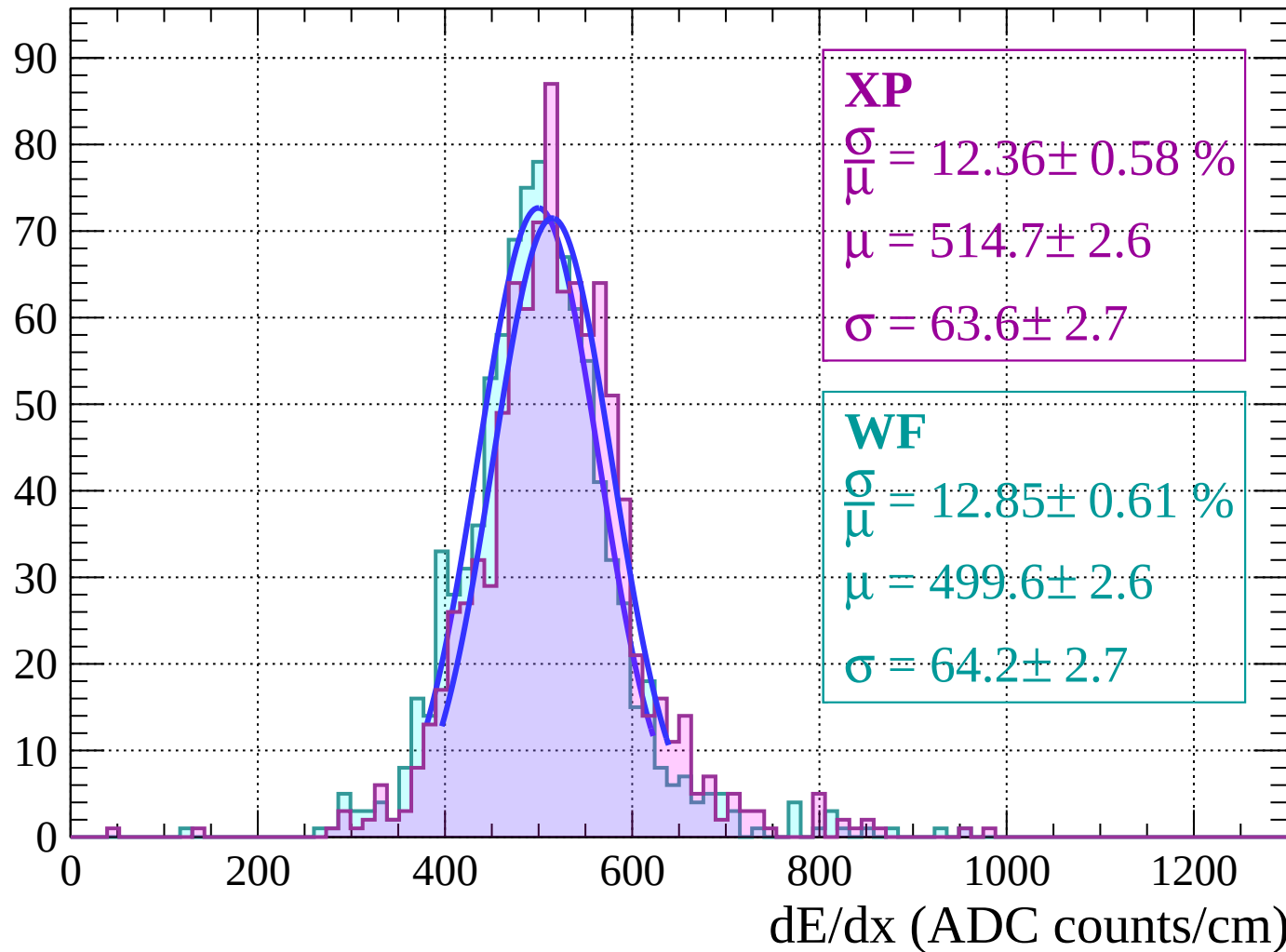
Energy loss | $40 < \theta < 42$

Count



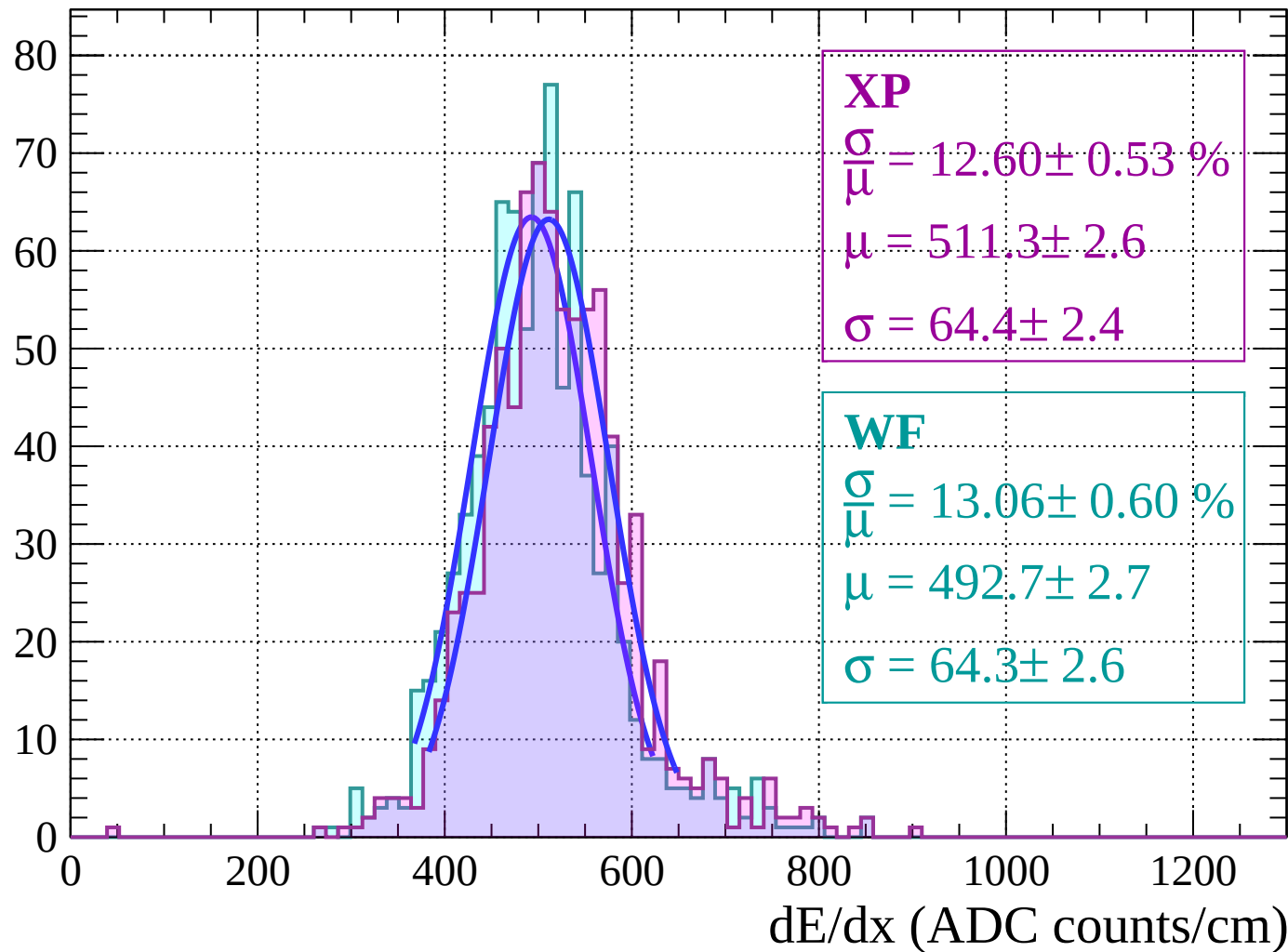
Energy loss | $42 < \theta < 44$

Count



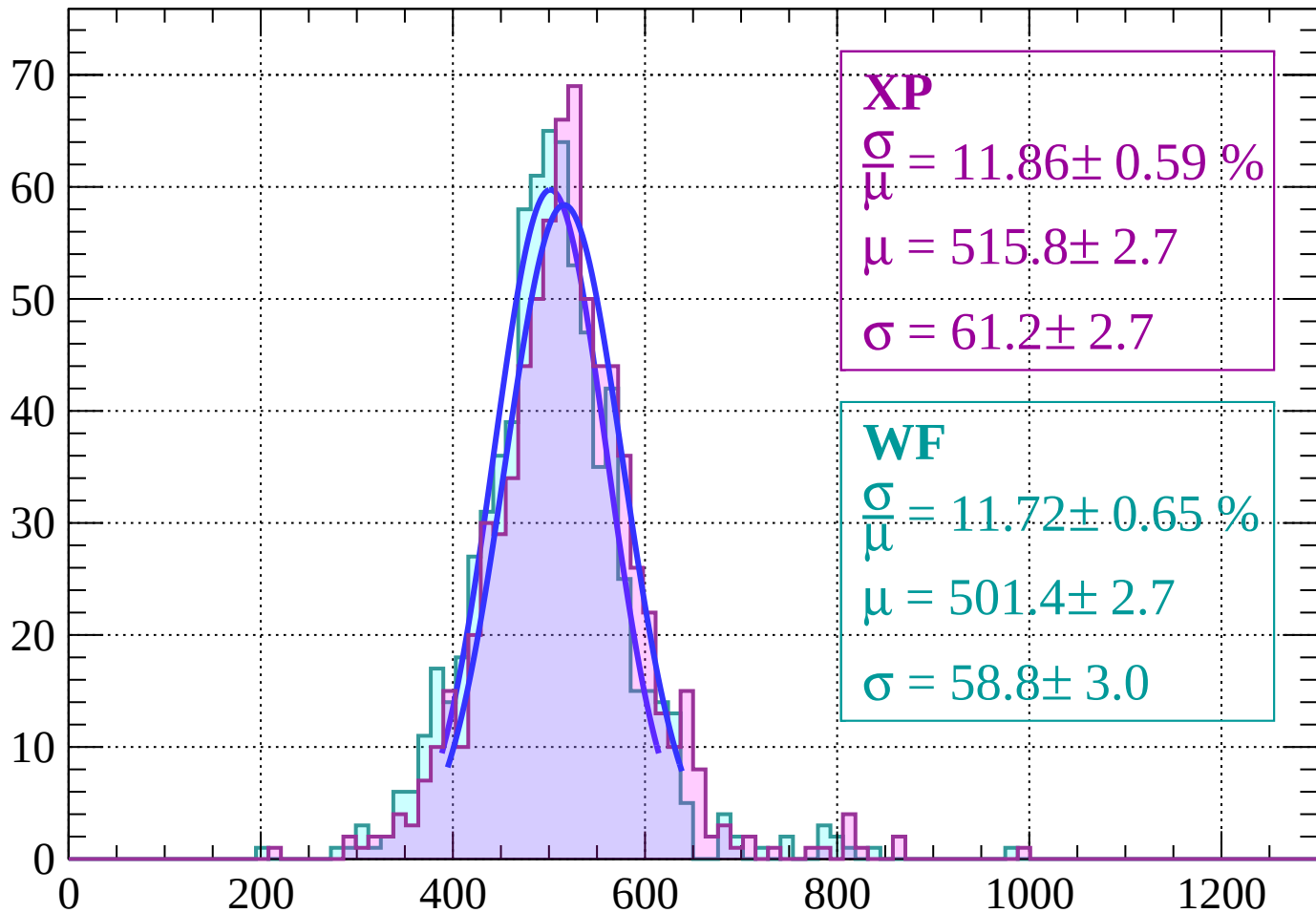
Energy loss | $44 < \theta < 46$

Count



Energy loss | $46 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 11.86 \pm 0.59 \%$$

$$\mu = 515.8 \pm 2.7$$

$$\sigma = 61.2 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 11.72 \pm 0.65 \%$$

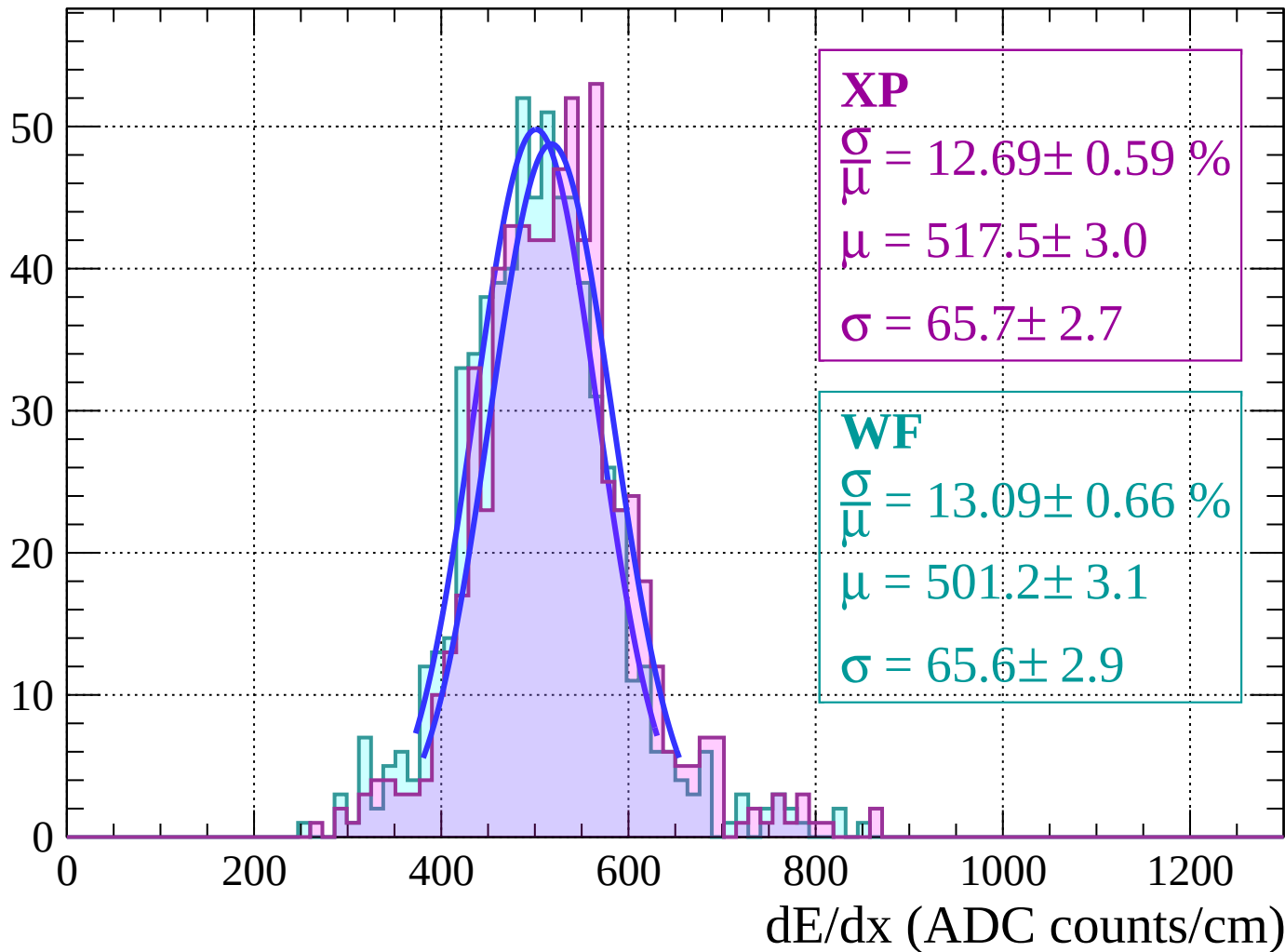
$$\mu = 501.4 \pm 2.7$$

$$\sigma = 58.8 \pm 3.0$$

dE/dx (ADC counts/cm)

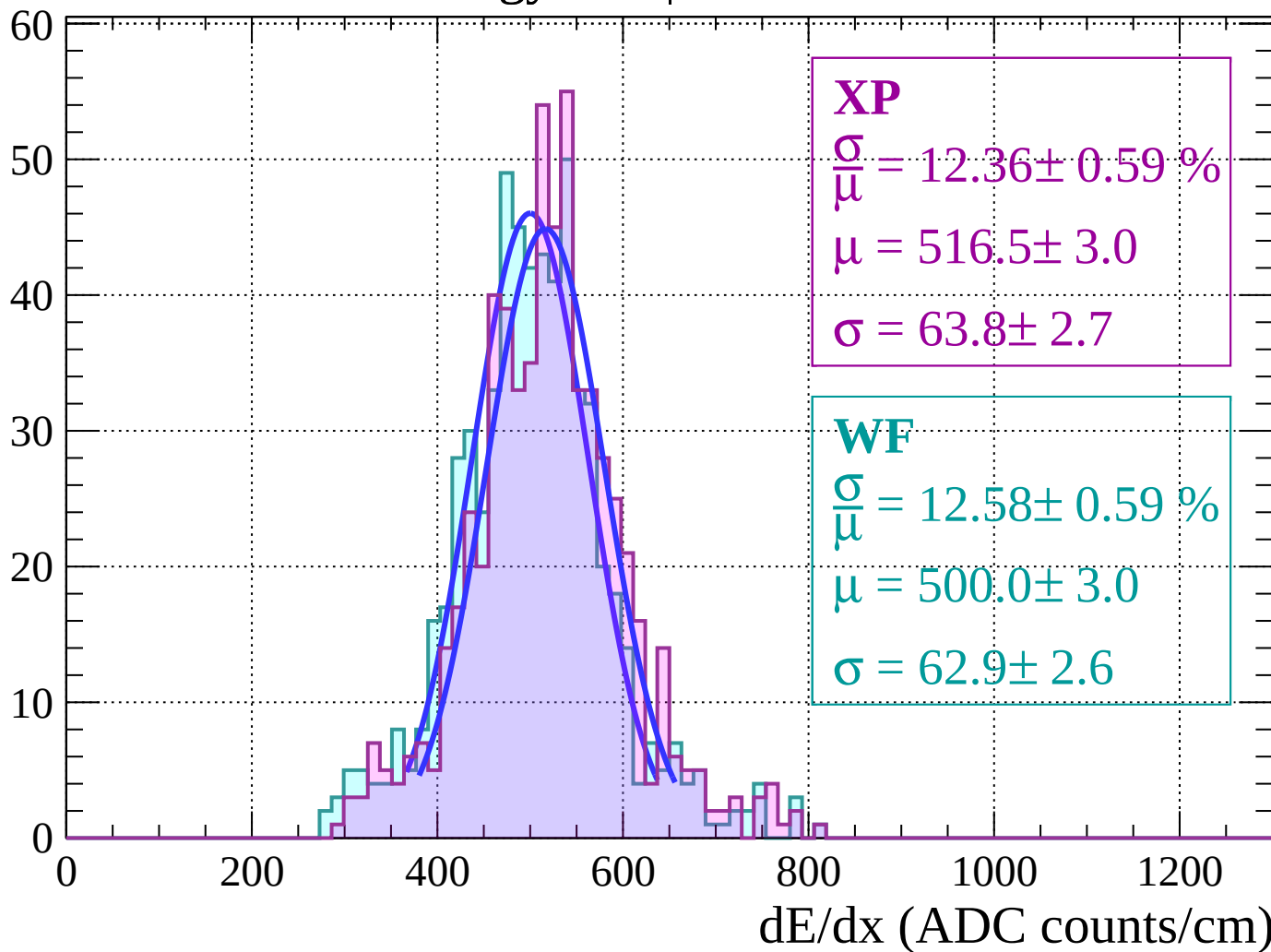
Energy loss | $48 < \theta < 50$

Count



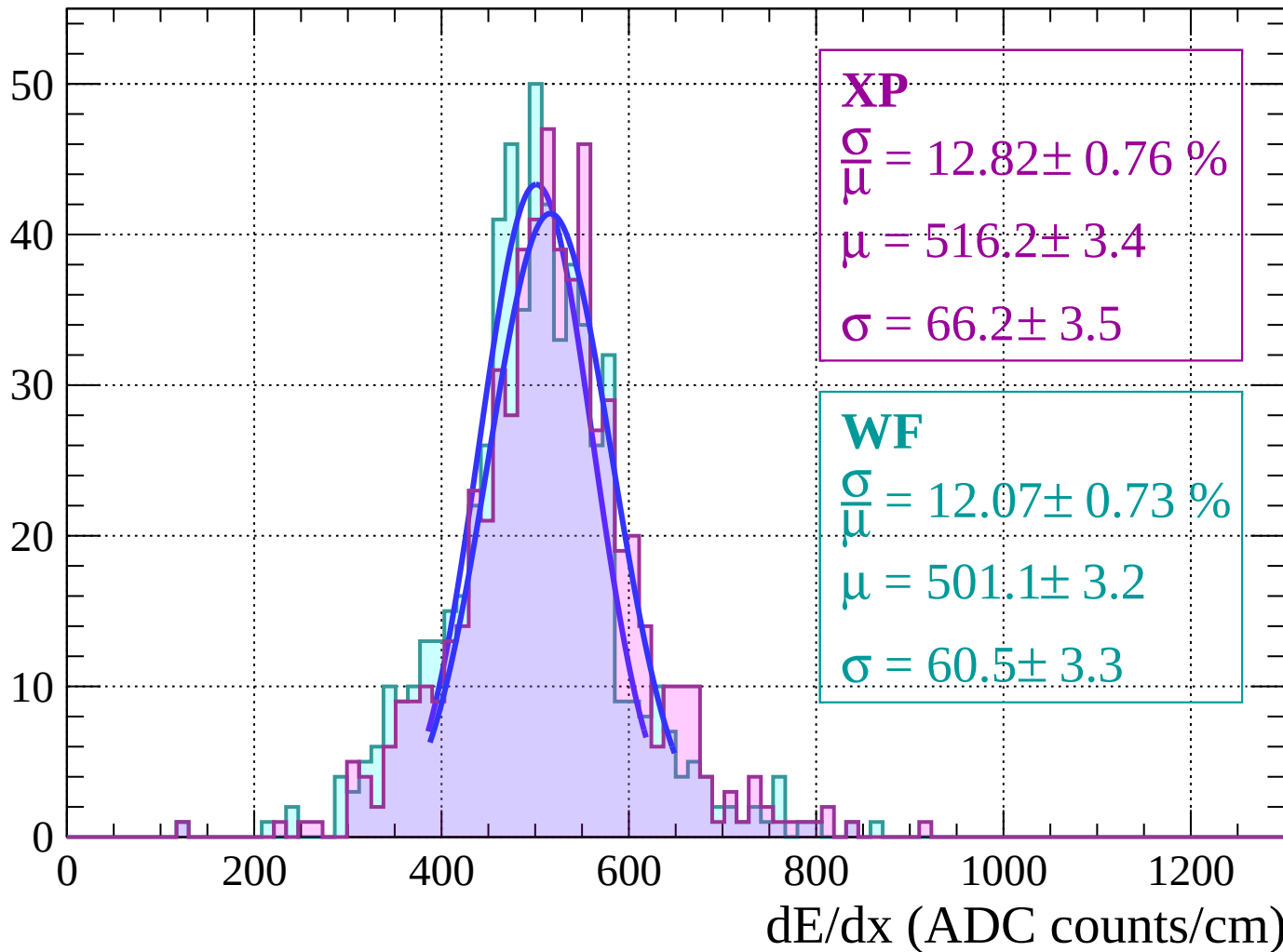
Energy loss | $50 < \theta < 52$

Count



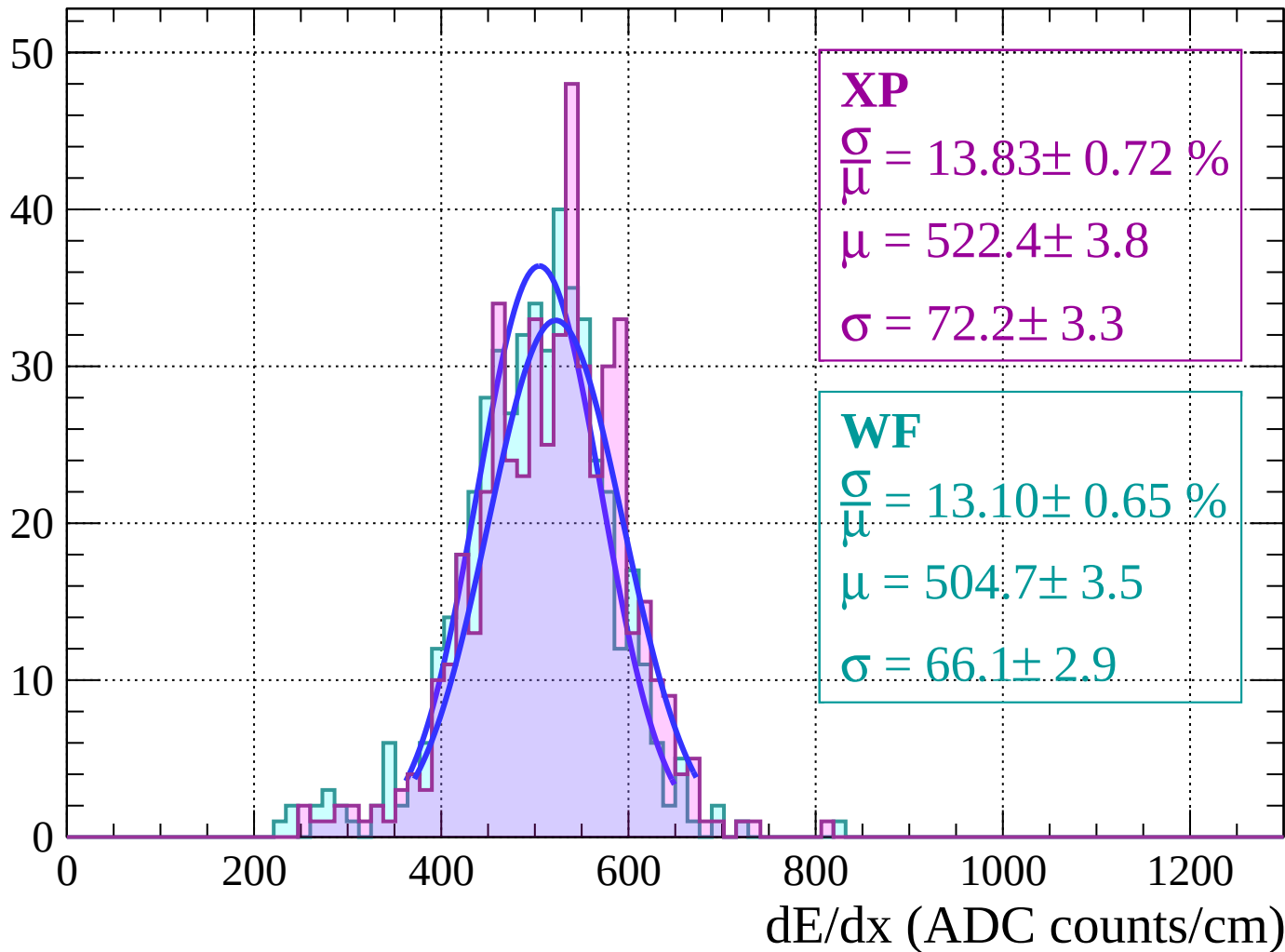
Energy loss | $52 < \theta < 54$

Count



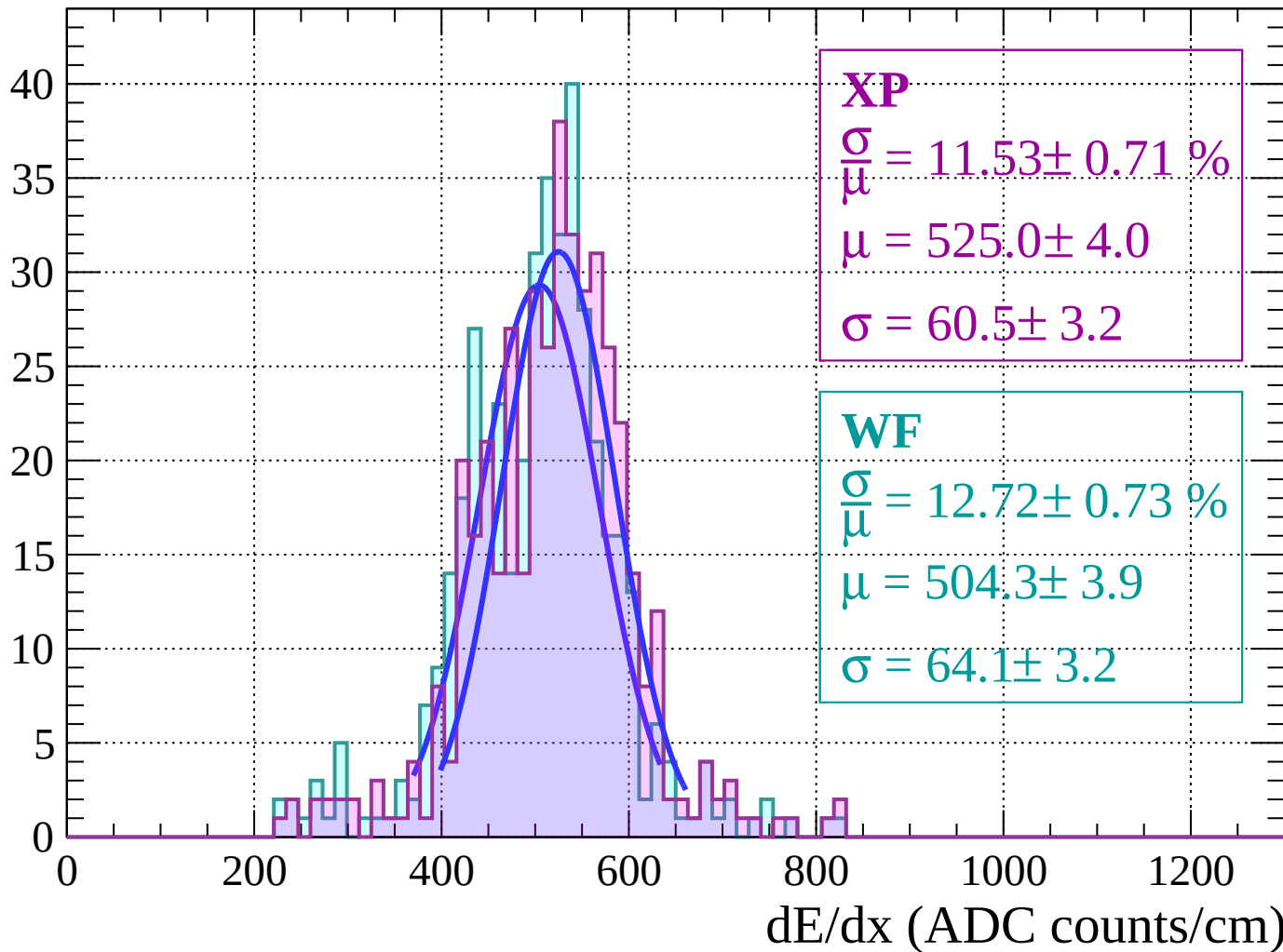
Energy loss | $54 < \theta < 56$

Count



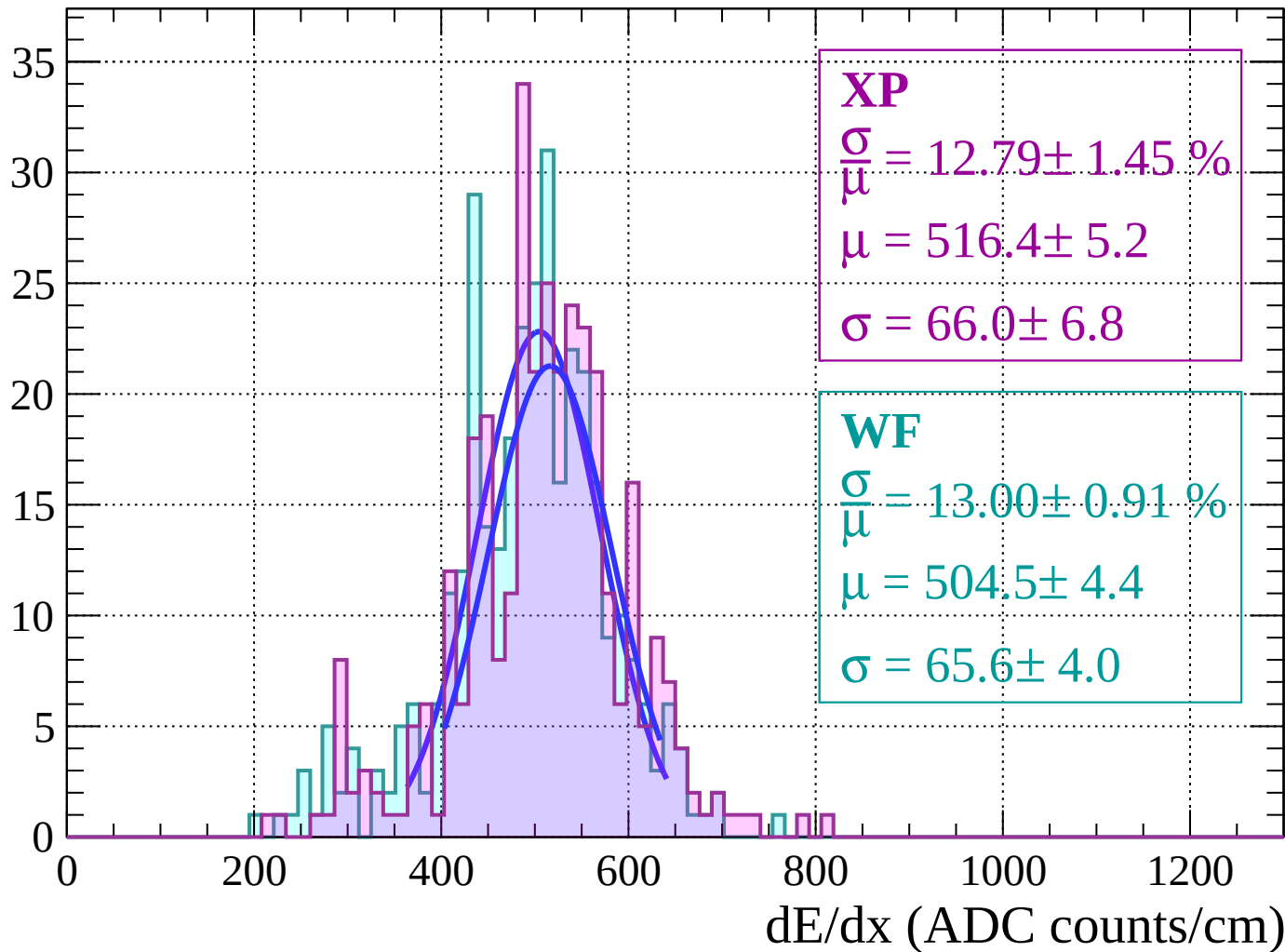
Energy loss | $56 < \theta < 58$

Count



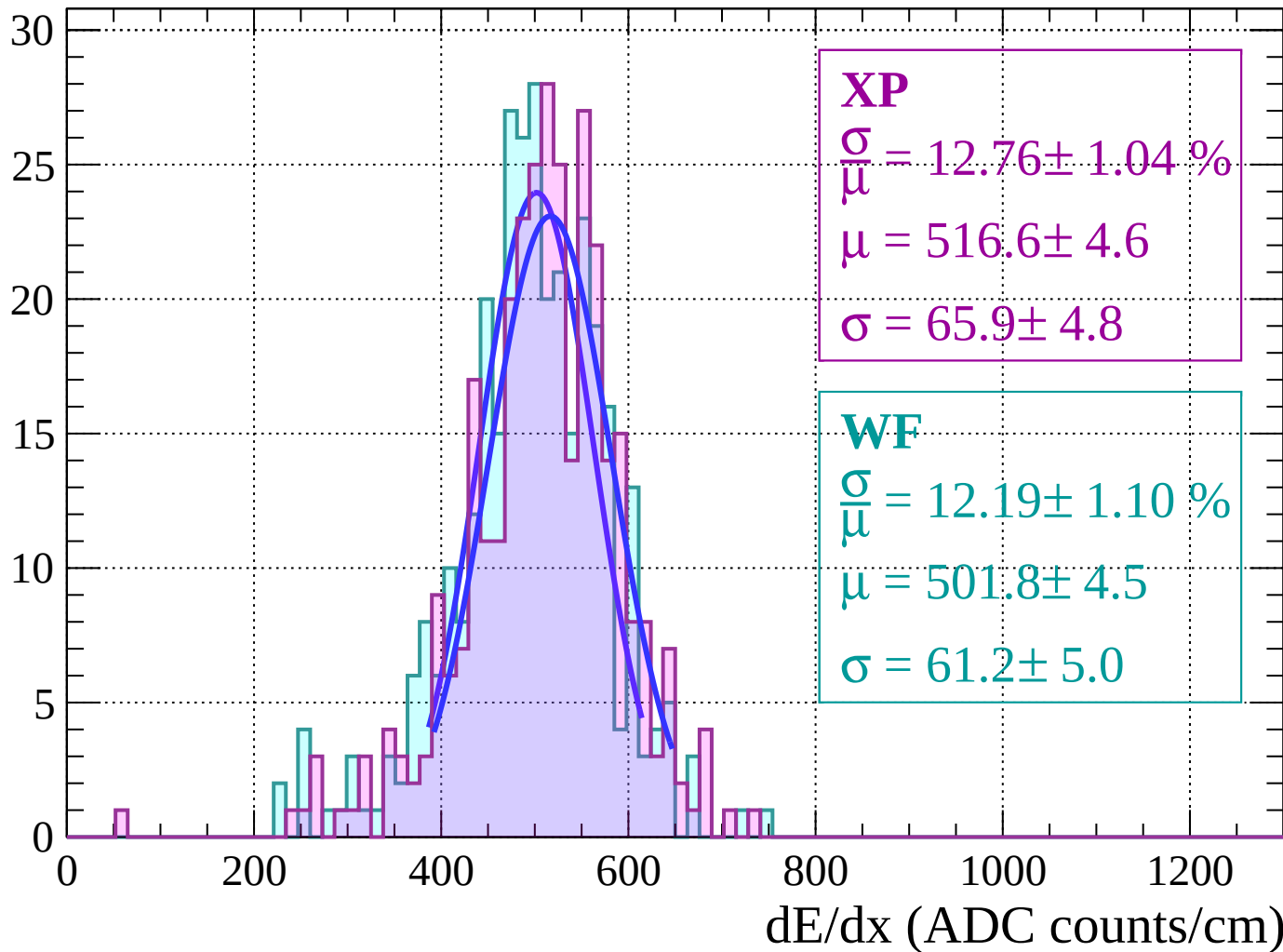
Energy loss | $58 < \theta < 60$

Count



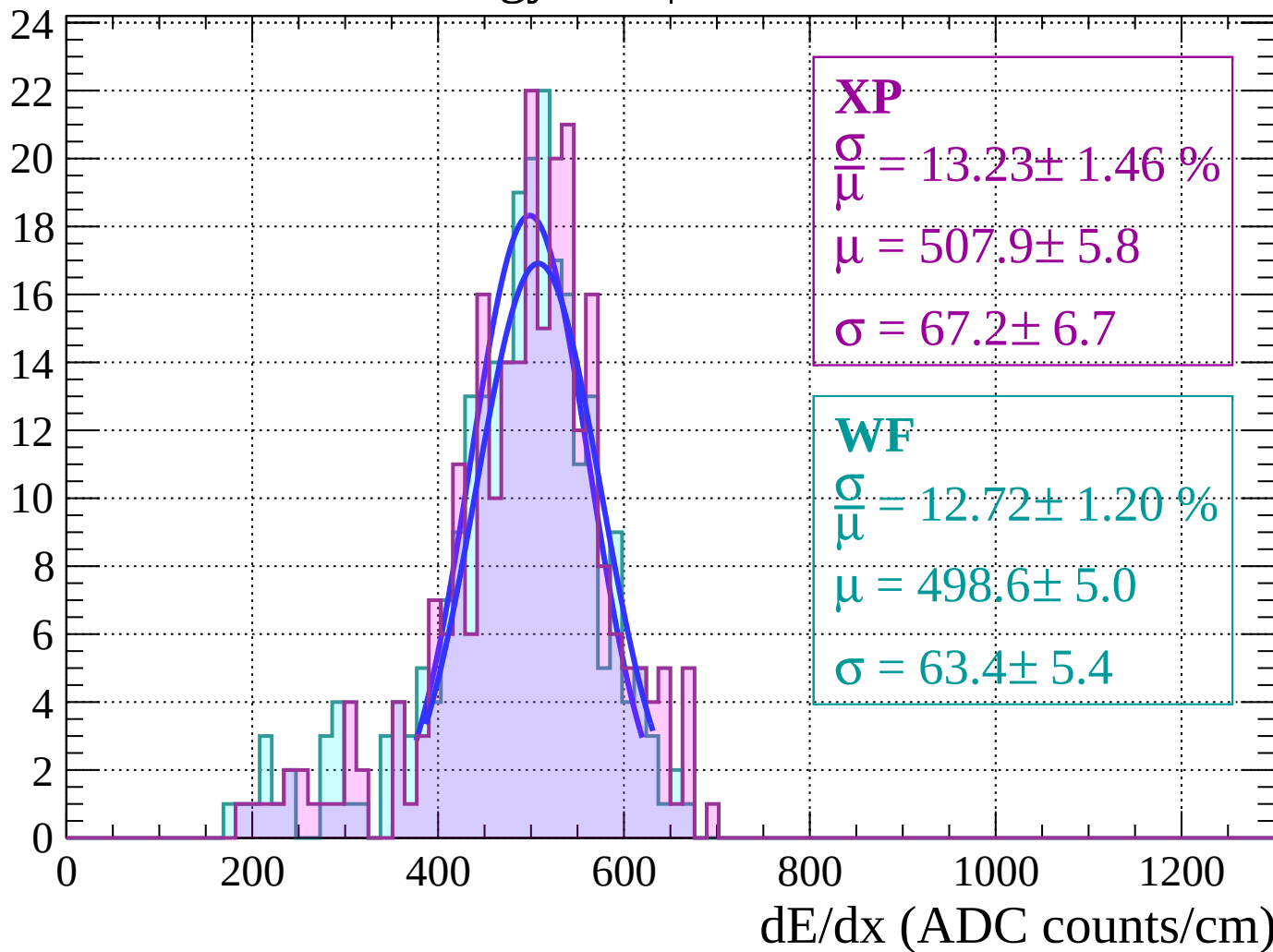
Energy loss | $60 < \theta < 62$

Count



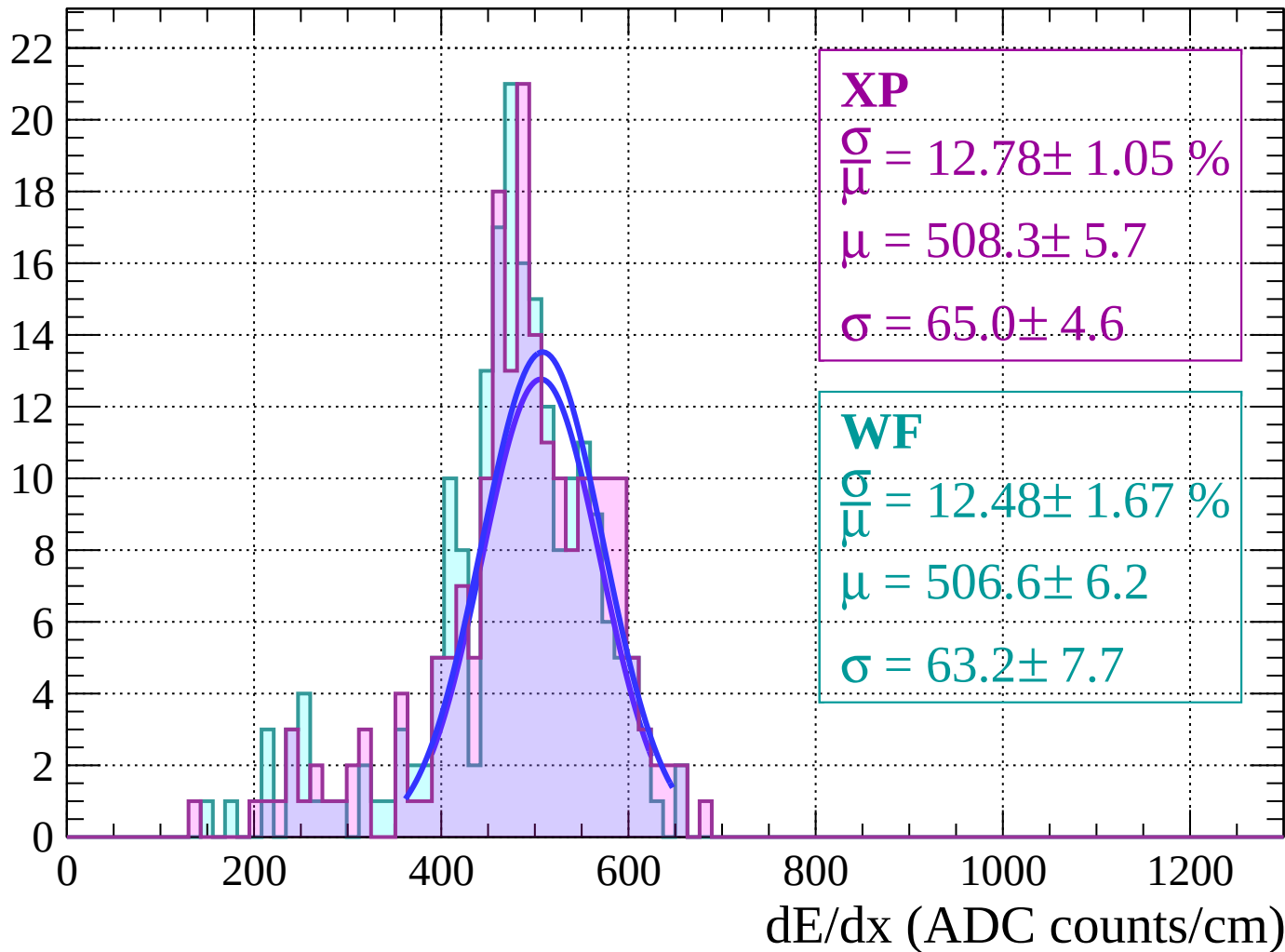
Energy loss | $62 < \theta < 64$

Count



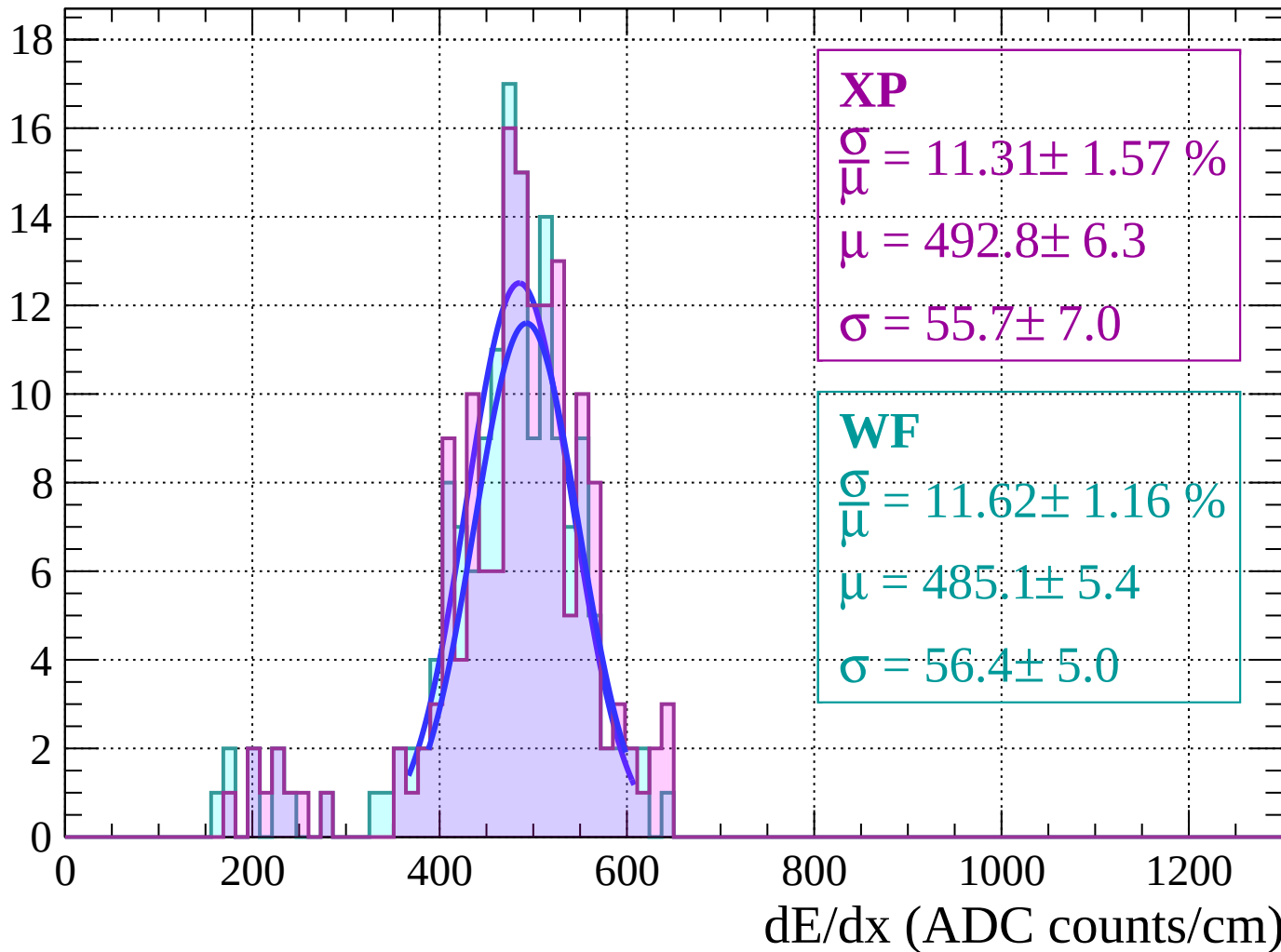
Energy loss | $64 < \theta < 66$

Count



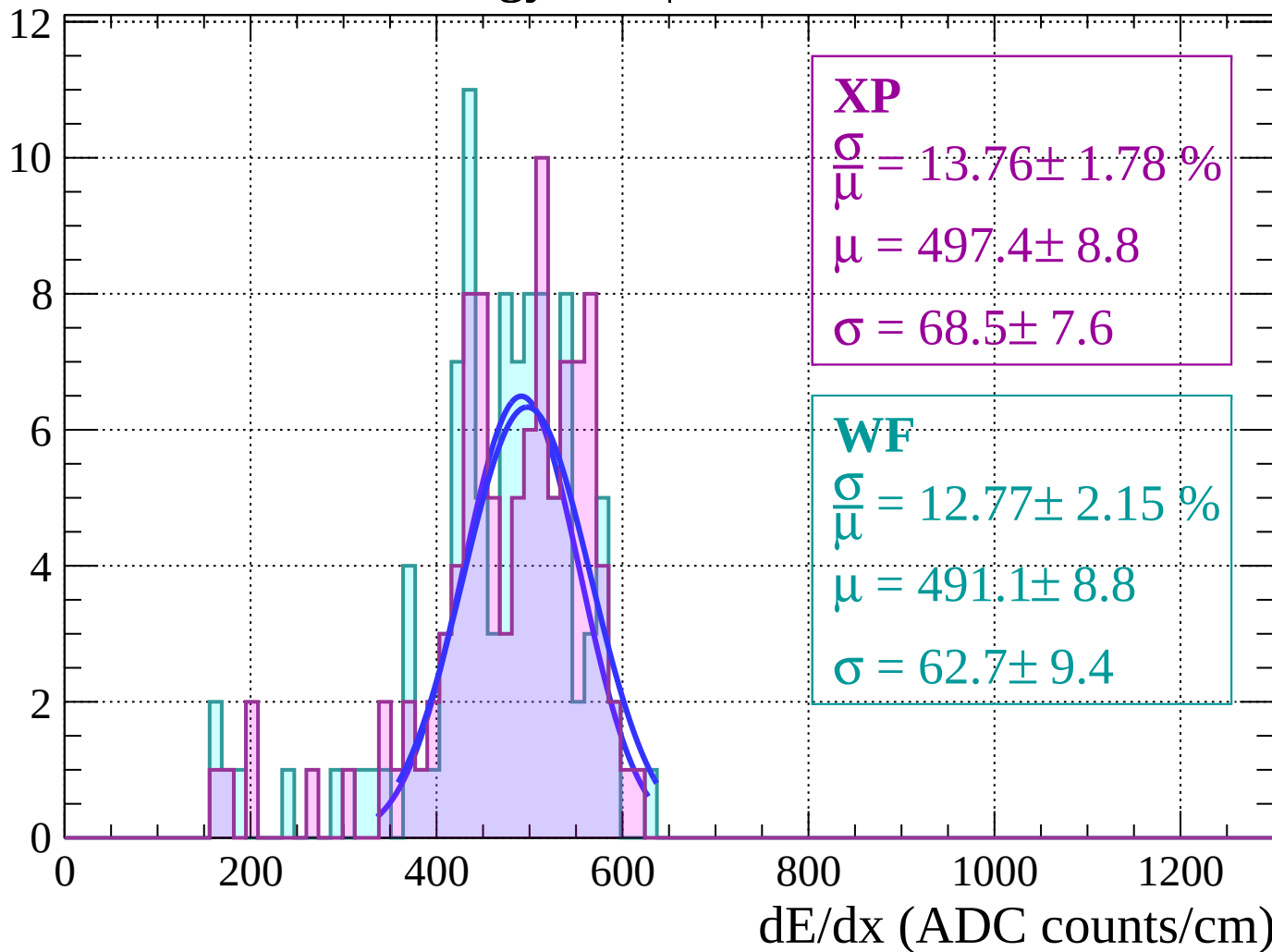
Energy loss | $66 < \theta < 68$

Count



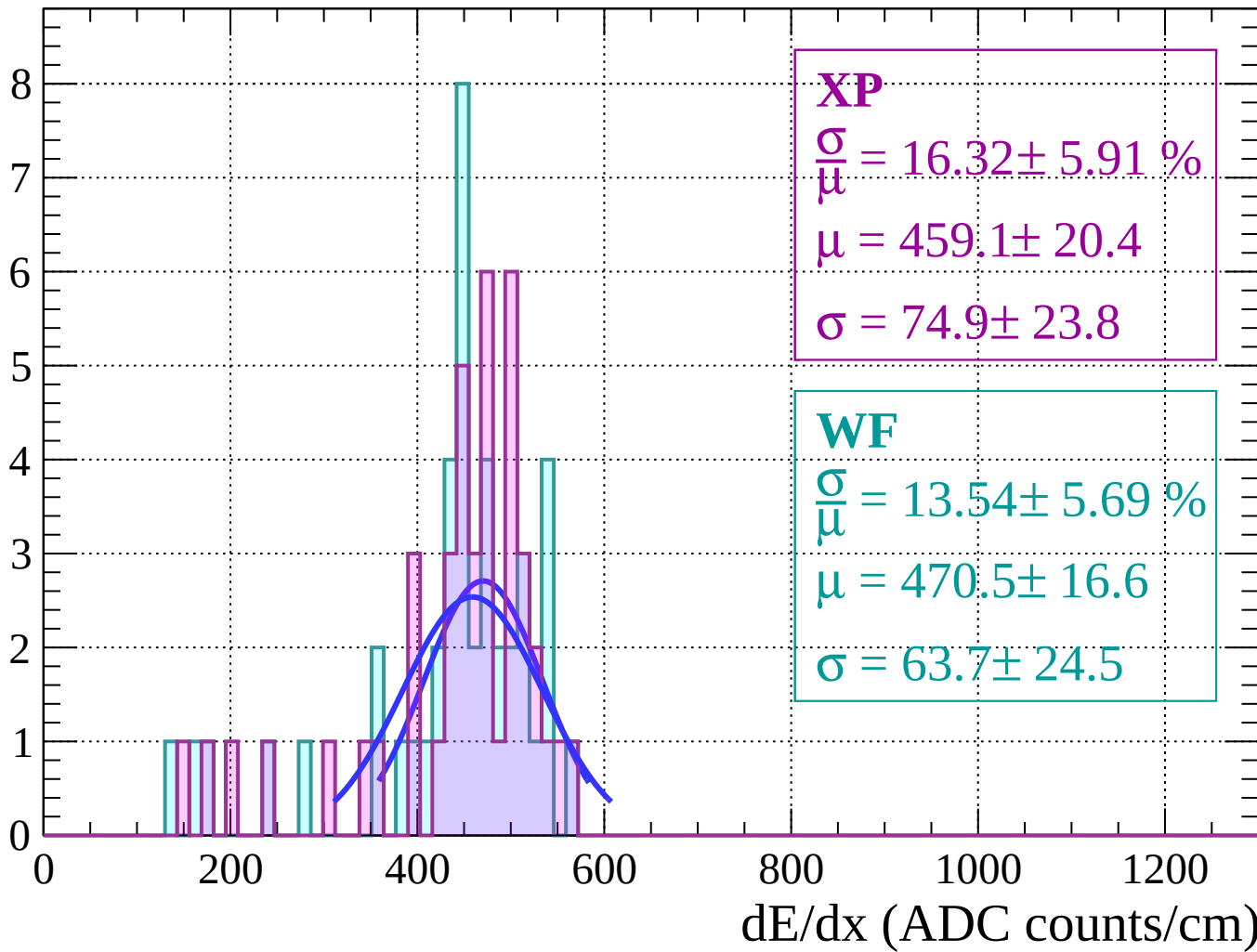
Energy loss | $68 < \theta < 70$

Count



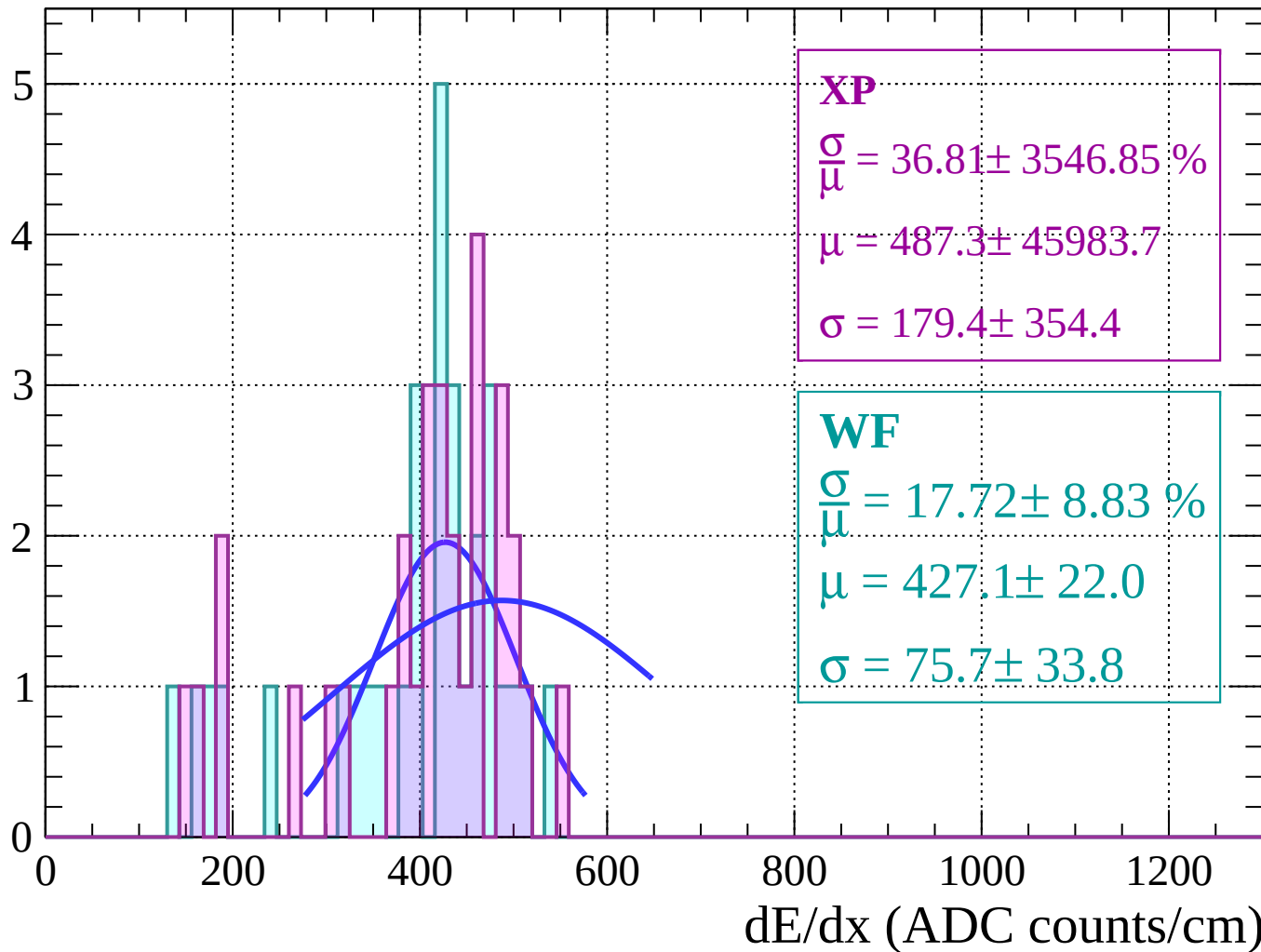
Energy loss | $70 < \theta < 72$

Count



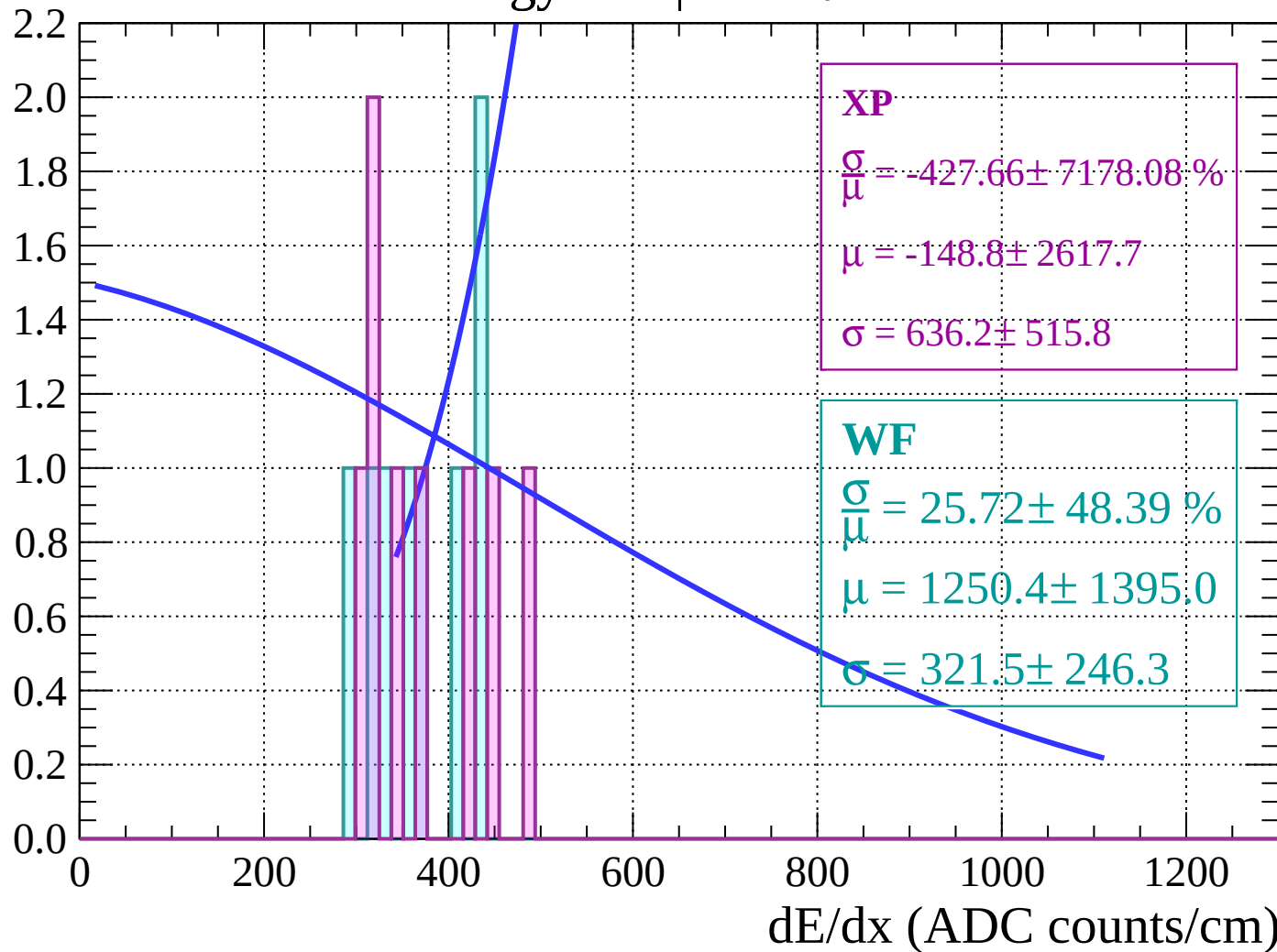
Energy loss | $72 < \theta < 74$

Count



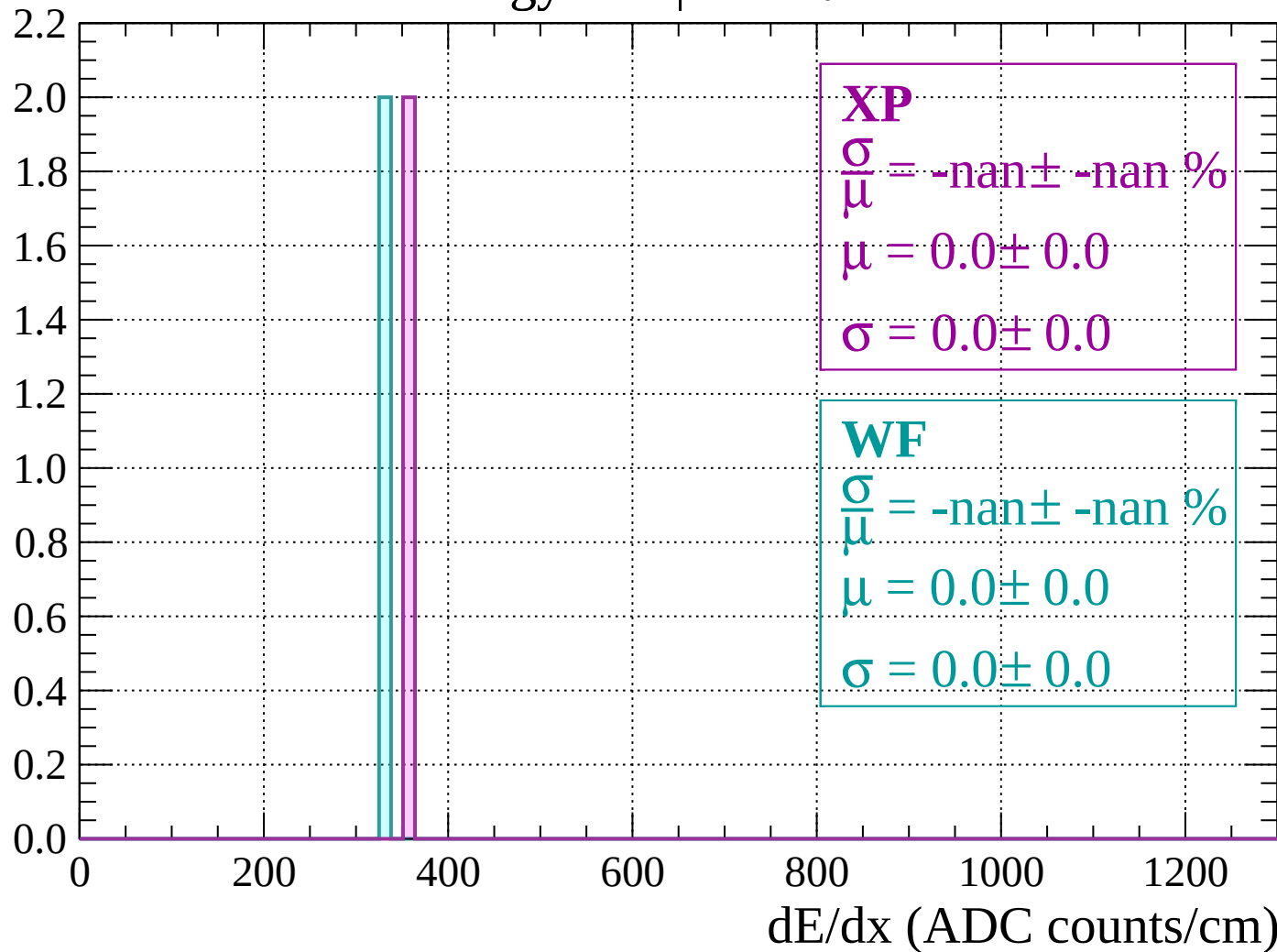
Energy loss | $74 < \theta < 76$

Count



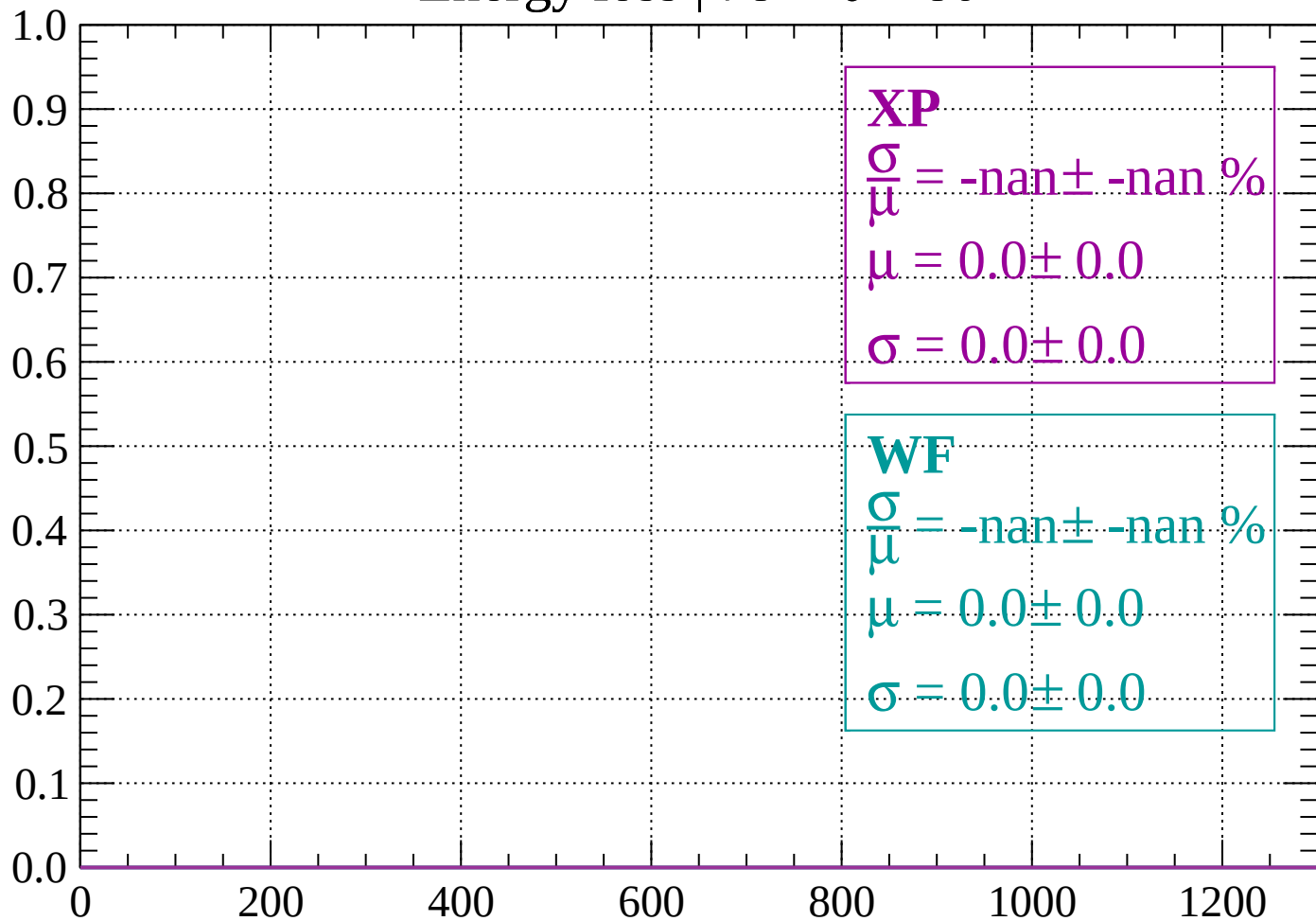
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

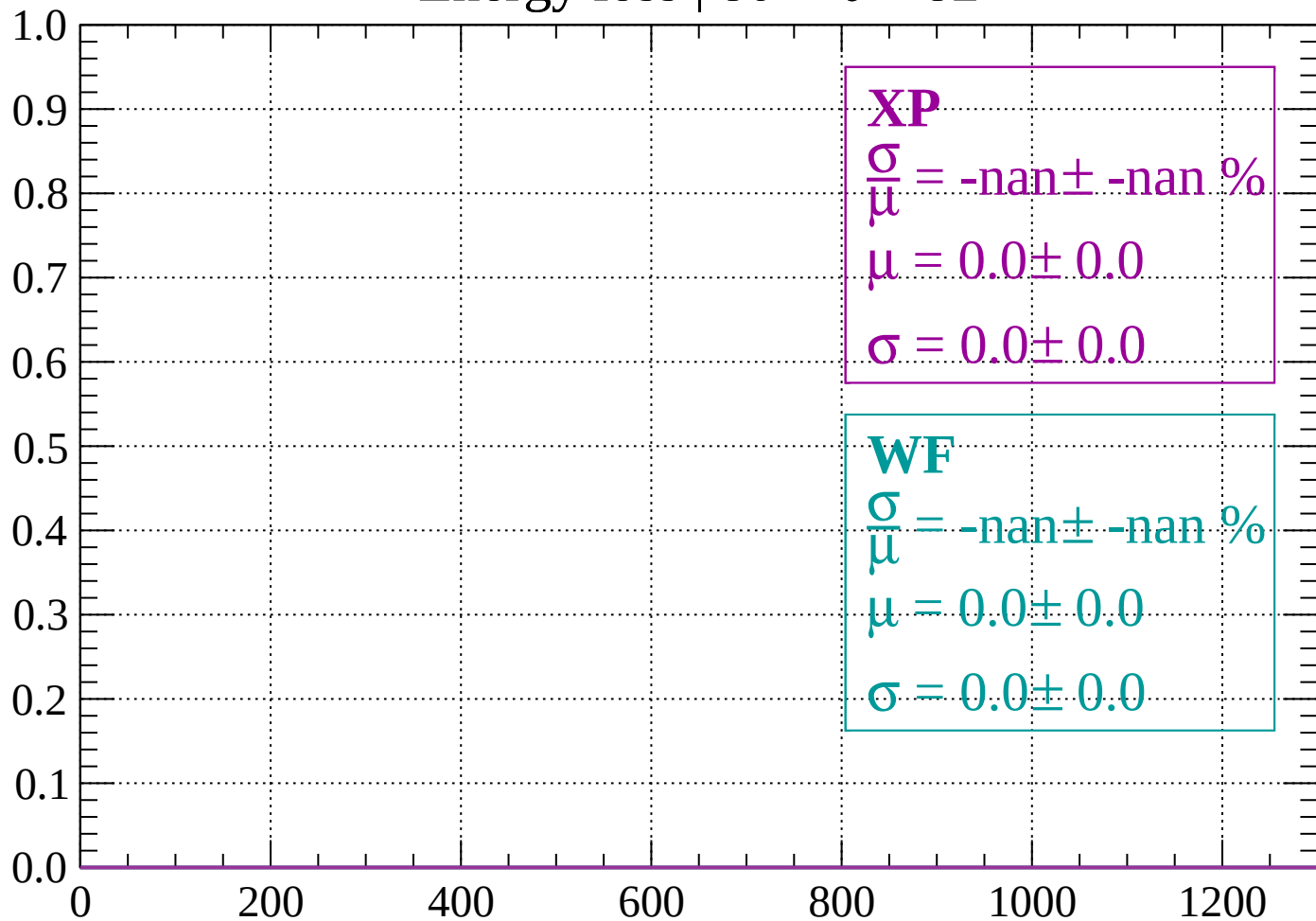
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

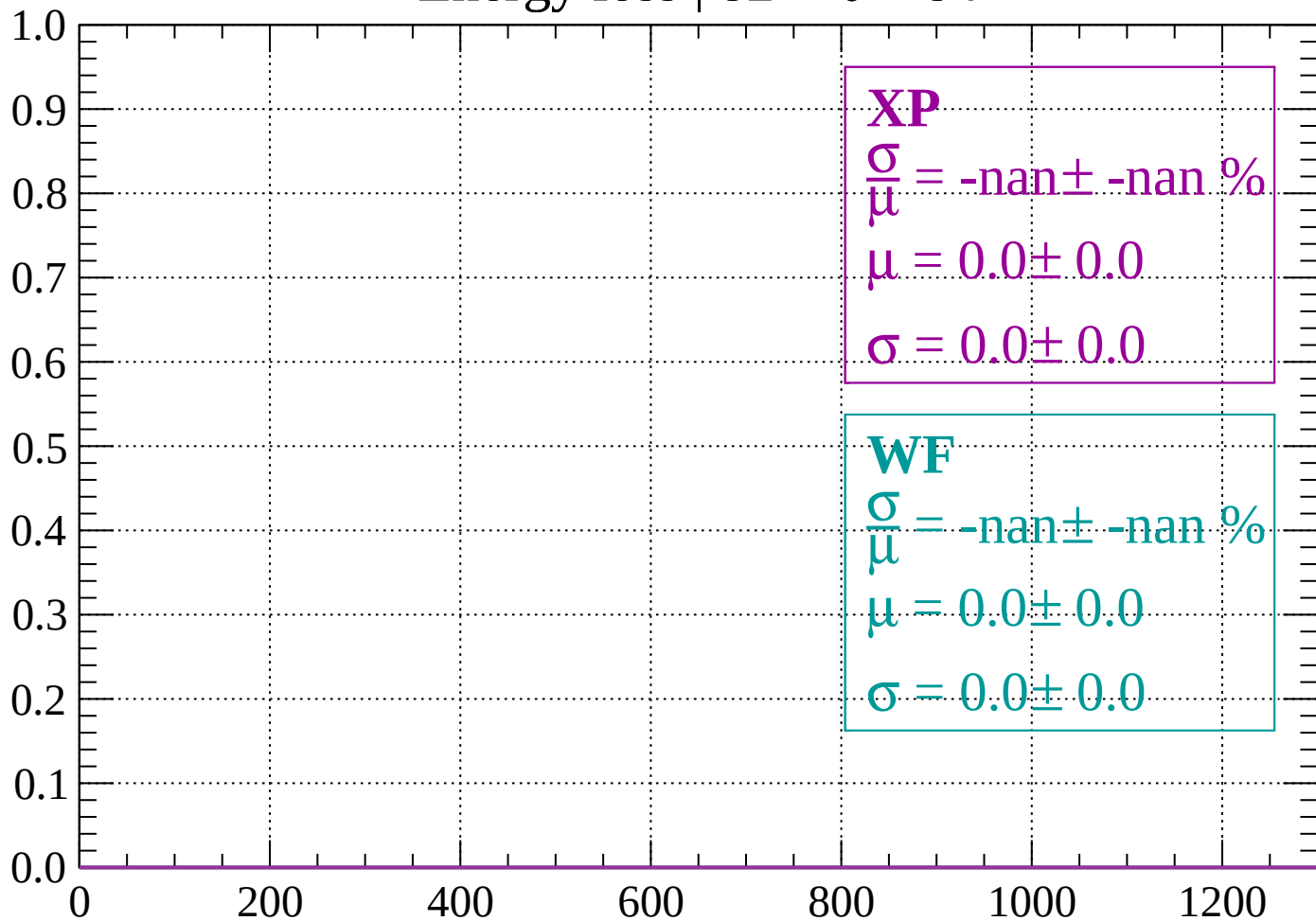
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

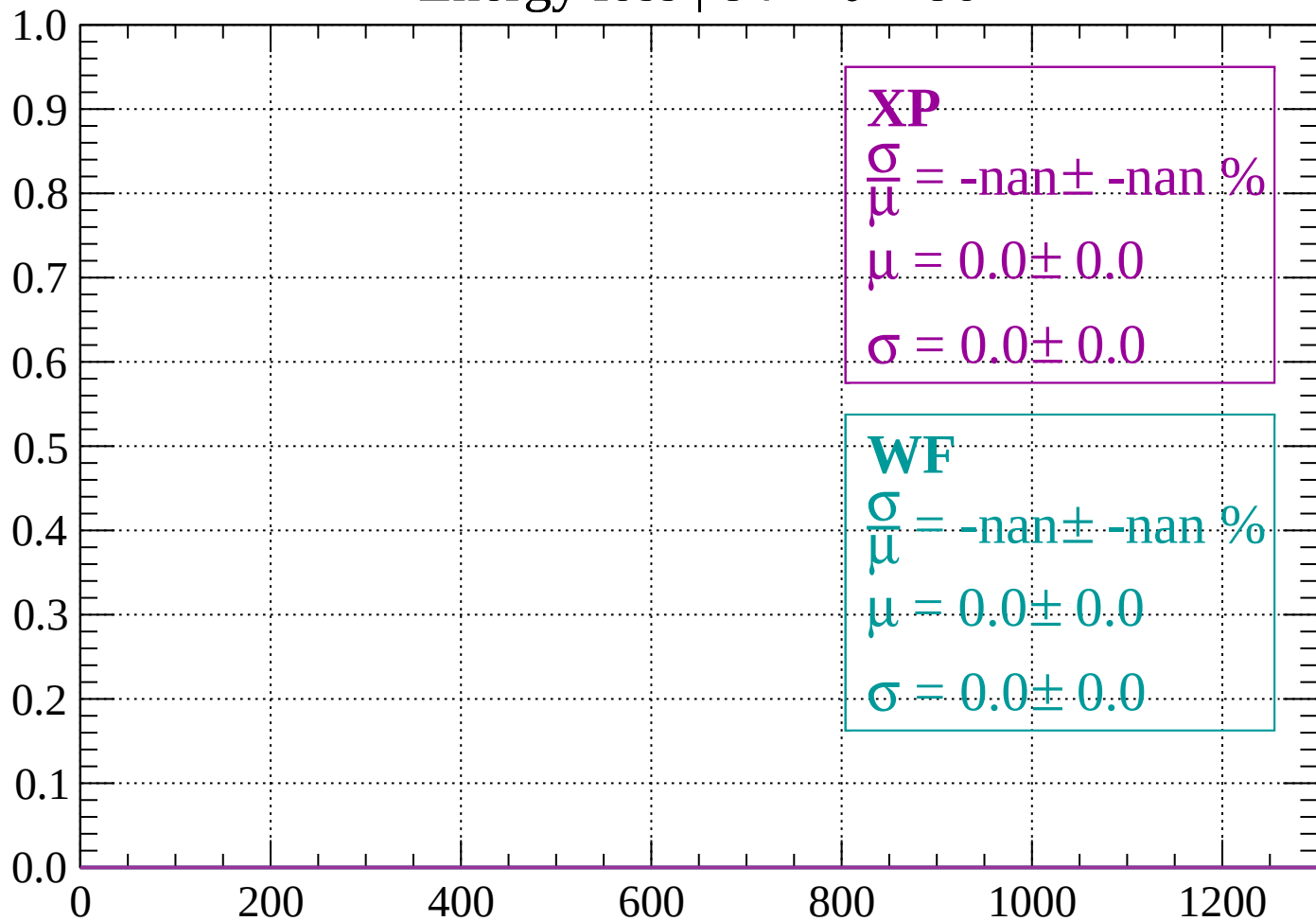
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

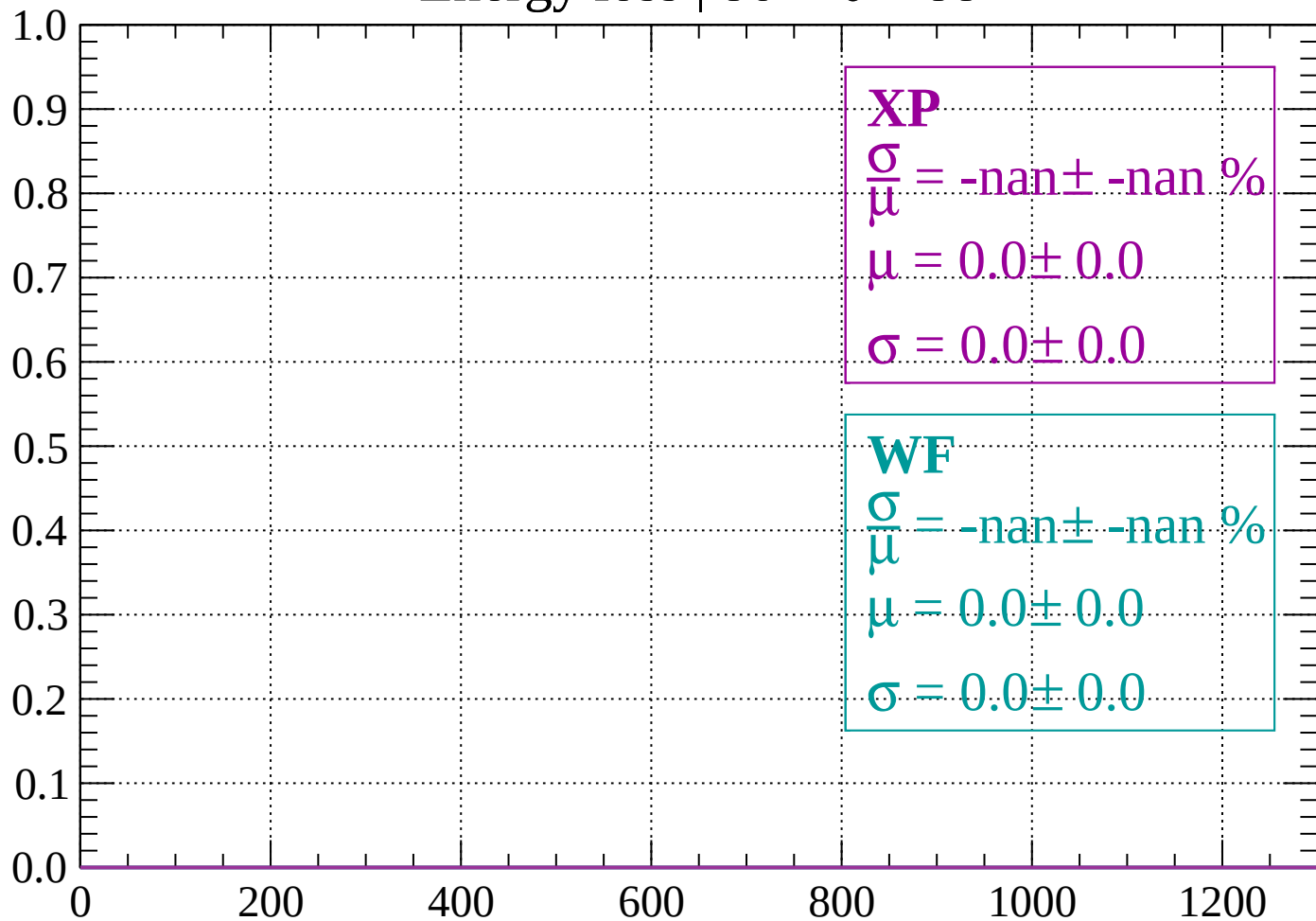
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

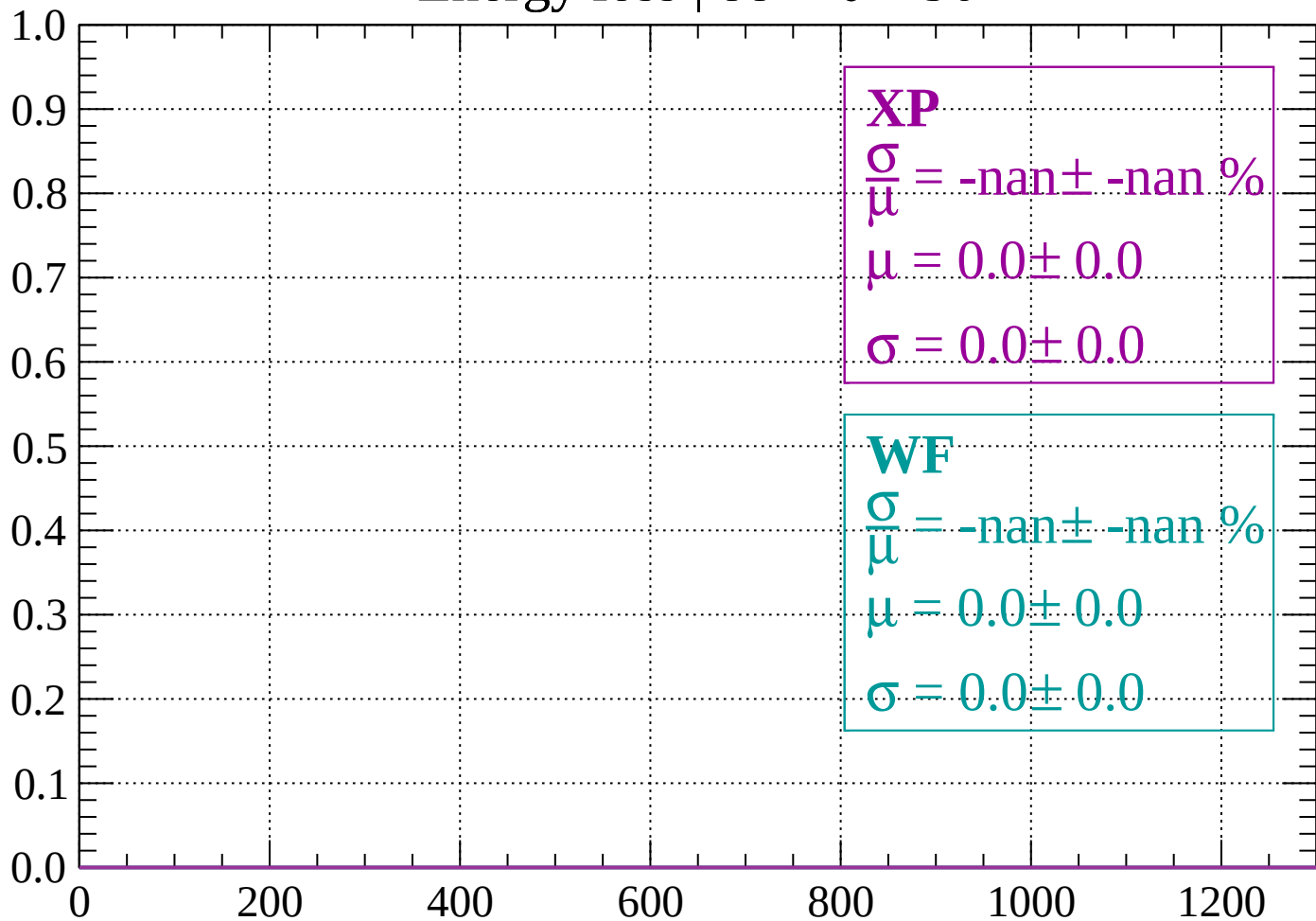
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

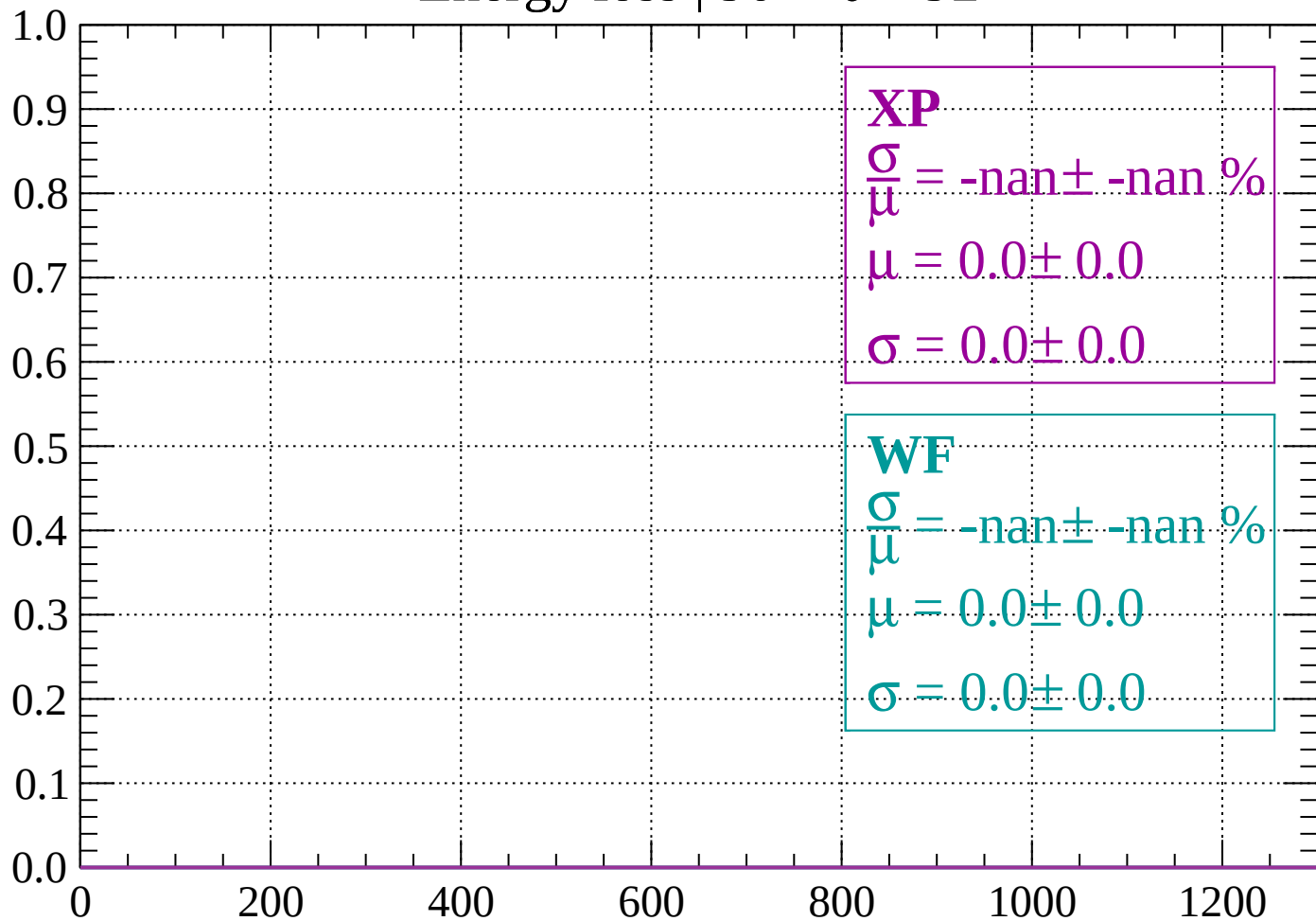
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

